BUCKET FOUNDATION · NUCLEUS BRAIN · PHYSICS BRANCH

The Quantum Atlas

A graded research-and-industry map of the whole field — the physics, the machines, the algorithms, every industry, the money, the history, and the honest frontier.

Built by Nucleus Brain (AI orchestrator) · AG Farms Venture Studio

Sources: arXiv · Nature/Science · NIST · vendor filings · the Bucket physics canon

Doctrine: index all · grade everything · a press release and a peer-reviewed result never weigh the same

184 graded source nodes · 8 chapters · 15 conflict objects · built 2026-07-10

Preface

What this is

The Quantum Atlas maps the whole of quantum — the physics that makes it possible, the machines built to exploit that physics, the algorithms that run on them, the adjacent technologies that ride alongside, and every industry now placing a bet. It is written to be two things at once: a map you can stand over and orient yourself with, and a textbook you can read from the postulates upward.

The field it covers is loud. Much of the noise comes from vendors with a valuation to defend, governments with a budget to justify, and analysts with a forecast to sell. So the atlas is built on one discipline that runs through every page: every claim is graded, and the grade travels with the claim. A century-old textbook result and a launch-day press release both appear here, and the atlas always says which is which. Where two credible sources disagree — "fault tolerance by 2029" against "not before 2040" — the disagreement is kept as a first-class object, written down in full, with a note on what experiment or event would settle it. The atlas never silently picks a side.

How to read it

There are two ways in. **To learn the field, read straight through** from Chapter 1 — the chapters are ordered so each earns the next, and nothing assumes you arrived knowing the vocabulary. **To use it as a reference, start with the Map** — `00-map/00-IDEAL-STATE-MAP.md` lays the entire territory on a single page, from the superposition postulate up through the machines and out into twenty-plus industries, and every entry is a node with a stable ID that keys back to the reference index at the back. When a term stops you, the glossary in Appendix D defines it in one or two plain sentences.

The book runs in eight chapters, in the order the physics builds:

1. **Foundations** — the twenty-six ideas that make any of this possible, from the state vector to entanglement to decoherence.
2. **Hardware** — the eight-or-nine competing bets on what a qubit should be made of, and the fridges, lasers, and foundries beneath them.
3. **The stack and algorithms** — how a noisy physical qubit becomes a useful answer: error correction, compilers, Shor and Grover, benchmarks.
4. **Adjacent technologies** — the quantum tech already commercial: key distribution, sensing, metrology, imaging.
5. **Industries** — where quantum actually lands in the economy, industry by industry, with the walls between "early" and "mature" drawn honestly.
6. **Ecosystem and geopolitics** — the money, the national programs, the standards bodies, the export controls, the talent race.
7. **History** — the full timeline, 1900 to today, told as the road that leads to the map.
8. **The honest frontier** — the open problems, named out loud, with who disagrees and what would resolve each one.

The appendices hold the machinery: the grading method, the conflict register, an evidence index of recent preprints, a glossary, and the node reference index that keys every citation anchor back to the map.

Read it front to back as a textbook and the chapters build on each other in sequence. Jump in as a map and each node stands on its own, cross-linked to the physics it depends on and the industry it serves.

How the grading works

Every claim carries an evidence tier, strongest to weakest. **T1** is established physics — textbook, reproduced for decades, like superposition or the Bell violation. **T2** is a peer-reviewed result, published and refereed. **T3** is a preprint or conference talk, real work not yet through review. **T4** is a vendor claim — a company announcing its own numbers, not independently checked. **T5** is an analyst forecast or a national-program dollar figure, the softest anchor in the book. **T6** is speculative: plausible, unproven, the room-temperature-qubit and killer-app timelines. The rule that does most of the work: a vendor announcing its own benchmark stays T4 until an independent group reproduces it, and never gets promoted to peer-reviewed T2 on the strength of a blog post. A press release and a *Nature* paper never weigh the same here. "Quantum advantage" claims are treated as contested by default and carry a note on whether a classical method later caught up.

A field kit — how to read a quantum headline

Two tools from later chapters, put here because they work on anything you read, in this book or out of it.

Four questions to ask of any quantum claim (Chapter 8): 1. Is this a demonstrated result or an announced one? A vendor's own number stays unverified until an independent group reproduces it. 2. Did a classical method later catch up? "Quantum advantage" is a moving target; several headline claims were matched on classical hardware within months. 3. What is the physical-to-logical qubit overhead? A useful computation needs error correction, and the honest all-in cost is often hundreds to a thousand physical qubits per logical one. 4. Is the timeline a roadmap or a result? A date on a slide is marketing until the machine hits the spec.

Two traps in the money numbers (Chapter 5): a *total-addressable-market* figure is a forecast, not booked revenue, and the same value gets double-counted across the shared chemistry-and-optimization pool. And *quantum-inspired* results run on classical computers — they borrow the vocabulary without the hardware.

Who it's for

The physicist can read Chapter 1 as a compact restatement of the foundations, then use the map to see how each postulate becomes load-bearing for a machine downstream — which physics a given qubit actually exploits, and where the open questions in measurement and interpretation still sit.

The builder and the investor can start at Chapters 2, 5, and 6, where the coordinates are qubit counts, fidelities, roadmaps, and budgets. The grading is the tool that matters most here: it separates the demonstrated from the announced, and it charges honestly for the classical counterattack before crediting any speedup.

The newcomer can read straight through. The chapters are ordered so that each one earns the next, and nothing assumes you arrived already knowing the vocabulary. The frontier chapter at the end is the honest reward — the field's real open problems, stated plainly.

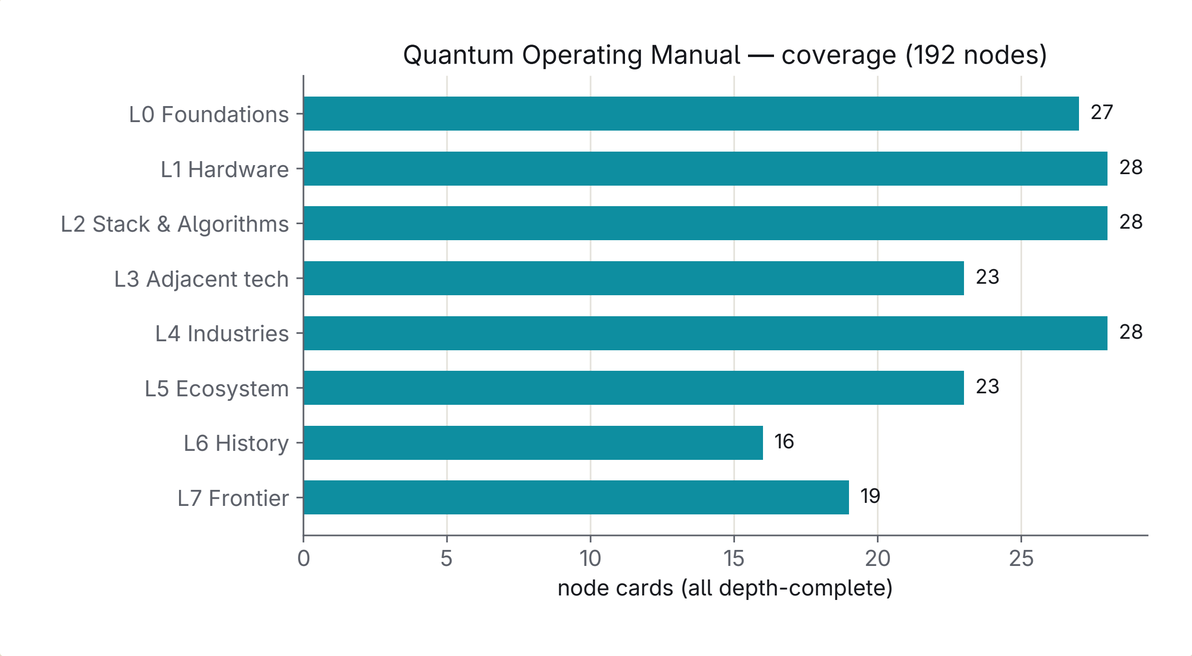
A note on honesty

The atlas tries to do one thing well: state what is settled, what is contested, and what is marketing, and always say which is which. Settled physics is marked settled. A live disagreement is written down as a disagreement, with both sides intact. A vendor's timeline is called a vendor's timeline. This is the whole method, and it is why the grades and the conflict register are load-bearing rather than decorative. The goal is a map you can trust because it never hides where it is unsure.

Contents

ORIENTATION

The Map



The territory — 184 source nodes across 8 layers.

The entire territory on one page — from the postulate that a system can be in two states at once, up through the machines that exploit it, and out into every industry that will be rebuilt on top of them.

It's a map, not a front-to-back read. Two things travel with every node: which chapter covers it (`§`), and how strong the evidence behind it is (graded per `evidence/SCHEMA.md`). A press release and a peer-reviewed threshold demo do not weigh the same, and the manual always says which is which.

Every node below is a coverage unit. `NETWORK-CAPACITY.md` counts how many are filled; `FRONTIER.md` is the live worklist. Node IDs are stable citation anchors.

Layer 0 — Foundations (the physics that makes any of this possible) · §01

NODE	WHAT	ID
Superposition & the state vector	A system holds a weighted combination of basis states until measured	`F-superpos`
Measurement & the Born rule	Probabilities as squared amplitudes; collapse / update	`F-measure`
Entanglement & nonlocality	Correlations with no classical explanation	`F-entangle`
Bell inequalities & their experimental violation	The test that ruled out local hidden variables (Aspect → 2022 Nobel)	`F-bell`
Decoherence & open quantum systems	Why the quantum world looks classical; the enemy of every qubit	`F-decoher`
The qubit & Bloch sphere	The unit of quantum information	`F-qubit`
No-cloning theorem	You cannot copy an unknown quantum state — the root of quantum crypto	`F-noclon-ing`

NODE	WHAT	ID
Quantum information theory	Von Neumann entropy, channels, fidelity, quantum capacity	`F-qinfo`
Uncertainty & complementarity	Conjugate observables; the Heisenberg bound	`F-uncertainty`
Interpretations	Copenhagen, many-worlds, Bohmian, QBism, relational, objective-collapse	`F-interp`
QFT / second-quantization bridge	Where quantum mechanics meets fields and particles	`F-qft`
Quantum thermodynamics	Work, heat, and information at the quantum scale	`F-qthermo`
Quantum tunneling	The physics inside every Josephson junction, flux qubit, STM	`F-tunneling`
Contextuality (Kochen-Specker)	The resource behind magic-state computation; bridges to QEC	`F-contextuality`
Identical particles & spin-statistics	Exchange symmetry — underlies quantum chemistry, fermionic sim, anyons	`F-statistics`
Adiabatic theorem	The foundation quantum annealing rests on	`F-adiabatic`
Teleportation & entanglement swapping	Foundational protocols under the quantum internet & modular QC	`F-teleport`
Geometric / Berry phase	Geometric gates, topological band theory, links to quantum materials	`F-berry`
Generalized measurement (POVMs)	Positive-operator measurements, Naimark dilation, weak measurement & instruments — the real measurement model	`F-povm`
Purification & Stinespring dilation	Every mixed state is a pure state, every channel a unitary, on a larger space — the church of the larger Hilbert space	`F-purification`
Lieb-Robinson bounds	An emergent finite speed limit / effective light cone in non-relativistic quantum systems	`F-liebrobinson`
PBR theorem	Pusey-Barrett-Rudolph: the reality status of the quantum state (ψ-ontic vs ψ-epistemic)	`F-pbr`
Wigner functions & quasiprobability	Phase-space QM; negativity as a nonclassicality / computation resource	`F-wigner`
Stabilizer formalism & Gottesman-Knill	Why a large class of quantum circuits is classically simulable; the backbone of QEC	`F-stabilizer`
Quantum discord	Quantum correlations that survive beyond entanglement	`F-discord`
Quantum entropy inequalities	Strong subadditivity, monogamy, data-processing — the bookkeeping laws of quantum information	`F-entropy-ineq`

Layer 1 — Hardware: the qubit modalities & the machines · §02

NODE	WHAT / WHO	ID
Superconducting (transmon)	IBM, Google, Rigetti, IQM, OQC, SEEQC	`H-supercon`
Trapped ions	IonQ, Quantinuum, Oxford Ionics, Alpine, eleQtron	`H-ion`

NODE	WHAT / WHO	ID
Photonic	PsiQuantum, Xanadu, QuiX, ORCA, TundraSystems	`H-photon-ic`
Neutral atoms	QuEra, Pasqal, Atom Computing, Infleqtion, planqc	`H-neutral`
Spin qubits in silicon	Intel, Diraq, Quantum Motion, SQC, Equal1	`H-silicon`
Topological / Majorana	Microsoft (Majorana 1), Nokia Bell	`H-topo`
Bosonic / cat qubits	Alice & Bob, AWS, Nord Quantique	`H-bosonic`
Quantum annealing	D-Wave	`H-anneal`
NV centers / diamond	Sensing + compute; Quantum Brilliance	`H-nv`
Cryogenics & dilution refrigerators	Bluefors, Oxford Instruments, Maybell	`H-cryo`
Control electronics & I/O	Qblox, Quantum Machines, Zurich Instruments, Keysight	`H-control`
Fabrication & supply chain	Foundries, wafers, cabling, rare materials, TLS defects	`H-fab`
Interconnects & modular QC	Chip-to-chip, photonic links, networked processors	`H-inter-con`
Microwave ↔ optical transduction	Bridging mK microwave qubits to optical fiber; gates fridge-to-fridge modularity	`H-trans-duce`
Single-photon detectors / SNSPDs	Photon counting for photonic QC & QKD; Single Quantum, ID Quantique, Photon Spot	`H-detect`
Cryogenic control ASICs	Cryo-CMOS control/readout in the fridge; Intel Horse Ridge, SemiQon, Equal1	`H-cryoc-mos`
Laser & photonics subsystems	Stabilized lasers/PICs for ion/atom/NV machines; TOPTICA, M Squared, Vescent	`H-lasers`
Deterministic photon sources	Single-photon / entangled-pair emitters; Quandela quantum dots, SPDC, Sparrow	`H-photon-source`
Optical frequency combs	Shared ruler-for-light under clocks/timing/atomic-machine lasers; microcombs, NIST	`H-frequen-cycomb`
Quantum-limited amplifiers	JPA/TWPA at mK + HEMT at 4 K for single-shot readout; Silent Waves, Low Noise Factory	`H-paramp`
Packaging & 3D integration	Flip-chip, TSVs, interposers, chiplets; MIT-LL, IBM, Google, QuantWare	`H-package`
Ion-trap chip fabrication	Microfabricated surface/QCCD traps; Sandia HOA/Enchilada, Honeywell, Infineon	`H-iontrap`
UHV & atomic-source systems	Ultra-high-vacuum + cryogenic chambers for ion/atom machines; sets atom-loss floor	`H-uhv`
300 mm foundry ecosystem	Who manufactures qubits at wafer scale; imec, GlobalFoundries, Intel, SkyWater	`H-foundry`
Quantum-dot vs donor spin fork	The silicon-spin strategic split; SQC (donor) vs Diraq/imec/Intel (dots)	`H-spin-split`
		`H-wiring`

NODE	WHAT / WHO	ID
Cryogenic microwave wiring	Passive coax / attenuators / ferrite isolators from 300 K to mK; the line-count scaling wall (split from H-cryo/H-fab)	
Spin-photon interfaces	Emitters entangling a matter spin with a flying photon; silicon T-centers, diamond SiV/SnV, erbium — the matter-qubit networking node	`H-spin-photon`

Layer 2 — The stack: from a noisy qubit to a useful answer · §03

NODE	WHAT	ID
Gates & circuits	Universal gate sets, native gates, connectivity	`S-gates`
Noise, NISQ & error mitigation	Where the field is today; ZNE, PEC	`S-nisq`
Quantum error correction	Surface code, qLDPC, magic-state distillation, threshold theorem	`S-qec`
Logical qubits & fault tolerance	The line between demos and utility	`S-logical`
Algorithms — Shor	Factoring; the reason cryptography must migrate	`S-shor`
Algorithms — Grover	Unstructured search, quadratic speedup	`S-grover`
Algorithms — VQE / QAOA	Near-term variational chemistry & optimization	`S-variational`
Algorithms — HHL & quantum linear algebra	Speedups (and the fine print)	`S-hhl`
Quantum simulation	Simulating quantum systems — Feynman's original pitch	`S-qsim`
Quantum machine learning	QML, kernels, hybrid; the hype-heavy corner	`S-qml`
Compilers, transpilers, middleware	Qiskit, Cirq, PennyLane, tket, Braket SDK	`S-software`
Benchmarks & "advantage" claims	Quantum volume, CLOPS, algorithmic qubits, supremacy/advantage history	`S-bench`
Cloud QPU access	IBM Quantum, AWS Braket, Azure Quantum, Google	`S-cloud`
QFT & phase estimation	The primitive under Shor/HHL/simulation; QPE depth is a FT problem	`S-qft`
Quantum walks & element distinctness	O(N^{2/3}) distinctness, BQP-universal walks, algorithmic framework	`S-walk`
Quantum Monte Carlo / amplitude estimation	Quadratic speedup for expectation values; feeds finance	`S-qmc`
Real-time QEC decoding	Decoder as its own discipline — Riverlane, ASIC/FPGA, the throughput wall	`S-decoders`
Quantum RAM (qRAM)	The load-bearing, largely-unproven data-loading assumption under HHL/QML	`S-qram`
Certified randomness	First delivered advantage application — JPMorgan/Quantinuum, Nature 2025	`S-certrand`
QSVT — quantum singular value transformation	The modern unifying lens: HHL/Grover/Hamiltonian-sim as one framework (Gilyén et al. 2019)	`S-qsvt`
Hamiltonian simulation methods	Trotter / qDRIFT / LCU / qubitization — the how behind digital quantum simulation	`S-hamsim`

NODE	WHAT	ID
Tensor-network classical simulation	The classical counterfactual that decides every advantage claim (MPS, contraction)	`S-tensor-net`
Classical shadows	Predict many properties from few measurements; size-independent sample complexity	`S-shadows`
Error mitigation as a discipline	ZNE/PEC/symmetry/virtual-distillation taxonomy; exponential sampling ceiling	`S-errmit`
Circuit knitting & cutting	Run big circuits on small QPUs via quasiprobability; exponential cut overhead	`S-circuitcut`
Pulse-level & optimal control	GRAPE/CRAB/Krotov/DRAG — where a gate becomes an analog waveform	`S-optcontrol`
Quantum advantage complexity theory	BQP/BPP, oracle & sampling separations — what "advantage" formally means	`S-complexity`

Layer 3 — Adjacent quantum technologies (not compute) · §04

NODE	WHAT	ID
Quantum key distribution (QKD)	Provably secure key exchange; BB84, E91	`A-qkd`
Quantum internet & repeaters	Entanglement distribution, memories, the future network	`A-qinternet`
Satellite QKD	Micius, space-based entanglement	`A-satqkd`
Post-quantum cryptography (PQC)	NIST standards (Kyber/ML-KEM, Dilithium), migration, HNDL threat	`A-pqc`
Quantum sensing & metrology	The nearest-term commercial quantum tech	`A-sensing`
Atomic clocks & timing	Optical clocks, PNT, GPS-free timing	`A-clocks`
Magnetometry	SQUID, OPM, brain imaging (MEG), mineral exploration	`A-magneto`
Gravimetry & inertial nav	Cold-atom gravimeters, GPS-free navigation	`A-gravimetry`
Quantum imaging & radar	Ghost imaging, quantum illumination, LiDAR	`A-imaging`
Quantum materials	Topological insulators, high-Tc superconductors, 2D materials	`A-materials`
Quantum RNG	Certified randomness from quantum sources	`A-qrng`
Quantum memories (component industry)	Rack-mounted/room-temp memories for repeaters; Qunnect, Aliro	`A-qmemory-hw`
Squeezed light (cross-cutting resource)	Sub-shot-noise light; LIGO production use, Xanadu CV/GKP compute	`A-squeezed`
Quantum/optical time-transfer networks	Comparing distant clocks over fiber/free-space; supports SI-second redefinition	`A-timedist`
NV-diamond sensing	Room-temp atom-scale sensing; Element Six/Bosch, SBQuantum, QDTI	`A-nvsensing`
		`A-pnt`

NODE	WHAT	ID
Assured PNT / GPS-denied navigation	System-level fusion of clock + cold-atom IMU + magnetometry; Q-CTRL, SandboxAQ, DARPA RoQS/PINS	
Rydberg RF electrometry / quantum antennas	Atom-based, self-calibrated RF receivers; Rydberg Technologies, Infleqtion	`A-rydberg`
Atom interferometry (platform)	Matter-wave interferometry under gravimetry/IMU + fundamental physics; MAGIS-100, AION	`A-atominterf`
Entanglement-based clock networks	GHZ-entangled clocks toward Heisenberg-limited timekeeping (contested advantage)	`A-entclock`
Quantum-secured / -resistant blockchain	PQC signatures on-chain (Algorand Falcon, QRL) + QKD-secured ledgers (niche)	`A-qblockchain`
Quantum-enhanced MRI/NMR & biomagnetism	Hyperpolarization (NVision) + NV/atomic magnet-free NMR + OPM MEG/MCG	`A-qmri`
Quantum-dot displays (boundary case)	QLED as quantum *material* not quantum tech; the QD single-photon source is the real thread	`A-qdisplay`

Layer 4 — Industries: where quantum actually lands · §05

INDUSTRY	USE CASES · ANCHOR PLAYERS	ID
Finance	Portfolio optimization, Monte Carlo pricing, risk, fraud — JPMorgan, Goldman, HSBC, Wells Fargo	`I-finance`
Pharma & healthcare	Drug discovery, protein folding, molecular sim — Boehringer, Roche, Cleveland Clinic, Moderna	`I-pharma`
Chemicals & materials	Catalysts, batteries, nitrogen fixation, carbon capture — BASF, Dow, Mitsubishi Chemical	`I-chem`
Energy & utilities	Grid optimization, battery chemistry, fusion, oil & gas — ExxonMobil, E.ON, EDF	`I-energy`
Logistics & supply chain	Routing, scheduling, traffic — DHL, Volkswagen, Airbus, DB	`I-logistics`
Automotive	Materials, batteries, aerodynamics, autonomous — BMW, Mercedes, Ford, Hyundai	`I-auto`
Aerospace & defense	Sensing, nav, sim, codebreaking — Airbus, Boeing, Lockheed, DARPA, DoD	`I-aerospace`
Telecom & networking	Quantum-secure networks, optimization — BT, Verizon, SK Telecom, NTT	`I-telecom`
Cybersecurity	PQC migration, "harvest now decrypt later," QKD networks	`I-cyber`
AI & machine learning	Quantum ML, hybrid pipelines, sampling	`I-ai`
Climate & sustainability	Carbon capture, fertilizer (Haber-Bosch) chemistry, emissions optimization	`I-climate`
Insurance & risk	Catastrophe modeling, actuarial optimization	`I-insurance`
Manufacturing & industrials	Process optimization, digital twins, quality	`I-manufacturing`
Oil & gas	Seismic inversion, reservoir sim, LNG logistics, materials	`I-extractive`
Space & Earth observation	Orbit optimization, quantum sensors on satellites	`I-space`

INDUSTRY	USE CASES · ANCHOR PLAYERS	ID
Government services	Tax/fraud detection, customs, census, public-sector optimization (distinct from defense)	I-gov
Agriculture & food science	Precision ag, fertilizer/nitrogen-fixation chemistry, agri sensing	I-agri
Retail	Demand forecasting, pricing, assortment/shelf optimization, recommendation	I-retail
Media & entertainment	Procedural generation, rendering, QRNG content, ad/recommendation optimization	I-media
Construction & built environment	Structural materials sim, project scheduling, urban microclimate	I-construction
Air-traffic management	Traffic-flow optimization, trajectory deconfliction, gate scheduling	I-atm
Semiconductors & EDA	Device-materials sim, computational lithography, quantum-chip EDA; the fab base	I-semiconductor
Weather & climate modeling	NWP / fluid-dynamics PDEs, quantum linear-solvers, QML nowcasting (distinct from I-climate chemistry)	I-weather
Nuclear & fusion	Plasma dynamics + stability (QAOA), nuclear-structure sim, reactor/fuel optimization	I-nuclear
Healthcare imaging & diagnostics	OPM-MEG brain imaging, NV/hyperpolarized MRI contrast, quantum-enhanced imaging (sensing, near-term)	I-healthimaging
Mining & mineral exploration	Diamond-magnetometer + cold-atom gravimeter critical-minerals exploration; mine-plan optimization	I-mining
Intelligence & cryptanalysis	Offensive SIGINT: Shor-driven decryption, harvest-now-decrypt-later, intercept analytics	I-intelligence

Layer 5 — Ecosystem, money & geopolitics · §06

NODE	WHAT	ID
National programs — US	National Quantum Initiative, NQI reauthorization, DOE, NSF, NIST	E-us
National programs — China	Reported ~$15B+ commitment, USTC, Micius, Jiuzhang	E-china
National programs — EU	Quantum Flagship (€1B+), EuroQCI, per-member strategies	E-eu
National programs — UK	National Quantum Strategy (£2.5B), NQCC	E-uk
National programs — others	India, Japan, Australia, Canada, Israel, South Korea, Germany	E-others
Private investment & VC	Funding rounds, SPACs, market size, the 2024–26 capital cycle	E-vc
Standards bodies	NIST, ISO/IEC JTC1, ETSI, IEEE, ITU	E-standards
Talent & workforce	The skills gap, university programs, quantum education	E-talent
Market forecasts	BCG, McKinsey, IDC, Gartner hype cycle — with skepticism	E-market
Patents & IP landscape	Who owns what; national patent races	E-patents
Export controls	US 2024 quantum export rule + allied plurilateral coordination (NL, JP, UK, FR)	E-export
Supply-chain chokepoints	Dilution fridges/He-3, quantum-grade lasers, specialty fab (geopolitics vs H-fab technical)	E-supply-chain

NODE	WHAT	ID
Open-source ecosystem	Qiskit/Cirq/PennyLane governance as soft power / lock-in; OpenQASM/QIR portability	`E-oss`
Defense funding channel	DARPA QBI (quantum benchmarking) + defense procurement, distinct from civilian NQI	`E-defense`
China's parallel software stack	Origin/QPanda, Baidu Paddle Quantum, Huawei HiQ as a fork / soft-power risk vs Qiskit/Cirq	`E-china-stack`
Consolidation wave (M&A / IPO)	IonQ roll-ups, D-Wave/Xanadu/IQM/Quantinuum exits 2025–26; distinct from primary VC	`E-mna`
Quantum-safe migration market	PQC migration as its own economic market (HNDL-driven), distinct from A-pqc crypto + E-market	`E-pqcmarket`
Insurance & liability for quantum risk	Cyber/D&O underwriting of the quantum threat; mirror of I-insurance	`E-insurance`
Sovereign compute & national champions	Gov-hosted machines (EuroHPC six-site fleet) + equity stakes; on-prem sovereignty	`E-sovereign`
Metrology & SI-redefinition governance	BIPM/CIPM/CGPM, second-redefinition to optical clocks (CGPM 2026), Kibble balance	`E-metrology-gov`
Workforce immigration & clearance	Visa/deemed-export/clearance filter on who can fill the talent gap; distinct from E-talent	`E-immigration`
Academic-lab leaderboard	USTC/MIT/Harvard/Delft/Oxford research base; the industry map one generation early	`E-labs`

Layer 6 — History: the full timeline · §07

ERA	MILESTONES	ID
Old quantum theory (1900–1925)	Planck, Einstein photoelectric, Bohr atom, de Broglie	`T-old`
The formalism (1925–1935)	Heisenberg matrix mechanics, Schrödinger, Dirac, EPR paradox	`T-formalism`
Foundations era (1935–1980)	Bell's theorem (1964), Bell tests, Aspect experiments	`T-foundations`
Birth of quantum computing (1980–1994)	Manin, Feynman (1981), Deutsch, Shor's algorithm (1994)	`T-birth`
Early experiments (1995–2010)	Cirac-Zoller, first gates, NMR Shor demo, DiVincenzo criteria	`T-early`
The engineering race (2011–2019)	D-Wave, IBM/Google scale-up, Google "supremacy" (2019)	`T-race`
The error-correction era (2020–now)	Logical qubits, QuEra 48-logical, Google Willow, utility-scale claims	`T-ecera`
Loophole-free Bell tests (2015)	Delft/NIST/Vienna close detection + locality together — the foundations capstone	`T-bell-tests`
The NISQ era (2018)	Preskill's coinage — the vocabulary of the noisy near-term machine	`T-nisq`
1927 Solvay Conference & Bohr–Einstein debates	The interpretation argument goes public; entanglement's opening act	`T-solvay`
EPR paradox & Bohr's reply (1935)	The field's most fertile question — locality vs. completeness — as its own milestone	`T-epr`

ERA	MILESTONES	ID
Birth of quantum information theory (1970–84)	Wiesner conjugate coding → Holevo bound → no-cloning → BB84	`T-qin-fobirth`
"Second quantum revolution" framing (2003)	Dowling–Milburn coinage — passive understanding → active single-system engineering	`T-2ndrev`
Nobel Prizes as the quantum spine (1918→2025)	The establishment's ratification timeline; Planck to Clarke/Devoret/Martinis	`T-nobel`
The "quantum winter" question	AI-winter analogy — the funding-downturn / unmet-promise risk, graded honestly	`T-winter`

Layer 7 — The honest frontier & open problems · §08

NODE	WHAT'S UNSETTLED	ID
Fault-tolerant scaling	Millions of physical qubits per useful computation — how far off, really	`O-scaling`
Error-correction overhead	The brutal physical:logical ratio	`O-overhead`
Decoherence & materials defects	Two-level systems, the materials science bottleneck	`O-materials`
Quantum advantage in practice	Which claimed speedups survive classical counterattack	`O-advantage`
Hype vs reality	Vendor timelines, "quantum-washing," the valuation question	`O-hype`
Room-temperature & photonic scaling	The paths that would change the economics	`O-roomtemp`
The killer app question	Is there a near-term application that pays for itself	`O-killerapp`
Interpretational & foundational openness	Measurement problem still unsolved	`O-foundations`
Real-time QEC decoding throughput	The classical co-processor bottleneck (~1 MHz/logical qubit); Riverlane	`O-decoder`
Chip-to-chip entanglement fidelity	Whether modular interconnects stay below threshold across module boundaries	`O-interconnect-loss`
Verifying a too-large computation	Certifying a result no classical machine can re-check (Mahadev)	`O-verification`
Funding-winter & talent attrition	Whether a capital downturn drains the field before roadmaps mature	`O-talent-attrition`
The classical-simulation frontier	Where the quantum-classical boundary actually sits vs the best classical method (tensor nets, belief propagation)	`O-classical-sim`
Benchmarking standardization crisis	No agreed cross-vendor yardstick; QV vs #AQ vs "logical qubits"; spoofable metrics	`O-benchmark-standard`
Business-model & ROI question	Who pays, at what margin — unit economics distinct from technical feasibility	`O-roi-business`
Energy consumption vs classical	Fixed cryo/control overhead vs joules-per-useful-answer; energetic advantage before computational	`O-energy`
The "quantum utility" definitional debate	Whether "utility" is a real milestone or a rhetorical hedge (IBM 2023 → classically simulated)	`O-utility-definition`

NODE	WHAT'S UNSETTLED	ID
Cryptographically-relevant QC timeline	When Shor breaks RSA-2048; HNDL threat; Mosca's inequality (distinct from general FTQC)	`O-crqc-timeline`

The honest edge of the map

Named out loud so nothing hides: (1) vendor roadmaps are marketing until a peer reviews them — every timeline node carries that caveat; (2) "quantum advantage" is a moving target as classical algorithms improve; (3) QML is the most hype-inflated corner and gets graded hardest; (4) national-program dollar figures are frequently inflated or double-counted; (5) quantum biology (`F`-adjacent, covered in the biophysics canon) is contested and lives in that branch, cross-linked here rather than duplicated.

"Network capacity" = fraction of `00-IDEAL-STATE-MAP.md` nodes with a filled, graded research card. Breadth ran to 100%; the loop then flipped to DEPTH, and depth is now complete across every layer.

Meter

```
TOTAL NODES:       184
FILLED (stub+):    184
DEPTH-COMPLETE:    184
NETWORK CAPACITY: 100.0%   ← breadth
DEPTH CAPACITY:   100.0%   ← every card deepened with primary sources + graded claims
CONFLICT OBJECTS:  15      (evidence/CONFLICTS.md)
RENDERING:         LaTeX/mhchem math → SVG · WeasyPrint PDF · figures via matplotlib+RDKit
```

Per-layer breakdown (counted from disk)

LAYER	NODES	FILLED	DEPTH-COMPLETE
L0 Foundations	26	26	26 (100%)
L1 Hardware	27	27	27 (100%)
L2 Stack & algorithms	27	27	27 (100%)
L3 Adjacent tech	22	22	22 (100%)
L4 Industries	27	27	27 (100%)
L5 Ecosystem & geopolitics	22	22	22 (100%)
L6 History	15	15	15 (100%)
L7 Frontier & open	18	18	18 (100%)
Total	**184**	**184**	**184 (100%)**

Trajectory

- **Cycle 1** (2026-07-08): 0→72%. Filled 94 original nodes; random-walk found 36 → map 94→130.
- **Cycle 2** (2026-07-08): 72→100% breadth. Filled the 36 gap nodes (125 cards).

- **Cycle 3 — DEPTH** (2026-07-08): every card deepened (stub → ~450–700 words with primary derivations, exact 2025–26 numbers, per-claim tiers). Random-walk during depth added **52 more nodes** (map 130→182): L0+8, L1+8, L2+8, L3+7, L4+6, L5+8, L6+6, L7+6. Conflict register grew 8→15.
- **Cycle 3b — RENDERING** (2026-07-08): math/chemistry marked up in LaTeX and rendered to SVG (`reports/render_math.py` via latex+dvisvgm), figures generated (`reports/gen_figures.py`), manual compiled to HTML (`build_manual.py`) and PDF (`build_pdf.py`, WeasyPrint).

Thin-node flags (honest edge)

`I-gov`, `I-retail`, `I-media` remain T5/T6-heavy (forecasts / classical-AI mislabeled as quantum). Deepened but honestly thin — the manual says so.

Still-open threads (queued, from depth-agent "what I'd chase" notes)

RESOLVED in the polish pass: QuEra 96-logical confirmed peer-reviewed (Nature s41586-025-09848-5 → T2); Duke/IonQ networking primary found (arXiv:2606.17173 → T3); Majorana 2 — no independent replication (kept contested, Legg critique + Microsoft Nature reply); Kalai↔Aaronson Mar-2026 exchange sourced verbatim both sides; Flatiron vs D-Wave — no third-party adjudication (kept contested). Node splits landed: +H-wiring, +H-spinphoton. SKIPPED (no real dated pilot): I-water (desalination/membrane), I-foodbev — only speculative coverage exists, so not added. Remaining candidate: QEC-theory-history (§07).

Last recomputed from disk: polish pass — 184/184 depth, 2026-07-08.

PART I

The Physics and the Machines

PART I · THE PHYSICS AND THE MACHINES

1 The Physics

Every machine in this atlas — the transmon fridges of Chapter 2, the logical qubits of Chapter 3, the key-distribution links and cold-atom gravimeters of Chapter 4 — is an engineering answer to a physical fact: nature, at small scales, does not store or process information the way a classical world does. This chapter lays down the substrate, the twenty-six ideas that make quantum technology possible, in the order they build on each other. Most of what follows is settled physics, reproduced for a century and graded T1 in the atlas's scheme. A few threads — what a measurement *is*, what the wavefunction *is* — remain open, and the chapter flags them where they arise.

The chapter has a spine and branches. The spine runs from a single linearity axiom up through the qubit, entanglement, and the limits on copying and communicating quantum information. The branches matter because some ideas are load-bearing for computing while others carry only sensing or cryptography, and knowing which is which tells you what physics a given machine actually exploits.

Learning objectives. After this chapter you can:

- State the superposition principle and the Born rule, and read measurement probabilities off a state vector.
- Place a qubit on the Bloch sphere and connect its rotations to the two-to-one map from SU(2) to SO(3).
- Explain why entanglement together with the Bell/CHSH bound rules out local hidden variables, and compute the Tsirelson bound 2√2.
- Say what no-cloning forbids and what it forces — quantum key distribution, error correction that protects without reading, and repeaters that swap entanglement.
- Name the single resource that separates classically simulable circuits from hard ones, and its three faces: contextuality, Wigner negativity, and the stabilizer/magic boundary.
- State the measurement problem precisely and tell apart the interpretations that bear on hardware from those that leave the machines unchanged.

The state vector and the superposition principle

Start with the one axiom everything else rests on. In plain terms, a quantum state is a list of numbers — one amplitude for each outcome the system could show. Formally, a quantum system is described by a state vector $|\psi\rangle$ in a Hilbert space, and any normalized linear combination of valid states is itself a valid state:

$$|\psi\rangle = \sum_i c_i\,|i\rangle, \qquad \sum_i |c_i|^2 = 1,$$

with complex amplitudes c_i. This is Dirac's superposition principle, and its content is deceptively small: add any two allowed states with complex weights and you get another. The weights carry a phase, and phases add and cancel — that cancellation is interference, the whole difference between quantum and classical probability. A classical mixture sums probabilities; a quantum superposition sums amplitudes and squares the total, so cross terms $2\,\mathrm{Re}(c_i^* c_j)$ appear that no classical distribution can produce. Interference fringes, entanglement, and every algorithmic speedup trace back to this linearity.

The principle is not a small-particle curiosity. Single electrons build up a two-slit pattern one at a time (Tonomura 1989, T1), C_{60} buckyballs diffract through a nanograting (Arndt 1999, T1), and molecules beyond 25 kDa — roughly two thousand atoms — have been placed in spatial superposition (Fein 2019, T2). Where superposition breaks for macroscopic objects is an open empirical frontier, and we return to it with objective-collapse models.

For computing, the payoff is exponential bookkeeping: an n-qubit register in superposition holds 2^n amplitudes at once, and that space is where quantum algorithms live. The catch, paid in full later, is that you cannot simply read those amplitudes out.

Measurement and the Born rule

Superposition is several of those amplitudes being nonzero at once. Measurement is what happens when you look, and it collapses that spread to a single outcome. Measuring an observable — a physical quantity you can read off, like energy, spin, or position — A returns one of its eigenvalues — the discrete values a measurement of it can produce — and the probability of outcome a is the squared magnitude of the corresponding amplitude — the Born rule:

$$P(a) = |\langle a|\psi\rangle|^2, \qquad \langle A\rangle = \langle\psi|A|\psi\rangle.$$

For a general (mixed) state the same rule reads $P(a) = \mathrm{Tr}(\rho P_a)$ with P_a the projector onto the a-eigenspace. After the measurement the state updates to the outcome's eigenstate, the state for which that outcome is now certain. This is where the theory touches experiment: every prediction quantum mechanics makes is, in the end, a Born-rule probability.

Two things about the rule are worth pausing on. It is quadratic in the amplitude, and that specific power is testable: a three-slit photon experiment bounded any third-order interference term to $\kappa = 0.0064 \pm 0.0119$, consistent with zero (Sinha 2010, T1) — the world squares amplitudes rather than cubing them. And the squared-modulus form is close to forced: Gleason's theorem shows that in dimension three or more, the only probability measure additive over orthogonal outcomes is $\mathrm{Tr}(\rho P)$ (T1).

The projective picture is an idealization. Real detectors are lossy, noisy, and coupled to ancillas, and the honest description of any of them is a POVM: a set of positive operators $\{E_k\}$ with $\sum_k E_k = I$, giving $p(k) = \mathrm{Tr}(\rho E_k)$. Naimark's dilation theorem (1940, T1) proves this is nothing exotic — every POVM is a sharp projective measurement on the system plus an ancilla you brought along and ignored. This matters downstream: dispersive readout of a transmon and fluorescence readout of an ion are finite-fidelity POVMs, and modeling them that way makes the readout-error mitigation and soft-information decoding of Chapter 3 possible. In the limit of vanishing coupling it becomes weak measurement, where single quantum trajectories of a monitored qubit are reconstructed in real time (Murch 2013, T2).

Why the Born rule holds, and what physically happens during the jump, is the measurement problem — unsolved, and treated at the end of this chapter.

The qubit as the unit of information

Specialize the state vector to two levels and you get the object the rest of the atlas is built from. In plain terms, a qubit is the two-outcome case — one amplitude for 0 and one for 1. A qubit is any two-level quantum system with pure states $\alpha|0\rangle + \beta|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$. Up to an unobservable global phase, every pure state maps to a point on a globe of one-qubit states:

$$|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\varphi}\sin(\theta/2)\,|1\rangle,$$

the Bloch sphere, with the two angles (θ, φ) as latitude and longitude. Mixed states fill the interior, written $\rho = \frac{1}{2}(I + \mathbf{r}\cdot\boldsymbol{\sigma})$ with Bloch vector $\mathbf{r}$; the center $\mathbf{r} = 0$ is maximally mixed. A single-qubit gate rotates this sphere rigidly. The rotation of the sphere is an element of SO(3); the gate acting on the state vector is an element of SU(2), the double cover of SO(3), which is why a 360° turn of the sphere leaves the state with a minus sign and only a 720° turn brings it back.

The qubit is the right abstraction because it makes hardware interchangeable. A transmon's two supercurrent directions, an ion's two hyperfine levels, a neutral atom's ground-and-Rydberg pair, a photon's polarizations, a spin in silicon — all implement the same Bloch-sphere object, so algorithms and codes are written once against it, and the DiVincenzo criteria (2000, T1) codify what a system must supply. Control can be extraordinary: a trapped-ion qubit has reached single-qubit gate error near 1.5×10^{-7} (Oxford 2024, T2), six-plus nines on one Bloch rotation. Whether the two-level unit always wins is an open engineering question — bosonic hardware (Chapter 2) discards it for an infinite-dimensional mode.

Uncertainty and complementarity

Some pairs of quantities cannot both be sharp at once; pin one down and the other spreads out. Because non-commuting observables cannot share a full set of eigenstates, no state can make two conjugate quantities both sharp. Robertson's inequality states it exactly:

$$\Delta A \cdot \Delta B \geq \tfrac{1}{2}\,|\langle [A,B] \rangle|,$$

and for position and momentum, where $[x,p] = i\hbar$, this is the familiar $\Delta x\,\Delta p \geq \hbar/2$. The bound comes from the algebra of the operators, with no reference to clumsy measurement — it is a fact about how states can be prepared (Heisenberg 1927, Robertson 1929; T1). For discrete observables the sharper statement is entropic, $H(A) + H(B) \geq \log(1/c)$ (Maassen–Uffink 1988, T1), and this is the form QKD security proofs in Chapter 4 use — an eavesdropper's information about one basis is bounded by their ignorance of the conjugate one.

Uncertainty sets the standard quantum limit, the noise floor of quantum sensing, and defines what squeezing can buy: push noise below the limit in one quadrature by paying it back in the other. LIGO's O4 run injected frequency-dependent squeezed vacuum and cut quantum noise by up to 5.8 dB, beating the standard quantum limit by as much as 3 dB in the 35–75 Hz band (LIGO 2023, T2). The measurement–disturbance version of the relation, by contrast, is contested — Ozawa's and Busch–Lahti–Werner's reformulations are each correct under their own definition of "measurement error" (T2/T3), and the dispute is about which definition is right.

Entanglement and the end of local realism

Two systems can end up sharing a joint state that neither one holds on its own, one that no product of individual states reproduces. That is entanglement, and Schrödinger called it *the* characteristic trait of quantum mechanics. The canonical case is the singlet,

$$|\Psi^-\rangle = \tfrac{1}{\sqrt{2}}\big(|01\rangle - |10\rangle\big)\ :$$

measure either qubit and you get 0 or 1 at random, but the two outcomes are perfectly anti-correlated however far apart the qubits sit. Locally each side sees only its reduced state $\rho_A = \mathrm{Tr}_B(|\Psi\rangle\langle\Psi|)$, which carries no message — this is why entanglement never signals, coexisting with relativity through the no-communication theorem (T1). The amount of entanglement in a pure bipartite state is the entropy of that reduced state, $S = -\mathrm{Tr}(\rho_A \log \rho_A)$: zero for a product state, one bit — "one ebit" — for a maximally entangled pair.

Entanglement is the workhorse resource of the field. A two-qubit gate is precisely the operation that creates it; a quantum computer's register *is* multipartite entanglement; teleportation, entanglement-based QKD, and the distributed protection of error-correcting codes all spend it. Every modality is judged partly on how well it entangles — Mølmer–Sørensen gates for ions, tunable couplers for transmons, Rydberg blockade for atoms, fusion for photons. And it travels: the Micius satellite distributed polarization-entangled pairs to ground stations 1,203 km apart and still violated a Bell inequality at $S = 2.37 \pm 0.09$ under Einstein locality (Yin 2017, T1).

That number carries the whole argument. John Bell proved in 1964 that any local hidden-variable theory caps the correlations between separated measurements. The CHSH form builds one number from four settings, S = E(a,b) − E(a,b′) + E(a′,b) + E(a′,b′), and every local-realist theory obeys $|S| \leq 2$. Quantum mechanics, using a Bell state and detectors 45° apart, reaches $S = 2\sqrt{2} \approx 2.828$ — the Tsirelson bound, the most quantum theory itself allows. The gap between 2 and $2\sqrt{2}$ is the experimental target, and six decades of measurements have closed it. The loophole-free trio of 2015 (Hensen, Giustina, Shalm) violated the bound with locality and detection loopholes shut simultaneously (T1), the last escape routes for local realism. The 2022 Nobel to Aspect, Clauser, and Zeilinger cites exactly this. The test now runs on the same technology used to build computers: Storz et al. entangled two superconducting qubits across a 30 m cryogenic link and measured $S = 2.0747 \pm 0.0033$ over a million trials, $p < 10^{-108}$ (Storz 2023, T1). Bell violation is the operational certificate that a device holds real entanglement, which is what device-independent QKD and certified randomness (Chapter 4) turn into security. The only ways around the theorem — superdeterminism and retrocausality — survive by denying that experimenters choose settings freely, and are widely regarded as unfalsifiable.

The deeper no-go structure

On a first read, this can wait. This section and the open-system dynamics that follow are dense, and so is the reference layer further on — the information-theory accounting, the classical-simulability boundary, and the physics under specific machines. You can skip ahead to the measurement problem at the end and come back for any of them; no-cloning and teleportation in between are short and worth reading straight through.

Bell nonlocality is one face of a larger fact: quantum outcomes cannot be pre-assigned as if they existed independently of what you choose to measure. The Kochen–Specker theorem proves that in dimension three or more, no assignment of definite $0/1$ values to all projectors is consistent with quantum algebra — a state-independent contradiction needing no special input state and no entanglement, sharpest in the Peres–Mermin square of nine two-qubit observables (demonstrated with trapped ions, Kirchmair 2009, T2). Bell is the special case where the contexts are spacelike-separated. And contextuality is a resource: Howard et al. (2014, T1) proved that for qudits it is exactly what "magic states" supply — Clifford operations on contextuality-free states are classically simulable, and the onset of contextuality coincides with the states magic-state distillation must manufacture to reach universality. A 1967 no-go theorem turns out to name the single most expensive component of a fault-tolerant machine (Chapter 3).

A third pillar concerns the wavefunction itself. The PBR theorem (Pusey–Barrett–Rudolph 2012) shows that if independently prepared systems have independent physical states, then distinct pure states must correspond to disjoint underlying realities — the state cannot be mere ignorance about a deeper variable, so the wavefunction is ψ-ontic, real. The result leans on preparation independence; drop it and ψ-epistemic models survive (Lewis 2012, T2), so PBR constrains a subclass rather than closing the question. Together, Bell, Kochen–Specker, and PBR fence off classical hidden-variable accounts from three directions, and the ψ-ontic reading is the ground under treating unknown states as real, uncopyable information.

Open systems: why the world looks classical

No system is perfectly isolated, and coupling to an environment changes everything. When a superposition entangles with its surroundings, the phase relations between its components leak into environmental degrees of freedom and become unrecoverable. The system's own state — the reduced density matrix $\rho_S = \mathrm{Tr}_E(|\Psi\rangle\langle\Psi|)$ — loses its off-diagonal coherences and looks like a classical mixture. This is decoherence, and it is why the everyday world appears classical without any need to modify quantum mechanics. For a Markovian environment the dynamics obey the Lindblad master equation,

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \sum_k \left(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \right),$$

where the first term is ordinary unitary evolution and the sum is the irreversible leak, each jump operator L_k modeling a channel. For a qubit two rates govern the race: T_1, energy relaxation, and $T_2 \le 2T_1$, dephasing. Decoherence was watched in real time in cavity QED (Brune 1996, T2) and turned into a knob by heating fullerenes until they radiated which-path information (Hackermüller 2004, T2). It is the enemy of every qubit: it caps circuit depth at roughly T_2/t_{gate} operations, and it dictates the architecture of error correction, which spreads one logical qubit across many physical ones so syndrome measurements catch errors before they pile up. Coherence times are the headline spec of every modality — transmons and fluxonium now reach the millisecond range (Somoroff 2023, $T_2^* \approx 1.48$ ms, T2), trapped ions hold for seconds to minutes — and pushing them up is the central materials problem of the field.

The formal frame beneath decoherence recurs throughout the stack. Purification and Stinespring dilation say mixedness and irreversibility are never fundamental. Every mixed state is the shadow of a pure state on a larger space, and every physical process — every completely positive trace-preserving map — is a unitary on a bigger space followed by discarding the addition, $\mathcal{E}(\rho) = \mathrm{Tr}_E[V\rho V^\dagger] = \sum_k K_k \rho K_k^\dagger$. Smolin nicknamed this the "church of the larger Hilbert space": noise is entanglement with something you stopped tracking. This Kraus/Stinespring structure is how decoherence channels get simulated and how threshold theorems get proved — the channel-side twin of the Naimark dilation we met for measurements.

What you cannot do: no-cloning, and what it forces

Linearity builds superposition and also forbids copying. The no-cloning theorem is three lines: a universal copier would act as $U(|\psi\rangle|b\rangle) = |\psi\rangle|\psi\rangle$ for every $|\psi\rangle$; apply it to two states and take inner products, and unitarity forces $\langle\psi|\varphi\rangle = \langle\psi|\varphi\rangle^2$, so the states are identical or orthogonal. No machine can copy a set containing any non-orthogonal pair. The best universal cloner reaches fidelity $5/6 \approx 0.833$ (reached in the lab by Lamas-Linares 2002, T2), the yardstick for any intercept-resend attack on deployed QKD.

No-cloning is the root of quantum cryptography — an eavesdropper cannot copy qubits in flight without disturbing them, which is what BB84 and E91 detect and Chapter 4 turns into quantum key distribution. It forbids naive backup of quantum data, which is why error correction protects information without reading or copying it, and it blocks classical signal amplification, so long-haul links need repeaters that swap entanglement rather than copy-and-amplify.

The workaround for moving quantum states is teleportation. Using one shared entangled pair and two classical bits, Alice transfers an unknown state to Bob while destroying her original — consistent with no-cloning, since α and β are never measured, only moved. The mechanism is an identity: expand $|\psi\rangle|\Phi^+\rangle_{AB}$ in the Bell basis of Alice's two qubits and it factors into four terms, each pairing one of Alice's outcomes with $U_i|\psi\rangle$ on Bob's side, U_i a Pauli. Her Bell measurement collapses the sum and tells her which i; Bob applies U_i^{-1} and recovers $|\psi\rangle$. The two classical bits are mandatory — before Bob hears them his qubit is $I/2$, carrying nothing — which is why teleportation never beats light and never clones. Its cousin, entanglement swapping, runs the same identity to entangle two particles that never interacted, the primitive a repeater uses to stitch short links into long ones. Both are demonstrated across real distances: 143 km free-space with feed-forward (Ma 2012, T2), and between non-neighboring nodes of a three-node network (Hermans 2022, T2). Gate teleportation applies the same trick to logical information — how magic states are consumed in fault-tolerant circuits, tying forward to Chapter 3.

The accounting: quantum information theory

If states carry information, there must be laws for how much, how compressibly, and how much survives a noisy channel. Quantum information theory is the Shannon theory of quantum states, built on the von Neumann entropy $S(\rho) = -\mathrm{Tr}(\rho \log \rho)$ — Shannon's formula with the density matrix's eigenvalues as the probabilities. Schumacher's theorem compresses N copies of ρ into about $N\,S(\rho)$ qubits and no fewer. Reading *classical* messages out of quantum states is capped by the Holevo bound, $I \le \chi = S(\rho) - \sum_i p_i S(\rho_i)$, with the blunt corollary that n qubits carry at most n classical bits. For quantum states down a noisy channel the capacity is the regularized coherent information (LSD theorem, T2) — and here the theory turns strange: two zero-capacity channels can combine to give positive capacity (Smith–Yard 2008, T2), and no single-letter formula for a general channel's quantum capacity is known. These laws set the hard limits for the quantum internet, and fidelity and trace distance are how every gate, memory, and teleportation demo is scored — ground-to-satellite teleportation over a 1,400 km uplink hit fidelity 0.80 ± 0.01, above the $2/3$ classical limit (Ren 2017, T2).

Underneath the capacities sit the structural laws. The deepest is strong subadditivity (Lieb–Ruskai 1973, T1), $S(\rho_{ABC}) + S(\rho_B) \le S(\rho_{AB}) + S(\rho_{BC})$. From it follow the data-processing inequality — local operations never increase quantum information — and the CKW monogamy inequality: entanglement cannot be freely shared, so if A is maximally entangled with B, none is left for C. Monogamy is the mathematical reason QKD is secure — an eavesdropper's correlation with the key is bounded by how much the legitimate parties already share, so listening in is detectable. Not every quantum correlation is entanglement. Quantum discord measures correlations that survive in separable states — the gap $D(A{:}B) = I(A{:}B) - \max_{\{\Pi_k\}} J(A{:}B)$ between two classically-identical expressions for mutual information. It is nonzero for generic separable states and was proposed as the resource behind mixed-state models like DQC1, which show speedup with vanishing entanglement (Datta 2008, T2), though whether it is an operational resource remains contested.

The classical-simulability boundary

A practical question hangs over the whole enterprise: which quantum circuits actually need a quantum computer, and which can a laptop shadow for free? The stabilizer formalism draws the sharpest known line. A stabilizer state on n qubits is the unique common $+1$ eigenstate of n commuting Pauli operators — an exponential compression, since n generators pin down a state that would otherwise need 2^n amplitudes. The Gottesman–Knill theorem (T1) says any circuit of preparations, Clifford gates (Hadamard, phase, CNOT), and Pauli measurements is efficiently simulable classically, because Clifford gates map Paulis to Paulis and you track the $\sim\ n$ generators instead of the full state (Aaronson–Gottesman sharpened this to $O(n^2)$ per gate, T1). The lesson is inverted and load-bearing: Clifford operations are *free*, and the non-Clifford resource — magic, the T gate — is what must be distilled at great cost to reach universality. Every mainstream error-correcting code is a stabilizer code, and syndrome extraction is just measuring the generators.

The continuous-variable version of the same boundary is Wigner-function negativity. The Wigner function W(x, p) represents a state as a real phase-space distribution whose marginals recover the right statistics, and its defining quantum feature is that it can go negative. Hudson's theorem pins the boundary: a pure state has a non-negative Wigner function exactly when it is Gaussian (T1). Gaussian states and operations alone are classically simulable, so in photonic and bosonic hardware a non-Gaussian, Wigner-negative element — photon subtraction, a cubic phase gate, a GKP state — is what lifts the platform to universality. These are three views of one fact. Contextuality is the Kochen–Specker resource that magic states supply; Wigner-function negativity is the phase-space signature of non-classicality; the stabilizer/magic boundary is the gate-level line between free Clifford operations and the costly non-Clifford resource. All three pick out the same divide between the circuits a classical computer can shadow efficiently and the circuits that force you to build a quantum one. In odd prime dimension the identification is exact: the non-negative discrete Wigner functions are precisely the stabilizer states, so a circuit is classically tractable exactly when it stays non-contextual, Wigner-positive, and magic-free — one resource wearing three faces. The qubit case is subtler and still being sharpened, an open question at the boundary rather than settled bookkeeping.

The physics under specific machines

Several foundations matter because they are the mechanism inside a class of hardware. Quantum tunneling is the first: a wavefunction has nonzero amplitude past a barrier it cannot classically climb, with transmission $T \approx e^{-2\kappa d},\ \kappa = \sqrt{2m(V_0 - E)}/\hbar$ — exponential in barrier width. Cooper pairs tunneling coherently across a Josephson junction obey $I = I_c \sin\varphi$ (Josephson 1962, T1), and the resulting nonlinear inductance is what makes a transmon a usable anharmonic oscillator. The same physics runs the scanning tunneling microscope and supplies whatever advantage annealing gets from tunneling-through-barriers over thermal hopping-over.

That annealing rests on the adiabatic theorem: start in the ground state of an easy Hamiltonian, deform slowly to a hard one, and the system stays in the instantaneous ground state provided the runtime scales as $T \sim 1/\Delta^2$, with Δ the smallest spectral gap. Adiabatic computation is polynomially equivalent to the gate model (Aharonov 2007, T1), so it is a legitimate paradigm. The catch is the gap: for hard instances it can shrink exponentially and erase any speedup, the unresolved crux over D-Wave's machines. A coherent anneal at 5,000+ qubits beat a matched classical model (King 2023, T2), the advantage claim contested and unresolved.

The geometric or Berry phase is the phase a state acquires around a closed loop in parameter space, $\gamma = \oint_C \mathbf{A} \cdot d\mathbf{R}$, depending only on the path and not on how fast it is traversed. Because it ignores timing, it grounds holonomic gates that resist pulse-duration errors, though whether they beat optimized dynamical gates is contested. Integrate the Berry curvature over a filled band and you get a quantized Chern number — the invariant behind the quantum Hall effect, whose resistance defines a metrology standard to parts in 10^9 (von Klitzing 1980, T1). This is the mathematics under topological band theory and, non-abelian, under braiding anyons.

Which brings in identical particles and spin-statistics. Exchange two identical particles and the wavefunction picks up $+1$ for bosons, -1 for fermions, the minus sign being the whole Pauli exclusion principle. This is why chemistry exists, so any molecular simulation must enforce fermionic antisymmetry — mapping fermionic modes to qubits via Jordan–Wigner or Bravyi–Kitaev is a core chemistry primitive and a real circuit-depth cost driver (Chapter 3, and Chapter 5's pharma and chemicals verticals). In two dimensions the exchange phase sits anywhere between the boson and fermion values, giving anyons; non-abelian anyons carry a unitary matrix, and braiding them is the entire premise of topological quantum computing (Microsoft's Majorana program), whose device-level existence remains contested.

Three foundations complete the substrate. Quantum field theory promotes fields to operators and reads particles as their quanta, with creation and annihilation operators on Fock space letting particle number change — the native language of photonic and bosonic hardware, where qubits are Fock, squeezed, and coherent states of light. Circuit QED, the backbone of transmon processors, is applied field quantization (Blais 2021, T1), and QED is the most precisely verified theory in physics (electron g−2 to 0.13 ppt, T1). Quantum thermodynamics supplies the energy floor: Landauer's principle says erasing one bit dissipates at least $kT \ln 2$ (T1), the hard floor beneath every syndrome reset in an error-correction cycle, and fluctuation theorems like Jarzynski's $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$ (T1) recover equilibrium free energies from irreversible drives. Lieb–Robinson bounds show that even without relativity, short-range lattice Hamiltonians have an emergent finite speed — information stays exponentially small outside a linear light cone. That velocity clocks how fast entanglement spreads across a chip, underpins the area law that makes tensor-network classical simulation efficient (a recurring counterweight to advantage claims in

Chapter 3), and, through gap-implies-clustering, explains why topological order and local error correction are stable.

Interpretations and the measurement problem

The formalism predicts probabilities perfectly. What it says about reality is unsettled, and the dispute sits on a mismatch between two rules the standard theory runs together. Between measurements a state evolves smoothly and deterministically by the Schrödinger equation $i\hbar\,\partial_t|\psi\rangle = H|\psi\rangle$, which is linear and never destroys a superposition. At a measurement the Born rule takes over and the state jumps, probabilistically, to one eigenstate. That discontinuous jump is nowhere in the linear equation, and specifying when and why it happens is the measurement problem — the field's oldest open question. Physicist polls show no consensus.

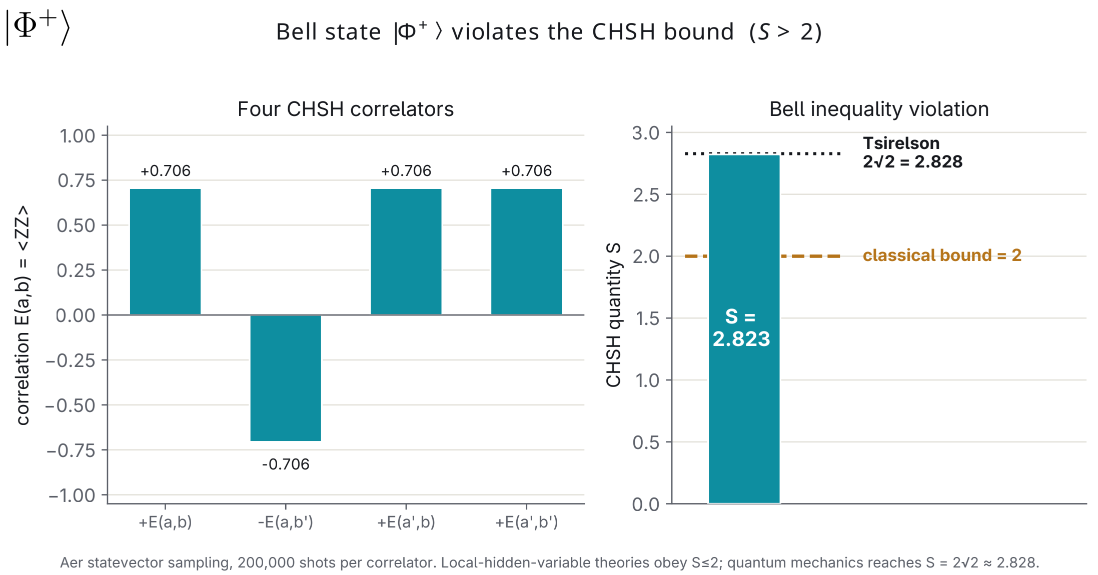
The interpretations are answers to it. Copenhagen takes the classical measurement context as primitive; many-worlds keeps only the linear equation and denies the jump is real, every outcome on its own branch; de Broglie–Bohm adds definite particle trajectories guided by the wave; QBism reads states as an agent's credences; relational QM makes them observer-relative. All of these are empirically identical to standard quantum mechanics and to each other, so they change nothing about how you run a gate or a code. The one family with teeth is objective collapse — GRW/CSL, Diósi–Penrose — which adds a small stochastic nonlinear term so that superpositions of large masses self-destruct on a physical timescale. That family makes numbers, and experiments are closing on them: an underground search excluded the parameter-free version of Diósi–Penrose collapse radiation (Donadi 2021, T2), and every heavier matter-wave superposition (Fein 2019) narrows the window collapse can occupy. This is the one place the interpretation debate rides on the same isolation-and-readout engineering sensing and bosonic programs push — otherwise, whichever interpretation you hold, the machines behave the same.

Key takeaways

- The whole framework builds from one linearity axiom: states are vectors, closed evolution is unitary, and measurement follows the Born rule.
- A qubit is the unit of quantum information — one two-level system carrying a continuum of amplitude, collapsing to one classical bit on readout.
- Entanglement plus the loophole-free Bell tests ends local realism, and the CHSH correlation is capped at the Tsirelson bound $2\sqrt{2}$.
- No-cloning is load-bearing: it underwrites quantum key distribution, forces error correction to protect information without copying it, and rules out classical signal amplification.
- Contextuality, Wigner negativity, and the stabilizer/magic boundary are three views of the one resource that makes a circuit hard to simulate classically.
- The core is T1 settled physics; the two open edges are the measurement problem and how large a superposition can survive.

Where the foundations stand

The substrate is solid. Superposition, the Born rule, entanglement, Bell violation, decoherence, the uncertainty relations, no-cloning, and the entropy laws are textbook physics, reproduced for decades and graded T1 — the 2015 loophole-free Bell tests and the 2023 superconducting reproduction closed the last serious doubt about local realism, and the rest has been secure far longer. The resource-theoretic layer — contextuality, Wigner negativity, and the stabilizer boundary as three views of the "magic" that makes quantum computing hard — is established but still being sharpened, especially for qubits, and that sharpening is live research rather than loose ends. What remains open sits at two edges: the measurement problem, which no experiment has yet touched and perhaps none can, and the frontier of how large a superposition survives, where objective-collapse models and matter-wave interferometers push toward each other. Everything downstream in this atlas — every qubit modality, every algorithm, every sensor and key-distribution link — is engineering built on that settled core, and the open questions at the edges change none of it.



Simulated CHSH violation: S = 2.823, just under the Tsirelson bound 2√2 (Aer, 200k shots/correlator).

Exercises and discussion

1. **(Derivation — Tsirelson bound.)** For the CHSH operator $B = A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1$ built from observables that square to the identity, show that $B^2 = 4I + [A_0, A_1][B_0, B_1]$. Bound the commutator term to conclude $\|B\| \le 2\sqrt{2}$, and check this against the simulated S = 2.823 in the figure above. Explain why a local hidden-variable model caps the same quantity at 2.

2. **(Derivation — Bloch sphere and SU(2).)** Write a pure qubit state as cos(θ/2)|0⟩ + e^{iφ} sin(θ/2)|1⟩ and verify it corresponds to the point (θ, φ) on the unit sphere. Show that a 2π rotation of the state vector returns the Bloch point to itself while multiplying the state by −1, and say what that means for the two-to-one map from SU(2) to SO(3).

3. **(Resource estimation — complexity.)** A Clifford circuit on n qubits is classically simulable in polynomial time (Gottesman–Knill). Estimate how the classical cost grows as you inject t non-Clifford (T) gates, using stabilizer-rank scaling of roughly 2^{αt}. For a circuit needing 50 T gates, is direct stabilizer-rank simulation still feasible? Tie the answer to the "magic" resource in the classical-simulability section.

4. **(Spec reading — sensing trade-off.)** Uncertainty sets the standard quantum limit. Using LIGO's 5.8 dB squeezing figure from the O4 run, work out the factor by which noise drops in the squeezed quadrature and the factor by which it rises in the conjugate one. What does that trade cost a broadband detector, and why is frequency-dependent squeezing the fix?

5. **(Seminar — the measurement problem.)** Copenhagen, many-worlds, de Broglie–Bohm, QBism, and relational QM are empirically identical yet disagree about what a measurement is. If no experiment can separate them, on what grounds should a practitioner prefer one? Does "it changes nothing about how you run a gate" settle the question or set it aside?

6. **(Seminar — the collapse frontier.)** Objective-collapse models (GRW/CSL, Diósi–Penrose) make numbers, and the Donadi 2021 underground search already excluded the parameter-free Diósi–Penrose case. Debate whether the measurement problem is a physics question closing in on data or an interpretive question data cannot reach, and locate matter-wave interferometry (Fein 2019) in that debate.

Further reading

- **Nielsen & Chuang, *Quantum Computation and Quantum Information* (10th anniversary ed., 2010).** The standard graduate text; Chapters 1–2 cover the state vector, Born rule, qubit, and density matrices used throughout this chapter.
- **Hensen et al., "Loophole-free Bell inequality violation using electron spins separated by 1.3 km," *Nature* 526 (2015).** The first Bell test to close the detection and locality loopholes at once — the experimental basis for the end of local realism.
- **Cirel'son (Tsirelson), "Quantum generalizations of Bell's inequality," *Letters in Mathematical Physics* 4 (1980).** Derives the $2\sqrt{2}$ ceiling on quantum CHSH correlations that Exercise 1 reconstructs.
- **Howard, Wallman, Veitch & Emerson, "Contextuality supplies the 'magic' for quantum computation," *Nature* 510 (2014).** Identifies contextuality as the resource behind the classical-simulability boundary discussed here.
- **Zurek, "Decoherence, einselection, and the quantum origins of the classical," *Reviews of Modern Physics* 75 (2003).** The standard review of how environmental interaction selects the classical states of the open-systems section.
- **Pusey, Barrett & Rudolph, "On the reality of the quantum state," *Nature Physics* 8 (2012).** The PBR theorem constraining ψ-epistemic models, referenced in the no-go structure section.

PART I · THE PHYSICS AND THE MACHINES

2 The Machines

Chapter 1 ended with a unit of information — the qubit, a point on the Bloch sphere holding a superposition until it is measured. This chapter is about the physical objects that hold that point still long enough to compute with it. There is no single quantum computer. There are eight or nine distinct bets on what a qubit should be made of, each with its own physics, its own supply chain, its own temperature, and its own set of companies staking valuations on it. This chapter maps that competition, and then maps the far less visible layer beneath it — the fridges, lasers, amplifiers, wires, and foundries without which none of the headline machines run.

Every qubit count and fidelity number below is a coordinate, and most of them come from the vendor that benefits from the number. Under the evidence schema, a company announcing its own benchmark is a T4 claim until an independent group reproduces it — never promoted to a peer-reviewed T2 on the strength of a blog post. That rule does most of the work in this chapter. When a number is a Nature paper someone else validated, it is said. When it is a roadmap, it is called a roadmap. The gap between those two categories is where the whole field's honesty lives.

Learning objectives. After this chapter you can:

- Name the DiVincenzo criteria and say what each one demands of a physical qubit.
- Compare the major modalities — superconducting, trapped-ion, neutral-atom, silicon-spin, photonic, topological, bosonic, and annealing — on coherence, gate fidelity, and connectivity.
- Read a vendor spec sheet and separate a reproduced T2 result from an unverified T4 claim.
- Explain the two hardware walls — wiring/interconnect and materials/two-level-system defects — and why neither is solved.
- Trace the support stack beneath the qubit: dilution fridges, control electronics, lasers, and foundries.
- Judge which modality leads on which axis, and which condition each one needs to hold that lead.

The criteria a qubit must meet

Before comparing the modalities, it helps to have the checklist they are all graded against. In 2000, David DiVincenzo wrote down five requirements any physical system needs to satisfy to be a quantum computer, plus two more for quantum communication. Twenty-five years later the list is still the cleanest lens on the field:

1. **A scalable physical system with well-characterized qubits.** You must be able to make many of them, and know each one's parameters. "Scalable" is the word that quietly kills most candidates.
2. **The ability to initialize** the qubits to a known simple state, like $|00\ldots0\rangle$. You cannot compute from an unknown starting point.
3. **Long coherence times** relative to the gate time. The ratio is what matters: how many operations you can run before the qubit forgets, which depends on both the lifetime (T_1, T_2) and the gate speed. A slow, long-lived qubit and a fast, short-lived one can land in the same place.
4. **A universal set of gates** — enough one- and two-qubit operations to approximate any computation.
5. **Qubit-specific measurement** — the ability to read out each qubit's state, fast and with high fidelity.

The two communication criteria — interconverting stationary qubits with "flying" photonic ones, and transmitting those photons faithfully — looked optional in 2000. They are now central, because every modality's endgame is modular: many processors linked together, which turns criteria 6 and 7 into the interconnect problem that dominates the back half of this chapter.

No modality wins all five (or seven) at once. Each one trades a strength on some criteria against a weakness on others. The comparison that follows is essentially a tour of those trade-offs.

Superconducting: the incumbent

The most mature gate-based modality by deployed-system count is built from electrical circuits cooled until they superconduct. A Josephson junction — two superconductors separated by a thin insulator — behaves as a nonlinear inductor, and the two lowest energy levels of the resulting circuit form an artificial atom. The dominant design, the **transmon**, shunts the junction with a large capacitor to flatten its sensitivity to charge noise. Qubits are driven with shaped microwave pulses around $4\text{–}6\,\mathrm{GHz}$ and read out through a coupled resonator, with the whole chip sitting at $\sim 15\,\mathrm{mK}$ in a dilution refrigerator.

This is the modality of IBM, Google, Rigetti, and IQM, and its selling point is speed. Gates run in tens of nanoseconds, roughly a thousand times faster than a trapped ion. The chips are lithographed, so they inherit some of the semiconductor industry's tooling. Google's **Willow** (105 qubits) delivered the result that mattered most: below-threshold surface-code error correction — below-threshold meaning that making the error-correcting code larger drives its error rate down, and a surface code being the leading 2D grid of physical qubits that together protect one logical qubit. It was published in Nature (2024) and is independently the strongest T2 evidence in the field that error correction can suppress errors as you scale a code up. Logical error fell by a factor $\Lambda \approx 2.14$ per step in code distance (the grid's size, which sets how many errors the code can catch), and the distance-7 logical qubit — one protected qubit built from many physical ones — outlived its best single physical qubit by $2.4\times$. IBM runs the largest roadmap: **Nighthawk** (120 transmons, Nov 2025) targeting 5,000-gate circuits, chained toward a **Starling** fault-tolerant system in 2029 claiming 200 logical qubits — a T4 roadmap number, years out. Rigetti reports 99.1% median two-qubit fidelity on a 108-qubit chiplet-tiled system (T4, Apr 2026); IQM reports 99.91% on its 150-qubit Radiance (T4, Mar 2026).

The trade-offs are equally sharp. Coherence sits in the $0.1\text{–}1.7\,\mathrm{ms}$ range — a Princeton tantalum-on-silicon transmon reached a T_1 of about 1.68 ms in 2025, roughly triple the prior best, which is direct evidence that the materials bottleneck is still yielding to engineering. Connectivity is fixed and sparse, and the exact lattice is a design fork worth naming. Google uses a **square grid**, each qubit coupled to four neighbors — the natural geometry for the surface code, whose data and measure qubits tile that same degree-4 lattice. IBM uses a **heavy-hex** lattice, where each qubit couples to only two or three others, deliberately thinned to suppress the frequency collisions and crosstalk that come with denser wiring. That sparseness makes heavy-hex a poor host for the vanilla surface code, and it is part of why IBM pivoted to **qLDPC** error correction: its bivariate-bicycle "gross" code (Nature 2024) protects 12 logical qubits inside 144 data qubits — roughly a tenfold overhead cut against the surface code — at the price of a handful of longer-range couplers per qubit, the c-couplers IBM's Loon processor now layers on top of the heavy-hex grid. And the machine's real enemy is the wiring, more than the qubit itself. Every transmon needs several coaxial lines running from room temperature down to the mixing chamber, and the heat they carry, times the number of qubits, is what caps a fridge. Past a few thousand qubits the superconducting roadmap stops being about qubits and becomes about cryo-CMOS, 3D packaging, and fridge-to-fridge links — the supply-chain nodes covered below.

Trapped ions: the quality leader

Take single atomic ions — $^{137}Ba^{+}$, Yb^{+}, Ca^{+} — strip off an electron, and hold them in an electromagnetic trap in ultra-high vacuum. The qubit lives in two internal energy levels of the atom, which means every qubit is identical by the laws of physics, with no fabrication variation to tune out. Gates are driven through the shared vibrational modes of the ion chain, giving effectively all-to-all connectivity: any ion can interact with any other.

Ions hold the records that matter for quality. **Quantinuum's Helios** (Nov 2025) runs 98 $^{137}Ba^{+}$ qubits at 99.921% two-qubit and 99.9975% single-qubit fidelity, with coherence in the seconds-to-minutes range — and the result was characterized in a preprint (arXiv:2511.05465) and independently validated by Sandia in Nature (June 2026), which lifts it from a vendor claim toward a reproduced one. That is the cleanest high-fidelity, many-qubit demonstration in the field. **IonQ**, having closed a $1.075\,\mathrm{B}$ acquisition of Oxford Ionics, claims a 99.99% two-qubit record (T4) and publishes a roadmap running to 2 million physical qubits by 2030 — aggressive well past any demonstrated trap-scaling result.

The cost of all that quality is speed and scale. Gates take microseconds to milliseconds, a thousand times slower than transmons, and a single trap chain saturates around 50–100 ions before the shared motional modes get too crowded to gate cleanly. Growing past that forces one of two paths: physically shuttling ions between zones on a chip (Quantinuum's QCCD architecture), or linking separate traps with photons (IonQ's increasing bet). Which one wins is still an open, actively tracked conflict. And the whole apparatus is a precision laser instrument, so its uptime tracks laser-lock stability — one unlocked tone halts the processor.

Neutral atoms: the scaling surprise

Neutral atoms — Rb, Sr, Yb, Cs — held in arrays of laser "optical tweezers" were the fastest-moving modality of 2024–26. The qubit lives in atomic states; two-qubit gates come from briefly exciting atoms to enormous **Rydberg** states whose strong interaction blockades their neighbors. The feature that changed the field is that the tweezers can be physically moved mid-computation, so connectivity is reconfigurable and logical qubits can be shuttled around. The whole apparatus runs at room temperature inside a vacuum chamber; only the atoms are cold.

This modality holds the logical-qubit record. **QuEra** (with Harvard and MIT) published 48 logical qubits with logical-layer magic-state distillation in Nature (Bluvstein et al., 2024), then in January 2026 reported **96 logical qubits from 448 physical atoms** using a high-rate [[16,6,4]] code, below-threshold across all 96 and peer-reviewed in Nature (s41586-025-09848-5) — the current T2 logical-qubit high mark. A companion result kept a $\sim 3{,}000$-atom array running for over two hours by reloading atoms mid-computation, converting atom loss from a hard stop into a managed error. **Pasqal** runs 1,000+ atom arrays and targets 10,000 (T4 roadmap); **Atom Computing** runs a 1,180-atom Sr array; Caltech loaded 6,100 atoms in a single array in September 2025, and vendors project 100,000 atoms per chamber within a few years.

The catch is that two-qubit fidelities ($\sim 99.5\%$) still trail ions and the best superconducting systems, atom loss is a distinctive error channel, and the optical-table laser subsystem is the practical scaling wall — the neutral-atom analog of superconducting wiring. Rearrangement and cooling cycles also set an effective clock speed that has to fit inside the error-correction budget. The open question is whether the logical-qubit lead survives the move from batch-mode demos to continuous, repeatable computation with fast mid-circuit measurement.

Photonics: room temperature, probabilistic gates

Encode the qubit in light itself — single photons in dual-rail, path, or time-bin form, or continuous-variable squeezed states — and interfere them through integrated waveguide circuits. Photons barely decohere and travel telecom fiber, so networking is native and much of the machine runs warm. The problem is structural: photons do not interact, so two-qubit gates are **probabilistic** (the leading "fusion" gate succeeds about half the time), and **photon loss** is the dominant error. The architecture leans hard on two other nodes: deterministic photon sources and near-perfect detectors.

PsiQuantum is the boldest bet in the field — skip the noisy near-term era entirely and go straight for a million-qubit fault-tolerant machine. Its Omega chipset (Feb 2025) integrates sources, fusion circuits, and cryogenic detectors, manufactured at GlobalFoundries on 300 mm silicon photonics; the company raised a $1\,\mathrm{B}$ Series E at a $7\,\mathrm{B}$ valuation and announced utility-scale datacenter builds in Brisbane and Chicago. Its million-qubit timeline is a T6 roadmap — plausible, unproven, with no intermediate-scale public benchmark to grade against, which is the modality's grading problem: the approach is close to all-or-nothing. **Xanadu's** Aurora (Jan 2025) is a modular networked photonic computer — 12 qubits across 35 chips and 13 km of fiber — and the company IPO'd in March 2026. Its earlier Borealis showed a Gaussian-boson-sampling advantage (Nature 2022) that later classical algorithms partly matched, which is the standard fate of advantage claims. **QuiX** and **ORCA** round out the field. No photonic machine has yet run a meaningful gate-based algorithm at scale, and the loss budget for million-photon operation is unproven end to end.

Silicon spin: the density bet

Store the qubit in the spin of a single electron confined in silicon. The pitch is manufacturing: these devices are $\sim 100\,\mathrm{nm}$ across, roughly a million times smaller in area than a transmon, so in principle the entire semiconductor industry becomes the quantum fab. Operating temperature is $\sim 100\,\mathrm{mK}$ to $1\,\mathrm{K}$, and because the devices are silicon, they are the natural home for putting the control electronics on the same chip.

The modality carries a strategic fork sharp enough to define its two camps: **gate-defined quantum dots** trap electrons in a lithographic well and optimize for manufacturability, while **donor qubits** put the spin on a single precisely-placed phosphorus atom and optimize for intrinsic uniformity. Both crossed the threshold that mattered in 2025. **Diraq and imec** demonstrated $>99\%$ two-qubit fidelity on gate-defined dots made in a standard 300 mm CMOS foundry flow, with four random devices from one wafer all under 1% error (Nature 2025) — the first foundry-made qubits at error-correction-grade fidelity. **SQC** (Michelle Simmons) took the donor route to 99.99% two-qubit fidelity and ran Grover's algorithm at 98.9% without error correction (Nature, Dec 2025), and patterned 250,000-qubit registers in eight hours. **Intel**, **Quantum Motion**, and **Equal1** pursue the dot route into the foundry.

Fidelity is no longer the question; count is. Silicon has tens of qubits where neutral atoms have thousands. The open questions are whether uniformity holds across 1,000+ dots on a wafer, and whether anyone builds a mid-scale (100-qubit) silicon system before the other modalities reach fault tolerance. Short-range connectivity and per-qubit cryogenic wiring are the unsolved pieces, and cryo-CMOS is the bet that closes them.

Topological: the highest risk, highest reward

Every other modality bolts error protection on top with a code. The topological approach builds it into the physics. A qubit is stored non-locally across a pair of **Majorana zero modes** — quasiparticles at the ends of a topological superconductor — so local noise cannot read or corrupt it. If it worked, the hardware error floor would be so low that the crushing error-correction overhead of every other modality would collapse.

After two decades, no independently verified topological qubit exists. **Microsoft** is effectively alone in productizing it: Majorana 1 (Feb 2025) claimed measurement-based topological qubits, but the accompanying Nature paper carried an editorial note that the results "do not represent evidence for the presence of Majorana zero modes." Majorana 2 (Jun 2026) claimed ~ 20-second state lifetimes and pulled the fault-tolerance timeline to 2029. As of mid-2026 there is still no independent replication; Henry Legg's formal Nature comment (Jun 2026) flagged flawed tune-up routines and analysis-code errors, and Microsoft filed a formal reply standing by its results. This is the modality with the widest gap between claim and independent verification, and for good reason — the same research lineage already produced one Nature paper retracted in 2021. The conflict resolves only when an independent lab reproduces topological-qubit *operation*.

Bosonic / cat qubits: attacking the overhead

Rather than fight the error-correction overhead with better qubits, the bosonic approach attacks it with clever encoding. Store the qubit in the many levels of a harmonic oscillator — a superconducting cavity — so one physical mode carries built-in redundancy. **Cat qubits** encode information in two coherent states $|\alpha\rangle$ and $|-\alpha\rangle$; a two-photon drive suppresses bit-flips *exponentially* in the photon number $|\alpha|^2$, leaving only phase-flips for a simple 1D repetition code to catch. A 2D error-correction problem becomes a 1D one.

Alice & Bob (Paris) reached a bit-flip lifetime over one hour (Boson 4, Sep 2025, up from seven minutes the year before) and claims a $\sim 15{:}1$ physical-to-logical ratio in an "Elevator Codes" preprint (Jan 2026, T3). **AWS's** Ocelot chip (Nature, Feb 2025) combined cat stabilization with a distance-5 repetition code and claimed up to $\sim 90\%$ overhead reduction versus surface codes. This modality reuses the entire transmon fab and cryogenics stack, so it inherits that supply chain for free. The load-bearing assumption is whether the noise bias survives *fast gates* on the cat rather than just storing it — if the bias holds only in memory, the overhead numbers, which are mostly extrapolations from single-digit-qubit demos, do not hold up.

Quantum annealing: the one that shipped

Quantum annealing is a distinct model of computation, and one company owns the category. There are no gates and no universal computation. You encode an optimization problem in the couplings of a network of superconducting flux qubits, start in the ground state of a simple Hamiltonian, and slowly interpolate to the problem Hamiltonian so the system settles into a low-energy answer — the adiabatic theorem made into a machine. **D-Wave's** Advantage2 reached general availability in May 2025 with 4,400+ qubits on the Zephyr topology (20-way connectivity), and hybrid solvers push problems to millions of variables. It is the only quantum computer that shipped a usable product years ahead of any gate-model rival.

The honest asterisk is large. Annealing is restricted to optimization and sampling, has no error correction and no threshold theorem, and shows no proven scaling of solution *quality* with qubit count. D-Wave's March 2025 Science paper claimed a quantum advantage in simulating spin-glass dynamics; within days, groups at EPFL and in the US reproduced key results classically with tensor-network and belief-propagation methods. Per the evidence schema's default, the advantage stays contested. No annealing workload yet holds a durable, third-party-audited win against tuned classical heuristics.

A ninth modality, **NV centers in diamond** (Quantum Brilliance), deserves a mention: an electron spin in a diamond defect works as a qubit at *room temperature*, with no laser-vacuum-cryo stack, in a package small enough to deploy at the edge or in a vehicle. Qubit counts are the lowest of any modality — single digits per node — and its strongest near-term value is clearly in sensing. It sits at the boundary of this chapter and the sensing chapter.

The field on one page

MODALITY	LEADERS	BEST 2025–26 NUMBER	COHERENCE	GATE SPEED	CONNECTIVITY	TEMPERATURE
Superconducting	IBM, Google, Rigetti, IQM	Willow below-threshold QEC (T2); IQM				

MODALITY	LEADERS	BEST 2025–26 NUMBER	COHERENCE	GATE SPEED	CONNECTIVITY	TEMPERATURE
2Q (T4)						
–						
	tens of ns	fixed, near-neighbor				
Trapped ions	Quantinuum, IonQ	Helios				
2Q, 98 qubits (T2, validated)	seconds–minutes	µs–ms	all-to-all (in chain)	UHV, near room temp		
Neutral atoms	QuEra, Pasqal, Atom Computing	96 logical qubits / 448 atoms (T2)				

MODALITY	LEADERS	BEST 2025–26 NUMBER	COHERENCE	GATE SPEED	CONNECTIVITY	TEMPERATURE
	sub-μs + overhead	reconfigurable	room temp (UHV)			
Photonic	PsiQuantum, Xanadu	Omega wafer-scale fab (T3)	n/a (loss-limited)	probabilistic gates	native fiber	warm (detectors 2–4 K)
Silicon spin	Diraq/imec, SQC, Intel					
2Q donor;						
foundry dot (T2)	ms-scale	~ns–μs	short-range exchange			
–						
Topological	Microsoft	claimed				

MODALITY	LEADERS	BEST 2025–26 NUMBER	COHERENCE	GATE SPEED	CONNECTIVITY	TEMPERATURE
lifetime (T4, contested)	—	—	—	mK		
Bosonic / cat	Alice & Bob, AWS					
bit-flip lifetime (T4); Ocelot (T2)	~1 s bit-flip	tens of ns	fixed			
Annealing	D-Wave	4,400+ qubits, Zephyr (T4)	short	analog	20-way	

Read across the rows and the picture resolves. Ions and superconducting circuits lead on demonstrated gate fidelity; neutral atoms lead on logical-qubit count; silicon leads on density and manufacturability; annealing leads on qubits shipped; photonics and topological lead on nothing demonstrated but promise the most if their bets land. The modality-comparison heatmap at the end of this chapter plots exactly this trade space — fidelity against count against coherence-to-gate ratio — and its first lesson is that no modality dominates the others; each leads on one axis and trails on another. Its second lesson is in the shading: most of the headline cells are still vendor T4 claims, and only a handful — Willow, Helios, the QuEra logical qubits, the Diraq/imec and SQC silicon results — have been reproduced to T2. They are climbing different faces of the same mountain.

How the numbers get measured

Every fidelity in the tables above is the output of a characterization protocol, and it helps to know which one produced it. **Randomized benchmarking (RB)** runs random sequences of Clifford gates that should compose back to the identity, then reads an average gate error from how fast the return probability decays — cheap, and largely immune to state-preparation and measurement error. **Interleaved RB** inserts one target gate into each sequence to isolate *its* error. **Mirror RB** runs a circuit forward and then in reverse, giving a scalable, volumetric benchmark that keeps working on many qubits at once where plain RB stops being practical. When a full error budget is needed — coherent errors, over-rotations, and all — **gate-set tomography (GST)** reconstructs the actual process matrices of a whole gate set, at far higher experimental cost. These protocols are the instruments behind the coordinates, and which one a vendor quotes says a lot about what the number does and does not capture.

The operational truth underneath them is that a quantum processor is not a fixed object. TLS defects wander, bias points settle, and laser locks drift, so fidelities move with them — devices are recalibrated once or twice a day, and yesterday's best qubit can be today's worst. This is why a practitioner re-reads the live calibration map before every run rather than trusting last week's numbers, and why any single headline fidelity is a snapshot of a moving target, not a fixed property of the machine.

The stack beneath the qubit

The named machines are the visible layer. Underneath sits a supply chain that determines which roadmaps are physically possible, and it is where the real bottlenecks live. This is modality-agnostic "picks and shovels" territory — every architecture needs most of it.

Cryogenics. Superconducting, silicon-spin, and bosonic qubits live at $10\text{–}20\,\mathrm{mK}$, reached by dilution refrigerators exploiting the finite solubility of ^{3}He in ^{4}He. **Bluefors** and **Oxford Instruments** hold over 70% of the market between them (T5); Bluefors' KIDE platform runs IBM's Quantum System Two. The fridge sets three hard ceilings: cooling power at the mixing chamber (a few hundred μW to $\sim 1\,\mathrm{mW}$ at $20\,\mathrm{mK}$), internal volume, and the number of wires you can route down without dumping heat. Each milliwatt of cooling at $20\,\mathrm{mK}$ costs roughly $25\,\mathrm{kW}$ at the wall — the thermodynamic tax on superconducting computing. And ^{3}He itself, a byproduct of tritium decay in nuclear weapons stockpiles, is a strategic chokepoint rarely priced into roadmaps.

Control electronics and cryo-CMOS. The classical layer generates shaped microwave pulses, digitizes readout, and — in the fault-tolerant era — runs a real-time error-correction decoder inside a latency budget under a microsecond per cycle. **Quantum Machines** (co-building DGX Quantum with Nvidia), **Zurich Instruments**, **Qblox**, and **Keysight** dominate room-temperature control at a cost on the order of $1\mathrm{k}+$ per qubit-channel — which drowns in cabling past a few thousand channels. The escape is **cryo-CMOS**: control ASICs (Intel's Horse Ridge, SemiQon, Equal1) fabricated in standard silicon that run *inside* the fridge and multiplex many qubits per line. The core tension is power versus proximity — the closer the ASIC sits to the qubits, the less heat it is allowed to dissipate.

Lasers, detectors, and photon sources. Ion, atom, and NV machines are laser instruments needing many sub-kHz-linewidth stabilized tones; **TOPTICA**, **M Squared**, and **Vescent** quietly sit under most of them, and the move to photonic integrated circuits (IonQ × imec) is the bet that keeps the optics table from becoming the scaling wall. Photonic computing and QKD rest on **single-photon detectors** — superconducting nanowire SNSPDs (**Single Quantum**, **ID Quantique**, **Photon Spot**) now exceed 99% system efficiency with picosecond jitter and sub-Hz dark counts (T2) — and on **deterministic photon sources** — **Quandela's** quantum dots hit $>30\%$ brightness, $g^2(0) < 0.05$, and Hong-Ou-Mandel visibility above 90% (a vendor spec, T4). Both detectors and dot sources still want cryogenics, and making many emitters *mutually* indistinguishable is the unsolved requirement for photonic scale-up. Under all the atomic machines runs a shared ruler: the **optical frequency comb**, now shrinking onto silicon-nitride chips.

Readout amplifiers. Reading a superconducting qubit means measuring a signal carrying a fraction of a photon. A **quantum-limited parametric amplifier** (JPA or broadband TWPA) at $\sim 10\,\mathrm{mK}$ amplifies it while adding the minimum noise physics allows — the half-photon standard quantum limit — before a HEMT amplifier at $4\,\mathrm{K}$ takes over. Without it, fast single-shot readout is impossible. **Low Noise Factory** supplies most of the HEMTs; TWPA yield and uniformity at scale is the open problem.

Fabrication and foundry. The materials base carries the field's most stubborn villain: the **two-level system (TLS)**, an atomic-scale defect in amorphous oxides and interfaces that resonantly absorbs qubit energy and drifts over hours, capping and destabilizing coherence. Raising T_1 is mostly a war against TLS — tantalum films and high-resistivity silicon substrates drove the recent millisecond-transmon record, and 2025 mapping work (arXiv:2511.05365) localized most surface TLS to the junction leads. The **foundry** question is separate: who can make qubits at wafer scale with yield. **imec** is the anchor; **GlobalFoundries** serves photonics (PsiQuantum) and cryo-CMOS (Equal1); **Intel** runs its own 300 mm line. The unresolved question is whether foundry uniformity holds across thousands of qubits, or whether device-to-device spread — the historic silicon-spin killer — reappears at scale.

Packaging, traps, and vacuum. Superconducting chips past a few hundred qubits cannot route every wire in from the plane's edge, so they borrow **flip-chip bonding, through-silicon vias, and interposers** from classical packaging (MIT Lincoln Laboratory, IBM, Google, QuantWare) — each lossy interface a new home for TLS. Ion machines migrated to **microfabricated surface traps** (Sandia's HOA and Enchilada, Honeywell's Helios chip), where the cost of lithographic scalability is anomalous heating from the nearby surface. Atomic machines depend on **ultra-high vacuum** — a single background-gas collision ejects an atom — with cryogenic chambers now reaching $3{,}000$-second trap lifetimes.

The two walls

Two bottlenecks matter more than any qubit count. Both are hard physical limits.

The first is the **wiring and interconnect wall**. A superconducting qubit needs several coaxial lines, each running through four temperature stages with attenuators and bulky ferrite isolators. A million-qubit machine implies millions of coax lines in a single fridge — physically impossible. The whole endgame therefore forks toward cryo-CMOS multiplexing, photonic readout, and multi-fridge modularity. But modularity across fridges needs **microwave-to-optical transduction** — coherently converting a single $\sim 5\,\mathrm{GHz}$ microwave photon into a $\sim 200\,\mathrm{THz}$ optical one and back — and the best 2025 devices reach only $\sim 2\%$ efficiency at near-single-photon added noise (arXiv:2509.26349). Theory says high efficiency with low noise is allowed, so this is an engineering wall rather than a fundamental one; until it falls, superconducting machines stay confined to single-fridge scale. Ion and neutral-atom machines sidestep the problem, networking natively over fiber — Oxford ran a teleported gate between two ion modules (Nature 2025), and a Duke/IonQ collaboration built a three-node GHZ state across remote ion memories (2026, T3). Whether any interconnect stays *below the error-correction threshold* across module boundaries is one of the quietest, most decisive open questions in the field.

The second is **materials and TLS**. The two-level-system bath caps coherence and drifts it over hours, and there is no field consensus that process control alone can drive TLS density down the order of magnitude that fixed-frequency architectures want. Junction-parameter spread sits at $1\text{–}2\%$ where sub-0.1% is wanted. Yield at wafer scale is the number that gates manufacturability, and it is the number vendors publish least — single-device records do not answer it.

Key takeaways

- No modality dominates. Each leads on one axis — fidelity, logical-qubit count, density, or qubits shipped — and trails on another.
- Trapped ions and superconducting circuits are the two most credible gate-based platforms, set apart by having their headline results reproduced by outside groups.
- Neutral atoms hold the logical-qubit record at 96; silicon spin is the long-term manufacturing threat; photonics and topological are asymmetric bets with nothing yet demonstrated.
- A vendor benchmark stays a T4 claim until an independent group reproduces it, and only a handful of results have cleared that bar.
- The two unsolved walls — wiring/interconnect and materials/two-level-system defects — are where the field's real physics sits.
- Timing matters as much as the axis: which modality leads depends on the window and on which enabling condition holds.

Where the hardware stands

There is no winner, and anyone who tells you otherwise is selling a modality. As of mid-2026, the race is a set of leaders on different axes, and timing decides as much as the axis — every horizon below is a roadmap read, a T4 judgment about the out-years rather than a demonstrated result. The useful question is less "which modality wins" than "which conditions have to hold for each to lead, and over what window."

Trapped ions and superconducting circuits are the two most credible gate-based platforms today, and they lead by having their headline results reproduced: Quantinuum's 99.921% Helios fidelity validated by Sandia in Nature, and Google's below-threshold Willow demonstration. Ions have the best fidelity and coherence; superconducting has the speed, the deepest industrial base, and the biggest roadmaps — along with the worst wiring problem. Neutral atoms are the fastest-improving platform and hold the logical-qubit record at 96, but their gate fidelities still trail and their continuous-operation story is young. Silicon spin crossed into error-correction-grade fidelity from a real foundry in 2025, which makes it the long-term manufacturing threat, if uniformity holds at scale — and it has almost no qubits yet. Photonics and topological are the asymmetric bets: photonics reuses the telecom and CMOS supply chains and could jump straight to a million qubits, but has no intermediate benchmark to grade; topological would rewrite every roadmap in this chapter, but has zero independently verified qubits after twenty years and one retracted paper in its lineage. Bosonic cat qubits are the most interesting overhead play, riding on whether noise bias survives fast gates. Annealing already ships and already sells, inside a problem class whose quantum advantage remains contested.

Timing sorts these leaders into windows, and each window rests on one enabling condition. Superconducting owns the late 2020s if cryo-CMOS and multi-fridge modularity arrive before the wiring wall caps it near single-fridge scale; if microwave-to-optical transduction stalls, its lead has a hard ceiling. Trapped ions hold the near-to-mid term for small, high-fidelity circuits where every gate has to count, contingent on photonic interconnect between traps staying below the error-correction threshold. Neutral atoms are the mid-decade wildcard — the fastest route to hundreds of logical qubits in roughly the 2027–2030 window, if continuous operation, fast mid-circuit measurement, and better two-qubit fidelities mature together, which means turning batch-mode demos into repeatable machines. Photonics is a late, binary bet: around the 2030 horizon it either wins outright — room temperature, telecom-native networking, a straight shot to a million qubits — or settles into being the interconnect fabric for everyone else, depending almost entirely on whether the end-to-end photon-loss budget closes. Silicon spin is the 2030s manufacturing play, decided by whether foundry uniformity holds across thousands of dots or device-to-device spread — the historic silicon killer — reappears at scale. Each horizon is a roadmap read, not a demonstrated result.

Every vendor number in this chapter stays a claim until an outside group reproduces it, and the handful that have been reproduced earn their weight precisely because someone did; the rest are coordinates on a roadmap. The two walls, wiring/interconnect and materials/TLS, are where the field's real physics lives, and neither is solved. The next chapter turns from the qubit to the stack that sits on top of it — how you take these noisy, imperfect machines and coax a correct answer out of them.

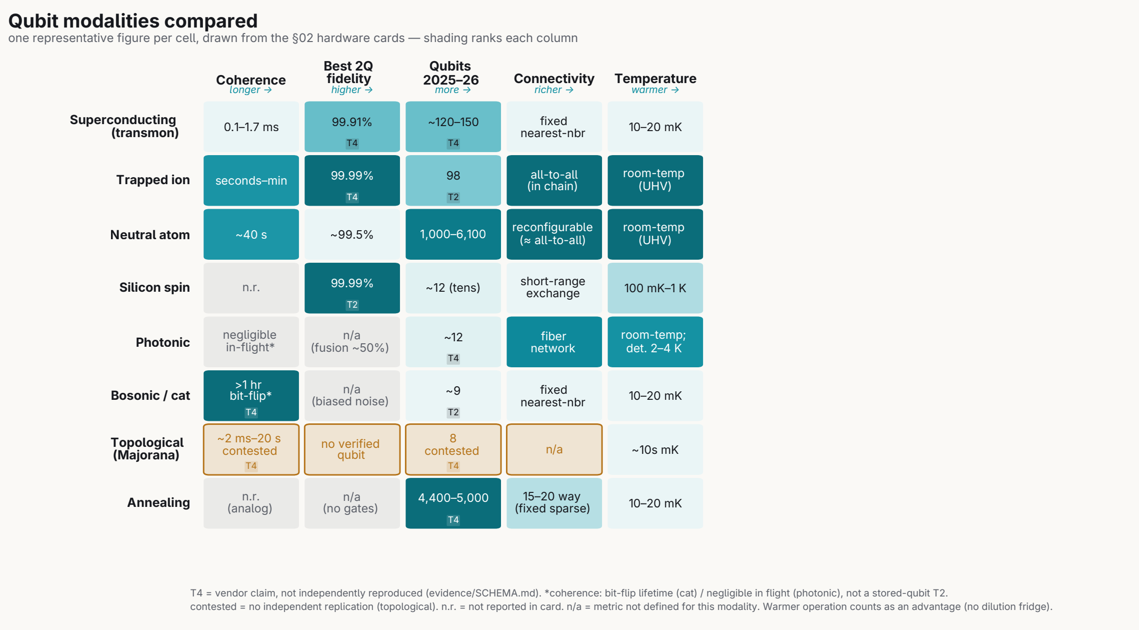
Exercises and discussion

1. **Modality comparison (all levels).** Pick two modalities from the field-on-one-page table — for example trapped ions and neutral atoms — and write a one-paragraph comparison on three axes: coherence time relative to gate time, demonstrated two-qubit gate fidelity, and native connectivity. State which axis each modality wins and what that implies for the kind of circuit it runs best.
2. **DiVincenzo audit (physics).** Take one modality and grade it against all five DiVincenzo criteria plus the two networking criteria. Identify the single criterion where it is weakest and give the physical reason.
3. **Error-budget estimate (CS).** A surface code needs physical two-qubit error rates below roughly 1%. For a modality at 99.5% two-qubit fidelity, targeting one logical qubit at code distance $d = 7$, estimate the physical-qubit count (a distance-d surface code uses _d_^{2} data qubits and _d_^{2} − 1 measurement qubits, so 2_d_^{2} − 1 in total) and say whether that modality sits above or below threshold. Redo the estimate at 99.9% fidelity and comment on how the error margin changes the overhead you would need.
4. **Spec-reading trade-off (engineering).** You are handed two vendor spec sheets: one claims 99.9% gate fidelity on 50 qubits, the other 99.5% on 1,000 qubits, and neither number is independently reproduced. List the questions you would ask — connectivity, measurement fidelity, crosstalk, T2, whether the figure is a best-pair or full-array result — before ranking them. Which would you fund for a fault-tolerant demonstration, and which for a near-term utility experiment?
5. **Seminar — the Majorana conflict.** Trace the record from the 2018 Nature Majorana paper (retracted 2021) to Microsoft's 2023–2025 topological-qubit claims. Debate: what evidence should the field require before it counts a topological qubit as demonstrated, and why does a single retraction cast such a long shadow over an entire modality?
6. **Seminar — the annealing-advantage dispute.** D-Wave has shipped annealers for more than a decade and claims a problem-class advantage that other groups contest. Debate: what would a defensible demonstration of quantum advantage for optimization look like, and how does "advantage" here differ from the below-threshold error-correction results claimed by gate-based platforms?

Further reading

- Google Quantum AI, "Quantum error correction below the surface code threshold," *Nature* **638**, 920–926 (2025). The Willow result — the first demonstration that adding physical qubits lowers the logical error rate.
- D. Bluvstein et al., "Logical quantum processor based on reconfigurable atom arrays," *Nature* **626**, 58–65 (2024). The neutral-atom demonstration that set the logical-qubit record.
- D. P. DiVincenzo, "The physical implementation of quantum computation," *Fortschritte der Physik* **48**, 771–783 (2000). The five criteria (plus two for networking) every physical qubit is graded against.

- P. Krantz et al., "A quantum engineer's guide to superconducting qubits," *Applied Physics Reviews* **6**, 021318 (2019). The standard circuit-QED and transmon review for the incumbent modality.
- C. D. Bruzewicz et al., "Trapped-ion quantum computing: progress and challenges," *Applied Physics Reviews* **6**, 021314 (2019). The companion review for the quality-leading platform.
- A. Chatterjee et al., "Semiconductor qubits in practice," *Nature Reviews Physics* **3**, 157–177 (2021). Silicon-spin qubits from a device and manufacturing standpoint.



Qubit modalities compared

one representative figure per cell, drawn from the §02 hardware cards — shading ranks each column

	Coherence (longer →)	Best 2Q fidelity (higher →)	Qubits 2025–26 (more →)	Connectivity (richer →)	Temperature (warmer →)
Superconducting (transmon)	0.1–1.7 ms	99.91% T4	~120–150 T4	fixed nearest-nbr	10–20 mK
Trapped ion	seconds–min	99.99% T4	98 T2	all-to-all (in chain)	room-temp (UHV)
Neutral atom	~40 s	~99.5%	1,000–6,100	reconfigurable (≈ all-to-all)	room-temp (UHV)
Silicon spin	n.r.	99.99% T2	~12 (tens)	short-range exchange	100 mK–1 K
Photonic	negligible in-flight*	n/a (fusion ~50%)	~12 T4	fiber network	room-temp; det. 2–4 K
Bosonic / cat	>1 hr bit-flip* T4	n/a (biased noise)	~9 T2	fixed nearest-nbr	10–20 mK
Topological (Majorana)	~2 ms–20 s contested T4	no verified qubit	8 contested T4	n/a	~10s mK
Annealing	n.r. (analog)	n/a (no gates)	4,400–5,000 T4	15–20 way (fixed sparse)	10–20 mK

T4 = vendor claim, not independently reproduced (evidence/SCHEMA.md). *coherence: bit-flip lifetime (cat) / negligible in flight (photonic), not a stored-qubit T2.
contested = no independent replication (topological). n.r. = not reported in card. n/a = metric not defined for this modality. Warmer operation counts as an advantage (no dilution fridge).

The qubit-modality landscape — modalities × key metrics. Vendor numbers labelled T4; contested cells marked.

PART II

From Qubit to Answer

PART II · FROM QUBIT TO ANSWER

3 From Qubit to Answer

The two chapters before this one built the raw material. Chapter 1 gave the physics — superposition, entanglement, the Born rule, decoherence as the enemy. Chapter 2 gave the machines — transmons, trapped ions, neutral atoms, and the cryostats and control lines that keep them barely coherent. This chapter is the bridge. It answers one question in stages: how does a single noisy physical qubit, which loses its state in microseconds, become a computation you can trust?

The path runs uphill in a fixed order, and the order is the argument. First you need a language for computation — gates and circuits, and a guarantee that a small set of them can express anything. Then you hit the wall the hardware imposes: today's devices are too noisy to run deep circuits and too small to correct themselves. From there the field splits into two responses — cheap, near-term error *mitigation* that reduces the bias in an estimate, and expensive, asymptotically-sound error *correction* that protects the quantum state itself. Correction, pushed far enough, yields the logical qubit: the unit the famous algorithms are actually written for. Most of those algorithms assume a machine that does not yet exist, so the second half of the chapter grades them by what they promise and how much survives contact with the overhead below.

Keep one habit from Chapter 0 in mind throughout: every claim carries an evidence tier, and a vendor announcing its own benchmark is T4 until someone independent reproduces it. Speedups are graded the same way. The interesting question is never whether something is faster in principle. It is whether it beats the best *classical* method once you charge honestly for loading the data, correcting the errors, and reading the answer out.

Learning objectives. After this chapter you can:

- Explain why a small set of gates is universal, and read a circuit as the program a quantum computer actually runs.
- Distinguish error *mitigation* from error *correction*, and state what the threshold theorem promises and what it demands in return.
- Estimate the physical-qubit cost of one logical qubit at a target logical error rate, given a code distance and a physical error rate.
- Grade an algorithm by the *kind* of speedup it claims — exponential, quadratic, or heuristic — and say how much survives fault-tolerant overhead and honest data loading.
- Name the load-bearing assumptions — qRAM, an efficient block-encoding, a good initial state — that decide whether a paper speedup ever reaches a real workload.
- Apply the evidence tiers to a vendor benchmark or speedup claim, and explain why a self-reported number sits at T4 until someone independent reproduces it.

Gates, circuits, and the meaning of universality

A quantum computation is a circuit: a sequence of unitary gates applied to qubits, ending in measurement. A finite set of gates is **universal** if composing them approximates any unitary to arbitrary precision. The canonical universal set is Clifford — the Hadamard H, phase gate S, and CNOT — plus the single non-Clifford T gate. Two theorems fix the meaning of this. The Solovay–Kitaev theorem says synthesis is efficient: approximating any single-qubit unitary to precision ε costs only $O(\log^c(1/\varepsilon))$ gates, polylogarithmic overhead. The Gottesman–Knill theorem draws the line that matters for cost: Clifford-only circuits are classically simulable in polynomial time. So the T gate is where quantum hardness lives, and the true currency of a fault-tolerant resource estimate is **T-count** and **T-depth** — how many non-Clifford gates the circuit needs, and how many must run in sequence. Raw gate count barely tracks the real cost. A circuit that looks short can be enormous in T-count, and that number is what decides whether an algorithm is affordable.

Universality speaks to *reachability*: it guarantees any unitary is expressible and says nothing about whether the circuit is cheap. Every hardware platform also exposes its own small native set — CZ or iSWAP on superconducting chips, Mølmer–Sørensen on ions, Rydberg-blockade CZ on neutral atoms — and the compiler rewrites everything into it, paying SWAP-routing overhead on chips with limited connectivity while all-to-all platforms like trapped ions avoid routing entirely. The binding constraint on everything above is two-qubit gate fidelity. As of late 2025 the best reported is Quantinuum's Helios at 99.921% across all pairs; leading superconducting devices sit near 99.5–99.9%. Those decimal places decide whether error correction works at all.

The NISQ reality: noise as the governing fact

Preskill's 2018 coinage, **NISQ** (Noisy Intermediate-Scale Quantum), named the era we are still in: 50 to a few thousand physical qubits, gate error rates around 10^{-3}, too noisy for deep circuits and too few for full correction. Because circuit fidelity decays roughly exponentially in the number of two-qubit gates, a NISQ device has a hard depth budget. Run past it and the output is indistinguishable from noise.

This single fact governs the rest of the chapter. It is why a whole family of near-term algorithms (variational methods, most of QML) was designed to stay shallow, and why the honest verdict on the NISQ era is sobering. The high-water mark was IBM's 2023 "utility" experiment — a 127-qubit kicked-Ising circuit run with error mitigation, published in *Nature* — reproduced within weeks on a laptop by classical tensor-network methods. That exchange set the field's expectations: if a circuit is shallow enough for near-term tricks to work, it is usually shallow enough for a competent classical team to simulate.

Error mitigation: accuracy without protection

The first response to noise is **error mitigation**: spend extra circuit runs to reduce the bias in an estimated expectation value, without encoding anything. Mitigation reduces the bias in an estimate while leaving the quantum state itself untouched; protecting the state is the work of correction, which comes later. The two workhorses (Temme, Bravyi, Gambetta 2017) are **Zero-Noise Extrapolation (ZNE)**, which deliberately amplifies the noise by folding gates or stretching pulses, measures the observable at several noise levels, and extrapolates back to the zero-noise limit; and **Probabilistic Error Cancellation (PEC)**, which learns a noise model and inverts it by sampling from a quasiprobability decomposition. PEC is unbiased, but the sampling overhead grows exponentially in circuit size. The wider taxonomy adds symmetry verification, virtual distillation — which suppresses errors as O(1/M) in the number of copies M — and learning-based methods.

The defining theoretical result is a hard ceiling: the sampling overhead of *any* mitigation grows exponentially with circuit depth and fault rate (Takagi et al.; Quek et al., *Nat. Phys.* 2024). Mitigation is therefore a constant-to-polynomial extension of reach, never an asymptotic fix. Its honest role, in 2025–26, is to buy accuracy on small *logical* devices during the early fault-tolerant window — layered on top of correction — rather than to rescue bare NISQ. Any advantage claim resting on mitigation alone is exposed to classical counterattack by construction. The reference implementation at the end of this chapter names ZNE and readout calibration as its next lever, and measures the gain: on its modeled noise channel the two together cut mean error by 77%.

Quantum error correction and the threshold theorem

Mitigation manages noise. **Quantum error correction (QEC)** defeats it. QEC encodes one *logical* qubit into many physical qubits so that errors are detected and corrected faster than they accumulate. The theoretical license for the whole enterprise is the **threshold theorem** (Aharonov–Ben-Or; Kitaev;

Knill–Laflamme–Zurek, 1996–98): if the physical error rate p sits below a code-dependent threshold p_{th}, then arbitrarily long computation is possible with only $\mathrm{polylog}(1/\varepsilon)$ overhead per logical operation. The condition $p < p_{th}$ is the hinge of the field. Above threshold, adding qubits makes things worse; below it, adding qubits makes things better, and scaling becomes an engineering problem rather than a physics one.

The **surface code** (Fowler et al. 2012) has been the default: a nearest- neighbor 2D layout with threshold near 1%, where a distance-d code uses d^2 data-plus-measure qubits and corrects $\lfloor (d-1)/2 \rfloor$ errors. In the smallest useful instance, a rotated distance-3 code spends 17 physical qubits — 9 data plus 8 measure — on a single logical qubit. Its weakness is encoding rate — one logical qubit costs many physical qubits. **qLDPC codes** promise a far better rate, packing more logical qubits per physical qubit, at the price of long-range connectivity the hardware must supply. Non-Clifford gates (T) cannot be done transversally and need **magic-state distillation**, historically the dominant cost of fault-tolerant computing; the newer **magic-state cultivation** cuts it further.

Two 2024–25 results anchor where this stands. Google's **Willow** (Nature 2024) is the era-defining demo: a distance-7 surface-code memory whose logical error rate *halves* with each increase in distance — a suppression factor $\Lambda \approx 2.14$, the first clear below-threshold memory, meaning bigger codes finally help. IBM pivoted to qLDPC with its [[144,12,12]] "gross code," protecting 12 logical qubits in 288 physical versus roughly 3,000 for equivalent surface-code protection — about a $10\times$ encoding-rate win — and its Loon chip (2025) demonstrated the long-range couplers qLDPC requires. Gidney's magic-state cultivation (2024) cut projected T-gate costs roughly $10\times$, feeding the falling Shor estimates below. Overhead is the whole game: ~ 100–$1{,}000$ physical qubits per logical qubit at useful error rates, and correction *slows* the logical clock because each logical gate spans many syndrome cycles plus decode latency. To make the overhead concrete: at a physical error rate of 10^{-3}, driving the logical error rate down to 10^{-12} takes code distance 23, about 1,057 physical qubits for one logical qubit. Surface code versus qLDPC is a live architectural fork, and whether qLDPC's connectivity stays below threshold across real couplers is unproven.

Logical qubits, decoders, and the substrate beneath them

A logical qubit is the corrected unit that algorithms actually consume, and fault tolerance means every operation — including the correction step itself — is done so that a single physical fault cannot cascade into a logical error. This is the line between demos and utility. Useful chemistry or cryptanalysis needs *hundreds to thousands* of logical qubits at logical error rates of 10^{-9} or better, executing *millions to billions* of logical gates — a regime entirely unlike the shallow error-detected circuits shown so far.

The logical-qubit count became the headline metric, which invites cherry- picking. The honest reading requires three numbers alongside any count: code distance d, measured logical error rate per cycle, and whether a *universal* gate set was exercised or the device merely post-selected on error *detection*. A distance-2 detection-only "logical qubit" and Willow's distance-7 corrected memory are different animals. With that caution, the 2025–26 milestones are real: Quantinuum's Helios produced 48 error-corrected logical qubits at a 2:1 physical:logical ratio plus a 94-logical-qubit GHZ state, both beating break-even (logical outperforms physical on the same task); QuEra demonstrated 96 logical qubits from 448 neutral atoms via a high-rate [[16,6,4]] code with below-threshold suppression across all 96, peer-reviewed in *Nature* (Jan 2026). Break-even is the meaningful 2025 threshold; deep universal computation across many logical qubits at once is the next, and nobody has shown it.

Two supporting disciplines make fault tolerance physically possible. **Real-time decoding** is the classical co-processor that reads the syndrome stream and infers the correction before the next cycle. The throughput wall is brutal — a large machine produces on the order of ~ 100 TB/s of syndrome data, and if the decoder falls behind, the error backlog diverges. Below even that sits **quantum optimal control** — GRAPE, Krotov, CRAB, DRAG — shaping the analog waveform that realizes each gate. The

2025 finding there is a note of restraint: well- calibrated DRAG already runs near the decoherence floor for single-qubit gates, so heavy numerical optimization pays off mainly for two-qubit and leakage-heavy gates and for resilience against drift.

That is the full ladder from a noisy qubit to a trustworthy logical operation. Everything below the algorithms — gates, mitigation, correction, decoding, control — exists to manufacture one thing: a logical qubit good enough and plentiful enough to run the programs the field was invented for. Now the programs.

The algorithms, graded by what they promise

Read the algorithm zoo by the *kind* of speedup claimed and how much survives the overhead just described. Three tiers matter: the provable-exponential family, the quadratic family, and the heuristic near-term family — with a modern unifying lens sitting over all of them.

The exponential promises: Shor, simulation, phase estimation, HHL

Shor's algorithm (1994) is the reason cryptography must migrate. It factors integers and computes discrete logarithms in polynomial time, $O((\log N)^2 \log\log N)$, by reducing factoring to period-finding and reading the period off with the quantum Fourier transform. Against the best classical method — the general number field sieve at sub-exponential $\exp(O((\log N)^{1/3}(\log\log N)^{2/3}))$ — this is a *superpolynomial* separation, and it breaks RSA, finite-field Diffie–Hellman, and elliptic-curve cryptography. Two honesty notes. No cryptographically relevant number has ever been factored quantumly — the honest records are 15 and 21, and larger "factorings" used shortcuts that presuppose the answer's structure and do not scale. And the separation is over the best *known* classical algorithm; factoring has no proven classical lower bound, so Shor is not a complexity-theoretic separation the way sampling tasks aim to be. The live action is resource estimation, and it keeps falling: Gidney (May 2025) cut the projected RSA-2048 machine from his own 2019 figure of ~ 20M noisy qubits over 8 hours to under 1M noisy qubits running under a week. That $\sim 20\times$ drop in six years is why the harvest-now-decrypt-later threat is a present-day problem even though the machine is a 2030s prospect.

Quantum simulation is the best-founded advantage in the whole portfolio, and Feynman's original 1981 pitch. Simulating quantum dynamics on a quantum machine is exponentially hard for known classical methods on *generic* systems, and Lloyd (1996) proved local-Hamiltonian dynamics is simulable in polynomial time on a quantum computer — the theorem that makes the class rigorous. The digital route implements e^{-iHt} by one of four method families: **Trotter–Suzuki** product formulas, simple and ancilla-free, which enjoy commutator error scaling that often beats worst-case bounds; **qDRIFT**, whose cost scales with the Hamiltonian's L1 norm $\lambda = \sum_k |h_k|$ and is independent of the number of terms; **LCU**, reaching $O(\log(1/\varepsilon))$ precision; and **qubitization**, achieving the provably optimal query complexity $O(\lambda t + \log(1/\varepsilon))$. Which wins is instance-dependent and actively contested — second-order Trotter is often competitive on real molecules because commutator scaling beats the λ-scaling, while LCU and qubitization win at high precision. The fine print: classical tensor-network and neural-quantum-state methods keep absorbing the low-entanglement and short-time instances, so the advantage lives in *long-time, high-entanglement* dynamics. Static ground-state chemistry, the real commercial prize, is QMA-hard in general; the machine helps with specific instances without a blanket guarantee, and it inherits a state-preparation problem — you must load a good approximate ground state before phase estimation refines it. The fault-tolerant showcase target, the FeMoco cofactor of nitrogen fixation, anchors the resource-estimate story: the 2021 Google/Lee study put it near ~ 4M physical qubits (~2,100 logical) and days of runtime, and Alice & Bob's 2025 cat-qubit estimate cut that to ~99,000 physical qubits — a ~27× reduction against the 2.7-million-qubit configuration they benchmarked (the same figures Chapters 5 and 8 carry).

The primitive under both Shor and chemistry is **Quantum Phase Estimation (QPE)**. The QFT itself uses only $O(n^2)$ gates on n qubits versus $O(n \cdot 2^n)$ for the classical FFT, but on its own it is not a speedup — you cannot read the coefficients out. QPE turns it into an engine: given $U\left|\psi\right\rangle = e^{2\pi i\varphi}\left|\psi\right\rangle$, it estimates the phase φ to n bits with exponential precision, at the cost of $O(2^n)$ controlled-U applications and a deep coherent circuit. That depth is exactly why QPE belongs to the fault-tolerant era, and why the field invented low-depth variants (iterative QPE, Lin–Tong early-fault-tolerant ground-state estimation, statistical phase estimation) that trade depth for repetitions. Its recurring hidden cost is eigenstate preparation — the same assumption that haunts the next algorithm.

HHL (Harrow–Hassidim–Lloyd, 2009) is the canonical read-the-fine-print algorithm. It solves Ax = b by preparing a state $\left|x\right\rangle \propto A^{-1}\left|b\right\rangle$ in time $O(\log(N) \cdot s^2 \cdot \kappa^2/\varepsilon)$ — polylogarithmic in dimension N, an *exponential* improvement over classical $O(N^3)$ elimination. The exponential survives only if four assumptions hold *together*: (1) A is sparse and well-conditioned, since runtime is quadratic in the condition number κ; (2) $\left|b\right\rangle$ can be prepared efficiently — a qRAM assumption; (3) A can be Hamiltonian-simulated efficiently; and (4) you want a scalar property $\left\langle x\right| M \left|x\right\rangle$ rather than the full vector, because reading out all N amplitudes costs O(N) tomography and erases the speedup. Ewin Tang's 2018 dequantization of quantum recommendation systems, and the quantum-inspired classical algorithms that followed, removed the exponential gap entirely for *low-rank* instances. Matrix inversion is BQP-complete, so a provably hard regime exists — sparse, high-rank, well-conditioned, with efficient state prep and property readout — but fifteen years on, no practical problem has been shown to live there.

The load-bearing weak link across this whole tier is **quantum RAM**: the assumed device that loads N classical numbers into superposition in $\operatorname{polylog}(N)$ time. If you cannot beat O(N) loading, the exponential advantage of an algorithm that then processes the data in polylog time evaporates. The bucket-brigade design keeps only $O(\log N)$ switches active per query but needs O(N) physical components, and if each must be error-corrected the overhead can swallow the gain. The first demonstrations arrived in 2025–26 at 4- and 8-bit scale with query fidelities of 0.81 and 0.60 — far from the millions of cells the speedup algorithms assume. This is the single biggest asterisk on data-heavy quantum advantage.

The quadratic promises: Grover, amplitude estimation, walks

Grover's algorithm (1996) finds a marked item among N unstructured possibilities in $O(\sqrt{N})$ oracle queries versus O(N) classically, by amplitude amplification, and the $\Omega(\sqrt{N})$ bound is provably optimal for black-box search. On 3 qubits the mechanism is visible: the probability of the marked item climbs from 0.125 to 0.945 at the optimal two iterations, then over-rotates back down if the circuit runs past that optimum. The 2020s reassessment is where the atlas's accounting matters most. Careful architecture-level accounting (Babbush et al. 2021; Hoefler–Häner–Troyer 2023) shows that quadratic speedups almost certainly deliver *no practical advantage* on foreseeable fault-tolerant hardware. Three things eat the $\sqrt{N}$ gain: the QEC overhead, the slow $\sim$ kHz–MHz logical clock, and the cost of implementing the oracle as a reversible circuit rather than a cheap classical function call. Worse, Grover needs $\sim \sqrt{N}$ *sequential* coherent iterations and admits no parallel speedup (Zalka 1999), so more qubits do not buy wall-clock time. Its symmetric-key-crypto consequence is modest — double the key length, AES-128 to AES-256 — which is why NIST treats Grover as minor next to Shor. Nobody credible pitches bare Grover search as a near-term product.

Amplitude estimation is the most likely survivor of the quadratic family. QAE estimates an amplitude to additive error ε with $O(1/\varepsilon)$ oracle calls versus classical Monte Carlo's $O(1/\varepsilon^2)$ — a quadratic speedup for expectation values, the core operation in option pricing and risk (where JPMorgan, Goldman, and HSBC have published implementations). The same two caveats bite: canonical QAE needs deep circuits (hence the iterative and maximum-likelihood variants), and the speedup assumes an efficient oracle that loads the target distribution into amplitudes — the qRAM problem again. Full resource accounting pushes useful financial QAE firmly into the fault-tolerant era, and whether the quadratic gain survives it is unresolved. **Quantum walks** round out the tier: element distinctness in $O(N^{2/3})$ queries — a *sub*-quadratic improvement over the classical O(N): the speedup factor is only N^(1/3), short of the square-root factor a true quadratic (Grover) buys, though it is provably optimal in the query model — and multi-particle continuous-time walks are BQP-universal, so walks are a full model of computation. But the speedups are query-model and polynomial, so the same overhead accounting applies.

The lesson is structural: a quadratic speedup is a factor of $\sqrt{N}$, and fault-tolerant overhead plus a slow logical clock is a large factor working the other way, so the crossover where quantum wins sits beyond the sizes real workloads reach. Exponential speedups have room to absorb the overhead; quadratic ones rarely do.

The near-term promises: variational methods and QML

The **variational family** — VQE for chemistry, QAOA for combinatorial optimization — pairs a shallow parameterized circuit $U(\theta)$ with a classical optimizer, designed explicitly for NISQ hardware. The theory turned against the program on three fronts. **Barren plateaus** (McClean et al. 2018): for expressive random ansätze the gradient variance vanishes exponentially in qubit count, so training needs exponentially many shots to see a signal. **Soft dequantization** (Cerezo et al. 2025): the circuit families that provably *avoid* barren plateaus turn out to be classically simulable — if a landscape is trainable it is likely also classically tractable, squeezing the useful regime from both sides. And **classical competition**: fixed-depth QAOA is provably beaten by classical algorithms on broad instance classes, and no VQE/QAOA run has beaten best-in-class classical chemistry (DMRG, CCSD(T)) or optimization (Gurobi-class solvers) on any real instance. Both are heuristics with no proven speedup, and every empirical claim has been matched classically once a competent classical team responded. The surviving role is as a *state-preparation subroutine* feeding fault-tolerant phase estimation.

Quantum machine learning is the most hype-inflated corner of the stack, and this atlas grades it hardest. The clearest object is the quantum kernel: encode each datum x into a state $|\phi(x)\rangle$ and use $k(x,y) = |\langle\phi(x)|\phi(y)\rangle|^2$ as the kernel a classical SVM consumes. The record on *classical* data is negative. Tang's dequantization killed the flagship; random-Fourier-feature methods reproduce quantum-kernel performance classically; and a Dec 2025 peer-reviewed benchmark found plain logistic regression beating quantum SVMs, quantum k-NN, and variational classifiers on 4 of 5 realistic tabular datasets, one classifier scoring below random guessing. What survives is narrow and real: a *provable* kernel separation exists for contrived, cryptographically structured data (discrete-log), and an *exponential* advantage exists for learning from *quantum* data when a quantum memory is available (Huang et al., *Science* 2022). The defensible position: QML on classical data has no evidence of practical advantage; QML on quantum data is a real but niche direction awaiting hardware.

The unifying lens and the honest measurement tools

Modern algorithm design speaks one language. **Quantum singular value transformation (QSVT)** (Gilyén–Su–Low–Wiebe 2019) takes a matrix A block-encoded inside a larger unitary and applies a polynomial transform to its singular values using $O(\deg p)$ calls. Choose the polynomial and you recover most known algorithms: $p \approx \mathrm{sign}$ gives amplitude amplification / Grover; $p \approx 1/x$ gives matrix inversion / HHL, at *linear* rather than quadratic κ-dependence; $p \approx e^{-ixt}$ gives Hamiltonian simulation at the qubitization optimum; a rectangular window gives phase estimation. QSVT is not itself a speedup — it is the design framework yielding near-optimal versions of the speedups already in this atlas, and it inherits their caveats. Its edifice sits on an efficient block-encoding of the input, and where that is only "sampling access" to a low-rank matrix, Gharibian–Le Gall showed QSVT is classically dequantizable. It sharpens *what* is provably fast and *where* the classical counterattack bites, in one language.

Classical shadows (Huang–Kueng–Preskill 2020) is the honest measurement tool. Apply random Pauli rotations, measure, invert the channel, and from $O(\log M/\varepsilon^2)$ snapshots you can predict M different local observables — a sample complexity *independent of system size*, saturating information-theoretic lower bounds, with the target properties chosen after the data is collected. It buys measurement efficiency rather than a computational speedup, and its variance blows up as 3^k for k-local observables, so it shines for low-weight, many-observable workloads. It is also the rigorous route that makes "learning from quantum data" provable.

A worked example: correct, hardware-validated, and not a speedup

The reference implementation is the chapter in miniature — a canonical quantum primitive that is correct and offers no speedup. It amplitude-encodes a normalized vector x into a state $|\psi_x\rangle = \frac{1}{\|x\|}\sum_i x_i\,|i\rangle$, so a 1024-dim embedding fits in 10 qubits. The whole engine is one identity: the overlap of two encoded states equals the cosine similarity of the originals, $\langle\psi_u|\psi_v\rangle = \cos(u,v)$. It measures that overlap two ways — the **swap test**, yielding $|\cos|^2$ via $P(0) = \frac{1}{2} + \frac{1}{2}|\langle\psi_u|\psi_v\rangle|^2$, and the **Hadamard test**, yielding the *signed* cosine from $P(0) = \frac{1}{2} + \frac{1}{2}\,\mathrm{Re}\,\langle 0|U\rangle\,0$ in fewer qubits and a shallower circuit. Stacked into a Gram matrix $K_{ij} = k(x_i, x_j)$, the quantum-estimated kernel drives a classical SVM to 100% on Iris (kernel RMSE 0.007), and the signed estimator runs on IBM's `ibm_fez` with mean error 0.009 across the full range, at the shot-noise floor that falls as $1/\sqrt{S}$.

The accounting is the point. Overlap estimation is a real subroutine — it underlies quantum kernels and HHL-style linear algebra — but exact amplitude encoding of an arbitrary vector costs up to $O(2^n)$ gates, while classical cosine similarity is O(d). The encoder dominates, so classical wins today. It is the whole chapter compressed: a correct quantum primitive, validated on hardware, with a measured error budget, whose speedup lives or dies on the data-loading assumption that no scalable device yet satisfies.

The counterfactual, the theory, and the machinery around it

Every advantage claim is adjudicated against a moving classical baseline. **Tensor-network simulation** is that baseline: a state or circuit represented as a graph of small tensors, efficiently contractible when entanglement is low. Circuits that stay shallow, geometrically local, noisy, or short-time are classically simulable, sometimes on a laptop; volume-law entanglement across a hard-to-contract geometry is where a quantum device might win. Tensor networks erased Sycamore's "10,000 years" (cut to hours then minutes) and matched IBM's 2023 kicked-Ising utility on a laptop. Google's Willow RCS (Dec 2024) and "quantum echoes" OTOC (Oct 2025, a claimed verifiable $13{,}000\times$) are the current standing claims tensor networks have *not yet* matched — the live test.

Complexity theory (safe to skip on a first read) says why no exponential quantum speedup is unconditionally proven in the standard model. Since $\mathrm{BPP} \subseteq \mathrm{BQP} \subseteq \mathrm{PSPACE}$, proving $\mathrm{BQP} \neq \mathrm{BPP}$ would resolve P versus PSPACE — beyond current mathematics. So the honest hierarchy of claims runs: unconditional-but-shallow (Bravyi–Gosset–König constant-depth separation) $>$ relativized (Raz–Tal, Simon) $>$ assumption-based (sampling advantage, hard only if the polynomial hierarchy does not collapse) $>$ best-known-classical (Shor, and most of what people mean by "quantum advantage"). **Benchmarks** are where this plays out publicly, and the field's fragmentation — Quantum Volume, CLOPS, Algorithmic Qubits, layer fidelity, all measured differently — is itself a finding. Put those metrics on one axis and they scatter; the gap between them is the point, because they do not measure the same thing. A benchmark controlled by the vendor reporting it is T4 by rule.

The rest of the machinery makes any of it usable. **Compilers and middleware** — Qiskit, Cirq, PennyLane, tket, CUDA-Q, interoperating through OpenQASM 3 and QIR — multiply hardware without creating asymptotic advantage; a $2\times$ depth reduction is worth roughly a hardware generation because fidelity decays exponentially in gate count. **Cloud access** sets the economics: at $\sim\$96$/QPU-minute on IBM, algorithm development happens on simulators and only validation touches hardware —

the workflow the reference impl documents. **Circuit cutting** trades quantum width for exponential classical sampling, useful only when cuts are few. And **certified randomness** is the one delivered *product* of quantum advantage: the JPMorgan–Quantinuum result (*Nature*, Mar 2025) generated 71,313 provably-random bits on a 56-qubit device, verified at ~ 1.1 exaFLOPS — a supremacy benchmark turned into a narrow service.

The practitioner layer: the daily surface beneath the map

Everything above reads the stack from altitude. The engineer's day happens at the surface, where four realities decide whether a correct algorithm returns a usable number.

Transpilation is where the circuit meets a specific chip. The logical circuit is not what runs. The transpiler maps virtual qubits onto physical ones by reading the backend's *live* calibration data — per-qubit T1/T2, per-gate error, and readout error, all measured hours ago and already drifting — then routes two-qubit gates across limited connectivity by inserting SWAPs, the dominant source of added depth on heavy-hex and grid layouts alike. At `optimization_level=3` the pass is stochastic: the standard practice is transpile-N-keep-best, running the layout-and-route stack many times against the current calibration snapshot and shipping the shallowest, highest-fidelity result. Dynamical-decoupling pulses fill idle windows to fight dephasing, and Pauli twirling converts coherent error into a more benign stochastic form. This is the machinery behind the earlier claim that a 2× depth cut is worth a hardware generation: on a real backend, depth is set by routing and calibration, not by the abstract gate count.

> **Box — the SDK you actually type into (2026).** Qiskit 1.0 (Feb 2024) was a hard break: `execute()` is gone, and you compose an explicit `transpile` step with a **primitive**. The primitives moved V1 → V2 — you call `Sampler` (for bitstrings) or `Estimator` (for expectation values), submitting an array of parameterized circuits rather than one job at a time; IBM Runtime dropped V1-primitive support in August 2024. Error mitigation is no longer hand-rolled research code: ZNE, PEC, and dynamical decoupling are exposed as a single `resilience_level` knob on the Estimator. And OpenPulse is largely closed on new hardware — `qiskit.pulse` was deprecated in v1.3 and removed in v2.0, and Heron-class chips expose fractional gates instead of arbitrary pulse shapes, so the pulse-level control that defined the early NISQ literature is unavailable where it would matter most.

From a demo to a running system. A hero circuit in a paper hides the wall-clock truth: for a hybrid loop, runtime is dominated by *queue wait and per-job submission latency*, not by quantum gate time. A naive VQE that submits one circuit, waits in the fair-share queue, reads a result, and resubmits can spend orders of magnitude more time waiting than computing. **Runtime sessions** exist to fix exactly this — they hold a device slot across many iterations so the classical optimizer and the quantum evaluations interleave without re-queuing, which is what makes iterative VQE and QAOA practical at all. And the feedback that QEC needs — measure a syndrome, decode it, apply a correction *within the coherence window* — is not only a physics requirement: it is programmed as a **dynamic circuit** with real-time classical feedforward, expressed through OpenQASM 3 control flow (`if`, `for`, mid-circuit measurement). The construct under error correction is a software construct as much as a hardware one.

The real error model. Textbook analysis models noise as a depolarizing channel — a clean, memory-less, uniform probability of a random Pauli after each gate. Real devices violate every part of that picture. Errors are *coherent* (systematic over- and under-rotations that accumulate quadratically instead of averaging out), *crosstalk*-ridden (always-on ZZ coupling means a gate on one qubit shifts its neighbors), and *leakage*-prone (population escaping to non-computational levels the code cannot see). Drift is *non-Markovian*: calibration wanders over hours, and cosmic-ray and two-level-system events produce *correlated bursts* that strike many qubits at once. That last property is the practitioner's version of Kalai's correlated-noise argument in Chapter 8 — the same spatial and temporal correlations that threaten the fault-tolerance threshold in principle show up every day as the crosstalk, leakage, and drift the calibration pipeline is forever chasing.

> **Key takeaways**
>
> - Universality is cheap; the wall is noise. Today's devices are too noisy to run deep circuits and too small to correct themselves.
> - Mitigation buys accuracy on an estimate without protecting the state. Correction protects the state and charges for it in overhead the threshold theorem makes worthwhile only once the physical error rate sits below threshold.
> - The logical qubit is the unit the famous algorithms are written for, and at today's error rates it costs on the order of a thousand physical qubits each at a cryptographically-relevant target.
> - Exponential speedups — Shor, quantum simulation — have room to absorb the overhead. Quadratic ones — Grover, amplitude estimation — mostly do not, and the near-term variational and QML family has no proven speedup at all.
> - Every non-oracle exponential is a separation over the best *known* classical method, and the classical frontier keeps moving: tensor networks and dequantization have erased more advantage claims than any quantum device has defended.
> - Data-heavy advantage rests on qRAM, which no one has built at scale. Loading the data honestly is where most speedups die.

Where the algorithms stand

Sort the portfolio by the strength of its promise, and the picture comes into focus.

Provable and structurally sound, pending the machine. Shor's superpolynomial factoring is real over the best known classical algorithm, and its falling resource estimates make post-quantum migration urgent now even though the machine is a 2030s prospect. Quantum simulation of long-time, high-entanglement dynamics is the best-founded advantage in the field — exponential over known classical methods, with the strongest claim on being the first application that pays for itself, most likely in chemistry, materials, and energy. Both need full fault tolerance, which is why the first half of this chapter matters as much as the second.

Real but eroded. The quadratic family — Grover, amplitude estimation, walks — is provably correct and mostly optimal in the query model, and mostly washed out in practice: fault-tolerant overhead and a slow logical clock consume a $\sqrt{N}$ gain at every problem size a real workload reaches. Amplitude estimation for finance is the likeliest survivor, and even it waits on fault tolerance and an honest data-loading oracle. QSVT and classical shadows sharpen the other claims — a design language and a measurement tool — rather than adding new speedups.

Unproven, and graded hardest. The near-term variational family (VQE, QAOA) and QML on classical data have no proven speedup and no empirical win that survived a classical response. Barren plateaus and soft dequantization squeeze the useful regime from both sides. QML on *quantum* data is the honest exception — narrow, real, awaiting hardware. Certified randomness is the one thing shipping, and its market is small.

Two caveats run under all of it. Every non-oracle exponential speedup is over the best *known* classical algorithm, and the classical frontier keeps advancing — tensor networks have erased more advantage claims than any quantum device has defended. And every data-heavy speedup rests on qRAM, which no one has built at scale. The load-bearing unknowns are the overhead beneath the algorithms and the assumptions hidden inside them — which is exactly why the path from a noisy qubit to a trustworthy answer runs through error correction first, and reaches the algorithms last.

Exercises and discussion

Derivation (physics). Grover's algorithm rotates the state vector toward the marked item by a fixed angle 2θ per iteration, where $\sin\theta = 1/\sqrt{N}$. Show that the marked-state probability after k iterations is $\sin^2((2k+1)\theta)$, that the optimum is $k^* \approx (\pi/4)\sqrt{N}$, and that running past k^* drives the probability back down. For $N = 8$ — the 3-qubit case the reference implementation plots — compute P(marked) at $k = 0$, 1, 2 and confirm the 0.125 → 0.781 → 0.945 climb and the over-rotation the figure shows.

Complexity (CS). (a) Finding one marked item among N takes O(N) classical queries and $\Theta(\sqrt{N})$ quantum ones. State the BBBV lower bound that makes $\sqrt{N}$ optimal for black-box search, and explain why this counts as a quadratic speedup. (b) Element distinctness runs in $\Theta(N^{2/3})$ quantum queries against a classical $\Theta(N)$. Show that the improvement is a factor of only $N^{1/3}$ — sub-quadratic — and say in one sentence why a smaller exponent buys a *weaker* speedup than Grover's, even though the walk algorithm is provably optimal in the query model.

Resource estimation (CS / engineering). The surface code suppresses logical error roughly as $p_L \approx A (p / p_{th})^{(d+1)/2}$ for physical error rate p below threshold p_{th}, using $\approx 2d^2 - 1$ physical qubits per logical qubit at code distance d. (a) Confirm that distance 3 needs 17 physical qubits — the 9 data + 8 measure the reference implementation draws. (b) Take $p = 10^{-3}$, $p_{th} \approx 10^{-2}$, and $A \approx 0.1$, and find the smallest odd distance d that reaches $p_L = 10^{-12}$; check it against the chapter's figure of $d = 23$ at ~1057 physical qubits per logical. (c) A cryptographically-relevant machine needs a few thousand logical qubits — estimate the physical-qubit count and compare it to the sub-1M-qubit RSA-2048 estimate in the text.

Spec-reading / trade-off (engineering). A vendor press release claims "a 100× speedup on a real-world optimization problem." Using the chapter's tools, write the five questions that decide whether the claim survives: which classical baseline, and was it best-in-class or a strawman; does the accounting charge for data loading, error correction, and readout; is the number self-reported (T4) or independently reproduced; does the speedup scale, or is it a fixed-size demo; and is the metric one anyone else uses, or a bespoke axis. Grade the claim.

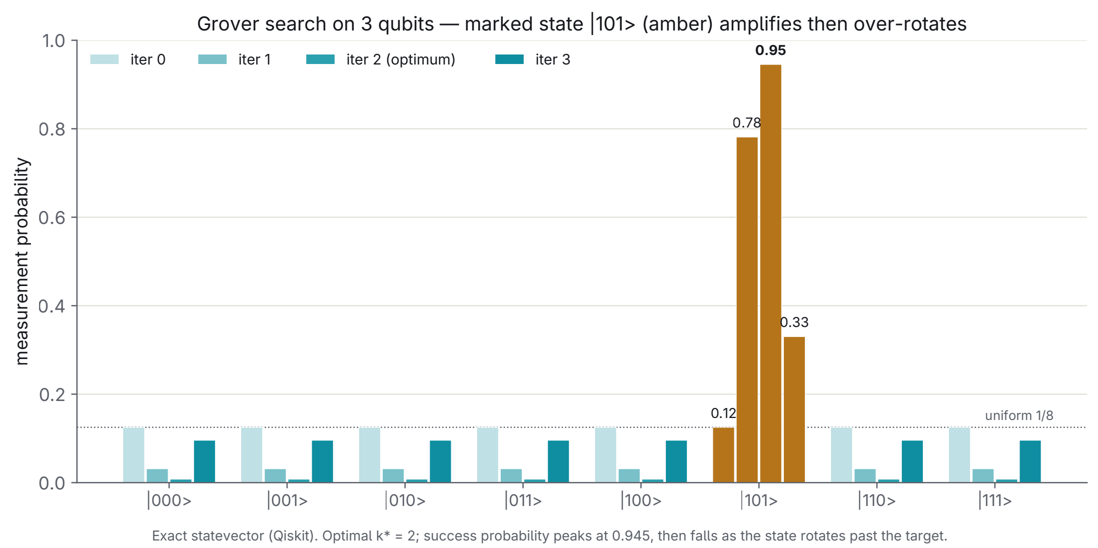
Seminar — the advantage-survival conflict. Chapter 8 records Gil Kalai's argument that correlated noise threatens the fault-tolerance threshold in principle, and this chapter notes the same correlations show up daily as crosstalk, leakage, and drift. Meanwhile the quadratic family is provably correct yet washed out by overhead, and tensor networks and dequantization keep erasing exponential claims. Take the motion "most published quantum-advantage results will not survive the decade," argue both sides using the evidence tiers, and decide which speedups you would bet on.

Seminar — the load-bearing assumption. Pick one of qRAM, an efficient block-encoding, or good initial-state preparation. Trace one algorithm whose exponential speedup depends on it — HHL, a QSVT-based inversion, or phase-estimation chemistry — and argue whether the assumption is an engineering delay or a permanent barrier. The reference implementation is a concrete anchor: its overlap primitive is correct and hardware-validated, yet exact amplitude encoding of an arbitrary vector costs up to $O(2^n)$ gates, which is why it still loses to O(d) classical cosine similarity.

Further reading

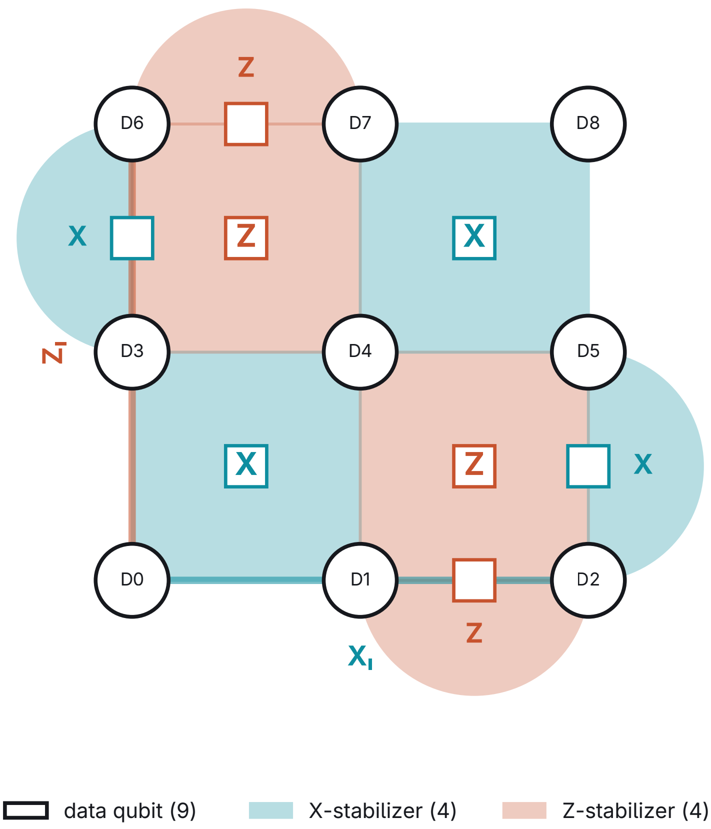
- **Shor, "Algorithms for quantum computation: discrete logarithms and factoring," FOCS 1994** (expanded in *SIAM J. Comput.* 26(5), 1997). The factoring and discrete-log algorithm that makes post-quantum migration urgent — the primary source behind this chapter's exponential tier.
- **Fowler, Mariantoni, Martinis, Cleland, "Surface codes: towards practical large-scale quantum computation," *Phys. Rev. A* 86, 032324 (2012).** The standard reference for the surface code and its resource overhead; the source of the distance-to-qubit-count arithmetic in the resource-estimation exercise.

- **Google Quantum AI, "Quantum error correction below the surface code threshold,"** ***Nature*** **638, 920 (2025)** — the Willow result showing logical error fall as code distance grows ($\Lambda > 1$), the first below-threshold demonstration on real hardware.
- **Gilyén, Su, Low, Wiebe, "Quantum singular value transformation and beyond," STOC 2019.** The qubitization/QSVT framework that recovers search, matrix inversion, and Hamiltonian simulation as choices of one polynomial — the unifying lens of the chapter.
- **Tang, "A quantum-inspired classical algorithm for recommendation systems," STOC 2019** (Ewin Tang, on a challenge from Scott Aaronson). The dequantization result that erased HHL-style exponential speedups for low-rank data and reset the classical baseline.
- **Hoefler, Häner, Troyer, "Disentangling hype from practicality: on realistically achieving quantum advantage,"** ***Comm. ACM*** **66(5), 82 (2023).** The resource-accounting argument that quadratic speedups deliver no practical advantage on foreseeable fault-tolerant hardware.



Grover on 3 qubits: P(marked) climbs 0.125 → 0.781 → 0.945 at the optimum k*=2, then over-rotates.

Rotated surface code, distance d = 3

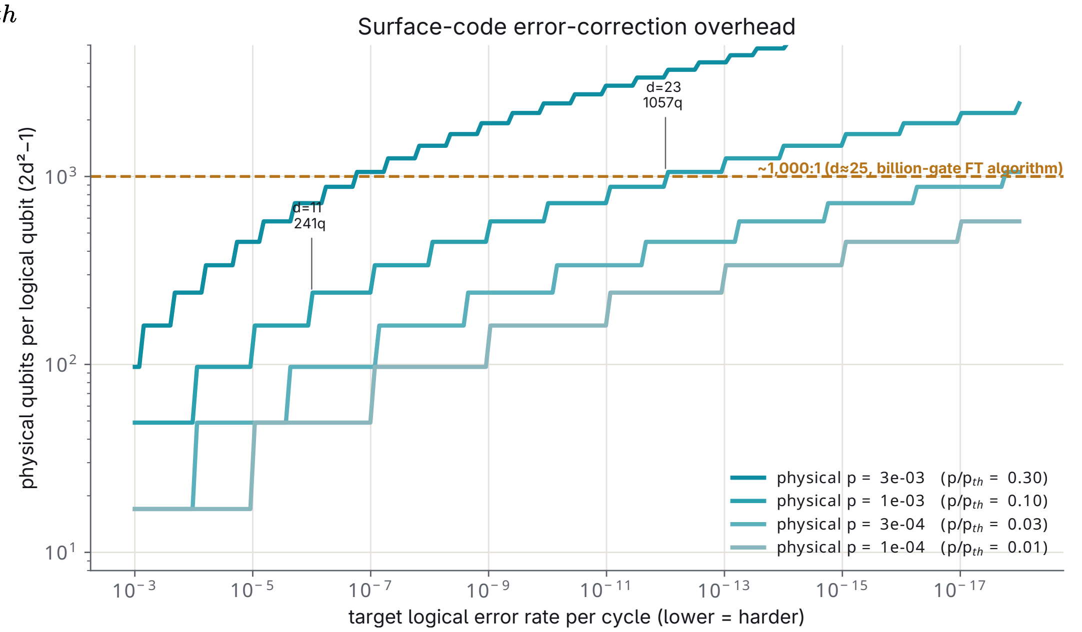


9 data + 8 measure qubits = 17 physical qubits encode 1 logical qubit ($2d^2-1 = 17$).

d = 3 detects up to 2 and corrects any single error. Logical X_l / Z_l are shortest error chains crossing the code (length 3).

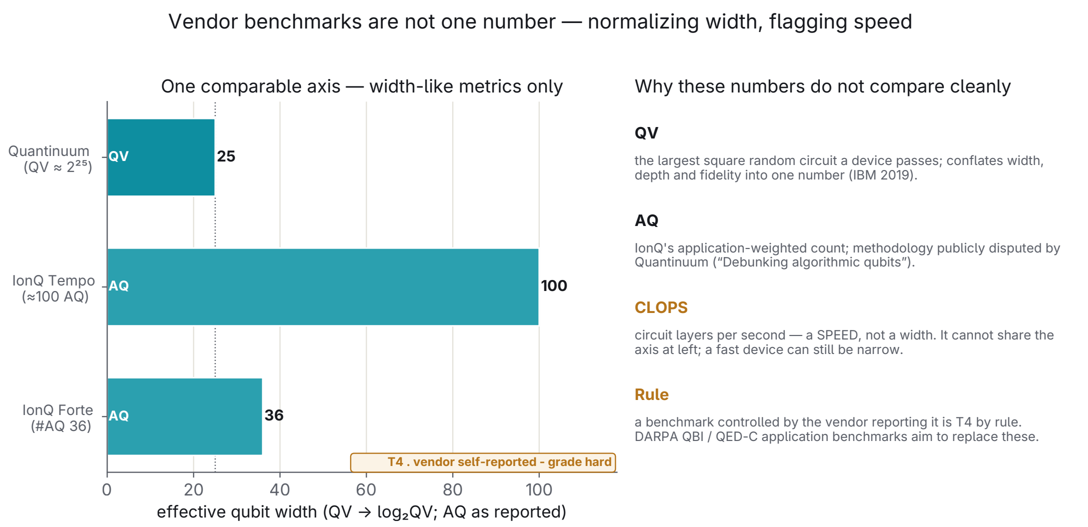
Rotated distance-3 surface code: 9 data + 8 measure = 17 physical qubits for one logical qubit.

th



Memory/Clifford estimate (Fowler et al. 2012 scaling, p_{th}= 1%). A full algorithm adds routing + magic-state factories on top — see O-overhead.

Surface-code overhead: at p=10^{-3}, reaching p_L=10^{-12} needs d=23, ~1057 physical qubits per logical.



Representative vendor-reported values (S-bench.md, H-supercon.md, H-ion.md). All T4 — not independently reproduced.

Vendor benchmarks on one axis — the gap is the point: they don't measure the same thing (all T4).

PART II · FROM QUBIT TO ANSWER

4 Beyond Computing

The first three chapters climbed toward a machine that mostly does not exist yet — a fault-tolerant quantum computer, still years of error-correction overhead away from paying for itself. This chapter turns the other way. The same physics that makes a qubit fragile also makes it useful the moment you stop trying to compute with it. A state that decoheres at the faintest field is a superb detector of that field. A photon that cannot be copied is a courier that cannot be wiretapped without leaving a mark. These are the adjacent technologies: quantum applied to communication, security, sensing, timing, and materials. Several of them ship today, carry revenue today, and have done so for a decade or more. This is where the forecast stops and the inventory begins.

Read it as a maturity gradient. Some nodes are commodity products you can order. Some are advanced R&D with a defense customer and a field-trial press release. Some are beautiful physics with no buyer. The job here is to say which is which, in prose, with the numbers attached, and to grade every vendor claim by the evidence schema — because in this corner the loudest voices are companies with something to sell.

Learning objectives. After this chapter you can:

- Sort the adjacent quantum technologies by maturity — commodity product, defense-led R&D, unbought physics — and place a given claim in the right band.
- Explain why QKD's information-theoretic security proof does not survive contact with trusted relays, and why four signals agencies steer national-security systems to post-quantum cryptography instead.
- Lay out a PQC migration in execution order, and use Mosca's inequality to decide which secrets move first.
- Read a quantum-sensor spec — sensitivity, operating temperature, size — and say why an optically pumped magnetometer competes with a SQUID for magnetoencephalography while an NV-diamond sensor does not.
- Tell a real quantum advantage (entangled or squeezed light) from a "quantum-inspired" classical system, and know why fielded "quantum radar" is still vaporware.
- Grade a vendor's field-trial numbers against the evidence schema, and know which navigation and imaging claims sit in the T4 column pending an independent referee.

Communication and security: QKD and the counterpoint that beat it

Start with the oldest promise. **Quantum key distribution** lets two parties grow a shared secret key whose security rests on physics rather than on the difficulty of a math problem. The no-cloning theorem forbids copying an unknown quantum state, and any measurement disturbs it, so an eavesdropper on the line unavoidably raises the quantum bit error rate and is caught. BB84 (Bennett–Brassard, 1984) encodes bits in non-orthogonal photon polarizations; E91 (Ekert, 1991) draws the key from entangled pairs and certifies it through a Bell violation. In the ideal model the security proof is information-theoretic — the strongest guarantee cryptography offers, graded T1.

The ideal model is where the trouble starts. Every real deployment inherits two hard walls. Fiber loss is exponential at ≈ 0.16–$0.2\,\mathrm{dB/km}$, and no-cloning forbids optical amplification, so unrepeatered reach caps near a few hundred kilometres. And QKD distributes only *keys* — it still needs a classically-authenticated channel and a symmetric cipher to carry any traffic. QKD is a real product line, two decades old, with governments and banks as the buyers. Its measured envelope: twin-field QKD reaches $111.74\,\mathrm{kbit/s}$ of secure key at $202\,\mathrm{km}$, and the same protocol has been pushed to $1002\,\mathrm{km}$ of ultralow-loss fiber (USTC/Pan, PRL 130, 2023, T2) — but at only $\sim 0.0034\,\mathrm{bit/s}$, a headline distance rather than a usable rate. Practical metro systems from Toshiba and ID Quantique run tens of kbit/s to Mbit/s over $10\text{–}100\,\mathrm{km}$. Toshiba launched a Paris service in 2025 and demonstrated cross-state QKD over live commercial fiber from Chicago into Indiana (T4, vendor, Dec 2025). China runs the largest fielded network: a $\sim 2000\,\mathrm{km}$ Beijing–Shanghai backbone (Chen et al., Nature 589, 2021, T2). Europe is building EuroQCI, a continent-wide quantum-secure network with national testbeds from Madrid to Budapest.

Then comes the counterpoint that reframes the whole node. Every backbone in service today depends on **trusted relays** — intermediate nodes that decrypt and re-encrypt because no quantum repeater yet exists to carry entanglement across the gap. A trusted relay knows the key. That single fact collapses the "unhackable" marketing into something narrower: end-to-end security only between adjacent nodes, physical trust everywhere in between. It is why four signals-intelligence agencies — the US NSA, the UK's NCSC, France's ANSSI, and Germany's BSI — advise *against* QKD for national-security systems and steer critical infrastructure toward post-quantum cryptography instead. Their objections are cost, the trusted-relay hole, no defense against store-and-forward attacks at the relays, and the awkward fact that QKD still needs classical (ideally post-quantum) crypto to authenticate the line. QKD vendors and China's program answer that physics-based security is worth the price on the highest-value links. This debate is a first-class analytical thread that runs through the entire communication cluster, so hold it in mind: **QKD is commercial and structurally niche.** It sells real boxes and secures real links. Its addressable market is capped by distance, by the relay compromise, and by the four-agency advice — and it is not the coming default for internet security.

The fix QKD needs is a **quantum internet**: a network that distributes entanglement itself, removing the trusted relay. Because loss is exponential and amplification is forbidden, long links require **quantum repeaters** — quantum memories that store entanglement in short segments, then fuse the segments by entanglement swapping and purification. The single figure of merit is delivered entanglement rate versus distance, and the field's watershed is the moment a repeater beats direct transmission on that metric. That crossover **has not happened anywhere.** Everything fielded so far is a building block toward it. QuTech linked Delft and The Hague over $\sim 25\,\mathrm{km}$ of deployed fiber with heralded entanglement between processor nodes (Stolk et al., Sci. Adv. 2024, T2) — the first multi-node network on installed fiber. A preprint reports memory–memory entanglement over $420\,\mathrm{km}$ (arXiv:2504.05660, T3); USTC reported memory–memory entanglement plus device-independent QKD across $100\,\mathrm{km}$ (Feb 2026, T2/T3). Germany's TD.QR repeater project (started Jan 2026) targets a viable repeater demo by **2028**. Startups sell early gear: Qunnect (memories) and Aliro (network orchestration). This node is **research-stage** — a decade-plus infrastructure bet whose killer application (distributed computing, blind computing, sensor networks) is real in principle and has no paying customer yet.

The near-term shortcut around the fiber wall is space. **Satellite QKD** sends photons through vacuum, where loss beats fiber's exponential attenuation over continental distances. China demonstrated it decisively. **Micius** (2016) distributed entanglement over $1200\,\mathrm{km}$ (Yin et al., Science 356, 2017, T2) and secured the 2017 Beijing–Vienna intercontinental video call — but it was a $\sim 600\,\mathrm{kg}$ spacecraft needing large, expensive ground stations. **Jinan-1** (2022) is the practicality milestone: it shrank the payload to $\sim 22.7\,\mathrm{kg}$, cut spacecraft cost roughly $45\times$, raised key rates by two to three orders of magnitude, and ran *real-time* on-board QKD with portable ground stations, generating up to $1.07\,\mathrm{Mbit}$ of secure key in a single pass and encrypting a China–South Africa image transfer over $12{,}900\,\mathrm{km}$ (Li et al., Nature 2025, T2). Europe is a launch or two behind — EAGLE-1 (ESA/SES) is slated for 2026. The catch is the same one that shadows all of QKD: a single satellite acting as relay is a trusted node that knows the key; the end-to-end-secure version beams entangled pairs to two stations at once and pays orders of magnitude in rate. Satellite QKD is **demonstrated and near-commercial, led decisively by**

China — a strategic-sovereignty capability for states and critical infrastructure rather than a mass-market technology.

Harvest now, decrypt later — and the standard that answers it

The reason cryptography must migrate at all is a threat that does not wait for the quantum computer to arrive. **Harvest now, decrypt later** (HNDL): an adversary records encrypted traffic today and stores it, planning to decrypt once a cryptographically-relevant quantum computer (CRQC) can run Shor's algorithm against RSA and elliptic-curve keys. Any secret with a shelf life past the CRQC date — state secrets, health records, genomic data, long-term IP — is *already* exposed, regardless of when the machine actually ships. Consensus estimates put the CRQC at roughly $5–15$ years out, which is inside the lifetime of data being transmitted now. That is the whole reason to act before the machine ships; the migration playbook below turns the pressure into a dated schedule.

The answer is **post-quantum cryptography**, and it is the most important node in this chapter — even though it is not quantum hardware at all. PQC is classical software built on problems believed hard for quantum computers too: structured lattices, hash functions, error-correcting codes. It runs on the machines we already own. NIST finalized the core standards in **August 2024** — FIPS 203 ML-KEM (Kyber), FIPS 204 ML-DSA (Dilithium), FIPS 205 SLH-DSA (SPHINCS+) — after years of public cryptanalysis (T2), and selected code-based HQC in March 2025 as a math-diverse backup KEM. Deployment is already at internet scale: Signal (PQXDH) and Apple iMessage (PQ3) shipped PQC messaging, with PQ3 doing continuous rekeying against HNDL, and Algorand put Falcon-signed accounts on mainnet (Nov 2025, T2). The web browsers that already default to hybrid post-quantum TLS, and the legal deadlines that make this a schedule rather than a suggestion, are laid out in the migration box below.

Grade PQC as **the single most real, most deployed, most economically important item in the whole adjacent-tech chapter.** If any node in Chapter 4 pays for itself this decade, it is this one. Two honest cautions remain. Lattice hardness has no unconditional security proof, and cryptanalysis can surprise you, which is why HQC exists as a math-diverse hedge and why hybrid modes run a classical and a post-quantum scheme together. And the gating work is not new math at all; it is operational, and the migration box below walks it in the order you execute it — where the hard step is the first one. The quantum-resistant blockchain story is a special case of exactly this: swapping ECDSA for a PQC signature is classical software (Algorand's Falcon, QRL's XMSS since 2018), with the interesting wrinkles being signature bloat (Falcon $\sim 0.7\,\mathrm{kB}$ and Dilithium $\sim 2.4\,\mathrm{kB}$ against ECDSA's $64\,\mathrm{bytes}$) and the unmovable-coins overhang — exposed public keys already on an immutable ledger cannot be retroactively protected. The "quantum-secured" (QKD-based) blockchain variant inherits every QKD limitation and adds little; treat it as a solution in search of a problem.

> **PQC migration: what to do and when.** The pieces are scattered through this section; here they are in the order you execute them. This is a five-step sequence, and the hard part is the first step, not the last.
>
> 1. **Inventory your cryptography.** You cannot migrate what you cannot find, and most organizations cannot enumerate where RSA and elliptic-curve keys actually live — across application code, protocols, firmware, certificates, and third-party dependencies. This discovery work is the real bottleneck, so start here.
> 2. **Prioritize by data shelf-life (Mosca's inequality).** For each secret, add the years it must stay confidential to the years your migration will take. If that sum reaches into the CRQC window (roughly 5–15 years out, so the early-to-mid 2030s), the data is already exposed to harvest-now-decrypt-later. Anything that must stay secret past ~2035 — state secrets, health and genomic records, long-term IP — is at risk today and goes to the front of the queue.

3. **Build crypto-agility.** Re-architect so the algorithm is swappable without touching the applications above it. SIKE's collapse in a single afternoon in 2022 is the standing reminder that any one scheme can fall, which makes the ability to rotate matter more than any single choice.
4. **Deploy hybrid now (classical + ML-KEM).** Run a classical key exchange and a post-quantum one together, so the session holds if either survives. Hybrid X25519 + ML-KEM-768 is already the default in Chrome, Firefox, and Cloudflare, and with the core standards final, there is nothing left to wait for.
5. **Hit the CNSA 2.0 deadlines.** For US national-security systems the schedule is legal rather than optional, and commercial buyers should treat it as the market's timetable too:

MILESTONE	WHAT CNSA 2.0 / NIST REQUIRE
Jan 1, 2027	New national-security acquisitions must be PQC-capable
2030	RSA and elliptic-curve deprecated (NIST IR 8547); migration of signing, networking, and infrastructure well underway
2030–2031	CNSA 2.0's key milestone window — PQC becomes the default across covered system classes
2035	Vulnerable algorithms disallowed; migration complete

Sensing and metrology: the corner that already earns

Here is where quantum technology has revenue today. **Quantum sensing** uses atomic transitions, spin states, entanglement, and squeezing as measurement instruments, reading a physical field off an interferometric phase at or below the standard quantum limit — and, with entanglement, pushing toward the Heisenberg limit where sensitivity scales as $1/N$ rather than $1/\sqrt{N}$. Chip-scale atomic clocks have shipped for over a decade. LIGO has run squeezed light in production since 2019. OPM magnetometers sell into neuroimaging now. Analysts size the market at roughly $\$0.4–0.9\mathrm{B}$ in 2025 growing at $\sim 15–23\%$ CAGR toward $\sim \$1.5\mathrm{B}$ by the mid-2030s (T5, forecast) — small but real, and mostly defense-led. Treat sensing as "real and shipping in niches," with each modality carrying its own maturity call.

Atomic and optical clocks are the most precise instruments humans have built, and their maturity is layered. At the bottom, chip-scale atomic clocks — the rubidium CSAC at $\sim 120\,\mathrm{mW}$ and $\sim 10^{-10}$ stability — are a decades-old commercial commodity in defense, telecom, and undersea gear. At the top, lab optical clocks reach systematic uncertainties near $10^{-18}–10^{-19}$. NIST's Al^+ quantum-logic ion clock set the record in July 2025 at $\sim 8.1 \times 10^{-19}$ — about 19 decimal places, 41% better than the prior best (T2). Portable optical clocks are just now leaving the lab: Infleqtion's **Tiqker** was deployed on the UK Royal Navy's uncrewed submarine XV Excalibur in October 2025 (T3), a first-of-kind at-sea optical clock. The institutional event underneath all this is the **redefinition of the SI second**. BIPM's roadmap conditions it on optical standards beating cesium by $\geq 100\times$, on ≥ 5 independent systems running continuously in different labs, and on more than one year of traceable agreeing data; the CGPM (2026) is expected to endorse the roadmap with the redefinition targeted around **2030**. The near-term commercial pull is defense/PNT holdover timing and geodesy — a 10^{-18} clock senses a $1\,\mathrm{cm}$ height change through gravitational redshift — rather than the 19-digit record itself.

The clocks are now better than the links that compare them, which is why **time-transfer networks** are load-bearing. This is the plumbing that carries an optical clock's stability across cities without degrading it: stabilized fiber, free-space laser links, frequency-comb transfer. A six-country optical-clock comparison (June 2025, T2) and ultrastable transfer over deployed multicore fiber at $\sim 3 \times 10^{-19}$ instability alongside live telecom traffic (Optica 2025, T2) are both done and published. Note the naming trap: despite the "quantum" framing, the record-setting transfer is classically comb-stabilized photonics serving quantum clocks. The **entanglement-based** version — networking clocks into a shared GHZ state to reach Heisenberg-limited timekeeping (Kómár et al., Nat. Phys. 2014, T2 theory) — is a beautiful idea nowhere near a network. Two Sr ion clocks were entangled at benchtop distance in 2022 (T2), and entanglement-based clock syntonization held $< 12\,\mathrm{ps}$ over $48\,\mathrm{km}$ of deployed fiber in 2025 (T2), but a 2026 critical assessment (arXiv:2604.10243) argues that classical comb-stabilized transfer already saturates the useful precision and that fragile GHZ states may not survive network loss. Grade the entangled version research/speculative; the near-term "quantum clock network" is classically-linked quantum clocks.

Magnetometry is the most clinically consequential sensing modality. The brain's magnetic signals are $\sim 50\text{–}500\,\mathrm{fT}$, roughly a hundred million times weaker than Earth's field, and three platforms reach into that range. SQUIDs, the cryogenic gold standard since the 1960s, hit $\sim 1\text{–}5\,\mathrm{fT}/\sqrt{\mathrm{Hz}}$ but need liquid helium and fixed dewars. Optically pumped magnetometers (OPMs) run near room temperature at $\sim 7\text{–}15\,\mathrm{fT}/\sqrt{\mathrm{Hz}}$ (QuSpin QZFM Gen-2), approaching SQUID sensitivity while being small and wearable. NV-diamond reaches $\sim 9.4\,\mathrm{pT}/\sqrt{\mathrm{Hz}}$ — three orders worse, at atom-scale resolution. The disruption is **OPM-MEG**: wearable magnetoencephalography helmets a patient can move in, replacing the fixed helmet inside a $\$2\text{–}3\mathrm{M}$ shielded room. Cerca Magnetics sells OPM-MEG and ran epilepsy and Parkinson's clinical work through 2024–25; FieldLine Medical installed 64-channel wearable systems at two US academic medical centres for pediatric epilepsy mapping without sedation. SandboxAQ's CardiAQ (with Mayo Clinic) applies OPMs to magnetocardiography. The honest gate here is **reimbursement**. The sensitivity is proven; whether OPM-MEG displaces SQUID suites this decade turns on FDA/CE clearance and insurance pace. This is the clinical face of a broader node, **quantum-enhanced MRI/NMR**, which bundles two different maturities: hyperpolarization (parahydrogen/DNP raising the MR signal $10{,}000\text{–}100{,}000\times$ to make real-time metabolic MRI feasible — NVision's POLARIS, human $^{13}\mathrm{C}$-pyruvate trials, close to clinic) and NV/atomic-sensor NMR (magnet-free micro-scale NMR, still lab). Grade each strand separately and resist the umbrella hype.

NV-diamond sensing deserves its own line because the nitrogen-vacancy center works at room temperature and ambient pressure with nanoscale, vector resolution. Its niche is anywhere you must put the sensor *at* the sample — current mapping inside a live chip, single-cell biology, scanning-probe imaging — where raw field noise matters less than proximity. Element Six (De Beers) supplies the quantum-grade diamond, and formed the "Bosch Quantum Sensing" JV in 2025 to chase mass-market volume. SBQuantum flies diamond vector magnetometers (~400 pT/√Hz, heading error <5 nT, T4) for GPS-free navigation; QDTI's NV "magnetic microscope" claims $< 1\,\mathrm{pg/mL}$ biomarker detection (T4). Grade NV as commercial-in-niches, mass-market-aspirational — the automotive/consumer bet is still a roadmap short of a deployment.

Gravimetry and inertial navigation rest on cold-atom interferometry — dropping a cloud of laser-cooled atoms and reading gravity or acceleration off the interference phase. Because the atom is the test mass and the laser wavelength is the ruler, the measurement is **absolute and drift-free**, unlike a spring gravimeter that needs recalibration. Here the split is sharp. Survey gravimeters are a shipping product: **Exail's AQG** has sold for years (continuous monitoring on Mt Etna), and an independent evaluation (J. Geodesy 2025, T2) measured $\sim 500\,\mathrm{nm{\cdot}s^{-2}}/\sqrt{\mathrm{Hz}}$ sensitivity reaching $\sim 1\,\mu\mathrm{Gal}$ after one hour, drift-free, with no active vibration isolation. Birmingham's cold-atom gradiometer located a buried structure in an open field (Stray et al., Nature 602, 2022, T2). Quantum *inertial navigation*, though, is not yet fielded — it is well-funded DoD prototyping. This is where the atom interferometer as a **platform** lives its double life: down into rugged, small navigation sensors (engineering-hard), and up into kilometre-scale physics detectors (MAGIS-100, AION, searching for dark matter and mid-band gravitational waves — science-speculative). The instrument is established, Nobel-lineage physics that measured the fine-structure constant to sub-ppb; grade each application on its own card.

That double life converges at **assured PNT**, the systems-level problem of knowing where and when you are without GPS. GPS is jammable, spoofable, and useless underwater, underground, or in contested airspace. The quantum answer combines several instruments into a **fusion**: a drift-free clock, a drift-free cold-atom IMU, and a map-matching magnetometer. A quantum clock plus a quantum IMU lets you dead-reckon for hours; a magnetic or gravity map fixes your absolute position against Earth's own signatures, and an adversary can jam neither. This is where the defense money flows, and where the claims deserve the hardest scrutiny. **Q-CTRL's Ironstone Opal** claims up to $\sim 100–111\times$ better GPS-denied accuracy than the best conventional inertial system, $\sim 4\,\mathrm{m}$ over $700\,\mathrm{km}$ flights, and 144+ hours unattended at sea (TIME Best Inventions 2025). Every one of those numbers is a vendor field trial graded T4, run against a specified, often unnamed baseline. No peer-reviewed head-to-head against classified military ring-laser or fiber-optic INS is public, and the "$111\times$" is measured against a stated alternative rather than the state of the art in a black program. AOSense cold-atom IMUs target $\sim 5\,\mathrm{m/hour}$ unaided drift (T3). Grade assured PNT as **advanced R&D nearing first operational deployments, with the flashiest multipliers unverified**. The pieces are individually real. The fused, ruggedized, independently-benchmarked navigator on a fleet of moving platforms does not yet exist in the open literature.

Rydberg RF electrometry points quantum sensing at the electromagnetic spectrum. An atom excited to principal quantum number $n \sim 50–100$ has a huge dipole moment; read the Autler–Townes splitting of an EIT line optically and you have an RF receiver whose sensing element is atoms, self-calibrated against Planck's constant, tiny, and broadband from MHz to sub-THz in one vapor cell. Rydberg Technologies and Infleqtion (SqyWire) sell into defense spectrum awareness. The honest caveat: standard Rydberg receivers have historically not beaten a good cryogenic low-noise-amplifier on raw sensitivity — the advantage is self-calibration, single-device bandwidth, small size, and jamming immunity. Grade it as a real product in a niche, with contested advantage in the mainstream RF use case.

Quantum imaging and radar is the node that most needs its label policed. The load-bearing distinction is *quantum* (entangled/squeezed light) versus *quantum-inspired* (classical correlations with quantum detectors) — and almost every commercial win is the latter. Single-photon SPAD LiDAR ships in automotive and remote sensing; a compact quantum-inspired LiDAR demonstrated $> 100\,\mathrm{dB}$ in-band-noise rejection with classical time-frequency correlations (Nat. Commun. 2023, T2). Entanglement-based quantum illumination — the basis of "quantum radar" — stays in the lab: its $6\,\mathrm{dB}$ theoretical error-exponent advantage (Tan et al., 2008) showed only $\sim 4\,\mathrm{dB}$ in a cryogenic demo, and fielded long-range quantum radar is implausible because idler storage over a realistic round trip would need a lossless quantum memory, and the advantage evaporates the moment you can simply transmit more classical power. **Quantum-inspired imaging is commercial and shipping; entanglement radar is, as of 2026, vaporware** (T6, contested).

Two cross-cutting resources close the sensing cluster. **Quantum random number generation** is the oldest commercial quantum product line. ID Quantique's Quantis chips have sold for two decades and shipped inside Samsung "Quantum" phones. Integrated devices now reach $\sim 3\,\mathrm{Gbit/s}$ (Toshiba/Optica 2025, T2) and a photonic record of $18.8\,\mathrm{Gbit/s}$. The 2025 headline was **certified randomness**. JPMorgan and Quantinuum expanded verifiable random bits on a 56-qubit trapped-ion processor, checked at exascale on Frontier (Nature 640, 2025, T2). It may be the first useful thing a beyond-classical machine has done. Aaronson himself flags the limits. The guarantee is computational, and the verification needs a machine only a handful of labs own. So hardware QRNG is a mature commodity. Certified randomness is a landmark demonstration that is not yet a business. **Squeezed light** is the clearest case here of a quantum resource doing paid work every day. LIGO injects squeezed vacuum into its interferometer on every observing run, cutting shot noise by $\sim 2–3\,\mathrm{dB}$ and raising the gravitational-wave detection rate $\sim 40–50\%$ (Phys. Rev. X 13, 041021, T2). This is established, deployed, and reproduced. The compute use (Xanadu's CV/GKP photonic machines, $18\,\mathrm{dB}$ on-chip squeezing) is real hardware progress that remains research. Sensing use, production; compute use, frontier. The picks-and-shovels layer under a future quantum internet is **quantum memories as a component industry**. Qunnect's room-temperature rubidium-vapor memories demonstrated 90.2% photon–memory entanglement fidelity at $\sim 1{,}200$ pairs/second (T3). This is real hardware built ahead of a market that depends on the unconfirmed quantum-internet gold rush.

Quantum materials as the substrate

Underneath everything sits **quantum materials**: topological insulators, high-T_c superconductors, and 2D materials like magic-angle twisted bilayer graphene that superconducts at $\sim 1.1°$ (Cao et al., Nature 556, 2018, T2). They play two roles at once — a physics frontier, and the substrate under qubits, Majorana platforms, Josephson junctions, and the two-level-system defects that cap coherence. The question is which of them ship. Low-T_c superconductors (NbTi, $\mathrm{Nb_3Sn}$) are fully industrial in MRI magnets, SQUIDs, and transmons. High-T_c REBCO tape crossed into product as the $\sim 20\ \mathrm{T}$ magnets of Commonwealth Fusion's SPARC — arguably the biggest commercial deployment of a high-T_c material to date (T2). Topological insulators and 2D-material devices remain research — no volume product ships on topological protection yet, which is exactly the risk in Microsoft's Majorana bet. And the recurring hype magnet, room-temperature superconductivity, has collapsed under replication every time (LK-99 in 2023, the Dias retractions); grade every new claim T6-contested until independent labs confirm. **Commercial at the boring end, research at the exciting end** — separate "superconductor that ships" from "topological phase we hope powers a qubit," and treat every room-temperature headline as guilty until replicated. One node in this chapter, **quantum-dot displays**, is included precisely to police the word: a QLED TV uses quantum confinement to tune color, which makes it quantum *physics*. It falls outside quantum *technology* in the strict sense, because it exploits no superposition, entanglement, or single-quantum readout. The quantum thread there is the QD single-photon source.

> **Key takeaways**
>
> - Post-quantum cryptography is the most important node in this chapter and the one certain to pay for itself this decade — classical software on a legally mandated clock, with cryptanalysis its only real risk.
> - QKD's security is strong in the ideal model and narrow in the field: trusted relays hold the key, so the "unhackable" claim shrinks to link-by-link physical trust, which is why the signals agencies prefer PQC.
> - The PQC migration bottleneck is inventory, not algorithms; crypto-agility and Mosca's inequality set the order of work.
> - Sensing is the revenue-bearing corner today — QRNG, squeezed light, atomic clocks, OPM magnetometers, and survey gravimeters all ship, several gated by reimbursement or benchmarks rather than physics.
> - The loudest numbers are in GPS-denied navigation and quantum radar; grade them T4 until an independent referee reproduces them, and separate quantum from quantum-inspired every time.
> - Beyond computing, quantum is already a business — the job is saying which parts, with the numbers attached.

Where the adjacent tech stands

Read the chapter as three bands.

Commercial today, earning revenue now. Post-quantum cryptography is the standout — classical software, shipping at internet scale, on a legally mandated clock, and the one node in Chapter 4 certain to pay for itself this decade; its only real risk is cryptanalytic. Hardware QRNG has sold for twenty years. Squeezed light does paid work in LIGO every observing run. Chip-scale atomic clocks are a commodity. Survey gravimeters (Exail) and OPM-MEG magnetometers (Cerca, FieldLine) are shipping instruments, the latter gated by reimbursement rather than physics. QKD sells real boxes into a real but capped and contested market. QLED displays are a multi-billion-dollar consumer business built on a quantum material.

Near-term — demonstrated, defense-led, nearing deployment. Satellite QKD is near-commercial and led by China. Lab and portable optical clocks are established metrology heading toward the 2030 SI-second redefinition. Assured PNT and quantum inertial navigation are advanced R&D with impressive field trials — and with the loudest numbers, the "$111\times$" and "$\sim 4\,\mathrm{m}$ over $700\,\mathrm{km}$" claims, graded T4 pending independent benchmarks against black-program INS. NV-diamond sensing, Rydberg electrometry, hyperpolarized metabolic MRI, and quantum memories as components are real in their niches with mass-market or network-scale bets still unproven.

Research — real physics, unconfirmed market. The quantum internet has not reached the repeater-beats-direct crossover and has no paying application beyond relay-free QKD. Entanglement-based clock networks are elegant theory that credible physicists doubt beats classical optical transfer once loss is included. Entanglement-based quantum radar is, as of 2026, vaporware. Topological materials remain lab physics with a credibility overhang from retractions.

The through-line is the QKD-versus-PQC debate, and it settles cleanly enough to state plainly: the deployed answer to the quantum threat is classical software that four signals agencies endorse, while the physics-based alternative stays a strategic niche for the highest-value links. The sensing cluster is the revenue-bearing corner of quantum technology and the part most likely to matter this decade — concentrated in defense PNT, neuroimaging, timing, and metrology, with the flashiest navigation multipliers still awaiting an independent referee. Beyond computing, quantum is already a business. The job is to say which parts, and to keep the vendor field trials in the T4 column where they belong until someone reproduces them.

Exercises and discussion

1. **(Sensitivity and units.)** The brain's magnetic signals sit around 100 fT–1 pT, roughly a hundred million times weaker than Earth's ~50 µT field. A SQUID reaches a few fT/√Hz but needs liquid helium and a fixed dewar; a QuSpin QZFM Gen-2 OPM reaches ~7–15 fT/√Hz near room temperature and is wearable; an NV-diamond magnetometer sits three orders of magnitude worse at atom-scale resolution. For a wearable magnetoencephalography helmet a moving patient wears, rank the three platforms and justify the winner in terms of both sensitivity relative to the signal band and the physical constraints. Then explain why NV-diamond still wins for mapping current inside a live chip.

2. **(QKD key rate versus distance.)** Fiber attenuates at ~0.2 dB/km, so single-photon transmission through a channel of length *L* falls as roughly 10^(−0.2·L/10). Sketch how the secret-key rate scales with distance for a point-to-point QKD link, and explain why, with no quantum repeater available, real backbones insert trusted relays every ~100 km. State precisely what security property is lost at each relay and what remains.

3. **(Migration planning with Mosca's inequality.)** An organization holds genomic records that must stay confidential for 40 years. Its full crypto migration is estimated at 7 years. Using Mosca's inequality (confidentiality time + migration time versus time to a cryptographically-relevant quantum computer, taken here as ~10–15 years), determine whether these records are already exposed to harvest-now-decrypt-later, and where they belong in the migration queue. Then order the five steps of the migration box for this organization and name the step that gates the rest.

4. **(Reading a spec for the trap.)** A vendor markets a "quantum LiDAR" and a "quantum radar." Using the quantum versus quantum-inspired distinction, state what evidence you would demand to grade each as a real quantum advantage rather than classical correlations read by quantum detectors, and explain why, as of 2026, one ships in automotive while the other stays in the lab.

5. **(Seminar.)** QKD versus PQC as a national investment. Four signals-intelligence agencies (NSA, NCSC, ANSSI, BSI) advise against QKD for national-security systems and steer critical infrastructure toward PQC. A rival government funds satellite QKD at national scale. Argue both sides: when, if ever, does physics-based key distribution earn its place over classical post-quantum software, and what would have to change for the agencies' verdict to flip?

6. **(Seminar.)** The revenue in this chapter concentrates in defense PNT, neuroimaging, timing, and metrology, while the loudest performance numbers — GPS-denied navigation multipliers, quantum radar — sit in the T4 column pending independent benchmarks. Discuss what an honest referee for these claims would look like, and how a buyer should price a field-trial result that no third party has reproduced.

Further reading

- **Bennett & Brassard, "Quantum cryptography: Public key distribution and coin tossing" (1984).** The BB84 protocol — the origin of QKD and its no-cloning security argument.
- **Ekert, "Quantum cryptography based on Bell's theorem" (Phys. Rev. Lett., 1991).** E91, the entanglement-based protocol that certifies a key through a Bell violation.
- **NIST FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), FIPS 205 (SLH-DSA), August 2024.** The finalized post-quantum standards; read the security-category definitions and the parameter sets you would actually deploy.
- **Mosca, "Cybersecurity in an era with quantum computers: Will we be ready?" (IEEE Security & Privacy, 2018).** The timeline argument and the inequality that names when to start migrating.
- **Degen, Reinhard & Cappellaro, "Quantum sensing" (Rev. Mod. Phys., 2017).** The standard review of how spins, atoms, and squeezing become instruments, including the standard-quantum and Heisenberg limits.
- **Yin et al., "Satellite-based entanglement distribution over 1200 kilometres" (Science, 2017).** The Micius result — the state of the art in long-range, repeater-free QKD and its relay problem.

PART III

The World It Enters

PART III · THE WORLD IT ENTERS

5 The Industry Map

Every chapter before this one has been about physics and machines: the postulate that a system holds two states at once, the qubit modalities that exploit it, the stack that turns a noisy qubit into an answer, and the sensing and cryptography technologies that ride alongside the computer. This chapter asks the question the money cares about. Twenty-seven industries have quantum programs. Where does the technology land, how hard is the ground under each landing, and how much of what you read about it is real?

The honest answer is a map with sharp relief. The industries do not sit on one smooth gradient from "early" to "mature." They cluster into three bands with visible walls between them, and the wall that matters most is the *kind* of quantum technology each industry is actually using. Read the twenty-seven cards back to back and one pattern surfaces before any other: **the industries that have something deployed today are running a quantum *sensor* or a *cryptographic defense*, and the industries running a quantum *computer* are running a pilot.** The near-term economy of quantum is sensing and crypto-migration. The near-term economy of quantum *computing* is controlled proof-of-concept and contested annealing.

To keep the vendor optimism and the peer-reviewed result from weighing the same, every claim below carries an evidence tier from `evidence/SCHEMA.md`. T1 is textbook physics, T2 a refereed result, T3 a preprint or conference demo, T4 a vendor announcement not independently reproduced, and T5 an analyst forecast. A JPMorgan result in *Nature* and a market-research firm's "trillion-dollar impact by 2035" are separated by four tiers, and the press treats them as one number.

The **industry-readiness matrix** — the figure this chapter narrates — plots the twenty-seven on two axes: *is the near-term value a sensor/defense or a computer*, and *is the computational advantage proven, contested, or promised*. Every industry falls into one of the three bands below. The matrix has a heavy diagonal: readiness rises as you move from "quantum computer chasing an optimization advantage" toward "quantum sensor doing a measurement job no classical instrument does as well." That diagonal is the honest shape of the market.

Learning objectives. After this chapter you can:

- Sort any quantum industry into the three-band map — proven, contested pilot, promise — and name the kind of quantum technology (sensor, cryptographic defense, or computer) that puts it there.
- Read a vendor or analyst claim against the T1–T5 evidence tiers and assign it the tier it earns.
- Explain why the near-term quantum economy is sensing and cryptographic migration while quantum *computing* is still pilots and contested annealing.
- Spot the two market-sizing traps — TAM double-counting and quantum-inspired conflation — inside a market report or a press release.
- Price a pure-play quantum company on its real current revenue rather than its trillion-dollar TAM deck, and read a DARPA-QBI stage as a government filter, not a scoreboard.

The industry map at a glance

Read the dashboard first, then the evidence behind each card. Eight industries are investable in 2026, and every one of them runs a quantum sensor that out-measures the classical instrument or a mandated post-quantum-cryptography migration. Ten more are 2030 watches whose advantage over a good classical method is still contested. Nine are ignores for now, dominated by inflated market math or by "quantum-inspired" classical work relabeled as quantum. The Invest-read column is the one-line version of each verdict below.

#	INDUSTRY	BAND	NEAR-TERM REALITY	BEST EVIDENCE	VERDICT	INVEST READ
1	Aerospace & defense	1	Quantum inertial nav fielded (air/sea/space)	T3	Sensing arriving; compute a bet	Invest now (sensing)
2	Healthcare imaging	1	OPM-MEG shipped (research-only)	T2/T3	Real sensor; awaits regulatory clearance	Invest now (sensing)
3	Mining	1	Diamond magnetometer + gravimeter exploration	T3/T4	Sensing credible; compute 2030s	Watch 2030 (sensing)
4	Space & EO	1	ACES clocks, satellite QKD, inertial sensors flying	T2/T3	Sensing/timing real; compute long-horizon	Invest now (sensing/timing)
5	Cybersecurity	1	PQC standards final, migration mandated	T2	Threat certain; danger is underreacting	Invest now (PQC)
6	Intelligence	1	HNDL doctrine + PQC mandates	T3	Certain in kind, uncertain in timing	Invest now (PQC)
7	Telecom	1	PQC shipping; QKD deployed and niche	T2/T3	Defensive crypto is the real value	Invest now (PQC)
8	Finance	1/3	Certified randomness delivered; optimization POC	T2	One refereed app; rest is promise	Watch 2030
9	Automotive	2	Ford Otosan annealing in production	T3/T4	Real deployment, contested advantage	Watch 2030
10	Manufacturing	2	Annealing scheduling + paid optimization	T3/T4	Grounded; gate-model advantage end-of-decade	Watch 2030
11	Logistics	2	Routing/scheduling pilots (VW, Airbus, DB)	T3	Most demos, least proven advantage	Ignore (quantum-inspired trap)
12	Air-traffic mgmt	2	QUBO traffic/gate demos + Airbus challenges	T3/T4	Small, classically matchable, cert-gated	Ignore
13	Energy & utilities	2	EDF/Pasqal EV-charging pilot (pre-commercial)	T3/T4	Sensing likelier payoff than compute	Watch 2030
14	Chemicals & materials	2	VQE/QPE on tiny molecules	T1 theory / T3 demo	Strongest theory, ~zero advantage today	Watch 2030

#	INDUSTRY	BAND	NEAR-TERM REALITY	BEST EVID-ENCE	VERDICT	INVEST READ
15	Pharma	2	Q4Bio 100-qubit chemistry (competitively judged)	T2/T3	Path shown; late 2020s–2030s	Watch 2030
16	Nuclear & fusion	2	Light-nucleus simulation on hardware	T2/T3	Best physics pedigree, widest value gap	Watch 2030
17	Oil & gas	2	Annealing reservoir/seismic subproblems	T3/T4	Heavy TAM, light results — grade hard	Ignore (TAM trap)
18	AI & machine learning	3	QML classifies toy data; repeatedly dequantized	T3/T4	Skepticism maximal; near-term product no	Ignore
19	Government services	3	QUBO fraud method papers; PQC is the real spend	T3	Watch the AI-relabeled mislabel	Invest now (PQC)
20	Retail	3	QNN forecasting demos; TAM-report noise	T3	Thinnest pitch; grade TAM hardest	Ignore (TAM trap)
21	Insurance & risk	3	Academic capital/copula demos	T3	Finance's bet, one step behind	Ignore
22	Agriculture	3				
estimate + hype-y precision-ag	T3	Sensing near; chemistry 2030s+	Watch 2030 (sensing)			
23	Climate & sustainability	3				
	T3		Watch 2030			

#	INDUSTRY	BAND	NEAR-TERM REALITY	BEST EVID-ENCE	VERDICT	INVEST READ
~99k-qubit estimate		Textbook killer app, decade+ away				
24	Construction	3	Small QUBO scheduling; mostly materials science	T3	Early academic; FTQC-gated	Ignore
25	Weather modeling	3	HHL/QML surveys + OQI pilot	T3/T4	I/O bottleneck kills the speedup	Ignore
26	Semiconductors & EDA	3	Quantum-for-design early; GPU-classical wins	T3/T4	Fab base real; design app 2030s	Invest now (fab base)
27	Media & entertainment	3	QRNG-seeded PCG in toy form	T3	Curiosity node; low weight	Ignore

Band 1 — Deployed and real today

Eight industries have quantum technology doing paying or fielded work in 2026. Not one of them earns that place with a general-purpose quantum computer. They earn it two ways: with **quantum sensors** that measure fields, gravity, time, and magnetism better than any classical instrument, and with **cryptographic defense** — post-quantum cryptography (PQC) and, in narrow cases, quantum key distribution (QKD) and certified randomness. What unites the band is that none of this needs a fault-tolerant computer. The physics is mature, the products ship, and the bottlenecks are regulatory approval and unit cost.

Sensing: aerospace, health imaging, mining, space

Aerospace & defense holds the strongest "quantum is real now" case in the entire atlas, and the thing that is real is a sensor. Q-CTRL's April 2025 field trials put a quantum inertial navigation package (~180 W) on an aircraft and on the Royal Australian Navy's MV *Sycamore*, clearing military-adequate thresholds for GPS-free positioning (T3). In August 2025 DARPA awarded Q-CTRL two Robust Quantum Sensors (RoQS) contracts (~$24.4M, Lockheed Martin subcontracting) for navigation sensors on moving platforms without shielding (T4), and Boeing flew what it called the highest-performing quantum inertial sensor ever in space aboard the X-37B spaceplane (T3/T4). The strategic driver is GPS-denied environments. The payoff comes from gravimetry, magnetometry, and cold-atom interferometry. Shor's algorithm is nowhere in it. Quantum *computing* for defense — codebreaking, materials simulation — is the same fault-tolerance-gated future everyone else faces, plus a classified overhang: no one outside secure spaces knows what adversaries can do. The card's verdict is the band's motto. Sensing is arriving; compute is a bet.

Healthcare imaging & diagnostics tells the identical story in a hospital. The flagship is OPM-MEG: optically-pumped magnetometers, a quantum sensor, replacing cryogenic SQUIDs for magnetoencephalography and giving wearable, room-temperature brain imaging with SQUID-class sensitivity (T2/T3). Cerca Magnetics (a Nottingham spinout) ships a commercial wearable OPM-MEG helmet to research sites today (T4). The honest ceiling is written into the product: as of 2025 it sells as a research system with no medical-regulatory approval in any jurisdiction, and epilepsy pre-surgical mapping is the most advanced application. The NIHR's January 2025 survey notes only two healthcare quantum-sensing technologies are past TRL 6. This needs no fault-tolerant computer, which is exactly why it is nearer-term than the drug-discovery story in pharma. The blocker is regulatory clearance and the magnetically-shielded room. The physics is settled.

Mining & mineral exploration is sensing-led for a reason that matters to the energy transition. Cold-atom gravimeters and NV-diamond magnetometers detect ore bodies and density anomalies from the air, sharpening the hunt for lithium, copper, nickel, cobalt, and rare earths. The QUAMINEX program (SBQuantum + Silicon Microgravity, March 2024) fuses a diamond quantum vector magnetometer with a MEMS gravimeter on a drone for 3D deposit mapping, funded by Canada (CAD $500k) and the UK (£414k), with SBQuantum's magnetometer miniaturized to a handheld sub-pound device claiming ≥30% better resolution than the magnetic-mapping standard (T3/T4). The resolution number is vendor-stated and the systems are still commercializing (niche, expensive), so grade it accordingly. The compute side — QUBO haulage optimization, mineral-processing chemistry — sits at the same small-instance, no-advantage stage as logistics. Sensing has a credible near path; compute is a 2030s bet.

Space & Earth observation is real where it is sensing, timing, and communication. ESA's ACES atomic-clock ensemble reached the ISS in April 2025 — the most accurate clock package ever flown, targeting fractional stability ~1×10^{-16} (T2/T3) — and satellite QKD (China's Micius) plus quantum inertial sensors are flying and returning data now. The University of Vienna even launched a compact photonic quantum computer into orbit in 2025 (T3), but read that one carefully: it demonstrates that the hardware *survives* launch. It computes nothing useful. Orbit and constellation optimization by quantum computers is speculative and well-served by classical HPC. As with aerospace, the near-term space win is a sensor.

Cryptographic defense: cyber, telecom, intelligence, finance

The other half of Band 1 is the defensive scramble to survive the quantum computer that does not yet exist.

Cybersecurity is the one industry in this whole chapter where the honest caveat runs *backward*. Everywhere else the danger is over-claiming. Here the danger is under-reacting. A cryptographically relevant quantum computer (CRQC) running Shor breaks RSA and elliptic-curve cryptography — that is T1 established mathematics, and only the CRQC's arrival date is uncertain (T6). NIST finalized the first PQC standards (FIPS 203 ML-KEM, 204 ML-DSA, 205 SLH-DSA) in August 2024, triggering the largest mandated crypto migration in history (T2), with NIST IR 8547 setting 2030-deprecate / 2035-disallow timelines and NSA's CNSA 2.0 requiring quantum-safe algorithms for new national-security systems by January 2027. "Harvest now, decrypt later" (HNDL) makes the risk retroactive: adversaries can store today's encrypted traffic to break later, so migration must begin now regardless of when the CRQC lands. And the horizon is tightening — resource estimates published between May 2025 and March 2026 cut the qubits needed to break RSA-2048 from ~20M (Gidney/Ekerå 2019) toward <1M, possibly ~100k on newer architectures (T3). Yet a May 2025 survey of 1,000+ security managers found only ~5% had quantum-safe encryption deployed. The threat is certain, the standards are final, the deadlines are binding, and the market is barely moving. QKD is over-marketed for this; classical-software PQC is the answer NSA and the UK's NCSC actually recommend.

Intelligence & cryptanalysis is the offensive mirror of the same coin — quantum's impact most certain in *kind*, most uncertain in *timing*, and wrapped in classification. Shor's threat is established; HNDL bulk collection is warned-of doctrine from NSA, CISA, and NIST (T3); and CNSA 2.0's deadlines exist precisely because agencies must defend their own secrets against an adversary CRQC. No CRQC exists, and whether any adversary is close is classified — the honest overhang. Any "they already break encryption with quantum" claim is unverifiable and should be treated as such (T6). The rational response is identical to cybersecurity: migrate to PQC now.

Telecom is the substrate for both defensive tracks, and it runs at two speeds. PQC is software crypto shipping now — mandated, underway, and where carriers spend seriously. QKD is hardware: deployed, working, and niche. BT stood up the UK's first commercial quantum-secured metro network in London in 2022 (with Toshiba; customers included HSBC and EY) (T3/T4), and SK Telecom launched a hybrid

QKD-PQC service in October 2024 with Nokia and ID Quantique in Seoul data centers (T3/T4). Both are real commercial footprints. Both are small, expensive, and distance-limited. The quantum internet is a research decade away. Telecom's honest near-term quantum value is defensive crypto, mostly the classical-software PQC kind.

Finance is the flagship "quantum use case" and the sharpest lesson in reading the tiers. Its one deployed, refereed, beyond-classical result is **certified randomness**. JPMorganChase, with Quantinuum, Argonne, ORNL, and UT Austin, delivered it on the 56-qubit Quantinuum H2-1 in *Nature* (26 March 2025) (T2), a cryptographic primitive with privacy and audit uses. That is the single delivered "advantage application" on the entire industry map, and it produces no alpha. Everything else finance chases is promise wearing a POC costume. HSBC and IBM claimed up to 34% improvement predicting corporate-bond-trade fills on real data using a Heron processor in a hybrid workflow (September 2025) — a bank-plus-vendor co-announcement that has not been independently reproduced (T4). A JPMorgan+AWS decomposition reported a modest ~12% runtime reduction on a constrained optimization (T4). The honest counter-signal lives inside the sector: Goldman Sachs scaled back its Monte Carlo pricing effort after concluding practical advantage remains far off. No bank runs a quantum computer in live operations. The "100x speed" figures are controlled tests, and the finance slice of McKinsey's ~$2.7T-by-2035 value estimate is a T5 forecast, double-counted across verticals and unadjusted for inflation. Certified randomness is Band 1. Portfolio optimization and derivative pricing are Band 3 promise, and the amplitude-estimation speedup finance has chased for years needs fault-tolerant depth that erases the near-term gain.

Band 2 — Pilots with contested advantage

Nine industries have real quantum programs on real hardware, producing dated results — and every one of those results is either a controlled proof-of-concept matched by classical methods, or a quantum-annealing deployment whose advantage over a good classical solver is contested. This is the band where the word "advantage" earns its scare quotes. The workflows run. Whether a quantum machine *beats* the classical alternative on the same problem is the open question in almost every case.

Annealing in production: automotive and manufacturing

Automotive and **manufacturing** share the field's most-cited "quantum in production" claim, and it is worth stating precisely because it is both real and narrower than the headline. In 2024 Ford Otosan deployed a hybrid-quantum application in production on the Ford Transit line, cutting vehicle sequencing time roughly 50% (about 30 minutes to under 5 per ~1,000-vehicle run) (T3/T4). That is a legitimate deployed workflow in a live plant. It is also *quantum annealing* (D-Wave), solving a scheduling problem where the annealer competes head-to-head with classical solvers and the advantage is contested. D-Wave reported its first meaningful optimization revenue at its March 2025 user conference — real dollars, contested advantage (T4). The rest of automotive's quantum work — battery and electrolyte simulation (Hyundai + IonQ, Ford + Quantinuum, VW + IQM), fuel-cell catalyst modeling (BMW + Airbus + Quantinuum, oxygen reduction on platinum) — is small-molecule exploratory study far from design-grade accuracy (T4), sharing the fault-tolerance wall of chemistry. Manufacturing under-claims relative to finance and pharma, which is precisely what makes it one of the more grounded verticals. The near-term story is quantum-flavored optimization services that sometimes help. Gate-model manufacturing advantage is an end-of-decade proposition.

Optimization pilots: logistics, air-traffic, energy

Logistics produces the most demos and the least proven advantage of any industry, because small routing and scheduling instances run on today's hardware and photograph well. Volkswagen ran a traffic-routing pilot in Lisbon (2019). Airbus and IonQ demonstrated aircraft cargo loading via the MAL-VQA algorithm on trapped-ion hardware (T3). Deutsche Bahn and Cambridge Quantum studied train rescheduling with F-VQE on realistic timetables (T3). Every result is a small instance where Gurobi, CPLEX, and OR-Tools still win at production scale. The real near-term wins here come from *quantum-inspired* algorithms — classical methods borrowing quantum structure — which is honest classical progress with no quantum computer in the loop. The press blurs the two constantly.

Air-traffic management is the same optimization template with a higher wall in front of it: safety certification. Airbus and QC Ware challenges report eye-catching numbers ("~400% faster analysis," "~70% higher airspace utilization") that are competition results on constrained problems, frequently quantum-inspired, and not independently reproduced (T4). Method papers demonstrate QUBO traffic-flow, trajectory, and gate-scheduling on mini instances (T3). No air navigation service provider runs a quantum computer operationally, and aviation's certification bar is high and slow even after a proven advantage appears.

Energy & utilities has credible, dated, openly pre-commercial pilots. EDF, Pasqal, and GENCI demonstrated EV smart-charging and demand forecasting on neutral atoms at 100+ scale in January 2025, stated plainly as not commercially viable (T3/T4). ExxonMobil (the first energy major in the IBM Quantum Network, 2019) explores LNG shipping-route optimization; E.ON works grid and pricing algorithms with IBM (T4). Grid dispatch is a real, hard, valuable problem — and classical plus quantum-inspired solvers and D-Wave annealing currently match or beat gate-model quantum on the instances tried. No utility runs quantum in dispatch. The nearer-term energy payoff more plausibly comes from quantum *sensing* (fault detection, magnetometry) than from compute.

Simulation POCs: chemistry, pharma, nuclear, oil & gas

The simulation industries carry the strongest *theoretical* case in the whole map and, today, essentially zero production advantage. This is the most important distinction in the chapter to hold cleanly.

Chemicals & materials is where simulating quantum systems is literally what the machine is *for* — VQE and quantum phase estimation map onto electronic structure directly (T1). BASF works homogeneous catalysis with SEEQC and optimization with Kipu; Mitsubishi Chemical simulates excited states of photochromic molecules with PsiQuantum (T3/T4). But every named pilot is a research collaboration probing algorithms. None is a deployed design tool. Today's devices simulate only tiny molecules ($\mathrm{H_2}$, LiH, small actives) that classical DFT and coupled-cluster already match or beat, and GPU-accelerated classical chemistry keeps raising the bar (NVIDIA's cuEST — a CUDA-X electronic-structure library launched for chip-materials work — reports up to ~50x faster calculations for adopters including TSMC (T4), a reminder that quantum chemistry's near-term competition is classical HPC). A FeMoco-class catalyst needs error-corrected machines with thousands of logical qubits. Strong theory, near-zero present-day advantage.

Pharma & healthcare is chemistry with a bigger prize and the same wall. The strongest signal to date is the Wellcome Leap Q4Bio prize (~April 2026, $2M) won by Algorithmiq, Cleveland Clinic, and IBM for simulating photodynamic-therapy processes on up to 100 qubits (T2/T3) — real chemistry circuits at scale, competitively judged, demonstrating a *path* rather than advantage over classical methods. Cleveland Clinic's on-site IBM System One (live 2023) anchors 50+ joint projects. AstraZeneca and IonQ reported a ~20x speedup on a drug-discovery *subroutine* (T4) — narrow, vendor-framed. No approved drug has been discovered by a quantum computer, and 2026 pharma reviews state plainly that no quantum computer yet simulates a drug-relevant molecule better than classical chemistry on the same system. Design-grade accuracy on hard active sites (FeMoco, cytochrome P450) needs fault tolerance and millions of gates. Late 2020s into the 2030s.

Nuclear & fusion has the best *physics* pedigree of any young node and the widest gap to engineering value. Quantum computers have simulated light nuclei (deuteron, $^4\mathrm{He}$-scale binding energies) on superconducting and trapped-ion hardware since ~2018 — solid nuclear-physics results (T2/T3). QAOA is framed for tokamak plasma-stability optimization (T3). But fusion's bottleneck is engineering and confinement, its modeling is elite classical HPC (M3D-C1, JOREK, NIMROD), and quantum methods sit at the review-and-small-demo stage. Real science, no engineering payoff yet.

Oil & gas is the band's cautionary tale on TAM. Shell and D-Wave applied annealing to North Sea reservoir mapping (framed weeks-to-hours) (T4); ExxonMobil and bp explore reservoir simulation with IBM (T4); seismic inversion appears as QUBO formulations on toy instances (T3). No operator runs quantum in its production geophysics pipeline. Against this sits "$2.6T oil-and-gas quantum impact by 2035, ~30% CAGR" (T5) — among the most inflated forecasts in the atlas. Grade it hardest. The nearer-term extractive thread is quantum *sensing* (cold-atom gravimeters, NV magnetometers), which is why mining was split out into its own Band 1 node.

Band 3 — Promise only

Ten industries have quantum programs that are, honestly, method papers, consortium memberships, vendor SDKs, and forecasts. The hardware runs toy problems. Classical methods match or beat every result. No deployment exists. These nodes belong on the map for completeness and for what they teach: this is where the optimization-and-simulation advantage is *entirely* promise, and where the TAM reports do the loudest talking.

Gate-model optimization, spread thin: government, retail, insurance, construction

Four industries run the same QUBO/QAOA optimization pitch — combinatorial matching, anomaly detection, scheduling — that finance runs, one or more adoption steps behind and with no deployment.

Government services has no quantum computer running in any tax, customs, or census agency. The substantive work is QUBO fraud detection via community detection in transaction graphs, on toy data (T3). The instructive trap is the mislabel: HMRC's £175M Quantexa contract to hunt tax fraud is classical AI despite the "Quant-" branding, and much "quantum government" press is exactly this pattern — AI relabeled, or PQC work that belongs in cybersecurity. The real government quantum program is defensive PQC migration.

Retail is one of the thinnest pitches on the map. Demand forecasting, dynamic pricing, and assortment reduce to problems where classical ML and modern MILP are strong and cheap, so the quantum bar is high and unmet. QNN forecasting demos run on benchmark data with no advantage (T3). The visible "activity" is dominated by market-sizing reports (a "quantum-enhanced demand forecasting" market projected ~$2B→~$2.64B, ~32% CAGR (T5)) and one vendor acquisition. No retailer runs quantum hardware. Grade the TAM hardest here.

Insurance & risk is finance's cousin, one adoption step behind with fewer named pilots. Monte Carlo catastrophe modeling and reinsurance capital optimization appear as peer-reviewed numerical demonstrations on small instances (T3), plus Allstate's Chicago Quantum Exchange membership (T4). The theoretical hook — quadratic amplitude-estimation speedup — is the same one finance never realized in production, because it needs fault-tolerant depth. No deployment.

Construction & built environment is an early academic node with no industrial pilot running quantum hardware in the loop. Its own literature admits ~42% of "quantum in construction" research is actually quantum *materials* science, and compute applications are "predominantly theoretical and confined to small-scale simulations." QUBO scheduling demos exist; classical MILP still beats them (T3).

Quantum machine learning: the hype-inflated corner

AI & machine learning is where skepticism should be maximal, and the map grades it hardest by design. Quantum ML circuits run and classify toy datasets — feasibility only, no advantage (T3/T4). The decisive history is *dequantization*: Tang's 2019 classical algorithm inspired by quantum recommendation systems matched the claimed speedup, and successors retroactively erased several QML "advantages."

Loading classical data into quantum states (the qRAM I/O bottleneck) often erases any theoretical gain before it starts. The dominant tooling — NVIDIA CUDA-Q, cuQuantum — is *classical GPU simulation* of quantum circuits, which tells you where the compute actually runs. Business leaders who bet on "quantum AI" in 2025 largely walked it back. The science is worth following. The near-term product is not. Realistic advantage: unproven, possibly never for mainstream ML, plausibly a niche for quantum-native data.

Distant killer apps: climate, agriculture

Two industries are downstream of quantum chemistry, and they hold the textbook "killer app" — nitrogen fixation — at a distance that the honest number makes concrete.

Climate & sustainability is largely applied chemistry: catalysts for $\mathrm{CO_2}$ capture, and cracking the FeMoco nitrogenase problem to replace energy-hungry Haber-Bosch ($\mathrm{N_2 + 3\,H_2 \longrightarrow 2\,NH_3}$), a reaction that consumes ~1-2% of world energy. In October 2025 Alice & Bob published a resource *estimate* for simulating FeMoco on cat qubits: ~99,000 physical qubits, a ~27x reduction against the 2.7-million-qubit configuration Alice & Bob benchmarked. That 2.7M baseline sits inside the few-million-physical-qubit range of the 2021 Google/Lee study, whose headline estimate is ~4 million physical qubits (~2,100 logical) — the same numbers Chapter 8 uses (T3). Read both halves of the result. The 27x reduction is worth reporting. The ~99,000 physical qubits, thousands of logical qubits under error correction, describe hardware that does not exist and is years to a decade-plus away. Present-day climate value from quantum computers is essentially zero.

Agriculture & food science shares the same FeMoco verdict for fertilizer chemistry, and adds a hype-heavy precision-ag corner: VQC crop-yield papers claiming ~30% better prediction and ~25% less water on small, lightly refereed trials that classical ML matches (T3, treat skeptically). The one thread with a plausible short horizon is quantum *sensing* for soil, nutrient, and water measurement, which needs no fault-tolerant computer. Deployed agricultural value from quantum computing today is zero.

The classical-HPC walls: weather, semiconductors, media

Weather & climate modeling is one of the hardest places to get a quantum win. The HHL-style linear-solver speedup is real on paper and dies on the input/output problem — loading a petabyte-scale state and reading out a full field — which the field's own 2025 surveys flag plainly (T3). Numerical weather prediction is mature, world-class classical HPC (ECMWF, NOAA), so the bar is very high. The CERN Open Quantum Institute's fluid-dynamics forecasting pilot is the honest bright spot: organized, dated, explicitly pre-commercial (T4). QML nowcasting is small and classically matchable. 2030s+, gated on fault tolerance and a qRAM-class data-loading solution.

Semiconductors & EDA carries a useful reality check. Quantum computing *applied to* chip design — device-materials simulation, computational lithography, QUBO place-and-route — is early and fault-tolerance-gated (T3/T4). The concrete 2025-26 wins in "compute for semiconductors" are GPU-classical: NVIDIA's cuLitho (20-50% better cost/cycle-time on lithography) and cuEST (~50x faster materials chemistry) (T4). That is the bar quantum must clear, and those wins are classical GPU compute. The semiconductor industry's load-bearing role in quantum is as its *fab base* — real and central, covered in the hardware chapter, distinct from this application node.

Media & entertainment is a curiosity node kept for completeness. The one thing quantum offers that classical cannot fake is certified true randomness as a generative seed for procedural content — real, shipped in toy form (James Wootton's quantum wave-function-collapse PCG (T3)), and a novelty resource with no performance advantage, since classical PRNGs are indistinguishable for entertainment. Rendering, recommendation, and ad optimization are the usual pitches with no media-specific hook and no deployed pilots. Honest weight: low.

Key takeaways

- The industries with something deployed today run a quantum sensor or a cryptographic defense; the industries running a quantum computer are running a pilot.
- The map has three bands — proven (Band 1), contested pilot (Band 2), promise (Band 3) — and readiness rises along the sensing-over-computing diagonal.
- Every claim carries an evidence tier: a refereed *Nature* result (T2) and an analyst forecast (T5) are four tiers apart and should never read as one number.
- Certified randomness (JPMorganChase × Quantinuum) is the single delivered beyond-classical application on the whole industry map, and it produces no alpha.
- TAM decks inflate by double-counting one chemistry advantage across five verticals and by relabeling quantum-inspired classical work as quantum.
- The only T4-or-better revenue today is sensing and software; treat pure-play quantum computing as a 2029–2030 option priced on the ~$1B of real revenue that exists now.

Where the industry stands

The throughline is one sentence, and every card above is evidence for it. **The near-term money in quantum is sensing and cryptographic defense. Every "optimization advantage" and "simulation advantage" is today a controlled proof-of-concept or a contested annealing result, and cybersecurity is the inverse case where the honest risk is under-reaction.** The industries in Band 1 got there by running a quantum sensor that measures better than any classical instrument, or by scrambling to migrate their cryptography before a computer that does not yet exist arrives to break it. The industries in Band 2 run real hardware and real pilots, and they are honest about it — Ford Otosan's annealing app is an actual production workflow whose advantage over a good classical solver is an open question. The industries in Band 3 are, for now, promise.

Two economic distortions follow from this shape — TAM figures inflated by double-counting one chemistry advantage across five verticals, and the quantum-inspired conflation that relabels classical work as quantum. Both are unpacked in the buyer-and-investor section below.

The distance to Band-1 status for the compute industries is a physics distance: logical qubits and gate fidelities, the fault-tolerance wall of Chapter 3. Until that wall comes down, the map will keep its shape: a quantum computer is a research instrument that runs paying pilots, a quantum *sensor* is a product, and a quantum-*safe* migration is a deadline. That is where the industry stands in 2026. The sensing is real, the crypto-defense is mandatory, and the computer is still the most important machine the economy is waiting on.

What this means for buyers and investors

Buy the revenue that already exists. Today's quantum income is a sensor that out-measures the classical instrument — inertial navigation, OPM-MEG brain imaging, atomic clocks, ore-body magnetometers — and the mandated migration to post-quantum cryptography. Both ship now, both have contracts or deprecation deadlines behind them, and neither waits on a fault-tolerant computer.

For lower variance, buy the picks and shovels. Every architecture in this atlas, sensing or compute, funnels through a short list of physical chokepoints: dilution refrigerators, helium-3, synthetic diamond and NV material, and single-photon detectors. A firm that owns one of those chokepoints earns whether the winning qubit is superconducting, trapped-ion, or photonic, and whether the killer app lands in 2030 or 2035.

Treat pure-play quantum computing as a 2029–2030 option. Price it on the roughly $1B of real sector revenue that exists today — hardware access, cloud time, contested annealing deployments — rather than on the trillion-dollar TAM decks. The compute advantage is still a controlled proof-of-concept or a contested annealing result, and closing the gap is a physics problem measured in logical qubits, covered in Chapter 3 and Chapter 8.

Discount two traps on sight. The first is TAM double-counting: one fault-tolerant chemistry advantage is booked separately in the pharma, chemicals, climate, agriculture, and energy forecasts, so the summed market numbers count the same machine five times. The second is quantum-inspired conflation: classical algorithms that borrow quantum structure and run on classical hardware get sold as "quantum optimizes X today." When a vendor cannot name the qubits, assume there are none.

The names, graded

The table below is the decision-grade companion to the map — the notable builders across this atlas, with where they trade, what actually books revenue this year, an honest valuation flag, and DARPA's Quantum Benchmarking Initiative (QBI) filter, which advanced eleven of its Stage-A performers to Stage B in November 2025. The four public pure-plays all trade at price-to-sales multiples in the hundreds; that is the definition of a T5 valuation. The diversified incumbents (IBM, Alphabet, Microsoft) carry no meaningful quantum multiple — quantum is a rounding error against their market cap, so the flag is not applicable. Private venture-stage names are pre-revenue unless a sensing or software line is already shipping.

COMPANY	PUBLIC / PRIVATE	MODALITY OR SEGMENT	NEAR-TERM REVENUE SOURCE	VALUATION FLAG	DARPA-QBI STAGE
IonQ	Public — NYSE: IONQ	Trapped-ion (absorbed Oxford Ionics)	Cloud access, government contracts, networking	Price/sales ~100×; T5	Stage B
D-Wave	Public — NYSE: QBTS	Quantum annealing	Annealing cloud + on-prem, optimization pilots	Price/sales ~700×+; T5	Not selected
Rigetti	Public — NASDAQ: RGTI	Superconducting	Cloud access, foundry, government R&D	Price/sales ~800×+; T5	Stage A (not advanced)
Quantinuum	Public — NASDAQ: QNT (IPO Jun 2026)	Trapped-ion	H-series cloud, Honeywell channel, Quantum Origin PQC	New listing at a high multiple; T5	Stage B
IBM	Public — NYSE: IBM (diversified)	Superconducting, heavy-hex → qLDPC pivot	Quantum Network subscriptions, cloud	Quantum immaterial to cap; **n/a**	Stage B
Alphabet (Google)	Public — NASDAQ: GOOGL (diversified)	Superconducting (Willow)	No direct quantum revenue — R&D program	Quantum immaterial to cap; **n/a**	Self-funded (absorbed Atlantic Quantum)
Microsoft	Public — NASDAQ: MSFT (diversified)	Topological (Majorana) + Azure Quantum	Azure Quantum cloud brokerage	Quantum immaterial to cap; **n/a**	US2QC lineage (QBI precursor)
	Private			~$6B pre-/$7B post-money; ~$2B raised	

COMPANY	PUBLIC / PRIVATE	MODALITY OR SEGMENT	NEAR-TERM REVENUE SOURCE	VALUATION FLAG	DARPA-QBI STAGE
PsiQuantum		Photonic, fault-tolerance bet	Pre-revenue; government/utility partnerships	(incl. $1B Series E, Sep 2025); pre-revenue; T5	Evaluation track (ex-US2QC)
QuEra	Private	Neutral-atom	Cloud access, government R&D	Venture-stage, pre-revenue; T4	Stage B
Pasqal	Private	Neutral-atom	On-prem installs, HPC integration	Venture-stage; T4	Not in named QBI cohort
Alice & Bob	Private	Superconducting cat qubits	Pre-revenue; resource-estimate R&D	Venture-stage; T4	Stage A
Atom Computing	Private	Neutral-atom	Pre-revenue; government R&D	Venture-stage; T4	Stage B
Infleqtion	Private	Neutral-atom + atomic clocks/ sensing	Sensing and clock hardware (shipping)	Revenue-backed; T4	Stage A
SandboxAQ	Private	PQC + quantum sensing (software)	PQC migration software, magnetometry	Large private raise; T4	—
Q-CTRL	Private	Control software + quantum sensing	Inertial-nav sensors, error-suppression SW	Venture-stage; T4	—

Read the QBI column as a vendor-neutral government filter, not a scoreboard: advancing to Stage B means a credible utility-scale plan survived review, and being absent (D-Wave's annealer, the diversified incumbents' self-funded programs) means the company opted out or sits outside the fault-tolerant race the program is scoring. The valuation flags are the sharper signal. A public pure-play priced at hundreds of times sales is a T5 lottery ticket on a machine that does not yet exist; a private neutral-atom builder with no revenue is the same bet without a daily quote. The only T4-or-better income in the table is sensing and software — Infleqtion's clocks, SandboxAQ's migration tooling, Q-CTRL's inertial-nav units — which is the same conclusion the map reaches from the other direction.

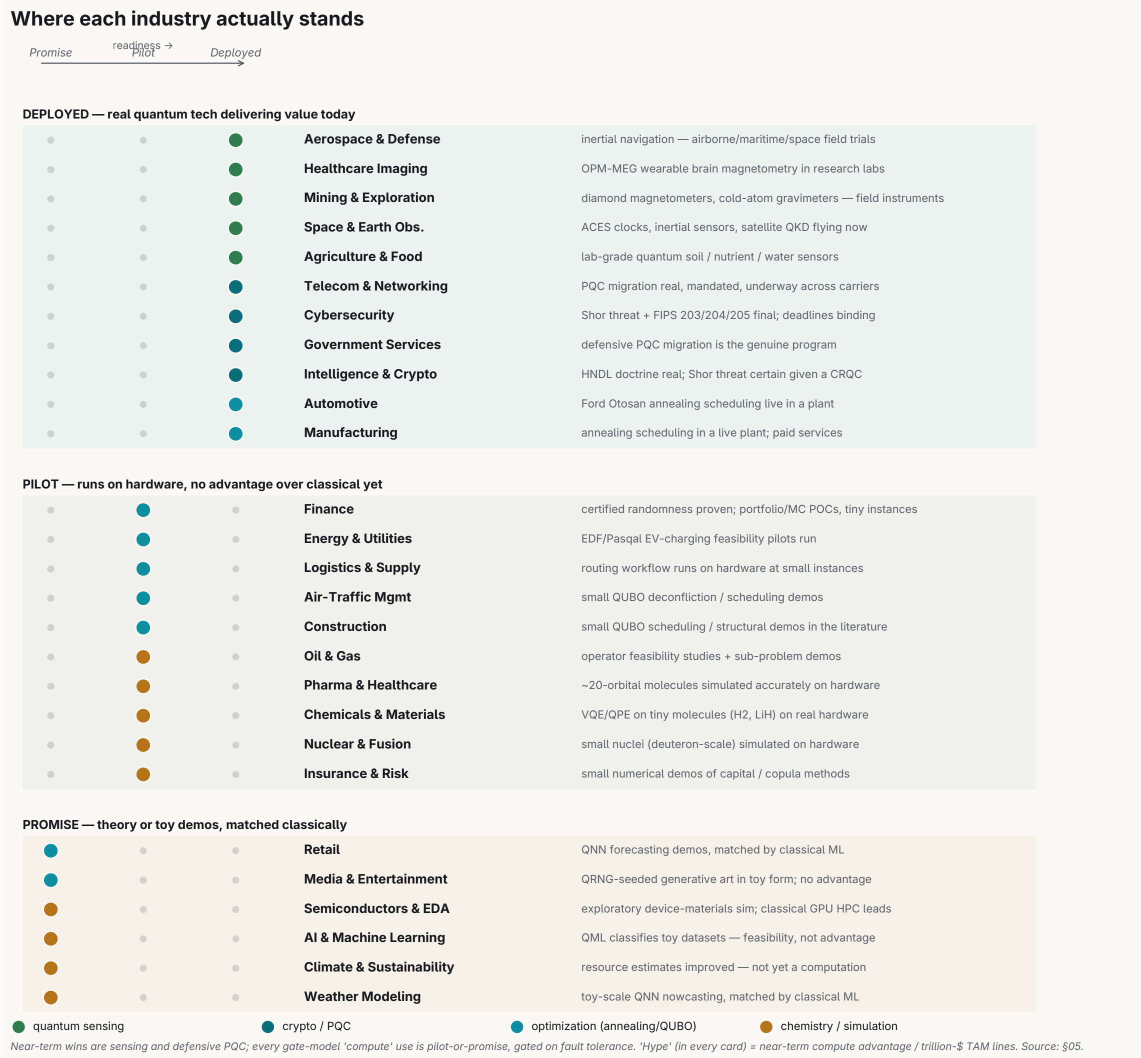
Exercises and discussion

1. **Classify and defend (all readers).** Pick three industries from the map, one from each band. For each, write a short paragraph defending its placement: name the specific quantum technology it depends on, cite the best evidence and its tier, and state the single thing that would have to change to move it up one band.
2. **Grade a vendor case study (engineers, analysts).** Take one vendor claim from the chapter — HSBC–IBM's "up to 34% improvement" on bond-trade fills (Heron, Sep 2025), or Airbus–QC Ware's "~400% faster" air-traffic result. Read the announcement as if you were the buyer, assign it a T1–T5 tier, and list what would have to be published for it to move up a tier and whether a quantum computer is provably in the loop.
3. **TAM versus revenue estimation (CS, quant).** The chapter sets ~$1B of real 2026 sector revenue against a ~$2.7T-by-2035 TAM. Treating the TAM as a target and assuming smooth compounding, compute the implied CAGR. Then identify which of the two traps most inflates that TAM and re-estimate a defensible 2035 figure with the double-count removed.

4. **Resource estimation (physics, CS).** A FeMoco-class catalyst simulation needs error-corrected machines with thousands of *logical* qubits. Using the surface-code overhead from Chapter 3, estimate the *physical*-qubit count at a plausible physical error rate, and explain why "today's devices simulate only tiny molecules" is a fault-tolerance statement rather than a software one.
5. **Seminar — where the money actually is.** The chapter argues the near-term economy is sensing and crypto-migration and that quantum computing is a 2029–2030 option. Build the opposite case: what would have to be true for compute revenue to overtake sensing revenue before 2030? Treat D-Wave's contested annealing as the test case — is it the exception that breaks the thesis, or evidence for it?
6. **Seminar — reading DARPA-QBI.** The QBI advanced eleven Stage-A performers to Stage B in November 2025, and several notable builders are absent from the cohort (D-Wave, the diversified incumbents, Pasqal). Debate what an investor should and should not infer from absence: is it a signal about a company's prospects, or an artifact of what the program is scoring?

Further reading

- **M. Liu, R. Shaydulin, P. Niroula, et al., "Certified randomness using a trapped-ion quantum processor," *Nature* 640, 343–348 (2025)** (arXiv:2503.20498). The one refereed beyond-classical *application* on the whole industry map — read it to see what a T2 result, and its narrow scope, actually looks like.
- **McKinsey & Company, *Quantum Technology Monitor* (annual).** Source of the ~$2T-by-2035 value figures the press repeats; read it against the chapter's caveat that these forecasts are unadjusted for inflation and double-counted across chemistry- and optimization-driven verticals.
- **Alice & Bob, "Quantum Resource Estimation for Ground State Energy of FeMoco and P450 on Cat Qubits" (2025).** Estimates ~99,000 physical cat qubits for FeMoco, a ~27× cut from Google's 2021 figure of ~2.7M — a concrete look at why quantum chemistry advantage is fault-tolerance-gated, and how fast the estimates move.
- **A. D. King, et al., "Beyond-classical computation in quantum simulation," *Science* (2025)**, alongside **J. Tindall, et al., "Dynamics of disordered quantum systems with two- and three-dimensional tensor networks," *Science* (2025)** (arXiv:2503.05693). The D-Wave Advantage2 spin-glass claim and the Flatiron Institute tensor-network rebuttal — the live "is annealing beyond classical?" exchange behind Band 2.
- **NIST, FIPS 203 / 204 / 205 (August 2024).** The finalized post-quantum cryptography standards (ML-KEM, ML-DSA, SLH-DSA) that make the Band-1 crypto-migration a dated deadline rather than a forecast.



The industry map — 27 industries on the proven / pilot / promise axis, coloured by the kind of quantum tech furthest along. Readiness rises along the diagonal toward sensing: the eight industries with something deployed today all run a quantum sensor or a cryptographic migration, and none runs a profitable quantum computer.

PART III · THE WORLD IT ENTERS

6 Money, Nations, and Standards

Chapter 5 mapped where quantum meets working sectors. This chapter climbs one level up, to where quantum stops being physics and becomes budget lines, patent counts, export rules, and hiring plans. Every previous chapter graded claims about nature. This one grades claims about money and power, and the rules of evidence get harder here. A qubit either holds coherence for a measured time or it does not. A national program's "$15 billion commitment" has no such anchor — it is a number a government, an analyst, or a vendor chose to say, and the person saying it usually gains something from the size of it. Read this chapter the way an auditor reads a pitch deck: find the denominator, find the motive, separate money spent from money *announced*.

Under the evidence schema, almost everything here is T5 — analyst forecast or program figure — and the schema flags national-program dollars for double-counting by default. That flag is not a formality: the single most-quoted number in the field, China's "$15 billion," has no primary source at all.

Learning objectives. After this chapter you can:

- Distinguish the three quantities that get quoted as one number — economy-wide value created, total addressable market, and booked revenue — and say which one a given headline figure is.
- Audit a quantum dollar figure by finding its denominator and the motive of whoever chose it, and separate money announced from money spent.
- Compare national strategies as instruments — verification, demand-pull procurement, government equity, multi-modality hedging — and name what each one buys and what it trades away.
- Explain why the two most-quoted 2025 investment totals differ by 3x, and show that the gap is a definitional artifact — each total counts a different scope under the same word.
- Read the post-quantum cryptography migration as the one quantum market with mandated near-term revenue, and trace how a timeline estimate and a deadline drive the spend.
- Locate the physical chokepoints — dilution fridges, helium-3, enriched silicon, quantum-grade diamond — and reason about why an export cordon keeps colliding with foreign ownership.

How to read a quantum dollar figure

Three quantities get quoted as if they were interchangeable, and they differ by one to three orders of magnitude:

$$\underbrace{\$2.7\text{T economic value}}_{\text{McKinsey, 2035}} \gg \underbrace{\$43\text{–}71\text{B vendor market}}_{\text{TAM, 2035}} \gg \underbrace{\sim \$1\text{B}}_{\text{actual sector revenue, 2025}}$$

Read left to right, the figure is a ladder: economy-wide value created towers over the total addressable market, which towers over the revenue the field actually books. The leftmost number is "value created across the economy," a modeled downstream benefit no company books. The middle is a total addressable market, a forecast of what vendors might one day sell. The rightmost is what the industry actually earned last year — McKinsey's own figure, roughly a billion dollars from hardware R&D contracts, cloud access, and government programs. Most press quotes the trillion and implies the billion, a 2,000x sleight of hand. The honest anchor for the whole chapter is that real revenue: about $1B/yr, which the same forecasters project reaching perhaps $4.4B by 2028. Every grander number bets on machines that do not yet exist arriving on their vendors' schedule.

The second habit: track a forecaster's **revision history** over its point estimate. McKinsey's 2026 Monitor put the 2035 computing market at $43–71B, revised up from about $28B a year earlier — a greater than

2x self-correction in twelve months, which says more about forecast elasticity than about the market. BCG runs on a different horizon — $450–850B in economic value by 2040, a $90–170B vendor market — so the routine press comparison to McKinsey's 2035 figure sets two different years side by side. Gartner parks quantum at the ten-years-to-plateau end of its hype cycle; IDC keeps cutting near-term numbers. The revision trail beats any single headline.

The national map

Five centers of gravity shape public quantum money: the United States, China, the European Union, the United Kingdom, and a pack of roughly a dozen mid-tier national programs.

The United States built its program on the National Quantum Initiative Act of 2018, which stood up a coordination office in the White House OSTP, tasked NIST, NSF, and DOE with research centers, and authorized about $1.2B over five years (2019–2023). Note the word *authorized*: authorization is permission to spend, which is a different thing from money out the door. Actual annual QIS budgets ran roughly $900M–1B/yr, and even that number folds in pre-existing agency programs relabeled as "quantum" — the double-counting the schema warns about, live in the flagship program. The authorization lapsed on 30 September 2023, and the field ran on annual appropriations for over two years. A reauthorization bill (S.3597) surfaced in January 2026 to extend the initiative to 2034.

Then the model changed. On **22 June 2026** two executive orders redefined the US posture from grant-making toward industrial policy. The first launched a DOE effort (QC-ADDS) to deploy a discovery-capable quantum computer by 2028 — a program *target* rather than a peer-reviewed roadmap. The second, a companion post-quantum-cryptography order, set hard federal migration deadlines: key establishment by 31 December 2030, digital signatures by 31 December 2031. Alongside these, a reported ~$2B Commerce Department package (May 2026) awarded money to nine firms, roughly half to IBM, reportedly taking **equity stakes** — a turn from vendor-neutral grants toward government-as-shareholder that the NQI Act's authors never designed for. How much of that $2B is new money versus repackaged CHIPS or DOE funds is not public — and folding the separate DARPA/IARPA defense channel into "US quantum spending" inflates the total further.

China runs the largest state-directed program in the world, centered on USTC in Hefei, the Chinese Academy of Sciences, and a National Laboratory for Quantum Information Sciences whose construction has been reported in the billion-dollar range. Its signature results are peer-reviewed at the physics layer: the Micius QKD satellite (2016), the Jiuzhang photonic-sampling series (Jiuzhang 4.0, 3,050 photons), the Zuchongzhi superconducting line (3.0, 105 qubits), and in March 2025 the Jinan-1 microsatellite's ~13,000 km China-to-South-Africa quantum-encrypted link. The 15th Five-Year Plan (adopted early 2026) ranked quantum as the top future industry, ahead of AI and semiconductors.

And yet the money is the least verifiable in the field. The famous "$15B+" commitment traces, when you follow it, to circular citation with no primary Chinese budget document behind it; a USTC physicist estimated real spending at "maybe 25 percent of that," roughly $3.75–5B. **This figure stays flagged as uncitable** — an upper bound on rhetoric rather than a budget line. Adjacent numbers (a ~$17.5B national guidance venture fund, a framing of $15B in 2024 alone) either span many deep-tech sectors with quantum's slice unknown or use opaque methodology that folds in provincial and venture-guidance money. A telling counter-signal: China's *corporate* quantum effort retreated even as the state advanced — Baidu donated its lab to a research institute, Alibaba shuttered its DAMO group. State labs and Origin Quantum now carry the momentum, the mirror image of the US private-capital surge.

The European Union built its base on the Quantum Flagship (2018, €1B over ten years), then layered EuroQCI (a quantum-secure communication network across all 27 member states), EuroHPC quantum-computer procurements, and — since 2 July 2025 — the Quantum Europe Strategy, a five-pillar plan aimed at 2030. A binding EU Quantum Act is slated for 2026. The recurring accounting trap: the €1B

Flagship gets *added* to member-state programs in "Europe spends €X" claims — the same euros counted at both the EU and national level. The commonly cited ~€11B combined figure sums multi-year pledges of different vintages and reads as order-of-magnitude only. Underneath it: Germany at roughly €5B (€2B stimulus plus a €3B action plan), France at ~€1.8B, the Netherlands at ~€615M. Europe's diagnosed weakness is its own framing — strong science, weak scale-up capital — and the VC data agrees. Its best startups (IQM, Pasqal, Quandela, Alice&Bob) raise less than US peers, and some list via US SPACs.

The United Kingdom has run a continuous national program since 2014, the earliest sustained Western bet, with about £1B through its first decade. The March 2023 National Quantum Strategy committed £2.5B over ten years around five missions, with the National Quantum Computing Centre at Harwell as hub. In 2026 a ~£2B scaling-and-procurement package added a distinctive move, **demand-side pull**: its centerpiece, ProQure, uses state procurement as an advanced market commitment — a promise to *buy* a billion-quantum-operation machine by 2031 — with QuOps milestones (1 MQuOp by 2030, 1 GQuOp by 2033, 1 TQuOp by 2035). The catch is sharp: no vendor can deliver that spec today, and the "QuOp" unit is not standardized, so buyers write contracts against undefined units. Whether the £2B sits inside or on top of the £2.5B envelope is ambiguous in the reporting — flag it until a budget document settles it. The UK's champions are strong and small, which is why Oxford Ionics being acquired by IonQ (2025, ~$1.08B) reads as a sovereignty question as much as a business one.

The rest of the pack follows one template: a multi-year headline pledge mixing new money with relabeled budgets, one flagship lab, and a niche bet. India's National Quantum Mission runs ₹6,003.65 crore (~$730M) to 2031 on indigenous full-stack hardware. Japan declared 2025 its "first year of quantum industrialization" behind a ¥1.05T (~$7.4B) headline that covers semiconductors *and* quantum — severe double-counting if quoted whole; the quantum-specific slice is closer to $900M. Australia has bet ~AU$2.3B cumulatively, dominated by ~AU$940M for one US-headquartered company's Brisbane machine (PsiQuantum), with no public plan B. Canada layers four programs (~C$360M strategy plus defence and champions tranches) on a deep startup bench. Israel runs a small headline (~$380M) over high density and deep defense overlap that hides the real total. South Korea is a late, large, chaebol-backed entrant (~$2.3B to 2035). The near-universal caveat: **pledge does not equal disbursement**, and almost none of these countries publish the gap.

Laid side by side, the five centers resolve into a scorecard, and reading across the rows exposes what each program actually is once the headline number is set aside. Every dollar figure below is T5, double-counting flagged by default; China's "$15B" stays uncitable.

COUNTRY / BLOC	HEADLINE PLEDGE	ESTIMATED REAL SPEND	INSTRUMENT MODEL	MODALITY SPECIALIZATION	KEY WEAKNESS
United States	~$1.2B NQI authorized (2019–23); June 2026 executive orders + ~$2B Commerce package	~$0.9–1B/yr QIS with relabeled agency programs folded in; the Commerce package's new-vs-repackaged-CHIPS split is not public (T5)	Grants pivoting to industrial policy + **equity stakes** (government-as-shareholder)	Broad; superconducting (IBM), plus neutral-atom and trapped-ion via the private surge	Authorization is not appropriation; the equity turn is untested; the DARPA/IARPA defense channel inflates any total
China	**"$15B+" — uncitable, flagged**	~$3.75–5B (a USTC physicist's "~25%"); no primary budget	State-directed national labs + guidance and venture-	QKD infrastructure (Micius, Jinan-1), photonic sampling (Ji-	No primary budget document; a corporate retreat (Baidu, Alibaba exits) be-

COUNTRY / BLOC	HEADLINE PLEDGE	ESTIMATED REAL SPEND	INSTRUMENT MODEL	MODALITY SPECIALIZATION	KEY WEAKNESS
		document exists (T5, stays uncitable)	guidance funds	uzhang), superconducting (Zuchongzhi)	hind the state advance
European Union	~€11B combined (order-of-magnitude only); €1B Flagship over ten years	Double-counts EU and member-state euros; Germany ~€5B, France ~€1.8B, Netherlands ~€615M (T5)	Grants + **EuroHPC procurement** (multi-modality hedge) + the EuroQCI network	Deliberately spans five modalities — superconducting (IQM), photonic (Quandela), and others	Strong science, weak scale-up capital; the best startups raise less and list via US SPACs
United Kingdom	£2.5B/10yr (2023 strategy) + ~£2B scaling & procurement (2026)	~£1B through 2014–24; whether the £2B sits inside or atop the £2.5B envelope is ambiguous (T5)	Grants + **demand-pull procurement** (the ProQure advanced market commitment)	Trapped-ion strength (Oxford Ionics, now IonQ-owned); broad otherwise	Buys against an undefined "QuOp" unit; champions are strong, small, and foreign-acquired
Others tier (India, Japan, Australia, Canada, Israel, Korea, Germany)	Japan ¥1.05T (~$7.4B, covers semiconductors *and* quantum); Korea ~$2.3B; Australia ~AU$2.3B; Germany ~€5B; India ₹6,003.65 cr (~$730M); Canada ~C$360M+; Israel ~$380M	Well below headline — Japan's quantum-specific slice is ~$900M; pledge is not disbursement and the gap goes unpublished (T5)	Grants + one flagship lab + a niche bet (procurement rare)	National niche bets — Australia → PsiQuantum photonic (Brisbane), India → indigenous full-stack, Japan → broad	Pledge is not disbursement; single-vendor concentration (Australia ~AU$940M on one firm, no plan B); Germany's ~€5B is *also* counted inside the EU aggregate above — the double-count made visible

Which policy lever buys what

Strip the money down to instruments and four distinct bets emerge. Each answers "how does a state turn quantum spend into use" a different way, and each buys a different thing at a different cost.

INSTRUMENT	MECHANISM	WHAT IT BUYS	THE TRADEOFF
DARPA QBI (Quantum Benchmarking Initiative)	Vendor-neutral verification — funds independent, staged testing of vendor claims against a hardware-agnostic utility metric	Signal: which roadmaps are real, without picking a modality winner	Verification only — it filters, it does not build capacity or guarantee a machine; the gatekeeping is slow
UK ProQure	Demand-pull procurement — an advanced market commitment to *buy* a billion-quantum-operation machine by 2031	A guaranteed customer that pulls vendors toward one concrete spec	The spec no vendor can hit today, written in "QuOp" units nobody has standardized — contracts against undefined quantities
US equity stakes	Government-as-shareholder — ~$2B Commerce, minority non-controlling equity in nine firms (~$1B to an IBM foundry)	Taxpayer upside plus onshore industrial capacity (a domestic foundry)	Conflates regulator and investor, dents vendor-neutrality, and folds in opaque CHIPS-versus-new money the NQI Act never designed for
EuroHPC	Multi-modality hedge — procures government-hosted ma-	Sovereign hosting plus insurance	12-to-54-qubit NISQ machines are symbolic sovereignty until fault tol-

INSTRUMENT	MECHANISM	WHAT IT BUYS	THE TRADEOFF
	chines spanning five modalities, each coupled to a supercomputer	against betting on the wrong technology	erance; capital spreads thin, no scale-up

The money cycle

Private capital ran through three phases. The **2021 SPAC wave** took IonQ, Rigetti, D-Wave, and Arqit public years before product-market fit; all traded more than 75% below peak during the **2022–23 trough**. Then came the **2024–26 boom**, in which generalist mega-capital — BlackRock, Nvidia, Temasek, sovereign funds — replaced the specialist quantum VCs.

Here the definitional games matter most, because the two most-quoted 2025 totals disagree by 3x for a reason:

$$\underbrace{\$3.9\mathrm{B}}_{\substack{\text{PitchBook}\\ \text{QC-only VC, 127 deals}}} \qquad \neq \qquad \underbrace{\$12.6\mathrm{B}}_{\substack{\text{McKinsey, all quantum-tech}\\ \text{startups+M\&A, } 6.3\times\text{ 2024}}}$$

The figure's point is that the 3x gap between these two totals is a definitional artifact: each firm counted a different thing under the same word. PitchBook's $3.9B counts pure quantum-*computing* venture rounds. McKinsey's $12.6B counts *all* quantum-technology startup investment, including sensing and communications, and folds in acquisition value. **Do not add them and do not compare them directly** — different denominators, same word. Either way, a striking shift held across both: government's share of quantum investment fell from roughly a third in 2024 to about 3% in 2025 as private capital took over, with BlackRock (~$1.7B) and Nvidia (~$1.6B) the largest deployers. The headline rounds tell the story. PsiQuantum raised a $1B Series E led by BlackRock, at a valuation press reports variously as $6B pre-money and $7B post. Quantinuum raised a ~$600–800M upsized round at a $10B pre-money valuation. A caution rides alongside them. IonQ traded up more than 700% in a trailing year at revenue multiples with no historical analog outside the dot-com era. About $1B of sector revenue cannot explain a valuation that size.

By H1 2026 the action rotated from primary rounds to **exits** — the consolidation wave, running on two engines. Roll-ups: IonQ ran the most aggressive acquisition spree in the sector's history — Oxford Ionics (~$1.08B, the year's largest deal), Lightsynq, Qubitekk, Capella Space, Vector Atomic, a majority stake in ID Quantique, and an announced ~$1.8B purchase of the US foundry SkyWater. D-Wave bought a gate-model developer (Quantum Circuits, $550M). And a listings wave: Xanadu via SPAC (~$3.1B enterprise value cited), IQM as the first publicly traded European quantum-computing company (~$233.5M injected), Quantinuum filing for a Nasdaq IPO. H1 2026 exits totaled ~$5.7B across four listings, roughly 15x the prior three years combined. Two skeptic's notes: many deal values are stock-heavy, so richly valued acquirer shares inflate headline M&A totals in a circular way; and consolidation reads as strength *or* distress — a field maturing toward a few integrated players, or one where standalone economics do not work at ~$1B total revenue. The whole money layer resolves the same way: post-merger revenue at the acquirers through 2027.

The standards war

Standards are where quantum becomes procurement, and the front lines are uneven. The mature one is **post-quantum cryptography**. NIST finalized three standards in August 2024 — FIPS 203 (ML-KEM) for key exchange, FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) for signatures — and selected HQC as a backup KEM in March 2025. These are the hardest artifacts in this layer: published, testable, mandated by executive order with deadlines attached. Everything else runs years behind. ISO/IEC's JTC 1/WG 14 published a terminology standard (ISO/IEC 4879) and QKD security requirements (ISO/IEC

23837); a new top-level committee, ISO/IEC JTC 3, stood up in 2024 as the first dedicated to quantum. ETSI's QKD group has run since 2008 with thin deployment adoption. ITU produced network terminology. IEEE's benchmarking and terminology efforts (P7130, P7131) remain drafts with limited pull.

The live fault line is **QKD versus PQC**. The US National Security Agency holds that quantum key distribution is unsuitable for national-security systems and favors PQC; China builds QKD backbones and dominates QKD standardization the way it dominates QKD patents. This split will be settled by a decade of deployment economics and by whether any PQC scheme is ever broken. The other open wound is the **benchmarking vacuum**: what counts as a "logical qubit" or a "QuOp" is unstandardized while procurement targets (the UK's ProQure) already cite those units in contracts. Buyers are purchasing against undefined quantities.

There is a second, quieter standards front most coverage misses: the governance of the SI units themselves, run by BIPM, CIPM, and CGPM. The 2019 redefinition anchored the kilogram, ampere, and kelvin to fundamental constants (Planck, elementary charge, Boltzmann), realized through instruments like the Kibble balance. The live 2025–26 question is redefining the **second** onto an optical atomic-clock transition, headed for CGPM 2026 — geopolitically loaded because whoever's clocks anchor the new second holds metrological soft power that ripples into timing, navigation, finance, and telecom.

The PQC migration market, and who insures the risk

Post-quantum migration deserves its own line because it is the one quantum-adjacent market with **real, mandated, near-term revenue**. It sells classical software and services — crypto discovery, crypto-agility, PKI replacement — that must ship regardless of whether a fault-tolerant quantum computer ever arrives. Its demand driver is "harvest now, decrypt later" (HNDL) — adversaries archiving encrypted traffic today to read once a capable quantum computer exists — captured by a Mosca-style inequality:

$$X + Y > Z \quad \Longrightarrow \quad \text{migrate now}$$

where X is how long your secrets must stay secret, Y is how long migration takes, and Z is the time until a capable quantum computer exists. If a secret's lifetime plus its migration time exceeds the time to "Q-day," you are already exposed — which is why long-lived secrets justify migrating today even under optimistic hardware-slip scenarios.

The timing term in that inequality — *time to a CRQC*, the cryptographically-relevant machine that can run Shor against RSA-2048 — is the one input nobody can pin down, so state it as a range. Point estimates cluster around **2030 ± 3 years**; expert-elicitation surveys (Global Risk Institute) put roughly **~25% odds on RSA-2048 falling by 2030 and about 50% by 2035**, with the heaviest weight in the mid-2030s, a tail running toward 2040, and a minority (Kalai) holding that a cryptographically-relevant machine never arrives (T5). Two forces pull against each other inside that band: falling resource estimates drag it earlier — Gidney's 2025 figure put RSA-2048 under ~1M noisy physical qubits, down from ~20M in 2019 — while the largest machines today run only ~1,000–6,000 physical qubits at the fidelity Shor needs, a two-to-three-order-of-magnitude gap that keeps it grounded. An estimate is not a machine. Chapter 8 carries the full timeline; the operational point here is that the federal migration deadlines (key establishment 2030, signatures 2031) are set *inside* the CRQC error bars, which is exactly why migration starts now rather than on Q-day.

The market forecasts show the definitional games at their worst: a **30x spread for the same year**. "PQC market" (products) runs ~\$0.5–0.8B for 2025; "PQC migration market" (adding services) ~\$1.9B; "migration solutions" (adding consulting and managed services) ~\$15B. Naming the definition is the whole game. The NIST finalization alone is estimated to trigger ~\$4.2B in net-new federal and regulated-industry spending across 2025–2026 — and this classical-cryptography spend is exactly what inflates "quantum market" tallies.

Insurers read the same risk from the other side and split their verdict. A crypto break is a **single correlated event across an entire book of policies** — the classic uninsurable-catastrophe shape, unlike the idiosyncratic risk cyber underwriting is built for — so long-run exposure is real, but near-term claims are considered unlikely. So insurers fold PQC-readiness questions into underwriting rather than reprice today; no "quantum insurance" product exists yet, and no case law says who is liable when crypto breaks. Vendors sell urgency, insurers price patience, and which is right depends on data lifetime versus Q-day.

Talent, patents, and the frontier of who owns what

The binding human constraint is **hybrid practitioners** — people who can run a dilution fridge, write control software, or turn a chemistry problem into circuits. PhD physicists exist; cryo technicians and quantum-literate domain scientists are scarce. The repeated ratio is one qualified candidate per three open roles, on a global workforce estimated around 30,000 — though both numbers rest on small, self-reported survey bases from employers with hiring incentives, and "quantum workforce" is definitional soup (does a PQC migration engineer count?). The pipeline is thin at the applied end: 176 university programs worldwide, but only 29 offer a graduate degree specifically in quantum computing. Senior engineers command $200k–500k+, which is the price signal of the shortage.

The shortage literature has a blind spot the ecosystem layer exposes. Roughly 50% of advanced-STEM-degree holders in the US defense industrial base are foreign-born, and 82% of those firms report trouble finding qualified workers, yet the technology is export-controlled and clearance-gated, which shrinks the eligible pool. The "deemed export" rule makes sharing controlled quantum technology with a foreign national *inside* the US a licensable export, so hiring a non-citizen into a controlled-tech role can itself require a license. No dataset separates "open quantum roles" from "roles a security-cleared citizen could actually fill," so the true domestic shortfall is unquantified.

Patents split on a **volume-versus-quality** axis, and the lead flips with the denominator:

$$\underbrace{\text{China } 60\% > \text{US } 19\%}_{\text{raw filings, 2024}} \qquad \text{vs} \qquad \underbrace{\text{US } 48\% > \text{China } 11\%}_{\text{international patent families}}$$

The figure's takeaway is a ranking that flips with the axis: China leads on raw filing volume, and the US leads on international patent families. China files the most quantum patents by far, but many are domestic-only, never examined abroad, and partly driven by subsidy programs that reward filing volume. The US dominates international patent families — patents protected across multiple jurisdictions, the proxy for inventions someone expects to monetize globally. China owns ~80% of QKD patent families, mirroring its infrastructure bet; IBM holds the largest single corporate computing portfolio. "Who leads quantum IP" is printed both ways from the same data — always state the denominator. And a growing shadow: error-correction decoders, control calibration, and fabrication recipes are increasingly held as **trade secrets**, so every patent landscape systematically undercounts the real frontier.

Upstream of talent and patents sits the academic base, in the same US/China/Europe tripole. USTC tops raw quantum-physics output in the Nature Index; the US clusters around MIT, Harvard, Caltech, Chicago, and Maryland's JQI; Europe around Delft's QuTech, Oxford, Innsbruck, and Paris. Read these rankings like patent filings — informative about volume, silent about breakthrough: the landmark error-correction demos came from *corporate* labs (Google, IBM, Quantinuum). The academic map is the industry map one generation early: Maryland and Innsbruck seeded the ion-trap firms, Harvard-MIT the neutral-atom ones, and Oxford fed IonQ — the same arrow that leaks talent to foreign champions.

The physical chokepoints and the export cordon

Behind the software and the money sit physical inputs with inelastic supply. A short list turns each into a strategic chokepoint: dilution refrigerators (build rate roughly one system per day, dominated by Bluefors and Oxford Instruments), helium-3 for their coldest stage (sourced mainly from nuclear-weapons tritium decay, priced $2,000–15,000/L, in structural deficit), quantum-grade lasers, silicon-28 enrichment, CVD quantum-grade diamond (near-monopoly at Element Six), and SNSPD detectors. A NATO study reportedly found more than 90% of high-purity quantum material processing happens outside NATO territory. Policy responds at small scale — Kiutra raised €13M for helium-3-free cooling, NIST launched a ~$20M center targeting the cryostat bottleneck — but $20M is thin against the capital intensity of the problem, which is why industrial-policy money and the chokepoint map are connected. Triaged for reshoring, the binding chokepoint is the cryogenic base: **dilution refrigerators, the helium-3 that feeds their coldest stage, and CVD quantum-grade diamond** gate everything downstream, so they are the first reshoring priority — silicon-28 enrichment, quantum-grade lasers, and SNSPD detectors rank behind them.

Export controls draw the cordon around all of it. On 6 September 2024 the US Bureau of Industry and Security controlled quantum computers and their enabling equipment, materials, software, and technology under new export classifications. The defining feature is that it is **plurilateral**: a license exception rewards allies who adopt harmonized controls, so coordinating states face fewer barriers than non-aligned ones. The Netherlands issued matching controls the next day; the UK, France, Spain, Australia, and Japan aligned — the first time quantum became a first-class multilateral export category. Its effectiveness is unproven — controls slow but do not stop a determined state building domestic supply to route around them — and its cohesion is untested: when IonQ (US) absorbs UK-based Oxford Ionics and Swiss ID Quantique, "allied national champion" and "foreign-owned, US-export-controlled" become the same entity, echoing the earlier ASML-Netherlands semiconductor tensions.

Software soft power and sovereign iron

Two final threads decide how the ecosystem coheres. The first is the open-source stack — soft power dressed as free tooling. Qiskit (IBM), Cirq (Google), and PennyLane (Xanadu) are permissively licensed and free, and each is a funnel toward its sponsor's cloud and hardware — whoever's framework a developer learns first shapes which machine they later reach for. The counter-force is portability: OpenQASM and the QIR Alliance (under the Linux Foundation, the one durable neutral-governance layer) keep code movable across backends. Nvidia's CUDA-Q is emerging as the hybrid GPU-QPU orchestration layer, giving Nvidia a grip on the classical side of every quantum workflow. The open geopolitical question is whether the world converges on this Western toolchain or **forks**.

Because China has built a full parallel stack. Origin Quantum's QPanda framework, its Origin Pilot operating system (open-sourced February 2026 as a deliberate soft-power move), its Tianji measurement-and-control system, plus Baidu's Paddle Quantum and Huawei's HiQ, form an end-to-end toolchain that never touches a US-governed component — Origin arguably the only company anywhere building the entire stack (processor, cryogenics, control, OS, framework, cloud) in-house. The software has already outlived the hardware that spawned it: Baidu and Alibaba exited quantum hardware, yet Paddle Quantum persists. Whether the two worlds keep honoring each other's intermediate representations through 2027–28, or export controls drive a hard bifurcation, is the fork question in concrete form.

The second thread is **sovereign compute**: states no longer only fund research, they acquire government-hosted machines on national soil. EuroHPC turned procurement pledges into six machines inaugurated across Europe — Euro-Q-Exa (IQM superconducting, 54 qubits scaling to 150, Munich), Lucy (Quandela 12-qubit photonic, 80% EU-sourced components, France), plus machines in Poland, Czechia, Spain, and Italy — deliberately spanning five modalities as a hedge against technological uncertainty, each coupled to a supercomputer. The US equity turn is the sharper version of the same instinct:

government as shareholder in national champions. The honest question underneath: does hosting a 12-to-54-qubit NISQ machine confer real sovereignty, or is it symbolic until fault tolerance arrives?

The DARPA QBI verification ladder from the instrument table has a concrete shape worth naming: it asks whether an industrially useful quantum computer is buildable by 2033, evaluating ~20 companies through staged funding —

$$\$1\text{M (concept)} \rightarrow \$15\text{M (roadmap)} \rightarrow \$300\text{M (build/demo)}$$

— with 11 at the middle stage as of November 2025. Because DARPA runs it as a filter with no prize at the end, being cut is a rare *public negative signal* in a promotional field, the opposite of the national programs now picking winners with equity checks.

Key takeaways

- Every dollar figure in this layer is T5 evidence: find the denominator, name the definition, and separate money announced from money spent before you trust it.
- The field's most-quoted number — China's "$15B" — has no primary source; the largest, loudest public figures are also the softest.
- National strategies are distinct instruments: verification (DARPA QBI), demand-pull procurement (UK ProQure), government equity (US stakes), and multi-modality hedging (EuroHPC) each buy a different thing at a different cost.
- The two headline 2025 investment totals differ by 3x because each counts a different scope under the same word — a definitional artifact, with the underlying facts in agreement.
- Post-quantum cryptography migration is the one quantum market with mandated near-term revenue — classical software sold against a future threat, driven by hard 2030–2031 deadlines.
- The physical base (fridges, helium-3, enriched silicon, quantum-grade diamond) and the "national champions" sit largely outside the alliance most trying to control them.

Where the ecosystem stands

The honest geopolitical-economic map looks like this. Public money is large, loud, and soft: the biggest single figure in the field — China's $15B — remains uncitable, and even the well-documented programs (the US NQI, the EU Flagship, the UK strategy) mix new money with relabeled budgets and count the same euros twice across national and supranational ledgers. Private money is real and rotating — from the 2021 SPAC wave through the 2022–23 trough to a 2024–26 record that is now flowing into exits — but it is priced against ~$1B of actual annual revenue and trillion-dollar TAMs that assume machines arriving on vendor schedules. That gap is the funding-winter risk, stated plainly: if fault tolerance slips and revenue does not grow into the valuations by 2027–28, the capital that generalist funds rotated in can rotate out, and a downturn could drain the field's scarce hybrid talent before the roadmaps mature.

The power map has three centers with distinct shapes. The US leads in high-quality IP, private capital, and independent verification, and has just pivoted toward industrial policy and equity stakes. China leads in state direction, raw patent volume, QKD infrastructure, and a self-sufficient parallel software stack, with its real spending unknowable and its corporate sector in retreat behind the state. Europe leads in public science and sovereign-compute hosting while lagging in scale-up capital. Around them, a dozen mid-tier programs place niche bets, and export controls draw an allied cordon that keeps colliding with the reality that "national champions" are increasingly foreign-owned.

The one market with mandated near-term revenue belongs to post-quantum cryptography migration — classical software sold against a future quantum threat, driven by hard 2030–2031 deadlines and harvest-now logic. The standards that make procurement possible are mature only for cryptography; "logical qubit" and "QuOp" stay undefined while governments write purchase contracts citing them. And the physical base — fridges, helium-3, enriched silicon, quantum-grade diamond — sits mostly outside the alliance that most wants to control it.

Read the whole layer with the auditor's reflex intact. Every dollar figure here is T5: find the denominator, name the definition, separate announced from spent, and anchor the trillion-dollar forecasts to the billion dollars the field actually earned. The physics in earlier chapters gets settled by experiment. This layer gets settled by revenue, by deployment, and by which promises the next three years keep.

Exercises and discussion

1. **Find the denominator, find the motive.** Take one real figure from this chapter — China's "$15B," McKinsey's $12.6B, or the $43–71B 2035 computing market. State which of the three quantities it is (value created, TAM, or booked revenue), name the denominator it hides, and say who benefits from the number being that large. Then write the one sentence you would append to the figure before citing it.

2. **Reconcile the 3x gap.** PitchBook reported ~$3.9B and McKinsey ~$12.6B for 2025 quantum investment. Using only the scope each firm counts, show that these numbers agree once you account for what each one measures. What single piece of metadata, if published alongside each figure, would end the confusion?

3. **Compare two national strategies (scorecard).** Pick two rows from the national map — say, the US equity-stakes turn and the UK ProQure advance market commitment. For each, name the instrument, what it buys, and its central tradeoff, then argue which one better survives a two-year slip in fault tolerance. Defend your answer against the opposite case.

4. **Resource estimation — the PQC spend.** The NIST finalization alone is estimated to trigger ~$4.2B in net-new federal and regulated-industry spending across 2025–2026, while the "PQC market" for products runs ~$0.5–0.8B for the same year. Explain how one PQC number can be 5–8x another for the same period, and identify which services and consulting layers separate the two definitions.

5. **Seminar — PQC vs. QKD national investment.** The US bets on post-quantum cryptography (software, math-based, standardized by NIST); China has built out QKD infrastructure (hardware, physics-based, a national fiber-and-satellite network). Stage a seminar debate: which is the better sovereign investment against a CRQC arriving "2030 ± 3 years"? Bring the Mosca inequality, the harvest-now-decrypt-later threat, and the deployment cost of each into your argument.

6. **Supply-chain chokepoint reasoning.** Dilution refrigerators, helium-3, enriched silicon-28, and quantum-grade diamond are the physical base, and most of it sits outside the alliance that most wants to control it. Choose one chokepoint, trace who supplies it and who depends on it, and reason about why an export cordon aimed at a "national champion" keeps colliding with the fact that the champion is often foreign-owned. What is the second-order effect of tightening that control?

Further reading

- **National Quantum Initiative Act of 2018** (Public Law 115-368), congress.gov — the statute that created the US NQI, the coordination structure, and the funding authorizations later relabeled and double-counted; read it to see what "new money" the program was actually designed to appropriate.
- **EU Quantum Flagship**, digital-strategy.ec.europa.eu — the €1B, ten-year European program and its EuroHPC procurement arm; the primary reference for the sovereign-compute and multi-modality-hedge instruments discussed here.

- **Global Risk Institute, *Quantum Threat Timeline Report*** (Mosca and Piani, annual), globalriskinstitute.org — the expert-elicitation survey behind the "~25% by 2030, ~50% by 2035" odds on RSA-2048; note that it is a poll of opinion, so read the spread and the dissenters alongside the median.
- **NIST Post-Quantum Cryptography project** (FIPS 203/204/205), csrc.nist.gov — the finalized standards that convert the quantum threat into a procurement deadline; the primary source for every PQC-migration market figure.
- **McKinsey *Quantum Technology Monitor*** (annual), mckinsey.com — a representative analyst monitor for TAM and investment totals. Caveat: treat every figure as T5, track its year-over-year revision history (the 2035 computing market was revised from ~$28B to $43–71B in twelve months), and never add its totals to another firm's.
- **BCG, *The Long-Term Forecast for Quantum Computing***, bcg.com — the $450–850B-by-2040 economic-value forecast; useful as the far end of the value-created ladder, and a clean example of why economy-wide value must never be compared to booked revenue.

PART IV

Time and the Frontier

PART IV · TIME AND THE FRONTIER

7 How We Got Here

The rest of this atlas is a map of the present: the physics that makes qubits possible, the machines that hold them, the software stack above them, and the industries and governments betting on all of it. This chapter is the road that leads to that map. Read the timeline figure as the spine, and read this as the story it tells — how a reluctant physicist's arithmetic trick in 1900 became, over 126 years, a global race to build a machine that has not yet paid for itself. The arc has three long movements: discovering the rules, arguing about what they mean, and learning to engineer with them. Each set up the one after it, and every topic elsewhere in this atlas has an ancestor on this line.

One habit travels with the whole chapter. The evidence schema grades every claim from T1 established physics down to T6 speculation, and the farther we walk toward the present, the more the grades matter. The early eras are almost entirely T1 — textbook, reproduced for a century. The recent era mixes Nature papers and launch-day press releases in the same month. The story is real; a growing number of its superlatives are provisional.

Learning objectives. After this chapter you can: - Place the field's major milestones on a timeline and name the three movements it divides into — discovering the rules, arguing over what they mean, and learning to engineer with them. - Explain why the "old quantum theory" of 1900–1925 produced correct spectra without any governing equation, and what forced the full formalism of the decade that followed. - Distinguish an established result (T1–T2) from a contested vendor claim (T4–T6) when both appear in the same month of the recent era. - Trace how a foundations-era idea once treated as a career risk — entanglement, Bell nonlocality, no-cloning — became a working component of today's engineering layer. - State the two questions Feynman and Shor left open in the 1980s and say why neither has yet been closed.

The rules arrive without an equation (1900–1925)

Classical physics broke on two experimental fronts, and the field began as a quarter-century of brilliant patching. In December 1900 Max Planck, a conservative 42-year-old thermodynamicist, quantized the exchange of energy into units of $E = h\nu$ to fit the blackbody spectrum — and disliked his own result enough to call it "an act of desperation." He treated the constant h as a mathematical device. The whole field is now named after it. Five years later Einstein, a patent clerk in Bern, took the quantum literally as a particle of light, a step so radical that Planck resisted it for a decade — and, nominating Einstein for the Prussian Academy in 1913, added a note excusing his "occasional" overreach on the light-quantum. In 1913 Bohr — then 27, working from Rutherford's Manchester lab — quantized the atom's orbits and computed the hydrogen spectrum's Rydberg constant from first principles. De Broglie closed the loop in his 1924 doctoral thesis by giving *matter* a wavelength; his examiners found the idea so strange they forwarded it to Einstein, who approved.

This was physics done by very young people under an old guard's skeptical eye, and its founders did not believe their own equations meant what they said. The "old quantum theory" produced correct spectra while lacking any governing equation. Sommerfeld's elliptical orbits and the Bohr–Sommerfeld rule $\oint p\, dq = nh$ were the high-water mark, and by 1925 the patchwork was visibly failing — it could not handle helium's two electrons or the anomalous Zeeman effect. Along the way came results this atlas still leans on: Stern and Gerlach's 1922 space-quantization experiment, now read as the prototype of a projective qubit measurement; Bose–Einstein statistics in 1924, the seed of every bosonic mode; Pauli's 1925 exclusion principle, the reason atoms have shell structure. The failure of the patchwork is exactly what forced the full formalism of the next decade. Nothing about quantum mechanics was chosen for elegance. It was dragged out of the laboratory.

One decade builds the whole formalism (1925–1935)

Then, in a single astonishing decade, the patches were replaced by a complete mathematical framework — twice, independently, from opposite instincts. Heisenberg, 23 and recovering from hay fever on the island of Helgoland in June 1925, built matrix mechanics by refusing to talk about anything unobservable and keeping only measurable transition amplitudes. Schrödinger, 38 and unusually old for a revolution, took de Broglie's matter waves literally and wrote his wave equation over the Christmas holidays of 1925–26. The two formalisms looked nothing alike, and their authors disliked each other's approach — Heisenberg called wave mechanics "disgusting," Schrödinger found matrix mechanics "repellent" — yet within months they were proven mathematically equivalent. Born supplied the piece that made the whole thing physics: $|\psi|^2$ is a probability, entered in a footnote added in proof, and Einstein spent the rest of his life objecting to it. Dirac, the quietest of them, unified the formalism, married it to special relativity in 1928, and predicted antimatter from the mathematics alone — Anderson found the positron in 1932. Von Neumann handed the field its Hilbert-space axioms in 1932, still the machinery in use today.

Notice how much of the present atlas is already present in this decade. The Born rule, Heisenberg's 1927 uncertainty relation, Dirac's relativistic equation as the bridge to quantum field theory, and von Neumann's density operator are all foundations the rest of this atlas leans on. The speed was the story: from Heisenberg's notebook to von Neumann's axioms was seven years, most of it among people in their twenties at Göttingen, Copenhagen, and Munich. The credit fell unevenly — Born, whose interpretation is arguably the deepest single idea of the decade, was left off Heisenberg and Jordan's Nobel and waited until 1954.

The decade did not close on completion. Its own architects flagged what the theory left unresolved, and those loose ends became the next movement's raw material. Two of them deserve their own place on the timeline.

The argument goes public: Solvay and EPR (1927–1935)

In October 1927 twenty-nine physicists gathered in Brussels for the Fifth Solvay Conference on "Electrons and Photons"; seventeen were or would become Nobel laureates. The group portrait — Einstein front-center, Curie the only woman, Bohr, Planck, Dirac, Heisenberg, Schrödinger, de Broglie, Born, and Pauli around them — is often called the most intelligent photograph ever taken. It is the moment quantum mechanics, barely two years old, was presented as a finished theory, and the moment its interpretation split the field. Bohr and Heisenberg's Copenhagen view — the wavefunction is complete, and a system has no definite properties until measured — met Einstein's refusal to accept that physics had become irreducibly probabilistic. The popular story compresses a week of hard argument into one quip, but the real exchange ran through breakfast and dinner: Einstein built ingenious thought experiments to beat the uncertainty relation, and Bohr answered each by showing that the very apparatus needed to make the measurement would blur the conjugate quantity by exactly enough to preserve the bound. "God does not play dice" captures the temperature of the debate, though it is a paraphrase. Einstein conceded the internal consistency of quantum mechanics. He never conceded its completeness.

He shifted ground to completeness and locality in May 1935, and it produced the most fertile question in the field's history. The EPR paper — Einstein, Podolsky, Rosen — ran four pages under a title posed as a question: "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" Their answer was no. Using two particles that had interacted and separated, they argued that measuring one instantly fixes a property of the other; if no signal can travel between them and the outcome is definite, that property was real all along — an "element of reality" the wavefunction omits. Either quantum mechanics is incomplete, or nature is nonlocal. Bohr replied five months later that the two particles form a single indivisible system. Schrödinger, energized by the paper, coined *Verschränkung* — entanglement — to name precisely the correlation EPR had isolated, and wrote his cat into the record the same year.

The word "paradox" is a misnomer; there is no logical contradiction, only a forced choice between locality and completeness. Einstein intended EPR as a critique. It became the seed of entanglement, teleportation, quantum key distribution, and the speedups quantum computers run on. For thirty years the exchange looked like unresolvable philosophy. Then someone made it an experiment.

The foundations era and its career risk (1935–1980)

For three decades the EPR question sat in philosophical exile while physics built QED and the bomb. Asking what the wavefunction "really" meant was, as John Clauser later put it, a good way to end a career. The renormalization of QED moved the mainstream to fields, leaving foundations to a marginalized few. David Bohm reopened it in 1952 with a working hidden-variable theory, pilot waves, that the orthodoxy had declared impossible by citing von Neumann's 1932 no-go proof; Bohm's counterexample showed the proof assumed too much. Everett's 1957 relative-state ("many-worlds") interpretation offered a collapse-free alternative and was ignored for a decade, after which Everett left academia for defense work.

The pivot of the whole era came from John Stewart Bell, a particle physicist at CERN who did foundations "on the side." In 1964 he proved that *any* local hidden-variable theory obeys an inequality that quantum mechanics violates — turning a metaphysical dispute into a laboratory measurement. He published it in an obscure, short-lived journal that paid its authors, partly so he could keep his real job. This is the founding act of Bell nonlocality, and everything downstream in the atlas — device-independent cryptography, certified randomness, the security of E91 — descends from it. CHSH recast the inequality in 1969 into a form real, inefficient detectors could test. Freedman and Clauser ran the first experiment in 1972 on borrowed equipment at Berkeley; Clauser had been warned it would sink his career, ran it anyway, and got the "wrong" answer, quantum mechanics being right and disappointing his own realist hopes. Aspect's 1981–82 experiments in Orsay switched their analyzer settings while the photons were in flight, closing the locality loophole of the day. Quantum mechanics won every round. In parallel, Zeh's 1970 decoherence program explained why the classical world emerges from the quantum — the mechanism now the enemy of every qubit. The people who built the conceptual foundation of a trillion-dollar field mostly did it against professional incentives.

Information before computation (1970–1984)

Before there was quantum computing, there was quantum information — the recognition that a quantum state carries information under its own rules, distinct from Shannon's. The patient zero is Stephen Wiesner, a Columbia graduate student who around 1970 wrote "Conjugate Coding," describing how to encode two messages in conjugate observables so a receiver can read one or the other but never both, and how to mint banknotes that cannot be counterfeited because an unknown quantum state cannot be copied. It was so far ahead of its time that it was rejected and sat unpublished for thirteen years; Wiesner spent years as a construction laborer in Israel while the field he seeded became a global enterprise. Two people who took the manuscript seriously, Charles Bennett and Gilles Brassard, turned it in 1984 into BB84 — the first quantum key distribution protocol, and quantum information's first working technology, still deployed. In parallel, Holevo proved in 1973 that a qubit carries at most one bit of accessible classical information, and Wootters, Zurek, and Dieks proved the no-cloning theorem in 1982 that makes the whole edifice secure.

This lineage corrects a common simplification: the field learned to *protect* information with quantum mechanics before it learned to *process* it. By 1984 the conceptual toolkit — qubits as carriers, no-cloning as a security primitive, channel capacities, conjugate coding — was in place, waiting for computation to arrive.

The birth of quantum computing (1980–1994)

It arrived from several directions at once. Yuri Manin (1980, in a Russian book) and Paul Benioff (1980, with an explicit quantum Turing machine) asked whether computation itself could be quantum-mechanical. Richard Feynman flipped the question at a May 1981 MIT keynote: classical computers choke on simulating quantum systems because the state space grows exponentially, so build a machine to simulate quantum physics natively — "Nature isn't classical, dammit." He was pitching a simulator, skeptical it could be built. David Deutsch made it rigorous in 1985 with a universal quantum computer and the first quantum algorithm, motivated partly by wanting to test the many-worlds interpretation. The Deutsch–Jozsa algorithm (1992) gave the first provable exponential separation, and quantum teleportation (1993) turned entanglement into a communication resource.

Then came the detonation. In November 1994 Peter Shor, at Bell Labs, factored integers in polynomial time — threatening RSA and the entire public-key infrastructure. The algorithm reportedly grew out of Simon's problem, which Shor heard about at a talk and generalized within weeks. Overnight, quantum computing went from a physicists' curiosity to a national-security priority. Every funding line in the ecosystem layer of this atlas — the US, China, EU, and defense programs — traces back to that one result, and the post-quantum cryptography migration it forced is still underway thirty years later. Shor's algorithm remains mathematically proven and hardware-starved, which is the tension the rest of this history is about.

Learning to build one (1995–2010)

After Shor, the question stopped being "is it useful?" and became "can you build one?" This era answered "in principle, yes," while revealing how hard the engineering would be. Cirac and Zoller published a realistic trapped-ion gate in May 1995, and Wineland's NIST group demonstrated a working logic gate on a single beryllium ion within seven months. The deepest results were theoretical: Shor (1995) and Steane (1996) proved that quantum errors could be corrected *without measuring, and thereby destroying, the encoded data*, removing the single strongest objection to the whole enterprise. The threshold theorem (Aharonov–Ben-Or, Knill–Laflamme–Zurek, Kitaev, 1996–98) then showed that below a critical error rate, arbitrarily long computation is possible. That theorem is the license under which the entire error-correction layer of this atlas operates, and its experimental realization would not come for another 28 years.

Grover added a second headline algorithm in 1996, the quadratic search speedup. NMR machines ran the first real algorithms — including Shor's algorithm factoring 15 in 2001 — before hitting a wall: Braunstein and colleagues showed in 1999 that room-temperature liquid-state NMR states are so mixed they are arguably never entangled, and the signal decays exponentially with qubit count. The early NMR lead was real and then it evaporated, a first lesson in how a demonstration can be true and un-scalable at once. DiVincenzo wrote down the five-item checklist any real platform must satisfy (2000), still the field's scorecard. And superconducting circuits went from the first coherent charge qubit (NEC 1999) through circuit QED (Yale 2004) to the transmon (Yale 2007), the charge-noise-insensitive design that still dominates superconducting hardware today. The transmon is built on a Josephson junction, the same device whose 1984–85 macroscopic-quantum-tunneling experiments later earned Clarke, Devoret, and Martinis the 2025 Nobel — the Nobel spine section below tells that story. This era is where the modern hardware map was drawn: ions and superconductors as the two front-runners, error correction as the organizing goal.

Naming the eras: the second revolution and NISQ

Two acts of naming shaped how the field was funded and pursued, and both belong on the timeline because a name changes what gets built. In 2003 Jonathan Dowling and Gerard Milburn crystallized the phrase "second quantum revolution" in a Royal Society paper that opened flatly: "We are currently in the midst of a second quantum revolution." The distinction is substantive. The first revolution (1900–1935) discovered the rules and used them passively — the transistor, the laser, the atomic clock all exploit quantum mechanics in bulk without addressing a single electron. The second revolution engineers individual quantum systems: trap one ion, entangle two photons on demand, hold a superconducting qubit coherent long enough to run a gate. That capability is exactly what the 2012 Nobel to Haroche and Wineland honored, and it is the precondition for every node in the hardware and stack layers. The framing also set an honest expectation bar: if the first revolution's payoff was the transistor, the second's advocates are implicitly promising something comparable, which the skeptics say has not yet arrived.

The second naming was narrower and more sober. By 2017 the field needed a word for the machines it was actually building: 50 to 100 physical qubits, imperfect gates, no error correction — powerful enough to be hard to simulate classically, too noisy to run a deep useful circuit. John Preskill supplied it in a December 2017 keynote: NISQ, Noisy Intermediate-Scale Quantum. The coinage stuck instantly and organized a research program around variational algorithms and error mitigation. Preskill gave hardware groups a realistic near-term target while warning that NISQ technology "will not change the world by itself." Eight years on, that warning reads as prescient: no NISQ algorithm has produced a clear, durable practical advantage, and the field's own answer has been to shift focus from noisy circuits to early fault tolerance.

The engineering race and the first "supremacy" (2011–2019)

Quantum computing left the physics lab and became an industrial race. D-Wave sold the first commercial machine in 2011 — an annealer rather than a gate-model computer — kicking off a decade-long argument about whether it delivered any quantum speedup at all; Rønnow and Troyer's 2014 benchmark found no generic speedup, and annealing settled into niche uses without ever demonstrating the general exponential speedup its early marketing implied. IBM put a real 5-qubit processor on the open cloud in 2016 and normalized publishing qubit counts as press events, birthing the cloud-QC business model. Google hired John Martinis's UCSB group in 2014 and pointed it at a single target: beat a classical supercomputer at *something*. Governments answered with billion-dollar programs — the EU Flagship (2016) and the US National Quantum Initiative Act (2018).

The era peaked in October 2019 with Google's Sycamore claim: a 53-qubit chip sampling random circuits in about 200 seconds against an estimated 10,000 classical years — the first claimed "quantum supremacy." IBM published its rebuttal the same week the Nature paper ran, arguing a well-configured Summit supercomputer could do the task in 2.5 days, and over the next three years tensor-network methods shrank the gap to hours. The demo stands; the "10,000 years" figure does not. That claim-and-counterattack cycle — a headline number, a rapid classical response, a milestone that survives while its superlative does not — became the template governing every advantage announcement since, and it is why the evidence schema treats "advantage" and "supremacy" claims as contested by default.

The error-correction era (2020 → 2026)

The scoreboard metric changed. Through the 2010s the field measured itself in physical qubit counts and one-off supremacy stunts; from about 2023 it measures itself in *logical* qubits and error suppression. Two results reset expectations. In a 2024 Nature result (Bluvstein et al.) a Harvard/QuEra/MIT collaboration operated 48 logical qubits on 280 neutral atoms — logical qubits at scale, credibly, for the first time. Then in December 2024 Google's Willow achieved the first *below-threshold* surface code: making the code larger made the logical error rate exponentially *smaller* ($\Lambda \approx 2.14$ across code distances 3, 5, and 7). This is the 1996 threshold theorem realized in hardware rather than proven on paper, and it is peer-reviewed (Nature 2025, T2). Fault tolerance became an engineering trajectory instead of a hope, and the record kept climbing — QuEra reported 96 logical qubits on 448 atoms in January 2026, a peer-reviewed Nature result that roughly doubled the count in about two years.

The rest of the era is where the grading discipline earns its keep, because peer-reviewed landmarks and contested vendor claims now appear in the same month. IBM's 2023 "utility" experiment on 127-qubit Eagle was matched by tensor-network methods within weeks. Google's 2025 "Quantum Echoes" verifiable-advantage claim is contested by default per the schema. Quantinuum launched Helios in November 2025 (98 barium-ion qubits, 99.921% two-qubit fidelity, up to 48 logical qubits) and filed confidentially for a ~$20B IPO in January 2026 — a T4 launch and a T5 valuation, neither an independent computational result. In June 2026 a Duke/IonQ team distributed a tripartite GHZ state across three networked trapped-ion nodes linked by photonic interconnects, a real step toward modular quantum computing, graded T3 as a preprint.

Three recent claims show exactly where the honest edges are. Sycamore's 2019 superlative fell to classical simulation. Willow's below-threshold QEC is peer-reviewed, but its companion random-circuit-sampling claim ("under 5 minutes vs. 10^{25} classical years") is a T4 marketing figure riding alongside a T2 result in the same press release. And Microsoft's topological-qubit program — Majorana 1 (Feb 2025), Majorana 2 progress (June 2026) — remains the era's sharpest credibility test: Nature's referees noted the 2025 data "does not, on its own, determine whether Majorana zero modes are present," and in June 2026 Henry Legg published a formal Nature critique citing flawed tune-up routines and software errors, to which Microsoft replied that it stands by its results. Two years of announcements without independent confirmation is the honest status. The trajectory is real; the specific superlatives are provisional.

The Nobel spine and the winter question

Two threads run underneath the whole timeline. The first is the Nobel spine, a reliable record of what is *established* that, by construction, lags the frontier by decades. The early prizes canonized the discovery of the rules — Planck (1918), Einstein (1921), Bohr (1922), Heisenberg (1932), Schrödinger and Dirac (1933). A middle cluster honored the deep theory and materials physics that became hardware — Born's probability rule (1954, a 28-year wait), BCS superconductivity (1972), the Josephson effect (1973). The recent prizes reward the second revolution directly — individual-quantum control to Haroche and Wineland (2012), the Bell-test verdict to Aspect, Clauser, and Zeilinger (2022), and macroscopic quantum tunneling in Josephson-junction circuits to Clarke, Devoret, and Martinis (2025), the direct ancestor of every superconducting qubit. The 2025 prize honored circuit experiments from 1984. No gate-model quantum *computer* has yet earned a Nobel, which is itself an honest signal about where the field is versus where the press releases say it is.

The second thread is the "quantum winter" question, the AI-winter analogy made into a live risk. The bear case is concrete: private quantum funding reportedly fell sharply in 2023 before recovering; the generative-AI boom pulled capital and talent toward a technology with immediate revenue; and every beyond-classical claim from Sycamore forward has drawn a classical counterattack. The bull case is also concrete: below-threshold QEC and doubling logical-qubit counts are peer-reviewed, and national programs provide funding floors that do not exit on a bad quarter. This is a scenario graded T5/T6 rather than an established fact, and it is the honest frame for reading every vendor roadmap in the ecosystem layer.

> **Key takeaways** - Quantum mechanics was dragged out of the laboratory: its founders distrusted their own equations, and correct spectra arrived before any single governing equation did. - The rules are settled to T1 — superposition, the Born rule, entanglement, the Bell violation — while the interpretation argument Einstein opened at Solvay is still open. - The scoreboard shifted around 2023 from physical-qubit counts to logical qubits and error suppression; below-threshold QEC (Willow, 2024) and doubling logical counts (QuEra) are peer-reviewed. - The Nobel spine lags the frontier by decades, and no gate-model quantum computer has yet earned one — an honest signal about where the field is versus where press releases say it is. - The "quantum winter" is a live scenario graded T5/T6,

> not a fact, and it is the frame for reading every vendor roadmap. - No quantum computer has yet produced a verified, durable advantage that survives classical counterattack.

Where the story stands

The three movements of this history hand the present a specific inheritance. The rules were dragged out of the laboratory and are settled to T1 — superposition, the Born rule, entanglement, and the Bell violation are not in dispute, and the loophole-free tests of 2015 closed the last experimental door. The interpretation argument that Einstein opened at Solvay is *not* settled; the measurement problem remains open, and the field built a trillion-dollar engineering program on top of a question it never answered. The foundations era's career-risky work became the resource layer of the whole atlas: entanglement, no-cloning, teleportation, and Bell nonlocality are now components.

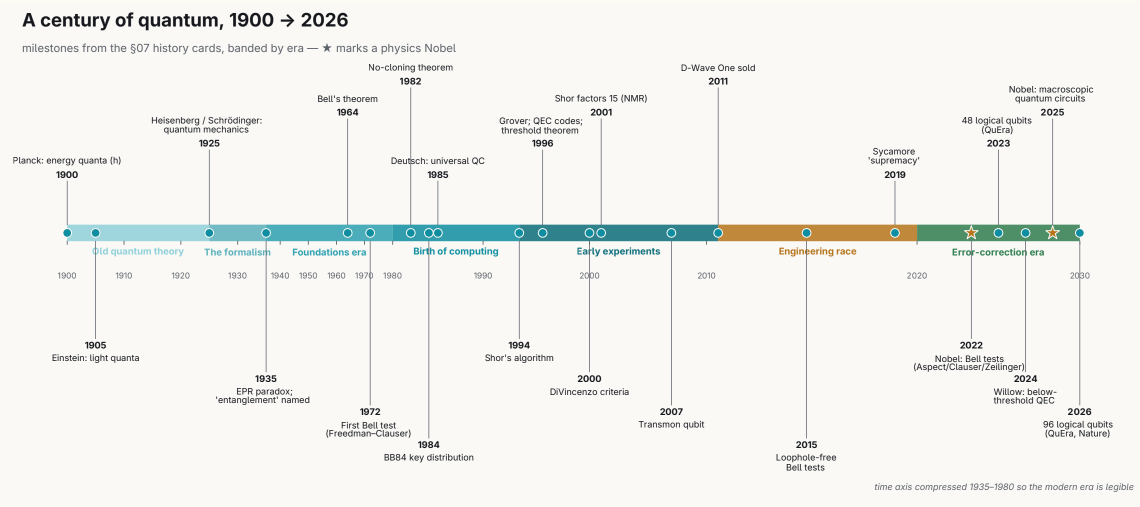
The engineering movement is the one still in motion, and it is where this atlas is most alive. The field has crossed from proving fault tolerance on paper to demonstrating it in hardware, the reason the error-correction layer is the center of gravity today. But the two questions Shor and Feynman left open in the 1980s are still open. Shor's algorithm is proven and waiting on millions of physical qubits it does not yet have. Feynman's simulation pitch is the application most likely to pay off first, and it has not yet paid off. Between them sits the verdict the timeline forces: the trajectory from foundations to engineering is real and accelerating, and no quantum computer has yet produced a verified, durable advantage that survives classical counterattack. That gap — between a physics that is finished and a machine that is not — is the present the rest of this atlas maps. The history explains why the gap exists, why it is narrower than it was, and why every claim about closing it should be read with its grade attached.

Exercises and discussion

1. **(Timeline.)** From the milestones in this chapter, pick eight and place them on a single timeline. Tag each with the movement it belongs to — discovering the rules, arguing over meaning, or engineering with them — and write one sentence defending every boundary you draw. Where does one movement bleed into the next, and which milestone was hardest to file?
2. **(Physics.)** The Bohr–Sommerfeld quantization rule $\oint p\, dq = nh$ reproduced the hydrogen spectrum but broke on helium and the anomalous Zeeman effect. Explain, in your own words, why a rule that fits a one-electron atom fails on a two-electron atom, and name what the 1925–1926 formalism supplied that the old quantum theory lacked.
3. **(Resource estimation, for CS.)** Shor's algorithm is proven and "waiting on millions of physical qubits it does not yet have." Given a below-threshold surface code with $\Lambda \approx 2$, sketch how logical error rate falls as code distance grows, and argue qualitatively why a cryptographically relevant factoring run needs so many physical qubits per logical one. No exact figure is required — reason about the scaling.
4. **(Critical reading.)** Choose one milestone from 2019–2026 that was over-claimed — Sycamore's 2019 "supremacy," IBM's 2023 "utility" run, or Google's 2025 "Quantum Echoes" — and reconstruct how the field learned the claim's limits. What was asserted, who counterattacked and with what method, and what grade (T1–T6) does the result deserve now?
5. **(Seminar.)** The Bohr–Einstein debate that opened at the 1927 Solvay conference was never resolved: the measurement problem stays open while a trillion-dollar engineering program was built on top of it. Is an unresolved interpretation a problem for the engineering or irrelevant to it? Argue both sides, then take a position.
6. **(Seminar.)** "Second quantum revolution" and "NISQ" are two names the field gave itself. What work does each framing do, and what does each hide? Consider who coined each term, when, and what they were arguing against.

Further reading

- **R. P. Feynman, "Simulating Physics with Computers," *International Journal of Theoretical Physics* 21 (1982).** The lecture that framed quantum simulation as the field's first natural application — the pitch this chapter calls still-unpaid.
- **J. S. Bell, "On the Einstein Podolsky Rosen Paradox," *Physics* 1, 195 (1964).** Six pages that turned the Solvay interpretation argument into an experiment, producing the inequality whose violation is now T1.
- **A. Aspect, "Closing the Door on Einstein and Bohr's Quantum Debate," *Physics* 8, 123 (2015).** A short retrospective by a principal of the Bell-test program on the 2015 loophole-free experiments; a readable history of how the tests closed each loophole.
- **J. Preskill, "Quantum Computing in the NISQ Era and Beyond," *Quantum* 2, 79 (2018).** The paper that named the present hardware era and set honest expectations for what noisy machines can and cannot do.
- **D. Kaiser, *How the Hippies Saved Physics* (Norton, 2011).** A popular history of how foundations work survived the "shut up and calculate" decades to become today's resource layer.
- **J. Gribbin, *In Search of Schrödinger's Cat* (Bantam, 1984).** A durable popular narrative of the 1900–1935 discovery movement for readers who want the story before the formalism.



The arc, 1900 → 2026, banded by era; Nobel Prizes starred. The ribbon's shape is the point: a quarter-century to find the rules, a long quiet middle, then a dense burst of hardware milestones after 2019 — with the Nobel stars trailing the frontier by decades every time.

PART IV · TIME AND THE FRONTIER

8 The Honest Frontier

Every chapter before this one described something that works. Superposition and the Born rule (Chapter 1) are textbook physics, reproduced for a century. Transmons, trapped ions, and neutral-atom arrays (Chapter 2) are real machines you can rent by the minute. Shor's algorithm, the surface code, and variational chemistry (Chapter 3) are written down and, at small scale, demonstrated. Sensing, QKD, and post-quantum cryptography (Chapter 4) already ship. This chapter is different. It collects the questions the field has not answered, and it is the chapter that decides whether the rest of the atlas is a map of a real territory or a map of a promise.

An immature field is graded by how it treats its own open problems. The mature move is to name them out loud, state who disagrees, and say exactly what experiment or event would settle each one. That is the whole method here: every claim carries a tier from T1 (established physics) down to T6 (speculative), and every unresolved dispute is kept as a first-class object with both sides intact. A vendor announcing its own benchmark is T4 until someone independent reproduces it; a market forecast is T5; a roadmap date is T4 or T6 on principle, since a roadmap is a marketing document until a referee sees the result. Hold those rules and the frontier stops being a fog of press releases and becomes a set of sharp, bettable questions.

The open problems cluster into a small number of threads: a **scaling gap** between the machines that exist and the machines the applications need; an **advantage attrition record**, a running scoreboard of claimed quantum wins and how many survived classical counterattack; a **money-and-hype** question that runs on a faster clock than the physics; a layer of **hard engineering bottlenecks** hiding under the headline qubit counts; and, underneath all of it, the **oldest open problem in the field**, the measurement problem, unsolved since 1925. Take them in that order.

Learning objectives. After this chapter you can: - State the scaling gap in numbers — why thousands of logical qubits and billions of gates imply hundreds of thousands to millions of physical qubits, and why the best independently verified count today is 96 logical qubits (QuEra, *Nature*, January 2026). - Distinguish a physical qubit from a logical one, and estimate error-correction overhead from a code's distance and a target logical error rate. - Grade a quantum-advantage claim on the T1–T6 scale and read the attrition scoreboard — which headline results were matched or dequantized classically, and which (D-Wave's spin-glass claim, Quantum Echoes) still stand. - Apply the Mosca inequality to decide when cryptographic migration must begin, independent of the exact date a cryptographically relevant quantum computer arrives. - Separate valuations, market forecasts, and roadmap dates (T4–T6) from refereed results (T1–T2), and name the falsification condition that would settle each open dispute — including Gil Kalai's in-principle objection. - Explain why the measurement problem remains open a century after the formalism was written down, and what experiment could still move it.

The scaling gap: how far

Start with the single number that governs the whole enterprise. A cryptographically or chemically useful quantum computer needs thousands of logical qubits running billions of gates. Under today's architectures that translates to hundreds of thousands to millions of physical qubits held at fidelity for the length of the computation. As of mid-2026 the largest devices sit at roughly 1,000–6,000 physical qubits, and the best independently verified logical-qubit demonstration counts 96 (QuEra, *Nature*, January 2026). The distance from "96 logical qubits, a few thousand operations" to "thousands of logical qubits, 10^9 gates" is five orders of magnitude in qubit count while *holding* fidelity — and holding fidelity as you scale is the part nobody has shown. Picture the scaling-gap figure as two curves on a log axis:

demonstrated physical-qubit counts creeping up year by year, and the required count for a useful algorithm sitting flat and far above. The open question is whether those curves meet in 2029, in 2040, or never. This is the scaling gap, and it is the spine the rest of the chapter hangs on.

The optimist camp has a concrete plan. IBM published a modular, milestone-by-milestone roadmap in June 2025: **Loon** (2025), **Kookaburra** (2026, the first fault-tolerant module joining logic and memory), **Cockatoo** (2027, entanglement between modules), and **Starling** (2028–29) at roughly 200 logical qubits and 100 million gates, which IBM calls large-scale fault tolerance. **Blue Jay** (2033) targets 2,000+ logical qubits and a billion gates, and IBM claims its move to qLDPC codes cuts physical-qubit overhead up to 90% versus surface codes. Every item on that list is a roadmap, graded T4 until a refereed result lands. Google targets a useful error-corrected machine on a similar 2029–2030 horizon, also T4. What lifts the optimist case above pure marketing is that 2025 delivered: Scott Aaronson, no one's idea of a hype man, wrote in March 2026 that the year "met or exceeded my expectations on hardware, with multiple platforms now boasting >99.9% fidelity two-qubit gates," and said he had updated "in favor of taking more seriously the aggressive pronouncements ... about where they could be in 2028 or 2029" (T3, expert commentary).

The skeptic camp splits into two very different objections, and honest reading keeps them separate. The first is a timing objection. Jensen Huang (Nvidia, January 2025) put "very useful" quantum computers 15–30 years out, a remark that cratered quantum stocks before he softened it at GTC's Quantum Day (T5). The timing skeptic grants the physics and doubts the calendar. The second objection is deeper. Gil Kalai argues fault-tolerant scaling is impossible *in principle*: noise in engineered many-body systems is inherently correlated, correlated noise violates the independent-error assumption behind the threshold theorem, and the correlations become fatal before you reach a million qubits. This is T3, a published minority position, and it is the most important single disagreement in the field because it is the only one that says the destination does not exist.

The March 2026 exchange between Kalai and Aaronson is worth reading in full because it models how to argue about a frontier. Aaronson's line: Kalai "keeps saying [the 2019 Sycamore] experiment was fallacious, but meanwhile a dozen other experiments by other companies have reached the same conclusion, so he's fighting a losing battle," and Kalai's "path has been getting narrower and narrower." Kalai's reply (gilkalai.wordpress.com, 10 March 2026) concedes the framing honestly — he "starts with quantum computation being impossible as [an] axiom" — and then makes a falsifiable claim: the measured exponential decay of error with code distance "does not contradict my conjectures regarding correlated errors," which he expects to surface at "hundreds ... and thousands of gates," not millions. That is a testable boundary. The correlated noise he points at also names a daily practitioner phenomenon: crosstalk and ZZ coupling, leakage out of the computational subspace, and calibration drift that device teams fight on every run, well below any scaling limit. Kalai's position dies the day any machine sustains fault tolerance across thousands of gates; it strengthens with every correlated-noise surprise at scale. Aaronson keeps "a little voice ... that asks 'but what if Gil Kalai turned out to be right,'" while adding that in 2026 it "can barely be heard." Here the call is editorial rather than a neutral finding: adopt Aaronson's posture, assign the skeptic a low but nonzero probability, and name the exact observation that would move it. A reader who weighs Kalai's in-principle argument more heavily would land on a higher probability; the evidence does not fix the number, so the weight is a judgment. What settles the scaling gap is a verified result: Kookaburra (2026) and Starling (2028–29) hitting spec independently, or missing it.

The tax you pay per logical qubit

The scaling gap is set by the **error-correction overhead**, and the numbers here need care because the same field reports ratios that differ by two orders of magnitude, all technically true. A standard surface code at useful code distance implies roughly

$$\frac{n_{\text{physical}}}{n_{\text{logical}}} \approx 1{,}000{:}1$$

once you include the magic-state factories that supply the non-Clifford T-gates. Recent demonstrations report far friendlier ratios: QuEra's 96 logical qubits came from 448 physical atoms, about $4.7{:}1$, using a high-rate [[16,6,4]] qLDPC code; Quantinuum reported 48 error-corrected logical qubits at roughly $2{:}1$ with logical error below 10^{-4}. On the same Helios machine, running its 98 physical qubits without error correction, Quantinuum also prepared a 94-qubit GHZ state — a single entangled state spanning 94 qubits, which is a state, not 94 logical qubits. Both numbers are real. The catch is code distance. A [[16,6,4]] code has distance $d = 4$. A billion-gate algorithm needs logical error near 10^{-12}, which takes distance around $d \approx 25$, and at that distance realistic architectures project back to hundreds-to-$1{,}000{:}1$ all-in once routing, ancillas, and T-factories are counted. Both positions stay on record: qLDPC papers claiming $\sim 2{:}1$ to $13{:}1$ demonstrated or roadmapped, and surface-code resource analyses insisting the all-in cost at target error stays in the hundreds. What settles it is stated precisely: an end-to-end fault-tolerant algorithm *including T-gates* at $d \geq 11$ with the full published budget. A sustained sub-$100{:}1$ ratio at algorithm-relevant distance hands the win to the qLDPC optimists; deep-circuit demos that quietly revert to $\sim 1{,}000{:}1$ once T-gates enter hand it to the pessimists. The tell, every time, is whether a "logical qubit" count reports a full universal gate set or only memory. Vendor counts often mean memory. Grade them T4 and ask.

The clock on cryptography

The scaling gap has one sub-question with a deadline attached: the **cryptographically-relevant quantum computer**, a machine that can run Shor at RSA-2048 scale and break the public-key crypto securing nearly all internet traffic (Chapters 3, 4, and 5). It is a specific fault-tolerant milestone, and it carries its own urgency through **harvest-now-decrypt-later**: an adversary records encrypted traffic today and decrypts it whenever the CRQC arrives, so any secret with a decade of shelf-life is exposed now. The governing relation is **Mosca's inequality**:

$$\text{if} \quad T_{\text{migrate}} + T_{\text{shelf-life}} \; > \; T_{\text{CRQC}}, \qquad \text{you are already exposed.}$$

The camps mirror the scaling-gap debate. Craig Gidney (Google, arXiv:2505.15917, 2025) cut the RSA-2048 estimate about 20×, from ~20M physical qubits in 2019 to **under 1M noisy qubits at ~1 week runtime** (T3 — a resource estimate on paper, with no machine behind it). Expert-elicitation surveys (Global Risk Institute) put ~~25% of cryptographers expecting RSA-2048 to fall by 2030 and ~50% by 2035, clustering around 2030 ± 3 (T5). NIST already standardized ML-KEM and ML-DSA in 2024 (T2, a standards action) because the threat is near enough to act on. The counter-case is the same five-orders gap in different clothing: a sub-1M-qubit *estimate* is not a sub-1M-qubit *machine*, resource estimates improve on paper faster than hardware improves in the lab, and the defensive move — finish PQC migration before any CRQC exists — neutralizes the threat regardless of the exact date. The dial to watch is physical-qubit count *at Shor-relevant fidelity*, since raw count alone says little.

The materials floor

Under the qubit-count story sits a materials-science story that the roadmaps tend to skip. The dominant coherence killer in superconducting qubits is the **two-level system**: microscopic defects in amorphous oxides, interfaces, and Josephson-junction barriers that couple to qubits, drift in frequency, and cause both decoherence and readout failures (Chapters 1 and 2). Coherence times have climbed from nanoseconds in 1999 to the millisecond range, an extraordinary run of engineering. TLS make individual qubits *unpredictable*: a qubit that is fine today can be lossy tomorrow as a defect diffuses onto resonance.

The suppressible-engineering camp reads TLS the way the 1960s semiconductor industry read yield: a fatal-looking problem that becomes a controlled discipline. Recent high-throughput junction studies correlate TLS density with electrode thickness and grain size and show a two-thirds reduction through fabrication changes alone (arXiv:2602.11469, T3); site-specific frequency tuning steers defects off resonance (arXiv:2503.04702, T3); ML-accelerated characterization speeds the search. The scaling-wall camp reads the same data and sees Kalai's regime arriving physically: TLS-induced dropouts scale with system size, so on a 1,000+ qubit chip some fraction is always out of spec, and correlated bursts — quasiparticle poisoning, cosmic-ray impacts that knock out many qubits at once — add a failure mode surface codes handle badly. This is where Kalai's abstract claim about correlated noise meets a concrete physical mechanism. The resolving experiment is precise: a 10,000-qubit chip holding every qubit within QEC spec across weeks, with published defect statistics showing the out-of-spec fraction does *not* grow with area. The negative signal is an out-of-spec fraction that scales with chip size — exactly Kalai's prediction, rendered in aluminum and sapphire.

The same "does it hold across a boundary" question recurs one level up, at the **interconnect**. No single chip holds a million qubits, so every serious roadmap is modular: many processors linked by quantum interconnects (Chapters 1 and 2). Gates *across* a link are slower and lower-fidelity than gates on a chip. IBM's 2024 Flamingo demo linked two Heron R2 chips and reported roughly **3.5% error** across the coupler — about two orders of magnitude worse than the best on-chip two-qubit gates (T4). The whole IBM modular roadmap (Cockatoo's cross-module entanglement, Starling's cross-module magic-state injection) assumes that number falls fast. What resolves it: a logical qubit whose stabilizer measurements straddle two modules, kept below threshold across the interconnect over many rounds, reproduced. The negative signal: cross-link error stuck near percent-level while on-chip gates keep improving, forcing architectures back toward monolithic chips nobody can actually build.

The advantage scorecard

Here is the chapter's hardest-nosed section. "Quantum advantage" is a moving target, and the historical record favors the classical attackers. Picture the advantage-scorecard figure as a table: each row a claimed quantum win, each with a date, a headline speedup, and a verdict column that reads *matched classically*, *contested*, or *still standing*. Most rows read *matched classically*.

The canonical row: Google's 2019 Sycamore random-circuit-sampling result, advertised as "10,000 years for a classical supercomputer," was matched within roughly three years by tensor-network methods on GPUs. The lesson generalized. IBM's 2023 "utility" experiment on a 127-qubit Eagle processor was reproduced within days by 2D tensor networks (gPEPS, Tindall et al.) and sparse Pauli-dynamics methods (Sandia, Caltech). Ewin Tang's **dequantization** program (2018 onward) erased the exponential speedup of quantum recommendation systems and much of the QML linear-algebra family by writing classical sampling algorithms that match them up to polynomial factors — this is the single biggest reason the QML corner (Chapter 3) is graded hardest in the whole atlas. HHL's fine print (state preparation, condition number, readout, and the unproven qRAM assumption) hollows out most quantum-linear-algebra applications, and Grover's quadratic speedup is widely expected to be eaten by error-correction overhead. Row after row, *matched classically*.

Two live rows are still open, and they are the interesting ones. The first is **D-Wave's spin-glass** claim. D-Wave's March 2025 *Science* paper claimed beyond-classical spin-glass dynamics on 5,000+ qubits, estimating classical MPS would need ~1M years on Frontier. In May 2026 the Flatiron Institute (Tindall et al., *Science* adx2728) computed comparable annealing dynamics classically with belief-propagation tensor networks, some instances on a laptop, and EPFL's Dries Sels showed the original baselines were weak. D-Wave rebutted: the classical method never attempted the hardest 3D lattice geometries, the largest simulations, the low-precision ensembles where correlations grow fastest, or the fourth-order observables of the original paper, and those specific instances remain unmatched. The status is that **no neutral body has adjudicated the dispute** — the contested benchmark instances stay unmatched, which leaves the claim *contested* while it is unresolved. It would be easy to write "D-Wave debunked" or "D-Wave vindicated," and both would be false.

The second live row is Google's **Quantum Echoes** (October 2025, 105-qubit Willow), an OTOC-based algorithm run 13,000× faster than the best known classical method and pitched as the first *verifiable* quantum advantage, cross-checkable on another quantum device. It is T4→T3: vendor-led, published, young. What makes it the field's most important current experiment is that the usual counterattack has, so far, failed: a March 2026 preprint (arXiv:2604.15427) argues belief-propagation tensor networks *cannot* feasibly simulate it. What resolves it is a clean bet: a verifiable advantage claim that stands roughly **3+ years** against motivated classical attack. If tensor-network attacks keep failing on Quantum Echoes through 2027–28, it becomes the first durable useful-ish advantage. If a classical method matches it — as happened to Sycamore — the attrition pattern holds. Watch this row.

Where the boundary actually sits

Underneath the scorecard is a theory question: the quantum-classical boundary moves, set by the *best classical algorithm anyone has written*, and clever people keep writing better ones. So the useful question skips brute-force state-vector simulation — a quantum computer beats that at ~50 qubits — and asks where the boundary sits against tensor networks, belief propagation, neural quantum states, and Clifford-plus-few-T sparsification, for the *structured* problems quantum machines actually run. Most advantage demos sit *near* that boundary, on low-entanglement, shallow-depth, geometrically-local circuits that classical heuristics exploit. A 2025 result (arXiv:2510.06324) gives quasi-polynomial classical sampling of noisy circuits under approximate Markovianity, squeezing the noisy-sampling case further. The counter-position: volume-law-entangled, deep, non-local circuits are believed hard for all known classical methods, and fault-tolerant Shor and honest quantum simulation live provably beyond the boundary given a working machine (Chapter 3). The frontier moves, but it may be converging on a stable line that quantum machines are starting to clear. The general resolution is a complexity-theoretic characterization of which structured problems admit efficient classical simulation, converting an empirical arms race into settled theory.

The words themselves

Two questions police the vocabulary, because loose words let a T4 press release read like a T2 result. The first takes on IBM's 2023 coinage of "quantum utility" — a noisy 127-qubit processor producing reliable results beyond brute-force classical simulation. IBM was careful: utility meant "useful and competitive with classical approximations," explicitly *below* the "advantage" bar. The question is whether utility names a real transition or a goalpost that moved once supremacy claims kept getting matched. The deciding fact is on the record: the 2023 utility problem was classically reproduced within days. Per the atlas rule, a result reproduced classically within weeks counts as a *classical-simulation-frontier data point* and stays below the advantage bar. The second generalizes the complaint. There is no agreed way to say one quantum computer is better than another: IBM reports Quantum Volume and CLOPS, IonQ and QED-C report Algorithmic Qubits, others report raw counts or "logical qubits," and each metric flatters one architecture. Cross-entropy benchmarking, behind the supremacy claims, is not a proof and has been spoofed classically. Classical computing converged on SPEC, LINPACK, and MLPerf; quantum has no equivalent. The resolution is a "quantum MLPerf": application-oriented, spoof-resistant benchmarks reporting *logical* performance at a stated code distance, run by a neutral body. Until it exists, treat every single-metric headline as T4 and ask which subsystem the number measures.

Money, hype, and the killer app

The physics runs on a decade clock. The capital runs on a quarterly one, and the mismatch between the two clocks is the money question. As of May 2026, IonQ, Rigetti, and D-Wave traded at price-to-sales ratios of roughly 109, 836, and 791; IonQ carried a $21B+ market cap on ~$187M trailing revenue and a $711M EBITDA loss. Total industry computing revenue sat near $1.4B (QED-C 2026). Two readings sit side by side. The substance camp: four independent roadmaps converge on 2029–2030 for early fault tolerance, real milestones keep landing (below-threshold QEC confirmed by Google, Quantinuum, Harvard/QuEra, and USTC; the QuEra 96-logical record; Quantum Echoes), and Aaronson updated toward the aggressive timelines. The bubble camp reads the same facts as a hype cycle running ahead of the physics: insiders at IonQ, Rigetti, and D-Wave were net sellers of ~$931M of stock over five years, and "quantum-washing" rebrands classical or hybrid products as quantum (Quantum Computing Inc.'s Q1 2026 "9,000% revenue surge" to $3.6M came largely from acquisitions). Its most cited voice is Sabine Hossenfelder, and her critique earns a full hearing here. Both readings can be true at once, and the two must stay separate: a company can be overpriced while the physics is on track. The valuation question is T5; the technical question is the scaling gap. What resolves the valuation debate is revenue — one customer choosing quantum on cost grounds — or a funding winter after missed 2029–2030 dates.

Hossenfelder's case deserves stating at full strength, because it is the sharpest version of the bubble reading. Her argument is that quantum computing sits in a hype bubble, and that the commercial applications used to justify the money — optimization, finance, logistics, machine learning — have been quietly eroded year after year even as investment climbed. Her core technical claim is checkable: the number of quantum algorithms with a proven advantage over the best classical methods is small, and as of 2026 none has been shown delivering that advantage on real hardware, because the gap between what the algorithms need and what the machines provide stays enormous. She grants the exceptions the atlas grants — quantum sensing and metrology are real and already in use, and quantum chemistry is the likeliest first payoff — and rests her skepticism on error correction remaining hard at scale. Her timing has been wrong, and she says so: she forecast the hype would break by 2024, and in late 2025 acknowledged plainly that it had not. Her return-on-investment critique grades T5 as a market forecast, and this chapter's own advantage scorecard corroborates it: claim after claim was matched classically, and no durable useful advantage has yet survived a serious classical counterattack. On the business question she is aligned with the evidence. On the physics question she holds the strong view that fault-tolerant scaling may never pay off commercially, a bet the hardware has not yet settled.

That funding winter is its own risk. Q1 2026 quantum-startup funding fell to ~$240M, down 51% quarter-over-quarter and ~68% year-over-year (Crunchbase), and funding concentrates hard — in 2024 two firms took roughly half of all VC. A downturn strands the long tail of startups and disperses their teams into AI and quant finance, where they rarely return, and the workforce is thin: roughly one qualified candidate per three open roles, ~250k professionals needed by 2030 (McKinsey), and PhDs that take years to grow (Chapter 6). The no-winter camp points to counter-cyclical government money (Chapter 6) that private downturns do not touch, and to the real milestones as evidence the field is earning its capital. The 2026 funding drop is the first real stress test; whether it is a blip or a trend is live. History has a name for the risk — the AI winters of the 1970s and late 1980s — and the atlas grades it in Chapter 7 rather than waving it off.

Both money threads ultimately point at one technical question: is there a **killer app**? Every platform technology needed one application valuable enough to fund the buildout, and the quantum candidates each have a defect. Chemistry and materials simulation (Chapter 5) is the strongest case — the Aspuru-Guzik line that "chemistry is quantum computing's killer app" is T2 theory — but the resource estimates for the flagship targets keep climbing as the analysis deepens. The FeMoco (nitrogen-fixation cofactor) problem is now estimated at ~1,137 logical qubits and $\sim 3.4 \times 10^8$ logical T-gates, far beyond 2026 hardware; a 2021 Google study (Lee et al.) put the cost at ~4M physical qubits (~2,100 logical), and Alice & Bob's 2025 cat-qubit estimate reached ~99,000 physical qubits — a 27× cut against the ~2.7M-qubit configuration they benchmarked. Chapter 5 carries the same figures. Optimization and finance are softer still: Babbush et al. show quadratic speedups evaporate under error-correction overhead. Shor destroys value once (breaking RSA), then the market migrates to PQC and evaporates. Aaronson's standing answer is that the killer app may be simulating quantum physics itself — a market whose dollar size nobody knows. A nearer-term perspective (arXiv:2506.19337) maps applications reachable with 25–100 logical qubits. What resolves it: one audited, repeated commercial workload where quantum beats the best good-faith classical baseline on cost or quality.

The return-on-investment question sharpens that into unit economics, because a real speedup can still be a bad business. The QPU-hour has to cost less than the classical alternative, and the value has to recur rather than accrue to a one-time result. D-Wave has a real commercial book — a $20M Florida Atlantic system sale, a ~$10M QCaaS deal — and IonQ guided to $225–245M for 2026, but most reported "quantum revenue" across the industry comes from acquisitions, government contracts, and peripheral consulting rather than customers paying for computations that beat classical. The dial to watch is the fraction of revenue that is *recurring quantum-compute consumption* versus one-time hardware, grants, and services.

The engineering bottlenecks under the headline

Three more problems hide under the qubit-count roadmaps, and each could cap a machine that is otherwise on track.

Decoder throughput. Error correction only works if a classical co-processor decodes each round of syndrome measurements and issues a correction *before the next round arrives*. Superconducting qubits emit a syndrome roughly every microsecond, so the decoder must sustain about **1 MHz per logical qubit** at sub-microsecond latency, for thousands of logical qubits at once. If it lags, syndromes pile up — the backlog problem (Terhal 2015) — and the effective clock speed collapses exponentially. A machine can have perfect qubits and still be useless if it cannot decode fast enough. Riverlane's December 2025 *Nature Communications* result hit a MegaQuOp (10^6 reliable operations) with ~4× fewer qubits (T2), and frames the path to TeraQuOp (10^{12}) as an engineering scaling problem (Chapter 3). The unsolved part is the accuracy/latency/bandwidth trilemma at scale — accurate decoders are slow, fast decoders raise the logical error rate — and qLDPC codes (which IBM now bets on) are *harder* to decode fast than surface codes because their checks are non-local.

Energy. Runtime is half of a fair comparison; energy is the other half. A dilution refrigerator draws roughly 10 kW continuously, and a full system runs near ~18 kW — a fixed cost paid whether the machine computes or idles. For sampling tasks the quantum machine wins enormously (Google's supremacy analysis found classical simulation on Summit would burn ~7 orders of magnitude more energy). Recent theory identifies an "energetic advantage before computational advantage" — in boson sampling and cat-qubit computation the quantum machine can win on energy at a problem size smaller than where it wins on wall-clock time (arXiv:2601.08068, 2605.19854). The counter: at fault-tolerant scale, millions of physical qubits plus real-time decoders could push total draw far above a single fridge, and no *useful* workload has shown a net energy win. What resolves it: a full-stack joules-per-useful-answer measurement against the best classical method.

Verification. Once a machine solves a problem no classical computer can redo, how do you know the answer is right? For factoring you multiply the factors back; for simulation, sampling, or optimization there may be no efficient classical check. Urmila Mahadev's 2018 theorem (FOCS Best Paper) proved a *classical* verifier can certify an arbitrary quantum computation using a single quantum prover, assuming the device cannot break post-quantum crypto (T2) — a landmark, though it needs a fault-tolerant device well beyond today's. Certified-randomness protocols already run on hardware: Quantinuum's 56-qubit H2 generated *certified* random bits verified classically (*Nature* 2025), the first delivered verification-flavored application (Chapter 3). The gap: certifying random *bits* is a long way from certifying a chosen drug-discovery *answer*. Quantum Echoes leans on the weaker but real form — cross-checkable on another quantum device — which is why verification and advantage are the same conversation.

And the path that would change all of the above economics: **room-temperature and photonic scaling.** Superconducting and spin qubits live at millikelvin inside dilution refrigerators, a wiring and I/O ceiling that gets brutal at a million qubits (Chapter 2). Photonic qubits run gates at room temperature, ship through ordinary fiber, and can be fabbed in existing foundries (Chapter 2). Xanadu's Aurora (*Nature* 2025) is a modular photonic machine — 4 racks, 35 chips, 13 km of fiber, 12 physical qubits — presented as scalable in principle to millions, targeting ~1,000 logical qubits by 2029 (T2 for the demo, T4/T6 for the extrapolation); PsiQuantum partners with GlobalFoundries to fab million-qubit photonic systems (T4). The asterisk: the single-photon detectors are cryogenic, so the cryoplant shrinks rather than vanishes; photon loss sits orders of magnitude above fusion thresholds; and Aurora's 12 physical qubits against a million-qubit roadmap is the largest extrapolation ratio in the industry. What resolves it: a photonic machine with error-corrected logical qubits and published end-to-end loss budgets.

The problem that predates the machine

The oldest open problem in the field predates the machines entirely: the **measurement problem**, unsolved since the formalism was written down in 1925 (Chapters 1 and 7). Quantum states evolve two ways: unitary and reversible between measurements, discontinuous and probabilistic at them. Nobody agrees on what happens at the second one, or on what the state *is*. Quantum computing sharpens the question rather than answering it — every working machine maintains coherent superposition at ever-larger scale, probing exactly the regime where interpretations differ, and a fault-tolerant machine would be the most macroscopic superposition ever engineered.

The camps line up cleanly. Everettians (Deutsch) argue unitarity all the way down dissolves the problem and read a scalable quantum computer as evidence for many-worlds. Objective-collapse theorists (GRW, CSL, Diósi–Penrose) make the strongest *testable* claim: collapse is physical, and their parameter space is being squeezed by underground experiments looking for the spontaneous X-ray emission collapse models predict (T2 for the constraints, T6 for the models). A 2025 *Nature* centenary survey of 1,100+ physicists found no consensus — Copenhagen leading at ~36%, QBist and epistemic views next, many-worlds after — with respondents split on whether the question is even physics. What resolves it, one way: detection of collapse-model radiation or heating ends the debate for objective collapse. Continued null results plus verified coherence at ever-more-macroscopic scale — including a working FTQC machine — squeeze collapse models toward irrelevance *without* adjudicating between Copenhagen, QBism, and many-worlds. Those three may be distinguishable by no experiment at all, which is why the problem may be permanent rather than merely hard. It is included here, at the end, because the physics the field stands on has an unsolved question at its foundation, and the machines being built are, among other things, the largest test of that question ever run.

> **Key takeaways** - The physics is settled to T1 — superposition, entanglement, and the Bell violations, ratified by a century of experiment and the 2022 Nobel. - The hardware is real and improving faster than skeptics expected — multiple platforms past 99.9% two-qubit fidelity, below-threshold error correction confirmed by four independent teams — yet a scaling gap of roughly five orders of magnitude in qubit count still separates today's machines from cryptographic or chemical usefulness. - The advantage scoreboard favors the classical attackers: Sycamore, the 2023 utility run, and the QML linear-algebra family were matched or dequantized; D-Wave's spin-glass result is contested-not-

> overturned, and Quantum Echoes is the live test. - Vendor roadmaps (T4) and market forecasts (T5) run ahead of both refereed results and revenue; the valuations price a business that does not yet exist, and the killer app has candidates without a confirmed winner. - Kalai's in-principle objection is narrowing without being closed, and the measurement problem stays open a century on — the machines are the largest test of it ever run. - The durable habit: ask four questions of any quantum headline — who claims it and what they gain, what evidence tier it is, whether anyone independent reproduced it against a fair classical baseline, and what future event would prove it wrong.

Where quantum stands

Put the whole atlas back together and the frontier resolves into a clear picture, stated plainly.

The physics is settled (Chapter 1). Superposition, entanglement, and the Bell violations that ruled out local hidden variables are T1, ratified by a century of experiment and the 2022 Nobel. The hardware is real and improving faster than the skeptics expected (Chapter 2): multiple platforms cleared 99.9% two-qubit gate fidelity in 2025, and below-threshold error correction was confirmed by four independent teams. The algorithms with proven asymptotic speedups — Shor, quantum simulation — remain T1/T2 theory waiting on a machine (Chapter 3). The adjacent technologies already ship revenue (Chapter 4): sensing, timing, QKD, and the PQC migration that the CRQC threat makes urgent today regardless of when the CRQC arrives.

Everything past that carries a grade and a caveat. Vendor roadmaps to 2029–2030 are T4 — convergent, milestone-structured, and unrefereed until Kookaburra and Starling ship. Market forecasts are T5 and flagged for optimism. "Quantum advantage" is contested by default, and the scorecard still favors the classical attackers: Sycamore, the 2023 utility experiment, and the QML linear-algebra family were all matched or dequantized, while D-Wave's spin-glass claim sits *contested-not-overturned* and Quantum Echoes is the live test of whether any useful advantage can stand three years against motivated attack. The scaling gap is five orders of magnitude in qubit count at held fidelity, and closing it depends on error-correction overhead ratios, TLS materials physics, interconnect fidelity, and decoder throughput that are each open questions rather than solved engineering. The valuations price a business that does not yet exist. The killer app has candidates and no confirmed winner. And Gil Kalai's in-principle objection, though narrowing, is not yet closed — it earns a low probability and a precise falsification condition, an editorial call the evidence permits rather than compels.

The single most useful habit this chapter can leave a reader is a simple one. When a headline says a quantum computer did something, ask four questions. Who is making the claim, and what do they gain? What tier is the evidence — refereed, preprint, or press release? Has anyone independent reproduced it, and did the classical baseline get a fair try? And what specific future event would prove it right or wrong? Every open problem in this chapter comes with the fourth answer already written down — the experiment or the date that settles it — because a frontier you can bet on is a frontier you understand.

Quantum computing in 2026 is a real physics, a real and improving set of machines, a hard scaling problem with an uncertain resolution, a valuation running well ahead of revenue, and an open foundational question older than any of it. That is not a verdict of success or failure. It is the state of the field as the evidence has it, graded and dated.

Exercises and discussion

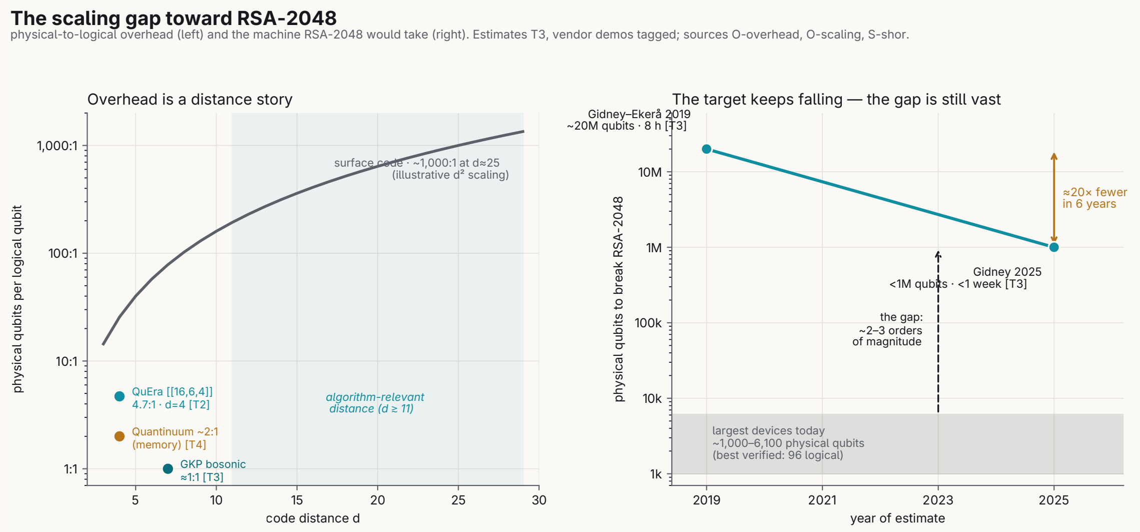
1. **(Critical reading.)** Take one quantum-computing headline from the past month and run the four questions on it: who is making the claim and what do they gain; what tier is the evidence (refereed, preprint, or press release); has anyone independent reproduced it against a fair classical baseline;

and what specific future event would prove it right or wrong. Assign the claim a grade from T1 to T6 and defend it in two sentences.

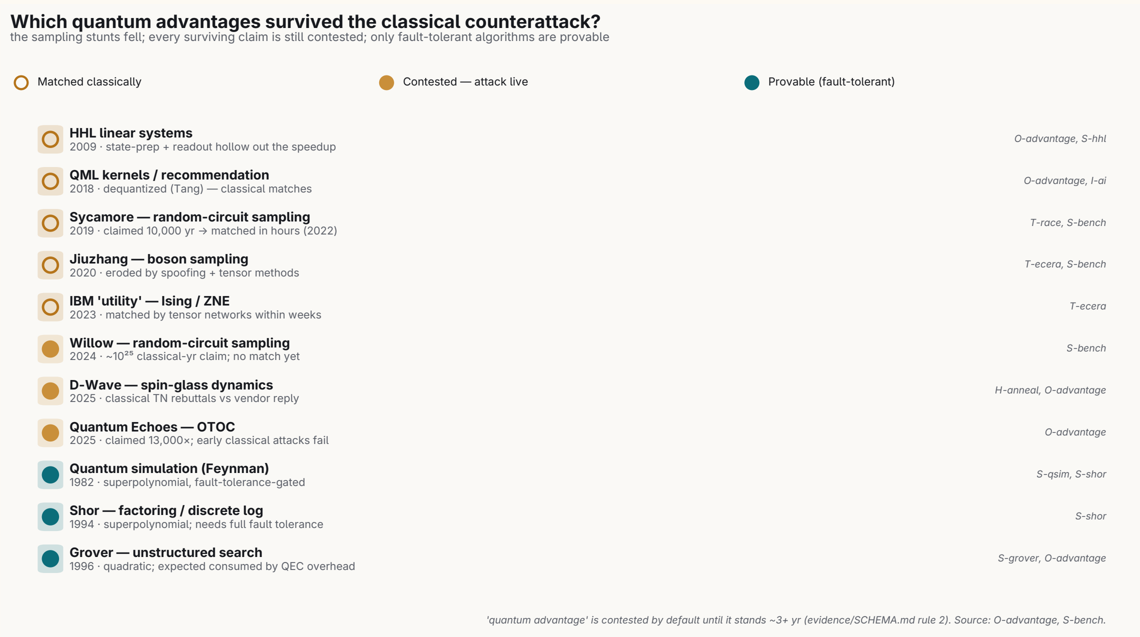
2. **(Resource estimation, for CS.)** A billion-gate algorithm needs a logical error rate near 10^{-15}. Using a surface code, sketch how logical error falls with code distance below threshold and argue why reaching that target takes a distance around 25–30 and therefore hundreds to thousands of physical qubits per logical qubit. Then explain why QuEra's demonstration packed 96 logical qubits into 448 physical atoms at a far friendlier ratio — what does the [[16,6,4]] code's distance buy, and what does it cost.
3. **(Cryptography / timeline.)** State the Mosca inequality with its three quantities: X, how long your data must stay secret; Y, how long migration takes; Z, time until a cryptographically relevant quantum computer. Explain why $X + Y > Z$ means the migration must start now, and use it to argue why the decision to migrate is largely independent of the exact date the machine arrives.
4. **(Physics.)** Objective-collapse models (GRW, CSL, Diósi–Penrose) make a testable prediction that Copenhagen and many-worlds do not: spontaneous radiation and heating. Explain why a verified fault-tolerant machine — the most macroscopic engineered superposition to date — squeezes collapse models toward irrelevance while leaving Copenhagen, QBism, and many-worlds mutually undecided. Which of these positions, if any, could an experiment ever separate?
5. **(Seminar.)** Steelman Gil Kalai, then steelman Scott Aaronson. Kalai argues noise is correlated in a way that forbids scalable fault tolerance in principle; Aaronson has moved toward the aggressive timelines as below-threshold results landed. Argue each position as strongly as its holder would, then state the single experimental result that would most move you toward one side.
6. **(Spec reading, for engineers.)** Read a current vendor roadmap (IBM's Kookaburra/Starling or Quantinuum's Helios/Sol line) and separate the T4 roadmap dates from the T1–T2 refereed results it rests on. Which single milestone, if it slipped by two years, would most change your grade of the whole roadmap, and why?

Further reading

- **G. Kalai, "The Argument Against Quantum Computers" (2019/2020), and G. Kalai, Y. Rinott, T. Shoham, "Google's Quantum Supremacy Claim: Data, Documentation, and Discussion" (arXiv:2210.12753, 2022).** The in-principle noise objection stated cleanly, then applied to the Sycamore data — the skeptic case this chapter grades and keeps open.
- **S. Aaronson, *Shtetl-Optimized* (scottaaronson.blog).** A principal's running commentary on the supremacy claims and the Kalai debate; the clearest public record of why the aggressive and skeptical timelines disagree.
- **C. Gidney, "How to factor 2048-bit RSA integers with less than a million noisy qubits" (arXiv:2505.15917, 2025).** The estimate that pushed the RSA-2048 requirement below one million physical qubits — the low end of the scaling-gap figure in this chapter.
- **R. Acharya et al. (Google Quantum AI), "Quantum error correction below the surface code threshold," *Nature* 638, 920 (2025).** The Willow result: logical error suppressed as code distance grows, the T1/T2 anchor for the claim that the hardware is improving faster than skeptics expected.
- **M. Mosca, "Cybersecurity in an era with quantum computers: Will we be ready?" *IEEE Security & Privacy* 16 (2018).** The source of the Mosca inequality used in the cryptography timeline exercise.
- **"A century on, physicists still can't agree on what quantum theory means," *Nature* centenary survey of 1,100+ physicists (2025).** The poll behind the measurement-problem section: no consensus, Copenhagen leading near 36%, with respondents split on whether the question is even physics.



The scaling gap: the RSA-2048 resource estimate fell from ~20M physical qubits (2019) to under 1M (2025), while today's largest machines hold only a few thousand — the curves remain three to four orders of magnitude apart.



The advantage scorecard: of the headline quantum-advantage claims, most were matched classically; only D-Wave's spin-glass result (contested) and Google's Quantum Echoes (still live) remain unbeaten.

APPENDIX A

How every claim is graded

Inherited from the health manual's neutrality mechanism, adapted for a field where the loudest claims come from vendors with something to sell.

Every claim carries

- `claim`: the statement, in plain language
- `tier`: evidence strength (below)
- `source`: primary citation (arXiv ID, DOI, journal, or named press release)
- `provenance`: who is making the claim and what they gain from it
- `status`: `established` · `demonstrated` · `claimed` · `contested` · `roadmap`

Tiers (strongest → weakest)

TIER	MEANING	EXAMPLE
T1 Established physics	Textbook, reproduced for decades	Superposition, Bell violation
T2 Peer-reviewed result	Published, refereed, ideally reproduced	Google Willow below-threshold QEC (Nature 2024)
T3 Preprint / conference	arXiv or talk, not yet refereed	Most fresh hardware milestones
T4 Vendor claim	Company announcement, not independently verified	Roadmap qubit counts, "advantage" PRs
T5 Analyst / forecast	Market projection, punditry	BCG/McKinsey TAM numbers
T6 Speculative	Plausible but unproven	Room-temp qubit timelines, killer-app dates

Rules

1. A vendor announcing its own benchmark is **T4 until independently reproduced** — never promote to T2 on the strength of a blog post.
2. "Quantum advantage / supremacy" claims are **contested by default** and must carry the classical-counterattack status (was it later matched classically?).
3. National-program dollar figures are **T5** and flagged for double-counting.
4. Conflicts are first-class objects — see `evidence/CONFLICTS.md`. When two credible sources disagree (e.g. "FTQC by 2029" vs "not before 2040"), keep both.
5. Timeline / roadmap nodes are **T4/T6** and say so in the manual text.

Conflict object format

```
id: C-<slug>
topic: <one line>
positions:
  - who: <source>  claim: <...>  tier: T?
```

```
  - who: <source>  claim: <...>  tier: T?
what_would_resolve_it: <the experiment/event that settles it>
```

APPENDIX B

Conflict register

Per `SCHEMA.md` rule 4: when two credible sources disagree, keep both. This file is the registry. Cards link here; the manual never silently picks a side.

C-ftqc-timeline — When (if ever) does large-scale fault-tolerant quantum computing arrive

WHO	POSITION	TIER
IBM (modular roadmap, Jun 2025)	Loon 2025 → Kookaburra 2026 (first FT module) → Cockatoo 2027 → Starling 2028-29 (~200 logical / 100M gates) → Blue Jay 2033 (2000+ logical / 1B gates); qLDPC cuts overhead up to 90%	T4
Google Quantum AI (roadmap)	useful error-corrected machine ~2029-2030	T4
Scott Aaronson (Mar 2026)	2025 met/exceeded hardware expectations (>99.9% 2Q gate fidelity on multiple platforms); takes 2028-29 pronouncements more seriously; declines a firm date	T3 (expert)
Jensen Huang (Nvidia, Jan 2025)	very useful QC is 15-30 years out (~20 central); walked back at GTC Quantum Day	T5
Gil Kalai	FTQC impossible in principle (correlated noise defeats threshold theorem); correlations observable in noisy-circuit simulations. Mar-2026 reply (gilkalai.wordpress.com, 10 Mar 2026): concedes he "starts with quantum computation being impossible as [an] axiom" but insists the observed exponential error decay "does not contradict my conjectures regarding correlated errors," which he expects at "hundreds… and thousands of gates"	T3 (contested minority)
Manifold prediction markets (2026)	steady engineering gains, no breakthrough	T5

Resolves when: IBM Kookaburra (2026) then Starling (2028-29) hitting spec, independently verified — or missing it. Aaronson (2026) notes a dozen experiments now reach the same below-threshold conclusion, calls Kalai's path "narrower and narrower"; Kalai's position dies with any sustained large-scale fault tolerance and strengthens with each correlated-noise surprise at scale. See O-scaling.

C-overhead-ratio — The realistic all-in physical:logical qubit ratio for useful algorithms

WHO	POSITION	TIER
QuEra / Quantinuum / qLDPC papers	ratios of ~2:1 to ~13:1 demonstrated or roadmapped; qLDPC cuts surface-code overhead 10-20x	T3/T4
surface-code resource analyses	at target logical error 1e-9..1e-12 with T-factories and routing, all-in cost stays hundreds-to-1000:1	T3

Resolves when: an end-to-end fault-tolerant algorithm (incl. T-gates) at code distance >=11 with the full published physical budget; sustained <100:1 settles it for qLDPC optimists.

C-tls-scaling — Whether materials defects (TLS) are an engineering nuisance or a scaling wall

WHO	POSITION	TIER
fabrication groups (arXiv:2602.11469 etc.)	TLS density is controllable — 2/3 reduction via electrode/grain engineering; tuning + ML mitigation handle the rest	T3
TLS-dropout studies (arXiv:2605.02755) + Kalai	dropouts and correlated noise scale with system size and violate QEC's error assumptions	T3 (Kalai position contested)

Resolves when: a 10k-qubit chip holding all qubits within QEC spec for weeks, defect statistics published — or persistent out-of-spec fractions that grow with chip size.

C-advantage-survival — Whether any claimed quantum speedup survives classical counterattack

WHO	POSITION	TIER
Google (Quantum Echoes, Oct 2025)	first verifiable quantum advantage, 13,000x vs best classical OTOC method on 105-qubit Willow	T4→T3 (published, vendor-led, young)
arXiv:2604.15427 (2026)	belief-propagation tensor networks CANNOT feasibly simulate Quantum Echoes — early sign the claim holds	T3
D-Wave (Science, Mar 2025)	beyond-classical spin-glass simulation on 5,000+ qubits; classical MPS would need ~1M yrs on Frontier	T2 paper / contested claim
Flatiron Institute (Tindall et al., Science adx2728, May 2026) + Sels (EPFL)	belief-propagation tensor networks compute the same annealing dynamics classically; some instances on a laptop; baselines were weak	T2/T3
D-Wave rebuttal (May 2026)	the classical framework fails to scale across the most complex topologies/ measurements; specifically Tindall et al. did not attempt the hardest 3D lattice geometry, the largest 3D simulations, the low-precision ensembles where correlations grow fastest, or the full-state/fourth-order observables of the original Science paper; result stands	T4
third-party adjudication status (mid-2026)	NONE — no neutral body has adjudicated the D-Wave/Flatiron spin-glass dispute; the specific contested benchmark instances remain unmatched classically, so it is contested, not overturned	T3 (absence of adjudication)
Ewin Tang (dequantization program)	quantum recommendation systems + much of QML linear algebra have no exponential advantage	T2
arXiv:2510.06324	noisy circuit sampling classically simulable in quasi-polynomial time under approximate Markovianity	T3

Resolves when: a verifiable advantage claim standing ~3+ years against motivated classical attack. Quantum Echoes is the live test case — if tensor-network attacks keep failing on it through 2027-28 it is the first durable useful-ish advantage. See O-advantage, O-classical-sim.

C-hype-valuation — Whether 2024-26 quantum valuations reflect substance or a bubble

WHO	POSITION	TIER
vendors + analysts (BCG/ McKinsey)	convergent 2029-30 roadmaps + real QEC milestones justify multi-decade value; tens of billions TAM	T4/T5
Hossenfelder	hype will crash; real-world logistics/finance value dissolves under practical constraints	T5
Kalai	the entire FTQC premise fails, so valuations rest on nothing	T3 (contested minority)
market analysts (2025-26)	150x-3,000x sales multiples are unjustifiable pending revenue	T5

Resolves when: revenue — a paying customer choosing quantum on cost/performance grounds; or a funding winter after missed 2029-30 roadmap dates.

C-photonic-scaling — Whether photonics' room-temperature path actually changes the economics

WHO	POSITION	TIER
Xanadu (Aurora, Nature 2025)	modular photonic architecture scales in principle to millions of qubits; ~1,000 logical by 2029	T2 demo / T4 extrapolation
PsiQuantum + GlobalFoundries	million-qubit photonic systems manufacturable in existing fabs, utility-scale sites underway	T4
photonics skeptics (CACM etc.)	SNSPD detectors stay cryogenic, photon loss sits orders of magnitude above fusion thresholds, and 12 qubits → 1M is the field's largest extrapolation	T3

Resolves when: any photonic machine demonstrating error-corrected logical qubits with published end-to-end loss budgets; PsiQuantum's first full-stack public benchmark.

C-killer-app — Whether a near-term application exists that pays for the machine

WHO	POSITION	TIER
chemistry camp (Aspuru-Guzik line, Quantinuum, pharma partners)	molecular simulation is the killer app, commercial in 5-10 years given hundreds of logical qubits	T2 theory / T4-T5 timeline
feasibility skeptics (2025-26 studies, New Scientist)	industrially relevant chemistry needs resources that stay daunting; chemistry may miss killer-app status	T3
optimization skeptics (Hossenfelder; quadratic-speedup analyses)	optimization/finance value evaporates under QEC overhead and real-world constraints	T2/T5
Aaronson	the honest killer app is simulating quantum physics itself; market size unknown	T3

Resolves when: one audited, repeated commercial workload where quantum beats the best good-faith classical baseline on cost or quality.

C-measurement-problem — Whether the measurement problem is solved, dissolvable, or permanently open

WHO	POSITION	TIER
Everettians (Deutsch et al.)	unitarity all the way down; scalable QC is evidence for many-worlds	T3 (philosophical)
objective-collapse camp (GRW/DP)	collapse is physical and testable; parameter space shrinking under experiment	T2 (constraints) / T6 (the models)
Nature centenary survey (2025)	no consensus — Copenhagen ~36%, QBist/epistemic next, many-worlds after; community split on whether it's even physics	T3 (survey)
foundations reviews (arXiv:2502.19278)	clarified by decoherence, still unsolved	T3

Resolves when: detection of collapse-model radiation/heating (ends it for objective collapse); continued unitarity at macro scale squeezes collapse out without adjudicating the no-collapse interpretations.

C-psi-ontology — Whether the quantum state is a real physical property (ψ-ontic) or an observer's information (ψ-epistemic)

WHO	POSITION	TIER
Pusey, Barrett & Rudolph (Nat. Phys. 8, 475, 2012; arXiv:1111.3328)	under "preparation independence," no ψ-epistemic model reproduces QM — the state is ontic	T2 (theorem, assumption-dependent)
PBR experimental tests (Ringbauer et al., Nat. Phys. 11, 249, 2015; Nigg et al.)	data rule out broad classes of ψ-epistemic models within stated assumptions	T2
ψ-epistemic defenders (Leifer review 2014; Spekkens toy model; Harrigan-Spekkens framework)	dropping preparation independence keeps ψ-epistemic models alive; PBR constrains but does not close them	T3 (contested)
instrumentalists / QBists	the ontic-vs-epistemic framing itself is optional; the state is a Bayesian credence	T3 (philosophical)

Resolves when: a theorem removing the preparation-independence assumption (would settle it either way), or an experiment excluding all maximally-ψ-epistemic models without auxiliary assumptions. See F-pbr.

C-discord-resource — Whether quantum discord (correlation beyond entanglement) is an operationally useful resource or a mathematical artifact

WHO	POSITION	TIER
DQC1 / mixed-state-power camp (Knill-Laflamme; Datta, Shaji, Caves, PRL 100, 050502, 2008)	discord tracks the speedup of the "power-of-one-qubit" model where entanglement vanishes — it is the resource	T2/T3
discord-consumption protocols (quantum state merging, encoding — Madhok-Datta; Cavalcanti et al.)	discord has an operational interpretation as consumed/ generated in named tasks	T3
skeptics	nonzero discord is generic (almost all states have it), it is basis/measurement-dependent, and no clean end-to-end advantage is pinned to it alone	T3 (contested)

Resolves when: an audited task where discord (with zero entanglement) is provably the sole resource behind a real speedup, reproduced independently. See F-discord.

C-majorana-existence — Whether Microsoft's topological hardware actually hosts Majorana zero modes / a topological qubit

WHO	POSITION	TIER
Microsoft (Majorana 1, Nature, Feb 2025)	InAs/Al topoconductor 8-qubit-capacity architecture; topological gap protocol supports a path to topological qubits	T4 (Nature editors: "results do not represent evidence for Majorana zero modes")
Henry Legg critique (formal Nature comment, d41586-026-01788-y, Jun 24 2026)	the presented data (only Z measurements) can be explained by trivial phenomena; identifies flawed tune-up routines, software errors in the analysis code, and omitted transport measurements; "nothing proves the existence of a topological qubit or Majoranas in these devices"	T3
prior retraction (2021, Delft/Microsoft)	earlier quantized-Majorana-conductance Nature paper retracted after data-analysis scrutiny	T1 (established retraction)
Microsoft response (formal Nature reply, 2026)	defends the topological gap protocol; Nayak "stands by our results and our roadmap"; Majorana 2 (Jun 2026) claims ~20 s lifetimes / 1,000× stability	T4
independent reproduction status (mid-2026)	NONE — no lab has independently replicated the topological-qubit claim; a de-facto verification vacuum persists (only a handful of labs can fabricate the InAs/Al, now Pb/Al, devices)	T3 (absence of evidence)

Resolves when: a demonstrated topological qubit with fusion-rule / braiding measurements (not just Z readout), reproduced independently, showing the predicted non-Abelian statistics. Verified mid-2026: still no independent replication; Legg's Nature critique stands and Microsoft's Nature reply defends the result — the dispute is unresolved and stays contested. See H-topo.

C-crqc-timeline — When a cryptographically-relevant quantum computer (Shor at RSA-2048 scale) arrives

WHO	POSITION	TIER
Gidney (Google, arXiv:2505.15917, 2025)	RSA-2048 in <1M noisy physical qubits, ~1 week — down ~20x from 20M (2019)	T3 (resource estimate, not a machine)
Global Risk Institute expert survey	~22.7% of cryptographers expect RSA-2048 to fall by 2030, ~50% by 2035; point estimates ~2030±3	T5
NIST (2024 PQC standards)	threat is near enough to standardize ML-KEM/ML-DSA now and migrate	T2 (standards action)
hardware skeptics / Kalai	estimates improve on paper faster than hardware in the lab; largest devices are ~1k-6k physical qubits at insufficient fidelity; Kalai says never	T3

Resolves when: a Shor factorization of a growing genuine semiprime (RSA-1024 then 2048) with fault tolerance — or, defensively, completion of PQC migration before any CRQC appears (Mosca's inequality). Watch physical-qubit count AT Shor-relevant fidelity. See O-crqc-timeline, A-pqc, S-shor.

C-quantum-utility — Whether "quantum utility" is a meaningful milestone or a rhetorical hedge

WHO	POSITION	TIER
IBM (Nature 618:500, 2023)	127-qubit error-mitigated processor produced reliable results beyond brute-force classical simulation — "utility before fault tolerance"	T2 paper / T4 framing
Tindall et al. (PRX Quantum 5, 010308, 2024) + Sandia/Caltech	the same kicked-Ising problem was classically simulated within days (2D tensor networks, sparse Pauli dynamics) — it was not beyond classical reach	T2
skeptics (O-hype)	"utility" became a movable goalpost with no agreed operational definition or independent adjudicator — invites quantum-washing	T3

Resolves when: a community-adopted taxonomy fixing utility (useful/competitive, may be classically reproducible) vs advantage (beats best classical) vs supremacy (beats all classical), with independent adjudication. Absent that, a result reproduced classically within weeks is a classical-simulation-frontier data point, not an advantage. See O-utility-definition, O-benchmark-standard.

C-energy-advantage — Whether quantum computing delivers a net energy advantage over classical for useful work

WHO	POSITION	TIER
energetic-advantage theory (arXiv:2601.08068, 2605.19854, 2026)	for boson sampling and cat-qubit computation, quantum wins on energy at a problem size SMALLER than where it wins on runtime	T3
Google supremacy energy analysis	classical simulation on Summit used ~7 orders of magnitude more energy than the QPU run	T3 (sampling task, no practical use)
full-system accounting (arXiv:2605.09580, 2026)	fixed cryo (~10 kW/fridge) + control + real-time decoders must be counted; at FT scale (millions of qubits) total draw could dwarf a single fridge	T3

Resolves when: a full-stack (fridge + control + classical co-processor) joules-per-useful-answer measurement for a USEFUL computation, vs the best classical method's energy for the same answer. Does the energetic-advantage crossover really precede the computational one on a problem someone cares about? See O-energy.

C-benchmark-metrics — Whether quantum computing can standardize honest cross-vendor benchmarks

WHO	POSITION	TIER
QED-C + standards bodies	application-oriented open benchmark suite (15+ apps) + Quantum Volume + ISO/IEEE/ETSI working groups are converging on standards	T3/T4
metrics-are-a-mess camp	no common yardstick exists; vendor metrics (#AQ, QV, raw counts, "logical qubits") each flatter one architecture; XEB is spoofable; results rarely independently reproduced	T3

Resolves when: broad adoption of application-oriented, independently-reproduced, spoof-resistant benchmarks reported by a neutral body (a "quantum MLPerf"), reporting LOGICAL performance with a defined universal gate set at stated code distance. See O-benchmark-standard, S-bench.

APPENDIX C

Recent preprint evidence

Window: submissions 2026-04-10 → 2026-07-09 (last ~90 days). **Categories:** quant-ph + cond-mat.mes-hall. **Method:** 25 claim-keyed arXiv queries → 413 unique papers in window → per-paper LLM assessment against `evidence/CONFLICTS.md` → strict filter to empirical / resource-estimate / classical-sim results that materially move a registered claim (61 papers).

Tier discipline (per `evidence/SCHEMA.md`): an arXiv posting is **T3 (preprint)** until refereed. Every row below upgrades a manual claim from **T4 (vendor announcement)** to at least **T3 (independent preprint)** — a real tier gain, since an independent preprint outranks a vendor blog post. Rows marked **T2** carry a verified `journal_ref` (published, refereed). No row was promoted to T2 on abstract wording alone.

> Caveat: tiers reflect *publication status verified from arXiv metadata*, not independent reproduction. A T3 preprint that agrees with a vendor claim corroborates it; it does not by itself retire the conflict. Replication status is unchanged for every `what_would_resolve_it` line in CONFLICTS.md.

Totals: 61 claim-moving papers | C-ftqc-timeline=23, C-overhead-ratio=15, C-majorana-existence=7, C-tls-scaling=7, C-photonic-scaling=3, C-advantage-survival=2, C-energy-advantage=2, C-quantum-utility=2

C-majorana-existence — Microsoft topological-qubit / Majorana claim (T4, no independent replication)

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
H-topo	Josephson spectroscopy study of kagome superconductors toward the deep point-contact regime	2605.06150v1	2026-05-07	T2 (Phys. Rev. B 113, 174502 (2026))	Demonstrates zero-bias conductance saturation in point-contact regime mimics Majorana signatures; cautions against misinterpretation in topological-qubit searches.
H-topo	Superconducting PdTe Thin Film Via Topotactic Transformation, Toward Topological Superconductors	2605.20437v1	2026-05-19	T2 (ACS Appl. Nano Mater. 2026, 9, 10684)	Demonstrates high-quality superconducting PdTe thin films via MBE; enables experimental platforms for Majorana detection but does not yet evidence MZMs.
H-topo	Exposing impostor Majorana zero modes through atomic-scale shot-noise	2604.26002v1	2026-04-28	T3	Demonstrates shot-noise method to distinguish trivial zero-bias peaks from true Majorana modes; directly addresses impostor problem in Fe(Se,Te).
H-topo	Experimental Evidence of Fractional Entropy in Critical Kondo Systems	2605.00669v1	2026-05-01	T3	Independent thermodynamic evidence of non-Abelian anyons (Majorana, Fibonacci) via fractional entropy in Kondo systems.
H-topo	Probing PbTe-Pb nanowire devices with radio-frequency reflectometry	2606.04544v2	2026-06-03	T3	Demonstrates rf reflectometry compatibility with PbTe-Pb nanowires under magnetic fields; advances measurement capability for topological-qubit platforms.

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
H-topo	Fabry-Perot Interference, g-factor Anisotropy, and Gate-Tunable Quantum dot in Chiral Tellurium Nanowires	2606.10001v1	2026-06-08	T3	Demonstrates Majorana-platform prerequisites (spin-orbit gap, g-factor anisotropy, quantum coherence) in elemental tellurium; not Majorana detection but materials foundation.
H-topo	Distinguishing Majorana zero modes from trivial defect states on the surface of the iron-based superconductor Fe(Te,Se)	2606.17499v1	2026-06-16	T3	Independent experimental replication showing Fe(Te,Se) zero modes are trivial Yu-Shiba-Rusinov states, not Majoranas—directly challenges topological-superconductor claim.

C-advantage-survival — Does any quantum speedup survive classical counterattack (Google Quantum Echoes, D-Wave spin-glass)

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
O-advantage;S-bench	Tensor network surrogate models for variational quantum computation	2604.20180v1	2026-04-22	T3	Tensor-network classical simulation successfully benchmarks QAOA on spin-glass; demonstrates parameter-transfer limits and feasibility scaling.
O-advantage;S-bench	Truncated Wigner dynamics of biclique quantum spin glasses	2606.20187v1	2026-06-18	T3	TWA classical method reproduces quantum spin-glass dynamics and critical exponents; scales to tens of thousands of qubits.

C-quantum-utility — IBM 'utility before fault tolerance' vs classical reproduction

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
O-utility-definition;S-bench	Ground-state energies of Ising models calculated using the samples from a quantum computer that simulates short-time evolution	2604.25715v1	2026-04-28	T3	Demonstrates Ising ground-state calculation at 63 qubits on real hardware; quantifies error scaling and utility boundary.
O-utility-definition;S-bench	Utility-scale quantum experiments using dynamic circuits to address collective dissipation in interacting qubits	2605.25830v1	2026-05-25	T3	Demonstrates utility-scale dissipative-dynamics simulation on 86 qubits with validated classical comparison; addresses classical reproducibility dispute.

C-ftqc-timeline — IBM/Google FTQC roadmaps (T4): below-threshold QEC & logical-qubit demos

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
S-qec;S-logical			2026-04-14	T3	

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
	Fast and accurate AI-based pre-decoders for surface codes	2604.12841v1			Demonstrates scalable real-time surface-code decoding at O(1 μs) with LER improvements over standard decoders.
S-qec;S-logical	Fault-Tolerant Error Detection Above Break-Even for Multi-Qubit Gates	2604.13219v2	2026-04-14	T3	Demonstrates fault-tolerant error detection above break-even on trapped-ion hardware; advances real-device QEC validation.
S-qec;S-logical;H-ion	Fault-Tolerant Quantum Computing with Trapped Ions: The Walking Cat Architecture	2604.19481v1	2026-04-21	T3	LDPC-based trapped-ion FT architecture with 110 logical qubits, 2,514 physicals, Hamiltonian-sim utility in ~1 month.
S-qec;S-logical	High-performance cellular automaton decoders for quantum repetition and toric code	2604.21866v1	2026-04-23	T3	Demonstrates scalable real-time QEC decoder with threshold ~7.5% and subthreshold scaling; advances practical error-correction architecture feasibility.
H-neutral;S-qec;S-logical	High-fidelity entangling gates and nonlocal circuits with neutral atoms	2604.25987v1	2026-04-28	T3	99.85% two-qubit gate fidelity on neutral atoms advances hardware baseline for below-threshold QEC demonstrations.
S-qec;S-logical	MCMit: Mid-Circuit Measurement Error Mitigation	2604.25863v2	2026-04-28	T3	Demonstrates measured QEC logical error reduction (1.2–9.4×) via mid-circuit measurement error mitigation on hardware.
S-qec;S-logical	Real-time Surface-Code Error Correction Using an FPGA-based Neural-Network Decoder	2605.04892v1	2026-05-06	T3	Demonstrates real-time surface-code QEC (distance-3) with closed-loop feedback on superconducting hardware; validates hardware-integrated decoder architecture for scalable fault tolerance.
S-qec;S-logical;H-silicon	Surface-Code Thresholds and Qubit Footprints in Shuttling-Based Spin-Qubit Railways	2605.05881v1	2026-05-07	T3	Demonstrates surface-code threshold and footprint scaling on silicon spin qubits with circuit-level noise; achieves Megaquop distance-7 code at p=10^−3 error rate.
S-qec;H-ion	Error Correction of Beam-splitter-Generated Entangled GKP States	2605.08009v1	2026-05-08	T3	First experimental demonstration of entangled GKP qubits with QEC extension; advances trapped-ion bosonic code roadmap.
H-ion;S-qec	Demonstration of a Multiplexing Trapped Ion Quantum Processing Unit	2605.16010v1	2026-05-15	T3	Demonstrates multiplexing scalability for trapped-ion QPUs; addresses wiring bottleneck to fault-tolerant scale.
S-qec;S-logical	Benchmarking a machine-learning differential equations solver on a neutral-atom logical processor	2605.21276v1	2026-05-20	T3	Demonstrates logical-qubit advantage in end-to-end ML application; validates fault-tolerant overhead trade-offs experimentally.
S-qec;S-logical	Real-Time Quantum Error Correction System Stack: Architecture, Algorithms, and Engineering Practice	2605.30765v1	2026-05-29	T3	Quantifies real-time QEC engineering gaps; benchmarks decoders for surface/qLDPC; identifies system-level bottlenecks beyond algorithm speed.
S-qec;S-logical;H-neutral	Quantum error correction with the toric code	2606.04079v1	2026-06-02	T3	First scalable toric-code logical-qubit demo on neutral atoms with 90 syndrome cycles, distance-dependent error suppression.

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
S-qec;S-logical	A superconducting surface-code processor with lattice-surgery logical operations	2606.06598v1	2026-06-04	T3	First experimental demonstration of below-threshold surface-code logical operations with multi-cycle error suppression on real hardware.
H-supercon;S-qec	Ultra-high Q-factor superconducting tantalum resonators on 300 mm Si wafers	2606.10719v1	2026-06-09	T3	Demonstrates industrial-scale ultra-high-Q superconducting resonators (>40M median, >60M peak) on 300mm wafers, enabling hardware-efficient bosonic error correction and scaling.
S-qec;S-logical	Efficient Magic State Factory Via Transversal Non-Clifford Gate	2606.16199v1	2026-06-15	T3	End-to-end magic-state factory simulations at distance d=5 with code switching; resource comparison against cultivation methods.
S-qec;S-logical	Cultivating logical catalysts for fault-tolerant dyadic phase rotations	2606.27358v1	2026-06-25	T3	Demonstrates fault-tolerant logical-qubit phase gates with error-corrected scaling; advances offline magic-state cultivation toward FTQC resource efficiency.
H-silicon;S-qec	High-Fidelity Hole Spin Qubits Reveal Quadrupolar Nuclear-Bath Dynamics in Isotopically Purified Planar Germanium	2606.28695v1	2026-06-27	T3	Demonstrates 99.9% single-qubit Ge-hole fidelity via isotopic purification, advancing hardware records for spin-qubit platform.
H-supercon;S-qec	High-Precision Calibration Workflow Achieves Above 99.9% CZ Gate Fidelity on a Scalable Superconducting Processor	2607.01422v1	2026-07-01	T3	Demonstrates 99.9% CZ fidelity on 84-qubit processor; advances two-qubit gate benchmark supporting FTQC feasibility.
H-neutral;S-qec;S-logical	Fault-tolerant quantum computation with static atomic buses	2607.02804v1	2026-07-02	T3	Demonstrates logical-qubit error correction and gate operations on neutral-atom hardware with simulated >10× improvement over prior architectures.
S-qec;S-logical	LUCI on IBM Hardware: Error Suppression with Almost Half Syndrome Density	2607.01887v1	2026-07-02	T3	Demonstrates logical-qubit error suppression on real hardware with reduced syndrome overhead; validates dynamic-code QEC beyond theory.
S-qec;S-logical	Genuine Multipartite Entanglement between Logical Qubits via Cross-Code Lattice Surgery	2607.04227v1	2026-07-05	T3	First experimental demonstration of logical-qubit universal gates via lattice surgery on real hardware; advances fault-tolerant QEC building blocks.
S-qec;S-logical;H-supercon	Quantum error correction of a grid-state qubit with state preparation and measurement errors below 10^{-3}	2607.06718v1	2026-07-07	T3	Demonstrates below-threshold QEC performance in grid-state qubits; SPAM errors reduced 100× to sub-10^{-3}, enabling practical logical-qubit operation.

C-overhead-ratio — physical:logical ratio; qLDPC '10–20× overhead cut' (T3/T4)

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
H-neutral;S-qec;O-overhead	Towards Ultra-High-Rate Quantum Error Correction with Reconfigurable Atom Arrays	2604.16209v2	2026-04-17	T3	Demonstrates ultra-high-rate qLDPC codes (>1/2 encoding) with sub-threshold logical error rates on neutral atoms, directly addressing overhead reduction.
O-overhead;S-qec	Networked Realization of Quantum LDPC Codes	2604.25026v1	2026-04-27	T3	First circuit-level noise study of networked bivariate bicycle qLDPC codes; quantifies overhead under practical constraints.
O-overhead;S-qec;H-silicon	CAbLECAR: efficiently scheduling QLDPC codes on a tileable spin qubit chip with shuttling	2604.24739v2	2026-04-27	T3	Circuit-level QLDPC scheduling on silicon spin qubits; identifies codes reducing logical error rates vs surface codes; quantifies overhead via shuttling range extension.
S-qec;O-overhead	A Scalable FPGA Architecture for Real-Time Decoding of Quantum LDPC Codes Using GARI	2605.01035v1	2026-05-01	T3	First multi-core FPGA decoder for qLDPC under correlated errors; 6× resource reduction advances practical overhead benchmarking.
O-overhead;S-qec	Mitigating Classical Resource Costs in Quantum Error Correction via Generalized qLDPC Predecoding	2605.03180v1	2026-05-04	T3	Demonstrates practical predecoding + hardware design reducing qLDPC decoder utilization 3,963x; scales to 36k–360k logical qubits on cryogenic ASIC.
O-overhead;S-qec	Space-Time Tradeoffs of Pauli-Based Computation in Distributed qLDPC Architectures	2605.03854v1	2026-05-05	T3	Quantifies qLDPC execution-time overhead vs surface code in distributed architecture; demonstrates order-of-magnitude speedup with large blocks.
O-overhead;S-qec;H-ion;H-neutral	Distributed Quantum Error Correction with Bivariate Bicycle Codes in a Modular Architecture	2605.04663v1	2026-05-06	T3	Demonstrates distributed BB code logical error rates and pseudo-threshold on modular hardware; measures practical qLDPC overhead scaling.
S-qec;O-overhead	A Resource Comparison of Logical T-State Preparation	2605.26522v1	2026-05-26	T3	Systematizes T-state preparation costs across distillation/cultivation/code-switching; clarifies overhead trade-offs under standardized comparison.
H-silicon;O-overhead;S-qec	Hardware-Tailored Resource Estimation for Magic-State Distillation on Silicon Spin Qubits	2605.28936v1	2026-05-27	T3	Quantifies magic-state distillation overhead on silicon; biased codes achieve 3× footprint reduction vs surface code.
O-overhead;S-qec	Evolutionary Discovery of Bivariate Bicycle Codes with LLM-Guided Search	2606.02418v1	2026-06-01	T3	Discovers new high-performing qLDPC codes; indecomposable [[288,16,12]] and codes with k=50, d=8 improve finite-length overhead data.
O-overhead;S-qec	Full Extractors for Logical Processing in Hypergraph Product Codes	2606.03507v1	2026-06-02	T3	Demonstrates full logical processing on HGP qLDPC codes with 50–80% size vs. base code; achieves 10^{-6} logical error at distance 10.

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
O-overhead;S-qec	Breakeven demonstration of quantum low-density parity-check codes	2606.06455v1	2026-06-04	T3	First experimental breakeven qLDPC code; 9× logical error-rate improvement over prior solid-state implementation.
O-overhead;S-qec	Large-Language-Model Discovery of Quantum LDPC Codes through Structured Concept Evolution	2606.24808v1	2026-06-23	T3	Demonstrates new qLDPC code families with measured overhead via BP+OSD decoding under realistic noise.
O-overhead;S-qec;H-neutral	Fast and Parallel High-Rate STAR Architecture for Megaquop Quantum Simulation	2606.25011v1	2026-06-23	T3	High-rate QEC code integration achieves ~5.5× space reduction vs surface code; demonstrates co-designed algorithm-code-hardware overhead optimization.
O-overhead;S-qec	Strictly Local Tile-Code Architectures on Two-Dimensional Planar Lattices	2607.05897v1	2026-07-07	T3	Demonstrates routed tile-code overhead <surface-code at p<0.08%; validates qLDPC scaling under realistic connectivity constraints.

C-tls-scaling — Are TLS/material defects an engineering nuisance or a scaling wall

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
H-supercon;H-fab	Tantalum-Encapsulated Niobium Superconducting Resonators: High Internal Quality Factor and Improved Temporal Stability via Surface Passivation	2604.09050v1	2026-04-10	T3	Demonstrates fab control of TLS density via Ta encapsulation; shows scaling path and temporal stability data.
H-supercon	Operating a bistable qubit	2605.03187v1	2026-05-04	T3	Demonstrates adaptive mitigation of TLS-induced dephasing in superconducting qubits with 77% error reduction; addresses whether TLS defects are a fundamental scaling wall.
H-supercon;H-fab	Readout failures in superconducting qubits due to TLS-defects in tunnel junctions	2605.02755v1	2026-05-04	T3	Demonstrates TLS-readout resonator coupling as a scaling mechanism limiting qubit fidelity; supports TLS as a fundamental scaling wall.
H-supercon;H-fab	Interface Piezoelectric Loss in Superconducting Qubits	2605.15554v1	2026-05-15	T3	Identifies interface piezoelectricity as distinct loss channel competing with/exceeding TLS at high frequencies; experimental evidence for new scaling constraint.
H-silicon;S-qec	Multi-Qubit Entanglement of Unit Cell Pairs in SiMOS	2605.20781v1	2026-05-20	T3	SiMOS 4-qubit multi-unit-cell coupling with 99%+ fidelities and preserved entanglement lifetime advances silicon scalability beyond isolated DQD units.
H-supercon;S-qec	Non-Local and Non-Markovian Effects of a Microscopic Two-Level Defect in Superconducting Quantum Circuits	2605.23385v2	2026-05-22	T3	Demonstrates tunable TLS coupling in scalable architecture; non-Markovian dynamics complicate error-correction models.

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
H-supercon;H-fab	Surface Platinum Alloying for Passivation of Oxide Interfaces on Superconducting Niobium Films	2607.00429v1	2026-07-01	T3	Demonstrates fab-level control of TLS density via Pt alloying; directly addresses surface-oxide scaling wall.

C-photonic-scaling — Xanadu/PsiQuantum million-qubit photonic roadmaps (T4)

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
H-photonic;S-qec	Picosecond Schrödinger cat states for ultrafast optical quantum processing	2606.24002v1	2026-06-22	T3	Demonstrates high-rate non-Gaussian state generation at picosecond timescales, addressing temporal-mode bottleneck for fault-tolerant photonic QC.
H-photonic	Industry-ready spin-photon interfaces for hybrid photonic quantum computing	2606.27787v1	2026-06-26	T3	Demonstrates production-line QD devices with near-unity photon purity, multi-photon entanglement, and path toward fault-tolerance loss thresholds.
H-photonic;S-qec	Integrated Photon-Memory Entanglement Generation using Dual Photonic Resonators	2607.01324v1	2026-07-01	T3	First integrated photon-memory entanglement on chip; demonstrates scalable quantum-repeater building block with published loss budgets and multimode storage capacity.

C-energy-advantage — Net energy advantage over classical for useful work

NODE ID(S)	PAPER	ARXIV	DATE	TIER	WHAT IT CHANGES
O-energy	Estimating The Energy Consumption of Quantum Computing from A Full System Aspect	2605.09580v2	2026-05-10	T3	First full-stack joules-per-answer model for NISQ and FTQC; quantifies energy breakdown by regime.
H-supercon;O-energy	Unveiling Energetic Advantage in Superconducting Cat-Qubits Quantum Computation	2605.19854v1	2026-05-19	T3	Provides full-stack energy consumption model and scaling analysis for superconducting cat-qubit systems with error correction; estimates quantum energetic advantage threshold at >26 qubits.

Screened and excluded (not claim-movers)

- **QKD / quantum-networking cluster (~30 papers):** active field (telecom quantum memories, satellite QKD, repeater protocols) but the manual registers **no T4 vendor conflict** for QKD/repeaters, so none upgrades a graded claim. Node activity for `A-qkd`/`A-qinternet`/`H-transduce`, not a tier change.
- **Majorana *detection-method theory* (~5 papers):** new signatures/diagnostics for distinguishing true from trivial zero modes. `C-majorana-existence` needs an *independent experimental* braiding/fusion replication; proposals don't move it.

- **Incremental algorithm/theory & reviews:** flagged as topically relevant but not providing a new experimental result, resource estimate, or classical-simulation attack/defense on a registered claim.

Reproduce

```
# 25 claim-keyed queries over the 90-day window, then assess vs evidence/CONFLICTS.md
python3 arxiv_sweep_S4.py   # (query set embedded; regenerates arxiv_sweep_S4.csv)
```

Raw per-paper assessments: `arxiv_assess.json` (413 papers). Curated table: `arxiv_sweep_S4.csv`.

APPENDIX D

Glossary

Plain-language definitions of the technical terms a reader meets in *The Quantum Atlas*. Chapter cross-references point to where a term is explained most fully. Where a term carries a graded-evidence nuance (an "advantage" that may not survive classical counterattack, a vendor metric, a forecast), the entry says so.

Adiabatic theorem — If you start a system in the lowest-energy state of an easy problem and change the problem slowly enough, it stays in the lowest-energy state of the hard problem you end on. This is the physical principle behind quantum annealing; "slowly enough" is set by the smallest energy gap along the way, which can shrink and erase any speedup (see Ch. 1, Ch. 2).

Algorithmic qubits — A vendor benchmark (IonQ, QED-C) that reports how many qubits a machine can use in a real algorithm rather than raw count. Like all single-number benchmarks it flatters the architecture that defined it; treat it as a T4 claim (see Ch. 3, Ch. 8).

Amplitude (probability amplitude) — The complex number attached to each possible outcome of a quantum state. Squaring its magnitude gives the probability of that outcome (see Born rule). Amplitudes can add and cancel, which is what makes interference possible (see Ch. 1).

Amplitude amplification — The engine inside Grover's algorithm: repeatedly nudging probability toward the answer you want so it becomes likely to be measured (see Ch. 3).

Amplitude encoding — Packing a list of classical numbers into the amplitudes of a quantum state, so a 1,024-number vector fits in 10 qubits. Cheap to store, expensive to load — the loading cost is where many "speedups" quietly die (see Ch. 3, qRAM).

Amplitude estimation (QAE) — A quantum routine that estimates an average or probability quadratically faster than classical Monte Carlo sampling. The most likely survivor of the quadratic-speedup family, and the basis of quantum finance pitches, though it needs fault-tolerant hardware and an efficient way to load the data (see Ch. 3).

Ancilla — A helper qubit added to a circuit to assist a measurement, gate, or error-correction step, then discarded or reset. Real detectors and error-correction cycles are full of them (see Ch. 1, Ch. 3).

Annealing (quantum annealing) — A non-gate model of computation that solves optimization problems by slowly settling into a low-energy state (see adiabatic theorem). D-Wave shipped the only commercial annealers; their advantage over tuned classical solvers is contested and unproven at scale (see Ch. 2, Ch. 5).

Anyon — An exotic particle possible only in two dimensions, whose quantum state changes when you move one around another. "Non-abelian" anyons are the basis of topological quantum computing (see Majorana zero mode; Ch. 1, Ch. 2).

Area law — A rule of thumb that entanglement in many natural systems grows with the boundary of a region rather than its volume. When it holds, classical tensor-network methods can simulate the system efficiently — a recurring reason some "quantum advantage" claims fall to classical computers (see Ch. 1, Ch. 3).

Assured PNT (positioning, navigation, timing) — Knowing where and when you are without GPS, which can be jammed or spoofed. The quantum answer fuses a drift-free clock, a cold-atom accelerometer, and a magnetic map. Defense-funded, nearing first deployments; the flashiest vendor accuracy numbers are unverified T4 field trials (see Ch. 4, Ch. 5).

Atom interferometry — Splitting and recombining a cloud of laser-cooled atoms so their wave nature reveals gravity, acceleration, or rotation. The instrument under cold-atom gravimeters and quantum inertial navigation (see Ch. 4).

Barren plateau — A training failure in variational quantum algorithms: for expressive circuits the optimization landscape flattens exponentially as qubits are added, so gradients vanish and learning stalls. One of three theoretical results that turned the field against near-term variational methods (see Ch. 3).

BB84 — The first quantum key distribution protocol (Bennett–Brassard, 1984). It encodes key bits in photon polarizations that an eavesdropper cannot copy or measure without leaving a detectable trace (see no-cloning; Ch. 4, Ch. 7).

Bell inequality / CHSH — A bound that any "local realist" theory (definite properties, no faster-than-light influence) must obey. Quantum mechanics violates it, and experiments confirm the violation. CHSH is the four-setting version whose value tops out at 2 classically and reaches $2\sqrt{2}$ quantumly (see Tsirelson bound). Violation is the operational certificate that a device holds real entanglement (see Ch. 1, Ch. 7).

Bell state — A maximally entangled pair of qubits, perfectly correlated no matter how far apart. The basic resource for teleportation and entanglement-based cryptography (see Ch. 1).

Below-threshold — The regime where making an error-correcting code larger makes the logical error rate smaller. Google's Willow (2024) was the first clear demonstration; it means scaling error correction is now an engineering problem rather than a physics gamble (see threshold theorem; Ch. 2, Ch. 3).

Berry phase (geometric phase) — A phase a quantum state picks up when its parameters are cycled around a loop, depending only on the path's shape and not on speed. Its speed-independence grounds error-resistant "holonomic" gates (see Ch. 1).

Block encoding — Embedding a matrix inside a larger unitary so a quantum computer can operate on it. The input format that QSVT and modern algorithms assume; where only weak "sampling access" is available, classical methods can often match the quantum routine (see Ch. 3).

Bloch sphere — The globe used to picture a single qubit: every pure state is a point on the surface, and a single-qubit gate is a rotation of the sphere (see Ch. 1).

Born rule — The recipe for turning a quantum state into predictions: the probability of an outcome is the squared magnitude of its amplitude. Every prediction quantum mechanics makes is, in the end, a Born-rule probability (see Ch. 1).

Bosonic qubit — A qubit stored in the many energy levels of a single oscillator (a superconducting cavity) rather than a two-level system, so one physical mode carries built-in redundancy (see cat qubit, GKP state; Ch. 2).

Break-even (error correction) — The milestone where an error-corrected logical qubit outlives or outperforms the best physical qubit doing the same job. The meaningful 2025 threshold; deep computation across many logical qubits at once is the next bar and has not been shown (see Ch. 2, Ch. 3).

CAGR (compound annual growth rate) — The smoothed year-on-year rate at which a market or figure would grow to reach a projected value, assuming steady compounding. Market forecasts often headline a large CAGR; it is a modeling assumption, not measured revenue, and reads as T5 (see TAM; Ch. 6, Ch. 8).

Cat qubit — A bosonic qubit encoded in two opposite-phase states of light ("Schrödinger-cat" states). A two-photon drive suppresses bit-flip errors exponentially, leaving only phase-flips for a simple 1D code to catch — trading a hard 2D error-correction problem for an easier 1D one. The open question is whether the noise bias survives fast gates (see Ch. 2).

Certified randomness — Provably unpredictable random bits generated and verified using a quantum computer. The JPMorgan–Quantinuum result (2025) is arguably the first useful thing a beyond-classical machine has done, though the market for it is small (see Ch. 3, Ch. 4).

Circuit (quantum circuit) — A sequence of quantum gates applied to qubits, ending in measurement. The standard way to describe a quantum computation (see Ch. 3).

Circuit cutting — Splitting a circuit too wide for the hardware into smaller pieces, at the cost of exponentially more classical post-processing. Useful only when the cuts are few (see Ch. 3).

Circuit QED — The physics of coupling superconducting qubits to microwave resonators on a chip — applied field quantization, and the backbone of transmon processors (see Ch. 1, Ch. 2).

Classical shadows — A measurement technique that predicts many properties of a quantum state from surprisingly few random measurements, with a sample cost independent of system size. It buys measurement efficiency rather than a computational speedup (see Ch. 3).

Clifford gates / Clifford+T — Clifford gates (Hadamard, phase, CNOT) are the "free" gates a classical computer can simulate efficiently. Adding the single non-Clifford T gate makes the set universal — and the T gate is where quantum hardness and cost live (see Gottesman–Knill; Ch. 3).

CLOPS (Circuit Layer Operations Per Second) — An IBM speed benchmark measuring how fast a machine runs circuit layers. One of several incompatible vendor metrics; T4 by rule (see Ch. 3, Ch. 8).

CNSA 2.0 — The US National Security Agency's requirement that new national-security systems adopt post-quantum cryptography, capable by 1 January 2027 and fully migrated by 2030–2035 (see Ch. 4, Ch. 6).

Coherence time (T1 / T2) — How long a qubit keeps its quantum information. **T1** is energy relaxation (the qubit decays toward its ground state); **T2** is dephasing (it loses the phase relationship that carries superposition). The headline spec of every hardware modality (note: T1/T2 here are coherence times, unrelated to the evidence tiers T1–T6) (see decoherence; Ch. 1, Ch. 2).

Coherent state — The most "classical" state of light, closest to a steady wave. The building block of cat qubits and a native state of photonic hardware (see Ch. 1).

Complementarity — The principle that some pairs of properties (position and momentum, or two measurement bases) cannot both be sharply defined at once; measuring one blurs the other (see uncertainty principle; Ch. 1).

Contextuality (Kochen–Specker theorem) — A no-go result proving quantum outcomes cannot be pre-assigned as if they existed independently of what you choose to measure. It is also a resource: contextuality is exactly what "magic states" supply to make quantum computing hard (see Ch. 1).

Cooper pair — Two electrons bound together that move without resistance in a superconductor. Cooper pairs tunneling across a Josephson junction are what make a transmon qubit work (see Ch. 1, Ch. 2).

Cross-entropy benchmarking — The statistical test behind random-circuit "supremacy" claims. It is not a proof of correctness and has been spoofed classically, which is part of why advantage claims are contested by default (see Ch. 8).

CRQC (cryptographically relevant quantum computer) — A future machine large enough to run Shor's algorithm against real RSA or elliptic-curve keys and break today's public-key encryption. Point estimates cluster around 2030 ± 3 years, with roughly 50% odds by 2035 (Global Risk Institute) and weight trailing toward 2040; a minority holds it will never arrive. The uncertainty is why migration must start now (see harvest now decrypt later; Ch. 4, Ch. 8).

Cryo-CMOS — Control electronics fabricated in standard silicon that run cold, inside the refrigerator next to the qubits, to reduce the crippling number of wires. The bet that lets superconducting and silicon machines scale past a few thousand qubits (see Ch. 2).

Cryostat / dilution refrigerator — The refrigerator that cools superconducting, silicon-spin, and bosonic qubits to near absolute zero by exploiting a mixture of helium-3 and helium-4. It sets hard ceilings on cooling power, volume, and wiring. Bluefors and Oxford Instruments dominate the market (see Ch. 2, Ch. 6).

Crypto-agility — Designing systems so their cryptographic algorithms can be swapped out quickly, without re-architecting. The practical requirement behind post-quantum migration: an agile system can adopt new standards as they land and retire broken ones fast (see post-quantum cryptography; Ch. 4, Ch. 6).

Decoherence — The loss of quantum behavior when a system leaks its phase information into its environment. It is why the everyday world looks classical, and it is the central enemy of every qubit — the reason error correction exists (see Ch. 1).

Dequantization — Writing a classical algorithm that matches a claimed quantum speedup, removing the advantage. Ewin Tang's 2018 work started a program that erased the exponential edge of several quantum machine-learning and linear-algebra methods (see Ch. 3, Ch. 8).

Deemed export — Under US export-control law, giving a foreign national access to controlled technology inside the country counts as an export to their home country. It means quantum hiring and lab access can trigger the same license rules as shipping hardware abroad (see export controls; Ch. 6).

Density matrix — A more general description of a quantum state that also covers "mixed" states — statistical blends arising from noise or ignorance. Its off-diagonal terms carry the coherence that decoherence destroys (see Ch. 1).

Dephasing — Loss of the phase relationship between the parts of a superposition, without energy loss. The T2 process (see coherence time; Ch. 1).

Discord (quantum discord) — A measure of quantum correlations that can persist even in states with no entanglement. Proposed as the resource behind certain mixed-state speedups; whether it is a usable resource is contested (see Ch. 1).

DiVincenzo criteria — David DiVincenzo's 2000 checklist of five requirements any physical system must meet to be a quantum computer (scalable qubits, initialization, long coherence, a universal gate set, and readout), plus two more for communication. Still the field's scorecard (see Ch. 2).

DRAG — A standard pulse-shaping technique that suppresses leakage into unwanted energy levels during a gate. Well-calibrated DRAG already runs near the physical noise floor for single-qubit gates (see optimal control; Ch. 3).

Dual-rail — A way of encoding a photonic qubit across two paths or modes, with the photon in one or the other. One of several photonic encodings (see Ch. 2).

E91 — Ekert's 1991 quantum key distribution protocol, which draws the key from entangled pairs and certifies security through a Bell-inequality violation (see Ch. 4, Ch. 7).

Ebit — One unit of entanglement: the amount contained in one maximally entangled pair of qubits (see Ch. 1).

Eigenvalue / eigenstate — For a given operation, an eigenstate is a state the operation leaves pointing the same way, and its eigenvalue is the number it gets scaled by. Measuring an observable can only return one of its eigenvalues, and the state collapses to the matching eigenstate (see observable, Born rule; Ch. 1).

Evidence tiers (T1–T6) — The atlas's grading scale for how much a claim can be trusted: T1 established physics (textbook, reproduced for decades); T2 peer-reviewed result; T3 preprint or conference report, not yet refereed; T4 vendor claim, not independently reproduced; T5 analyst or market forecast; T6 speculative. A vendor press release (T4) and a peer-reviewed threshold demonstration (T2) never weigh the same. "Quantum advantage" claims are treated as contested by default. See the Preface and Appendix A.

EIT (electromagnetically induced transparency) — An optical effect used to read out Rydberg atoms in RF sensing; the atoms become transparent to a probe laser in a way that shifts measurably with an applied field (see Rydberg; Ch. 4).

Entanglement — A shared quantum state of two or more systems that no description of the parts alone can reproduce. Measuring one instantly correlates with the other however far apart, yet it carries no message on its own. The workhorse resource of quantum computing, communication, and error correction (see Ch. 1).

Entanglement swapping — Entangling two particles that never interacted, by using a joint measurement on intermediaries. The primitive a quantum repeater uses to stitch short links into long ones (see Ch. 1, Ch. 4).

EPR (Einstein–Podolsky–Rosen) — The 1935 argument that quantum mechanics is either incomplete or nonlocal, built on an entangled pair. Intended as a critique, it became the seed of entanglement, teleportation, and quantum cryptography (see Ch. 1, Ch. 7).

Error correction (QEC) — Encoding one protected "logical" qubit across many physical qubits so errors are detected and fixed faster than they accumulate. The path from noisy hardware to trustworthy computation (see threshold theorem, surface code; Ch. 3).

Error mitigation — Cheaper near-term techniques (ZNE, PEC, and others) that reduce the bias in a measured value by running extra circuits, without protecting the quantum state. Its cost grows exponentially with circuit depth, so it extends reach but is never an asymptotic fix (see Ch. 3).

EuroQCI — A planned continent-wide quantum-secure communication network across all 27 EU member states (see Quantum Flagship; Ch. 4, Ch. 6).

Export controls — Government restrictions on selling quantum computers and enabling equipment abroad. The 2024 US controls are "plurilateral," rewarding allies who adopt matching rules; effectiveness and cohesion are unproven (see Ch. 6).

Fault tolerance — Building a computer so that a single physical fault — even one during the correction step itself — cannot cascade into a logical error. The regime useful algorithms need, and the goal the whole error-correction stack serves (see Ch. 3, Ch. 8).

FeMoco — The iron-molybdenum cofactor that lets bacteria fix nitrogen at room temperature. The textbook "killer app" target for quantum chemistry; recent estimates put it beyond a thousand logical qubits and billions of gates, far past today's hardware (see Ch. 5, Ch. 8).

Fidelity — A score from 0 to 1 for how close a produced quantum operation or state is to the ideal. Two-qubit gate fidelity is the binding constraint on whether error correction works; "three nines" (99.9%) is the rough entry point (see Ch. 2, Ch. 3).

Fock state — A state of light with a definite number of photons. The native language of photonic and bosonic hardware (see Ch. 1).

Foundry — A semiconductor fabrication plant. The open question for silicon and photonic qubits is whether a foundry can make thousands of qubits uniformly enough — yield at wafer scale, which vendors publish least (see Ch. 2).

Gate (quantum gate) — A basic reversible operation on one or more qubits — the quantum analog of a logic gate. Sequences of gates form circuits (see Ch. 3).

Gaussian states — A well-behaved class of light states (including coherent and squeezed states) that are, on their own, classically simulable. Reaching universality in photonic hardware requires a non-Gaussian element (see Wigner negativity; Ch. 1).

GHZ state — A maximally entangled state of three or more qubits. Used in tests of quantum mechanics and as a resource in networked and clock protocols (see Ch. 1, Ch. 4).

GKP state — A "grid" bosonic code state that protects a qubit against small shifts in an oscillator. A sibling of the cat code and one route to fault-tolerant photonic and cavity hardware (see Ch. 2).

Gottesman–Knill theorem — The result that circuits built only from Clifford gates and measurements can be simulated efficiently on a classical computer. It pins down why the non-Clifford T gate is the expensive, load-bearing resource (see Ch. 1, Ch. 3).

GRAPE / Krotov / CRAB — Numerical methods for shaping the control pulses that realize each gate ("quantum optimal control"). They pay off most for two-qubit and leakage-heavy gates and for robustness against drift (see Ch. 3).

Gravimetry — Measuring gravity precisely, often to find what is underground. Cold-atom gravimeters are absolute and drift-free because the atom is the test mass; survey instruments (Exail's AQG) already ship (see Ch. 4, Ch. 5).

Grover's algorithm — A quantum search that finds a marked item among N possibilities in about √N tries instead of N. Provably optimal for unstructured search, but careful accounting shows its quadratic speedup likely delivers no practical advantage once error-correction overhead and a slow logical clock are charged (see Ch. 3).

Hadamard test — A short circuit that estimates the overlap between two quantum states, including its sign. Used in the atlas's reference implementation to measure cosine similarity (see swap test; Ch. 3).

Hamiltonian — The mathematical object describing a system's energy and how its state evolves in time. Encoding a problem in a Hamiltonian is the starting point for annealing and simulation (see Ch. 1, Ch. 3).

Hamiltonian simulation — Using a quantum computer to model how a quantum system evolves. The best-founded quantum advantage, strongest for long-time, high-entanglement dynamics. Method families include Trotter–Suzuki, qDRIFT, LCU, and qubitization (see Ch. 3).

Harvest now, decrypt later (HNDL) — The threat that an adversary records encrypted traffic today and stores it to decrypt once a capable quantum computer exists. Any secret with a long shelf life is already exposed, which is the whole argument for migrating to post-quantum cryptography now (see Mosca's inequality; Ch. 4).

HEMT (high-electron-mobility transistor) — A low-noise amplifier stage that boosts a qubit's faint readout signal after the first cryogenic amplifier. Low Noise Factory supplies most of them (see paramp/TWPA; Ch. 2).

Heisenberg limit — The ultimate precision floor for a measurement, where sensitivity improves with the number of resources N (rather than √N). Reaching it needs entanglement; most sensors sit at the easier standard quantum limit (see Ch. 4).

HHL algorithm — The Harrow–Hassidim–Lloyd routine for solving linear systems, exponentially faster in principle — but only if four fine-print assumptions hold together (sparse well-conditioned matrix, efficient data loading, efficient simulation, and reading out just a summary). The canonical "read the fine print" algorithm; no practical problem has been shown to live in its sweet spot (see Ch. 3).

Hilbert space — The mathematical space of all possible states of a quantum system. An n-qubit register lives in a space of 2^n dimensions, which is where the "exponential" of quantum computing comes from (see Ch. 1).

Holevo bound — The limit that n qubits can deliver at most n classical bits of readable information, however much they seem to hold internally (see Ch. 1).

Hyperpolarization — Boosting a nuclear magnetic-resonance signal enormously (parahydrogen or DNP methods) to make real-time metabolic MRI feasible. The near-clinic strand of quantum-enhanced MRI (see Ch. 4).

Interconnect — The link that carries quantum information between separate chips or modules. Gates across a link are slower and far noisier than on-chip gates (IBM reported ~3.5% error across one), and whether any interconnect stays below the error-correction threshold is a quiet, decisive open question (see transduction; Ch. 2, Ch. 8).

Interference — The adding and canceling of quantum amplitudes, the effect with no classical analog that underlies every quantum speedup (see Ch. 1).

Jordan–Wigner / Bravyi–Kitaev — Two standard recipes for mapping electrons (fermions) onto qubits so a quantum computer can simulate chemistry, enforcing the antisymmetry that fermions require. A real driver of circuit depth in chemistry problems (see Ch. 1, Ch. 3).

Josephson junction — Two superconductors separated by a thin barrier, across which Cooper pairs tunnel. Its nonlinearity is what turns a superconducting circuit into a usable qubit (the transmon) (see Ch. 1, Ch. 2).

Kraus operators — The mathematical pieces that describe how noise or any physical process transforms a quantum state. The machinery used to simulate error channels and prove correction theorems (see purification; Ch. 1).

Lieb–Robinson bound — A speed limit on how fast information and entanglement can spread through a lattice of interacting particles, even without relativity. It underpins why tensor-network simulation works and why local error correction is stable (see Ch. 1).

Lindblad master equation — The standard equation for how an open quantum system evolves while leaking into its environment — unitary evolution plus irreversible noise terms. The formal frame for decoherence (see Ch. 1).

Logical qubit — A single protected qubit built from many physical qubits through error correction, and the unit real algorithms actually consume. Honest counts come with a code distance, a measured error rate, and whether a universal gate set was used or the device merely detected errors (see Ch. 3).

Loophole-free Bell test — A Bell experiment that closes the escape routes (locality and detection) local-realist theories could hide in. The 2015 trio of experiments shut the last serious doubt about entanglement being real (see Ch. 1, Ch. 7).

Magic state / magic-state distillation — A special resource state needed to run the non-Clifford T gate fault-tolerantly, manufactured by "distilling" many noisy copies into a few clean ones. Historically the dominant cost of a fault-tolerant computer; newer "cultivation" methods cut it (see Ch. 1, Ch. 3).

Magnetometry / OPM — Measuring tiny magnetic fields. Optically pumped magnetometers (OPMs) run near room temperature and approach the sensitivity of cryogenic SQUIDs, enabling wearable brain scanners (OPM-MEG). The clinical bottleneck is reimbursement, not physics (see Ch. 4, Ch. 5).

Majorana zero mode — An exotic quasiparticle predicted to appear at the ends of a topological superconductor, able to store a qubit non-locally so local noise cannot corrupt it. The basis of Microsoft's topological program; after two decades no independently verified topological qubit exists (see Ch. 2, Ch. 8).

Many-worlds — An interpretation of quantum mechanics that keeps only the smooth Schrödinger evolution and denies any collapse, with every outcome realized on its own branch. Empirically identical to the alternatives (see measurement problem; Ch. 1).

Measurement problem — The unresolved question of what physically happens, and why, when a smoothly evolving quantum state "jumps" to a definite outcome on measurement. The field's oldest open problem, unsolved since 1925 and untouched by any experiment so far (see Ch. 1, Ch. 8).

ML-KEM / ML-DSA / SLH-DSA — The core post-quantum cryptography standards NIST finalized in August 2024: ML-KEM (FIPS 203, formerly Kyber) for key exchange, and ML-DSA (FIPS 204, Dilithium) and SLH-DSA (FIPS 205, SPHINCS+) for digital signatures. Already deployed at internet scale (see Ch. 4, Ch. 6).

Mølmer–Sørensen gate — The standard entangling two-qubit gate for trapped ions, driven through the ions' shared vibrational motion (see Ch. 2, Ch. 3).

Monogamy of entanglement — The rule that entanglement cannot be freely shared: if two systems are maximally entangled, neither has any left for a third. The mathematical reason quantum key distribution is secure (see strong subadditivity; Ch. 1).

Mosca's inequality — A rule of thumb for cryptographic urgency: if the time your secrets must stay secret plus the time to migrate exceeds the time until a capable quantum computer arrives, you are already exposed. The formal case for acting on the quantum threat today (see harvest now decrypt later; Ch. 4, Ch. 6).

Naimark dilation — The theorem showing that any realistic (noisy, lossy) measurement is just a clean projective measurement on the system plus an extra ancilla. It makes real detector modeling and soft-information decoding possible (see POVM; Ch. 1).

National Quantum Initiative — The 2018 US law that organized federal quantum research through a coordination office and NIST, NSF, and DOE centers, authorizing about $1.2B over five years. It lapsed in 2023 and ran on annual appropriations; a 2026 executive-order turn shifted US policy toward industrial policy and equity stakes (see Ch. 6).

Native gate set — The small set of gates a given hardware platform physically performs (CZ or iSWAP on superconducting chips, Mølmer–Sørensen on ions, Rydberg CZ on atoms). Compilers rewrite everything into it (see Ch. 3).

Neutral atom — A qubit modality using neutral atoms held in movable laser "tweezers," with two-qubit gates driven by exciting atoms to large Rydberg states. The fastest-moving modality of 2024–26 and holder of the logical-qubit record; gate fidelities still trail ions (see Ch. 2).

NISQ (Noisy Intermediate-Scale Quantum) — Preskill's 2017 name for today's machines: tens to a few thousand physical qubits, noisy gates, no full error correction — powerful enough to be hard to simulate, too noisy to run deep useful circuits. No NISQ algorithm has produced a durable practical advantage (see Ch. 3, Ch. 7).

No-cloning theorem — The impossibility of copying an unknown quantum state. It is the root of quantum cryptography, the reason error correction cannot simply back up data, and why long-haul quantum links need repeaters rather than amplifiers (see Ch. 1).

No-communication theorem — The result that entanglement alone cannot send a message faster than light, which is how quantum mechanics coexists with relativity (see Ch. 1).

Normalized (state) — A quantum state scaled so its outcome probabilities add up to exactly one. Every physical state must be normalized, since some outcome has to happen; the Born rule only gives valid probabilities for a normalized state (see Born rule, state vector; Ch. 1).

NV center — A nitrogen-vacancy defect in diamond that acts as a qubit and sensor at room temperature and ambient pressure, with nanoscale resolution. Best near-term value is sensing; qubit counts are the lowest of any modality (see Ch. 2, Ch. 4).

Objective collapse (GRW / CSL / Diósi–Penrose) — A family of theories that add a tiny physical mechanism causing large superpositions to collapse on their own. Unlike other interpretations, they make testable predictions, and experiments are steadily narrowing their allowed range (see measurement problem; Ch. 1, Ch. 8).

Observable — Any physical quantity you can measure — position, energy, spin along an axis. In the formalism each observable is an operator whose eigenvalues are the possible readings; measuring one yields an eigenvalue with a Born-rule probability and leaves the state in the matching eigenstate (see eigenvalue, Born rule; Ch. 1).

Optical clock — The most precise instruments ever built, keeping time on an atomic transition at optical frequencies (uncertainties near 19 decimal places). Heading toward a redefinition of the SI second around 2030 (see Ch. 4, Ch. 6).

Optical tweezers — Tightly focused laser beams that trap and move single neutral atoms, forming the reconfigurable arrays of the neutral-atom modality (see Ch. 2).

OTOC (out-of-time-order correlator) — A quantity measuring how quantum information scrambles through a system. The basis of Google's 2025 "Quantum Echoes" claim of a verifiable quantum advantage — a live test of whether the result survives classical attack (see Ch. 3, Ch. 8).

Overhead (error-correction) — The many physical qubits and extra operations needed per logical qubit. Estimates span two orders of magnitude depending on code and target error rate — from roughly 2:1 in recent demos to hundreds-to-1,000:1 once T-gate factories and full error rates are counted. The whole game (see Ch. 3, Ch. 8).

Paramp / TWPA (parametric / traveling-wave amplifier) — The first, near-noiseless amplifier stage that boosts a superconducting qubit's readout signal while adding the minimum noise physics allows. Without it, fast single-shot readout is impossible; TWPA yield at scale is an open problem (see HEMT; Ch. 2).

Paul trap — The oscillating-electric-field trap that holds ions in place. Modern machines use microfabricated "surface traps" rather than bulk versions (see Ch. 2).

PBR theorem — The Pusey–Barrett–Rudolph result (2012) arguing that the quantum wavefunction is physically real rather than mere ignorance about a deeper variable, given a preparation-independence assumption (see Ch. 1).

Photonic qubit — A qubit encoded in light. Photons barely decohere and travel fiber natively, so networking is easy and much of the machine runs warm — but photons do not interact, so two-qubit gates are probabilistic and photon loss is the dominant error (see Ch. 2).

Post-quantum cryptography (PQC) — Classical encryption software built on math problems believed hard even for quantum computers, running on the machines we already own. The most deployed, most economically important item in the whole adjacent-tech chapter, on a legally mandated timeline (see ML-KEM; Ch. 4, Ch. 6).

POVM — The honest, general description of a real measurement — a set of positive operators that account for loss, noise, and ancillas — rather than an idealized sharp measurement. What dispersive readout and ion fluorescence actually are (see Naimark dilation; Ch. 1).

Probabilistic gate (fusion) — A photonic two-qubit gate that only succeeds part of the time (the leading "fusion" gate about half). The structural challenge of photonic computing, worked around with many attempts and heralding (see Ch. 2).

Pure state vs mixed state — A pure state is a single definite quantum state, described by one state vector. A mixed state is a statistical blend of pure states, arising from noise or ignorance, and needs a density matrix to describe it. Decoherence turns pure states into mixed ones (see density matrix, purification; Ch. 1).

Purification / Stinespring dilation — The principle that every noisy, mixed state is the shadow of a pure state on a larger space, and every physical process is a clean operation on a bigger system followed by discarding part. "Noise is entanglement with something you stopped tracking" (see Ch. 1).

Q-day — The informal name for the day a quantum computer can break today's public-key encryption (see CRQC). Used as a planning horizon; point estimates cluster around 2030 ± 3 years, roughly 50% by 2035, which is why migration starts now (see harvest now decrypt later; Ch. 4, Ch. 6).

QAOA (Quantum Approximate Optimization Algorithm) — A near-term variational algorithm for combinatorial optimization. Provably beaten by classical algorithms on broad instance classes, with no demonstrated win on real problems (see variational algorithms; Ch. 3, Ch. 5).

qLDPC codes — Quantum low-density parity-check codes that protect many more logical qubits per physical qubit than the surface code, at the price of long-range connectivity the hardware must supply.

IBM's bet; whether their connectivity stays below threshold and decodes fast enough is unproven (see Ch. 3, Ch. 8).

QKD (quantum key distribution) — Sharing a secret key whose security rests on physics (an eavesdropper unavoidably disturbs the photons) rather than on math. A real two-decade-old product, but structurally niche: limited by distance, by "trusted relay" nodes that know the key, and by four signals agencies advising post-quantum cryptography instead (see Ch. 4).

QRNG (quantum random number generation) — Hardware that produces true randomness from quantum measurements. The oldest commercial quantum product line, shipping for twenty years (see Ch. 4).

qRAM (quantum RAM) — An assumed device that loads N classical numbers into superposition in about log-N time. The single biggest asterisk on data-heavy quantum advantage: if loading costs O(N), the speedup evaporates. Only tiny 4- and 8-bit demos exist (see Ch. 3).

QSVT (quantum singular value transformation) — A unifying framework that recovers most known quantum algorithms (search, matrix inversion, simulation, phase estimation) as special cases by applying a chosen polynomial to a block-encoded matrix. It sharpens what is provably fast rather than adding new speedups, and inherits the same caveats (see Ch. 3).

Quantum advantage / supremacy — A quantum computer doing a task no classical computer feasibly can. Graded as **contested by default**: it is defined against the best *known* classical method, which keeps improving, and the historical scoreboard favors the classical attackers (Sycamore and IBM's 2023 utility experiment were both matched classically). "Supremacy" usually means an artificial benchmark; "advantage" a useful task (see Ch. 3, Ch. 8).

Quantum Flagship — The EU's €1B, ten-year quantum research program launched in 2018, later joined by EuroQCI, EuroHPC procurements, and a 2025 Quantum Europe Strategy (see Ch. 6).

Quantum internet — A network that distributes entanglement itself, removing the trusted-relay weakness of QKD. It needs quantum repeaters and has not yet reached the watershed where a repeater beats direct transmission — a decade-plus infrastructure bet with no paying application yet (see Ch. 4).

Quantum kernel — A method that encodes data into quantum states and uses their overlaps as a similarity measure for a classical classifier. The clearest quantum machine-learning object; on classical data the record is negative, matched or beaten by ordinary classical methods (see Ch. 3).

Quantum machine learning (QML) — Using quantum circuits for learning tasks. The most hype-inflated corner of the stack: on classical data there is no evidence of practical advantage (repeatedly dequantized); on genuinely quantum data a real but niche advantage exists, awaiting hardware (see Ch. 3, Ch. 5).

Quantum phase estimation (QPE) — The core subroutine that reads an eigenvalue (a "phase") to high precision, powering Shor's algorithm and quantum chemistry. Its deep circuits put it firmly in the fault-tolerant era, and it hides a state-preparation cost (see QFT; Ch. 3).

Quantum Fourier transform (QFT) — The quantum version of the Fourier transform, using far fewer gates than the classical FFT. On its own it is not a speedup (you cannot read the result out); it becomes powerful as the engine inside phase estimation and Shor (see Ch. 3).

Quantum simulation — See Hamiltonian simulation. Feynman's original 1981 pitch and the best-founded advantage in the field (see Ch. 3, Ch. 5).

Quantum supremacy — See quantum advantage. The older term for a quantum computer doing a task no classical computer feasibly can, now largely deprecated in favor of "quantum advantage"; where it survives it usually flags an artificial benchmark rather than a useful task. Contested by default (see Ch. 3, Ch. 8).

Quantum Volume — An IBM whole-machine benchmark combining qubit count, connectivity, and fidelity into one number. Like all vendor metrics, it flatters its own architecture; T4 (see Ch. 3, Ch. 8).

Quantum walk — The quantum analog of a random walk, giving speedups for problems like element distinctness and serving as a full model of computation. Its speedups are query-model and polynomial, so overhead accounting applies (see Ch. 3).

Qubit — The basic unit of quantum information: any two-level quantum system that can hold a superposition of 0 and 1. Every hardware modality implements the same abstract qubit, so algorithms are written once (see Bloch sphere; Ch. 1).

QUBO (quadratic unconstrained binary optimization) — A standard way of writing an optimization problem as binary variables with a quadratic cost function. It is the native input format for quantum annealers and many optimization pitches, since a QUBO maps directly onto a physical energy landscape (see annealing, Hamiltonian; Ch. 2, Ch. 5).

QuOp (quantum operation) — A proposed unit of quantum computing throughput used in UK procurement targets (MQuOp, GQuOp, TQuOp). It is not standardized, so buyers are writing contracts against an undefined quantity (see Ch. 6).

Random circuit sampling (RCS) — Running a random circuit and sampling its output, the artificial task behind "supremacy" claims. Chosen because it is hard classically, not because it is useful (see cross-entropy benchmarking; Ch. 7, Ch. 8).

Readout — Measuring a qubit's state, fast and accurately, at the end of a computation. A finite-fidelity POVM in practice, and one of the DiVincenzo criteria (see Ch. 1, Ch. 2).

Repeater (quantum repeater) — A device that extends entanglement over long distances by storing it in short segments and fusing them, since amplification is forbidden by no-cloning. The missing piece of the quantum internet; none yet beats direct transmission (see Ch. 4).

Rydberg state / Rydberg blockade — A highly excited atomic state with a huge dipole, so one excited atom prevents its neighbors from being excited too — the "blockade" that drives neutral-atom two-qubit gates. The same big dipole makes Rydberg atoms sensitive RF sensors (see Ch. 2, Ch. 4).

Satellite QKD — Distributing quantum keys through space, where loss beats fiber over continental distances. Demonstrated and near-commercial, led decisively by China (Micius, Jinan-1). A single satellite acting as relay still knows the key (see Ch. 4).

Shor's algorithm — Peter Shor's 1994 algorithm that factors large numbers and computes discrete logarithms in polynomial time, breaking RSA and elliptic-curve cryptography. No cryptographically relevant number has been factored yet; the live action is falling resource estimates (now under ~1M noisy qubits for RSA-2048), which make post-quantum migration urgent today (see Ch. 3, Ch. 7).

Silicon spin qubit — A qubit stored in the spin of a single electron in silicon, roughly a million times smaller than a transmon. The manufacturing bet — the semiconductor industry as quantum fab. Crossed into error-correction-grade fidelity in 2025 but has very few qubits so far (see Ch. 2).

SNSPD — Superconducting nanowire single-photon detectors, now exceeding 98% efficiency with picosecond timing. Essential to photonic computing and QKD (see Ch. 2, Ch. 4).

Solovay–Kitaev theorem — The guarantee that any single-qubit operation can be approximated efficiently from a finite universal gate set, with only modest (polylogarithmic) overhead. It makes universality practical (see Ch. 3).

Squeezing / squeezed light — Light engineered to have less noise in one property at the cost of more in the conjugate one, pushing a measurement below the standard quantum limit. LIGO injects squeezed vacuum on every observing run — a quantum resource doing paid work today (see uncertainty principle; Ch. 1, Ch. 4).

Stabilizer formalism / stabilizer code — A compact way to describe an important class of states and every mainstream error-correcting code by the operators that leave them unchanged, rather than by all their amplitudes. It draws the sharpest line between classically simulable (Clifford) circuits and the "magic" that makes quantum computing hard (see Gottesman–Knill; Ch. 1, Ch. 3).

Standard quantum limit (SQL) — The noise floor a conventional measurement hits, set by the uncertainty principle, with precision improving as √N. Squeezing and entanglement push below it toward the Heisenberg limit (see Ch. 1, Ch. 4).

State vector — The mathematical description of a pure quantum state, living in Hilbert space. The starting axiom of the whole formalism (see superposition; Ch. 1).

Strong subadditivity — The deepest structural law of quantum information (Lieb–Ruskai, 1973), from which the data-processing inequality and the monogamy of entanglement follow (see Ch. 1).

Superconducting qubit — A qubit made from a superconducting circuit built around a Josephson junction, operated at millikelvin. The most industrially mature gate-based modality (IBM, Google), fast but wiring-limited (see transmon; Ch. 2).

Superposition — A quantum system being in a combination of states at once, with complex weights that can interfere. The one axiom the whole field rests on; an n-qubit register in superposition holds 2^n amplitudes at once (see Ch. 1).

Surface code — The default error-correcting code: a 2D nearest-neighbor grid of qubits that has been the workhorse of fault-tolerance demos. Its weakness is a poor encoding rate — one logical qubit costs many physical qubits (see qLDPC; Ch. 3).

Swap test — A circuit that estimates how similar two quantum states are by measuring one ancilla qubit. Used with the Hadamard test in the atlas's reference implementation (see Ch. 3).

Syndrome (measurement) — The error-detection readout in error correction: measuring certain operators reveals *that* an error happened and where, without disturbing the protected information. A large machine produces a torrent of syndrome data a decoder must keep up with (see decoder; Ch. 3).

T gate / T-count — The single non-Clifford gate that makes a gate set universal, and the true currency of fault-tolerant cost. A circuit's T-count and T-depth — not its raw gate count — decide whether it is affordable, because T gates need expensive magic states (see Clifford+T; Ch. 3).

TAM (total addressable market) — The full revenue a product could earn if it captured its entire market. Quantum market reports headline large TAM figures years out; a TAM is a forecast of potential demand, not booked revenue, and reads as T5 (see CAGR; Ch. 6, Ch. 8).

Teleportation — Moving an unknown quantum state from one place to another using a shared entangled pair and two classical bits, destroying the original. It never beats light and never clones. The same trick ("gate teleportation") is how magic states are consumed in fault-tolerant circuits (see Ch. 1).

Tensor network — A classical method that represents a quantum state or circuit as a graph of small tensors, efficient when entanglement is low. The moving classical baseline every advantage claim is judged against — it erased Sycamore's "10,000 years" and matched IBM's 2023 utility experiment (see Ch. 3, Ch. 8).

Threshold theorem — The foundational result that if the physical error rate sits below a code-dependent threshold, arbitrarily long computation is possible with manageable overhead. Below threshold, adding qubits helps; above it, adding qubits hurts. The license under which all error correction operates (see below-threshold; Ch. 3, Ch. 7).

Time-bin — A photonic encoding that stores a qubit in whether a photon arrives in an early or late time slot (see Ch. 2).

TLS (two-level system) — Atomic-scale defects in the oxides and interfaces of a chip that absorb qubit energy and drift over hours, capping and destabilizing coherence. The materials floor under superconducting qubits, and whether it can be suppressed enough at scale is an open question (see Ch. 2, Ch. 8).

Topological qubit — A qubit that builds error protection into its physics by storing information non-locally (see Majorana zero mode). The highest-risk, highest-reward modality; no independently verified topological qubit exists after two decades (see Ch. 2, Ch. 8).

Transduction (microwave-to-optical) — Converting a single microwave photon (superconducting qubits' language) into an optical one and back, to link fridges over fiber. The best 2025 devices reach only modest efficiency; until it improves, superconducting machines stay confined to a single fridge (see interconnect; Ch. 2).

Transmon — The dominant superconducting qubit design: a Josephson junction shunted by a large capacitor to reduce charge-noise sensitivity. The workhorse of IBM, Google, Rigetti, and IQM (see Ch. 2).

Trapped ion — A qubit modality using single atomic ions held in electromagnetic traps, addressed by lasers. Every qubit is identical by physics, giving the best gate fidelities and coherence, at the cost of slower gates and a scaling limit per trap (see Ch. 2).

Trotter–Suzuki / qDRIFT / LCU / qubitization — The four method families for digital Hamiltonian simulation. Trotter is simple and ancilla-free; qDRIFT's cost is independent of the number of terms; LCU and qubitization reach optimal precision scaling. Which wins is instance-dependent (see Ch. 3).

Trusted relay / trusted node — An intermediate node in a QKD network that decrypts the key on one link and re-encrypts it on the next, because no quantum repeater yet exists to pass entanglement through untouched. The relay holds the key in the clear, so anyone who controls the node — or compromises it — sees the secret. Real QKD networks are chains of these nodes, and a satellite acting as relay is one too. It is the structural hole that collapses QKD's "unhackable" marketing into security only as strong as every node along the path, and part of why security agencies point to post-quantum cryptography instead (see quantum internet, repeater; Ch. 4).

Tsirelson bound — The maximum Bell-inequality value quantum mechanics itself allows ($2\sqrt{2}$ for CHSH), sitting above the classical limit of 2. The gap between them is the experimental target that six decades of tests have closed (see Ch. 1).

Tunneling (quantum tunneling) — A particle passing through a barrier it could not classically climb, because its wavefunction extends past the barrier. The mechanism inside the Josephson junction, the scanning tunneling microscope, and any advantage annealing gets over thermal hopping (see Ch. 1).

Twin-field QKD — A QKD protocol that extends secure key distribution over unusually long fiber (hundreds of kilometers) by interfering fields from both parties at a midpoint (see Ch. 4).

Uncertainty principle — The rule that certain pairs of quantities (position and momentum) cannot both be sharp in any state — a fact about how states can be prepared, not about clumsy measurement. It sets the standard quantum limit and defines what squeezing can buy (see Ch. 1).

Unitary (operation) — The kind of transformation that describes every closed-system quantum evolution and every quantum gate: reversible, and preserving total probability so a normalized state stays normalized. Measurement is the one non-unitary step. Block encoding exists precisely because a quantum computer can only apply unitaries directly (see normalized, gate; Ch. 1, Ch. 3).

Universal gate set — A finite set of gates that can approximate any quantum operation. Universality guarantees *reachability*, not cheapness — a circuit can be expressible yet astronomically expensive in T-count (see Clifford+T, Solovay–Kitaev; Ch. 3).

Utility (quantum utility) — IBM's 2023 term for a noisy processor producing reliable results beyond brute-force classical simulation, explicitly set below the "advantage" bar. Contested, since the 2023 utility problem was reproduced classically within days (see Ch. 8).

Variational algorithms / VQE — Near-term algorithms (VQE for chemistry, QAOA for optimization) that pair a shallow tunable circuit with a classical optimizer, designed for NISQ hardware. Undermined by barren plateaus, soft dequantization, and strong classical competition; no win has survived a classical response. Their surviving role is as a state-preparation subroutine (see Ch. 3).

Von Neumann entropy — The quantum version of Shannon entropy, measuring the uncertainty or mixedness of a quantum state and the amount of entanglement in a pure bipartite state. The foundation of quantum information theory (see Ch. 1).

Weak measurement — A gentle measurement that extracts a little information while barely disturbing the state, letting the moment-by-moment path of a monitored qubit be reconstructed (see Ch. 1).

Wavefunction — The full quantum description of a system. Whether it is physically real or a bookkeeping tool is part of the unresolved interpretation debate (see PBR theorem, measurement problem; Ch. 1).

Wigner function / negativity — A way to picture a quantum state of light as a phase-space distribution. Its ability to go *negative* is the tell of nonclassicality; a Wigner-negative element is what lifts photonic hardware to universality (see Gaussian states; Ch. 1).

Zero-noise extrapolation (ZNE) / probabilistic error cancellation (PEC) — The two workhorse error-mitigation methods. ZNE deliberately amplifies noise, measures the result at several noise levels, and extrapolates back to zero; PEC learns a noise model and inverts it by sampling. Both extend reach without protecting the state, and both cost grows exponentially with depth (see error mitigation; Ch. 3).

APPENDIX E

Math primer

Prerequisites: comfort with vectors, matrices, and matrix multiplication over the real numbers, plus basic probability. Everything specific to quantum — complex amplitudes, bra-ket notation, unitaries, density matrices — is built here from that starting point.

This appendix gives the linear algebra over the complex numbers $\mathbb{C}$ that you actually need to read Chapter 1 and the rest of the atlas. It is deliberately narrow. The aim is that when a later page writes $|\psi\rangle$, $\langle\phi|\psi\rangle$, $U^\dagger U = I$, or ρ, you know exactly what each symbol means and can compute with it.

Complex amplitudes and the unit circle

A complex number is $z = a + bi$ with a, b real and $i^2 = -1$. Its **conjugate** is $z^* = a - bi$, and its **magnitude** is $|z| = \sqrt{z^* z} = \sqrt{a^2 + b^2}$. Every complex number can be written in polar form:

$$z = re^{i\theta} = r(\cos\theta + i\sin\theta), \qquad r = |z|.$$

The numbers with $r = 1$ — the ones of the form $e^{i\theta}$ — lie on the **unit circle** in the complex plane. They are called **phases**. A phase has magnitude 1, so multiplying by $e^{i\theta}$ rotates a number without changing its length.

This matters because a quantum amplitude is a complex number, and two facts about it come up constantly. First, only $|z|^2$ shows up in measurement probabilities, so the phase is invisible to a single measurement. Second, phases are exactly what interference manipulates — quantum algorithms arrange for unwanted amplitudes to have opposite phases so they cancel. Hold both ideas: the phase is hidden from one look, and it is the whole game across many.

The state vector and normalization

A quantum state is a **column vector of complex amplitudes**, one entry per outcome the system could show when measured. A single qubit has two outcomes, labeled 0 and 1, so its state is a length-2 complex vector:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \alpha|0\rangle + \beta|1\rangle, \qquad |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The amplitudes obey one constraint, **normalization**:

$$|\alpha|^2 + |\beta|^2 = 1.$$

The reason is coming in the Born rule: $|\alpha|^2$ and $|\beta|^2$ are probabilities, and probabilities of the exhaustive outcomes sum to 1. Geometrically, the state is a **unit vector**. Physical operations keep it a unit vector, which is why they turn out to be a restricted class of matrices.

Bra-ket notation

Bra-ket (Dirac) notation is bookkeeping for column and row vectors and their products. A **ket** $|\psi\rangle$ is a column vector — the state. A **bra** $\langle\psi|$ is its conjugate-transpose, a row vector:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \implies \langle\psi| = \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix}.$$

Put a bra next to a ket and you multiply a row by a column, which gives a single complex number — the **inner product**:

$$\langle\phi|\psi\rangle = \begin{pmatrix} \gamma^* & \delta^* \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \gamma^*\alpha + \delta^*\beta.$$

The inner product measures overlap. Some facts to keep:

- $\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1$ for a normalized state — a state fully overlaps itself.
- $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$ — swapping the order conjugates the value.
- If $\langle\phi|\psi\rangle = 0$ the states are **orthogonal**: perfectly distinguishable in one measurement. $|0\rangle$ and $|1\rangle$ are orthogonal.

The inner product $\langle\phi|\psi\rangle$ is the cosine-similarity of the two state vectors, and the reference implementation in `reference-impl/` is built entirely on estimating it. A worked example: for $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|\phi\rangle = |0\rangle$, the overlap is $\langle\phi|\psi\rangle = \frac{1}{\sqrt{2}}$.

The Born rule

The **Born rule** connects the vector to what a measurement shows. Measure $|\psi\rangle$ in the computational basis $\{|i\rangle\}$, and outcome i appears with probability

$$P(i) = |\langle i|\psi\rangle|^2.$$

The bra $\langle i|$ picks out the i-th amplitude, so $\langle i|\psi\rangle$ is that amplitude and $|\langle i|\psi\rangle|^2$ is its squared magnitude. For the qubit, $P(0) = |\alpha|^2$ and $P(1) = |\beta|^2$, and normalization guarantees they sum to 1. You never read an amplitude directly; you estimate these probabilities by repeating the preparation many times — the repeats are called **shots** — and the phase of each amplitude drops out of any single measurement.

Tensor products for multi-qubit states

Two qubits together are described by the **tensor product** (Kronecker product) of the single-qubit spaces. If register A is in $|a\rangle$ and register B in $|b\rangle$, the joint state is $|a\rangle \otimes |b\rangle$, often written $|a\rangle|b\rangle$ or $|ab\rangle$. Concretely,

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} \otimes \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} \alpha_0\beta_0 \\ \alpha_0\beta_1 \\ \alpha_1\beta_0 \\ \alpha_1\beta_1 \end{pmatrix}.$$

So n qubits live in a space of dimension 2^n, with basis the 2^n bit strings $|00\ldots0\rangle, \ldots, |11\ldots1\rangle$. A general n-qubit state is

$$|\psi\rangle = \sum_{i=0}^{2^n-1} c_i\,|i\rangle, \qquad \sum_i |c_i|^2 = 1.$$

That exponential dimension is the resource quantum computing spends. A state that **cannot** be written as a single tensor product of individual qubit states is **entangled** — the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is the standard example. Entanglement is the subject of Chapter 1's central sections.

Unitary operators and why gates are unitary

An operation on a state is a matrix U acting as $|\psi\rangle \mapsto U|\psi\rangle$. Physical operations must send unit vectors to unit vectors, and the matrices that preserve length and inner products are exactly the **unitary** matrices, defined by

$$U^\dagger U = UU^\dagger = I,$$

where $U^\dagger$ is the conjugate-transpose. Unitarity has two consequences worth stating plainly. Length is preserved, so a normalized state stays normalized. Overlaps are preserved, $\langle U\phi|U\psi\rangle = \langle\phi|\psi\rangle$, so a unitary never makes two distinct states more or less distinguishable. And because $U^\dagger$ is also unitary and undoes U, every gate is **reversible** — a hard constraint on how quantum circuits are built. The one-qubit **Hadamard** gate,

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

builds an equal superposition and is its own inverse ($H^2 = I$). Nearly every interference trick in the atlas is H, a controlled operation, then H again.

The Pauli matrices and the Bloch sphere

Three 2×2 unitaries, the **Pauli matrices**, generate single-qubit dynamics:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

X is the bit flip ($X|0\rangle = |1\rangle$), Z is the phase flip ($Z|1\rangle = -|1\rangle$), and $Y = iXZ$. Each is Hermitian and unitary, and each squares to I.

Any single-qubit pure state can be parameterized by two angles,

$$|\psi\rangle = \cos\tfrac{\theta}{2}\,|0\rangle + e^{i\varphi}\sin\tfrac{\theta}{2}\,|1\rangle,$$

and mapped to the point $(\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta)$ on a unit sphere — the **Bloch sphere**. The north pole is $|0\rangle$, the south pole $|1\rangle$, and the equator holds the equal superpositions. Rotations about the x, y, z axes are generated by the Pauli matrices via $R_k(\theta) = e^{-i\theta k/2}$, so single-qubit gates are literally rotations of this sphere. The global phase — an overall $e^{i\gamma}$ on the whole state — moves no Bloch point and is physically undetectable, which is the geometric reason phase-only differences vanish under measurement.

Density matrices

A state vector describes a system you know completely. When you have a statistical mixture, or you are holding only part of an entangled pair, you need the **density matrix**. For a pure state it is the outer product

$$\rho = |\psi\rangle\langle\psi|,$$

a $d \times d$ matrix that is Hermitian ($\rho = \rho^\dagger$), positive semidefinite, and has trace 1. A **mixed** state — outcome $|\psi_k\rangle$ with classical probability p_k — is the weighted sum

$$\rho = \sum_k p_k \, |\psi_k\rangle\langle\psi_k|.$$

The clean test that separates the two cases is the **purity** $\mathrm{Tr}(\rho^2)$: it equals 1 for a pure state and is strictly less than 1 for a mixed one. Measurement probabilities read out as $P(i) = \langle i|\rho|i\rangle$, and expectation values as $\langle A\rangle = \mathrm{Tr}(\rho A)$.

The **partial trace** extracts the state of a subsystem. Given a joint ρ_{AB}, tracing out B gives $\rho_A = \mathrm{Tr}_B(\rho_{AB})$, which is the correct description of A alone. The key phenomenon: trace one qubit out of an entangled pure state and the remaining single-qubit ρ_A is **mixed**. Entanglement with an environment you cannot see is exactly what makes a subsystem look noisy — the mechanism behind decoherence, covered in Chapter 1.

Open systems: the Lindblad master equation

A closed quantum system evolves reversibly by the Schrödinger equation, equivalently $\dot\rho = -\frac{i}{\hbar}[H,\rho]$ for the density matrix. Real devices are **open**: they leak information to an environment, so their evolution is not reversible and cannot be a plain unitary. The **Lindblad master equation** is the standard description of that leak. It adds dissipative terms, built from **jump operators** L_k that encode each noise channel (energy loss, dephasing, and so on), to the reversible part:

$$\dot\rho = -\frac{i}{\hbar}[H,\rho] + \sum_k \gamma_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \Big).$$

The first term is the reversible Hamiltonian evolution; the sum is the irreversible drift toward a mixed state at rates γ_k. This is the equation that predicts T_1 (energy relaxation) and T_2 (dephasing) times, sets how fast a stored qubit degrades, and underwrites the noise models and error budgets you will run in the Lab track. When Chapter 2 quotes coherence times, this is the equation behind the numbers.

Back to Chapter 1 · Foundations, which builds the physics — superposition, the Born rule, entanglement and the Bell/CHSH bound, no-cloning, and decoherence — on top of exactly this linear algebra.

APPENDIX F

Lab track

A graded, runnable lab sequence. Every lab is built on the reference implementation shipped in `reference-impl/` — a hardware-validated quantum similarity-search project (swap test, Hadamard test, quantum kernels, error mitigation, real IBM runs). The labs move from a two-line physics demonstration to the project's own error-budget pipeline.

Read `reference-impl/README.md` and `reference-impl/MATH.md` first; the code reads like a transcription of the equations. Every lab except the paper exercise runs on a **classical simulator** and submits nothing to a real device. Real-QPU runs exist only in the last lab and are gated behind an explicit `--run` flag and saved IBM credentials.

Setup (once)

```
cd reference-impl
python3 -m venv .venv && . .venv/bin/activate   # skip the create step if .venv exists
pip install -r requirements.txt                  # qiskit 2.5, qiskit-aer, numpy, scikit-learn
```

Keep this virtual environment active for every lab. Labs 1–2 are short scripts you write yourself and run on the same environment; Labs 4–6 run modules that ship in `src/` and `tests/`.

Lab 1 — CHSH / Bell violation (simulator only)

Objective. Measure the CHSH correlation of an entangled pair and confirm it exceeds the classical bound of 2, approaching the quantum Tsirelson bound $2\sqrt{2} \approx 2.828$.

Background. A local hidden-variable theory obeys the CHSH inequality $|S| \leq 2$, where S combines four correlation measurements at two settings per party. Quantum mechanics violates it: a Bell pair measured at the right angles reaches $2\sqrt{2}$. This is the experiment that rules out local realism, and it is the T1 "settled physics" anchor for the whole atlas — see `01-foundations/F-bell.md` and the Tsirelson-bound objective in `01-foundations/_CHAPTER.md`.

Run. This script is not part of `src/` — you write it and run it on the lab venv. Save as `chsh.py` in `reference-impl/`, or paste after `python - <<'PY'`.

```
import numpy as np
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
sim = AerSimulator()
def E(a, b, shots=20000):
    qc = QuantumCircuit(2, 2)
    qc.h(0); qc.cx(0, 1)                 # Bell pair
    qc.ry(-2*a, 0); qc.ry(-2*b, 1)       # measurement-angle rotations
    qc.measure([0, 1], [0, 1])
    c = sim.run(qc, shots=shots).result().get_counts()
    return sum((-1)**(int(k[0]) + int(k[1])) * v for k, v in c.items()) / shots
a, ap = 0, np.pi/4
b, bp = np.pi/8, 3*np.pi/8
S = E(a, b) - E(a, bp) + E(ap, b) + E(ap, bp)
print("CHSH S =", round(S, 3), " (classical <= 2, Tsirelson 2.828)")
```

Observe. `S` lands near `2.82`, comfortably above `2`. Each `E(a,b)` is a correlation in $[-1, +1]$.

Questions. (1) Set both measurement angles to the same value everywhere (a product state instead of the tuned Bell angles) and confirm `S` drops to ≤ 2. (2) The result sits just under 2.828, not exactly at it. Which effect from the Math primer's Born-rule section accounts for the gap, and how would you shrink it?

Lab 2 — Grover search on 3 qubits (simulator only)

Objective. Amplify a single marked 3-bit string with Grover's algorithm and read it out with high probability after the optimal number of iterations.

Background. Grover search finds a marked item among $N = 2^n$ in $O(\sqrt{N})$ oracle calls, a quadratic speedup over classical $O(N)$. Each iteration is an oracle that phase-flips the marked state followed by a diffusion (inversion-about-the-mean) step. For $n = 3$ the optimal count is $\lfloor \frac{\pi}{4}\sqrt{8} \rfloor = 2$. Background: `03-stack-algorithms/S-grover.md`.

Run. Also a build-it-yourself script on the lab venv (target string `101`):

```
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
sim = AerSimulator(); n = 3; target = '101'
qc = QuantumCircuit(n, n); qc.h(range(n))
def oracle(qc):
    for i, b in enumerate(target):
        if b == '0': qc.x(i)
    qc.h(2); qc.ccx(0, 1, 2); qc.h(2)        # phase-flip |111>
    for i, b in enumerate(target):
        if b == '0': qc.x(i)
def diffuser(qc):
    qc.h(range(n)); qc.x(range(n))
    qc.h(2); qc.ccx(0, 1, 2); qc.h(2)
    qc.x(range(n)); qc.h(range(n))
for _ in range(2):                            # optimal iteration count for n=3
    oracle(qc); diffuser(qc)
qc.measure(range(n), range(n))
c = sim.run(qc, shots=4096).result().get_counts()
top = max(c, key=c.get)
print("top outcome:", top[::-1], " count", c[top], "of 4096")
```

Observe. The marked string `101` dominates — roughly 95% of shots (about 3900 of 4096).

Questions. (1) Change the loop to 1 iteration, then 3, and record the success probability each time. Why does 3 iterations do *worse* than 2? (2) Estimate how many oracle calls a classical search would need on average for $N = 8$, and compare to Grover's 2.

Lab 3 — Surface-code overhead (paper exercise + one-liner)

Objective. Count the physical qubits a rotated surface code needs to protect one logical qubit at a target logical error rate, and feel the overhead that dominates fault-tolerance planning.

Background. The rotated surface code of **distance** d uses d^2 data qubits plus $d^2 - 1$ measurement (ancilla) qubits, so $2d^2 - 1$ physical qubits per logical qubit. Below the threshold p_{th} (about 1% for the surface code), the logical error rate falls roughly as $p_L \approx A\,(p/p_{\text{th}})^{(d+1)/2}$. Higher distance buys exponentially lower error at quadratic qubit cost. Context: `03-stack-algorithms/S-decoders.md` and the QEC sections of `03-stack-algorithms/_CHAPTER.md`. The reference impl carries no QEC calculator, so this lab is paper plus a tiny arithmetic snippet.

Run. Solve on paper for the distance, then check the count:

```
import math
p, p_th, A, target = 1e-3, 1e-2, 0.1, 1e-12   # physical err, threshold, prefactor, goal
d = 1
while A * (p / p_th) ** ((d + 1) / 2) > target:
    d += 2                                      # surface-code distance is odd
phys = 2 * d**2 - 1
print(f"need distance d={d}: {phys} physical qubits per logical qubit")
```

Observe. With $p = 10^{-3}$ you reach 10^{-12} around $d = 11$–13, i.e. a few hundred physical qubits for one logical qubit — before any algorithm qubits.

Questions. (1) Recompute for a worse device, $p = 5 \times 10^{-3}$. How does the required distance and qubit count move? (2) A useful algorithm might need 1,000 logical qubits. Multiply through and compare your total to the physical-qubit counts quoted in Chapter 2's hardware cards.

Lab 4 — Cosine similarity via the swap and Hadamard tests (simulator only)

Objective. Reproduce classical cosine similarity, and a full kernel (Gram) matrix, from quantum overlap measurements. This is the core of the reference project.

Background. Amplitude-encode a vector x into $|\psi_x\rangle = \frac{1}{\|x\|}\sum_i x_i|i\rangle$; then the overlap of two encoded states equals the cosine similarity of the originals, $\langle\psi_u|\psi_v\rangle = \cos(u, v)$. The **swap test** estimates $|\cos|^2$; the **Hadamard test** estimates the signed $\cos$ directly. Stacking pairwise values gives a quantum kernel matrix. Full derivation: `reference-impl/MATH.md` §3–6, with code in `src/encode.py`, `src/swap_test.py`, `src/hadamard_test.py`, `src/kernel.py`.

Run.

```
python tests/test_estimators.py     # correctness proof: quantum == classical
python -m src.experiment            # single-pair + 5x5 kernel demo on the noiseless sim
python -m src.qsvm                  # quantum-kernel SVM on Iris, vs the exact kernel
```

Observe. `test_estimators.py` prints `ALL TESTS PASSED` (estimators match `numpy` to within shot tolerance; the kernel diagonal is exactly 1). `src.experiment` prints the classical, swap, and Hadamard estimates side by side with a $|\mathrm{err}|$ column near 0.01. `src.qsvm` writes `results/qsvm.json` and reports the quantum-kernel SVM matching the classical kernel's accuracy.

Questions. (1) In `MATH.md` §5 the Hadamard test uses $U = \mathrm{prep}_u^\dagger\, \mathrm{prep}_v$ in that order. What does `test_hadamard_matches_signed_cosine` catch if you reverse it? (2) The swap test returns only $|\cos|$ while the Hadamard test returns the sign too. Which pairs in the demo make that difference visible?

Lab 5 — Shot-noise $1/\sqrt{S}$ scaling (simulator only)

Objective. Show that a perfect (noiseless) device still has statistical error, and that it falls as $1/\sqrt{S}$ in the shot count S.

Background. Every quantum estimate reads probabilities off a finite number of measurement repetitions. A probability estimated from S Bernoulli trials has standard error $\sim \sqrt{P(1-P)/S}$, so the cosine error scales as $1/\sqrt{S}$ — halving the error costs $4\times$ the shots. This is the noiseless floor before any hardware imperfection enters. Code: `src/studies.py` (`shot_scaling`), theory in `MATH.md` §4.

Run.

```
python -m src.studies       # writes results/studies.json + results/shot_scaling.png
```

Observe. The console lists `mean|err|` for shots from 128 to 16384; it roughly halves each time shots go up $4\times$. `results/shot_scaling.png` plots the measured error against the $1/\sqrt{S}$ reference line on log-log axes — the two should track closely.

Questions. (1) From `studies.json`, take the errors at $S = 1024$ and $S = 4096$ and check the ratio is near 2. Does it match $1/\sqrt{S}$? (2) You need cosine accuracy 0.005. Extrapolate the shot budget. Why can shots alone never fix *hardware* noise (see Lab 6)?

Lab 6 — Readout calibration + ZNE, and the hardware gate (simulator; real QPU gated)

Objective. Turn on a hardware-like noise model, then recover accuracy with two error-mitigation levers — readout-error calibration and zero-noise extrapolation (ZNE) — and quantify the improvement as a number. Then see how a real-hardware run is staged without submitting one.

Background. Real qubits decohere and gates misfire (the open-system decay of the Math primer's Lindblad section). Two mitigations help: **readout calibration** inverts the measured confusion matrix of the ancilla, and **ZNE** deliberately amplifies gate noise by unitary folding then extrapolates back to zero. The reference impl runs a five-condition error budget (noiseless floor, raw noisy, +readout cal, +ZNE, +both) and reports the mean cosine error for each. Code: `src/mitigation.py`, `src/error_budget.py`, `src/noise_models.py`; deep-dive in `writeup/error-mitigation.md`.

Run.

```
python tests/test_mitigation.py     # proves each primitive in isolation (readout, folding, extrapolation)
python -m src.error_budget          # 5-condition budget -> results/error_budget.{json,png}
```

Real-hardware preview — the safe, QPU-free preflight (transpiles and predicts, submits nothing):

```
python -m src.angle_sweep --backend ibm --check    # free preflight: auth + transpile, NO job
# python -m src.angle_sweep --backend ibm --run     # ONLY this submits a real job (needs saved IBM creds)
```

Observe. `test_mitigation.py` prints `ALL MITIGATION TESTS PASSED`. `src.error_budget` shows the mean $|\cos\text{ err}|$ dropping from raw-noisy toward the noiseless floor as mitigations stack — roughly a 77% reduction with both — and writes `results/error_budget.png`. The `--check` path names the least-busy device, transpiles the sweep circuits, reports the two-qubit-gate cost, and submits no job. Without `--run` and saved credentials, nothing reaches a real device.

Questions. (1) In the `error_budget.json` `reductions_vs_raw` block, which single lever removes more error — readout calibration or ZNE — and does combining them beat either alone? (2) The `--check` preflight reports two-qubit-gate counts for `dim=2` versus `dim=4`. Why does the low-depth `dim=2` encoding give a cleaner first hardware result (see the rationale in `src/angle_sweep.py`)?

What ran where

LAB	MAPS TO	HARDWARE
1 · CHSH	you write `chsh.py`; runs on the lab venv (`qiskit-aer`)	simulator only
2 · Grover	you write the script; runs on the lab venv	simulator only
3 · Surface-code overhead	paper + arithmetic snippet; theory in `03-stack-algorithms/`	none (paper)
4 · Cosine / swap / Hadamard	`tests/test_estimators.py`, `src/experiment.py`, `src/qsvm.py`	simulator only
5 · Shot-noise scaling	`src/studies.py` → `results/shot_scaling.png`	simulator only
6 · Mitigation + gate	`tests/test_mitigation.py`, `src/error_budget.py`, `src/angle_sweep.py`	simulator; real QPU behind `--run`

Labs 1–3 are self-contained (CHSH and Grover as short scripts on the lab venv; the surface-code lab is arithmetic). Labs 4–6 exercise modules that ship in `reference-impl/src/` and `reference-impl/tests/`. Only Lab 6's `--run` path touches a real device, and only with credentials you supply.

APPENDIX G

Node reference index

The 184 graded source nodes behind the chapters — the raw knowledge base.

Chapter 1 · The Physics

Adiabatic Theorem · F-adiabatic

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

If a quantum system starts in an eigenstate of its Hamiltonian and the Hamiltonian is changed slowly enough, the system stays in the corresponding instantaneous eigenstate. "Slowly enough" is set by the spectral gap: the runtime must scale roughly as $1/\mathrm{gap}^2$ (Born & Fock, 1928; rigorous conditions later tightened by Kato and others). If the gap closes or nearly closes, the theorem fails and the system leaks to excited states (Landau-Zener transitions).

CORE IDEA / KEY EQUATION

The whole theorem hangs on one ratio. Call $\triangle$ the smallest energy gap between the eigenstate you are riding and its nearest neighbor over the entire schedule, and let the drive matrix element be roughly how fast the Hamiltonian changes, dH/dt. The system stays put as long as the drive is slow compared to the gap squared — the standard adiabatic condition is $(\text{rate of change of } H)/\Delta^2 \ll 1$, so the total runtime scales as $T \sim 1/\Delta^2$. Halve the gap and you must run four times slower. The failure mode is quantitative and named: at an **avoided crossing** the probability of jumping to the excited state is the Landau-Zener formula $P = e^{-2\pi\Gamma}$, with $\Gamma = \Delta^2/(4\hbar v)$, where $\mathcal{V}$ is the sweep speed through the crossing. Go slow enough and the jump probability is exponentially suppressed; sweep too fast through a tiny gap and you almost certainly leave the ground state. The exponent $\triangle^2$ is the same one that appears in the runtime — the gap is the single number that governs whether adiabatic evolution works.

WHY IT MATTERS FOR QUANTUM TECH

The adiabatic theorem is the load-bearing foundation under **quantum annealing** and **adiabatic quantum computation** (Farhi et al., 2000): encode a hard optimization problem's solution as the ground state of a final Hamiltonian, start in the easy-to-prepare ground state of an initial one, and interpolate slowly. If you stay adiabatic, you end in the answer. Adiabatic QC is polynomially equivalent to the gate model (Aharonov et al., 2004), which makes it a legitimate computing paradigm rather than a heuristic (see H-anneal). The catch is the gap: for hard instances it can shrink exponentially, erasing any speedup — the central open question hanging over D-Wave's machines (see C-dwave-advantage). The same $1/\mathrm{gap}^2$ control appears in quantum simulation, where adiabatic state preparation is a standard way to reach a many-body ground state before measuring it (see S-qsim), and the Landau-Zener physics is exactly what STIRAP and other geometric state-transfer protocols exploit (see F-berry). Slow, gap-limited ramps are also how experimentalists load qubits and cold-atom lattices into their target states across every hardware line (see H-ion, H-supercon).

KEY GRADED CLAIMS

- T1 A system remains in its instantaneous eigenstate under sufficiently slow Hamiltonian change, with error controlled by the inverse spectral gap — Born & Fock, Z. Phys. 51, 165 (1928) (status: established)
- T2 Adiabatic quantum computation is polynomially equivalent to the standard gate model — Aharonov, van Dam, Kempe, Landau, Lloyd, Regev, SIAM J. Comput. 37, 166 (2007); arXiv:quant-ph/0405098 (status: established)
- T2 Coherent quantum annealing demonstrated at scale — quantum critical dynamics in a 5,000+-qubit programmable spin glass, faster than the matched classical annealing model — King et al., Nature 617, 61 (2023) (status: demonstrated, classical-advantage claim contested)

CONFLICTS / OPEN QUESTIONS

- Whether the minimum gap for practically important optimization problems stays large enough for a real speedup is unresolved — it is the crux of the annealing-advantage debate.

GO DEEPER

- Farhi et al., "Quantum Computation by Adiabatic Evolution," arXiv:quant-ph/0001106 (2000)
- Albash & Lidar, "Adiabatic quantum computation," RMP 90, 015002 (2018)

SOURCES

- Born & Fock (1928)
- Aharonov et al., arXiv:quant-ph/0405098
- King et al., Nature 617, 61 (2023). doi:10.1038/s41586-023-05867-2

Bell Inequalities & Their Experimental Violation · F-bell

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Bell (1964) proved that any theory of local hidden variables puts a numerical ceiling on correlations between separated measurements; quantum mechanics predicts violations of that ceiling. The CHSH form (1969) made it testable. Six decades of experiments — Freedman–Clauser (1972), Aspect (1982), and the 2015 loophole-free trio — confirm the quantum prediction. Nature violates local realism.

CORE IDEA / KEY EQUATION

CHSH builds one number from four correlation measurements: $S = E(a,b) - E(a,b') + E(a',b) + E(a',b')$, where $E(x,y)$ is the average product of the ± 1 outcomes when Alice measures setting x and Bob setting y. Any local hidden-variable theory obeys $|S| \leq 2$ — the outcomes are fixed functions of a shared variable, so the four terms cannot all be large at once. Quantum mechanics, using a Bell state and measurement angles 45° apart, reaches $S = 2\sqrt{2} \approx 2.828$, the Tsirelson bound (the maximum quantum mechanics itself allows). The gap between 2 and 2.828 is the experimental target: measure $S > 2$ with the two stations space-like separated and no detection bias, and no local-realist theory can reproduce it. That gap is what six decades of increasingly airtight experiments have nailed down.

WHY IT MATTERS FOR QUANTUM TECH

Bell violation is the operational certificate that a device holds real entanglement (F-entangle) — the basis of device-independent QKD (A-qkd), where security follows from S > 2 alone without trusting the hardware, and of certified quantum randomness (A-sensing, related randomness-expansion protocols), where the violation guarantees the output bits are unpredictable to any adversary. The same self-testing logic lets you verify an untrusted quantum processor. The 2022 Nobel citation names Bell violation as the launch point of quantum information science, and the 2023 superconducting demonstration (below) shows the test now runs on the same chip technology (H-supercon) used to build quantum computers.

KEY GRADED CLAIMS

- T1 Local hidden-variable theories obey $|S| \leq 2$ (CHSH); quantum mechanics reaches $2\sqrt{2} \approx 2.828$ (Tsirelson bound) — Bell, Physics Physique Fizika 1, 195 (1964); Clauser–Horne–Shimony–Holt, PRL 23, 880 (1969) (status: established)
- T1 Bell inequalities are violated experimentally with all major loopholes (locality, detection) closed simultaneously — Hensen et al., Nature 526, 682 (2015); Giustina et al., PRL 115, 250401 (2015); Shalm et al., PRL 115, 250402 (2015) (status: established)
- T1 Loophole-free violation demonstrated with superconducting circuits: two qubits entangled across a 30 m cryogenic link, $S = 2.0747 \pm 0.0033$ over >1 million trials, $p < 10^{-108}$ — Storz et al. (Wallraff group, ETH Zurich), Nature 617, 265 (2023), doi:10.1038/s41586-023-05885-0 (status: established)
- T1 2022 Nobel Prize in Physics to Aspect, Clauser, Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science" — nobelprize.org/prizes/physics/2022 (status: established)

CONFLICTS / OPEN QUESTIONS

- Superdeterminism and retrocausality evade Bell's theorem by denying measurement independence; consistent but widely viewed as unfalsifiable. The freedom-of-choice loophole was pushed back by cosmic-photon and Big Bell Test (Nature 557, 212, 2018) experiments, never fully closable.

GO DEEPER

- Aspect, Dalibard, Roger, PRL 49, 1804 (1982); Brunner et al., "Bell nonlocality," RMP 86, 419 (2014)

SOURCES

- Bell (1964) doi:10.1103/PhysicsPhysiqueFizika.1.195 · CHSH (1969) doi:10.1103/PhysRevLett.23.880
- Hensen et al., Nature 526, 682 (2015). doi:10.1038/nature15759 · Giustina et al. doi:10.1103/PhysRevLett.115.250401 · Shalm et al. doi:10.1103/PhysRevLett.115.250402
- Storz et al., "Loophole-free Bell inequality violation with superconducting circuits," Nature 617, 265 (2023). doi:10.1038/s41586-023-05885-0
- Nobel press release: https://www.nobelprize.org/prizes/physics/2022/press-release/

Geometric / Berry Phase · F-berry

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

When a quantum system is driven adiabatically around a closed loop in parameter space, its state picks up a phase with two parts: the familiar dynamical phase ($\propto$ energy $\times$ time) and a **geometric phase** that depends only on the *path traced* and stays the same however fast or slow the loop is run. Berry (1984) gave the general formula; the phase is the integral of a "Berry connection," and its curvature is the "Berry curvature." Pancharatnam had found the optical version in 1956, and Aharonov-Bohm is a special case. The geometric phase is gauge-invariant and observable.

CORE IDEA / KEY EQUATION

The geometric phase is a loop integral. As the parameters R traverse a closed path C, the state acquires $\gamma = \oint_C \mathbf{A}(R) \cdot d\mathbf{R}$, where $\mathbf{A}(R) = i\langle n(R)|\nabla_R|n(R)\rangle$ is the **Berry connection** — the same object a vector potential is in electromagnetism. By Stokes' theorem this equals a surface integral of the **Berry curvature** $F = \nabla \times \mathbf{A}$ over any cap bounded by the loop: $\gamma = \iint F \cdot d\mathbf{S}$. The clean example is a spin-$\frac{1}{2}$ in a magnetic field swept around a cone: the Berry phase equals exactly half the solid angle the field vector sweeps out, $\gamma = -\Omega/2$, independent of how slowly you go — pure geometry. Two consequences follow. Timing errors cancel because γ carries no dependence on the dynamical phase (energy $\times$ time), which is why geometric gates resist certain pulse fluctuations. And when the parameter space is a material's Brillouin zone, the curvature integrated over a filled band is forced to be 2π **times an integer**, the Chern number — a topological invariant that cannot change under smooth deformation, so the response it controls is quantized to extraordinary precision.

WHY IT MATTERS FOR QUANTUM TECH

Because the geometric phase is set by geometry rather than timing, it is intrinsically resistant to certain control errors — the basis of **holonomic / geometric quantum gates**, where a qubit is steered around a loop to enact a gate that resists fluctuations in pulse duration and amplitude (see H-supercon, H-ion). Berry curvature is also the organizing quantity of modern band theory: it produces the anomalous Hall effect, defines topological invariants (Chern numbers) that classify topological insulators and the quantum Hall effect, and underlies the topological protection sought for qubits (see H-topo, A-materials, F-statistics). Non-abelian generalizations of the Berry phase — matrix-valued holonomies — are the mathematical content of braiding non-abelian anyons, so this node sits directly under topological quantum computing (see H-topo). The same curvature integrated over a Brillouin zone is what makes a material's response quantized and defect-tolerant, which is why classifying it is a target for materials-simulation algorithms (see S-qsim, A-materials).

KEY GRADED CLAIMS

- T1 Adiabatic cyclic evolution yields a path-dependent geometric phase separate from the dynamical phase — Berry, Proc. R. Soc. Lond. A 392, 45 (1984) (status: established)
- T1 Berry curvature integrated over a filled band gives a quantized Chern number = the quantum-Hall conductance — TKNN, Thouless, Kohmoto, Nightingale & den Nijs, PRL 49, 405 (1982) (Thouless, Nobel 2016) (status: established)
- T1 The quantized Hall resistance is reproducible to a few parts in 10^9, precise enough to define the resistance standard $R_K = 25812.807\ \Omega$ — von Klitzing, Dorda & Pepper, PRL 45, 494 (1980), Nobel 1985 (status: established)
- T2 Geometric/holonomic gates realized on superconducting and NV qubits — e.g. Abdumalikov et al., Nature 496, 482 (2013) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Whether geometric gates actually beat optimized dynamical gates in real error budgets is contested — the noise-suppression advantage can be eroded by the longer loop times.

GO DEEPER

- Xiao, Chang & Niu, "Berry phase effects on electronic properties," RMP 82, 1959 (2010)
- Wilczek & Shapere, *Geometric Phases in Physics* (1989)

SOURCES

- Berry, Proc. R. Soc. A 392, 45 (1984). doi:10.1098/rspa.1984.0023
- TKNN, PRL 49, 405 (1982)
- von Klitzing, Dorda & Pepper, PRL 45, 494 (1980). doi:10.1103/PhysRevLett.45.494

Contextuality (Kochen-Specker) · F-contextuality

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Quantum measurement outcomes cannot be assigned pre-existing values independent of which other compatible observables are measured alongside them. The Kochen-Specker theorem (1967) proves this: for a Hilbert space of dimension ≥ 3, no assignment of definite values $(0/1)$ to all projectors is consistent with the algebraic relations quantum mechanics requires. Contextuality generalizes Bell nonlocality — Bell inequalities are the special case where the "contexts" are spacelike-separated — and it is the sharper statement of what makes quantum theory non-classical. Spekkens later reformulated it operationally so it applies to noisy, real experiments.

CORE IDEA / KEY EQUATION

The cleanest witness is the **Peres-Mermin square**: a 3×3 grid of nine two-qubit observables, each ± 1-valued and each commuting with its row-mates and column-mates. Quantum mechanics forces the product of the three observables along every row to be $+1$ and along two columns $+1$ but the third column -1. Multiply all six line-products together: quantum mechanics gives -1. Now try to pre-assign a definite value ± 1 to each of the nine boxes independent of context. Each box sits in one row and one column, so it enters the grand product exactly twice, and any ± 1 assigned to it contributes its square, $+1$ — the noncontextual product is forced to be $+1$. The contradiction, $-1 \neq +1$, is state-independent: it holds for every input state, needs no inequality and no entanglement, and no assignment of hidden values escapes it. That algebraic clash is the whole content of Kochen-Specker, compressed to nine boxes.

WHY IT MATTERS FOR QUANTUM TECH

Contextuality is a computational *resource*. Howard, Wallman, Veitch & Emerson (2014) proved that for qudits, contextuality is precisely what "magic states" supply in the leading fault-tolerant model: Clifford operations plus contextuality-free (stabilizer) states are efficiently classically simulable (the Gottesman-Knill theorem), and the onset of contextuality coincides exactly with the states that magic-state distillation must produce to reach universal quantum computation. This draws a direct line from a 1967 no-go theorem to the resource that error-corrected machines spend most of their overhead manufacturing — magic-state factories are the dominant space and time cost in surface-code architectures (see S-qec, S-logical). The classical-simulability boundary it draws also tells simulator designers exactly which subroutines a classical machine can shadow for free and which demand real quantum hardware (see S-qsim). Contextuality further underlies some certified-randomness and self-testing arguments, where a measured contextuality violation is the certificate that a device is behaving quantumly (see A-qrng, O-verification).

KEY GRADED CLAIMS

- T1 No non-contextual hidden-variable model reproduces QM in dimension ≥ 3 — Kochen & Specker, J. Math. Mech. 17, 59 (1967) (status: established)
- T2 Contextuality is the resource enabling universal quantum computation via magic-state distillation (qudit model) — Howard, Wallman, Veitch & Emerson, Nature 510, 351 (2014), arXiv:1401.4174 (status: established)
- T2 State-independent contextuality demonstrated experimentally, violating a noncontextuality inequality with trapped ions and needing no special input state — Kirchmair et al., Nature 460, 494 (2009), arXiv:0904.1655 (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- The qubit ($d = 2$) case is subtler — Wigner-function negativity and contextuality diverge; the exact resource-theoretic statement for qubits is still refined.

GO DEEPER

- Spekkens, "Contextuality for preparations, transformations, measurements," PRA 71, 052108 (2005)
- Bermejo-Vega et al., PRL 119, 120505 (2017) (contextuality + qubit magic)

SOURCES

- Kochen & Specker (1967)
- Howard et al., Nature 510, 351 (2014). doi:10.1038/nature13460
- Kirchmair et al., Nature 460, 494 (2009). doi:10.1038/nature08172
- Mermin, "Hidden variables and the two theorems of John Bell," RMP 65, 803 (1993) (Peres-Mermin square)

Decoherence & Open Quantum Systems · F-decoher

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

No system is perfectly isolated. When a quantum system couples to its environment, the environment effectively "measures" it: phase relations between superposition components leak into environmental degrees of freedom and become unrecoverable. The reduced state decoheres into an apparent classical mixture in a preferred ("pointer") basis selected by the interaction. Decoherence explains why the everyday world looks classical, and it sets the clock every qubit races against (T1 energy relaxation, T2 dephasing).

CORE IDEA / KEY EQUATION

The state of the system alone is the reduced density matrix $\rho_S = \mathrm{Tr}_E(|\Psi\rangle\langle\Psi|)$, traced over the environment. Coupling entangles system with environment, and the off-diagonal terms of ρ_S (the "coherences" that encode superposition) decay — for a qubit, the $|0\rangle\langle 1|$ element shrinks as $\exp(-t/T_2)$. The continuous-time law for a Markovian environment is the Lindblad (GKS) master equation:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \sum_k \left(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \right)$$

where the jump operators L_k model each channel (relaxation, dephasing). The first term is ordinary unitary evolution; the sum is the irreversible leak. Two rates matter for a qubit: T_1, the energy-relaxation time ($|1\rangle \to |0\rangle$), and $T_2 \le 2\,T_1$, the total dephasing time. Decoherence picks out the pointer basis — the states the interaction leaves alone — which for most environments is position/energy, and that is why big warm objects never look superposed.

WHY IT MATTERS FOR QUANTUM TECH

Decoherence is the enemy: it caps circuit depth on NISQ hardware (S-nisq) — you get only $\sim T_2/t_{\mathrm{gate}}$ operations before the state is noise — and dictates the entire architecture of quantum error correction (S-qec), which spreads one logical qubit across many physical ones so syndrome measurements catch errors before they accumulate. Coherence times are the headline spec of every qubit modality: superconducting transmons and fluxonium (H-supercon) now reach the millisecond range, trapped-ion hyperfine qubits (H-ion) hold coherence for seconds to minutes, and neutral atoms (H-neutral) sit in between. The whole T1/T2 vocabulary comes straight from this section, and pushing those numbers up is the central materials-and-engineering problem of the field.

KEY GRADED CLAIMS

- T1 Environmental entanglement suppresses interference and selects pointer states — Zeh, Found. Phys. 1, 69 (1970); Zurek, PRD 24, 1516 (1981) and PRD 26, 1862 (1982); review: Zurek, RMP 75, 715 (2003) (status: established)
- T2 Decoherence of a mesoscopic superposition was watched in real time in cavity QED — Brune et al., PRL 77, 4887 (1996) (Haroche group; 2012 Nobel) (status: demonstrated)
- T2 Decoherence was turned on as a knob: heating C70 fullerenes to ~3,000 K in a Talbot-Lau interferometer made them emit thermal photons that carry away which-path information, washing out interference fringes in quantitative agreement with theory — Hackermüller, Hornberger, Brezger, Zeilinger, Arndt, Nature 427, 711 (2004), doi:10.1038/nature02276 (status: demonstrated)
- T2 Coherence-time frontier: a fluxonium qubit reached Ramsey coherence $T_2^* \approx 1.48 \pm 0.13$ ms with single-qubit gate fidelity above 0.9999 — Somoroff et al. (Manucharyan group), PRL 130, 267001 (2023), arXiv:2103.08578 (status: demonstrated)
- T2 The formal machinery is the theory of open systems: Lindblad/GKS master equations, CPTP maps — Lindblad, Commun. Math. Phys. 48, 119 (1976); Breuer & Petruccione (2002) (status: established)

CONFLICTS / OPEN QUESTIONS

- Decoherence explains the *appearance* of classicality without selecting a single outcome — whether it dissolves or merely reframes the measurement problem is contested (Schlosshauer, Phys. Rep. 831, 1, 2019).

GO DEEPER

- Zurek, "Decoherence, einselection, and the quantum origins of the classical," RMP 75, 715 (2003), arXiv:quant-ph/0105127
- Schlosshauer, *Decoherence and the Quantum-to-Classical Transition* (Springer 2007)

SOURCES

- Zeh (1970) doi:10.1007/BF00708656 · Zurek RMP 75, 715 (2003) doi:10.1103/RevModPhys.75.715

- Brune et al., PRL 77, 4887 (1996). doi:10.1103/PhysRevLett.77.4887
- Hackermüller et al., "Decoherence of matter waves by thermal emission of radiation," Nature 427, 711 (2004). doi:10.1038/nature02276
- Somoroff et al., "Millisecond coherence in a superconducting qubit," PRL 130, 267001 (2023). arXiv:2103.08578

Quantum Discord · F-discord

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Quantum discord measures quantum correlations that survive even when a state has no entanglement. It captures the gap between two expressions for mutual information that are identical in classical probability but diverge for quantum states. Harold Ollivier and Wojciech Zurek introduced it in 2001, with Leah Henderson and Vlatko Vedral independently defining the classical-correlation term the same year. The striking claim: many separable (unentangled) mixed states still carry nonzero discord, so entanglement does not exhaust the ways quantum systems can be correlated.

CORE IDEA / KEY EQUATION

Classically, mutual information can be written two ways — $I(A:B) = H(A) + H(B) - H(A,B)$ and $J(A:B) = H(A) - H(A|B)$ — and they always agree. Quantumly the conditional entropy $H(A|B)$ requires choosing a measurement on B, and the two expressions come apart. Discord is the minimized difference:

$$D(A:B) = I(A:B) - \max_{\{\Pi_k\}} J(A:B)$$

where the maximization runs over local measurements (POVMs) on B and I uses von Neumann entropies. $D \geq 0$ always, and $D = 0$ exactly for states that stay undisturbed by some local measurement on B ("classical-quantum" states). Because the set of zero-discord states is much smaller than the set of separable states, generic separable states have $D > 0$. A concrete example is the two-qubit Werner-like separable mixtures used in the DQC1 "power of one qubit" model, where entanglement is absent yet discord tracks the computational speed-up. Computing discord requires an optimization over measurements, so closed forms exist only for special families (e.g. two-qubit Bell-diagonal states).

WHY IT MATTERS FOR QUANTUM TECH

Discord matters where entanglement is scarce but quantum behavior persists — highly mixed states, room-temperature and noisy platforms. It was proposed as the resource behind the DQC1 model of mixed-state quantum computation, which shows apparent speed-up with vanishing entanglement (see S-nisq, S-qsim). Discord appears in analyses of quantum metrology, state merging, and remote state preparation, and it connects to the broader map of correlations that includes entanglement and Bell nonlocality (see F-entangle, F-qinfo). For security arguments, the sharper structural constraint comes from entanglement monogamy rather than discord (see A-qkd, F-entropy-ineq).

KEY GRADED CLAIMS

- T1 Some separable states have strictly positive discord; zero entanglement does not imply classical correlations — Ollivier & Zurek, PRL 88, 017901 (2001); Henderson & Vedral, J. Phys. A 34, 6899 (2001) (status: established)
- T2 The DQC1 model exhibits computational advantage with negligible entanglement but nonzero discord — Datta, Shaji & Caves, PRL 100, 050502 (2008), arXiv:0709.0548 (status: demonstrated, interpretation debated)
- T2 Comprehensive framework and measures for discord-type correlations — Modi, Brodutch, Cable, Paterek & Vedral, Rev. Mod. Phys. 84, 1655 (2012), arXiv:1112.6238 (status: established review)

CONFLICTS / OPEN QUESTIONS

- Whether discord is an *operational* resource — one that buys a task no cheaper protocol can — is still contested; several claimed advantages have alternative explanations, and no single agreed resource theory of discord has won out.
- Discord is measurement-optimization-dependent and generally hard to compute (NP-hard in general), so which of the many proposed measures is canonical remains unsettled.

GO DEEPER

- Modi, Brodutch, Cable, Paterek & Vedral, Rev. Mod. Phys. 84, 1655 (2012)
- Ollivier & Zurek, PRL 88, 017901 (2001)

SOURCES

- Ollivier & Zurek, PRL 88, 017901 (2001). doi:10.1103/PhysRevLett.88.017901
- Henderson & Vedral, J. Phys. A 34, 6899 (2001). doi:10.1088/0305-4470/34/35/315
- Datta, Shaji & Caves, PRL 100, 050502 (2008). arXiv:0709.0548
- Modi et al., Rev. Mod. Phys. 84, 1655 (2012). arXiv:1112.6238

Entanglement & Nonlocality · F-entangle

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Two or more systems are entangled when their joint state cannot be written as a product of individual states — the whole carries information that no part carries alone. Measurement outcomes on separated entangled particles show correlations stronger than any local hidden-variable model allows (quantified by Bell inequalities, see F-bell). Einstein, Podolsky and Rosen flagged it as a paradox in 1935; Schrödinger named it *Verschränkung* the same year and called it *the* characteristic trait of quantum mechanics.

CORE IDEA / KEY EQUATION

A pure two-party state is entangled when $|\Psi\rangle \neq |\varphi\rangle_A \otimes |\chi\rangle_B$ for any single-system states. The canonical example is the singlet Bell state $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$: measure either qubit and you get 0 or 1 at random, but the two results are perfectly anti-correlated no matter how far apart. For mixed states, ρ is separable only if $\rho = \sum_i p_i \rho_A^{(i)} \otimes \rho_B^{(i)}$; anything else is entangled. The amount of entanglement in a pure bipartite state is the entanglement entropy $S = -\mathrm{Tr}(\rho_A \log \rho_A)$, where $\rho_A = \mathrm{Tr}_B(|\Psi\rangle\langle\Psi|)$ is the reduced state of one side — $S = 0$ for a product state and $S = 1$ bit ("one ebit") for a maximally entangled qubit pair. Locally you see only the reduced state, which is why entanglement never lets you signal; the correlations only appear when the two measurement records are later compared.

WHY IT MATTERS FOR QUANTUM TECH

Entanglement is the workhorse resource. A two-qubit gate (S-gates) is the operation that *creates* it — a CZ or CNOT turns a product state into a Bell state, and multipartite entanglement across a register is what a quantum computer *is*. It powers teleportation (F-teleport), entanglement-based QKD (A-qkd, the E91 protocol), and the stabilizer entanglement that error-correcting codes (S-qec) distribute across physical qubits to protect one logical qubit. Every hardware modality is judged partly on how well it entangles: trapped ions (H-ion) via Mølmer-Sørensen gates, superconducting qubits (H-supercon) via tunable couplers, neutral atoms (H-neutral) via Rydberg blockade, photons (H-photonic) via measurement-induced fusion. Distributing entanglement between distant nodes is the physical layer of the quantum internet.

KEY GRADED CLAIMS

- T1 Entangled states exist and violate local realism — EPR, Phys. Rev. 47, 777 (1935); Schrödinger, Proc. Camb. Phil. Soc. 31, 555 (1935); experimental record in F-bell (status: established)
- T1 Entanglement cannot transmit signals — the no-communication theorem keeps quantum correlations consistent with relativity (status: established)
- T2 Entanglement is a quantifiable, interconvertible resource (entanglement entropy, distillation, monogamy) — Horodecki ×4, Rev. Mod. Phys. 81, 865 (2009), arXiv:quant-ph/0702225 (status: established)
- T1 Entanglement survives distribution over 1,200 km: the Micius satellite delivered polarization-entangled photon pairs (source ~5.9 million pairs/s) to two ground stations 1,203 km apart, violating a Bell inequality by $S = 2.37 \pm 0.09$ under Einstein locality — Yin et al., Science 356, 1140 (2017), doi:10.1126/science.aan3211 (status: established)
- T2 Many-body entanglement at high fidelity: a 60-atom neutral-atom array ran parallel two-qubit gates at 99.5% fidelity and produced three-qubit GHZ states at 90.9(6)% raw fidelity, above the surface-code threshold — Evered, Bluvstein et al. (Lukin group), Nature 622, 268 (2023), doi:10.1038/s41586-023-06481-y (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Deciding whether an arbitrary mixed state is entangled (the separability problem) is NP-hard in general; multipartite entanglement classification is unfinished mathematics.

GO DEEPER

- Horodecki et al., "Quantum entanglement," RMP 81, 865 (2009) — the canonical review
- Nielsen & Chuang §2.6; Preskill lecture notes ch. 4

SOURCES

- Einstein, Podolsky, Rosen, Phys. Rev. 47, 777 (1935). doi:10.1103/PhysRev.47.777
- Schrödinger, Proc. Camb. Phil. Soc. 31, 555 (1935). doi:10.1017/S0305004100013554
- Horodecki, Horodecki, Horodecki, Horodecki, RMP 81, 865 (2009). arXiv:quant-ph/0702225
- Yin et al., "Satellite-based entanglement distribution over 1200 kilometers," Science 356, 1140 (2017). doi:10.1126/science.aan3211
- Evered, Bluvstein et al., "High-fidelity parallel entangling gates on a neutral-atom quantum computer," Nature 622, 268 (2023). doi:10.1038/s41586-023-06481-y

Quantum Entropy Inequalities · F-entropy-ineq

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The structural laws that von Neumann entropy must obey across subsystems. The deepest is strong subadditivity (SSA), proved by Elliott Lieb and Mary Beth Ruskai in 1973 after being conjectured by Robinson, Ruelle, and Lanford in the late 1960s. From SSA follow the monotonicity of relative entropy, the data-processing inequality, the Araki-Lieb triangle bound, and — for entanglement — the Coffman-Kundu-Wootters monogamy inequality. Together they are the arithmetic of how quantum information distributes and degrades, and every capacity theorem and security proof leans on them.

CORE IDEA / KEY EQUATION

Strong subadditivity, for any tripartite state ρ_{ABC}:

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$$

Equivalently, conditioning on more never increases conditional entropy, and the mutual information $I(A:C|B) \geq 0$. This one inequality is the workhorse; several key bounds are corollaries: - **Data processing:** relative entropy is non-increasing under any CPTP channel, $D(\Lambda\rho \,\|\, \Lambda\sigma) \leq D(\rho\|\sigma)$ — you cannot make two states more distinguishable by processing them, so quantum information cannot be increased by local operations. Equivalent in strength to SSA. - **Araki-Lieb (triangle):** $|S(\rho_A) - S(\rho_B)| \leq S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B)$. The lower bound has no classical analog; it forces a global pure state ($S(\rho_{AB}) = 0$) to have $S(\rho_A) = S(\rho_B)$, the entropy of entanglement. - **CKW monogamy:** for three qubits, $C^2(A:B) + C^2(A:C) \leq C^2(A:BC)$ in the concurrence (tangle). Entanglement cannot be freely shared — if A is maximally entangled with B, it has none left for C.

WHY IT MATTERS FOR QUANTUM TECH

Monogamy is the mathematical reason quantum key distribution is secure: an eavesdropper's correlation with the key is bounded by how much the legitimate parties share, so listening in is detectable (see A-qkd). Data processing sets the ceiling on every quantum channel capacity and underwrites the LSD coherent-information formula and Holevo-type bounds (see F-qinfo). Entropy inequalities gate entanglement distillation and the resource accounting of teleportation and state merging (see F-entangle). In many-body physics and holography, SSA constrains entanglement entropy and area laws (see S-qsim).

KEY GRADED CLAIMS

- T1 Strong subadditivity holds for von Neumann entropy — Lieb & Ruskai, J. Math. Phys. 14, 1938 (1973) (status: established)
- T1 Relative entropy is monotone under CPTP maps (data processing), equivalent to SSA — Lindblad, Commun. Math. Phys. 40, 147 (1975); Uhlmann (1977) (status: established)
- T2 Three-qubit entanglement obeys the CKW monogamy inequality; generalized to n qubits by Osborne & Verstraete — Coffman, Kundu & Wootters, PRA 61, 052306 (2000), arXiv:quant-ph/9907047; Osborne & Verstraete, PRL 96, 220503 (2006) (status: established)
- T2 Equality in SSA characterizes quantum Markov chains (recoverable states) — Hayden, Jozsa, Petz & Winter, Commun. Math. Phys. 246, 359 (2004); strengthened by Fawzi & Renner, arXiv:1410.0664 (2015) (status: established)

CONFLICTS / OPEN QUESTIONS

- The remainder-term / approximate-recovery refinements of SSA (how much information a recovery map can restore when SSA is nearly saturated) are still being sharpened.
- Beyond qubits, the exact form of monogamy for general entanglement measures and higher-dimensional systems is not fully settled; monogamy can fail for some correlation measures (e.g. discord is not always monogamous — see F-discord).

GO DEEPER

- Nielsen & Chuang, ch. 11 · Wilde, *Quantum Information Theory* (2013), arXiv:1106.1445
- Lieb & Ruskai, J. Math. Phys. 14, 1938 (1973)

SOURCES

- Lieb & Ruskai, J. Math. Phys. 14, 1938 (1973). doi:10.1063/1.1666274
- Coffman, Kundu & Wootters, PRA 61, 052306 (2000). arXiv:quant-ph/9907047
- Lindblad, Commun. Math. Phys. 40, 147 (1975). doi:10.1007/BF01609396
- Fawzi & Renner, Commun. Math. Phys. 340, 575 (2015). arXiv:1410.0664

Interpretations of Quantum Mechanics · F-interp

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The formalism predicts probabilities perfectly; what it *says about reality* is unsettled. The main families: Copenhagen (Bohr/Heisenberg — the classical measurement context is primitive), many-worlds (Everett 1957 — no collapse, all branches real), de Broglie–Bohm (1952 — particles with definite trajectories guided by the wave), objective collapse (GRW/CSL, Diósi–Penrose — collapse is a physical process, and testable), QBism (states are agents' credences), and relational QM (Rovelli — states are relative to observers). All except objective collapse are empirically identical to standard QM.

CORE IDEA / KEY EQUATION

The whole dispute sits on a mismatch between two rules the standard formalism runs together. Between measurements a state evolves smoothly and deterministically by the Schrödinger equation, $i\hbar\,\partial_t|\psi\rangle = H|\psi\rangle$, which is linear and never destroys a superposition. At a measurement the Born rule takes over: outcome a appears with probability $P(a) = |\langle a|\psi\rangle|^2 = \mathrm{Tr}(\rho P_a)$, and the state jumps to the matching eigenstate. Superposition in, one definite outcome out — that discontinuous, probabilistic jump is nowhere in the linear equation, and specifying when and why it happens is the measurement problem. The interpretations are answers to it: many-worlds keeps only the linear equation and denies the jump is real (every outcome happens, on its own branch); de Broglie–Bohm keeps the wave and adds definite particle positions; objective-collapse theories add a small stochastic nonlinear term to the Schrödinger equation so that superpositions of large masses self-destruct on a physical timescale, which makes them the one family with numbers to test. The GRW/CSL collapse rate λ and the Diósi–Penrose gravitational timescale are the parameters experiments try to pin.

WHY IT MATTERS FOR QUANTUM TECH

Mostly none — the no-collapse interpretations agree on every number, so gates, memories, and algorithms behave the same whichever you believe. The exception has teeth: objective-collapse models predict that a superposition of a large enough mass decoheres on its own, and the platforms that build ever-heavier or ever-more-isolated superpositions are exactly the matter-wave interferometers, levitated nanoparticles, and mechanical oscillators shared with quantum sensing and bosonic hardware (H-bosonic). Each new mass or coherence-time record tightens the bound on the collapse rate. The interpretation debate does not change how you run S-gates or S-qec, but the collapse tests ride on the same isolation-and-readout engineering that H-photonic and sensing programs push.

KEY GRADED CLAIMS

- T1 All mainstream no-collapse interpretations reproduce identical predictions; no experiment to date distinguishes them (status: established)
- T2 Objective-collapse models are being squeezed experimentally: an underground test of Diósi–Penrose collapse-induced radiation excluded its natural parameter-free version — Donadi et al., Nat. Phys. 17, 74 (2021), doi:10.1038/s41567-020-1008-4 (status: demonstrated)
- T2 The superposition principle holds for objects far heavier than atoms: matter-wave interference in a 2-m Talbot–Lau interferometer was shown for oligoporphyrin molecules beyond 25,000 Da made of up to 2,000 atoms — the heaviest objects put into a spatial superposition to date, directly narrowing the window collapse models can occupy — Fein et al., Nat. Phys. 15, 1242 (2019), doi:10.1038/s41567-019-0663-9 (status: demonstrated)
- T3 Extended Wigner's-friend theorems sharpen what any observer-independent account must give up — Frauchiger & Renner, Nat. Commun. 9, 3711 (2018); Bong et al., Nat. Phys. 16, 1199 (2020) (status: contested)

CONFLICTS / OPEN QUESTIONS

- C-measurement-problem: the field's oldest open question. Physicist polls (e.g. Schlosshauer et al., arXiv:1301.1069) show no consensus — Copenhagen-ish pluralities, growing Everettian and QBist minorities. What would resolve it: a confirmed collapse-model deviation, or nothing ever.

GO DEEPER

- Everett, Rev. Mod. Phys. 29, 454 (1957) · Bohm, Phys. Rev. 85, 166 & 180 (1952) · Ghirardi, Rimini, Weber, PRD 34, 470 (1986)
- Fuchs, Mermin, Schack, Am. J. Phys. 82, 749 (2014) · Rovelli, Int. J. Theor. Phys. 35, 1637 (1996)

SOURCES

- As listed above; Donadi et al. (2021) doi:10.1038/s41567-020-1008-4; Bassi et al., "Models of wave-function collapse," RMP 85, 471 (2013)
- Fein et al., Nat. Phys. 15, 1242 (2019). doi:10.1038/s41567-019-0663-9

Lieb-Robinson Bounds (Emergent Locality) · F-liebrobinson

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Non-relativistic quantum mechanics has no built-in speed limit — the Schrödinger equation is not Lorentz-invariant, and a local perturbation could in principle be felt everywhere at once. Lieb and Robinson (1972) proved that for lattice systems with short-range interactions this does not happen: information propagates at most at a finite emergent velocity, and outside a linear "light cone" correlations are exponentially small. This speed limit is effective rather than fundamental — it emerges from the Hamiltonian's locality — yet it acts like relativity's light cone for spin chains and cold atoms. Hastings and Koma (2006) extended it and used it to prove that a spectral gap forces exponential decay of ground-state correlations, one of the most-used tools in modern many-body physics.

CORE IDEA / KEY EQUATION

For local operators A and B supported on regions separated by distance d, evolving A to time t under a local Hamiltonian gives a commutator bound of the form

$$\|[A(t), B]\| \leq C\,\|A\|\,\|B\| \cdot \exp\left(\frac{v|t| - d}{\xi}\right),$$

where v is the Lieb-Robinson velocity, ξ a length scale set by the interaction range, and C depends on the boundary of the supports. The commutator measures how much a probe at B can detect a disturbance at A. It stays exponentially small until $v|t| \approx d$ — the arrival time of the effective light cone — and only then becomes order one.

Plain version: poke one spin, and a spin a distance d away barely notices until a time d/v has passed. A local Hamiltonian — each term touching only nearby sites — is enough to manufacture an approximate causal structure with a group velocity, with no relativity in the theory.

WHY IT MATTERS FOR QUANTUM TECH

Lieb-Robinson velocity sets the clock for how fast entanglement and information spread, bounding how quickly a quantum computer can move data across a chip and how fast entangling gates over distance can be (H-supercon, H-ion, S-gates). It underpins the area law for entanglement entropy and the efficiency of tensor-network (MPS/DMRG) simulation, so it controls what classical machines can and cannot simulate (F-qinfo). The gap-implies-clustering result explains why local error correction and topological order are stable (S-qec, S-logical). It also bounds state-transfer rates in quantum networks and metrology protocols spread over many sites (A-qinternet, A-sensing).

KEY GRADED CLAIMS

- T1 Short-range lattice Hamiltonians have a finite group velocity; correlations vanish exponentially outside a linear light cone — Lieb & Robinson, Commun. Math. Phys. 28, 251 (1972) (status: established)
- T1 A nonzero spectral gap implies exponential decay of ground-state correlations, via Lieb-Robinson bounds — Hastings & Koma, Commun. Math. Phys. 265, 781 (2006); Nachtergaele & Sims, Commun. Math. Phys. 265, 119 (2006) (status: established)
- T2 A light-cone spread of correlations at a finite velocity was observed directly in a quantum gas — Cheneau et al., Nature 481, 484 (2012); trapped-ion light cones, Jurcevic et al. & Richerme et al., Nature 511, 202 & 198 (2014) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- The tight velocity and light-cone shape for long-range (power-law $1/r^{\alpha}$) interactions are still being pinned down; whether the cone is linear, polynomial, or logarithmic depends on α, and recent work keeps sharpening the boundaries.
- Optimal (tight) Lieb-Robinson velocities for generic interacting models remain open; known bounds are often loose by a constant or more.

GO DEEPER

- Nachtergaele & Sims, "Lieb-Robinson bounds in quantum many-body physics," Contemp. Math. 529, 141 (2010), arXiv:1004.2086
- Chen, Lucas & Yin, "Speed limits and locality in many-body quantum dynamics," Rep. Prog. Phys. 86, 116001 (2023), arXiv:2303.07386

SOURCES

- Lieb & Robinson, Commun. Math. Phys. 28, 251 (1972). doi:10.1007/BF01645779

• Hastings & Koma, Commun. Math. Phys. 265, 781 (2006). doi:10.1007/s00220-006-0030-4

• Cheneau et al., Nature 481, 484 (2012). doi:10.1038/nature10748

Measurement & the Born Rule · F-measure

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Measuring an observable returns one of its eigenvalues, with probability equal to the squared magnitude of the corresponding amplitude — the Born rule. After the measurement the state updates ("collapses") to the outcome's eigenstate (von Neumann's projection postulate), or more generally per the POVM/instrument formalism. The rule is where quantum mechanics touches experiment: everything the theory predicts is a Born-rule probability.

CORE IDEA / KEY EQUATION

$P(a) = |\langle a|\psi\rangle|^2$, the probability of outcome a when observable A is measured on state $|\psi\rangle$, where $|a\rangle$ is the eigenstate with eigenvalue a. Equivalently the expectation value is $\langle A\rangle = \langle\psi|A|\psi\rangle$, and for a general (mixed) state $P(a) = \mathrm{Tr}(\rho P_a)$ with P_a the projector onto the a-eigenspace. The rule is quadratic in the amplitude, which is why it kills the higher-order interference terms a linear or cubic rule would produce — three-path interference must vanish exactly. Gleason's theorem shows this squared-modulus form is forced: on any Hilbert space of dimension ≥ 3, the only probability measure over projectors that is additive on orthogonal outcomes is $\mathrm{Tr}(\rho P)$. So the "why squared" question has a partial answer — given the Hilbert-space structure and non-contextuality, no other rule is consistent.

WHY IT MATTERS FOR QUANTUM TECH

Every quantum computation ends in measurement — the whole game of algorithm design is arranging interference so the Born-rule statistics concentrate on the answer, which is why Grover (S-grover) and Shor (S-shor) both end with a projective readout that samples the amplified peak. Readout fidelity is a first-class hardware metric: dispersive readout of a transmon (H-supercon) probes the cavity to infer $|0\rangle$ vs $|1\rangle$, and state-dependent fluorescence reads an ion (H-ion) at >99.9%. Error correction (S-qec) runs on repeated *non-destructive* stabilizer measurements — the POVM/instrument formalism, projecting only the syndrome and leaving the logical state intact. Measurement statistics also underwrite security: QKD (A-qkd) turns an eavesdropper's unavoidable measurement disturbance into a detectable error rate.

KEY GRADED CLAIMS

- T1 Outcome probabilities are $|\langle a|\psi\rangle|^2$ — Born, Z. Phys. 37, 863 (1926); confirmed in a century of experiments (status: established)
- T1 The Born rule is the only consistent probability assignment on Hilbert spaces of dimension ≥ 3 given non-contextuality of the measure — Gleason's theorem, J. Math. Mech. 6, 885 (1957) (status: established)
- T1 Direct test of the rule's quadratic form: a three-slit photon experiment bounds genuine third-order (Sorkin) interference to below ~1% of the pairwise term, $\kappa = 0.0064 \pm 0.0119$, consistent with zero — Sinha, Couteau, Jennewein, Laflamme, Weihs, Science 329, 418 (2010), doi:10.1126/science.1190545 (status: established)
- T2 Weak measurements trade information gain against disturbance, allowing partial, non-projective readout — Aharonov, Albert & Vaidman, PRL 60, 1351 (1988); single quantum trajectories of a transmon reconstructed under continuous weak monitoring (readout cavity probed with ~1 photon, near-quantum-limited parametric amplifier) — Murch, Weber, Macklin, Siddiqi, Nature 502, 211 (2013), doi:10.1038/nature12539 (status: demonstrated)
- T2 Step-by-step ("progressive") collapse of a cavity field was watched via repeated QND photon-number measurement with non-absorbing atoms — Guerlin et al. (Haroche group), Nature 448, 889 (2007), doi:10.1038/nature06057; 2012 Nobel (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- *Why* the Born rule, and what physically happens during collapse, is the measurement problem — unsolved; interpretations disagree (see F-interp). Derivations (decision-theoretic, envariance) remain contested.

GO DEEPER

- von Neumann, *Mathematical Foundations of Quantum Mechanics* (1932), ch. VI
- Busch, Lahti & Mittelstaedt, *The Quantum Theory of Measurement* (1996)
- Wiseman & Milburn, *Quantum Measurement and Control* (2009)

SOURCES

- Born, "Zur Quantenmechanik der Stoßvorgänge," Z. Phys. 37, 863 (1926). doi:10.1007/BF01397477
- Gleason, J. Math. Mech. 6, 885 (1957)
- Aharonov, Albert, Vaidman, PRL 60, 1351 (1988). doi:10.1103/PhysRevLett.60.1351
- Sinha et al., "Ruling out multi-order interference in quantum mechanics," Science 329, 418 (2010). doi:10.1126/science.1190545

- Murch, Weber, Macklin, Siddiqi, "Observing single quantum trajectories of a superconducting quantum bit," Nature 502, 211 (2013). doi:10.1038/nature12539
- Guerlin et al., "Progressive field-state collapse and quantum non-demolition photon counting," Nature 448, 889 (2007). doi:10.1038/nature06057

No-Cloning Theorem · F-nocloning

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

No physical process can make a perfect copy of an unknown quantum state. The proof is three lines: a universal copier would have to act linearly, and linearity forbids cloning superpositions of states it can clone individually. Proved independently by Wootters & Zurek and by Dieks in 1982 (with an earlier, little-noticed version by Park in 1970), partly in response to a proposed FTL-communication scheme that cloning would have enabled.

CORE IDEA / KEY EQUATION

Suppose a unitary U and a blank register $|b\rangle$ could copy any state: $U(|\psi\rangle|b\rangle) = |\psi\rangle|\psi\rangle$ for every $|\psi\rangle$. Apply it to two different states, $U(|\psi\rangle|b\rangle) = |\psi\rangle|\psi\rangle$ and $U(|\varphi\rangle|b\rangle) = |\varphi\rangle|\varphi\rangle$. Unitaries preserve inner products, so taking the overlap of the two input equations against the two output equations gives $\langle\psi|\varphi\rangle = \langle\psi|\varphi\rangle^2$. A number equal to its own square is 0 or 1, which forces the states to be either identical or orthogonal. A machine cannot clone a set that contains any non-orthogonal pair — in particular it cannot clone the two poles and the equator of the Bloch sphere at once. The same one-line argument, run with density matrices, gives no-broadcasting for mixed states. What survives is approximate cloning: the best universal $1 \longrightarrow 2$ machine produces two copies each with fidelity $F = \langle\psi|\rho_{\text{out}}|\psi\rangle = 5/6 \approx 0.833$, a ceiling set by the same linearity that forbids the perfect copy.

WHY IT MATTERS FOR QUANTUM TECH

No-cloning is the root of quantum cryptography: an eavesdropper cannot copy qubits in flight without disturbing them, which is what BB84 and E91 detect and is the physical assumption behind the security proofs in S-qkd → A-qkd. It also forbids naive backup of quantum data — the reason error correction (S-qec) has to protect information without reading or copying it, spreading one logical qubit across many physical ones instead of duplicating it. It blocks classical-style signal amplification, so long-haul links on photonic hardware (H-photonic) need quantum repeaters and entanglement swapping rather than repeaters that copy-and-amplify. The $5/6$ optimal-cloner bound is also the yardstick against which any real intercept-resend attack on deployed QKD is measured.

KEY GRADED CLAIMS

- T1 An unknown pure state cannot be perfectly copied — Wootters & Zurek, Nature 299, 802 (1982); Dieks, Phys. Lett. A 92, 271 (1982) (status: established)
- T1 Corollaries: no-broadcasting for mixed states (Barnum et al., PRL 76, 2818, 1996) and no-deleting (Pati & Braunstein, Nature 404, 164, 2000) (status: established)
- T2 Imperfect cloning is possible up to a tight bound — the optimal universal $1 \longrightarrow 2$ cloner reaches fidelity $5/6 \approx 0.833$: Bužek & Hillery, PRA 54, 1844 (1996) (status: established)
- T2 The $5/6$ bound was reached in the lab: single-photon cloning by stimulated parametric down-conversion produced clones at near-optimal fidelity, universal across input states — Lamas-Linares, Simon, Howell & Bouwmeester, Science 296, 712 (2002), arXiv:quant-ph/0205149 (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- None at the theorem level; the live work is in how close practical eavesdropping/cloning attacks get to the optimal-cloner bound in deployed QKD (see A-qkd).

GO DEEPER

- Scarani et al., "Quantum cloning," RMP 77, 1225 (2005)
- Nielsen & Chuang, Box 12.1

SOURCES

- Wootters & Zurek, Nature 299, 802 (1982). doi:10.1038/299802a0
- Dieks, Phys. Lett. A 92, 271 (1982). doi:10.1016/0375-9601(82)90084-6
- Bužek & Hillery, PRA 54, 1844 (1996). doi:10.1103/PhysRevA.54.1844
- Lamas-Linares et al., Science 296, 712 (2002). arXiv:quant-ph/0205149

The PBR Theorem (Reality of the Quantum State) · F-pbr

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The PBR theorem (Pusey, Barrett & Rudolph, 2012) is a no-go result about what the wavefunction is. Foundations splits interpretations into ψ-ontic (the quantum state is a real property of an individual system) and ψ-epistemic (the state is only information about some deeper underlying "ontic" state λ, so two different states could correspond to the same reality). PBR proves that ψ-epistemic models are ruled out, given one natural assumption: systems prepared independently have independent physical states (preparation independence). Under that assumption, distinct pure quantum states must correspond to disjoint sets of ontic states — the state is real. The framework is the ontological-models formalism of Harrigan & Spekkens (2010).

CORE IDEA / KEY EQUATION

Model each preparation of $|\psi\rangle$ as sampling a hidden ontic state λ from a distribution $\mu_\psi(\lambda)$. "ψ-epistemic" means two non-orthogonal states can overlap: there exist $|\psi_0\rangle, |\psi_1\rangle$ with μ_0 and μ_1 sharing support. PBR take n independent copies, each prepared in $|\psi_0\rangle$ or $|\psi_1\rangle$, so the joint ontic state factorizes (preparation independence). If the single-system distributions overlap with probability $q > 0$, then with probability q^n all n systems land in the shared region and the model cannot tell which product state was prepared. PBR construct an entangled measurement on the n copies with an outcome that quantum mechanics assigns probability zero to every prepared product state. A model reproducing those zeros cannot have overlapping μ's — so distinct pure states are ontologically distinct.

Plain version: if the wavefunction were just our ignorance about a deeper reality, two different wavefunctions could secretly be the same underlying state. PBR show that a joint measurement on independently prepared copies would then sometimes give an outcome quantum theory forbids. Assuming independent preparations are independent, the wavefunction has to be a real feature of the system.

WHY IT MATTERS FOR QUANTUM TECH

PBR sets the terms for how we reason about the state as a resource (F-interp, F-qinfo). It joins Bell nonlocality (F-bell) and Kochen-Specker contextuality (F-contextuality) as the third pillar of quantum no-go theorems, together fencing off classical hidden-variable accounts. The ψ-ontic conclusion supports treating unknown states as carrying real, uncopyable information — the ground under no-cloning and the security proofs of QKD (F-nocloning, A-qkd), and under certified-randomness arguments where the state's reality constrains what an adversary can know (A-qrng, O-verification).

KEY GRADED CLAIMS

- T2 Under preparation independence, no ψ-epistemic model reproduces quantum statistics; distinct pure states are ontologically distinct — Pusey, Barrett & Rudolph, Nat. Phys. 8, 475 (2012), arXiv:1111.3328 (status: established, given the assumption)
- T2 Without preparation independence, ψ-epistemic models can still reproduce all of quantum theory in dimension $d \geq 2$ — Lewis, Jennings, Barrett & Rudolph, Phys. Rev. Lett. 109, 150404 (2012) (status: established)
- T3 Experimental tests bound the extent of ontic overlap, constraining epistemic explanations of indistinguishability — Ringbauer et al., Nat. Phys. 11, 249 (2015), arXiv:1412.6213 (status: demonstrated, model-dependent)

CONFLICTS / OPEN QUESTIONS

- The whole result hinges on preparation independence; drop it and epistemic models survive (Lewis et al. 2012), so critics argue PBR constrains a subclass of models rather than closing the question.
- The ontic/epistemic dichotomy assumes an underlying realist λ exists at all — interpretations that reject that framing (relational QM, QBism, Everett) sit outside PBR's target.

GO DEEPER

- Harrigan & Spekkens, "Einstein, incompleteness, and the epistemic view of quantum states," Found. Phys. 40, 125 (2010), arXiv:0706.2661
- Leifer, "Is the quantum state real? An extended review of ψ-ontology theorems," Quanta 3, 67 (2014), arXiv:1409.1570

SOURCES

- Pusey, Barrett & Rudolph, Nat. Phys. 8, 475 (2012). doi:10.1038/nphys2309, arXiv:1111.3328
- Lewis, Jennings, Barrett & Rudolph, Phys. Rev. Lett. 109, 150404 (2012). doi:10.1103/PhysRevLett.109.150404
- Ringbauer et al., Nat. Phys. 11, 249 (2015). doi:10.1038/nphys3233

Generalized Measurement (POVMs, Naimark, Instruments) · F-povm

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

A POVM (positive operator-valued measure) is the general description of what a quantum measurement can be. Projective (von Neumann) measurement is the special case where the outcome operators are orthogonal projectors. Real detectors — lossy, noisy, coupled to ancillas — are POVMs. Naimark's dilation theorem (1940) proves that every POVM is a projective measurement on a larger space: couple the system to an ancilla, measure sharply there, and the induced statistics on the system reproduce the POVM. A **quantum instrument** upgrades this to also give the post-measurement state, and **weak measurement** (Aharonov, Albert & Vaidman, 1988) is the regime of vanishing coupling where disturbance shrinks and the "weak value" can fall outside the eigenvalue spectrum.

CORE IDEA / KEY EQUATION

A POVM is a set of positive operators $\{E_k\}$, $E_k \geq 0$, with $\sum_k E_k = I$. Outcome k occurs with probability $p(k) = \mathrm{Tr}(\rho E_k)$. Each $E_k = M_k^\dagger M_k$ factors through measurement (Kraus) operators M_k; the post-measurement state is $\rho \to M_k \rho M_k^\dagger / p(k)$, and the collection $\{M_k\}$ defines the instrument. Naimark: for any POVM $\{E_k\}$ on $\mathcal{H}$ there is a larger space $\mathcal{H} \otimes \mathcal{H}_{\mathrm{anc}}$, a state of the ancilla, and a projective measurement $\{P_k\}$ such that $\mathrm{Tr}(\rho E_k) = \mathrm{Tr}((\rho \otimes |a\rangle\langle a|)\, P_k)$.

In plain terms: any fuzzy or partial readout you can build in the lab equals a perfect textbook measurement performed on your system plus a helper system you brought along. Nothing exotic happens — the extra outcomes and the softened back-action are bookkeeping for the ancilla you traced out.

WHY IT MATTERS FOR QUANTUM TECH

POVMs are the operational language of every real measurement in the stack. Dispersive readout of a superconducting qubit (H-supercon) and fluorescence readout of a trapped ion (H-ion) are POVMs with finite fidelity, and modeling them as POVMs is what makes readout-error mitigation and soft-information decoding possible (S-decoders, S-qec). Photon-counting detectors are POVMs (H-photonic). Unambiguous state discrimination, optimal measurements for QKD, and tomography all live in the POVM formalism (A-qkd, O-tomography). Weak measurement and continuous monitoring drive quantum feedback and metrology (A-sensing). Naimark dilation is the measurement-side twin of the Stinespring dilation for channels (F-purification), and both feed the decoherence account of how classical outcomes emerge (F-decoher, F-measure).

KEY GRADED CLAIMS

- T1 Every POVM is realized by a projective measurement on a system-plus-ancilla space (Naimark dilation) — Naimark 1940; textbook treatment Nielsen & Chuang §2.2.8 (status: established)
- T2 Weak measurement with pre- and post-selection yields weak values outside the eigenvalue range (proposed value 100 for a spin-$1/2$) — Aharonov, Albert & Vaidman, Phys. Rev. Lett. 60, 1351 (1988) (status: established, experimentally demonstrated)
- T2 Weak-value amplification measured a beam deflection of ~560 femtoradians — Hosten & Kwiat, Science 319, 787 (2008) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Whether weak values are a real physical quantity or a statistical artifact of post-selection is still debated; their utility for metrological amplification against technical noise is also contested.
- The instrument (which post-measurement state) is not fixed by the POVM alone — the same $\{E_k\}$ admits many Kraus decompositions, so the physical implementation carries extra content.

GO DEEPER

- Nielsen & Chuang, *Quantum Computation and Quantum Information*, §2.2.6–2.2.8 (POVMs, Naimark)
- Wiseman & Milburn, *Quantum Measurement and Control* (2010), Ch. 1 (instruments, continuous measurement)

SOURCES

- Naimark, Izv. Akad. Nauk SSSR Ser. Mat. 4, 277 (1940)
- Aharonov, Albert & Vaidman, Phys. Rev. Lett. 60, 1351 (1988). doi:10.1103/PhysRevLett.60.1351
- Hosten & Kwiat, Science 319, 787 (2008). doi:10.1126/science.1152697

Purification & Stinespring Dilation · F-purification

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Purification says every mixed state is the shadow of a pure state living on a larger space: for any density matrix ρ_A there is a pure state $|\psi\rangle_{AB}$ on a joint system such that $\rho_A = \mathrm{Tr}_B|\psi\rangle\langle\psi|$. Stinespring's dilation theorem (1955) is the channel-level version: every physical evolution — every completely positive trace-preserving (CPTP) map — is a unitary (isometry) into a bigger space followed by discarding the added part. Together they are the formal content of what John Smolin nicknamed the "church of the larger Hilbert space": mixedness and irreversibility are never fundamental; they are what remains after you throw away access to a purifying partner. Uhlmann (1976) sharpened purification into a formula for fidelity between mixed states.

CORE IDEA / KEY EQUATION

Purification: given $\rho_A = \sum_i p_i|i\rangle\langle i|$, define $|\psi\rangle_{AB} = \sum_i \sqrt{p_i}\,|i\rangle_A|i\rangle_B$ on $\mathcal{H}_A \otimes \mathcal{H}_B$ with $\dim \mathcal{H}_B \geq \mathrm{rank}\,\rho_A$; then $\mathrm{Tr}_B|\psi\rangle\langle\psi| = \rho_A$. The purification is unique up to a unitary on B.

Stinespring: any CPTP map $\mathcal{E}$ on $\mathcal{H}_A$ can be written $\mathcal{E}(\rho) = \mathrm{Tr}_E[V\rho V^\dagger]$, where $V : \mathcal{H}_A \to \mathcal{H}_A \otimes \mathcal{H}_E$ is an isometry ($V^\dagger V = I$) into a system-plus-environment space. Equivalently $\mathcal{E}(\rho) = \sum_k K_k\rho K_k^\dagger$ with $\sum_k K_k^\dagger K_k = I$ (Kraus form); the K_k are the isometry's components along an environment basis. Minimal dilation dimension is the Kraus rank, $\leq (\dim \mathcal{H}_A)^2$.

Plain version: any noisy or lossy process you run in the lab is a clean reversible rotation on your system plus a hidden meter, with the meter then ignored. Noise is entanglement with something you stopped tracking.

WHY IT MATTERS FOR QUANTUM TECH

This is the backbone of quantum error correction and open-system engineering. Modeling decoherence as a CPTP channel (via Stinespring/Kraus) is how amplitude damping, dephasing, and depolarizing noise get simulated and how threshold theorems are proved (S-qec, F-decoher). Purification underlies entanglement measures, thermofield-double states, and the environment "monitoring" picture of einselection (F-decoher, F-qinfo). It is the channel twin of Naimark dilation for measurements (F-povm) and the engine behind teleportation-based error correction and remote state preparation (F-teleport). Randomized benchmarking and process tomography assume the Kraus/Stinespring structure (O-tomography). In quantum thermodynamics and black-hole information debates, "purify the environment" is the standard move (F-interp).

KEY GRADED CLAIMS

- T1 Every CPTP map equals an isometry into a larger space followed by partial trace (Stinespring dilation) — Stinespring, Proc. Amer. Math. Soc. 6, 211 (1955) (status: established)
- T1 Every mixed state has a pure purification on a doubled space, unique up to a unitary on the ancilla — standard corollary; textbook Nielsen & Chuang §2.5 (status: established)
- T2 Fidelity between mixed states equals the maximal overlap over all purifications — Uhlmann, Rep. Math. Phys. 9, 273 (1976) (status: established)

CONFLICTS / OPEN QUESTIONS

- None mathematically. The interpretive question is whether the purifying partner is a real physical system or a formal device; the "church of the larger Hilbert space" is a modeling stance, and Stinespring's non-uniqueness means many physical environments realize the same channel.

GO DEEPER

- Nielsen & Chuang, §2.5 (purification) and §8.2 (quantum operations, Kraus/Stinespring)
- Paulsen, *Completely Bounded Maps and Operator Algebras* (2002), Ch. 4 (Stinespring, rigorous)

SOURCES

- Stinespring, Proc. Amer. Math. Soc. 6, 211 (1955). doi:10.1090/S0002-9939-1955-0069403-4
- Uhlmann, Rep. Math. Phys. 9, 273 (1976). doi:10.1016/0034-4877(76)90060-4
- Choi, Linear Algebra Appl. 10, 285 (1975). doi:10.1016/0024-3795(75)90075-0

QFT / Second-Quantization Bridge · F-qft

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Quantum field theory promotes fields, and the particles are their quanta: second quantization replaces "a wavefunction for N particles" with creation/annihilation operators acting on Fock space, letting particle number change and making quantum mechanics compatible with special relativity. Dirac quantized the electromagnetic field in 1927, giving the photon a rigorous home; QED, and later the Standard Model, followed. QFT is the deepest experimentally confirmed layer of physics.

CORE IDEA / KEY EQUATION

Second quantization writes each field mode as a quantum harmonic oscillator and reads its excitations as particles. A free field expands as $\varphi(x) = \sum_k [a_k f_k(x) + a_k^\dagger f_k^*(x)]$, where $a_k^\dagger$ creates a quantum in mode k and a_k destroys one, obeying the commutator $[a_k, a_{k'}^\dagger] = \delta_{k,k'}$ for bosons (and anticommutators for fermions, which forces the Pauli exclusion principle). The number operator $N_k = a_k^\dagger a_k$ counts quanta in mode k, and states live in Fock space built by acting with creation operators on the vacuum $|0\rangle$: $|n\rangle = (a^\dagger)^n/\sqrt{n!}\,|0\rangle$. The whole point is that particle number is now a dynamical variable — an interaction term in the Hamiltonian can turn one quantum into three — which is what a fixed-N wavefunction could never describe and what relativistic processes (pair creation, emission, decay) demand. The single-mode oscillator relation $E_n = \hbar\omega(n + \frac{1}{2})$ is the seed of everything, from the photon to the transmon.

WHY IT MATTERS FOR QUANTUM TECH

Photonic quantum computing *is* applied field quantization: the qubits and resources are Fock states, squeezed states, and coherent states of light modes, which is the native language of H-photonic and of the bosonic codes in H-bosonic. Circuit QED treats a microwave resonator as a quantized field mode coupled to an artificial atom, and it is the theoretical backbone of superconducting transmon processors (H-supercon), where the same $a/a^\dagger$ algebra sets qubit frequencies and dispersive readout in S-gates and S-bench. Simulating interacting field theories — scattering amplitudes, lattice gauge dynamics — is a headline target for fault-tolerant machines (S-qec), one of the few applications with a proven super-polynomial speedup argument.

KEY GRADED CLAIMS

- T1 The quantized EM field: emission/absorption from field quanta — Dirac, Proc. R. Soc. A 114, 243 (1927) (status: established)
- T1 QED is the most precisely verified physical theory: the electron g-2 measurement (0.13 ppt precision) agrees with QED prediction — Fan et al., PRL 130, 071801 (2023), doi:10.1103/PhysRevLett.130.071801 (status: established)
- T1 The muon magnetic anomaly is now measured to 127 ppb, the most precise muon g-2 result and one of the sharpest QFT tests: the Fermilab Muon g-2 final result (Runs 1–6) agrees with the revised Standard Model prediction — Muon g-2 Collaboration, final result announced June 2025 (submitted to PRL), improving on the 2021/2023 measurements (status: established)
- T2 Circuit QED — superconducting circuits realizing quantized field–atom physics — is the theoretical backbone of transmon processors: Blais et al., RMP 93, 025005 (2021), arXiv:2005.12667 (status: established)
- T3 Efficient quantum algorithms exist for simulating scattering in interacting QFTs — Jordan, Lee & Preskill, Science 336, 1130 (2012), arXiv:1111.3633 (status: demonstrated in theory)

CONFLICTS / OPEN QUESTIONS

- Rigorous mathematical construction of interacting 4D QFT (Yang–Mills mass gap) is an open Clay Millennium Problem; QFT + gravity remains unreconciled.
- The muon g-2 anomaly is unresolved rather than closed: the experimental value is now far more precise than theory, and the tension depends on which Standard Model hadronic-vacuum-polarization input (data-driven vs. lattice QCD) is used.

GO DEEPER

- Peskin & Schroeder, *An Introduction to QFT* · Weinberg, *The Quantum Theory of Fields* vol. 1

SOURCES

- Dirac (1927) doi:10.1098/rspa.1927.0039 · Fan et al. (2023) doi:10.1103/PhysRevLett.130.071801
- Blais et al., RMP 93, 025005 (2021). arXiv:2005.12667
- Fermilab Muon g-2 Collaboration, final result (127 ppb), June 2025 (submitted to PRL); prior: PRL 126, 141801 (2021) and PRL 131, 161802 (2023)

Quantum Information Theory · F-qinfo

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The Shannon theory of quantum states: how much quantum information a state carries (von Neumann entropy $S = -\mathrm{Tr}\,\rho \log \rho$), how well it can be compressed (Schumacher coding), how much classical information quantum states can convey (Holevo bound), and what noisy quantum channels can transmit (quantum capacity). Fidelity, trace distance, and CPTP channels are the working vocabulary of every hardware benchmark.

CORE IDEA / KEY EQUATION

The central quantity is the von Neumann entropy $S(\rho) = -\mathrm{Tr}(\rho \log \rho)$, which is Shannon's $H = -\sum p \log p$ rewritten with the density matrix's eigenvalues as the probabilities. It counts, in qubits, the irreducible size of a quantum source: Schumacher's theorem says N copies of ρ compress into about $N \cdot S(\rho)$ qubits and no fewer, the quantum echo of Shannon's compression bound. When you want to read *classical* messages out of quantum states, the ceiling is the Holevo χ quantity: for an ensemble $\{p_i, \rho_i\}$ with average state $\rho = \sum_i p_i \rho_i$, the accessible information obeys $I \le \chi = S(\rho) - \sum_i p_i S(\rho_i)$. A corollary is blunt — n qubits carry at most n classical bits of retrievable message. For sending quantum states down a noisy channel $\mathcal{N}$, the analogue of Shannon capacity is the coherent information $I_c(\rho, \mathcal{N}) = S(\mathcal{N}(\rho)) - S((\mathcal{N} \otimes I)(|\psi\rangle\langle\psi|))$, and the quantum capacity is its regularized maximum (the LSD theorem). Regularization is the sting in the tail: unlike the classical case, you generally cannot compute the capacity from a single use of the channel.

WHY IT MATTERS FOR QUANTUM TECH

Channel capacities set the hard limits for quantum communication and the quantum internet, and they are why long links on photonic hardware (H-photonic) need entanglement distillation and repeaters rather than amplification (see F-nocloning). Fidelity and trace distance are how every gate, memory, and teleportation demo is scored across H-supercon, H-ion, and H-neutral, and they are the figures of merit that S-bench randomized benchmarking and S-qec threshold estimates report. The Holevo bound underwrites the key-rate accounting in QKD security proofs (S-qkd → A-qkd). Von Neumann entropy is also the bookkeeping unit for entanglement as a resource, which is the currency spent by teleportation, superdense coding, and the whole S-gates → S-qec stack.

KEY GRADED CLAIMS

- T1 A quantum source is compressible to $S(\rho)$ qubits per signal — Schumacher, PRA 51, 2738 (1995); entropy formalism from von Neumann (1932) (status: established)
- T1 n qubits cannot convey more than n classical bits — Holevo, Probl. Inf. Transm. 9, 177 (1973) (status: established)
- T2 Quantum channel capacity is given by regularized coherent information (LSD theorem) — Lloyd, PRA 55, 1613 (1997); Shor (2002); Devetak, IEEE TIT 51, 44 (2005) (status: established)
- T2 Channel capacities are strange: additivity fails (Hastings, Nat. Phys. 5, 255, 2009) and quantum capacity is superactivatable — two zero-capacity channels can combine to positive capacity (Smith & Yard, Science 321, 1812, 2008) (status: demonstrated)
- T2 Quantum information survives transmission at continental scale: ground-to-satellite teleportation of single-photon qubits over a 1,400 km uplink to the Micius satellite reached average fidelity 0.80 ± 0.01 across six mutually unbiased input states, above the $2/3$ classical state-estimation limit — Ren et al., Nature 549, 70 (2017), arXiv:1707.00934 (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- No closed-form, single-letter formula for the quantum capacity of a general channel; whether it is even computable in practice remains open.

GO DEEPER

- Nielsen & Chuang chs. 11–12; Wilde, *Quantum Information Theory* (2013), arXiv:1106.1445
- Preskill notes ch. 10

SOURCES

- Holevo (1973); Schumacher, PRA 51, 2738 (1995) doi:10.1103/PhysRevA.51.2738
- Devetak, IEEE Trans. Inf. Theory 51, 44 (2005). arXiv:quant-ph/0304127
- Hastings, Nat. Phys. 5, 255 (2009). arXiv:0809.3972 · Smith & Yard, Science 321, 1812 (2008). arXiv:0807.4935
- Ren et al., Nature 549, 70 (2017). arXiv:1707.00934

Quantum Thermodynamics · F-qthermo

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Thermodynamics rebuilt for single quantum systems, where work and heat are stochastic, information is physical, and coherence itself enters the energy books. Anchors: Landauer's principle (erasing one bit costs at least $kT \ln 2$), Bennett's resolution of Maxwell's demon via erasure cost, fluctuation theorems (Jarzynski, Crooks) that hold arbitrarily far from equilibrium, and resource-theoretic "second laws" for quantum states.

CORE IDEA / KEY EQUATION

The founding law is **Landauer's principle**: erasing one bit of information forces the environment to absorb at least $kT \ln 2$ of heat — about 3×10^{-21} J, or 0.018 eV, at room temperature ($T = 300$ K, k = Boltzmann's constant). Reversible computation dodges this floor; the cost lands only when information is discarded. That is how Bennett exorcised Maxwell's demon: the demon must eventually reset its memory, and that erasure pays back exactly the work it appeared to conjure. Away from equilibrium the accounting sharpens into the **Jarzynski equality**, $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$, with $\beta = 1/kT$. Read it plainly: take the stochastic work W done in any drive of the system — fast or slow, gentle or violent — repeat the drive many times, average e to the minus-βW over those runs, and the result equals e to the minus-β times the equilibrium free-energy change ΔF. A free-energy difference, an equilibrium quantity, is recovered from irreversible non-equilibrium pulls. Crooks' theorem refines this into a symmetry between the forward and reverse work distributions.

WHY IT MATTERS FOR QUANTUM TECH

Sets the energetic floor of computation and the physics of cooling: dilution refrigerators, algorithmic cooling, and the (open) question of the true energy cost per logical qubit of a fault-tolerant machine. Landauer's $kT \ln 2$ is the hard floor beneath every erasure in a quantum error-correction cycle — each syndrome reset and each measurement discard has a thermodynamic price that scales with the code's overhead (see S-qec, S-logical). Trapped-ion and superconducting platforms run their whole logic inside milli-kelvin baths, so the wall-plug cost of pumping that heat out dominates system power budgets (see H-ion, H-supercon). Quantum heat engines and absorption refrigerators are candidate on-chip coolers for keeping qubits near their ground state, and fluctuation theorems give the estimation tools for reading free energies out of the noisy, driven dynamics that quantum simulators produce (see S-qsim).

KEY GRADED CLAIMS

- T1 Erasure of one bit dissipates $\geq kT \ln 2$ — Landauer, IBM J. Res. Dev. 5, 183 (1961); demon exorcised by erasure cost, Bennett, Int. J. Theor. Phys. 21, 905 (1982) (status: established)
- T2 Landauer's bound verified experimentally — classically with a colloidal particle (Bérut et al., Nature 483, 187, 2012) and in the quantum regime with a trapped ion (Yan et al., PRL 120, 210601, 2018) (status: demonstrated)
- T2 Single-atom heat engine operated with one trapped ion — Roßnagel et al., Science 352, 325 (2016), doi:10.1126/science.aad6320 (status: demonstrated)
- T2 Quantum coherence measurably increases the power output of a microscopic engine in the small-action regime — Klatzow et al., PRL 122, 110601 (2019), NV-center working fluid in diamond (status: demonstrated)
- T2 Fluctuation theorems: $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$ — Jarzynski, PRL 78, 2690 (1997); quantum versions established (status: established)

CONFLICTS / OPEN QUESTIONS

- Definitions of work and heat for coherent quantum systems remain multiple and inequivalent; whether quantum coherence/entanglement give thermodynamic advantage in engines is case-by-case, still contested.

GO DEEPER

- Goold et al., "The role of quantum information in thermodynamics," J. Phys. A 49, 143001 (2016), arXiv:1505.07835
- Vinjanampathy & Anders, Contemp. Phys. 57, 545 (2016), arXiv:1508.06099

SOURCES

- Landauer (1961) doi:10.1147/rd.53.0183 · Jarzynski (1997) doi:10.1103/PhysRevLett.78.2690
- Roßnagel et al., Science 352, 325 (2016) · Yan et al., PRL 120, 210601 (2018)
- Klatzow et al., PRL 122, 110601 (2019). doi:10.1103/PhysRevLett.122.110601

The Qubit & Bloch Sphere · F-qubit

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The qubit is the unit of quantum information: any two-level quantum system, whose pure states are $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. Up to global phase, every pure qubit state maps to a point on the surface of a unit sphere — the Bloch sphere — with mixed states filling the interior. Single-qubit gates are rotations of this sphere. The term "qubit" was coined by Benjamin Schumacher in 1995; the geometric picture descends from Bloch's treatment of spin-½ and the Poincaré sphere for polarization.

CORE IDEA / KEY EQUATION

A pure qubit state is $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$, with $0 \le \theta \le \pi$ and $0 \le \varphi < 2\pi$. The two angles (θ, φ) are the polar and azimuthal coordinates of a point on a unit sphere, so every pure state is one point on the surface. The general (possibly mixed) state is the density matrix $\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$, where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices and $\mathbf{r}$ is the Bloch vector; $|\mathbf{r}| = 1$ is the pure-state surface and $|\mathbf{r}| < 1$ is the mixed interior, with $\mathbf{r} = 0$ the maximally mixed center. This works because a 2×2 Hermitian, unit-trace, positive matrix has exactly three real free parameters, the components of $\mathbf{r}$. A unitary gate acts as $U \rho U^\dagger$ and rotates $\mathbf{r}$ rigidly about some axis — the group is SU(2), the double cover of the rotation group SO(3), which is why a 2π turn of the Bloch vector corresponds to a 4π turn in SU(2) and picks up a physical sign. Two real numbers describe the state; one global phase is unobservable and drops out.

WHY IT MATTERS FOR QUANTUM TECH

The qubit is the abstraction that makes hardware interchangeable: transmons (H-supercon), trapped ions (H-ion), photons (H-photonic), neutral atoms (H-neutral), and spins in silicon (H-silicon) all implement the same Bloch-sphere object, so algorithms and error correction are written once against it. Every single-qubit gate in S-gates is a named Bloch rotation (X, Y, Z, Hadamard, T), and a gate's fidelity is literally the accuracy of that rotation, extracted by randomized benchmarking in S-bench. Error correction (S-qec) is built on the same picture — a logical qubit is a Bloch sphere protected inside many physical ones. Continuous-variable hardware breaks the two-level frame and uses an infinite-dimensional mode instead (H-bosonic).

KEY GRADED CLAIMS

- T1 Any two-level system's pure states are parameterized by two real angles (θ, φ) on a sphere; unitaries are SU(2) rotations — Bloch, Phys. Rev. 70, 460 (1946); Feynman, Vernon & Hellwarth, J. Appl. Phys. 28, 49 (1957) (status: established)
- T1 "Qubit" and the qubit as the quantum information unit: Schumacher, "Quantum coding," PRA 51, 2738 (1995) (status: established)
- T2 What a physical system needs to serve as a qubit register is codified in the DiVincenzo criteria — DiVincenzo, Fortschr. Phys. 48, 771 (2000), arXiv:quant-ph/0002077 (status: established)
- T2 A single physical qubit can be controlled at the six-nines level: a $^{43}Ca^{+}$ trapped-ion hyperfine "clock" qubit reached average single-qubit gate fidelity 99.9999% (error $\approx 1 \times 10^{-6}$), memory coherence $T_2^* = 50$ s, and combined prep+readout 99.93%, all in a room-temperature surface trap — Harty et al., PRL 113, 220501 (2014), arXiv:1403.1524 (status: demonstrated)
- T2 The record has since improved by an order of magnitude: single-qubit gate error $1.5(4) \times 10^{-7}$ in a trapped-ion qubit — Oxford group, arXiv:2412.04421 (2024) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Qudits (d-level) and continuous-variable/bosonic encodings compete with the two-level abstraction in some hardware (see H-bosonic); which unit wins per modality is an engineering question, still open.

GO DEEPER

- Nielsen & Chuang §1.2, §4.2; Preskill notes ch. 2

SOURCES

- Schumacher, PRA 51, 2738 (1995). doi:10.1103/PhysRevA.51.2738
- DiVincenzo (2000). arXiv:quant-ph/0002077
- Bloch, Phys. Rev. 70, 460 (1946). doi:10.1103/PhysRev.70.460
- Harty et al., PRL 113, 220501 (2014). arXiv:1403.1524 · Oxford single-qubit 10^{-7} gate, arXiv:2412.04421 (2024)

Stabilizer Formalism & Gottesman-Knill · F-stabilizer

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The stabilizer formalism describes a large, structured class of quantum states not by writing out 2^n amplitudes but by listing the operators that leave them fixed. A stabilizer state on n qubits is the unique common $+1$ eigenstate of a commuting group of n independent Pauli operators. Daniel Gottesman developed the formalism in his 1997 PhD thesis, building on the Heisenberg picture. It is the mathematical backbone of quantum error correction and, through the Gottesman-Knill theorem, marks a sharp boundary between quantum circuits that classical computers can simulate and those they cannot.

CORE IDEA / KEY EQUATION

A state $|\psi\rangle$ is stabilized by a Pauli operator S when $S|\psi\rangle = |\psi\rangle$. Fix a group $\mathcal{S} = \langle S_1, \ldots, S_n\rangle$ of n commuting, independent Pauli operators (excluding $-I$); their joint $+1$ eigenspace is a single state — the stabilizer state. This is exponential compression: n generators, each specified by $\sim 2n$ bits, pin down a state that would otherwise need 2^n complex amplitudes.

Gottesman-Knill theorem: a circuit built from state preparations in the computational basis, Clifford gates (Hadamard, phase S, and CNOT), and measurements in the Pauli basis can be simulated efficiently on a classical computer. The reason is that Clifford gates map Pauli operators to Pauli operators under conjugation, so instead of tracking the state you track how a set of $\sim n$ Pauli generators transforms — a polynomial-size bookkeeping problem. Aaronson & Gottesman (2004) sharpened this to an $O(n^2)$-per-step tableau algorithm (their CHP program handles thousands of qubits) and proved the simulation task is complete for the class $\oplus L$, evidence that stabilizer circuits are not even universal for classical computation, never mind quantum.

WHY IT MATTERS FOR QUANTUM TECH

Every mainstream quantum error-correcting code is a stabilizer code: the surface code, color codes, and the toric code all define the protected logical subspace as the joint +1 eigenspace of a set of stabilizer generators, and error detection is just measuring those generators to read a syndrome (see S-qec, S-logical). Gottesman-Knill also defines what a quantum computer must add to escape classical simulability: Clifford operations are free, and the non-Clifford resource — magic states, T gates, Wigner negativity, contextuality — is what has to be distilled at great cost (see F-contextuality, F-wigner). Stabilizer simulation is the standard tool for validating error-correction circuits and estimating logical error rates on near-term devices (see S-nisq, S-qsim).

KEY GRADED CLAIMS

- T1 Clifford circuits on stabilizer states with Pauli measurements are efficiently classically simulable (Gottesman-Knill) — Gottesman, PhD thesis, arXiv:quant-ph/9705052 (1997); Heisenberg-representation form arXiv:quant-ph/9807006 (1998) (status: established)
- T1 Stabilizer circuits simulate in $O(n^2)$ per gate and the problem is $\oplus L$-complete — Aaronson & Gottesman, PRA 70, 052328 (2004), arXiv:quant-ph/0406196 (status: established)
- T2 Adding any single non-Clifford gate (e.g. T) to the Clifford group yields a universal gate set — Boykin et al., 2000; standard result (status: established)

CONFLICTS / OPEN QUESTIONS

- Where exactly the classical-to-quantum boundary sits as non-Clifford resources are added is quantified by stabilizer rank and magic monotones, and tight bounds remain an active research target.
- Efficient simulation degrades gracefully with a few T gates (extended stabilizer / stabilizer-decomposition methods); the precise scaling frontier is still being pushed.

GO DEEPER

- Gottesman, "Stabilizer Codes and Quantum Error Correction," arXiv:quant-ph/9705052 (1997)
- Nielsen & Chuang, §10.5 (stabilizer formalism and codes)
- Aaronson & Gottesman, PRA 70, 052328 (2004)

SOURCES

- Gottesman, arXiv:quant-ph/9705052 (1997) · arXiv:quant-ph/9807006 (1998)
- Aaronson & Gottesman, PRA 70, 052328 (2004). doi:10.1103/PhysRevA.70.052328 · arXiv:quant-ph/0406196

Identical Particles & Spin-Statistics · F-statistics

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Identical quantum particles are indistinguishable: the joint state must be symmetric or antisymmetric under exchange of any two. The spin-statistics theorem (Fierz 1939, Pauli 1940, from relativistic QFT) fixes which: integer-spin particles are **bosons** (symmetric, can share a state) and half-integer-spin particles are **fermions** (antisymmetric, forbidden from sharing — the Pauli exclusion principle). Exchanging two particles multiplies the wavefunction by $+1$ (bosons) or -1 (fermions). In two dimensions the exchange phase can take any value $e^{i\theta}$, giving **anyons**, and non-abelian anyons carry a whole unitary matrix.

CORE IDEA / KEY EQUATION

The rule is a single sign on exchange: $\psi(r_1, r_2) = \pm\psi(r_2, r_1)$, with $+1$ **for bosons and** -1 **for fermions**. The minus sign is the whole Pauli principle — put two fermions in the same single-particle state and $\psi = -\psi$ forces $\psi = 0$, so that state cannot be occupied twice. Written as occupation numbers, the average filling of a level of energy ε follows the two distributions $n(\varepsilon) = 1/(e^{\beta(\varepsilon-\mu)} \mp 1)$: the minus sign (Bose-Einstein) lets occupation diverge and pile particles into the ground state (condensation), the plus sign (Fermi-Dirac) caps every level at one and stacks fermions up to the Fermi energy. In two dimensions the exchange is no longer a bare sign but a phase $e^{i\theta}$ that can sit anywhere between the boson value $\theta = 0$ and the fermion value $\theta = \pi$ — that is what "any-on" names. For a fractional quantum Hall state at filling $\nu = 1/m$ the quasiparticles carry charge e/m and exchange phase $\theta = \pi/m$. Non-abelian anyons replace the phase with a unitary matrix, so exchanging them rotates the state inside a degenerate subspace.

WHY IT MATTERS FOR QUANTUM TECH

Exchange antisymmetry is why atoms have shell structure and why chemistry exists at all — so any quantum simulation of molecules or materials must enforce fermionic statistics. Mapping fermionic modes onto qubits (Jordan-Wigner, Bravyi-Kitaev transforms) is a core primitive of quantum chemistry algorithms, and the Jordan-Wigner string overhead is a real cost driver in the circuit depth of those simulations (see S-qsim, S-variational, I-pharma, I-chem). Bosonic statistics underlie photonic quantum computing, where the Hong-Ou-Mandel effect (two indistinguishable photons bunching at a beam splitter) is the basic two-photon gate ingredient, and boson sampling is a near-term quantum-advantage benchmark (see H-photonic, S-bench). Non-abelian anyons are the entire premise of topological quantum computing: braiding them enacts fault-tolerant gates protected by topology, which is what Microsoft's Majorana program pursues (see H-topo, F-berry).

KEY GRADED CLAIMS

- T1 Integer spin → symmetric states (bosons); half-integer spin → antisymmetric states (fermions), forced by relativistic locality — Pauli, Phys. Rev. 58, 716 (1940) (status: established)
- T1 Two-dimensional systems admit anyonic exchange statistics — Leinaas & Myrheim (1977); Wilczek (1982); confirmed via fractional quantum Hall interferometry, Nakamura et al., Nature Physics 16, 931 (2020) (status: established/ demonstrated)
- T2 Anyon collider directly measures fractional exchange statistics at filling $\nu = 1/3$ — anyon-anyon collisions give an exchange phase $\theta \approx \pi/3$, matching the predicted $1/3$ statistics — Bartolomei et al., Science 368, 173 (2020) (status: demonstrated)
- T2 Bose-Einstein condensation realizes bosons macroscopically sharing one quantum state — ~2000 rubidium-87 atoms condensed at ~170 nK, Anderson et al., Science 269, 198 (1995), Nobel 2001 (status: established)

CONFLICTS / OPEN QUESTIONS

- Non-abelian anyons (Majorana zero modes) as computational objects remain contested at the device level — see C-majorana-existence (evidence/CONFLICTS.md) and H-topo.

GO DEEPER

- Wilczek, "Quantum Mechanics of Fractional-Spin Particles," PRL 49, 957 (1982)
- Nayak et al., "Non-Abelian anyons and topological quantum computation," RMP 80, 1083 (2008)

SOURCES

- Pauli, Phys. Rev. 58, 716 (1940). doi:10.1103/PhysRev.58.716
- Nakamura et al., Nature Physics 16, 931 (2020)
- Bartolomei et al., Science 368, 173 (2020). doi:10.1126/science.aaz5601
- Anderson, Ensher, Matthews, Wieman & Cornell, Science 269, 198 (1995). doi:10.1126/science.269.5221.198

Superposition & the State Vector · F-superpos

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

A quantum system is described by a state vector $|\psi\rangle$ in a Hilbert space, and any normalized linear combination of valid states is itself a valid state. Before measurement the system holds all components of the combination at once, with complex amplitudes; the amplitudes interfere, which is what separates quantum from classical probability. This is the first postulate of quantum mechanics and the raw resource behind everything downstream.

CORE IDEA / KEY EQUATION

$|\psi\rangle = \sum_i c_i|i\rangle$ with complex amplitudes c_i obeying $\sum_i |c_i|^2 = 1$. For a single qubit this is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$ — a point on the Bloch sphere (see F-qubit). The content is Dirac's superposition principle: add any two allowed states with complex weights and you get another allowed state. The weights carry a phase, and phases add and cancel — that is interference. A classical mixture sums probabilities; a quantum superposition sums amplitudes and squares the total (see the Born rule, F-measure), so cross terms $2\,\mathrm{Re}(c_i^* c_j)$ appear that have no classical analog. Everything quantum — interference fringes, entanglement, algorithmic speedup — traces back to this one linearity axiom.

WHY IT MATTERS FOR QUANTUM TECH

An n-qubit register in superposition spans 2^n amplitudes simultaneously — the state space quantum algorithms compute in. A physical qubit realizes $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ as two circulating supercurrent directions on a transmon (H-supercon), two hyperfine levels of an ion (H-ion), or two Rydberg/ground states of a neutral atom (H-neutral). Interference between amplitudes is how algorithms extract answers: Shor's algorithm (S-shor) routes the right period into a constructive peak of the quantum Fourier transform (F-qft), and Grover's search (S-grover) amplifies the marked amplitude by repeated reflection. Single-qubit gates (S-gates) are exactly the operations that rotate a superposition on the Bloch sphere. The frontier question — how large a superposition survives — is set by decoherence (F-decoher), which is why coherence time bounds every architecture.

KEY GRADED CLAIMS

- T1 Physical states superpose linearly; interference of single particles is observed directly — Schrödinger, Ann. Phys. 384, 361 (1926); Dirac, *Principles of Quantum Mechanics* (1930); single-electron double-slit buildup, Tonomura et al., Am. J. Phys. 57, 117 (1989) (status: established)
- T1 Matter-wave interference holds for whole molecules: C60 buckyballs (720 amu, de Broglie wavelength ~2.5 pm, ~400× smaller than the molecule) diffracted through a 100 nm-period grating — Arndt, Nairz, Vos-Andreae, Keller, van der Zouw, Zeilinger, Nature 401, 680 (1999), doi:10.1038/44348 (status: established)
- T2 Superposition persists in ever-larger objects: matter-wave interference demonstrated for functionalized oligoporphyrin molecules beyond 25 kDa (~2,000 atoms, ~2×10^4 amu) in a Talbot-Lau interferometer — Fein et al., Nat. Phys. 15, 1242 (2019), doi:10.1038/s41567-019-0663-9 (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Where (if anywhere) superposition breaks for macroscopic objects is the empirical frontier — objective-collapse models predict a scale limit; interferometry keeps pushing it upward (see F-interp, F-decoher).

GO DEEPER

- Dirac, *Principles of QM*, ch. 1 (the superposition principle stated as the axiom)
- Nielsen & Chuang, *Quantum Computation and Quantum Information*, §2.1
- Arndt & Hornberger, Nat. Phys. 10, 271 (2014) — testing the limits review

SOURCES

- Fein et al., "Quantum superposition of molecules beyond 25 kDa," Nat. Phys. 15, 1242–1245 (2019). doi:10.1038/s41567-019-0663-9
- Arndt et al., "Wave–particle duality of C60 molecules," Nature 401, 680–682 (1999). doi:10.1038/44348
- Tonomura et al., Am. J. Phys. 57, 117 (1989). doi:10.1119/1.16104

Teleportation & Entanglement Swapping · F-teleport

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Quantum teleportation transfers an unknown state from Alice to Bob using one shared entangled (EPR) pair and two classical bits — no quantum system travels between them, and the original is destroyed (consistent with no-cloning, see F-nocloning). Alice performs a joint Bell-state measurement on her qubit and her half of the pair, sends the two-bit outcome, and Bob applies one of four Pauli corrections to recover the exact state. Bennett, Brassard, Crépeau, Jozsa, Peres & Wootters proposed it in 1993. **Entanglement swapping** (Żukowski, Zeilinger, Horne & Ekert, 1993) is teleportation applied to a member of another entangled pair, so two particles that never interacted end up entangled — the primitive that lets a repeater stitch short entangled links into a long one.

CORE IDEA / KEY EQUATION

The identity that makes it work is a rewriting of three qubits. Alice holds the unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and one half of a shared Bell pair; Bob holds the other half. Expand the joint three-qubit state in the **Bell basis** of Alice's two qubits and it factors exactly into four terms: $|\psi\rangle|\Phi^+\rangle_{AB} = \frac{1}{2}\sum_i (\text{Alice's Bell state})_i \otimes (U_i|\psi\rangle)_{\text{Bob}}$, where the four U_i are the Pauli operators I, X, Z, XZ. Alice's Bell measurement collapses the sum to one term at random and tells her which i occurred (two classical bits). Bob's qubit is then $U_i|\psi\rangle$, so applying U_i^{-1} — one of I, X, Z, or ZX — recovers $|\psi\rangle$ exactly. The two classical bits are mandatory: before Bob hears them his qubit is the maximally mixed state $I/2$, carrying no information, which is why teleportation transmits nothing faster than light and never clones (α and β are never measured, only moved). Entanglement swapping is the same identity with $|\psi\rangle$ itself half of another Bell pair.

WHY IT MATTERS FOR QUANTUM TECH

Teleportation is the workhorse of quantum networking and modular computing. Entanglement swapping is how quantum repeaters beat photon loss to build a long-range quantum internet (see A-qinternet). Teleporting gates ("gate teleportation") moves logical information between error-correction blocks and is how magic states are consumed in fault-tolerant circuits — the T-gate that universalizes the Clifford group is applied by teleporting through a magic state (see S-logical, S-qec, F-contextuality). Chip-to-chip and fridge-to-fridge links in modular architectures rely on teleported entanglement (see H-intercon, O-interconnect-loss), and the same photonic Bell-measurement primitive is the fusion operation in measurement-based photonic computing (see H-photonic). Distributed multi-node quantum computing over ion or NV registers stitches processors together by teleporting logical qubits between them (see H-ion). First photonic demonstrations came in 1997 (Innsbruck; Rome).

KEY GRADED CLAIMS

- T1 An unknown state is teleported with one EPR pair + two classical bits — Bennett et al., PRL 70, 1895 (1993) (status: established)
- T1 Entanglement swapping entangles independent, never-interacting particles — Żukowski, Zeilinger, Horne & Ekert, PRL 71, 4287 (1993) (status: established)
- T2 Deterministic teleportation between distant nodes demonstrated — Pfaff et al., Science 345, 532 (2014, NV centers 3 m); ground-to-satellite, Ren et al., Nature 549, 70 (2017) (status: demonstrated)
- T2 Free-space teleportation over 143 km with active feed-forward, fidelity above the $2/3$ classical bound — Ma et al., Nature 489, 269 (2012), La Palma to Tenerife (status: demonstrated)
- T2 Teleportation between non-neighboring nodes of a three-node quantum network, using entanglement swapping on the middle node plus a memory qubit — Hermans et al., Nature 605, 663 (2022), NV-center registers in Delft (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- None foundational. Engineering frontier is fidelity and rate over lossy links and quantum memories (see A-qmemory-hw, O-interconnect-loss).

GO DEEPER

- Bennett & Wiesner dense coding, PRL 69, 2881 (1992) (dual protocol)
- Pirandola et al., "Advances in quantum teleportation," Nature Photonics 9, 641 (2015)

SOURCES

- Bennett et al., PRL 70, 1895 (1993). doi:10.1103/PhysRevLett.70.1895
- Żukowski et al., PRL 71, 4287 (1993). doi:10.1103/PhysRevLett.71.4287
- Ma et al., Nature 489, 269 (2012). doi:10.1038/nature11472

- Hermans et al., Nature 605, 663 (2022). doi:10.1038/s41586-022-04697-y

Quantum Tunneling · F-tunneling

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

A particle described by a wavefunction has nonzero amplitude on the far side of an energy barrier it could not classically climb, so it can appear there with finite probability. The transmission amplitude falls off exponentially with barrier width and the square root of barrier height. Tunneling was worked out in 1928 by Gamow, and independently Gurney & Condon, to explain alpha decay — the alpha particle escapes the nucleus by tunneling through the Coulomb barrier — and by Fowler & Nordheim for field emission. It is a direct consequence of the Schrödinger equation, with no classical analogue.

CORE IDEA / KEY EQUATION

Inside a barrier taller than the particle's energy, the Schrödinger equation gives an evanescent (decaying) wave rather than an oscillating one, so the transmission probability through a barrier of height V_0 and width d for a particle of energy E is $T \approx e^{-2\kappa d}$, with the decay constant $\kappa = \sqrt{2m(V_0 - E)}/\hbar$. Two features do the work: the dependence is **exponential in width** (double the barrier and the tunneling rate does not halve — it drops by a huge factor), and κ scales with the **square root of the barrier height** above the particle's energy. This exponential sensitivity is why alpha-decay half-lives span more than twenty orders of magnitude for a narrow range of decay energies (the Geiger-Nuttall law), and why a scanning tunneling microscope resolves single atoms: the tunneling current changes by about a decade for every 1 Å the tip moves. The Josephson junction is the coherent version, carried by the tunneling amplitude of Cooper pairs rather than the incoherent rate of single particles.

WHY IT MATTERS FOR QUANTUM TECH

Tunneling is the physical mechanism inside the Josephson junction: Cooper pairs tunnel coherently across a thin insulating barrier, and the resulting nonlinear inductance is what makes a transmon or flux qubit a usable anharmonic oscillator (see H-supercon). The same $I = I_c \sin\varphi$ relation sets the qubit frequency, the coupling to readout resonators, and the parametric amplifiers that read the qubits out. It is also the readout physics of the scanning tunneling microscope (Binnig & Rohrer, 1981 Nobel 1986) used to image and place the individual dopants and defects that seed some qubit platforms, the operating principle of flash-memory Fowler-Nordheim programming, and the leakage that limits classical transistor scaling. In quantum annealing, tunneling through (rather than thermal hopping over) energy barriers is the proposed source of any advantage (see H-anneal, F-adiabatic), and tunneling matrix elements govern the reaction and electron-transfer rates that molecular quantum simulators aim to compute (see S-qsim, I-chem, I-pharma).

KEY GRADED CLAIMS

- T1 A quantum particle transmits through a classically forbidden barrier with exponentially suppressed probability — Gamow (1928); Gurney & Condon, Nature 122, 439 (1928) (status: established)
- T1 Cooper-pair tunneling across a Josephson junction obeys $I = I_c \sin\varphi$, $d\varphi/dt = 2eV/\hbar$ — Josephson, Phys. Lett. 1, 251 (1962), Nobel 1973 (status: established)
- T2 The scanning tunneling microscope resolves single atoms because the tunneling current decays ~1 order of magnitude per 1 Å of tip-sample gap, giving ~0.1 nm lateral and ~0.01 nm vertical resolution — Binnig, Rohrer, Gerber & Weibel, PRL 49, 57 (1982), Nobel 1986 (status: established)
- T3 The tunneling delay time in strong-field ionization of atomic hydrogen is zero within an uncertainty of 12 attoseconds (attoclock) — Sainadh et al., Nature 568, 75 (2019) (status: demonstrated, interpretation debated)

CONFLICTS / OPEN QUESTIONS

- Tunneling *time* — how long a particle "spends" in the barrier — remains debated (Hartman effect, attoclock experiments); no impact on device physics.

GO DEEPER

- Razavy, *Quantum Theory of Tunneling* (2003)
- Tinkham, *Introduction to Superconductivity*, ch. 6 (Josephson effects)

SOURCES

- Gamow, Z. Phys. 51, 204 (1928)
- Josephson, Phys. Lett. 1, 251 (1962). doi:10.1016/0031-9163(62)91369-0
- Binnig, Rohrer, Gerber & Weibel, PRL 49, 57 (1982). doi:10.1103/PhysRevLett.49.57 · Nobel Prize in Physics 1986
- Sainadh et al., Nature 568, 75 (2019). doi:10.1038/s41586-019-1028-3

Uncertainty & Complementarity · F-uncertainty

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Conjugate observables (position/momentum, phase/number, orthogonal spin axes) cannot both have sharp values in the same state: $\Delta x\,\Delta p \geq \hbar/2$. Heisenberg found the physics in 1927; Kennard and Robertson made it a theorem about state preparation — the bound follows from the operators failing to commute, with no reference to measurement clumsiness. Bohr's complementarity is the companion idea: wave and particle descriptions are mutually exclusive but jointly necessary, and which one shows up depends on the experimental arrangement.

CORE IDEA / KEY EQUATION

The sharp statement is Robertson's: for any two observables A and B in any state, $\Delta A \cdot \Delta B \geq \frac{1}{2}|\langle[A,B]\rangle|$, where ΔA is the standard deviation and $[A,B] = AB - BA$ is the commutator. For position and momentum $[x,p] = i\hbar$, so the right side is $\hbar/2$ and you recover $\Delta x\,\Delta p \geq \hbar/2$. The content is that the bound comes from the algebra — non-commuting operators cannot share a full set of eigenstates, so no state makes both spreads zero. This is a fact about how states are prepared, with no clumsy-measurement story attached; the Cauchy–Schwarz inequality applied to the two operators acting on the state is the whole proof. The entropic version replaces variances (which behave badly for periodic or discrete observables) with Shannon entropies of the measurement outcomes: $H(A) + H(B) \geq \log(1/c)$, where C is the largest squared overlap between the two observables' eigenbases. The entropic form is state-independent and is the one QKD proofs actually use.

WHY IT MATTERS FOR QUANTUM TECH

Uncertainty relations set the noise floor of quantum sensing — the standard quantum limit — and they define what squeezing can and cannot buy: pushing noise below the limit in one quadrature by paying it back in the conjugate one. This is the physics LIGO's squeezed-light injection exploits, and the same trade governs bosonic sensing and error correction (H-bosonic). The entropic form underwrites the security proofs behind QKD (S-qkd → A-qkd), where an eavesdropper's information about one basis is bounded by their ignorance of the conjugate basis. Complementarity — the impossibility of reading which-path information without killing the interference — is the operational lever behind interferometric sensors and photonic protocols (H-photonic).

KEY GRADED CLAIMS

- T1 Preparation uncertainty: $\Delta A\,\Delta B \geq \frac{1}{2}|\langle[A,B]\rangle|$ — Heisenberg, Z. Phys. 43, 172 (1927); Kennard, Z. Phys. 44, 326 (1927); Robertson, Phys. Rev. 34, 163 (1929) (status: established)
- T1 Entropic form: $H(A) + H(B) \geq \log(1/c)$ — Maassen & Uffink, PRL 60, 1103 (1988); extended with quantum memory, Berta et al., Nat. Phys. 6, 659 (2010) (status: established)
- T2/T3 Measurement error–disturbance versions of the relation are formalization-dependent: Ozawa's reformulation (PRA 67, 042105, 2003) was supported by neutron/photon experiments, while Busch–Lahti–Werner (PRL 111, 160405, 2013) prove a Heisenberg-type bound under different error definitions (status: contested)
- T2 Squeezing beats the quantum noise floor in a working instrument: frequency-dependent squeezed vacuum injected into the LIGO detectors during observing run O4 cut quantum noise by 5.8 dB (a factor of 1.9) at Livingston and 4.0 dB at Hanford, surpassing the standard quantum limit by up to 3 dB in the 35–75 Hz band — LIGO O4 Detector Collaboration, "Broadband Quantum Enhancement of the LIGO Detectors with Frequency-Dependent Squeezing," PRX 13, 041021 (2023) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- The error–disturbance dispute is about which operational definition of "measurement error" is the right one — both camps' theorems are correct under their own definitions.

GO DEEPER

- Coles et al., "Entropic uncertainty relations and their applications," RMP 89, 015002 (2017)

SOURCES

- Heisenberg (1927) doi:10.1007/BF01397280 · Robertson (1929) doi:10.1103/PhysRev.34.163
- Ozawa, PRA 67, 042105 (2003) · Busch, Lahti, Werner, PRL 111, 160405 (2013). arXiv:1306.1565
- LIGO O4 Detector Collaboration, PRX 13, 041021 (2023). doi:10.1103/PhysRevX.13.041021

Wigner Functions & Quasiprobability · F-wigner

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

A Wigner function $W(x,p)$ represents a quantum state as a real distribution over phase space, the closest quantum analog of a classical probability density. Eugene Wigner introduced it in 1932 to compute quantum corrections to classical statistical mechanics. Its defining feature: unlike a genuine probability density, W can take negative values. Those negative regions are the phase-space signature of nonclassicality. The construction generalizes to finite-dimensional (qudit) systems through the discrete Wigner function of Gross and Wootters, where the same negativity plays the same role.

CORE IDEA / KEY EQUATION

$$W(x,p) = \frac{1}{\pi\hbar} \int \langle x+y|\rho|x-y\rangle \, e^{-2ipy/\hbar} \, dy$$

a real function whose marginals recover the correct position and momentum distributions, so integrating W reproduces every measurable statistic. What breaks the classical picture is that $W(x,p) < 0$ in some regions for most states.

Hudson's theorem (1974) pins down the boundary: a pure continuous-variable state has a non-negative Wigner function if and only if it is a Gaussian (coherent, squeezed, or thermal-pure) state. Everything with negativity — Fock states, cat states, GKP states — sits outside that Gaussian island. In the odd-prime-dimension discrete case (Gross 2006), the non-negative pure states are exactly the stabilizer states, tying Wigner positivity to the stabilizer formalism (see F-stabilizer). Negativity is then a resource: Wigner-non-negative states, Gaussian/Clifford operations, and positive measurements are efficiently classically simulable, so any quantum speed-up must inject negativity somewhere.

WHY IT MATTERS FOR QUANTUM TECH

Wigner negativity is the continuous-variable currency of quantum advantage. In photonic and bosonic hardware (see H-photonic, H-bosonic), Gaussian states and operations alone are classically simulable; a non-Gaussian, Wigner-negative element (photon subtraction, a cubic phase gate, a GKP state) is what lifts the platform to universality. In the qudit circuit model, negativity of the discrete Wigner function is the same resource that magic-state distillation must manufacture (see F-contextuality, S-qec, S-logical). Wigner negativity is also a practical witness for benchmarking non-classical state preparation on NISQ and bosonic devices (see S-nisq, S-qsim).

KEY GRADED CLAIMS

- T1 A pure state has a non-negative Wigner function iff it is Gaussian — Hudson, Rep. Math. Phys. 6, 249 (1974); qudit extension Gross, J. Math. Phys. 47, 122107 (2006), arXiv:quant-ph/0602001 (status: established)
- T2 Wigner negativity is a necessary resource for super-polynomial quantum speed-up; non-negative states + operations are efficiently classically simulable — Mari & Eisert, PRL 109, 230503 (2012), arXiv:1208.3660; Veitch, Ferrie, Gross & Emerson, New J. Phys. 14, 113011 (2012), arXiv:1201.1256 (status: established)
- T2 In odd prime dimension, non-negative discrete Wigner functions characterize exactly the stabilizer states — Gross (2006) (status: established)

CONFLICTS / OPEN QUESTIONS

- The qubit ($d = 2$) case is subtle: the standard discrete Wigner construction fails to make stabilizer states non-negative, so the clean "negativity = magic" statement holds cleanly only for odd dimension. Rebit and qubit refinements tie negativity to contextuality instead (see F-contextuality).
- Whether Wigner negativity, contextuality, and magic are the *same* resource or merely coincident measures is still being sharpened.

GO DEEPER

- Veitch, Ferrie, Gross & Emerson, New J. Phys. 14, 113011 (2012)
- Kenfack & Życzkowski, "Negativity of the Wigner function as an indicator of nonclassicality," J. Opt. B 6, 396 (2004)

SOURCES

- Wigner, Phys. Rev. 40, 749 (1932). doi:10.1103/PhysRev.40.749
- Hudson, Rep. Math. Phys. 6, 249 (1974). doi:10.1016/0034-4877(74)90007-X
- Gross, J. Math. Phys. 47, 122107 (2006). arXiv:quant-ph/0602001
- Veitch et al., New J. Phys. 14, 113011 (2012). arXiv:1201.1256 · Mari & Eisert, PRL 109, 230503 (2012). arXiv:1208.3660

Chapter 2 · The Machines

Quantum annealing · H-anneal

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

A special-purpose analog approach that rests directly on the adiabatic theorem (`F-adiabatic`): encode an optimization problem in the couplings of a network of superconducting flux qubits, initialize in the ground state of a simple transverse-field Hamiltonian, then slowly interpolate to the problem Hamiltonian so the system stays near its instantaneous ground state and settles into a (near-)minimum-energy configuration. There are no gates and no universal computation — the machine natively solves Ising/QUBO problems and samples from hard distributions. It is not a gate-model computer wearing a costume; it is a distinct model of computation, and one company defines the commercial category.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **D-Wave**: Advantage2 reached general availability May 2025 — 4,400+ qubits on the Zephyr topology (20-way qubit connectivity), with higher per-qubit coherence and a larger energy scale than Advantage1 (5,000+ qubits, Pegasus, 15-way). Delivered via the Leap cloud; hybrid solvers push problems up to millions of variables by splitting work between the QPU and classical heuristics. Gate-model annealing-adjacent research (coherent annealing) is also on D-Wave's roadmap.
- **Advantage claim**: D-Wave's March 2025 Science paper reported quantum-annealer simulation of quantum magnetic phase-transition dynamics (spin glasses) it argued was beyond reach of classical supercomputers. Within days, groups in Switzerland (EPFL) and the US posted arXiv counter-papers reproducing key results with tensor-network / belief-propagation methods on classical hardware; D-Wave rebutted that the classical methods only cover part of the parameter regime and time scales. Per the schema's default, the advantage stays **contested**.

KEY GRADED CLAIMS

- T4 Advantage2 GA: 4,400+ qubits, Zephyr, higher coherence — D-Wave release + whitepaper (claimed)
- [T2/contested] Quantum advantage in spin-glass dynamics simulation — Science (Mar 2025); classical tensor-network counterattacks on arXiv (contested)
- T5 Commercial utility of annealing for production optimization — vendor case studies (claimed; few third-party-audited head-to-head wins)

TRADE-OFFS (VS OTHER MODALITIES)

Thousands of qubits available today and real paying deployments years ahead of any gate-model rival — the modality shipped a usable product first. Against that: it is restricted to optimization/sampling problem classes, has no error correction and no threshold theorem, shows no proven scaling of *solution quality* with qubit count, and every headline speedup so far has been at least partially matched by tuned classical heuristics (simulated/parallel tempering, tensor networks).

CONFLICTS / OPEN QUESTIONS

C-anneal-advantage (also **C-dwave-advantage**): does any annealing workload hold a durable quantum advantage once classical heuristics are tuned against it? Resolution would be a widely reproduced benchmark where the annealer wins on time-to-solution at fixed solution quality against best-known classical methods, audited by third parties — which has not yet happened.

SOURCES

dwavequantum.com GA release + 4,400-qubit whitepaper; Science (Mar 2025); EPFL/US arXiv counter-papers; Physics World + Scientific American + HPCwire contest coverage.

Bosonic / cat qubits · H-bosonic

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Qubits encoded in the many-level state space of a harmonic oscillator (a superconducting cavity or resonator mode) instead of a two-level system, so a single physical mode carries redundancy that a normal qubit does not. **Cat qubits** store information in superpositions of two coherent states $|\alpha\rangle$ and $|-\alpha\rangle$; a two-photon drive stabilizes them so that bit-flips are suppressed *exponentially in the photon number* $|\alpha|^2$, leaving only phase-flips for a simple 1D repetition code to catch — turning a 2D error-correction problem into a 1D one. GKP (grid) and binomial codes are sibling encodings that protect against small displacements. The goal across all of them: collapse the physical-per-logical overhead that makes the surface code so brutal.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Alice & Bob** (Paris): Boson 4 chip reached >1 hour bit-flip lifetime (Sep 2025, up from 7 minutes in 2024); the "squeezed cat" scheme cut correction cost (2025); the "Elevator Codes" preprint (Jan 2026) claims a ~15:1 physical:logical ratio at ~10,000× lower logical error. Raised €100M (Jan 2025); roadmap: 100 logical qubits by 2030.
- **AWS**: Ocelot chip (Nature, Feb 2025) — first integrated multi-cat-qubit chip combining cat stabilization with a distance-5 repetition code; bit-flip times approaching 1 s; AWS claims up to ~90% QEC-overhead reduction vs surface codes for a target logical error rate.
- **Nord Quantique** (Sherbrooke): multimode bosonic (GKP/binomial) encoding in 3D cavities; demonstrated error correction beyond break-even and pitches high logical density per physical unit — a single mode doing the work of a small code block.
- Yale (the modality's academic origin — Devoret/Schoelkopf lineage) continues to set GKP and cavity-coherence records.

KEY GRADED CLAIMS

- T2 Ocelot: cat-qubit + repetition-code chip, ~1 s bit-flip times — AWS, Nature (2025) (demonstrated)
- T4 Boson 4: >1 hr bit-flip lifetime — Alice & Bob release (claimed)
- T3 Elevator codes: ~15:1 overhead, 10,000× lower error — Alice & Bob preprint, Jan 2026 (claimed, theory + partial demo)
- T4 100 logical qubits by 2030 — Alice & Bob roadmap (roadmap)

TRADE-OFFS (VS OTHER MODALITIES)

Reuses the entire transmon fab/cryo stack (`H-fab`, `H-cryo`) but promises an order-of-magnitude smaller QEC overhead *if* the noise-bias holds under operation; against that, phase-flips remain and must still be corrected, gate sets on biased-noise qubits are awkward (bias-preserving CNOTs are the hard case), and no bosonic machine has yet run multi-logical-qubit algorithms. Most overhead numbers are extrapolations from single-digit-qubit demos.

CONFLICTS / OPEN QUESTIONS

Does the exponential bit-flip suppression survive once you do *fast gates* on the cat rather than just storing it — i.e., does the noise bias hold during computation, not only in memory? That is the load-bearing assumption behind every overhead claim in this node.

SOURCES

AWS Ocelot (Nature, Feb 2025); alice-bob.com newsroom (squeezed cat, Boson 4, Elevator Codes preprint); postquantum.com cat-qubit + Nord Quantique analyses; The Quantum Insider (Mar 2025).

Control electronics & I/O · H-control

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The classical layer that makes qubits do anything: arbitrary-waveform generators and FPGAs synthesizing shaped microwave pulses (typically upconverted from an IF), fast ADCs digitizing the readout, and real-time feedback engines that must decode errors and branch mid-circuit within qubit coherence times. In the NISQ era this is signal generation; in the fault-tolerant era the controller becomes a real-time computer running a QEC decoder inside a latency budget of hundreds of nanoseconds to a microsecond per cycle (the decoder-throughput wall — see `S-decoders`, `O-decoder`). The layer is modality-agnostic "picks and shovels" revenue: every architecture in this chapter needs it.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Quantum Machines** (Israel): OPX family with the QUA language for sub-microsecond real-time classical control flow; raised $170M Series D (Feb 2025); co-built **DGX Quantum** with Nvidia — a Grace-Hopper GPU tightly coupled to the controller for QEC-era feedback and calibration, targeting few-µs round-trip.
- **Zurich Instruments** (Switzerland, Rohde & Schwarz): SHFQC integrated controllers; launched the **ZQCS** platform (Mar 2026) explicitly aimed at operating long-lived logical qubits at large scale, with distributed synchronization across many units.
- **Qblox** (Delft): modular Cluster series; ~$32M raised; the "open architecture" stack via partnerships with Bluefors (cryo integration), Q-CTRL (autonomous calibration), and QuantWare. **Keysight**: QCS platform plus its test-and-measurement moat. **Zurich/Keysight/QM** dominate room-temperature control.
- Frontier direction: push the controller into the cold as cryo-CMOS (`H-cryocmos`) to escape one-coax-per-qubit; couple GPUs for decoding (Nvidia CUDA-Q, `S-software`).

KEY GRADED CLAIMS

- T4 ZQCS targets large-scale logical-qubit operation — Zurich Instruments launch (Mar 2026) (claimed)
- T4 Sub-µs real-time feedback + GPU-coupled decoding (DGX Quantum) — QM/Nvidia (demonstrated in deployments; benchmarks vendor-supplied)
- T5 Quantum funding context: ~$3.77B raised Jan–Sep 2025, ~3× all of 2024 — trade press (reported/forecast)

TRADE-OFFS (VS OTHER APPROACHES)

Room-temperature racks are flexible and vendor-agnostic but cost on the order of ~$1k+ per qubit-channel and drown in cabling past a few thousand channels; cryo-CMOS cuts the wire count but must run on a milliwatt heat budget with re-characterized transistor models. Real-time decoding forces a hard co-design between the controller, the decoder ASIC/FPGA, and the QEC cycle time — latency, not raw compute, is the binding constraint.

CONFLICTS / OPEN QUESTIONS

Can decoders and controllers keep up with surface-code cycle times (~1 µs) across 1,000+ logical qubits — the decoder-throughput wall? Will hyperscaler/GPU vendors (Nvidia) absorb this layer into their accelerated-computing stack, or does it stay a specialist market?

SOURCES

Zurich Instruments ZQCS (The Quantum Insider, Mar 2026); Quantum Machines / Nvidia DGX Quantum; qblox.com + PitchBook; postquantum.com enabling-technologies analysis.

Cryogenics & dilution refrigerators · H-cryo

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Superconducting, silicon-spin, and bosonic qubits live at 10–20 millikelvin — colder than deep space — reached by dilution refrigerators that exploit the finite solubility of ^3He in ^4He: pumping ^3He across the phase boundary in the mixing chamber absorbs heat continuously. Modern "dry" (cryogen-free) fridges reach the 4 K stage with pulse-tube coolers instead of a liquid-helium bath, which is what made mK an engineering commodity rather than a specialist art. The fridge sets three hard limits on qubit count: **cooling power** at the mixing-chamber stage (a few hundred µW to ~1 mW at 20 mK), **internal volume** for the sample and wiring, and the **number of coax lines** that can be routed from 300 K to the mixing chamber without dumping heat — the passive-wiring bottleneck (coax, attenuators, isolators — see `H-wiring`) that cryo-CMOS (`H-cryocmos`) exists to break.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Bluefors** (Finland): the market anchor; with Oxford Instruments it holds >70% of the dilution-refrigerator market (ICV 2025). The **KIDE** platform is purpose-built for large-scale QC — 1,000+ qubit support, hexagonal side-loading, and multi-unit clustering; IBM Quantum System Two runs on it. Ultra-Compact LD launched Feb 2025 for space-constrained labs.
- **Oxford Instruments NanoScience** (UK): ProteoxMX/LX line, the other half of the duopoly. **Maybell Quantum** (US): the "Icebox" packing dense wiring (thousands of lines) into one fridge. **Leiden Cryogenics**, **Zero-Point Cryogenics** (Canada), and Chinese entrants (e.g. domestic supply for USTC programs) round out the field.
- **Market**: laboratory dilution-fridge market ~$320M (2025), projected ~$520M by 2034 — quantum programs are the main growth driver.

KEY GRADED CLAIMS

- T5 Bluefors + Oxford Instruments >70% market share — ICV Global Dilution Refrigerator Report 2025 (claimed)
- T4 KIDE supports 1,000+ qubit systems — Bluefors (claimed; IBM deployment corroborates at current scales)
- T5 Market ~$320M (2025) → ~$520M (2034) — analyst forecasts (forecast)
- T1 Dilution refrigeration reaches ~10 mK continuously via $^3He/^4He$ mixing — established cryogenics

TRADE-OFFS (VS OTHER MODALITIES/COMPONENTS)

Dry fridges made mK routine, but each ~1 mW of cooling power at 20 mK costs on the order of ~25 kW at the wall — the thermodynamic tax on superconducting QC. Wiring heat-load per qubit forces cryo-CMOS control, photonic I/O, or aggressive multiplexing before a fridge can hold thousands of qubits. Ion, neutral-atom, photonic, and NV modalities skip dilution fridges entirely (photon detectors still want a 2–4 K stage — see `H-detect`).

CONFLICTS / OPEN QUESTIONS

Can a single fridge + its wiring scale to million-qubit superconducting machines, or does the endgame require chip-to-chip links across *many* fridges (`H-intercon`), which in turn needs microwave-optical transduction (`H-transduce`) that does not yet exist? **^3He supply** — a byproduct of tritium decay in nuclear-weapons stockpiles — is a strategic chokepoint rarely priced into roadmaps (see `E-supplychain`).

SOURCES

bluefors.com (KIDE, Ultra-Compact LD); The Quantum Insider (Feb 2025); ICV Global Dilution Refrigerator Report 2025; Maybell Icebox coverage; intelmarketresearch/valuates market reports.

Cryogenic control ASICs · H-cryocmos

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Every qubit needs control and readout wires running from room-temperature electronics down into the fridge. At thousands of qubits this **wiring bottleneck** turns fatal: heat load down the coax, cabling bulk, and connector count all explode past what a dilution fridge (`H-cryo`) can absorb. The fix is **cryo-CMOS** — control/readout ASICs fabricated in standard silicon that run *inside* the cryostat, either at the 4 K stage or, ideally, alongside the qubits at ~10 mK–100 mK — collapsing the wire count by multiplexing many qubits per line and moving the classical/quantum boundary down into the cold. It is a prerequisite for any million-qubit superconducting or silicon-spin machine, and it borrows the entire CMOS supply chain (`H-foundry`) rather than inventing new fab.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Intel — Horse Ridge I/II**: cryo-CMOS SoCs generating microwave qubit-drive pulses at ~4 K; Horse Ridge II added integrated readout and multi-gate pulsing. Intel's Pando Tree targets scale-up interconnect.
- **SemiQon** (Finland): cryo-CMOS transistors (launched late 2024) claiming control/readout of hundreds of qubits in one cooldown at ~100× lower power than room-temperature electronics; pursues co-locating electronics on the qubit chip.
- **Equal1** (Ireland): UnityQ / Bell-1 rack-mounted silicon-spin system with control integrated in GlobalFoundries 22FDX FD-SOI CMOS; validated the CMOS process April 2025; Bell-1 headed to ESA's Φ-lab (2025).
- **Diraq** (with Emergence Quantum) and **QuTech** (with Intel) also active. Silicon spin qubits (`H-silicon`) benefit most: same-die integration is native to their CMOS-compatible modality, whereas transmons must place the ASIC at a separate cold stage.

KEY GRADED CLAIMS

- T3 Cryo-CMOS control of multiple silicon quantum dots in a commercial CMOS process — Equal1, GlobalFoundries 22FDX (demonstrated/validated)
- T4 SemiQon cryo-CMOS controls hundreds of qubits per cooldown at ~100× lower power — company claim (claimed)
- T4 Horse Ridge II integrates drive + readout + multi-gate pulsing at 4 K — Intel (demonstrated on Intel's own stack)

TRADE-OFFS

The core tension is **power vs proximity**: dissipation in the ASIC heats the cold stage, so the nearer the qubits (mK, where cooling power is ~µW–mW) the tighter the power budget. Running at 4 K eases the thermal budget (cooling power ~W) but reintroduces a wiring hop back down to the qubits. Transistors misbehave at cryo — threshold shifts, flicker noise, and carrier freeze-out all require re-characterized device models. Yield and reliability of cryo-CMOS at scale are unproven, and a failure inside a cold fridge is expensive to reach.

CONFLICTS / OPEN QUESTIONS

Can cryo-CMOS dissipate little enough to sit at the mixing-chamber stage, or is 4 K the practical floor — leaving a residual wiring problem between 4 K and the qubits? Does same-die integration advantage silicon spin qubits enough to offset their smaller lead in qubit quality versus transmons and ions?

SOURCES

intc.com (Horse Ridge II); semiqon.com/technology; quantumzeitgeist.com (SemiQon); thequantuminsider.com 2025/04/17 (Equal1 CMOS validation); design-reuse.com; qutech.nl; arXiv:2604.16216.

Single-photon detectors / SNSPDs · H-detect

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Photonic quantum computing, QKD, and much of quantum networking rest on detecting one photon at a time — fast, with near-unity efficiency and near-zero false counts. The dominant technology is the **superconducting nanowire single-photon detector (SNSPD)**: a current-biased superconducting nanowire (NbN, NbTiN, WSi, MoSi) held just below its critical current, where a single absorbed photon breaks superconductivity locally, forces the current to divert, and produces a measurable voltage spike. As of 2026 the SNSPD is the fastest and most efficient single-photon detector available, outperforming avalanche photodiodes (SPADs) and transition-edge sensors (TES) on efficiency, jitter, and dark counts at once. It is the emitter's counterpart (`H-photonsource` makes the photons; SNSPDs catch them) and a quiet load-bearing chokepoint: photonic QC and QKD scale no faster than their detector supply.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Single Quantum** (Netherlands): multi-pixel free-space arrays; a 6×6 NbTiN array built for the NASA/JPL Psyche deep-space optical-comms terminal, >50% system efficiency at 1550 nm, <15 ns per-pixel dead time.
- **ID Quantique** (Switzerland): ships the **ID281** and **ID281 Pro** turnkey SNSPD systems in closed-cycle cryostats; used in QKD and lab research; parallel-nanowire photon-number resolution.
- **Photon Spot** (US): turnkey SNSPD systems and cryostats, common in US labs. Others: **Quantum Opus**, **Scontel**, **Photec**.
- **State of the art**: system detection efficiency demonstrated >99%; dark-count rates ~0.25 counts/hour; timing jitter <3 ps; recovery ~500 ps (giving ~100s of MHz count rates). Large-area devices now reach mid-infrared (to ~7.4 μm). Waveguide-integrated SNSPDs on the same chip as the photonic circuit are the scaling path.

KEY GRADED CLAIMS

- T2 SNSPD system detection efficiency >99% with picosecond jitter and sub-Hz dark counts — peer-reviewed device papers (demonstrated)
- T2 6×6 SNSPD array flown for NASA Psyche deep-space optical comms — SPIE 2025 / Single Quantum (demonstrated)
- T1 SNSPDs are the fastest/most-efficient single-photon detectors as of 2026 — review literature (established)

TRADE-OFFS

SNSPDs need a cryostat (~1–3 K, a smaller cooler than a dilution fridge but still a system-cost and integration burden) versus room-temperature SPADs, which are cheaper but far noisier and slower. Scaling to the thousands-to-millions of pixels a PsiQuantum/Xanadu-class machine needs strains cryogenic wiring and readout (links to `H-control`, `H-cryocmos`, `H-paramp`). Photon-number resolution — telling one photon from two — remains hard and is a live research front (parallel nanowires, TES hybrids).

CONFLICTS / OPEN QUESTIONS

Can SNSPD arrays scale to the pixel counts fusion-based photonic FTQC needs without a cryogenic-wiring blowup that recreates the superconducting-QC wiring problem? Does monolithic on-chip integration of detectors with the photonic circuit arrive before detector supply becomes the binding constraint on the whole modality?

SOURCES

Wikipedia SNSPD; Nature Light: Science & Applications s41377-023-01374-1; Science Advances adt0502; idquantique.com (ID281 Pro); ADS 2025SPIE13699E; arXiv:2101.05407.

Fabrication & supply chain · H-fab

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The materials science and manufacturing base under every qubit: Josephson-junction deposition (shadow-evaporated Al/AlOx, or foundry-compatible variants), substrate choice (sapphire vs high-resistivity silicon), superconducting films (Nb, Al, Ta, TiN), 300 mm foundry processes, and the specialty supply chain (coax, attenuators, isolators/circulators, ^{3}He, isotopically enriched ^{28}Si, single-photon detectors). The central villain is the **two-level system (TLS)** — atomic-scale defects in amorphous oxides and at interfaces that resonantly absorb qubit energy and drift over hours, capping and destabilizing coherence. Getting T1 up (see the 1.68 ms transmon in `H-supercon`) is mostly a war against TLS.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **TLS science**: 2025 mapping experiments (arXiv:2511.05365) localized most detectable surface TLS to the Josephson-junction leads, clustered where surface roughness and amorphous SiO_2 sit — actionable fab guidance rather than folklore. High-throughput junction studies (arXiv:2602.11469) tie TLS density to structural control; site-specific TLS frequency tuning has been demonstrated (arXiv:2503.04702). Titanium sacrificial layers cut TLS loss in tantalum CPW resonators (arXiv:2601.16369).
- **Materials**: tantalum films (Princeton/Houck lineage) cut microwave loss vs niobium and pushed transmon T1 past 0.3–0.5 ms; replacing sapphire with high-resistivity silicon then cut bulk-substrate loss further, yielding time-averaged Q ≈ 9.7×10^{6} across 45 qubits and the 1.68 ms record.
- **Foundries**: imec's 300 mm superconducting-qubit CMOS flow (Nature npj 2024); Diraq/imec spin qubits >99% from a standard 300 mm line (2025); SQC's 250k-register patterning (2025); GlobalFoundries manufacturing PsiQuantum's Omega photonic chipset; SkyWater, Intel's 300 mm line, and national fabs (SQMS at Fermilab) round out capacity. Related nodes: `H-package` (3D/TSV integration), `H-iontrap` (surface-trap fab), `H-foundry` (the 300 mm ecosystem).

KEY GRADED CLAIMS

- T3 Surface TLS concentrate on junction leads / rough, SiO_2-rich regions — arXiv:2511.05365 (demonstrated, awaiting reproduction)
- T2 300 mm CMOS flows yield error-correction-grade spin qubits and high-coherence transmons — Nature/npj papers (demonstrated)
- T6 TLS problem "engineerable away" by process control alone — no field consensus (speculative)

TRADE-OFFS

Foundry processes buy uniformity and volume but restrict the exotic materials and geometries that still hold coherence records in the lab (shadow-evaporated Al junctions). The specialty supply chain — ^{3}He, dilution fridges (Finnish/UK duopoly, `H-cryo`), SNSPDs (`H-detect`), isolators, ^{28}Si — concentrates in few hands and is a geopolitical single point of failure (see `E-supplychain`).

CONFLICTS / OPEN QUESTIONS

Junction-parameter spread (~1–2% today) vs the <0.1% wanted for fixed-frequency architectures; whether TLS density can be driven down ~10× by process control, or whether architectures must simply tolerate a fixed TLS bath. Yield at wafer scale, not single-device records, is the number that gates manufacturability.

SOURCES

arXiv:2511.05365, 2602.11469, 2503.04702, 2601.16369; PMC11446867 (imec 300 mm); Princeton Nature s41586-025-09687-4; Science Advances ado6240 (phonon engineering); quantumzeitgeist TLS coverage.

The 300 mm / wafer-scale foundry ecosystem · H-foundry

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The distinct-from-materials question of *who can actually manufacture qubits at wafer scale, on industrial tooling, with yield*. `H-fab` covers the materials science (junctions, films, TLS); this node covers the **foundry ecosystem** — the 200/300 mm semiconductor lines being retooled to make superconducting, silicon-spin, and photonic quantum chips, and the strategic fact that quantum is increasingly won or lost on manufacturing uniformity rather than single-device records. The pitch that pulled billions into silicon spin and photonics is precisely this: if a qubit can be built in a CMOS-compatible 300 mm flow, the entire semiconductor industry becomes the quantum fab, and Moore's-law-style volume follows.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **imec** (Belgium): the anchor R&D foundry. Ran a 300 mm superconducting-qubit CMOS flow (Nature npj 2024) and, with **Diraq**, produced gate-defined silicon spin qubits at >99% 2Q fidelity from a standard 300 mm line (Nature, 2025) — the strongest evidence to date that foundry-made qubits can hit error-correction-grade fidelity.
- **GlobalFoundries**: manufactures PsiQuantum's Omega **photonic** chipset on 300 mm silicon photonics, and hosts Equal1's cryo-CMOS control in its 22FDX FD-SOI process (`H-cryocmos`) — one commercial fab serving multiple modalities.
- **Intel**: its own 300 mm line makes Tunnel Falls spin-qubit chips with high-volume test infrastructure. **SkyWater** (US) runs quantum/superconducting programs; **national fabs** (SQMS at Fermilab, LPS/US-government lines) and **TSMC-adjacent** research round out capacity. **SQC** patterned 250k-register donor devices with industrial STM/CMOS tooling (2025).

KEY GRADED CLAIMS

- T2 Error-correction-grade qubits produced from a standard 300 mm industrial flow (spin) — Diraq/imec, Nature (2025) (demonstrated)
- T2 Commercial 300 mm silicon-photonics fab makes integrated photonic-QC chips — PsiQuantum/GlobalFoundries (demonstrated, component-level)
- T4 "Foundry compatibility ⇒ Moore's-law-style qubit scaling" — industry thesis (claimed)

TRADE-OFFS

A merchant foundry buys uniformity, volume, metrology, and yield learning that no lab can match — but it constrains you to the materials and geometries the line already qualifies (no exotic substrates, restricted junction processes), which is exactly where lab-scale coherence records still live (`H-fab`). Retooling a fab for quantum (superconducting films, mK-relevant dielectrics, ^{28}Si, single-photon devices) is expensive and slow, and the volume that justifies it does not exist yet — a chicken-and-egg problem.

CONFLICTS / OPEN QUESTIONS

Does foundry uniformity actually hold across thousands of qubits on a wafer, or does device-to-device spread reappear at scale (the historic silicon-spin killer)? Which modality's foundry bet pays off first — silicon spin (native CMOS), photonics (silicon photonics), or superconducting (needs bespoke films)? Whether a true merchant "quantum foundry" market emerges, or each vendor stays captive to one R&D fab (imec, GF, Intel).

SOURCES

PMC11446867 (imec 300 mm superconducting flow); Diraq/imec Nature (2025); PsiQuantum/GlobalFoundries Omega; Intel Tunnel Falls; SQC 250k-register release; postquantum.com silicon-spin + photonic ecosystem analyses.

Optical frequency combs · H-frequencycomb

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

An optical frequency comb is a laser whose spectrum is a set of perfectly evenly spaced sharp lines — a ruler for light. It coherently links optical frequencies (~hundreds of THz) to microwave frequencies (GHz) that electronics can count, which is what lets an optical atomic clock (`A-clocks`) actually output a usable signal, and what lets many stabilized laser tones in an atomic quantum computer share one absolute reference. The 2005 Nobel-winning technology (Hänsch/Hall) is shared infrastructure that sits under three otherwise separate nodes: **clocks** (`A-clocks`), **time/frequency transfer** (`A-timedist`), and **laser subsystems** for ion/atom machines (`H-lasers`). The frontier is shrinking a lab-bench mode-locked comb onto a chip.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Chip-scale microcombs (Kerr combs)**: Kerr-nonlinear microresonators (silicon nitride, on-chip) generate soliton combs at CMOS-compatible scale, replacing rack-sized mode-locked-laser combs. A Feb 2025 Nature Photonics result used a **Vernier dual-microcomb** scheme to divide a stabilized ~871 nm clock laser down to a countable ~235 MHz output — the key step toward a fully integrated optical clock.
- **NIST** and academic groups (Caltech, EPFL, Purdue, Chalmers) drive the physics; **Vescent**, **Menlo Systems**, **TOPTICA**, and **IMRA** supply commercial combs. NIST has demonstrated chip-scale atomic clocks and chip-scale optical frequency synthesizers at ~10^{-16} relative uncertainty.
- **Direction**: integrate the dual comb, pump laser, on-chip heaters/tuning, and spectral filters onto a single die — the enabling piece for μPNT (micro positioning/navigation/timing) and portable optical clocks (`A-pnt`).

KEY GRADED CLAIMS

- T2 Vernier dual-microcomb optical frequency division of a stabilized clock laser — Nature Photonics s41566-025-01617-0 (2025) (demonstrated)
- T2 Chip-scale optical frequency synthesizer at 2.7×10^{-16} relative uncertainty — Science Advances (established/demonstrated)
- T1 Frequency combs coherently link optical and microwave frequencies — established (2005 Nobel)

TRADE-OFFS

Bench mode-locked-laser combs are mature, low-noise, and turnkey but rack-sized and power-hungry. Microcombs promise mass-manufacturable, low-SWaP combs on a chip but currently struggle with octave-spanning bandwidth (needed for self-referencing), spectral flatness, and turnkey soliton initiation. Which you pick trades performance against integration — the same tension running through `H-lasers`.

TRADE-OFFS / WHY IT'S ITS OWN NODE

Because a comb is shared plumbing under clocks, timing networks, and atomic-computer lasers, keeping it as a separate node avoids duplicating the same hardware story three times across `A-clocks`, `A-timedist`, and `H-lasers`. Improvements here propagate to all of them at once.

SOURCES

Nature Photonics s41566-025-01617-0 (Vernier microcombs, 2025); Science Advances (chip-scale synthesizer); NIST optical-frequency-combs program page; Laser Focus World (microcomb clocks, 2025); arXiv:2602.05151, 2512.05005.

Interconnects & modular quantum computing · H-intercon

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

No single chip or trap reaches a million qubits, so every serious roadmap goes modular: multiple processors linked by quantum interconnects that distribute entangled Bell pairs between modules, then consume them for teleported gates (the protocol under `F-teleport`). Three flavors, at three length scales: **short cryogenic microwave/chip-to-chip couplers** inside one fridge (cm-scale, fast, high-fidelity); **optical-photonic links** over fiber between fridges/racks/buildings (km-scale, slow, lossy); and **microwave-to-optical transduction** (`H-transduce`) to let a superconducting qubit talk to a fiber at all. Interconnect *quality*, more than raw qubit count, now gates every modality's endgame — a bad link below the QEC threshold poisons the whole logical computation.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Oxford (trapped ions)**: distributed quantum computing across an optical network link — deterministic teleported CZ gates between two photonically linked ion-trap modules, executing Grover's algorithm across the boundary (Nature, 2025). The cleanest full-stack demonstration of the modular paradigm to date.
- **Duke Quantum Center + IonQ (Monroe group)**: first *fully-distributed* GHZ (tripartite) state across a **three-node** network of single trapped-ion memories linked by photonic interconnects — three modules ~2 m apart, 3 m single-mode fibers to a central free-space GHZ generator; bounded fidelity $0.841(17) \leq F \leq 0.881(17)$ at a generation rate $0.095(5)\,\mathrm{s}^{-1}$, with a Mermin-inequality violation closing the detection loophole for the first time in a distributed multipartite state (Jun 2026). Clears the two-node barrier toward genuine networked/modular QC.
- **IBM**: Nighthawk's 2026–27 plan chains up to three 120-qubit modules (360 qubits) with **l-couplers** (short-range chip-to-chip); **Loon**'s c-couplers add long-range on-chip connectivity for qLDPC codes; **Flamingo** targets ~1 m cryogenic links between modules.
- **Photonic Inc.** (silicon T-centers): distributed entanglement between modules over telecom fiber, using a spin-photon interface native to the T-center (see `H-spinphoton`); roadmap targets ~200 kHz remote-entanglement rate at 99.8% fidelity.
- **PsiQuantum / Xanadu**: chip-to-chip photonic-qubit interconnects are native to the photonic modality (`H-photonic`) rather than an add-on.
- **Quantinuum / IonQ**: photonic interconnects are the declared scaling path beyond single-trap limits. Silicon-photonic CNOT-gate teleportation between separate chips demonstrated (PRL 2025–26).

KEY GRADED CLAIMS

- T2 Teleported two-qubit gates between separate ion-trap modules; algorithm run across the link — Nature s41586-024-08404-x (demonstrated)
- T3 First fully-distributed three-node GHZ state of remote ion memories, F ≈ 0.84–0.88, Mermin violation with detection loophole closed — Duke + IonQ, Goetting et al., arXiv:2606.17173 (2026) (demonstrated; preprint)
- T4 IBM 3×120-qubit multi-module systems in 2026 — IBM roadmap (roadmap)
- T4 Photonic Inc. 200 kHz / 99.8% distributed entanglement — company target (roadmap)

TRADE-OFFS

Optical links carry entanglement kilometers but at kHz rates and ~90–97% Bell-pair fidelities — orders of magnitude below intra-module gates (99.9%+). Microwave couplers are fast and high-fidelity but reach only centimeters. Entanglement distillation buys fidelity back at a further rate cost. The core tension: the faster and cleaner the link, the shorter its reach; bridging fridges cheaply needs a transducer that does not yet exist (`H-transduce`).

CONFLICTS / OPEN QUESTIONS

C-interconnect-loss / **O-interconnect-loss**: whether modular interconnects can stay below the QEC threshold across module boundaries. Microwave-optical transducers remain far from the efficiency/added-noise needed for fridge-to-fridge superconducting links. Does modularity's overhead (slower inter-module gates) break QEC cycle budgets before it buys scale?

SOURCES

Nature s41586-024-08404-x (Oxford distributed ions); Goetting et al., arXiv:2606.17173 (Duke/IonQ three-node GHZ, 2026); photonic.com networking releases; IBM roadmap/newsroom (Nov 2025); PRL (silicon-photonic CNOT teleportation); arXiv:2412.18458.

Trapped-ion qubits · H-ion

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Individual atomic ions (Yb^+, Ba^+, Ca^+) held in electromagnetic (Paul) traps in ultra-high vacuum; qubits live in hyperfine or optical levels of identical atoms, so every qubit is perfect by construction and needs no fabrication tuning. Gates are driven with lasers (or on-chip microwaves/electronics) through shared motional modes of the ion chain, giving effectively all-to-all connectivity within a chain. Modern systems use microfabricated **surface traps** (see `H-iontrap`) rather than bulk blade traps; Quantinuum's QCCD (quantum charge-coupled device) architecture physically shuttles ions between dedicated storage, gate, and readout zones to sidestep the single-chain length limit.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Quantinuum**: Helios launched Nov 2025 — 98 physical $^{137}Ba^+$ qubits at 99.921% 2Q / 99.9975% 1Q fidelity, coherence in the seconds-to-minutes range, on a Honeywell-fabricated QCCD trap; results characterized in an arXiv preprint (2511.05465) and independently validated by Sandia in Nature (June 2026). Roadmap: Sol (2027), **Apollo** fault-tolerant system ~2029. Helios already runs dozens of logical qubits with real-time QEC.
- **IonQ**: closed its $1.075B acquisition of **Oxford Ionics** (Sep 2025); claims a 99.99% 2Q fidelity record (2025); Forte Enterprise and Tempo (~100 algorithmic qubits) systems shipping, Tempo contracted to KISTI (South Korea) and to AFRL. Combined roadmap: 256 physical qubits at 99.99% (2026), ~10,000 (2027), 2M by 2030 — aggressive relative to any demonstrated trap-scaling result.
- **Oxford Ionics** (now IonQ): electronic (laser-free) gate control delivered on chip traps; supplied the QUARTET system to the UK NQCC. **Alpine Quantum Technologies** (Innsbruck): Ca^+, rack systems. **eleQtron** (Germany): MAGIC microwave-driven gates.

KEY GRADED CLAIMS

- T2 Helios: 98 qubits, 99.921% 2Q fidelity, seconds-to-minutes coherence — arXiv:2511.05465 + Sandia/Nature validation Jun 2026 (demonstrated)
- T4 IonQ 99.99% 2Q gate record — company claim 2025 (claimed)
- T4 IonQ 2M physical qubits by 2030 — post-acquisition roadmap (roadmap)
- T1 Ion qubits are intrinsically identical (no fab variability) — atomic physics (established)

TRADE-OFFS (VS OTHER MODALITIES)

The best gate fidelities and by far the longest coherence (seconds to minutes) of any modality, all-to-all connectivity, and no dilution fridge (UHV chamber, room-temperature-adjacent — see `H-uhv`). Against that: gates are slow (μs–ms, ~1000× slower than transmons), a single trap chain saturates around ~50–100 ions before motional-mode crowding degrades gates, and scaling then demands ion shuttling (heating, speed cost), multi-zone chips, or photonic interconnects (`H-intercon`). The whole machine is a precision laser instrument (`H-lasers`), so uptime tracks laser-lock stability.

CONFLICTS / OPEN QUESTIONS

Whether QCCD shuttling or photonic networking wins the scaling race — Quantinuum bets on the former, IonQ (post-Oxford-Ionics) increasingly on the latter. Whether the μs–ms gate-speed penalty erases the fidelity advantage once algorithms reach millions of gates. IonQ's 2030 count is a roadmap number, not a demonstrated one.

CONFLICTS LOGGED

See `C-ion-scaling` (shuttling vs photonic networking as the trapped-ion scaling path).

SOURCES

Quantinuum Helios release + arXiv:2511.05465; HPCwire Sandia/Helios validation (Jun 2026); IonQ/Oxford Ionics acquisition filings; DCD KISTI Tempo coverage; postquantum.com trapped-ion ecosystem + Helios analyses.

Ion-trap chip fabrication (surface traps) · H-iontrap

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The trapped-ion modality (`H-ion`) has quietly migrated from macroscopic four-rod blade traps to **microfabricated surface-electrode traps**: a planar silicon or glass chip patterned with dozens to hundreds of gold or aluminum electrodes whose shaped RF and DC fields levitate a chain of ions ~50–100 µm above the surface, in ultra-high vacuum (`H-uhv`). The chip *is* the processor. Modern designs carve the surface into functional **zones** — storage (long chains wait), gate (2–4 ions isolated for entangling operations), junctions (X/Y intersections to shuttle ions between zones), and readout (fluorescence collection) — the QCCD architecture that lets ion machines grow past a single chain. Integrating optics (waveguides, grating couplers, on-chip photodetectors) and control electronics into the trap chip is the current fabrication frontier.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Sandia National Laboratories**: the reference research foundry. Its **High Optical Access (HOA)** platform (Phoenix, Peregrine traps) gives good laser access skimming the surface; the "Enchilada" trap stores up to ~200 ions across a branched layout — among the most complex surface traps ever fabricated. Sandia supplies traps to much of the US government-funded ion community.
- **Honeywell / Quantinuum**: the Helios QCCD trap chip (Nov 2025, 98 qubits at 99.921% 2Q — see `H-ion`) was fabricated in Honeywell's MEMS/microfabrication facility, a rare vertically integrated trap-fab-to-computer pipeline.
- **Infineon** (with academic partners) fabricates ion-trap chips at scale; **Oxford Ionics** (now IonQ) builds electronic (integrated-electrode) traps for laser-free gates. Sandia's TICTOC program integrates photonics onto trap chips for compact ion clocks.

KEY GRADED CLAIMS

- T2 Microfabricated surface-electrode traps run multi-zone QCCD processors at record fidelity — Helios/Honeywell + Sandia literature (demonstrated)
- T2 Branched surface traps store ~200 ions with multiple junctions — Sandia Enchilada (demonstrated)
- T3 On-chip integrated photonics for laser delivery/readout in trap chips — Sandia TICTOC + academic demos (early)

TRADE-OFFS

Surface traps trade the deep, symmetric confinement of blade traps for lithographic scalability, multi-zone layouts, and integrable optics/electronics. The cost: ions sit close to a surface, so **anomalous heating** from surface electric-field noise is worse (it scales steeply with ion-electrode distance), degrading gate fidelity unless the surface is cryogenically cooled or carefully treated. Junction shuttling adds motional heating and time overhead. Fabrication uniformity and dielectric charging are yield issues.

CONFLICTS / OPEN QUESTIONS

Can integrated photonics and control deliver every laser tone and readout on-chip fast enough to remove the external optics table (`H-lasers`) that otherwise bounds scaling? Does anomalous surface heating cap how small and dense trap electrodes can go? Whether commercial trap fabrication moves to a merchant-foundry model (like CMOS) or stays captive to Sandia/Honeywell/Infineon.

SOURCES

Sandia HOA / Enchilada trap literature; arXiv:2009.02398 (Phoenix/Peregrine); Quantinuum Helios (Honeywell fab); postquantum.com trapped-ion ecosystem; Sandia TICTOC project page; arXiv:1008.0990 (surface-trap demonstration).

Laser & photonics subsystems · H-lasers

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Trapped-ion, neutral-atom, and NV-diamond machines are, physically, laser instruments. They need many stabilized laser tones — for trapping/cooling, Rydberg excitation, qubit-state manipulation, repumping, and readout — each at the right wavelength with sub-kHz linewidth, mode-hop-free tuning, and long-term frequency stability locked to an optical reference (increasingly an optical frequency comb, see `H-frequencycomb`). As atomic machines scale from tens to thousands of atoms, the laser subsystem becomes the dominant cost, footprint, and reliability risk — the atomic-modality analog of superconducting wiring. It is an arms-supplier layer: a handful of photonics vendors sit quietly under most atomic/ionic quantum computers, and the push is toward low-SWaP (size, weight, power) modules and photonic integrated circuits (PICs).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **TOPTICA Photonics** (Germany): the dominant supplier of tunable diode-laser systems and frequency-stabilized references; serves virtually every major neutral-atom and ion lab and commercial system.
- **M Squared Lasers** (UK): high-performance Ti:sapphire and frequency-stabilized systems used for Rydberg excitation (the demanding blue/UV tones).
- **Vescent** (US): precision diode lasers, frequency combs, and servo/lock electronics widespread in atomic physics. Trapping power: **IPG Photonics**, **NKT Photonics**.
- **PIC integration**: IonQ + imec are developing photonic integrated circuits to route and deliver laser light on-chip for ion machines (2025), the same bet neutral-atom vendors are making — replacing free-space beam paths with waveguides to kill drift and shrink the footprint.

KEY GRADED CLAIMS

- T1 Neutral-atom/ion machines require multiple sub-kHz-linewidth stabilized laser tones — atomic physics literature (established)
- T4 TOPTICA/M Squared supply the Rydberg-excitation lasers for most neutral-atom systems — supply-chain analyses (claimed/reported)
- T3 Photonic integrated circuits can shrink ion-machine laser delivery (IonQ × imec) — 2025 collaboration (roadmap/early)

TRADE-OFFS

Free-space bulk-optics laser tables give the best performance but do not scale — large, drift-prone, labor-intensive to align, and sensitive to vibration and temperature. PIC integration promises SWaP and manufacturability but currently trades away optical power and linewidth, especially at the blue/UV wavelengths ion qubits need. Laser reliability (diode lifetime, lock stability) directly bounds machine uptime: a single unlocked tone can halt the whole processor mid-computation.

CONFLICTS / OPEN QUESTIONS

Does laser delivery integrate onto chips fast enough to keep pace with atom-count roadmaps (`H-neutral`, `H-ion`), or does the optics table become the scaling wall for atomic modalities the way wiring is for superconducting ones? Blue/UV PIC materials and on-chip high-power delivery are the specific unsolved pieces.

SOURCES

toptica.com (quantum technologies / LASER 2025); globenewswire.com 2025/07/08 (neutral-atom market, Toptica/Hamamatsu); postquantum.com (neutral-atom supply chain); ionq.com/news (IonQ × imec PICs); arXiv:2304.08402.

Neutral-atom qubits · H-neutral

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Neutral atoms (Rb, Sr, Cs, Yb) held in arrays of optical tweezers; qubits live in hyperfine or nuclear-spin states, and two-qubit gates come from exciting atoms to giant **Rydberg** states whose strong interaction blockades neighbors. Tweezers can be physically rearranged mid-computation, giving reconfigurable, effectively all-to-all connectivity and native support for shuttling logical qubits — the feature that made the modality the fastest-moving of 2024–26. Atoms are laser-cooled but the apparatus itself runs at room temperature inside an ultra-high-vacuum chamber (see `H-uhv`); the machine is fundamentally a laser instrument (`H-lasers`).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **QuEra / Harvard / MIT**: 2025 Nature results — 48 logical qubits on 280 atoms with below-threshold operation and logical-layer magic-state distillation; a separately reported continuously-operating array kept a ~3,000-atom system running for over two hours by reloading atoms mid-computation, addressing atom-loss as an error channel. **Jan 2026: 96 logical qubits from 448 physical atoms via a high-rate $[[16,6,4]]$ code (~4.7:1 encoding), below-threshold across all 96 — peer-reviewed in Nature (s41586-025-09848-5), the current logical-qubit record.** Roadmap: 100 logical qubits ~2026. Raised $230M (Feb 2025), Google/SoftBank participating.
- **Pasqal** (France): 1,000+ atom arrays; first neutral-atom demo of logical qubits outperforming physical ones on a differential-equation workload (2026); analog + digital modes; ~3 kW rack systems. Targets 10,000 qubits ~2026–27.
- **Atom Computing** (US): 1,180-atom Sr array (1,225 sites), ~40 s coherence; with Microsoft demonstrated 24 entangled logical qubits (2024); Magne (~50 logical from ~1,200 physical) targeted ~2027.
- **Infleqtion** (US): Sqale platform, UK NQCC deployment. **planqc** (Germany): Sr arrays, DLR/LRZ contracts.
- **Atom-count frontier**: Caltech loaded 6,100 atoms in a single tweezer array (Sep 2025); QuEra/Atom Computing project 100,000 atoms per chamber within a few years.

KEY GRADED CLAIMS

- T2 48 logical qubits + logical magic-state distillation — Harvard/QuEra, Nature (2025) (demonstrated)
- T2 96 logical qubits via a high-rate $[[16,6,4]]$ code on 448 atoms, below-threshold — QuEra, Nature s41586-025-09848-5 (2026) (demonstrated, peer-reviewed)
- T2 Continuous >2 hr operation via mid-computation atom reloading (~3,000 atoms) — Harvard/MIT, Nature (2025) (demonstrated)
- T3 6,100-atom single-array loading — Caltech (Sep 2025) (demonstrated, static array)
- T4 Pasqal 10,000 qubits by ~2026–27 — company roadmap (roadmap)

TRADE-OFFS (VS OTHER MODALITIES)

The largest qubit counts of any gate-based modality, reconfigurable connectivity, a room-temperature apparatus, and intrinsically identical atoms; against that, gates are slower than transmons (Rydberg gates ~sub-μs but with cooling/rearrangement overhead per cycle), atom loss is a distinctive error channel that forces continuous reloading, and 2Q fidelities (~99.5%) still trail trapped ions. The optical-table laser subsystem is the practical scaling wall (`H-lasers`), the analog of superconducting wiring.

CONFLICTS / OPEN QUESTIONS

Does the logical-qubit lead survive the move from batch-mode demos to continuous, repeatable computation with fast mid-circuit measurement and feed-forward at scale? Rearrangement time and cooling cycles set an effective clock speed that has to fit inside QEC budgets.

SOURCES

quera.com; Nature 2025 logical-qubit + continuous-operation papers; QuEra 96-logical Nature s41586-025-09848-5 (2026); Caltech 6,100-atom (Sep 2025); Pasqal newsroom; IEEE Spectrum "Neutral-Atom QC 2026"; postquantum.com neutral-atom guide.

NV centers / diamond qubits · H-nv

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

A nitrogen-vacancy (NV) center is a point defect in diamond — a substitutional nitrogen atom beside a missing carbon — whose electron spin works as a qubit that can be optically initialized and read out and stays coherent at *room temperature*, because the stiff, low-spin-density diamond lattice isolates it almost as well as a millikelvin vacuum would. Nearby ^{13}C nuclear spins act as long-lived ancilla qubits and quantum memory. The same physics drives quantum sensing, where NV is a commercial reality today (see `A-nvsensing`, `A-magneto`); for computing, the pitch is a cryogenics-free, rack- or edge-deployable "quantum accelerator."

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Quantum Brilliance** (Australia/Germany): the anchor company. 2025: three QB-QDK2.0 units deployed at Oak Ridge National Laboratory for hybrid quantum-classical work; Europe's first room-temperature NV accelerator went live at Fraunhofer IAF (Jun 2025). Roadmap: 25–100 qubit systems by 2026–27 via incremental qubit-count "drops," moving toward full chip production. Part of a €35M German Cyberagency portable-QC program (with ParityQC) targeting a transportable defense system by 2027.
- **Academic multi-qubit records** (Delft/QuTech, Stuttgart, Ulm, Harvard): ~10-qubit registers on a single NV node with fault-tolerant-grade control and demonstrated small logical operations; NV nodes double as the memory/repeater endpoints in early quantum-network experiments (a bridge to `A-qinternet`, `A-qmemory-hw`). Coherence: NV electron-spin T2 reaches milliseconds at room temperature and seconds with dynamical decoupling; ^{13}C nuclear memories exceed a minute.

KEY GRADED CLAIMS

- T4 QB-QDK2.0 accelerators operating at ORNL and Fraunhofer IAF — deployment announcements (demonstrated as installations; compute utility unproven)
- T4 25–100 qubits by 2026–27 — Quantum Brilliance roadmap (roadmap)
- T2 ~10-qubit NV registers with high-fidelity control and small logical ops — academic literature (demonstrated)
- T1 NV electron/nuclear spins are coherent qubits at room temperature — established physics

TRADE-OFFS (VS OTHER MODALITIES)

Room temperature, no laser-vacuum-cryo stack, car-battery-scale power, and a physically tiny package — the only modality plausibly deployable at the edge or in a vehicle. Against that: qubit counts are the lowest of any modality (single digits per node), entangling *many* NV centers is hard because each defect sits at a random lattice position, and deterministic placement of many high-quality NVs in an array is unsolved fabrication.

CONFLICTS / OPEN QUESTIONS

Is diamond compute a real computing modality, or a superb sensing technology wearing a compute badge? The 25–100-qubit roadmap requires deterministic NV placement and NV-to-NV entangling that nobody has demonstrated at those numbers. The strongest near-term value of the platform is clearly on the sensing side.

SOURCES

quantumbrilliance.com; OLCF/ORNL Q&A (Sep 2025); Fraunhofer IAF release (Jun 2025); The Quantum Insider; IOPscience 2633-4356/ade359 (NV registers); entangledfuture.com NV guide (2026).

Packaging, interposers & 3D integration · H-package

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Once a superconducting chip passes a few hundred qubits, you cannot route every control and readout line in from the edges of a single plane — wiring congestion, microwave crosstalk, and mechanical stress make 2D layout a dead end. The fix is borrowed from classical chip packaging and pushed into the superconducting, millikelvin regime: **flip-chip** bonding (a qubit chip bump-bonded face-to-face to a separate wiring/readout chip), **through-silicon vias (TSVs)** carrying signals vertically through the substrate, and **silicon interposers** hosting superconducting coplanar-waveguide routing between chiplets. This lets a machine partition into a **qubit chiplet + bus/wiring chiplet** stack — the quantum analog of 2.5D/3D heterogeneous integration — so qubit count can grow without the signal fan-out choking it.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **MIT Lincoln Laboratory** pioneered superconducting TSVs and 3D-integrated qubit stacks (high-coherence tileable 3D architectures; silicon hard-stop spacers for controlled flip-chip gaps). **IBM** and **Google** both run flip-chip processes in production (Google's Sycamore/Willow and IBM's Heron/Nighthawk use multi-layer wiring and bump bonds).
- **QuantWare** (Delft): explicitly markets chiplets + 3D packaging ("VIO" vertical I/O) to let smaller labs build high-density modules — packaging-as-a-product.
- **imec**, **Chalmers**, and foundry programs develop superconducting TSV and micro-bump processes; through-sapphire-substrate machining is an alternative route (arXiv:2406.09930). A 2026 review (arXiv:2602.04831) surveys large-scale superconducting integration; microwave-crosstalk characterization in planar/3D devices is an active measurement front (arXiv:2606.02440).

KEY GRADED CLAIMS

- T2 High-coherence qubits preserved through TSV/flip-chip 3D integration — MIT-LL, arXiv:2107.11140 / 1708.02226 (demonstrated)
- T4 Chiplet + interposer packaging enables high-density modular qubit stacks — QuantWare (claimed)
- T3 3D partition (qubit chiplet + bus chiplet) as the path past 2D wiring limits — IEEE/review literature (demonstrated at small scale)

TRADE-OFFS

3D integration buys signal density and routing headroom but adds lossy interfaces — every bump bond, via, and dielectric layer is a candidate home for TLS defects (`H-fab`) that can degrade the very coherence the packaging is meant to serve. Flip-chip gaps must be controlled to microns; TSV superconductivity must survive thermal cycling to mK. The engineering is a constant fight between more connectivity and preserved coherence.

CONFLICTS / OPEN QUESTIONS

Can 3D packaging scale line density fast enough to feed thousand-qubit chips without importing enough interface loss to cap coherence? Does the industry converge on a standard interposer/chiplet interface (a "quantum socket"), or stay bespoke per vendor? Packaging is emerging as a distinct chokepoint alongside the fridge and the controller.

SOURCES

arXiv:2107.11140, 1708.02226, 1907.12882 (MIT-LL 3D/TSV); arXiv:2602.04831 (2026 large-scale integration review); 2406.09930 (through-sapphire); 2606.02440 (microwave crosstalk); QuantWare VIO materials; NSF interposer-packaging report (par.nsf.gov/10340124).

Quantum-limited amplifiers (JPA / TWPA / HEMT) · H-paramp

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Reading out a superconducting qubit means measuring a microwave signal carrying, on average, a fraction of a photon — buried in thermal and amplifier noise. A **quantum-limited parametric amplifier** at the mixing-chamber stage (~10 mK) boosts that signal while adding the minimum noise quantum mechanics allows (half a photon for phase-insensitive gain), *before* the signal reaches the noisier high-electron-mobility-transistor (**HEMT**) amplifier at the 4 K stage and the room-temperature electronics (`H-control`). Without this first amplifier, single-shot dispersive readout — telling |0⟩ from |1⟩ in one measurement, fast enough for real-time QEC — is impossible. Two architectures dominate: the resonator-based **Josephson parametric amplifier (JPA)** (high gain, narrow band) and the **Josephson traveling-wave parametric amplifier (TWPA/JTWPA)** (near-quantum-limited over a wide band, enabling frequency-multiplexed readout of many qubits on one line).

KEY PLAYERS & STATE OF THE ART (2025–26)

- The canonical readout chain: a near-quantum-limited JPA or TWPA at 10 mK → a HEMT (~2–4 K, adding a few photons) → room-temperature amplifiers. TWPAs deliver high gain over GHz bandwidths at ~1 added photon, letting a single line read out many frequency-multiplexed qubits — the reason they matter for scaling.
- 2025 advances: low-intrinsic-loss coplanar lumped-element JTWPAs (arXiv:2503.07559); inverse-Kerr phase-matching for flatter gain (arXiv:2507.17039); in-operando S-parameter-calibrated characterization (APL 2024); two-mode-squeezing schemes that push readout fidelity past the standard quantum limit (arXiv:2603.15804).
- Suppliers/developers: academic groups (Berkeley/Siddiqi lineage, Chalmers, NIST), plus vendors bundling TWPAs into readout stacks (Silent Waves' "Argo" TWPA, Raytheon/Lincoln Lab, and control-vendor integrations). HEMTs come chiefly from **Low Noise Factory** (Sweden).

KEY GRADED CLAIMS

- T1 A phase-insensitive linear amplifier must add ≥ ½ photon of noise (standard quantum limit) — Caves theorem, established
- T2 JTWPAs provide near-quantum-limited gain over GHz bandwidth, enabling multiplexed single-shot readout — Science aaa8525 + follow-ons (demonstrated)
- T3 Two-mode-squeezed readout beats the SQL for qubit measurement — arXiv:2603.15804 (demonstrated, lab-scale)

TRADE-OFFS

JPAs give the cleanest noise but narrow bandwidth and low saturation power (few qubits per amp); TWPAs trade a little noise for wide bandwidth and higher saturation power (many qubits per line) at the cost of a long, fabrication-sensitive Josephson transmission line. The HEMT is unavoidable but sits at 4 K and its power dissipation and noise both scale with channel count — another line-count pressure feeding the `H-cryocmos` and `H-cryo` bottlenecks.

CONFLICTS / OPEN QUESTIONS

Can TWPA fabrication yield and uniformity reach the level needed to read out thousands of qubits without per-device tuning? Do parametric amplifiers get absorbed into cryo-CMOS readout, or stay a distinct superconducting-component layer? Photon-number-resolving and directional (nonreciprocal) amplifiers that drop the bulky isolators are a live front.

SOURCES

Science aaa8525 (near-quantum-limited JTWPA); arXiv:2503.07559, 2507.17039, 2603.15804; AIP APL 125/104001 (in-operando calibration); Low Noise Factory HEMT datasheets.

Photonic qubits · H-photonic

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Qubits encoded in light — single photons (dual-rail, path, polarization, time-bin) or continuous-variable squeezed states (see `A-squeezed`). Computation interferes photons in integrated waveguide circuits (thin-film lithium niobate, silicon, or silicon-nitride) and measures them; the leading architecture is measurement-based **fusion** computation, where small entangled resource states are stitched into a large cluster state by probabilistic fusion gates. Photons barely decohere and travel telecom fiber, so networking is native and much of the machine runs warm — the catch is that photons do not interact, so gates are probabilistic, and **photon loss** is the dominant error. The modality leans on two other hardware nodes: deterministic photon sources (`H-photonsource`) and near-unity single-photon detectors (`H-detect`).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **PsiQuantum**: Omega chipset (Feb 2025) manufactured at GlobalFoundries on 300 mm silicon photonics — single-photon sources, fusion-gate circuits, and cryogenic detectors integrated with chip-to-chip qubit interconnects. $1B Series E at $7B valuation (Sep 2025); ~$1B utility-scale datacenter build-outs announced for Brisbane and Chicago (Mar 2026). Bets on million-qubit fault tolerance, skipping NISQ entirely.
- **Xanadu**: Aurora (Jan 2025) — modular networked photonic computer: 12 qubits across 35 photonic chips + 13 km fiber, room-temperature apart from cryogenic detectors; uses GKP/CV encoding. IPO'd on Nasdaq/TSX Mar 2026 (~$302M raised). Earlier Borealis showed programmable Gaussian-boson-sampling advantage (Nature 2022).
- **QuiX Quantum** (Netherlands): €15M Series A to ship a first-generation ~8-qubit *universal* single-photon computer in 2026, on silicon-nitride chips. **ORCA Computing** (UK): PT-series time-bin/boson-sampling systems in national labs, fiber-based, rack-mounted. **TundraSystems** (UK) and **Quandela** (see `H-photonsource`) round out the field.

KEY GRADED CLAIMS

- T3 Omega chipset with integrated fusion + wafer-scale fab at GlobalFoundries — PsiQuantum (2025) (demonstrated, component-level)
- T4 Aurora: first scalable networked modular photonic QC — Xanadu press (claimed)
- T2 Gaussian-boson-sampling advantage (Borealis) — Xanadu, Nature (2022) (demonstrated, advantage contested by later classical algorithms)
- T6 Million-qubit photonic FTQC in datacenter form by late decade — PsiQuantum (roadmap)

TRADE-OFFS (VS OTHER MODALITIES)

Room-temperature circuits, native fiber networking, and reuse of the telecom/CMOS-photonics supply chain; against that, gates are probabilistic (fusion succeeds ~50%, so massive multiplexing and fast switching are required), photon loss compounds along every waveguide and coupler, detectors still need cryogenics (2–4 K), and no photonic machine has run a meaningful gate-based algorithm at scale. The loss budget for million-photon operation is unproven end-to-end.

CONFLICTS / OPEN QUESTIONS

Fusion-based FTQC has no intermediate-scale public benchmark — the approach is close to all-or-nothing, so there is little early signal to grade against a roadmap. Whether deterministic, indistinguishable single-photon sources and low-loss switches arrive fast enough to close the loss budget is the field's binding question (see `H-photonsource`, `H-detect`).

SOURCES

psiquantum.com/news (Omega); xanadu.ai press (Aurora) + Nature 606 (Borealis, 2022); quixquantum.com Series A; Optica OPN photonics roadmap (Jun 2025); The Quantum Insider.

Deterministic single-photon & entangled-pair sources · H-photonsource

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The emitter counterpart to the SNSPD (`H-detect`): a photonic quantum computer or QKD link is only as good as its supply of *single, identical, on-demand* photons. Two families compete. **Deterministic** sources use a solid-state quantum emitter — an InGaAs/GaAs semiconductor quantum dot in a micropillar or open cavity, Purcell-enhanced so it emits exactly one photon per trigger — approaching "push a button, get one photon." **Probabilistic** sources use nonlinear optics: spontaneous parametric down-conversion (SPDC) or four-wave mixing that occasionally splits a pump photon into a heralded pair; simple and room-temperature, but the emission time is random, so scaling many sources means heavy multiplexing. The three figures of merit are **brightness** (collection efficiency), **single-photon purity** ($g^2(0) \to 0$), and **indistinguishability** (Hong-Ou-Mandel visibility $\to$ 1).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Quandela** (France): the anchor deterministic-source company. InGaAs quantum dots in micropillar cavities grown by molecular-beam epitaxy; current devices hit in-fibre brightness >30%, $g^2(0) < 0.05$, and HOM visibility > 90% from a single dot. Its **Prometheus** is a plug-and-play single-photon-source appliance; Quandela also builds full photonic processors (see `H-photonic`) on this source.
- **2026 milestone**: Quandela + C2N demonstrated 88 ± 1% indistinguishability between photons from *two independent* quantum-dot sources *without spectral filtering* — a prerequisite for scaling beyond one emitter (arXiv 2602-series). Telecom-band (1550 nm) two-photon-resonant Purcell-enhanced dots also reported (arXiv:2602.06140).
- **SPDC/heralded**: the workhorse of QKD and boson-sampling; bulk-crystal and thin-film-lithium-niobate waveguide pair sources are commercial (e.g. from QKD vendors). **Sparrow Quantum** (Denmark) supplies quantum-dot single-photon chips.

KEY GRADED CLAIMS

- T2 Deterministic QD source: >30% in-fibre brightness, $g^2(0) < 0.05$, HOM > 90% — Quandela device papers (demonstrated)
- T3 88% indistinguishability between two independent QD sources, filter-free — Quandela/C2N, arXiv (2026) (demonstrated)
- T1 SPDC produces heralded single photons but at random times — established nonlinear optics

TRADE-OFFS

Deterministic dots give high brightness and on-demand timing but need cryogenics (~4 K), suffer chip-to-chip wavelength/efficiency spread (screening and binning, or on-chip tuning, required), and are hard to make *mutually* indistinguishable across many emitters. SPDC is room-temperature and dead-simple but fundamentally probabilistic, forcing multiplexing that reintroduces loss and switching complexity — the very loss budget that gates `H-photonic`.

CONFLICTS / OPEN QUESTIONS

Which source wins for million-photon FTQC: perfected deterministic dots, or heavily multiplexed SPDC? Cross-source indistinguishability at scale (many identical emitters feeding one interferometer) is the specific unsolved requirement, and 2026's filter-free two-source result is the first real signal it might be reachable.

SOURCES

quandela.com (Prometheus, single-photon-source technology); arXiv:2602.06140 (telecom QD) + 2026 two-source indistinguishability preprint; Nano Letters 5c01560 (remote QD-cavity indistinguishability); quantumzeitgeist.com single-photon-source coverage.

Silicon spin qubits · H-silicon

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Qubits stored in the spin of single electrons (or nuclei) confined in silicon, split into two families that share a substrate but not a philosophy: **gate-defined quantum dots**, where lithographic gates electrostatically trap electrons in a well (prioritizes manufacturability — the devices are CMOS transistor structures), and **donor qubits**, where the spin lives on a single phosphorus atom placed with atomic precision (prioritizes uniformity — every P atom is identical by nature). The pitch for both: qubits are ~100 nm scale (roughly a million times smaller in area than a transmon), so in principle the entire semiconductor industry becomes the quantum fab. Operates around 100 mK–1 K, and — because the devices are silicon — is the natural home for cryo-CMOS co-integration (`H-cryocmos`).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Diraq + imec** (Sep 2025, Nature): >99% two-qubit fidelity on **gate-defined** dots made in a standard 300 mm CMOS foundry flow — four randomly selected devices from one wafer all under 1% 2Q error, three at 99.9%+ single-shot readout. First foundry-made qubits at error-correction-grade fidelity, and the strongest evidence yet for the "uniformity via lithography" thesis.
- **SQC** (Silicon Quantum Computing, Michelle Simmons): **donor** route — 99.99% 2Q fidelity in atom-precision devices and Grover's algorithm at 98.9% without error correction (Nature, Dec 2025); patterned 250,000-qubit registers in 8 hours with industrial STM/CMOS tooling (Nov 2025).
- **Quantum Motion** (UK): full-stack silicon QC delivered to the UK NQCC (2025), built on 300 mm wafers. **Intel**: Tunnel Falls 12-qubit research chip distributed to labs; 300 mm line with high-volume test. **Equal1** (see `H-cryocmos`): control integrated in GlobalFoundries 22FDX. **SiMOS single-qubit** RB records sit near 99.9%.
- **2026 firsts**: quantum error detection in silicon (Jan 2026); first universal logical operations on silicon spins (Mar 2026).

KEY GRADED CLAIMS

- T2 >99% 2Q fidelity from a standard 300 mm industrial flow (gate-defined) — Diraq/imec, Nature (2025) (demonstrated)
- T2 99.99% 2Q + Grover at 98.9% in donor devices — SQC, Nature (Dec 2025) (demonstrated)
- T4 250k-register patterning ⇒ manufacturing scalability — SQC release (claimed)
- T1 Donor P atoms are intrinsically identical; gate-dots buy reproducibility from lithography — device physics (established)

TRADE-OFFS (VS OTHER MODALITIES)

Unmatched density, direct reuse of the CMOS supply chain, and solid-state coherence good enough for foundry-grade fidelity; against that, qubit counts are tiny (tens at most today versus thousands for atoms), historic device-to-device variability, short-range exchange connectivity, and cryogenic control wiring per qubit that is unsolved at scale — cryo-CMOS is the bet that closes it. The quantum-dot vs donor split is a live strategic fork (see `H-spinsplit`): manufacturability vs intrinsic uniformity.

CONFLICTS / OPEN QUESTIONS

Foundry-grade fidelity arrived in 2025; the open question is whether uniformity holds across 1,000+ dots on a wafer, and whether anyone stands up a mid-scale (100-qubit) silicon system before other modalities reach fault tolerance. Spin-shuttling fidelity across a chip is the connectivity bottleneck.

SOURCES

Diraq/imec Nature paper + The Quantum Insider (Sep 2025); SQC Nature (Dec 2025); postquantum.com silicon-spin fabrication + ecosystem analyses; PatSnap spin-qubit array review (2026).

Spin-photon interfaces — T-centers, SiV/SnV, rare-earth emitters · H-spinphoton

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The physical component that lets a stationary matter qubit (an electron or nuclear **spin**) emit a **photon** whose state — polarization, time-bin, or frequency — is entangled with the spin. That single-emitter primitive is what turns an isolated qubit into a network node: interfere photons from two remote emitters, herald on a detector click, and the two distant spins are entangled (the heralded-entanglement protocol under `F-teleport`, the building block of modular QC `H-intercon` and the quantum internet `A-qinternet` / repeaters). Unlike microwave↔optical transduction (`H-transduce`), which frequency-converts a photon with no memory attached, and unlike deterministic photon sources for photonic computing (`H-photonsource`), which want indistinguishable photons but no long-lived spin, a spin-photon interface must deliver **both** a coherent optical transition and a spin qubit that outlives many entanglement attempts.

The hard figures of merit: the **zero-phonon-line fraction** (Debye–Waller factor — how much emission lands in the coherent, interference-usable line rather than the phonon sideband), photon **indistinguishability**, spin **coherence** T_2, emission **wavelength** (telecom O/C-band beats visible for fiber reach), and the resulting remote-entanglement **rate × fidelity**. Nanophotonic cavities rescue weak emitters via the Purcell effect, $F_P = \frac{3}{4\pi^2}\left(\frac{\lambda}{n}\right)^3\frac{Q}{V}$, funnelling emission into the ZPL and raising cooperativity $C = g^2/(\kappa\gamma)$ toward the deterministic-interface regime.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Silicon T-centers** (Photonic Inc., Canada): spin-photon interface native to silicon, emitting at telecom **1326 nm** — CMOS-foundry-compatible and fiber-ready; distributed entanglement between modules demonstrated, roadmap ~200 kHz remote-entanglement rate at 99.8% fidelity (see `H-intercon`).
- **Group-IV diamond color centers — SiV / SnV** (Harvard/Lukin, QuTech, academic + Quantum Brilliance-adjacent): cavity-coupled SiV memory nodes and **tin-vacancy** centers combine a strong ZPL with better spin coherence than the NV (`H-nv`); high-fidelity spin-photon entanglement in nanophotonic cavities.
- **Rare-earth ions — erbium** in silicon or Y_2SiO_5: telecom **1.5 µm** emission and exceptional coherence, but weak dipoles that require strong cavity Purcell enhancement to emit usefully.
- **Quantum dots**: brightest, most indistinguishable photons, but short spin coherence and spectral diffusion limit them as memory nodes (they excel as sources instead — `H-photonsource`).

KEY GRADED CLAIMS

- T3 Telecom-band (1326 nm) silicon T-center spin-photon entanglement, foundry-compatible — Photonic Inc. (demonstrated; rate/fidelity targets still roadmap)
- T2/T3 Cavity-enhanced SiV/SnV spin-photon entanglement at high fidelity in diamond nanophotonics — Harvard/QuTech (demonstrated in lab)
- T4 200 kHz / 99.8% remote entanglement over telecom fiber — company target (roadmap)

TRADE-OFFS

Every candidate trades brightness against coherence against wavelength. Diamond centers give clean optics and good spin memory but need exotic fabrication and mostly emit in the visible; T-centers and erbium reach telecom fiber but start dim and lean on cavities; quantum dots are bright but forgetful. No single emitter yet wins all axes at once — which is why this is a distinct node rather than a footnote to `H-silicon` (electrically-controlled spin *compute*) or `H-nv` (NV sensing/compute). The spin-photon interface is the emitter node that makes matter-qubit networking possible at all.

CONFLICTS / OPEN QUESTIONS

Which emitter reaches deterministic, high-rate, high-fidelity, telecom-band operation first — and does cavity integration (`H-photonsource`, `H-package`) hold coherence while boosting brightness? Whether spin-photon links can stay above the entanglement-distillation threshold across real fiber decides if matter-qubit modality networks natively and sidesteps the microwave-transduction wall (`H-transduce`) that superconducting machines face.

SOURCES

Photonic Inc. T-center networking releases; Harvard/Lukin SiV cavity-QED memory-node papers (Nature/Science 2020–25); SnV spin-photon results (Nature Physics/PRX 2024–26); erbium-in-silicon single-photon emission (Nature 2024–26). Cross-links: `H-silicon`, `H-nv`, `H-photonsource`, `H-transduce`, `H-intercon`, `A-qinternet`, `F-teleport`, `H-detect`.

Quantum-dot vs donor: the silicon-spin fork · H-spinsplit

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Inside the silicon-spin modality (`H-silicon`) there is a strategic fork sharp enough to warrant its own node: two ways to hold a single-electron spin in silicon, optimizing for opposite things. **Gate-defined quantum dots** trap electrons in an electrostatic well shaped by lithographic top-gates (SiMOS or Si/SiGe heterostructures) — the qubit position and count are set by the mask, so the approach inherits CMOS manufacturability and prioritizes *scalability*. **Donor qubits** put the spin on a single dopant atom — usually phosphorus — placed with atomic precision by STM lithography, so every qubit is physically identical by nature, prioritizing *uniformity and raw qubit quality*. Both now clear 99% two-qubit fidelity, so the fork is no longer about which one works; it is about which one scales.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Donor camp** — **SQC** (Michelle Simmons, Australia): STM-placed phosphorus, 99.99% 2Q fidelity and Grover at 98.9% without error correction (Nature, Dec 2025); patterned 250,000-qubit registers in 8 hours (Nov 2025). The argument: identical atoms remove device-to-device variability at the source.
- **Quantum-dot camp** — **Diraq + imec**: gate-defined dots at >99% 2Q from a standard 300 mm CMOS flow, with four random wafer devices all under 1% error (Nature, 2025). **Intel** (Tunnel Falls), **Quantum Motion**, and **Equal1** all pursue the dot route because it plugs directly into the foundry (`H-foundry`) and cryo-CMOS (`H-cryocmos`).
- Hybrid ideas (flip-flop qubits, donor-dot hybrids, spin-shuttling buses) blur the line and try to take uniformity from donors and routing from dots.

KEY GRADED CLAIMS

- T2 Donor P devices: 99.99% 2Q, Grover 98.9% — SQC, Nature (Dec 2025) (demonstrated)
- T2 Gate-defined dots: >99% 2Q from a 300 mm CMOS flow — Diraq/imec, Nature (2025) (demonstrated)
- T1 Phosphorus donors are intrinsically identical; dots buy reproducibility from lithography — device physics (established)

TRADE-OFFS

Donors give the best intrinsic uniformity and coherence but atomic-precision STM placement is slow, serial, and hard to interface with industrial CMOS back-ends. Gate-defined dots ride the foundry and cryo-CMOS ecosystems and are naturally reproducible in *layout*, but electrostatic disorder and charge noise make each dot's exact working point drift, so tuning many dots (autotuning, ML calibration) is itself a research problem. Connectivity is short-range exchange in both, pushing spin-shuttling as the bus.

CONFLICTS / OPEN QUESTIONS

Which philosophy reaches a working 100-qubit silicon system first — the donor bet that uniformity beats the variability wall, or the dot bet that foundry volume and autotuning tame it? Does spin-shuttling across a chip preserve fidelity well enough to make either route's short-range connectivity a non-issue? The fork may resolve into a hybrid rather than a winner.

SOURCES

SQC Nature (Dec 2025); Diraq/imec Nature (2025); postquantum.com silicon-spin ecosystem; PatSnap spin-qubit-array review (2026); Intelligent Computing icomputing.0115 (single-electron spin review).

Superconducting (transmon) qubits · H-supercon

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Artificial atoms made from superconducting circuits — a Josephson junction gives a nonlinear inductance, so the two lowest energy levels of the circuit form a qubit. The transmon variant runs the junction shunted by a large capacitor to flatten its sensitivity to charge noise, trading anharmonicity for stability; the fluxonium variant shunts with a large inductance and reaches higher anharmonicity and longer coherence at the cost of a harder control problem. Qubits are driven with shaped microwave pulses at ~4–6 GHz and read out dispersively through a coupled resonator; the whole chip sits at ~10–20 mK in a dilution refrigerator (see `H-cryo`), with readout amplified by a near-quantum-limited parametric amplifier (see `H-paramp`). This is the most mature gate-based modality by deployed-system count and the one IBM, Google, and Rigetti build on.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **IBM**: Nighthawk (Nov 2025) — 120 transmons on a square lattice targeting 5,000-gate circuits in 2025, rising to 7,500 (2026), 10,000 (2027), 15,000 (2028), chained into three-module 360-qubit configs via l-couplers. Loon (2025) demonstrated c-couplers giving six-way non-nearest-neighbor connectivity for qLDPC codes while holding several-hundred-μs coherence. Roadmap: Kookaburra (2026, first qLDPC memory + logic-processing unit), Cockatoo (2027), **Starling** fault-tolerant system in 2029 (200 logical qubits, 100M gates), Blue Jay (2033, ~2,000 logical).
- **Google**: Willow, 105 qubits — below-threshold surface-code QEC (Nature 2024): logical error suppressed $\Lambda \approx 2.14\times$ per code-distance step ($d{=}3 \to 5 \to 7$), distance-7 logical memory outliving the best physical qubit by 2.4×, with a real-time decoder. Roadmap: a long-lived logical qubit, then ~1M physical qubits.
- **Rigetti**: Cepheus-1-108Q GA system (Apr 2026), 99.1% median 2Q fidelity via a modular 4×9-qubit chiplet tiling; targets 150+ qubits at 99.7% by late 2026, 1,000+ by end-2027.
- **IQM**: Radiance, ~150 qubits, 99.91% 2Q fidelity reported Mar 2026. **OQC** (UK): coaxmon 3D architecture. **SEEQC**: single-flux-quantum (SFQ) digital control on-chip.
- **Coherence records**: a Princeton 2D transmon on tantalum/high-resistivity-silicon reached $T_1 \approx 1.68$ ms ($Q \approx 2.5\times10^7$), roughly 3× the prior best (Nature 2025) — direct evidence the TLS/materials bottleneck (see `H-fab`) is still yielding to substrate engineering.

KEY GRADED CLAIMS

- T2 Below-threshold surface-code error correction on 105-qubit Willow — Nature s41586-024-08449-y (2024) (demonstrated)
- T2 2D transmon T1 up to 1.68 ms on tantalum/Si — Princeton, Nature (2025) (demonstrated)
- T4 IBM Starling: fault-tolerant, 200 logical qubits by 2029 — IBM roadmap (roadmap)
- T4 Rigetti 99.1% median 2Q on 108-qubit GA system — company release, Apr 2026 (claimed)
- T4 IQM 99.91% 2Q fidelity on Radiance — company claim, Mar 2026 (claimed)

TRADE-OFFS (VS OTHER MODALITIES)

Fast gates (tens of ns, ~1000× faster than ions), lithographic fab, and the deepest industrial base; against that, coherence measured in ~0.1–1.7 ms, mostly fixed nearest-neighbor connectivity, dilution cryogenics, and TLS defects that cap and drift coherence. Wiring heat-load and fridge volume dominate the scaling problem past ~1,000 qubits, which is why every roadmap turns to cryo-CMOS (`H-cryocmos`), 3D integration (`H-package`), and modular interconnects (`H-intercon`).

CONFLICTS / OPEN QUESTIONS

Can qLDPC codes (IBM) beat the surface code's overhead in real hardware, or is the surface code's higher threshold worth its worse encoding rate? Does the modality hit a wiring/cryo wall before millions of qubits, forcing fridge-to-fridge links that depend on unsolved microwave-optical transduction (`H-transduce`)?

SOURCES

IBM Quantum blog "large-scale FTQC" + roadmap 2025/2026; Google Quantum AI, Nature s41586-024-08449-y; Princeton Nature s41586-025-09687-4 (millisecond transmon); Rigetti investor release; IQM Radiance release; quantumzeitgeist.com superconducting guide (2026).

Topological / Majorana qubits · H-topo

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Qubits encoded in **Majorana zero modes** — quasiparticles predicted at the ends of a topological superconductor (a semiconductor nanowire proximitized by a superconductor, tuned into a topological phase by magnetic field and gate voltage). A single logical qubit is stored non-locally across a pair (or four) of these modes, so local noise cannot read or corrupt it: the error protection is built into the physics rather than bolted on by a code. Gates would be done by measurement (Microsoft's "measurement-based" scheme) or by braiding modes around each other. The prize is a qubit with a hardware error floor so low that QEC overhead collapses. The problem: after two decades, nobody has an independently verified working topological qubit.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Microsoft**: Majorana 1 (Feb 2025) — an 8-qubit "topoconductor" chip (InAs/Al), claiming measurement-based topological qubits with ~2 ms state lifetimes and an architecture that scales to 1M qubits on a palm-sized chip. The accompanying Nature paper's editors appended a note that the results "do not represent evidence for the presence of Majorana zero modes." Majorana 2 (Build milestone, Jun 2026): aluminum replaced with lead, claimed ~20-second state lifetimes (~1,000× improvement) and an FTQC timeline pulled from 2033 to 2029.
- Microsoft is effectively alone in productizing. **Nokia Bell Labs**, Delft/QuTech, Copenhagen, and other academic groups continue the underlying topological-matter physics; several have published careful null or ambiguous results on the same nanowire signatures.

KEY GRADED CLAIMS

- T4 Majorana 1 hosts measurable topological qubits — Microsoft (Feb 2025) (contested — Nature editorial note; APS-meeting skepticism; no independent reproduction)
- T4 Majorana 2: ~20 s lifetimes, 1,000× stability gain — Microsoft (Jun 2026) (contested — critics note the protocol-dependence and the still-missing independent demonstration of the topological phase). As of mid-2026 there is still **no independent replication**; the sharpest critique remains Henry Legg's formal Nature comment (24 Jun 2026, d41586-026-01788-y) flagging flawed tune-up routines, software errors in the analysis code, and omitted transport measurements — Microsoft filed a formal Nature reply (Nayak: "we stand by our results and our roadmap"). Reproduction is throttled by a de-facto verification vacuum: only a handful of labs can fabricate the InAs/Al (now Pb/Al) devices.
- T6 Scalable topological FTQC by 2029 — Microsoft roadmap (roadmap)

TRADE-OFFS (VS OTHER MODALITIES)

If real: intrinsic hardware error protection, digital voltage-based control (reusing the semiconductor toolset), a tiny footprint, and a dramatic cut in QEC overhead that would reshape every scaling roadmap in this chapter. Today: zero independently verified qubits, extreme sensitivity to disorder, and a research lineage that already produced one retracted Nature paper (the 2018 Delft/Microsoft-funded ballistic-Majorana quantized-conductance claim, retracted 2021). The modality is graded hardest in the atlas for exactly this reason.

CONFLICTS / OPEN QUESTIONS

C-majorana-exists: Microsoft (T4) vs. much of the condensed-matter community, which holds that disorder-induced Andreev bound states can mimic every reported signature. What would resolve it: an independent lab reproducing topological-qubit *operation*, or a braiding/fusion demonstration with published raw data that unambiguously passes the topological-gap protocol.

SOURCES

Nature d41586-026-01788-y (2026 skeptics piece) + Majorana 1 paper editorial note; Science/AAAS coverage (2025, 2026); Physics World expert roundup; postquantum.com Majorana 2 analysis; 2021 Nature retraction of the 2018 quantized-conductance paper.

Microwave⇔optical transduction · H-transduce

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Superconducting qubits live at ~10 mK and speak microwave photons (~5–10 GHz). Optical fiber carries quantum information between fridges at ~200 THz with almost no loss. Nothing bridges the ~10,000× frequency gap cheaply, so a **quantum transducer** — a device that coherently converts a single microwave photon into a single optical photon and back — is the missing link for fridge-to-fridge superconducting modularity (`H-intercon`) and for a superconducting-to-photonic quantum internet (`A-qinternet`). Three physical routes compete: **electro-optic** (χ^2 crystals, thin-film lithium niobate), **optomechanical** (a vibrating membrane or bulk-acoustic mode couples both fields), and **magneto-optic / atomic** (rare-earth ions like erbium, or magnons in YIG). The figures of merit are conversion **efficiency** η, **added noise** (junk photons per converted photon), and **bandwidth**.

KEY PLAYERS & STATE OF THE ART (2025–26)

- The target is the "quantum-enabled" regime: added noise below 1 photon *and* η above ~50% simultaneously. Theory says η > 1/2 with low noise is physically allowed, so this is an engineering wall, not a fundamental one.
- 2025 membrane-based **optomechanical** transducers reached input-referred added noise approaching single-photon levels in both directions; one platform reported external η ≈ 2.2% at added noise ≈ 0.94, with an efficiency-bandwidth product ~2 orders of magnitude above prior low-noise demos (arXiv:2509.26349).
- **Electro-optic** thin-film-lithium-niobate devices push bandwidth and integration; **erbium-ion-in-crystal** transducers coupled to planar photonic + superconducting resonators pursue the atomic route (Nature Communications).
- Academic groups (Caltech, JILA/NIST, Delft, Chicago, Yale) dominate; no commercial merchant transducer ships yet. IBM, PsiQuantum, and every superconducting-modularity roadmap treat it as an *open dependency*, not a solved input.

KEY GRADED CLAIMS

- T3 Membrane optomechanical transduction near single-photon added noise, η ~2% — APS SMT 2025 / arXiv:2509.26349 (demonstrated, low efficiency)
- T1 η > 1/2 with low added noise is physically allowed — transduction theory (established)
- T4 Efficiency high enough for fault-tolerant fridge-to-fridge links — no group claims this yet (roadmap)

TRADE-OFFS

The brutal three-way tension is efficiency vs added noise vs bandwidth: pumping harder raises η but heats the device and injects thermal noise; narrow bandwidth mismatches qubit lifetimes and throughput. Percent-level efficiency means most photons are simply lost — unusable for QEC, which needs high-fidelity links. This node, not qubit count, gates whether superconducting machines can ever go truly modular across fridges.

CONFLICTS / OPEN QUESTIONS

Will any of the three routes reach simultaneous high-η, sub-unity-noise, qubit-bandwidth operation — and on what timeline? Or do superconducting machines stay confined to single-fridge scale (`H-cryo` volume limits) while photonic (`H-photonic`) and ion (`H-ion`) modalities network natively over fiber and sidestep the problem entirely? This is one of the quietest but most decisive forks in the whole hardware map.

SOURCES

schedule.aps.org/smt/2025 (MAR-A09); arXiv:2509.26349; Nature Communications s41467-023-36799-0; arXiv:2503.01133; arXiv:2406.02704.

Ultra-high-vacuum & atomic-source systems · H-uhv

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Trapped-ion (`H-ion`) and neutral-atom (`H-neutral`) machines hold their qubits — bare atoms — in an **ultra-high or extreme-high vacuum** chamber (10^{-11} mbar and below). The vacuum *is* the qubit's isolation: a single collision with a stray background gas molecule knocks an atom out of its trap, which is the modality's characteristic error channel (atom/ion loss). The system comprises the chamber and viewports, ion/getter pumps, an atomic source (an oven or dispenser for ions; a 2D-MOT / dispenser cold-atom source for neutral atoms), and — increasingly — a **cryogenic shroud** that both improves the vacuum (cold surfaces cryopump residual gas) and blocks blackbody radiation that would otherwise limit Rydberg-state lifetimes. It is the atomic-modality counterpart of the dilution fridge (`H-cryo`): unglamorous plumbing that sets a hard ceiling on qubit lifetime and count.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Cryogenic vacuum for atoms**: a high-optical-access cryogenic chamber achieved a **3,000-second single-atom trap lifetime** for Rydberg arrays (arXiv:2412.09780) — using cryogenically cooled metal walls and windows to cryopump the background gas and suppress unwanted transitions. Trap lifetime, historically seconds, is now limited by design rather than by attainable vacuum.
- **Continuous operation**: Harvard/MIT ran a ~3,000-atom array for over two hours by reloading atoms mid-computation (2025), turning atom loss from a hard stop into a managed, replenished error — a vacuum/atom-source engineering result as much as a physics one (see `H-neutral`).
- **Scale**: Caltech loaded 6,100 atoms in one chamber (Sep 2025); QuEra and Atom Computing project ~100,000 atoms per vacuum chamber within a few years — which pushes chamber size, pumping speed, and source flux hard. Vendors like **ColdQuanta/Infleqtion** and specialist UHV-chamber makers supply the hardware.

KEY GRADED CLAIMS

- T2 3,000-second trap lifetime in a cryogenic high-optical-access chamber — arXiv:2412.09780 (demonstrated)
- T2 Continuous >2 hr operation via mid-computation atom reloading — Harvard/MIT, Nature (2025) (demonstrated)
- T1 A background-gas collision ejects a trapped atom (loss is vacuum-limited) — established atomic physics

TRADE-OFFS

Room-temperature UHV chambers are simpler and give full optical access but cap trap lifetime at seconds to minutes; cryogenic chambers extend lifetime by orders of magnitude and add blackbody suppression, at the cost of a cooling system and reduced optical access. Bigger chambers hold more atoms but are harder to pump and to keep field-uniform. The atomic source flux trades loading speed against vacuum degradation.

CONFLICTS / OPEN QUESTIONS

Does the path to 100,000-atom machines run through room-temperature chambers with fast reloading, or through cryogenic chambers with long-lived atoms — and can either keep vacuum-limited loss below the QEC threshold at that scale? Optical access vs cryogenic shrouding is a live chamber-design tension for the biggest neutral-atom roadmaps.

SOURCES

arXiv:2412.09780 (3,000 s cryogenic trap); Nature 2025 continuous-operation paper; Caltech 6,100-atom (Sep 2025); APS Physics v18/103 (longer atom trapping); postquantum.com building-a-neutral-atom-QC guide.

Cryogenic microwave wiring — coax, attenuators, isolators · H-wiring

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The passive microwave plumbing that carries signals between room-temperature electronics (`H-control`) and the qubits at the bottom of the fridge (`H-cryo`) — and the single least-glamorous scaling wall in superconducting and spin QC. Every superconducting qubit needs on the order of 2–4 lines (drive, flux, readout in/out), each a coaxial cable running through four temperature stages. Three passive families dominate: **coax** (semi-rigid CuNi or stainless above 4 K for low thermal conductance; superconducting NbTi below 4 K for near-zero loss), **attenuators** anchored and distributed across stages (a canonical drive-line budget is ~20 dB at 4 K, ~10 dB at the still, ~20–30 dB at the mixing chamber), and **isolators / circulators** — bulky ferrite, magnetically biased, non-reciprocal parts on the output line that protect the qubit from amplifier back-action (`H-paramp`) but do not shrink or integrate. Filters (IR/low-pass, eccosorb) and thermal anchoring round out the chain.

The attenuators are not there to weaken the signal for its own sake; they thermalize the Johnson–Nyquist noise riding down each line, so the qubit sees a cold photon bath. Residual thermal occupation is $\bar{n} = (e^{\hbar\omega/k_B T} - 1)^{-1}$; at 5 GHz and 20 mK, $\bar{n} \sim 10^{-5}$, but only if every stage is properly attenuated and anchored. Each coax also dumps conducted + radiated heat, so line count is capped by the fridge's cooling power — the bottleneck `H-cryocmos` and photonic I/O exist to break.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Coax / connectors**: HuberSuhner, Radiall, Maury Microwave; **cryo attenuators**: XMA/Omni-Spectra, Bluefors wiring kits.
- **Delft Circuits** (NL): flexible **Cri/oFlex** ribbon cabling puts dozens of lines in one flex, cutting the per-qubit space and heat-load footprint — the leading answer to the coax-count problem.
- **Low Noise Factory / Silent Waves / Quantum Microwave**: cryo isolators, directional couplers, and the output-chain components feeding the HEMT/TWPA readout.
- **Isolator replacement** is the hot research front: on-chip Josephson circulators, directional/nonreciprocal TWPAs, and multiplexed readout aim to delete the bulky ferrite parts entirely — none is yet a drop-in production replacement.

KEY GRADED CLAIMS

- T1 Distributed multi-stage attenuation + NbTi superconducting coax deliver $\bar{n} \ll 1$ thermal photons at the qubit — established cryo-microwave practice
- T3 Flexible ribbon cabling (Cri/oFlex-class) materially raises line density per fridge — Delft Circuits + lab deployments (demonstrated)
- T4 On-chip nonreciprocity will retire ferrite isolators at scale — research roadmap (not yet productized)

TRADE-OFFS

Every added line buys control fidelity and costs cooling power and volume; the ferrite isolators are the worst offenders on footprint and refuse to integrate. A million-qubit superconducting machine implies millions of coax lines — physically impossible in one fridge, which is exactly why the endgame forks toward cryo-CMOS multiplexing (`H-cryocmos`), photonic readout, and multi-fridge modularity (`H-intercon`, `H-transduce`). This node is the passive-supply-chain twin of the active control problem, split out from `H-cryo` (the fridge) and `H-fab` (the chip materials).

CONFLICTS / OPEN QUESTIONS

Does the wiring bottleneck get solved by better passive engineering (denser ribbons, integrated isolators) or does it force an architectural escape (cryo-CMOS in-fridge control, optical I/O) before superconducting qubit counts reach the tens of thousands? The ferrite-isolator replacement is the pivotal unsolved passive component. Supply concentrates in few vendors — a chokepoint mirrored in `E-supplychain`.

SOURCES

Krinner et al., "Engineering cryogenic setups for 100-qubit scale systems," EPJ Quantum Technology 6, 2 (2019); Delft Circuits Cri/oFlex documentation; Bluefors wiring guides; Low Noise Factory / Silent Waves component notes. Cross-links: `H-cryo`, `H-cryocmos`, `H-control`, `H-paramp`, `H-fab`, `H-intercon`, `E-supplychain`.

Chapter 3 · From Qubit to Answer

Benchmarks & "advantage" claims · S-bench

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

How the field measures machines and adjudicates beyond-classical claims. **Holistic metrics**: Quantum Volume (IBM 2019 — the largest square random circuit a device passes, 2^n; conflates width, depth, fidelity into one number), CLOPS (circuit layers per second, a speed metric), Algorithmic Qubits (IonQ, application-weighted — disputed methodology). **Component metrics**: gate fidelity via randomized benchmarking / cross-entropy benchmarking / cycle benchmarking (`S-gates`), plus IBM's "layer fidelity." **Advantage claims** rest on sampling tasks — random circuit sampling (RCS, scored by cross-entropy benchmarking / XEB) and boson sampling — where classical simulation cost is *conjectured* exponential, tying the claim to a complexity assumption rather than a proof (`S-complexity`, `S-tensornet`).

WHERE IT STANDS (2025–26)

Every advantage claim has drawn a classical counterattack; several fell. **Google 2019** (Sycamore, 53-qubit RCS, "10,000 years"): IBM answered 2.5 days immediately; Pan–Chen–Zhang tensor-network methods (2021–22) cut it to hours-then-minutes on GPU clusters — effectively matched (status: fell). **USTC Jiuzhang** boson sampling (2020): substantially eroded by spoofing and tensor methods. **Willow** (Dec 2024): 105-qubit RCS claimed at $\sim 10^{25}$ classical-years; no classical match to date, tensor-network teams still narrowing the gap — contested-by-default. **Willow "quantum echoes"** (Oct 2025): 13,000× on an OTOC measurement (`S-qsim`), marketed as *verifiable* advantage — under scrutiny, unresolved. Quantinuum holds the QV record ($\sim 2^{25}$ region, self-reported, 2025); IBM de-emphasized QV in favor of gate-count / layer-fidelity metrics; the resulting **benchmark fragmentation is itself a finding**. IBM staked "verified quantum advantage by end of 2026" on Nighthawk. The reference implementation models the honest small-scale version of this discipline: it reports kernel RMSE and diagonal-deviation as per-run noise gauges rather than a headline advantage number (`reference-impl/MATH.md` §6–7).

KEY GRADED CLAIMS

- T2 Sycamore 2019 RCS beyond-classical — Arute et al., Nature 574, 505 (2019), then matched classically: Pan et al., PRL 129, 090502 (2022) (status: fell)
- T3/T4 Willow 2024 RCS unmatched classically so far — Google blog + Nature 638 (2024) (claimed; contested by default, standing)
- T4 QV $\approx 2^{25}$ record — Quantinuum blog (vendor self-reported)
- T4 Algorithmic Qubits metric — IonQ; publicly disputed by Quantinuum ("Debunking algorithmic qubits") (contested)

SPEEDUP / CAVEAT

Sampling tasks have **no known application**; their hardness is conjectural and erodes as classical algorithms improve (`S-tensornet`, `O-advantage`). A benchmark controlled by the vendor reporting it is **T4 by rule** (`evidence/SCHEMA.md`). "Verifiable" advantage matters because RCS results are themselves hard to check — Google's OTOC claim leans on this (`O-verification`).

CONFLICTS / OPEN QUESTIONS

C-advantage: does Willow-class RCS stay unmatched? What replaces QV as the cross-platform standard — DARPA QBI / QED-C application benchmarks are the leading candidates for an application-grounded, vendor-neutral metric.

SOURCES

Nature 574, 505 (2019); PRL 129, 090502 (2022); arXiv:2408.13687; blog.google quantum-echoes; quantinuum.com/blog. Cross-links: `S-tensornet`, `S-complexity`, `S-qsim`, `S-gates`, `O-advantage`, `O-verification`, `reference-impl/`.

Certified randomness · S-certrand

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Certified randomness is a protocol that produces bits *provably* random — not merely unpredictable to an observer, but backed by a mathematical certificate that no adversary (not even the device manufacturer) could have known or biased them. The construction (Aaronson, 2018–20) piggybacks on random circuit sampling, the same task used for quantum-advantage claims: a classical server sends hard random-circuit challenges to a remote quantum computer, times the responses, and later spot-checks them on a classical supercomputer via cross-entropy benchmarking. Because a genuine quantum device is the only thing that could answer fast enough, passing responses certify that fresh entropy was generated. It is the first delivered *application* of quantum advantage — a working product built on top of the sampling experiments that were previously only benchmarks. It bridges to `A-qrng` (physical randomness sources): certified randomness adds the remote, adversary-proof *certificate* that a local quantum RNG cannot give.

WHERE IT STANDS (2025–26)

The landmark is JPMorgan × Quantinuum × Argonne/Oak Ridge × UT Austin, published in Nature (26 Mar 2025). Running on Quantinuum's 56-qubit trapped-ion H2, the team generated 71,313 certified-random bits; verification consumed ~1.1 exaFLOPS across supercomputers — the classical check itself is what makes the quantum step meaningful (only a real quantum device answers in time, only a classical HPC cluster can verify). Provenance matters: JPMorgan (a customer with a cryptography use case) and Quantinuum (the hardware vendor) co-authored, but this cleared peer review in Nature, so it grades above a vendor PR. The honest caveat is that certified randomness is a niche, if real, first application — its commercial pull (lotteries, cryptographic key material, auditable fairness) is genuine but narrow, and the verification cost is enormous.

KEY GRADED CLAIMS

- T2 Protocol: certified randomness expansion from random circuit sampling — Aaronson–Hung, STOC 2023 (arXiv:2303.01625) (established theory)
- T2 First experimental certified randomness: 71,313 bits on 56-qubit H2, verified at ~1.1 exaFLOPS — Nature 640 (2025), s41586-025-08737-1 (peer-reviewed)
- [T4→T2] "First commercial application of quantum computing" framing is vendor-sourced (Quantinuum/JPMorgan), but the underlying result is peer-reviewed (claimed marketing / demonstrated result)

SPEEDUP / CAVEAT

Advantage as a *product*, not a raw speedup. It exploits the fact that random circuit sampling is classically intractable to run but checkable, turning a supremacy benchmark into a service. Caveat: verification requires exaFLOPS-class classical compute per batch, and the application space (adversary-proof entropy) is narrow versus the general-purpose promise of quantum computing.

CONFLICTS / OPEN QUESTIONS

Whether the RCS foundation stays classically hard (spoofing/classical-simulation counterattacks target the same experiments — see `O-advantage`, `S-bench`), and whether the verification cost can drop enough to make this a routine service rather than a showcase.

SOURCES

Nature 640, s41586-025-08737-1 (2025); arXiv:2303.01625; JPMorganChase / UT Austin CNS / ORNL press (2025). Cross-links: `A-qrng`, `S-bench`, `O-advantage`, `I-finance`.

Circuit knitting & cutting · S-circuitcut

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Circuit knitting runs a circuit that is too large for one QPU by decomposing it into smaller subcircuits, executing them separately, and reconstructing the full result classically. Two cut types: **gate cuts** replace a non-local two-qubit gate with a quasiprobability sum of local operations, and **wire cuts** break a qubit's timeline into a measure-and-reprepare step. The reconstruction uses a **quasiprobability decomposition** — write the non-local operation as a signed combination of local channels, sample from it, and combine with the right signs in post-processing. This trades quantum width for classical sampling: it lets a 100-qubit problem run on smaller devices, or partitions a circuit across multiple QPUs (relevant to modular/interconnect hardware, `H-intercon`). IBM's Circuit Knitting Toolbox / qiskit-addon-cutting includes an automated cut-finder that searches for the cuts minimizing overhead.

WHERE IT STANDS (2025–26)

The technique is mature as software but bounded by a hard scaling law: each cut multiplies the sampling cost, and the **total overhead grows exponentially in the number of cuts** (γ^2 per gate cut, worse for wire cuts). So knitting is practical for a handful of cuts — separating weakly-entangled blocks, or shaving a few qubits past a device limit — and hopeless for cutting a densely entangled circuit into many pieces. The 2024–25 research pushed on the exponent: classical side-information from intermediate measurements improves post-processing (Phys. Rev. Research 2025), and a Dec 2025 result reduces the measurement overhead from exponential to **polynomial** in special cases via a quantum-tomography reformulation (arXiv:2512.19623) — a meaningful but still-narrow escape from the exponential wall. It shares a mathematical core with error mitigation (`S-errmit`): both are quasiprobability sampling, both pay exponential sampling cost, both estimate expectation values rather than states.

KEY GRADED CLAIMS

- T2 Gate/wire cutting via quasiprobability; overhead exponential in cut count — Bravyi–Smith–Smolin, PRX 6, 021043 (2016); Peng et al., PRL 125, 150504 (2020) (established)
- T2 Automated cut-finding minimizing sampling overhead — IBM Circuit Knitting Toolbox, IBM Research QCE 2024 (established, software)
- T2 Classical side-information improves reconstruction — Phys. Rev. Research (2025) (peer-reviewed)
- T3 Exponential → polynomial overhead in special cases via tomography reformulation — arXiv:2512.19623 (2025) (preprint)

SPEEDUP / CAVEAT

Not a speedup — a **space-for-time trade** (fewer qubits, exponentially more shots). It is a bridge technology for the pre-fault-tolerant / modular era, useful only when cuts are few and cross-cut entanglement is low; for a highly entangled cut it costs as much as the classical simulation it was trying to avoid (`S-tensornet`). Overlaps error mitigation's cost structure, so the same "shallow enough to knit ≈ shallow enough to simulate classically" caution applies.

CONFLICTS / OPEN QUESTIONS

How far the polynomial-overhead special cases generalize, and whether knitting across real chip-to-chip links (`H-intercon`, `O-interconnect-loss`) beats simply building a bigger monolithic device. Whether it earns a durable role once large fault-tolerant machines exist.

SOURCES

PRX 6, 021043 (2016); PRL 125, 150504 (2020); IBM Research QCE 2024; arXiv:2411.17756; arXiv:2512.19623. Cross-links: `S-errmit`, `S-tensornet`, `S-shadows`, `H-intercon`, `O-interconnect-loss`.

Cloud QPU access · S-cloud

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

How anyone actually touches a quantum computer: cloud platforms queueing circuits to remote QPUs. Three models dominate. **IBM Quantum** — own fleet, per-QPU-minute runtime, tiered Open/Pay-as-you-go/Premium. **AWS Braket** — an aggregator fronting IonQ, Rigetti, IQM, QuEra, and others, priced per-task plus per-shot. **Azure Quantum** — aggregator fronting IonQ and Quantinuum, plus the Azure resource estimator, priced per-program or by subscription. Google Quantum AI stays research-partner-gated. The compilation and submission all run through the software layer (`S-software`). The reference implementation is written to hit exactly these backends — Qiskit Runtime to IBM's least-busy device, Braket to IonQ — with only the backend flag changing (`reference-impl/README.md`).

WHERE IT STANDS (2025–26)

Access is a solved problem; the cost structure shapes research. Representative 2026 numbers: IBM PAYG ≈ **$96/QPU-minute** ($48–72 with Flex/Premium tiers); Braket ≈ **$0.30/task** plus per-shot fees that vary ~25× by hardware (a 100-shot circuit runs ~$8.30 on IonQ Forte versus ~$0.34 on Rigetti); a 200k-shot VQE run spans **$85–$6,000** depending on backend. Quantinuum sells H-series/Helios access directly and via Azure. NVIDIA's CUDA-Q and "quantum-centric supercomputing" partnerships (IBM–RIKEN, various HPC centers) are pulling QPUs into HPC job schedulers as accelerators alongside GPUs. On-prem purchases (IQM, Rigetti, IBM systems at national labs) grew as sovereignty concerns rose (`E-supplychain`, `E-us`). Free tiers shrank across the board as demand from error-correction experiments soaked up device time.

KEY GRADED CLAIMS

- T2 Multi-vendor cloud access operational since 2016 (IBM) / 2019 (Braket, Azure) — platform documentation (established)
- T4 Published pricing: IBM ~$96/min PAYG; Braket $0.30/task + per-shot — aws.amazon.com/braket/pricing, IBM docs (vendor-published, verifiable)
- T5 "Quantum cloud market" TAM projections — analyst reports (forecast; double-counting risk, `E-market`)

SPEEDUP / CAVEAT

Cloud access sets the **economics** of experimentation. At ~$100/minute, algorithm development happens on simulators and only validation touches hardware — exactly the workflow the reference impl documents (develop on Aer, validate on IBM/IonQ). Queue times and calibration drift make published fidelity numbers optimistic relative to what a random user session sees; a device benchmarked at 99.9% on a good day may deliver noticeably worse mid-drift.

CONFLICTS / OPEN QUESTIONS

Aggregator versus vertical model — do Braket/Azure margins survive if the winning hardware vendors (IBM, Quantinuum) keep their best machines first-party? How fast does HPC-integrated access displace pure cloud submission as QEC workloads demand tight classical-quantum coupling (`S-decoders`)?

SOURCES

aws.amazon.com/braket/pricing; learn.microsoft.com/azure/quantum/pricing; IBM Quantum docs. Cross-links: `S-software`, `S-decoders`, `E-market`, `E-supplychain`, `reference-impl/`.

Quantum advantage complexity theory — BQP, BPP & the separations · S-complexity

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The complexity-theory scaffolding that tells you *what* quantum computers can provably do faster, and what "quantum advantage" formally means. **BQP** (bounded-error quantum polynomial time) is the class of problems a quantum computer solves efficiently; **BPP** is its classical randomized counterpart. The central facts: $\mathrm{BPP} \subseteq \mathrm{BQP} \subseteq \mathrm{PSPACE}$, so quantum computers are no more powerful than classical ones up to polynomial space, and a proof that $\mathrm{BQP} \neq \mathrm{BPP}$ would imply $\mathrm{P} \neq \mathrm{PSPACE}$ — beyond current mathematics. This is why **every** quantum speedup is either (a) relative to the *best known* classical algorithm, not a proven separation (Shor, `S-shor`), or (b) proven only in a restricted **oracle / query model** (Grover, `S-grover`; Simon's problem, the first exponential query separation), or (c) proven under a **complexity-theoretic assumption** (sampling advantage rests on the non-collapse of the polynomial hierarchy). The map's `O-advantage` node is the empirical face of this node's theory.

WHERE IT STANDS (2025–26)

The strongest unconditional evidence that quantum beats classical is relativized. **Raz–Tal (STOC 2019)** built an oracle under which $\mathrm{BQP} \not\subset \mathrm{PH}$ (the polynomial hierarchy) — refuting the natural conjecture that quantum power stays inside the classical hierarchy, using the Forrelation problem and a deep bound against constant-depth (AC^0) circuits. **Shallow-circuit separations** (Bravyi–Gosset–König, Science 2018, and the noise-robust follow-ups) prove a *provable, unconditional* advantage of constant-depth quantum over constant-depth classical circuits — one of the few speedups needing no complexity assumption at all. **Sampling advantage** (random circuit sampling, boson sampling; `S-bench`) is proven hard for classical computers *only if* the polynomial hierarchy does not collapse — a widely believed but unproven assumption, and one that erodes empirically as tensor-network methods improve (`S-tensornet`). **Interactive verification** (Mahadev 2018) lets a classical verifier check a quantum computation it cannot itself run, underwriting the "verifiable advantage" framing (`O-verification`). Dequantization (Tang, `S-hhl`/`S-qml`) is the counter-current: several claimed exponential separations collapsed when the classical input model was matched.

KEY GRADED CLAIMS

- T1 $\mathrm{BPP} \subseteq \mathrm{BQP} \subseteq \mathrm{PSPACE}$; $\mathrm{BQP} \subseteq \mathrm{P}^{\#\mathrm{P}}$ — Bernstein–Vazirani, SIAM J. Comput. 26 (1997) (established)
- T1 Oracle separation $\mathrm{BQP} \not\subset \mathrm{PH}$ — Raz–Tal, STOC 2019 (established, relativized)
- T1 Unconditional constant-depth quantum > constant-depth classical — Bravyi–Gosset–König, Science 362, 308 (2018) (established)
- T2 Sampling advantage hard unless PH collapses — Aaronson–Arkhipov (boson sampling, 2011); Bouland et al., Nat. Phys. 15 (2019) (established, conditional)
- T2 Classical verification of quantum computation — Mahadev, FOCS 2018 (established)

SPEEDUP / CAVEAT

This node is the *definition* of the caveat. No exponential quantum speedup over classical is unconditionally proven in the standard (non-oracle) model — proving one would resolve open problems in classical complexity. So the honest hierarchy of claims is: unconditional-but-shallow (Bravyi–Gosset–König) > relativized (Raz–Tal, Simon) > assumption-based (sampling) > best-known-classical (Shor, and the algorithms most people mean by "quantum advantage"). Grading every speedup by which rung it sits on is exactly the atlas's job (`evidence/SCHEMA.md`).

CONFLICTS / OPEN QUESTIONS

Whether $\mathrm{BQP} \neq \mathrm{BPP}$ at all (unknown, and likely as hard as P vs PSPACE). Whether any *practical* problem sits in the proven-hard regime with an efficient input model (the running `S-hhl`/`S-qram` question). Whether sampling hardness assumptions survive continued classical progress (`S-tensornet`, `S-bench`).

SOURCES

SIAM J. Comput. 26, 1411 (1997); Raz–Tal, STOC 2019; Science 362, 308 (2018); Nat. Phys. 15, 159 (2019); Mahadev, FOCS 2018. Cross-links: `S-shor`, `S-grover`, `S-bench`, `S-tensornet`, `O-advantage`, `O-verification`.

Real-time QEC decoding · S-decoders

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Error correction (`S-qec`) is only half the job — something must read the stream of syndrome measurements and infer, in real time, which correction to apply before the next round of gates. That something is the *decoder*, and it has become its own engineering discipline. The throughput wall is brutal: a surface-code logical qubit emits a syndrome every QEC cycle (~1 µs on superconducting hardware), and the decoder must keep up or the error backlog grows without bound (the "decoding backlog problem," Terhal). At scale this is a firehose — Riverlane estimates a large machine produces on the order of ~100 TB/s of syndrome data. General software decoders (minimum-weight perfect matching, union-find, belief-propagation) are accurate but too slow, so the field has moved to dedicated FPGA and ASIC decoders co-located with the control electronics (`H-control`).

WHERE IT STANDS (2025–26)

Riverlane is the pure-play here (Deltaflow QEC system; Deltaflow 2 shipped 2025, Deltaflow 3 targeted late 2026 with continuous "streaming" decoding). In Dec 2025 they published a hardware decoder in Nature Electronics: a Local Clustering Decoder on FPGA that decodes a round in under 1 µs while adaptively modeling noise, cutting the physical-qubit count per logical qubit ~4× under a leakage-dominated model. Google's Willow result (2024) used a real-time-capable decoder to show below-threshold scaling, and its AlphaQubit work applied ML decoding for higher accuracy at the cost of speed — the central tension of the field: accuracy vs latency vs power. This is a named open problem in its own right (`O-decoder`): the classical co-processor bandwidth may bottleneck fault-tolerant scaling as hard as the qubits do.

KEY GRADED CLAIMS

- T2 Decoding must run inside the QEC cycle or the backlog diverges — Terhal, Rev. Mod. Phys. 87, 307 (2015) (established)
- T2 Sub-µs adaptive hardware (FPGA) decoder, ~4× qubit reduction under leakage noise — Riverlane, Nature Electronics (Dec 2025) (peer-reviewed)
- T2 Real-time decoding enabled Willow below-threshold demo; ML decoders (AlphaQubit) trade latency for accuracy — Google, Nature 638 (2024/25) (peer-reviewed)

SPEEDUP / CAVEAT

Not an algorithm speedup — an enabling constraint. Fault tolerance is impossible without a decoder that sustains syndrome throughput at scale; a slow decoder silently caps how large a computation can run regardless of qubit quality.

CONFLICTS / OPEN QUESTIONS

Whether ASIC/FPGA decoders scale to millions of qubits (100 TB/s-class data movement, cryo-adjacent power budgets) or become the true bottleneck — see `O-decoder`. Accuracy-vs-latency-vs-power has no settled optimum.

SOURCES

Rev. Mod. Phys. 87, 307 (2015); Riverlane Nature Electronics 2025 (HPCwire coverage); Google Willow, Nature 638 (2025); AlphaQubit, Nature 2024. Cross-links: `S-qec`, `S-logical`, `H-control`, `O-decoder`.

Error mitigation as a discipline · S-errmit

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The technique taxonomy that `S-nisq` names in passing, treated as its own layer because it now sits *underneath* early fault tolerance, not only on bare NISQ. Error mitigation reduces bias in an estimated **expectation value** by spending extra circuit runs; it does not protect a quantum state (that is correction, `S-qec`). The main families (Cai et al., Rev. Mod. Phys. 95, 045005, 2023): - **Zero-Noise Extrapolation (ZNE)** — run at amplified noise, extrapolate to zero. - **Probabilistic Error Cancellation (PEC)** — learn the noise, invert it via a quasiprobability sum (unbiased, exponential sampling cost). - **Symmetry verification / post-selection** — discard runs that violate a known conserved quantity. - **Virtual distillation / ESD** — prepare M copies and measure to suppress errors as the state's purity, cancelling incoherent noise to $O(1/M)$. - **Probabilistic Error Amplification (PEA)** — the learned-noise-model backbone of IBM's 2023 utility experiment. - **Learning-based / Clifford-data regression** — train the correction on classically simulable Clifford circuits.

WHERE IT STANDS (2025–26)

The defining theoretical result is a hard ceiling: the sampling overhead of *any* mitigation grows **exponentially** with circuit depth and fault rate (Takagi et al.; Quek et al., "exponentially tighter bounds," Nat. Phys. 2024). So mitigation is a constant-to-polynomial extension of reach, never an asymptotic fix — this is what separates it from correction as a matter of principle. The 2025 shift is that mitigation is being layered **on top of logical qubits**: demonstrations of error mitigation applied to already-error-corrected circuits (Nat. Commun. 2025) and "mitigation for early fault tolerance" show the discipline surviving into the FTQC transition rather than being abandoned with NISQ. PEC and PEA also power the standing (and contested) utility-scale expectation-value claims that tensor networks (`S-tensornet`) keep answering.

KEY GRADED CLAIMS

- T2 Comprehensive discipline review; ZNE/PEC/symmetry/virtual-distillation taxonomy — Cai et al., Rev. Mod. Phys. 95, 045005 (2023) (established)
- T2 Exponential sampling-overhead lower bound for generic mitigation — Quek et al., Nat. Phys. 20 (2024); Takagi et al., npj QI (2022) (established)
- T2 Virtual distillation suppresses errors as $O(1/M)$ in copy number — Huggins et al., PRX 11, 041036 (2021); Koczor, PRX 11, 031057 (2021) (established)
- T2 Mitigation demonstrated on logical qubits — Nat. Commun. (2025), s41467-025-67768-4 (peer-reviewed)

SPEEDUP / CAVEAT

Never a speedup — a bias-reduction technique bought with exponentially many shots in the worst case. The rule follows directly (`evidence/SCHEMA.md`): any advantage claim resting on mitigation alone is exposed to classical counterattack, because a circuit shallow enough for mitigation to converge is usually shallow enough to simulate (`S-tensornet`). Mitigation's honest role is to buy accuracy on small logical devices during the early-FT window.

CONFLICTS / OPEN QUESTIONS

Where the crossover sits — at what device quality does adding a logical layer beat spending the same resources on more mitigation shots? Whether learning-based mitigation generalizes beyond the Clifford-regression regime it is trained on.

SOURCES

Rev. Mod. Phys. 95, 045005 (2023); Nat. Phys. 20 (2024); PRX 11, 041036 (2021); PRX 11, 031057 (2021); Nat. Commun. (2025). Cross-links: `S-nisq`, `S-qec`, `S-tensornet`, `S-bench`, `O-advantage`.

Gates, circuits & universal sets · S-gates

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Quantum computation is expressed as circuits: sequences of unitary gates on qubits, followed by measurement. A finite gate set is **universal** if it approximates any unitary to arbitrary precision; the canonical example is Clifford (H, S, CNOT) plus the T gate. The Solovay–Kitaev theorem guarantees efficient synthesis — approximating any single-qubit unitary to precision ε costs $O(\log^c(1/\varepsilon))$ gates ($c \approx 2$–4 depending on the variant), i.e. polylogarithmic overhead. The Gottesman–Knill theorem draws the dividing line: **Clifford-only** circuits are classically simulable in polynomial time, so the T gate is the source of quantum hardness and the currency of fault-tolerant cost. Every hardware platform exposes a small **native** set — CZ or iSWAP-family on superconducting chips, Mølmer–Sørensen on trapped ions, Rydberg-blockade CZ on neutral atoms — into which the compiler (`S-software`) rewrites everything. The Hadamard gate that powers the interference in the atlas's swap/Hadamard-test reference implementation (`reference-impl/MATH.md` §1) is the simplest such primitive.

WHERE IT STANDS (2025–26)

Two-qubit gate fidelity is the binding constraint on everything upstream (`S-qec` thresholds, `S-logical` counts). Best reported: **Quantinuum Helios** at 99.921% two-qubit / 99.9975% single-qubit across all qubit pairs (Nov 2025, trapped ions, all-to-all connectivity); superconducting devices sit around 99.5–99.9%; neutral atoms are climbing past 99.5%. Connectivity matters as much as raw fidelity: heavy-hex lattices (IBM) and 2D grids (Google) pay SWAP-routing overhead to move qubits together, while trapped ions and reconfigurable atom arrays offer all-to-all connectivity that removes routing entirely. Dynamic circuits — mid-circuit measurement plus classical feed-forward — became standard on major platforms in 2024–25 and are prerequisite for real-time QEC (`S-decoders`).

KEY GRADED CLAIMS

- T1 Clifford+T is universal; Solovay–Kitaev gives polylog-overhead synthesis — Nielsen & Chuang; Dawson–Nielsen, quant-ph/0505030 (established)
- T1 Clifford-only circuits classically simulable in poly time — Gottesman–Knill, quant-ph/9807006 (established)
- T2 Helios 99.921% two-qubit fidelity, all-to-all, 98 qubits — Quantinuum (Nov 2025); postquantum.com analysis (demonstrated; vendor-built device, third-party analyzed)
- T4 Vendor "gate count" roadmaps (IBM Nighthawk 5,000→15,000 two-qubit gates by 2028) — IBM Quantum blog (roadmap)

SPEEDUP / CAVEAT

Universality is a **compilation** statement, not a speed statement — it says any unitary is reachable, not that it is cheap. T gates dominate fault-tolerant cost (each needs a distilled or cultivated magic state, `S-qec`), so **T-count** and **T-depth** are the real currency of algorithm resource estimates (`S-shor`, `S-qsim`). A circuit that looks short in physical gates can be enormous in T-count.

CONFLICTS / OPEN QUESTIONS

Cross-platform fidelity numbers are measured with different protocols — randomized benchmarking (RB), cross-entropy benchmarking (XEB), component/cycle benchmarking — and are only loosely comparable (`S-bench`). A single "99.9%" can mean different things on two vendors' datasheets.

SOURCES

Nielsen & Chuang (2000); quant-ph/0505030; quant-ph/9807006; Quantinuum Helios (2025); ibm.com/quantum/blog. Cross-links: `S-qec`, `S-software`, `S-bench`, `S-shor`, `reference-impl/`.

Grover's algorithm · S-grover

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Grover (1996) finds a marked item among N unstructured possibilities in $O(\sqrt{N})$ oracle queries, versus $O(N)$ classically. The mechanism is **amplitude amplification**: start in a uniform superposition, then repeat a two-reflection step (oracle phase-flip of the marked state, then reflection about the mean) $\sim (\pi/4)\sqrt{N}$ times, rotating amplitude toward the solution in the 2D subspace spanned by marked and unmarked states. The quadratic speedup is provably optimal for black-box search — the BBBV theorem (Bennett–Bernstein–Brassard–Vazirani, 1997) shows no quantum algorithm beats $\Omega(\sqrt{N})$ queries. Amplitude amplification generalizes Grover to boost the success probability of any subroutine that outputs a "good" flag with amplitude a from $O(1/a^2)$ to $O(1/a)$ repetitions, making it a workhorse inside counting, quantum Monte Carlo / amplitude estimation (`S-qmc`), and optimization heuristics.

WHERE IT STANDS (2025–26)

Grover is textbook-established and has been demonstrated on a handful of qubits on every major platform. The 2020s reassessment is sobering. Careful architecture-level accounting (Babbush et al. 2021; Hoefler–Häner–Troyer, CACM 2023) shows quadratic speedups almost certainly deliver **no practical advantage** on foreseeable fault-tolerant hardware: the QEC overhead, the ~kHz–MHz logical clock speed, and the cost of implementing the oracle as a reversible circuit eat the √N gain unless the classical runtime is already astronomically large (roughly, the crossover sits beyond problem sizes any real workload reaches). The reason is structural — Grover needs $\sim \sqrt{N}$ *sequential* coherent iterations and admits no parallel speedup (Zalka 1999), so throwing more qubits at it does not help wall-clock time. The field now treats Grover as a building block and a lower-bound landmark; nobody credible pitches bare Grover search as a near-term product.

KEY GRADED CLAIMS

- T1 $O(\sqrt{N})$ unstructured search, provably optimal — Grover, quant-ph/9605043; Bennett–Bernstein–Brassard–Vazirani, SIAM J. Comput. 26, 1510 (1997) (established)
- T2 Quadratic speedups washed out by fault-tolerance overhead for practical sizes — Babbush et al., PRX Quantum 2, 010103 (2021); Hoefler et al., CACM 66(5), 82 (2023) (established analysis, some parameter dependence)
- T1 No parallel advantage; $\sim \sqrt{N}$ iterations must be sequential — Zalka, PRA 60, 2746 (1999) (established)

SPEEDUP / CAVEAT

Proven quadratic query speedup, optimal in the query model. Fine print: (1) query complexity only — a cheap classical oracle call versus an expensive reversible quantum circuit is an unfair comparison; (2) the sequential-iteration requirement kills wall-clock advantage at realistic N and logical clock rates; (3) the symmetric-key-crypto consequence is modest — simply double key lengths (AES-128 → AES-256), which is why NIST treats Grover as a minor concern next to Shor.

CONFLICTS / OPEN QUESTIONS

Whether amplitude-amplification subroutines embedded in larger algorithms (Monte Carlo pricing in `S-qmc`, optimization) survive the same overhead accounting. Current evidence leans no for quadratic-only gains; near-quadratic amplitude estimation is the most likely survivor. The two-reflection structure is a special case of QSVT (`S-qsvt`), which is the modern lens for reasoning about its cost.

SOURCES

quant-ph/9605043; SIAM J. Comput. 26, 1510 (1997); PRX Quantum 2, 010103 (2021); CACM 66(5), 82 (2023); PRA 60, 2746 (1999). Cross-links: `S-qmc`, `S-qsvt`, `S-walk`, `O-advantage`.

Hamiltonian simulation methods — Trotter / qDRIFT / LCU / qubitization · S-hamsim

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The *how* behind digital quantum simulation (`S-qsim`): concrete algorithms that implement the evolution operator e^{-iHt} on a gate-based machine for a Hamiltonian $H = \sum_k h_k P_k$. Four method families, each with a different complexity profile. - **Trotter–Suzuki product formulas.** Approximate e^{-iHt} by interleaving $e^{-ih_k P_k t/r}$ short steps. Simple, ancilla-free, and it enjoys **commutator scaling** — error depends on how much the terms fail to commute, so it is often far better than worst-case bounds suggest (Childs–Su–Tran–Wiebe–Zhu 2021). A p-th order formula gives error $O((t/r)^{p+1})$ per step. - **qDRIFT.** Randomly compose terms sampled with probability $\propto |h_k|$, so cost scales with the L1 norm $\lambda = \sum |h_k|$ and is **independent of the number of terms** — good for molecular Hamiltonians with many small terms. Trades a random-sampling error for term-count independence (Campbell 2019). - **LCU (linear combination of unitaries).** Write the propagator (e.g. its Taylor series) as a weighted sum of unitaries and apply it with an ancilla "select/prepare" pair. Achieves $O(\log(1/\varepsilon))$ precision scaling — exponentially better in ε than Trotter (Berry–Childs–Cleve–Kothari–Somma 2015). - **Qubitization / quantum walk.** Block-encode H and walk its eigenphases; **optimal** query complexity $O(\lambda t + \log(1/\varepsilon))$, the provable lower bound (Low–Chuang 2017/2019). This is the QSVT (`S-qsvt`) instance for time evolution.

WHERE IT STANDS (2025–26)

The methods are complementary, and the 2024–25 literature is about hybridizing them. For a fixed accuracy, **second-order Trotter** is often competitive with qubitization/LCU/qDRIFT on real molecular systems because commutator scaling beats the λ-scaling those methods pay, especially for geometrically local Hamiltonians; but LCU/qubitization win at high precision. Recent work compensates Trotter error with an LCU correction using one ancilla, getting both good system-size scaling and high accuracy (PRX Quantum 6, 010359, 2025), and Richardson-extrapolated qDRIFT (Watson 2024) reduces the standard qDRIFT step count. Fault-tolerant chemistry resource estimates (the FeMoco line in `S-qsim`) now overwhelmingly use qubitization for its optimal T-count. Everything here is a **fault-tolerant** primitive: the deep coherent evolution requires `S-qec` .

KEY GRADED CLAIMS

- T1 Trotter with commutator error scaling — Childs–Su–Tran–Wiebe–Zhu, PRX 11, 011020 (2021); Lloyd, Science 273 (1996) (established)
- T1 qDRIFT: cost $\propto \lambda$, independent of term count — Campbell, PRL 123, 070503 (2019) (established)
- T1 LCU / Taylor-series simulation, $O(\log 1/\varepsilon)$ precision — Berry et al., PRL 114, 090502 (2015) (established)
- T1 Qubitization: optimal $O(\lambda t + \log 1/\varepsilon)$ query complexity — Low–Chuang, Quantum 3, 163 (2019) (established, matches lower bound)
- T2 Hybrid Trotter+LCU error compensation, one ancilla — PRX Quantum 6, 010359 (2025) (peer-reviewed)

SPEEDUP / CAVEAT

The *enabling* engine for the best-founded quantum advantage (quantum simulation, `S-qsim`) — exponential over classical for generic dynamics. Caveats: (1) all four assume you can already load / block-encode H and prepare a useful initial state; (2) the constant factors and T-counts are large, keeping useful runs in the fault-tolerant era; (3) classical tensor-network methods (`S-tensornet`) absorb the low-entanglement / short-time instances, so the advantage lives in long-time, high-entanglement regimes.

CONFLICTS / OPEN QUESTIONS

Which method wins for a given target is instance-dependent and actively contested (Trotter's commutator advantage versus qubitization's optimal asymptotics). Whether early-fault-tolerant machines favor the low-ancilla randomized methods (qDRIFT, single-ancilla LCU) over qubitization's block-encoding overhead.

SOURCES

PRX 11, 011020 (2021); PRL 123, 070503 (2019); PRL 114, 090502 (2015); Quantum 3, 163 (2019); PRX Quantum 6, 010359 (2025). Cross-links: `S-qsim` , `S-qsvt` , `S-qft` , `S-tensornet` , `S-qec` , `I-chem` , `I-pharma` .

HHL & quantum linear algebra · S-hhl

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

HHL (Harrow–Hassidim–Lloyd, 2009) solves the linear system $Ax=b$ by preparing a quantum state $|x\rangle \propto A^{-1}|b\rangle$ in time $O(\log(N) \cdot s^2 \cdot \kappa^2/\varepsilon)$ — polylogarithmic in the dimension N, where s is sparsity, κ the condition number, ε the target accuracy. Mechanism: phase-estimate (`S-qft`) the eigenvalues of A on $|b\rangle$, rotate an ancilla by $1/\lambda$ conditioned on each eigenvalue, then uncompute — inverting A eigenvalue-by-eigenvalue. Against classical Gaussian elimination's $O(N^3)$, or conjugate gradient's $O(Ns\kappa)$ for sparse systems, the $\log(N)$ scaling is an **exponential** improvement in dimension. HHL seeded a whole family: quantum linear algebra, quantum singular value transformation (`S-qsvt`), quantum recommendation systems, and the ML applications that fueled the 2015-era hype. The inner-product primitive at its core — estimating $\langle u|v\rangle$ between amplitude-encoded states — is exactly what the atlas's **reference implementation** builds and validates on real hardware via the swap and Hadamard tests (`reference-impl/MATH.md` §3, §5, §8).

WHERE IT STANDS (2025–26)

HHL is the canonical "read the fine print" algorithm (Aaronson 2015). The exponential speedup survives only if **four assumptions hold together**: (1) A is sparse and well-conditioned — runtime is quadratic in κ, disqualifying most engineering-grade (ill-conditioned) systems; (2) $|b\rangle$ can be prepared efficiently — a qRAM assumption (`S-qram`) that may itself hide exponential cost; (3) A can be Hamiltonian-simulated efficiently (`S-qsim`); (4) you want a scalar *property* of x ($\langle x|M|x\rangle$) rather than the vector itself — reading out all N amplitudes takes $O(N)$ tomography and kills the speedup. Ewin Tang's 2018 dequantization of quantum recommendation systems, and the quantum-inspired classical algorithms that followed (Chia et al., Gilyén et al.), removed the exponential gap entirely for **low-rank** instances. Matrix inversion is BQP-complete, so a hard regime provably exists — sparse, high-rank, well-conditioned, with efficient state prep and property readout — but no practical problem has been demonstrated to live there fifteen years on.

KEY GRADED CLAIMS

- T1 HHL $O(\log N \cdot s^2\kappa^2/\varepsilon)$ runtime under stated assumptions; matrix inversion BQP-complete — PRL 103, 150502 (2009) (established)
- T2 Caveat catalogue (state prep, κ, sparsity, output access) — Aaronson, Nat. Phys. 11, 291 (2015) (established)
- T2 Low-rank dequantization removes the exponential gap — Tang, STOC 2019, arXiv:1807.04271; Chia et al., STOC 2020 (established)
- T1/T2 QSVT subsumes HHL with improved κ, ε dependence — Gilyén–Su–Low–Wiebe, STOC 2019, arXiv:1806.01838 (established)

SPEEDUP / CAVEAT

Exponential in dimension, conditional on four assumptions that rarely hold together in applications. Where they fail, classical or quantum-inspired methods match. QSVT (`S-qsvt`) achieves linear rather than quadratic κ-scaling — the modern way to state the resource cost.

CONFLICTS / OPEN QUESTIONS

Does any end-to-end application (differential-equation solvers, finance PDEs, `I-finance`) satisfy all four assumptions with real data and beat classical? None demonstrated. The state-preparation bottleneck (`S-qram`) is the load-bearing unknown.

SOURCES

PRL 103, 150502 (2009); Nat. Phys. 11, 291 (2015); arXiv:1807.04271; arXiv:1806.01838. Cross-links: `S-qsvt`, `S-qft`, `S-qram`, `S-qsim`, `S-qml`, `reference-impl/`.

Logical qubits & fault tolerance · S-logical

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

A logical qubit is an error-corrected qubit (`S-qec`) — the unit that algorithms in `S-shor`/`S-qsim` actually consume. Fault tolerance means every operation (gates, measurement, and the correction step itself) is performed so that a single physical fault cannot cascade into a logical error. This node marks the line between demos and utility. Useful chemistry or cryptanalysis needs **hundreds to thousands of logical qubits** at logical error rates of 10^{-9} or better, executing **millions to billions** of logical gates — an entirely different regime from the shallow error-detected circuits shown so far. The key subtleties that make bare counts meaningless: code distance d, measured logical error rate per cycle, whether a *universal* (non-Clifford, `S-gates`) gate set was actually exercised, and whether the device did true correction or only post-selected error *detection*.

WHERE IT STANDS (2025–26)

The logical-qubit count became the field's headline metric, which invites cherry-picking. Milestones: Harvard/QuEra ran circuits on **48 logical qubits** of neutral atoms (Bluvstein et al., Nature 2024 — error-detected, shallow). Microsoft+Quantinuum demonstrated **12 logical qubits** with 22× error suppression (2024). **Quantinuum Helios** (Nov 2025) produced **48 error-corrected logical qubits at a 2:1 physical:logical ratio** — thought impossible a few years ago — plus a **94-logical-qubit GHZ state**, both with better-than-break-even fidelity (the logical qubits outperform physical qubits running the same task). QuEra demonstrated **96 logical from 448 physical** (a high-rate $[[16,6,4]]$ code, ~4.7:1 encoding) with below-threshold error suppression across all 96 logical qubits — peer-reviewed in Nature (Jan 2026), which makes it the strongest-graded logical-qubit record to date. Caution: a distance-2 detection-only "logical qubit" and Willow's distance-7 corrected memory are different animals, and none of these systems yet runs deep universal computation across many logical qubits simultaneously.

KEY GRADED CLAIMS

- T2 48-logical-qubit circuits on neutral atoms — Bluvstein et al., Nature 626, 58 (2024) (demonstrated; error-detected, shallow)
- T2 12 logical qubits, 22× error suppression — Microsoft/Quantinuum, arXiv:2404.02280 (demonstrated)
- T3/T4 Helios: 48 error-corrected logical at 2:1 encoding + 94-qubit GHZ, break-even fidelity — Quantinuum (Nov 2025) + Iceberg-code paper (claimed; partially peer-reviewed)
- T2 96 logical qubits via a high-rate $[[16,6,4]]$ code on 448 neutral atoms (~4.7:1), below-threshold across all logical qubits — QuEra, Nature s41586-025-09848-5 (doi:10.1038/s41586-025-09848-5, Jan 2026) (demonstrated, peer-reviewed)
- T4 IBM Starling: 200 logical qubits / 100M gates by 2029 — IBM roadmap (roadmap)

SPEEDUP / CAVEAT

Logical-qubit counts are **incomparable** across vendors without stating code distance, logical error rate, and gate-set completeness. Treat any bare "N logical qubits" headline as marketing until those three numbers appear (`S-bench`). Break-even (logical beats physical) is the meaningful 2025 threshold; deep universal computation on many logical qubits is the next.

CONFLICTS / OPEN QUESTIONS

"FTQC by 2029" (IBM roadmap, Google-adjacent optimism) versus "not before the mid-2030s" (much of academia) is the field's central unresolved timeline conflict (`C-ftqc-timeline`). Whether the physical:logical ratio keeps falling — Helios's 2:1 is architecture-specific and does not include the deep-circuit distillation overhead.

SOURCES

Nature 626, 58 (2024); arXiv:2404.02280; Quantinuum Helios (2025); QuEra 96-logical — Nature s41586-025-09848-5 (2026); ibm.com/quantum/blog. Cross-links: `S-qec`, `S-decoders`, `S-gates`, `S-bench`, `O-scaling`, `O-overhead`.

Noise, NISQ & error mitigation · S-nisq

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

NISQ — Noisy Intermediate-Scale Quantum, coined by Preskill (2018) — names the current era: 50–1,000+ physical qubits at error rates around 10^{-3}, too noisy for deep circuits and too few for full error correction (`S-qec`). Error **mitigation** trades extra circuit runs for accuracy without encoding logical qubits. The two workhorses (Temme et al. 2017): **Zero-Noise Extrapolation (ZNE)** runs the circuit at deliberately amplified noise (gate folding or pulse stretching), then extrapolates the observable back to the zero-noise limit; **Probabilistic Error Cancellation (PEC)** learns a noise model and inverts it by sampling from a quasiprobability decomposition, which is unbiased but carries a sampling overhead that grows exponentially in circuit size. Mitigation is not correction — it reduces bias in an *estimated expectation value*, it does not protect a quantum state. It sits distinct enough from correction to deserve its own discipline card (`S-errmit`).

WHERE IT STANDS (2025–26)

IBM's 2023 "utility" experiment — a 127-qubit kicked-Ising circuit with ZNE (Nature) — was the high-water mark of the mitigation era, and was matched within weeks by classical tensor-network methods running on a laptop (`S-tensornet`). That exchange set the field's expectations: mitigation extends the reach of noisy hardware by a constant factor, while theory shows generic mitigation costs grow **exponentially** with circuit depth and fault rate (Takagi et al.; Quek et al., Nat. Phys. 2024, giving exponentially tighter lower bounds). The field's center of gravity moved to early fault tolerance (`S-qec`, `S-logical`); mitigation persists as a complement layered on top of small logical devices, and 2025 work demonstrates mitigation *on logical qubits* (Nat. Commun. 2025). The reference implementation names ZNE and readout calibration as its own natural next lever against the hardware-noise floor (`reference-impl/MATH.md` §7).

KEY GRADED CLAIMS

- T2 Preskill's NISQ framing — arXiv:1801.00862, Quantum 2, 79 (2018) (established as terminology)
- T2 127-qubit kicked-Ising beyond brute-force simulation — Kim et al., Nature 618, 500 (2023) (demonstrated)
- T2 Same observables reproduced classically with tensor networks — Tindall et al., PRX Quantum 5, 010308 (2024) (contested → resolved: classical matched it)
- T2 Error mitigation has exponential sampling overhead in general — Quek et al., Nat. Phys. 20 (2024); Takagi et al. (established)
- T2 ZNE/PEC origins — Temme–Bravyi–Gambetta, PRL 119, 180509 (2017) (established)

SPEEDUP / CAVEAT

Mitigation buys accuracy at sampling cost and **never** creates asymptotic speedup. Any "utility" claim demonstrated with mitigation alone is exposed to classical counterattack by construction — if the hardware is shallow enough for mitigation to work, it is usually shallow enough for tensor networks to simulate (`S-tensornet`, `O-advantage`).

CONFLICTS / OPEN QUESTIONS

Whether any mitigated-NISQ computation will ever beat the best classical method on a problem anyone cares about. The 2023–25 record is uniformly negative once a competent classical team responds (conflict `C-nisq-utility`).

SOURCES

arXiv:1801.00862; Nature 618, 500 (2023); PRX Quantum 5, 010308 (2024); Nat. Phys. 20 (2024); PRL 119, 180509 (2017). Cross-links: `S-errmit`, `S-qec`, `S-tensornet`, `reference-impl/`, `O-advantage`.

Pulse-level & quantum optimal control · S-optcontrol

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The bottom of the stack, where an abstract gate becomes an actual analog waveform driving a qubit. Quantum optimal control (QOC) shapes the microwave, laser, or flux pulse that steers the system's Hamiltonian to realize a target unitary with maximum fidelity in minimum time, respecting hardware constraints (bandwidth, power, leakage to non-computational levels). It sits below the gate/compiler layer (`S-gates`, `S-software`) and directly on the control electronics (`H-control`). Core methods: - **GRAPE** (gradient ascent pulse engineering) — piecewise-constant controls, all updated together via gradient/L-BFGS. - **Krotov** — monotonically-convergent gradient method, updates controls sequentially within an iteration. - **CRAB** — expand the pulse in a truncated random basis and optimize the few coefficients (gradient-free, good with a physical simulator in the loop). - **GOAT / analytic-gradient** methods, and **DRAG** — an analytic correction that removes leakage for weakly-anharmonic transmons, the workhorse for single-qubit gates. - **Reinforcement-learning control** — increasingly used where a model is expensive or unknown.

WHERE IT STANDS (2025–26)

QOC is standard practice for squeezing the last fidelity out of a device and is how record two-qubit gates are calibrated. The honest 2025–26 finding is a note of restraint: a systematic transmon study concluded that **properly calibrated DRAG already operates near the decoherence floor** for single-qubit gates, so heavy numerical optimization (GRAPE) buys little there — the win from fancy control is real mainly for two-qubit gates, leakage-heavy or strongly-coupled systems, and robustness against calibration drift (arXiv:2511.12799, 2025). Reinforcement-learning control matured for robust perfect-entangling gates (npj QI 2025). The practical tension is the same everywhere: model-based optimal pulses are only as good as the device model, so closed-loop calibration against the live hardware (and periodic recalibration against drift) matters as much as the optimizer. This layer is also where analog quantum simulation (`S-qsim`) and pulse-efficient compilation of `S-hamsim` blocks live.

KEY GRADED CLAIMS

- T1 GRAPE gradient pulse optimization — Khaneja et al., J. Magn. Reson. 172, 296 (2005) (established)
- T1 DRAG leakage suppression for weakly-anharmonic qubits — Motzoi et al., PRL 103, 110501 (2009) (established)
- T1 Krotov / CRAB optimal-control frameworks — Krotov (1996); Caneva–Murphy–Calarco, PRA 84, 022326 (2011) (established)
- T3 Calibrated DRAG near the decoherence floor; numerical optimization helps mainly for hard cases — arXiv:2511.12799 (2025) (preprint)
- T2 Reinforcement-learning robust entangling gates — npj Quantum Information (2025), s41534-025-01065-2 (peer-reviewed)

SPEEDUP / CAVEAT

Not an algorithmic speedup — a **fidelity and time-optimality** layer. It raises gate quality (feeding the thresholds in `S-qec`) and can shorten gate duration to beat decoherence, but it is bounded by the quantum speed limit and by hardware (bandwidth, coherence T_1/T_2, control-line crosstalk). Over-optimizing to a stale device model can *reduce* real fidelity, so the practical caveat is that calibration and drift-tracking often dominate the theoretical pulse shape.

CONFLICTS / OPEN QUESTIONS

When numerical optimal control actually beats simple analytic pulses (DRAG/Gaussian) is instance-specific and debated. How to make optimal pulses robust to drift without constant recalibration; how well RL control transfers across devices.

SOURCES

J. Magn. Reson. 172, 296 (2005); PRL 103, 110501 (2009); PRA 84, 022326 (2011); arXiv:2511.12799 (2025); npj QI s41534-025-01065-2 (2025). Cross-links: `S-gates`, `S-software`, `H-control`, `S-qsim`, `S-qec`.

Quantum error correction · S-qec

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

QEC encodes one *logical* qubit into many physical qubits so errors are detected and corrected faster than they accumulate. The **threshold theorem** (Aharonov–Ben-Or; Kitaev; Knill–Laflamme–Zurek, 1996–98) proves that if the physical error rate p sits below a code-dependent threshold p_{th}, arbitrarily long computation is possible with only $\mathrm{polylog}(1/\varepsilon)$ overhead per logical operation. The **surface code** (Fowler et al. 2012; threshold $\approx 1\%$, nearest-neighbor 2D layout, distance d protects against $\lfloor (d-1)/2 \rfloor$ errors using d^2 data + measure qubits) has been the default. **qLDPC codes** promise far better encoding rate — more logical qubits per physical qubit — at the price of long-range connectivity. Non-Clifford gates need **magic-state distillation** (or newer **cultivation**, `S-gates`), historically the dominant cost of fault-tolerant computing. Syndrome extraction runs every cycle and must be decoded in real time (`S-decoders`).

WHERE IT STANDS (2025–26)

Google's **Willow** result (Nature 2024) is the era-defining demo: a distance-7 surface-code logical qubit whose error rate **halves** with each increase in code distance (suppression factor $\Lambda \approx 2.14$) — the first clear below-threshold memory, meaning bigger codes now help rather than hurt. IBM pivoted to qLDPC: the $[[144,12,12]]$ "gross code" protects 12 logical qubits with 288 physical qubits, versus roughly 3,000 physical for equivalent surface-code protection — about a 10× encoding-rate win; the **Loon** chip (2025) demonstrated the required long-range couplers. Gidney's magic-state **cultivation** (2024) cut projected T-gate costs by roughly 10×, feeding directly into the falling Shor estimates (`S-shor`). Quantinuum, QuEra, and Harvard demonstrated logical operations on trapped-ion and neutral-atom platforms (`S-logical`).

KEY GRADED CLAIMS

- T1 Threshold theorem — Aharonov–Ben-Or, STOC 1997; Knill–Laflamme–Zurek (established)
- T2 Below-threshold surface-code memory, $\Lambda = 2.14$, $d = 7$ — Google, Nature 638 (2024), arXiv:2408.13687 (demonstrated)
- T2 Gross code $[[144,12,12]]$ ~10× better encoding rate — Bravyi et al., Nature 627, 778 (2024) (demonstrated in theory + partial hardware)
- T3 Magic-state cultivation reduces distillation overhead ~10× — Gidney et al., arXiv:2409.17595 (claimed)
- T4 IBM Loon validates qLDPC architectural components — IBM blog, Nov 2025 (claimed, awaiting peer review)

SPEEDUP / CAVEAT

Overhead is the whole game (`O-overhead`): ~100–1,000 physical qubits per logical qubit at useful error rates, and QEC **slows** the logical clock because each logical gate spans many syndrome cycles plus decode latency. Encoding rate (surface) versus connectivity demand (qLDPC) is the central engineering tension.

CONFLICTS / OPEN QUESTIONS

Surface code versus qLDPC is a live architectural fork (Google and most of the field versus IBM); whether qLDPC's long-range connectivity is buildable at scale, and stays below threshold across the couplers, is unproven (`O-interconnect-loss`). Real-time decoding throughput at scale (`O-decoder`) is a separate open bottleneck.

SOURCES

arXiv:2408.13687; Nature 627, 778 (2024); arXiv:2409.17595; ibm.com/quantum/blog; PRA 86, 032324 (2012). Cross-links: `S-logical`, `S-decoders`, `S-gates`, `S-shor`, `O-overhead`, `O-decoder`.

Quantum Fourier Transform & Phase Estimation · S-qft

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The Quantum Fourier Transform (QFT) is the change of basis that maps amplitudes to their discrete-Fourier coefficients using only $O(n^2)$ gates on n qubits — exponentially fewer than the $O(n \cdot 2^n)$ of the classical FFT on 2^n points. On its own the QFT is not a speedup (you cannot read the coefficients out), but it is the engine of Quantum Phase Estimation (QPE): given a unitary U and an eigenstate $|\psi\rangle$ with $U|\psi\rangle = e^{2\pi i\varphi}|\psi\rangle$, QPE estimates the phase φ to n bits by applying controlled powers of U into a register and running an inverse QFT. QPE is the primitive that sits *underneath* the famous algorithms — Shor's period-finding (`S-shor`), HHL eigenvalue inversion (`S-hhl`), quantum-chemistry energy estimation (`S-qsim`), and amplitude estimation (`S-qmc`) are all QPE in different clothing.

WHERE IT STANDS (2025–26)

QFT (Coppersmith, 1994) and QPE (Kitaev, 1995) are textbook-established. The live engineering story is depth: textbook QPE demands $\sim O(1/\varepsilon)$ coherent applications of U plus a deep controlled-rotation ladder, which is hopeless on NISQ hardware. The response is a family of low-depth variants — iterative/Kitaev-style QPE (one ancilla, classical feedback), Bayesian and statistical phase estimation, Lin–Tong Heisenberg-limited ground-state energy estimation for *early* fault tolerance (PRX Quantum 3, 010318, 2022, shorter maximal circuit depth via a step-function filter), Wan–Berta–Campbell randomized statistical phase estimation (PRL 129, 030503, 2022), and the optimal-precision refinements (arXiv:2403.18927) — that trade circuit depth for repetitions. Approximate QFT (dropping small-angle rotations) cuts gate count with negligible error and is standard in fault-tolerant compilation. QPE is also a special case of QSVT (`S-qsvt`) — the eigenvalue-transform lens. It remains a fault-tolerant algorithm: the canonical demonstration of why long coherent circuits need `S-qec`.

KEY GRADED CLAIMS

- T1 QFT in $O(n^2)$ gates; QPE estimates eigenphase to n bits — Coppersmith 1994 (quant-ph/0201067); Kitaev 1995 (quant-ph/9511026) (established)
- T1 QPE is the shared subroutine under Shor, HHL, quantum chemistry, amplitude estimation — Nielsen & Chuang Ch. 5 (established)
- T2 Iterative/low-depth, statistical, and optimal-precision QPE variants for near-term and early-FT use — Lin–Tong, PRX Quantum 3, 010318 (2022); Wan–Berta–Campbell, PRL 129, 030503 (2022); arXiv:2305.04908; arXiv:2403.18927 (established/demonstrated)

SPEEDUP / CAVEAT

QPE delivers exponential precision in the phase (n bits from $O(2^n)$ total controlled-U applications) — the engine behind Shor's exponential factoring speedup. Caveat: it needs a good eigenstate to start from and deep coherent evolution of U, so it is a fault-tolerant, not NISQ, primitive; the reference implementation at `reference-impl/` estimates inner products via the shallower swap/Hadamard test precisely to sidestep full QPE depth.

CONFLICTS / OPEN QUESTIONS

How much of the eigenstate-preparation cost (the same qRAM-style assumption that haunts `S-hhl`/`S-qram`) is hidden inside "just run QPE" for real chemistry and finance instances.

SOURCES

quant-ph/0201067; quant-ph/9511026; Nielsen & Chuang §5.2; arXiv:2305.04908; arXiv:2403.18927. Cross-links: `S-shor`, `S-hhl`, `S-qmc`, `S-qsim`, `reference-impl/`.

Quantum Monte Carlo / amplitude estimation · S-qmc

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Quantum Amplitude Estimation (QAE) estimates the amplitude a in a state $\sqrt{1-a}\,|0\rangle\,|\psi_0\rangle + \sqrt{a}\,|1\rangle\,|\psi_1\rangle$ to additive error ε using $O(1/\varepsilon)$ calls to the state-preparation oracle — versus the $O(1/\varepsilon^2)$ samples any classical Monte Carlo needs, since classical error shrinks only as $O(1/\sqrt{M})$. That is a quadratic speedup for estimating expectation values, the core operation in Monte Carlo integration. Montanaro (2015) generalized it to arbitrary-variance mean estimation. Because expected values price options, aggregate risk (VaR/CVaR), and compute Greeks, QAE is the headline near-to-mid-term algorithm for `I-finance` (JPMorgan, Goldman Sachs, HSBC all have published implementations). Canonically QAE = amplitude amplification (`S-grover`) + phase estimation (`S-qft`); "Grover-free" iterative and maximum-likelihood variants remove the deep QPE register.

WHERE IT STANDS (2025–26)

The quadratic speedup is proven and demonstrated at small scale on trapped-ion and superconducting hardware (arXiv:2109.09685, arXiv:2201.06987). Two caveats dominate the honest picture. (1) Depth: canonical QAE needs very deep circuits — the reason iterative QAE (Grinko et al.) and maximum-likelihood QAE were invented, trading depth for repetitions to reach NISQ. (2) The oracle problem: the speedup assumes an efficient circuit that loads the target probability distribution into amplitudes without approximation error — a state-preparation / `S-qram` assumption that, as with `S-hhl`, can silently cost as much as it saves. Resource estimates (Chakrabarti et al., 2021) put useful financial QAE firmly in the fault-tolerant era, needing many logical qubits and long coherent runs.

KEY GRADED CLAIMS

- T1 QAE gives $O(1/\varepsilon)$ vs classical $O(1/\varepsilon^2)$ — quadratic Monte Carlo speedup — Brassard–Høyer–Mosca–Tapp 2002 (quant-ph/0005055); Montanaro, Proc. R. Soc. A 471 (2015) (established)
- T2 Grover-free / iterative / MLE amplitude estimation for shallower circuits — Grinko et al., npj QI 7 (2021); Suzuki et al. 2020 (established)
- T3/T4 Financial QAE (option pricing, risk) demonstrated small-scale; useful scale needs FTQC — arXiv:1905.02666 (JPMorgan); Chakrabarti et al., Quantum 5, 463 (2021) (demonstrated / roadmap)

SPEEDUP / CAVEAT

Proven quadratic speedup over classical Monte Carlo. Caveats: (1) only quadratic, so fault-tolerance overhead can wash it out for realistic runtimes (`S-grover`, `O-advantage`); (2) requires deep circuits and an efficient, exact distribution-loading oracle — the loading cost is the crux and links directly to `S-qram`.

CONFLICTS / OPEN QUESTIONS

Does the distribution-loading oracle for real market data admit an efficient circuit, and does the quadratic gain survive full FT resource accounting? Unresolved — the same fine print that dogs Grover.

SOURCES

quant-ph/0005055; Proc. R. Soc. A 471, 20150301 (2015); npj QI 7, 52 (2021); arXiv:1905.02666; Quantum 5, 463 (2021); arXiv:2109.09685. Cross-links: `S-grover`, `S-qft`, `S-qram`, `I-finance`, `O-advantage`.

Quantum machine learning · S-qml

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

QML covers quantum kernels, variational quantum classifiers, quantum neural networks, and hybrid pipelines — the proposal that quantum feature maps or quantum linear algebra could accelerate learning. The clearest instance is the **quantum kernel**: encode each datum $\mathcal{X}$ into a state $|\phi(x)\rangle$ and use $k(x, y) = |\langle\phi(x)|\phi(y)\rangle|^2$ as the kernel a classical SVM consumes. The atlas's **reference implementation** is exactly this object built honestly end to end — amplitude-encode vectors, estimate the overlap with a swap/Hadamard test, assemble the Gram matrix, feed it to a classical SVM (100% on Iris, kernel RMSE 0.007), with a full shots-versus-noise error budget (`reference-impl/MATH.md` §6–8). It is a correct, hardware-validatable quantum kernel that also names, in the same breath, why it is **not** a speedup (classical cosine similarity is $O(d)$). QML is the most hype-inflated corner of the stack and the atlas grades it accordingly.

WHERE IT STANDS (2025–26)

The record on **classical** data is negative. Tang's 2018 dequantization killed the flagship (quantum recommendation systems, `S-hhl`); random-Fourier-feature methods reproduce quantum-kernel performance classically; and a Dec 2025 peer-reviewed benchmark found plain logistic regression beating quantum SVMs, quantum k-NN, and variational classifiers on 4 of 5 realistic tabular datasets — one variational classifier scored below random guessing. Barren plateaus (`S-variational`) apply with full force to trainable models. What survives is narrow and real: **provable** kernel separations exist for contrived, cryptographically structured data (discrete-log — Liu–Arunachalam–Temme 2021), and an **exponential** advantage exists for learning from *quantum* data / experiments when a quantum memory is available (Huang et al., Science 2022). The defensible position: QML on classical data has no evidence of practical advantage; QML on quantum data is a real but niche research direction awaiting hardware.

KEY GRADED CLAIMS

- T2 Low-rank QML dequantized — Tang, STOC 2019, arXiv:1807.04271 (established)
- T2 Rigorous quantum-kernel advantage on a crypto-structured task — Liu–Arunachalam–Temme, Nat. Phys. 17, 1013 (2021) (established; artificial problem)
- T2 Exponential advantage learning from quantum experiments with quantum memory — Huang et al., Science 376, 1182 (2022) (demonstrated; quantum data only)
- T2 Classical baselines beat QML across realistic tabular benchmarks — Scientific Reports (Dec 2025) (demonstrated)
- T5 "18% of quantum-algorithm revenue from AI by 2026" — Hyperion Research (forecast; graded skeptically)

SPEEDUP / CAVEAT

No proven or empirical speedup on classical data. Provable advantages exist only where the problem is built from cryptographic hardness or the data is itself quantum. Even when a quantum kernel is *expressive*, the state-loading cost (`S-qram`) and the $O(m^2)$ pairwise-overlap cost of building the Gram matrix (as the reference impl makes concrete) dominate. Assume any commercial "quantum AI" pitch is unsubstantiated until it names its separation.

CONFLICTS / OPEN QUESTIONS

Whether "quantum data" applications (sensor streams `A-sensing`, simulation outputs) become a real market, and whether generative "GenQAI" vendor framings (Quantinuum) ever acquire evidence. Classical shadows (`S-shadows`) are the leading rigorous tool for turning quantum data into predictions with few measurements — a more defensible route than variational QML.

SOURCES

arXiv:1807.04271; Nat. Phys. 17, 1013 (2021); Science 376, 1182 (2022); Sci. Rep. (Dec 2025). Cross-links: `S-hhl`, `S-variational`, `S-qram`, `S-shadows`, `reference-impl/`, `O-hype`.

Quantum RAM (qRAM) · S-qram

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Quantum RAM is the assumed device that loads classical data into quantum superposition: given an address register in a superposition of indices, qRAM returns the corresponding data entries entangled with those addresses, ideally in time polylogarithmic in the number of entries N. It is the load-bearing, largely-unproven assumption under the exponential-speedup algorithms — HHL (`S-hhl`), quantum recommendation systems, and most quantum machine learning (`S-qml`). If you cannot get N classical numbers into the machine faster than $O(N)$, the exponential advantage of algorithms that then process them in polylog time evaporates. The canonical design is the bucket-brigade architecture (Giovannetti–Lloyd–Maccone, 2008), which routes an address through a tree of quantum switches so that only $O(\log N)$ of the $\sim N$ nodes are actively entangled per query, making it far more error-tolerant than a naive fanout.

WHERE IT STANDS (2025–26)

qRAM remains the field's most honest weak link. The cost debate: bucket-brigade buys gentle (polylogarithmic) query-infidelity scaling but at the price of $O(N)$ physical components / exponentially many ancillae, and if each must be error-corrected the fault-tolerant overhead can swallow the algorithmic gain (Arunachalam et al. analysis; the recurring theme of `S-hhl`). 2025 brought the first experimental demonstrations — a Zhejiang University superconducting team realized bucket-brigade qRAM addressing 4-bit and 8-bit data (query fidelities ~0.81 and ~0.60, Nature Physics 2026) — proof of principle at tiny scale, far from the millions of addressable cells the speedup algorithms assume. Dequantization results (Tang et al.) further showed that *if* you grant qRAM-style access, many "quantum" ML speedups can be matched classically, sharpening the question of what qRAM actually buys.

KEY GRADED CLAIMS

- T1/T2 Bucket-brigade qRAM: $O(\log N)$ active switches per query, more noise-tolerant than naive fanout — Giovannetti–Lloyd–Maccone, PRL 100, 160501 (2008) (established)
- T2 Fault-tolerant qRAM overhead can negate the algorithms it feeds; polylog infidelity vs $O(N)$ components tradeoff — arXiv analyses (Arunachalam et al. 2015; redundancy-repair, Sci. Rep. 2025) (established/contested)
- T3 First experimental bucket-brigade qRAM (4-/8-bit, fidelity 0.81/0.60) — Zhejiang, Nature Physics (2026), arXiv:2506.16682 (demonstrated, small scale)

SPEEDUP / CAVEAT

Enabling assumption, not a speedup. When qRAM is granted for free, HHL/QML claim exponential gains; when its real cost ($O(N)$ hardware, error correction, state-prep fidelity) is charged, most of those gains shrink or vanish. No practical, scalable qRAM exists — this is the single biggest asterisk on data-heavy quantum advantage.

CONFLICTS / OPEN QUESTIONS

Is scalable, fault-tolerant qRAM physically buildable at the sizes (millions–billions of cells) the speedup algorithms need, and does any application clear the bar once loading is charged honestly? Open — parallels the `S-hhl` fine print directly.

SOURCES

PRL 100, 160501 (2008); arXiv:2506.16682; Nat. Phys. (2026); Sci. Rep. 15 (2025), arXiv:2312.17483; Tang, STOC 2019. Cross-links: `S-hhl`, `S-qml`, `S-qmc`, `reference-impl/` (amplitude encoding = the atlas data-loading it abstracts).

Quantum simulation · S-qsim

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Using a controllable quantum system to simulate another quantum system — Feynman's original 1981–82 pitch and still the most credible application class. Two flavors. **Digital**: implement the evolution operator e^{-iHt} on a gate-based machine by Trotter–Suzuki product formulas, linear combination of unitaries (LCU), qubitization, or QSVT (the methods are unpacked in `S-hamsim` and unified by `S-qsvt`), then extract observables — often via quantum phase estimation (`S-qft`) for ground-state chemistry. **Analog**: build a purpose-made quantum system whose native Hamiltonian *is* the target (e.g. Rydberg-atom arrays realizing spin models). Lloyd (1996) proved that local-Hamiltonian dynamics is simulable in polynomial time on a quantum computer — the theorem that makes the whole class rigorous.

WHERE IT STANDS (2025–26)

Analog simulators lead on scale: QuEra/Harvard 256+ atom arrays probing quantum phase transitions, spin liquids, and quench dynamics that strain classical methods (`H-neutral`). Digital simulation is the target of the current advantage push. IBM's 2023 kicked-Ising "utility" claim on 127 qubits was matched within weeks by classical tensor-network methods on a laptop (see `S-nisq`, `S-tensornet`) — the canonical example of the classical counterattack. Google's Oct 2025 "quantum echoes" experiment on Willow measures an out-of-time-order correlator (OTOC) and claims a **verifiable 13,000× speedup** over the best classical estimate, with a molecular-structure (NMR) application hook — under active classical-counterattack scrutiny, unresolved. For fault-tolerant chemistry the showcase target is the FeMoco cofactor (biological nitrogen fixation, `I-chem`/`I-agri`): resource estimates sit around ~4M physical qubits and days of runtime, after a decade of refinement that cut earlier figures by orders of magnitude.

KEY GRADED CLAIMS

- T1 Local-Hamiltonian dynamics simulable in poly time on a quantum computer — Lloyd, Science 273, 1073 (1996) (established)
- T1 Feynman's simulation proposal — Int. J. Theor. Phys. 21, 467 (1982) (established, foundational)
- T2 256-atom analog simulation of quantum phases — Ebadi et al., Nature 595, 227 (2021) (demonstrated)
- T3/T4 "Quantum echoes" OTOC 13,000× beyond-classical, verifiable — Google, Nature (Oct 2025) + blog.google (claimed; contested-by-default, classical response pending)
- T3 FeMoco ground-state estimate ~4M qubits / ~days — Lee et al., PRX Quantum 2, 030305 (2021) lineage (claimed resource estimate)

SPEEDUP / CAVEAT

Exponential over known classical methods for **generic** quantum dynamics — the best-founded advantage in the portfolio (`O-advantage`, `O-killerapp`). Fine print: classical tensor-network and neural-quantum-state methods keep absorbing "hard" instances, especially anything with limited entanglement or short evolution time (`S-tensornet`); static ground-state estimation (chemistry's real prize) is QMA-hard in general, so the machine helps with *specific* instances without a blanket guarantee. Digital simulation also inherits the state-preparation problem — you must load a good approximate ground state before phase estimation refines it.

CONFLICTS / OPEN QUESTIONS

Is quantum simulation the first paying application (field consensus: most likely) and when — 2027 (vendor optimism) versus mid-2030s (conservative academia)? See conflict `C-killerapp-timing`.

SOURCES

Int. J. Theor. Phys. 21, 467 (1982); Science 273, 1073 (1996); Nature 595, 227 (2021); blog.google quantum-echoes (2025); PRX Quantum 2, 030305 (2021). Cross-links: `S-hamsim`, `S-qsvt`, `S-qft`, `S-tensornet`, `S-nisq`, `I-pharma`, `I-chem`, `O-killerapp`.

Quantum singular value transformation (QSVT) · S-qsvt

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

QSVT is the modern unifying lens for quantum algorithm design. Given a matrix A block-encoded inside a larger unitary (A sits in a corner of U, accessed by ancilla-controlled calls), QSVT applies a **polynomial transformation to the singular values of A** using $O(\deg(p))$ calls to U and its inverse, plus one ancilla and a sequence of single-qubit rotations whose angles encode the polynomial p. It generalizes **quantum signal processing** (Low–Chuang 2017), which does the same for the eigenvalues of a single qubit's rotation, up to arbitrary block-encoded operators (Gilyén–Su–Low–Wiebe, STOC 2019). The payoff is conceptual and practical: pick the right polynomial and you recover most known quantum algorithms as special cases. $p \approx$ sign function → **amplitude amplification / Grover** (`S-grover`); $p \approx 1/x$ on the spectrum → **matrix inversion / HHL** (`S-hhl`); $p \approx e^{-ixt}$ → **Hamiltonian simulation** (`S-hamsim`, `S-qsim`); a rectangular window → **phase estimation** (`S-qft`) and eigenstate filtering; a threshold → quantum walks (`S-walk`). Martyn–Rossi–Tan–Chuang (2021) laid this out as an explicit "grand unification of quantum algorithms."

WHERE IT STANDS (2025–26)

QSVT is now the standard framework in which fault-tolerant algorithms are designed and their costs are stated, because it gives near-optimal query complexity with transparent resource accounting: the cost is the polynomial degree, and Chebyshev-approximation theory tells you the degree needed for a target accuracy. For matrix inversion it achieves **linear** dependence on the condition number κ (versus HHL's κ^2) and additive $\log(1/\varepsilon)$ precision — the tightest statement of the linear-systems cost. It also delivers optimal-scaling Hamiltonian simulation matching the qubitization lower bound (`S-hamsim`). The practical friction is **phase-angle computation**: finding the rotation angles that realize a target polynomial was numerically unstable for high-degree polynomials; 2020–24 work (Haah; Dong–Meng–Whaley–Lin; Motlagh–Wiebe) produced stable O(deg) angle-finding, largely closing that gap. QSVT is a fault-tolerant tool — it assumes a block-encoding, which itself needs coherent access to A (a qRAM-style or sparse-oracle assumption, `S-qram`), so it inherits the same input-model caveats as the algorithms it unifies.

KEY GRADED CLAIMS

- T1 QSVT: polynomial transform of singular values of a block-encoded operator, $O(\deg)$ queries — Gilyén–Su–Low–Wiebe, STOC 2019, arXiv:1806.01838 (established)
- T1 Quantum signal processing (the single-qubit precursor) — Low–Chuang, PRL 118, 010501 (2017) + PRX 6, 041067 (2016) (established)
- T2 Grand unification: amplitude amplification, HHL, Hamiltonian simulation as QSVT instances — Martyn–Rossi–Tan–Chuang, PRX Quantum 2, 040203 (2021) (established, tutorial)
- T2 Stable phase-angle finding for high-degree polynomials — Dong et al., PRA 103, 042419 (2021); Motlagh–Wiebe, PRX Quantum (2024) (established)
- T2 QSVT is dequantizable in the low-rank / sampling-access regime — Gharibian–Le Gall, STOC 2022, arXiv:2111.09079 (established; bounds where it does NOT help)

SPEEDUP / CAVEAT

Not itself a speedup — a **design framework** that yields near-optimal versions of the speedups already in the map. Its honest limit is the same one that runs through `S-hhl`/`S-qram`: the whole edifice sits on top of an efficient block-encoding of the input, and where that block-encoding is only "sampling access" to a low-rank matrix, Gharibian–Le Gall showed QSVT is classically dequantizable. So QSVT sharpens *what* is provably fast and *where* the classical counterattack bites, in one language.

CONFLICTS / OPEN QUESTIONS

Which real applications admit an efficient block-encoding that is not itself the bottleneck (the recurring `S-hhl` question). Whether angle-finding and block-encoding constant factors are small enough for early-fault-tolerant machines, or whether QSVT stays a large-scale-only tool.

SOURCES

arXiv:1806.01838; PRL 118, 010501 (2017); PRX Quantum 2, 040203 (2021); PRA 103, 042419 (2021); arXiv:2111.09079. Cross-links: `S-hhl`, `S-grover`, `S-hamsim`, `S-qsim`, `S-qft`, `S-walk`, `S-qram`, `reference-impl/`.

Classical shadows · S-shadows

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Classical shadows (Huang–Kueng–Preskill, Nat. Phys. 2020) is a measurement protocol that builds a compact classical sketch of an unknown quantum state ρ from very few measurements, then predicts many properties from that one sketch. Recipe: apply a random unitary from a fixed ensemble (random global Cliffords, or — more practically — random single-qubit Pauli rotations), measure in the computational basis, and invert the measurement channel to get a classical "snapshot." Average $O(\log M/\varepsilon^2)$ snapshots and you can estimate M **different observables** to accuracy ε with high probability. The headline result: for local observables the number of measurements is **independent of system size** and saturates information-theoretic lower bounds — a fundamentally different scaling from full quantum state tomography, which needs exponentially many measurements. Crucially the target properties can be chosen *after* the data is collected.

WHERE IT STANDS (2025–26)

Classical shadows became one of the most-used primitives in near-term quantum computing and a rigorous alternative to variational readout. Applications: estimating energies and forces for chemistry, entanglement entropies, fidelities, two-point correlators, and — the deepest connection — as the efficient tool that makes "learning from quantum data" provable (the exponential advantage in `S-qml`, Huang et al. Science 2022, uses shadow-style measurements). The active research frontier is robustness and specialization: noise-robust shadows under gate-dependent noise, **fermionic** shadows with mode-independent sample complexity for chemistry (2026), qudit and magic-gate variants, and shadows combined with circuit cutting (`S-circuitcut`). The Pauli-measurement version is directly runnable on today's hardware, which is why it displaced heavier tomography for benchmarking and for the atlas's kind of kernel/overlap estimation (`reference-impl/` — shadows are the sample-efficient cousin of the swap/Hadamard-test readout, predicting many overlaps from one measurement budget).

KEY GRADED CLAIMS

- T2 Predict M observables from $O(\log M/\varepsilon^2)$ measurements, size-independent for local observables — Huang–Kueng–Preskill, Nat. Phys. 16, 1050 (2020), arXiv:2002.08953 (established)
- T2 Sample complexity saturates information-theoretic lower bounds — same, with matching lower bounds (established)
- T2 Shadows enable provable exponential advantage learning from quantum experiments — Huang et al., Science 376, 1182 (2022) (demonstrated; quantum data)
- T3 Noise-robust and fermionic shadows extend applicability — Chen et al., PRX Quantum (2021); fermionic-shadow work arXiv:2606.27254 (2026) (established/preprint)

SPEEDUP / CAVEAT

A **measurement-efficiency** result, not a computational speedup — it reduces how many times you must run and measure a circuit, saturating a fundamental lower bound. Caveats: the estimator variance can be large for global or high-weight observables (random-Pauli shadows scale as 3^k for k-local operators), so it shines for **low-weight, many-observable** workloads and degrades for global ones; and it presupposes you can prepare ρ repeatedly, so it inherits the state-preparation costs of whatever produced ρ.

CONFLICTS / OPEN QUESTIONS

Which measurement ensemble is optimal for a given task (global Clifford = strong but hard circuits; local Pauli = easy but weaker for global observables). How much shadow variance survives realistic hardware noise at scale — an active robustness question.

SOURCES

Nat. Phys. 16, 1050 (2020), arXiv:2002.08953; Science 376, 1182 (2022); arXiv:2011.09636 (robust shadows); arXiv:2606.27254. Cross-links: `S-qml`, `S-variational`, `S-circuitcut`, `S-bench`, `reference-impl/`.

Shor's algorithm · S-shor

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Shor (1994) showed a quantum computer can factor integers and compute discrete logarithms in polynomial time. The engine is **period-finding**: factoring N reduces to finding the period r of $f(x) = a^x \bmod N$, and the period is read off by applying the quantum Fourier transform (QFT, see `S-qft`) to a superposition of function evaluations, then measuring and post-processing with continued fractions. Gate cost is $O((\log N)^2 \cdot \log\log N)$ with fast integer multiplication, i.e. polynomial in the bit-length. The best classical factoring algorithm, the general number field sieve, runs in sub-exponential $\exp(O((\log N)^{1/3}(\log\log N)^{2/3}))$ time — so Shor is a **superpolynomial** separation and breaks RSA, finite-field Diffie–Hellman, and elliptic-curve cryptography. That is the entire reason the world is migrating to post-quantum cryptography (`A-pqc`) and the source of the harvest-now-decrypt-later threat (`I-cyber`).

WHERE IT STANDS (2025–26)

No cryptographically relevant number has ever been factored quantumly. Honest records are tiny: 15 (NMR, Vandersypen 2001) and 21; larger published "factorings" used variational or annealing shortcuts that presuppose structure in the answer and do not scale (survey arXiv:2410.14397). The live action is **resource estimation**. Gidney (May 2025) cut the projected machine for RSA-2048 from his own 2019 estimate of ~20M noisy qubits over 8 hours to **<1M noisy qubits running under a week**, via approximate residue arithmetic, yoked surface codes, and magic-state cultivation (see `S-qec`). That ~20× drop in six years is why NIST/NSA migration deadlines (2030–2035) keep tightening rather than relaxing. The cost is dominated by modular exponentiation depth and by the T-gate / magic-state budget (`S-gates`), not by the QFT.

KEY GRADED CLAIMS

- T1 Polynomial-time quantum factoring and discrete log — Shor, SIAM J. Comput. 26, 1484 (1997) (established)
- T2 Largest honest quantum factorization remains trivial (15, 21) — Vandersypen et al., Nature 414, 883 (2001); survey arXiv:2410.14397 (established)
- T3 RSA-2048 with <1M noisy qubits in <1 week — Gidney, arXiv:2505.15917 (claimed; widely cited, unrefuted)
- T3 2019 baseline: 20M qubits / 8 hours — Gidney–Ekerå, arXiv:1905.09749 (established as the prior estimate)
- T5/T6 "Q-Day" dates (2030–2035) — NIST IR 8547 migration timeline + analyst punditry (roadmap/speculative)

SPEEDUP / CAVEAT

Superpolynomial over the best *known* classical algorithm. Factoring has no proven classical lower bound, so a classical breakthrough is possible in principle and Shor is not a complexity-theoretic separation the way sampling tasks aim to be (`S-bench`, `O-advantage`). Practical fine print: it needs full fault tolerance — millions of physical qubits, deep coherent circuits — so Shor is a 2030s+ threat under every credible hardware roadmap. HNDL makes it a present-day data-security problem even though the machine does not exist yet.

CONFLICTS / OPEN QUESTIONS

Resource estimates keep falling — is <100k qubits reachable with better codes and arithmetic? (See conflict `C-shor-timeline`.) Whether any state actor is ahead of public estimates is unknowable from open sources.

SOURCES

SIAM J. Comput. 26, 1484 (1997); arXiv:2505.15917; arXiv:1905.09749; arXiv:2410.14397; NIST IR 8547. Cross-links: `S-qft`, `S-qec`, `S-gates`, `A-pqc`, `I-cyber`, `O-advantage`.

Compilers, transpilers & middleware · S-software

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The software layer between an abstract circuit and pulses on hardware: SDKs for authoring circuits, **transpilers** that rewrite them into native gates (`S-gates`) while routing around limited connectivity (inserting SWAPs), optimizers that cut depth and two-qubit count, and runtime/middleware for hybrid execution. Compilation quality changes what a chip can do — a 2× depth reduction is worth roughly a hardware generation, because circuit fidelity decays exponentially in two-qubit-gate count. The problems it solves are hard in their own right: qubit routing on a fixed coupling graph is NP-hard, so transpilers use heuristics (SABRE-style routing, template matching, ZX-calculus rewrites) with no guarantee of optimality.

WHERE IT STANDS (2025–26)

Qiskit (IBM, Apache-2.0) is the de-facto standard: v2.x rebuilt the core in Rust, added a C API, and claims best-in-class transpilation on the Benchpress suite. **Cirq** (Google) serves the Google-stack research community. **PennyLane** (Xanadu) owns the differentiable-programming/QML niche (`S-qml`) with autodiff through circuits. **tket/pytket** (Quantinuum) is the leading retargetable third-party compiler. **Braket SDK** (AWS) and **Q#/QIR** (Microsoft, with the Azure resource estimator) round out the cloud layer (`S-cloud`); NVIDIA **CUDA-Q** pushes GPU-hybrid workflows and drives the "quantum-centric supercomputing" framing. Interoperability runs through **OpenQASM 3** and **QIR** (an LLVM-based intermediate representation). The frontier: dynamic-circuit compilation (mid-circuit measurement / feed-forward), error-aware routing using live calibration data, and the first **fault-tolerant compilers** targeting a *logical* instruction set (lattice-surgery operations) rather than physical gates. The reference implementation leans on exactly this layer — Qiskit's exact state-preparation compiler builds the amplitude-encoding circuits it runs (`reference-impl/MATH.md` §2).

KEY GRADED CLAIMS

- T2 All major SDKs actively maintained; OpenQASM 3 / QIR as interchange — GitHub release records (established)
- T4 Qiskit "most performant transpiler" — IBM Benchpress benchmarking, arXiv:2409.08844 (vendor-run benchmark; methodology public)
- T2 Compilation reduces two-qubit counts 2–50% versus naive lowering across suites — tket paper, Quantum Sci. Technol. 6, 014003 (2021) + Benchpress (demonstrated)
- T4 Vendor "quantum-centric supercomputing" middleware roadmaps — IBM/NVIDIA announcements (roadmap)

SPEEDUP / CAVEAT

Software **multiplies** hardware; it never creates asymptotic advantage. Between-SDK benchmark claims are usually vendor-run — check who compiled the competitor's baseline, since a poorly configured rival transpiler is an easy strawman (`S-bench`).

CONFLICTS / OPEN QUESTIONS

Whether the field consolidates on Qiskit the way ML consolidated on PyTorch, or QIR-level interop keeps the layer plural (open-source governance is itself soft power, `E-oss`). How fault-tolerant ISAs (lattice-surgery instruction sets) get standardized across vendors.

SOURCES

github.com/Qiskit/qiskit; quantumai.google/cirq; pennylane.ai; github.com/CQCL/tket; arXiv:2409.08844; Quantum Sci. Technol. 6, 014003 (2021). Cross-links: `S-gates`, `S-cloud`, `S-qml`, `E-oss`, `reference-impl/`.

Tensor-network classical simulation — the counterfactual · S-tensornet

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The classical method that decides most quantum-advantage arguments (`S-bench`, `O-advantage`). A tensor network represents an n-qubit state or a whole circuit as a graph of small tensors; contracting the graph computes amplitudes or expectation values. The key resource is **entanglement**: a 1D state with bounded entanglement is captured exactly by a **matrix product state (MPS)** of small bond dimension χ, and cost grows only polynomially in n and χ. States with more entanglement need 2D networks (PEPS) or full amplitude-tensor contraction, whose cost is set by the network's treewidth. The upshot is a sharp dividing line: circuits that stay **low-entanglement** — shallow, geometrically local, noisy, or short-time — are classically simulable, sometimes on a laptop; circuits that generate volume-law entanglement across a hard-to-contract geometry are where a quantum device might win. This is the honest counterfactual the atlas holds every advantage claim against.

WHERE IT STANDS (2025–26)

Tensor networks have repeatedly erased "beyond-classical" claims. **Sycamore 2019** RCS was cut from "10,000 years" to hours-then-minutes by hyper-optimized contraction on GPU clusters (Pan–Chen–Zhang; and 2024 work ran 1,432 GPUs ~7× faster than Sycamore). **IBM's 2023 kicked-Ising "utility"** experiment (`S-nisq`) was matched within weeks by an MPS/belief-propagation tensor-network simulation on a laptop (Tindall–Fishman–Stoudenmire–Sels 2024) — the cleanest recent example, because the circuit's limited entanglement made it easy. Gaussian and lossy boson sampling fell to MPS methods too. The 2025 Nature Reviews Physics survey frames this as a productive arms race: each advantage claim forces better contraction heuristics, and each classical advance raises the bar for what counts as advantage. Google's Dec 2024 Willow RCS and Oct 2025 "quantum echoes" (`S-qsim`) are the current standing claims that tensor networks have **not** yet matched — the live test.

KEY GRADED CLAIMS

- T2 Sycamore RCS classically matched via tensor contraction — Pan–Zhang, PRL 129, 090502 (2022); "Leapfrogging Sycamore" arXiv:2406.18889 (2024) (established)
- T2 IBM kicked-Ising utility reproduced by laptop MPS — Tindall et al., PRX Quantum 5, 010308 (2024) (established)
- T2 MPS captures bounded-entanglement states in $\mathrm{poly}(n, \chi)$; DMRG foundation — Vidal, PRL 91, 147902 (2003); White DMRG (1992) (established)
- T2 Tensor networks as verification + counterattack tool for advantage — Nature Reviews Physics (2025), arXiv:2503.08626 (established survey)

SPEEDUP / CAVEAT

This is the classical baseline, so its "caveat" runs the other way: tensor networks are efficient **only** for low-entanglement or low-treewidth instances, and cost blows up exponentially with entanglement across the contraction cut. A quantum advantage survives precisely where no efficient contraction exists — which is why long-time, high-entanglement dynamics (`S-hamsim`) is the most defensible advantage target, and shallow/noisy circuits are the most vulnerable.

CONFLICTS / OPEN QUESTIONS

Whether Willow-class RCS and the OTOC "quantum echoes" experiment stay uncontracted, or fall like their predecessors (`C-advantage`). How far belief-propagation-augmented tensor networks push the simulable frontier for structured circuits.

SOURCES

PRL 129, 090502 (2022); PRX Quantum 5, 010308 (2024); PRL 91, 147902 (2003); Nature Rev. Phys. (2025), arXiv:2503.08626; arXiv:2406.18889. Cross-links: `S-bench`, `S-nisq`, `S-qsim`, `S-hamsim`, `O-advantage`, `O-verification`.

Variational algorithms — VQE & QAOA · S-variational

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Variational quantum algorithms (VQAs) pair a shallow parameterized circuit $U(\theta)$ with a classical optimizer in a feedback loop. **VQE** (Peruzzo et al., 2014) minimizes $\langle\psi(\theta)|\, H \,|\psi(\theta)\rangle$ for a molecular Hamiltonian H to estimate ground-state energy — the variational principle guarantees the estimate is an upper bound. **QAOA** (Farhi–Goldstone–Gutmann, 2014) encodes a combinatorial problem (MaxCut, etc.) in p alternating layers of problem-phase and mixer unitaries $e^{-i\gamma C}e^{-i\beta B}$, interpolating toward the optimum as p grows. They were designed for NISQ hardware (`S-nisq`): short circuits, hybrid by construction, partly noise-tolerant. Gradients are computed by the parameter-shift rule; the objective is estimated by repeated measurement (shots), so cost scales with the number of observables and the desired precision.

WHERE IT STANDS (2025–26)

The theory turned against the program on three fronts. **Barren plateaus** (McClean et al., 2018): for expressive random ansätze the gradient variance vanishes exponentially in qubit count, so training needs exponentially many shots to see a signal. **Soft dequantization** (Cerezo et al., Nat. Commun. 2025, arXiv:2312.09121): the known families of circuits that provably *avoid* barren plateaus turn out to be classically simulable in polynomial time — if a landscape is trainable it is likely also classically tractable, squeezing the useful regime from both sides. **Classical optimization competition**: fixed-depth QAOA is provably beaten by classical algorithms on broad instance classes (Bravyi et al. 2020; Farhi–Gamarnik–Gutmann obstruction bounds), and no VQE/QAOA run has beaten best-in-class classical chemistry (DMRG, coupled cluster CCSD(T)) or optimization (Gurobi-class solvers) on any real instance. Add measurement/optimization overhead (thousands of shots per energy evaluation, noisy gradients) and the near-term value proposition is thin. Community response: use variational circuits as *state-preparation subroutines* feeding fault-tolerant phase estimation, or move on.

KEY GRADED CLAIMS

- T2 VQE original demonstration — Peruzzo et al., Nat. Commun. 5, 4213 (2014) (demonstrated, toy scale)
- T2 Barren plateaus in random parameterized circuits — McClean et al., Nat. Commun. 9, 4812 (2018) (established)
- T2 Absence of barren plateaus implies classical simulability for known families — Cerezo et al., Nat. Commun. 16 (2025), arXiv:2312.09121 (established, with stated caveats)
- T2 Fixed-depth QAOA obstructed / classically matched on many instances — Bravyi et al., PRL 125, 260505 (2020) (established)

SPEEDUP / CAVEAT

No proven speedup exists for VQE or QAOA — both are heuristics. Every empirical claim to date has been matched or beaten classically once a competent classical team responds. The training cost (barren plateaus) and measurement cost (shot budget) compound. The escape hatches — quantum-data inputs, initial-state prep for phase estimation, problem-tailored non-random ansätze — remain unproven at advantage scale.

CONFLICTS / OPEN QUESTIONS

Whether *any* ansatz family is simultaneously trainable, classically hard, and noise-tolerant is the open question. Cerezo et al. conjecture essentially no for barren-plateau-free landscapes; parts of the field dispute how far the "simulable" argument generalizes to structured (non-random) circuits.

SOURCES

Nat. Commun. 5, 4213 (2014); arXiv:1411.4028; Nat. Commun. 9, 4812 (2018); arXiv:2312.09121; PRL 125, 260505 (2020). Cross-links: `S-nisq`, `S-qsim`, `S-qml`, `S-qft`, `I-pharma`, `O-advantage`.

Quantum walks & element distinctness · S-walk

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Quantum walks are the quantum analog of the classical random walk — a superposition spreading over a graph, in discrete-time (coined) or continuous-time (Hamiltonian) form. They are both an algorithmic framework and a universality result. Ambainis's element-distinctness algorithm (2004) walks on the Johnson graph of colliding subsets to find two equal items among N in $O(N^{2/3})$ queries — provably optimal, and a genuinely super-quadratic improvement over the $O(\sqrt{N})$ a naive Grover approach would give. The framework generalizes to subgraph/triangle finding, spatial search, and matrix-verification, and underpins the MNRS (Magniez–Nayak–Roland–Santha) quantum-walk search meta-algorithm.

WHERE IT STANDS (2025–26)

Two structural results anchor the node. First, universality: a multi-particle continuous-time quantum walk on an unweighted graph is BQP-complete (Childs–Gosset–Webb, Science 2013) — quantum walks alone can run any quantum computation, so they are a full model, not a trick. Second, unification: Childs (2010) showed discrete- and continuous-time walks are interconvertible, and recent "multidimensional quantum walk" work (Jeffery–Zur, STOC 2023, arXiv:2208.13492) finally gave a continuous-time $O(N^{2/3})$ element-distinctness algorithm and a time-efficient k-distinctness walk, closing a long-open gap. Walks also power the fastest known algorithms for several graph problems. Like Grover, the honest caveat is that most speedups are query-model and polynomial, so fault-tolerant overhead accounting (`S-grover`, `O-advantage`) applies.

KEY GRADED CLAIMS

- T1 Element distinctness in $O(N^{2/3})$ queries, optimal — Ambainis, SIAM J. Comput. 37 (2007); FOCS 2004 (established)
- T1 Multi-particle continuous-time quantum walk is universal for BQP — Childs–Gosset–Webb, Science 339, 791 (2013) (established)
- T2 Discrete/continuous-time walk equivalence; multidimensional walks give CT element distinctness + time-efficient k-distinctness — Childs 2010; Jeffery–Zur, arXiv:2208.13492 (established/preprint)

SPEEDUP / CAVEAT

Provable polynomial speedups (up to $O(N^{2/3})$ for distinctness, quadratic for many searches), often optimal in the query model. Caveat: these are query-complexity results — the same fault-tolerance overhead that erodes Grover's practical advantage applies here, and the walk oracle must be cheap to implement reversibly for any wall-clock win.

CONFLICTS / OPEN QUESTIONS

Whether any quantum-walk speedup on a real problem (not a black-box oracle) survives end-to-end resource estimation. As with `S-grover`, current evidence leans no for the polynomial-only cases.

SOURCES

SIAM J. Comput. 37, 210 (2007); Science 339, 791 (2013); arXiv:2208.13492; arXiv:1302.7316; Childs arXiv:0810.0312. Cross-links: `S-grover`, `S-qft`, `O-advantage`.

Chapter 4 · Beyond Computing

Atom interferometry (as its own platform) · A-atominterf

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Atom interferometry is the *technique* underneath several sensing cards, promoted here to its own node because it is a distinct platform, not an application. Laser-cool a cloud of atoms, then use light pulses to coherently split, redirect, and recombine each atom's matter-wave along two paths; the phase difference at recombination encodes whatever acted differently on the two arms — gravitational acceleration, rotation, or a fundamental constant. Because the de Broglie wavelength is tiny and the atom is a perfect, drift-free test mass referenced to atomic structure, atom interferometers are **absolute** instruments. The same hardware measures gravity (`A-gravimetry`), rotation (gyroscopes), acceleration (the IMU under `A-pnt`), and probes physics: the fine-structure constant, tests of the equivalence principle, and searches for dark matter and gravitational waves in a new band.

MATURITY & REAL DEPLOYMENTS (2025–26)

A mature laboratory platform; early-commercial as gravimeters; a physics tool at the frontier. - **Commercial instruments**: Exail's AQG absolute gravimeter (`A-gravimetry`) and AOSense inertial/gravity sensors are atom interferometers you can buy. - **Fundamental physics**: atom interferometers have measured the fine-structure constant α to ~0.2 parts per billion and run the tightest lab tests of the weak equivalence principle (Stanford 10 m tower, and the MICROSCOPE-successor concepts). **AION** (UK) and **MAGIS-100** (Fermilab, ~100 m baseline) are building long-baseline atom interferometers to search for ultralight dark matter and mid-band gravitational waves (~0.1–10 Hz, between LIGO and LISA). Space concepts (STE-QUEST, and Bose-Einstein-condensate interferometry on the ISS Cold Atom Lab) extend the free-fall time that sets sensitivity. - **The sensitivity lever**: interferometer phase sensitivity scales with the interrogation time squared and the momentum splitting — which is why the field pushes toward longer drop towers, microgravity, and large-momentum-transfer beam-splitters.

KEY GRADED CLAIMS

- T1 Light-pulse atom interferometry gives absolute, atomic-referenced measurement of acceleration/rotation — Kasevich–Chu lineage (established)
- T2 Fine-structure constant measured to sub-ppb by atom interferometry — Parker et al./Morel et al. (established)
- T2 Long-baseline atom interferometers (MAGIS-100, AION) under construction for dark-matter / GW search — collaboration reports (demonstrated construction, roadmap science)
- T3 BEC atom interferometry in microgravity (Cold Atom Lab, ISS) — NASA/JPL results (demonstrated)

CONFLICTS / OPEN QUESTIONS

- **The moving-platform problem** (shared with `A-pnt`/`A-gravimetry`): a launched cloud dislikes vibration and rotation, so turning a beautiful lab gravimeter into a strapdown navigation-grade 6-DoF sensor on a maneuvering vehicle is unsolved engineering.
- **Physics upside is speculative**: whether long-baseline interferometers actually *detect* dark matter or mid-band gravitational waves (rather than just set limits) is unknown — that is the science bet, not a guaranteed return.

THE HONEST CALL

A mature, foundational quantum-sensing platform — Nobel-recognized physics (1997/2005 laser cooling and precision measurement lineage), buyable as gravimeters, and the engine of the field's most precise fundamental-constant measurements. Its two frontiers are opposite in character: *down* into rugged, small navigation sensors (engineering-hard, `A-pnt`), and *up* into kilometre-scale physics detectors (science-speculative). Grade the instrument as established; grade each application on its own card.

SOURCES

- https://arxiv.org/abs/... (MAGIS-100 detector, Fermilab)
- https://aion-project.web.cern.ch/ (AION long-baseline atom interferometer)
- Parker et al., "Measurement of the fine-structure constant as a test of the Standard Model," Science 360 (2018)
- NASA JPL Cold Atom Lab BEC interferometry results

Atomic & Optical Clocks · A-clocks

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Clocks locked to atomic transition frequencies — the most precise instruments humans have built. Cesium fountains define the SI second today at $\sim 10^{-16}$ fractional uncertainty; optical clocks (trapped ions, optical lattices) run their reference transition $\sim 10^{5}$ times faster and reach systematic uncertainties near $\mathbf{10^{-18}}$–$\mathbf{10^{-19}}$, good to better than a second over the age of the universe. Two quality metrics matter: **accuracy** (systematic uncertainty — how close to the true transition frequency) and **stability** (Allan deviation — how well it averages down over time). Downstream: GPS itself, telecom/finance timestamping, geodesy (a 10^{-18} clock senses a $1\ \mathrm{cm}$ change in height via gravitational redshift), and GPS-free PNT (`A-pnt`).

MATURITY & REAL DEPLOYMENTS (2025–26)

Layered maturity — from mass-market chips to lab instruments defining the SI second. - **Chip-scale atomic clocks (CSACs)** — e.g. Microchip's CSAC (rubidium, $\sim 120\ \mathrm{mW}$, $\sim 10^{-10}$ stability) — are long-standing commercial products in defense, telecom, and undersea gear. - **Lab optical clocks** are established metrology. **NIST's** Al^{+} **quantum-logic ion clock** set the accuracy record in July 2025 at $\sim \mathbf{8.1 \times 10^{-19}}$ **systematic uncertainty (~19 decimal places), 41% better than the prior best**. A **six-country coordinated optical-clock comparison** (June 2025) is the largest cross-border agreement test yet — direct groundwork for redefining the second. - **Portable/deployed optical clocks** are commercializing: **QuantX Labs** (rubidium two-photon optical clock, operated at sea), Adelaide's warm-ytterbium-vapor clock, and **Infleqtion's Tiqker** — which was deployed on the **UK Royal Navy's XV Excalibur** uncrewed submarine in October 2025, a first-of-kind at-sea optical-clock deployment.

THE SI-SECOND REDEFINITION (THE BIG INSTITUTIONAL EVENT)

BIPM/CCTF's roadmap conditions redefinition on optical standards beating cesium by $\geq \mathbf{100\times}$, $\geq \mathbf{5}$ **independent systems** running continuously in different labs, and $> \mathbf{1}$ **year** of traceable agreeing data. The CGPM (2026) is expected to endorse the roadmap, with the actual redefinition targeted around **2030**. This is why time-transfer (`A-timedist`) is load-bearing: the clocks are now better than the links that compare them.

KEY GRADED CLAIMS

- T2 Al^{+} optical clock at $\sim 8.1 \times 10^{-19}$ systematic uncertainty, record accuracy — NIST, July 2025 (established)
- T2 Six-country coordinated optical-clock comparison as redefinition groundwork — Optica/NMI collaboration, June 2025 (demonstrated)
- T3 Infleqtion Tiqker optical clock deployed on Royal Navy XV Excalibur submarine — Oct 2025 (demonstrated, vendor+navy)
- T5/T6 SI second redefined on an optical transition ~2030 — CIPM/BIPM roadmap, arXiv:2307.14141 (roadmap)

CONFLICTS / OPEN QUESTIONS

- Which transition wins the redefinition — Sr lattice (most-run), Yb, Yb^{+} E3, Al^{+} (most accurate), or a weighted ensemble of several — is unsettled; a single-transition definition versus an ensemble is an open governance choice.
- Portable optical clocks trade 2–4 orders of accuracy for size, weight and power. Whether field PNT actually needs optical performance or just a better vapor-cell/CSAC holdover clock is an open market question — for many jam-resistant timing use cases a nanosecond-per-day holdover is enough.

THE HONEST CALL

Split maturity, all real. CSACs are a decades-old commercial commodity; lab optical clocks are the most accurate machines ever built and are about to redefine the second; portable optical clocks are just now leaving the lab (Tiqker at sea). The near-term commercial pull is defense/PNT holdover timing and geodesy, not the 19-digit lab record.

SOURCES

- https://www.nist.gov/news-events/news (Al^{+} clock record, July 2025)
- https://www.optica.org/about/newsroom/news_releases/2025/unprecedented_optical_clock_network_lays_groundwork_for_redefining_the_second/
- https://spaceinsider.tech/2026/04/10/quantum-sensing-for-pnt-nears-deployment/ (Tiqker on XV Excalibur)
- https://arxiv.org/pdf/2307.14141 (redefinition roadmap)

Entanglement-based clock networks · A-entclock

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Distinct from both the clocks (`A-clocks`) and the classical time-transfer plumbing (`A-timedist`): this node is the proposal to **entangle distant clocks** so a network of them behaves as one collective quantum sensor. The theory (Kómár et al., Nat. Phys. 2014) is clean: with K clocks of N atoms each linked by entanglement into a shared GHZ state, the frequency uncertainty scales toward the **Heisenberg limit ∝ 1/(K·N)** instead of the standard quantum limit ∝ **1/√(K·N)** — a network of 10 optical-lattice clocks of 1,000 atoms each could in principle reach an Allan deviation of **~2×10^{-18} at 1 s**, beating any single clock. It would also give a *distributed* reference immune to a single node being spoofed or lost, and enable clock comparisons whose precision beats the link noise.

MATURITY & REAL DEPLOYMENTS (2025–26)

Research — theoretically compelling, experimentally embryonic, and its practical advantage is contested. - **First realization (2022)**: two Sr^{88+} **ion clocks** separated by ~2 m were entangled via a photonic link, and entanglement reduced the number of measurements needed toward the Heisenberg-limit scaling — a proof of principle at benchtop distance, not a network. - **Metropolitan groundwork (2025)**: an entanglement-based clock **syntonization** demonstration synchronized two rubidium clocks to **<12 ps** at all times over **48–50 km** of deployed fiber across the Métropole Côte d'Azur, riding an entanglement-based QKD link (~7 kbit/s key). This shows entanglement infrastructure and clock timing can share a real fiber network — a step toward, not a demonstration of, Heisenberg-limited networked clocks. - **On paper**: multiple 2025–26 surveys/ roadmaps (arXiv:2604.04437, 2606.15421) map the architectures; long-baseline entanglement distribution (`A-qinternet`) is the missing enabler.

KEY GRADED CLAIMS

- T1 GHZ-entangled clock networks scale to Heisenberg-limited frequency precision ∝ 1/(KN) — Kómár et al., Nat. Phys. 10 (2014) (established theory)
- T2 Two remote ion clocks entangled via photonic link, sub-Heisenberg measurement reduction — 2022 experiment (demonstrated, ~2 m)
- T2 Entanglement-based clock syntonization, <12 ps over 48 km deployed fiber — Appl. Phys. Lett. 126, 174003 (2025) (demonstrated)
- T6 Continental Heisenberg-limited entangled clock network — roadmap only (speculative)

CONFLICTS / OPEN QUESTIONS

- **Does entanglement actually help in practice?** A 2026 critical assessment (arXiv:2604.10243) argues that for real time-synchronization tasks, classical comb-stabilized optical transfer already saturates the useful precision, and GHZ states are fragile (one lost photon collapses the whole network state), so the theoretical 1/N gain may not survive loss and decoherence over network distances. This is the same skepticism flagged in `A-timedist`.
- **Enabler dependency**: it cannot exist without long-distance entanglement distribution and quantum memories (`A-qinternet`, `A-qmemory-hw`), which themselves have not reached the repeater crossover.

THE HONEST CALL

A beautiful idea that is nowhere near a network. The Heisenberg-scaling theory is solid and the benchtop/metropolitan building blocks are real, but there is no fielded entangled clock network, and credible physicists question whether entanglement beats good classical optical transfer for actual timekeeping once loss is included. Grade as research/speculative — the near-term "quantum clock network" is classically-linked quantum clocks (`A-timedist`), not entangled ones.

SOURCES

- https://pubs.aip.org/aip/apl/article/126/17/174003/3345796/ (entanglement-based clock syntonization, 48 km, 2025)
- Kómár et al., "A quantum network of clocks," Nature Physics 10 (2014)
- https://arxiv.org/pdf/2604.04437 (Quantum Clock Synchronization Networks: A Survey)
- https://arxiv.org/pdf/2604.10243 (critical assessment of quantum time-sync)

Gravimetry & Inertial Navigation · A-gravimetry

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Cold-atom interferometry: drop or launch a cloud of laser-cooled atoms, split each atom's wavefunction with a sequence of light pulses (a π/2–π–π/2 Mach–Zehnder), and read gravity (or acceleration/rotation) from the accumulated interference phase. Because the atom is the test mass and the laser wavelength is the ruler, the measurement is **absolute and drift-free** — classical spring gravimeters and mechanical/FOG inertial units drift and need recalibration against a reference. Applications: subsurface surveying (voids, tunnels, aquifers, ore, volcano/CO_2-storage monitoring), and drift-free inertial navigation for the GPS-denied problem (`A-pnt`).

MATURITY & REAL DEPLOYMENTS (2025–26)

Early commercial for gravimetry; pre-deployment for navigation. - **Absolute gravimeters**: **Exail** (ex-Muquans) has sold its **AQG** (Absolute Quantum Gravimeter) for years — including continuous monitoring on Mt Etna. Independent evaluation (Journal of Geodesy 2025) of the AQG-A02/B10 against a reference measured **~500 nm·s⁻²/√Hz sensitivity**, reaching **~10 nm·s⁻² (≈1 µGal) after 1 h integration** with drift-free sub-µGal long-term stability — and notably **no active vibration isolation** (it measures microseismic noise with an accelerometer and compensates on the laser phase). - **Gradiometry**: the University of Birmingham's cold-atom gravity **gradiometer detected a buried structure in an open field** (Stray et al., Nature 602, 2022) — the landmark proof that differential (gradiometer) measurement cancels vibration noise outdoors, where a single gravimeter is swamped. - **Navigation**: the hot line. **Q-CTRL's Ironstone Opal** fuses quantum magnetometry/gravimetry with map-matching; airborne trials claim up to **111× better GPS-denied accuracy** than the best conventional alternative and **~4 m over 700 km flights**; a maritime trial logged **144+ hours** unattended at sea (2025–26). **AOSense** cold-atom IMUs target **~5 m/hour** navigation drift without external signals (DARPA PINS lineage). DARPA's **RoQS** funds ruggedization ($24.4M to Q-CTRL); **Lockheed Martin + Q-CTRL** won a March 2025 DoD prototype quantum-INS contract. France (Exail/ONERA) and the UK (Infleqtion, ship/aircraft trials) run parallel programs.

KEY GRADED CLAIMS

- T2 Exail AQG: ~500 nm·s⁻²/√Hz, ~1 µGal after 1 h, drift-free — independent eval, J. Geodesy 99 (2025); Ménoret et al., Sci. Rep. 2018 (demonstrated)
- T2 Cold-atom gradiometer located a buried structure in open-field conditions — Stray et al., Nature 602 (2022) (demonstrated)
- T3 Cold-atom IMU ~5 m/hour navigation drift, unaided — AOSense / DARPA PINS (demonstrated, program-level)
- T4 Ironstone Opal: GPS-denied navigation up to 111× better than conventional, ~4 m over 700 km — Q-CTRL, 2025 (claimed, vendor field trials)

CONFLICTS / OPEN QUESTIONS

- **No independent benchmark for the nav claims**: quantum-INS vendor numbers are self-reported field trials; there is no peer-reviewed head-to-head against classified military-grade ring-laser-gyro/FOG INS.
- **SWaP and dynamics** are the wall between trials and fleet use: size/weight/power, and robustness to vibration and hard maneuvers (a launched atom cloud dislikes a jinking aircraft). Gravimeters mostly measure stationary or slow platforms; strapdown atom-interferometer accelerometers/gyros for full 6-DoF navigation on a moving platform are the harder, less-mature problem.

THE HONEST CALL

Gravimetry is a real, shipping instrument (Exail sells them; independent labs verify the µGal numbers). **Quantum inertial navigation is not yet fielded** — it is in well-funded DoD prototyping with impressive but unverified trial claims, gated by SWaP and dynamic robustness. Split the two: survey gravimeters = commercial; GPS-denied quantum nav = advanced R&D nearing deployment (see `A-pnt`).

SOURCES

- https://link.springer.com/article/10.1007/s00190-025-01995-x (independent Exail AQG evaluation, 2025)
- https://q-ctrl.com/blog/q-ctrls-new-maritime-quantum-navigation-solution-successfully-undergoes-first-defense-trials-at-sea
- Stray et al., "Quantum sensing for gravity cartography," Nature 602 (2022)
- https://aosense.com/ (cold-atom IMU / gradiometer)

Quantum Imaging & Radar · A-imaging

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Imaging and ranging schemes that exploit photon correlations. **Ghost imaging** reconstructs an object from a beam that never touched it, via correlations with a reference beam that did. **Quantum illumination (QI)** (Lloyd 2008; Tan et al. 2008) sends one half of an entangled signal–idler pair at a target and keeps the idler; joint correlation detection gives a theoretical **6 dB advantage in the error exponent** in the high-noise, low-reflectivity, low-signal regime — the basis of "quantum radar" claims. Related: sub-shot-noise microscopy, NOON-state phase imaging, and single-photon LiDAR. A crucial taxonomic split runs through this node — **truly quantum** (entangled/squeezed light) vs **quantum-inspired** (classical correlations, quantum detectors) — because almost all the commercial wins are the latter.

THE QUANTUM VS QUANTUM-INSPIRED LIDAR SPLIT (THE LOAD-BEARING DISTINCTION)

- **Single-photon / quantum-inspired LiDAR is genuinely deployed.** SPAD arrays and photon-counting time-of-flight sensors ship in automotive and remote sensing for photon-starved conditions — but they use **quantum detection with classical light**, not entanglement. A compact all-fiber **quantum-inspired LiDAR** demonstrated **>100 dB rejection** of in-band noise with single-photon sensitivity using classical time-frequency correlations (Nature Communications 2023); a 2025 **photon-number-resolving** single-photon LiDAR approached the **standard quantum limit** in ranging precision (Light: Science & Applications 2025). These keep the noise-rejection benefit without needing entanglement or cryogenics — which is exactly why they, not QI radar, reach the field.
- **Entanglement-based QI** stays in the lab. Microwave QI was demonstrated in principle inside dilution refrigerators, showing **~4 dB** measured enhancement over a classical-noise-radar benchmark (below the 6 dB theoretical ceiling), but only over a narrow parameter range and requiring cryogenic idler storage.

MATURITY & REAL DEPLOYMENTS

Mostly research; the commercial wins are quantum-inspired. **Quantum radar** is the graded-hardest corner: fielded long-range quantum radar is implausible near-term because idler storage over realistic ranges is impossible (you'd need a lossless quantum memory for the round-trip light-time), entanglement generation rates are tiny, and the advantage evaporates the moment you can just transmit more classical power. Defense-lab and academic assessments are broadly skeptical despite recurring Chinese vendor claims. Ghost imaging and sub-shot-noise microscopy (Padgett, Genovese groups) are established lab techniques with niche microscopy use.

KEY GRADED CLAIMS

- T2 QI's 6 dB error-exponent advantage in the noise-limited regime — Tan et al., PRL 101, 253601 (2008); microwave lab demos ~4 dB (established theory, demonstrated small-scale)
- T2 Ghost imaging and sub-shot-noise imaging demonstrated repeatedly — Padgett group, Genovese reviews (established)
- T2 Noise-tolerant / photon-number-resolving LiDAR near the standard quantum limit — Light Sci. Appl. 2025; Nat. Commun. 2023 (demonstrated, quantum-inspired)
- T6 Fielded long-range quantum (entanglement) radar — vendor/press claims, no verified system (speculative; contested)

CONFLICTS / OPEN QUESTIONS

- **C-qradar**: proponents cite the QI advantage; skeptics (RAND-adjacent and academic reviews) note it assumes conditions — pre-shared entanglement surviving a lossy round trip, known target range, extreme background noise, no option to just raise transmit power — that no fielded radar meets. Resolution: any independently verified over-the-air quantum (not quantum-inspired) radar demo. None exists.
- Where entanglement-based quantum imaging beats cheap computational classical imaging is still case-by-case and mostly loses on cost.

THE HONEST CALL

Quantum-inspired imaging/LiDAR is commercial and shipping; entanglement-based quantum radar is not, and probably won't be soon. The single most useful thing this node does is police the label: a "quantum LiDAR" in a product is almost always a photon-counting classical system, and "quantum radar" as a fielded weapon is, as of 2026, vaporware.

SOURCES

- https://www.nature.com/articles/s41377-025-01880-4 (photon-number-resolving single-photon LiDAR, 2025)
- https://www.nature.com/articles/s41467-023-40914-6 (quantum-inspired LiDAR, >100 dB rejection)
- https://arxiv.org/abs/2211.05684 (quantum advantage in microwave quantum radar, ~lab)
- https://arxiv.org/pdf/2103.12548 (Quantum Technology for Military Applications — skeptical review)

Magnetometry · A-magneto

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Measuring magnetic fields with quantum systems, down to femtotesla (fT) — the brain's magnetic signals are ~50–500 fT, roughly a hundred-million-times weaker than Earth's ~50 µT field. Three platforms with different trade-offs: - **SQUIDs** — superconducting loops, the cryogenic gold standard since the 1960s; **~1–5 fT/√Hz**, but need liquid helium and fixed dewars. - **OPMs (optically pumped magnetometers)** — alkali (Rb/Cs) or ^{4}He vapor cells, near-room-temperature, small and wearable; commercial zero-field OPMs (QuSpin QZFM Gen-2) reach **~7–15 fT/√Hz** (1–100 Hz), approaching SQUID sensitivity without cryogenics. - **NV-diamond** — nanoscale, ambient, vector; best NV-MEG-oriented sensors reach **~9.4 pT/√Hz** (5–100 Hz) — three orders worse than OPMs, but at atom-scale spatial resolution (see `A-nvsensing`, `H-nv`).

MATURITY & REAL DEPLOYMENTS (2025–26)

Commercial today across several niches, with neuroimaging the standout disruption. - **MEG (magnetoencephalography)**: SQUID-MEG has imaged brain fields in hospitals for decades inside **$2–3M shielded rooms** with fixed helmets. The disruption is **OPM-MEG** — wearable helmets the patient can move in. **Cerca Magnetics** (Nottingham spin-out) sells OPM-MEG and ran epilepsy/Parkinson's clinical work through 2024–25; **FieldLine Medical** installed 64-channel wearable OPM-MEG at two US academic medical centres in 2024 for pediatric epilepsy pre-surgical mapping without sedation; **MAG4Health** (France) sells a ^{4}He-OPM system; an 80-sensor OPM-MEG study (NeuroImage 2025) marks research-grade adoption. - **Cardiac**: SandboxAQ's **CardiAQ** (with Mayo Clinic) applies OPMs to magnetocardiography. - **Navigation**: magnetic-anomaly map-matching (Q-CTRL Ironstone Opal, SandboxAQ AQNav — see `A-sensing`, `A-pnt`). - **Geophysics/defense**: SQUID transient-EM sensors (CSIRO LANDTEM) have found ore bodies; OPM/NV airborne surveys are emerging; magnetic-anomaly submarine detection (MAD) is a classified but active area.

KEY GRADED CLAIMS

- T1 SQUIDs (~1–5 fT/√Hz) and zero-field OPMs (~7–15 fT/√Hz) reach femtotesla sensitivity — decades of metrology + QuSpin datasheets (established)
- T2 Wearable OPM-MEG matches cryogenic MEG signal quality while allowing head motion — Boto et al., Nature 555 (2018); 80-sensor system, NeuroImage 2025 (demonstrated)
- T2 NV-diamond magnetometer at ~9.4 pT/√Hz enabling ambient-condition MEG-scale sensing — Phys.org/2024 diamond magnetometer result (demonstrated)
- T4 CardiAQ OPM cardiac diagnostics in clinical evaluation with Mayo Clinic; Cerca OPM-MEG clinical work 2024–25 — company announcements (claimed)

CONFLICTS / OPEN QUESTIONS

- **Reimbursement, not physics, gates OPM-MEG.** The sensitivity is there; whether it displaces SQUID suites this decade turns on FDA/CE clearance and insurance reimbursement pace, plus the still-needed magnetically shielded room (OPMs cut cost but don't fully eliminate shielding).
- **NV fit is unsettled**: NV's ambient operation and spatial resolution are unmatched, but it lags OPM/SQUID by 2–3 orders in raw sensitivity — so its commercial niche (current mapping inside chips, single-cell biology, scanning probe) is real but narrow.

THE HONEST CALL

Commercial and the most clinically consequential quantum-sensing modality. SQUID magnetometry is a mature medical/industrial product; OPM-MEG is a live, well-funded disruption already installed in hospitals and limited by regulatory/reimbursement timelines rather than performance; NV magnetometry is real but a specialist tool, not an OPM/SQUID replacement.

SOURCES

- https://www.sciencedirect.com/science/article/pii/S1053811925001843 (80-sensor OPM-MEG, 2025)
- https://entangledfuture.com/quantum-sensors/magnetometers/ (OPM/SQUID/NV sensitivity comparison)
- https://phys.org/news/2024-06-highly-sensitive-diamond-quantum-magnetometer.html (9.4 pT/√Hz NV)
- Boto et al., "Moving magnetoencephalography towards real-world applications," Nature 555 (2018)

Quantum Materials · A-materials

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Materials whose macroscopic behavior is set by quantum effects — **topological insulators** (insulating bulk, symmetry-protected conducting surface states), **high-Tc superconductors** (cuprates since 1986, iron-based since 2008), and **2D materials** (graphene, transition-metal dichalcogenides, and twisted bilayers where a "magic angle" $\sim 1.1°$ produces superconductivity). They sit in two roles at once: a physics frontier in their own right, and the **substrate layer under qubits** — Majorana platforms (`H-topo`), Josephson-junction materials, and the two-level-system (TLS) defects that cap coherence (`H-fab`, `O-materials`). This node treats quantum materials as an adjacent *technology*, so the grading question is: which of these ship in a product, and which are still lab physics?

MATURITY & REAL DEPLOYMENTS (2025–26)

Split — a spectrum from industrial commodity to contested lab claim. - **Low-Tc superconductors** (NbTi, $\mathrm{Nb_3Sn}$) are fully industrial: MRI magnets, SQUIDs (`A-magneto`), and the transmon circuits of `H-supercon` all run on them. - **High-Tc REBCO tape** crossed from lab to product: it is load-bearing in **Commonwealth Fusion's SPARC** high-field magnets (the $\sim 20\ \mathrm{T}$ magnets that make compact tokamaks viable) — arguably the biggest commercial deployment of a high-Tc material to date. - **Topological insulators and 2D-material devices remain research-stage** — no volume product ships *on topological protection* yet, which is precisely the risk in Microsoft's Majorana bet (`H-topo`). Recent motion: UChicago/WVU showed **topological superconductivity in $\mathrm{FeTe_{1-x}Se_x}$ can be tuned in/out by composition** (Nature Communications, 2025/26); 2D magnetic/topological heterostructures are engineered for ultralow-power spintronics; topological phase transitions were used to raise thermoelectric performance in Cr-doped PbSe.

KEY GRADED CLAIMS

- T1 Topological insulators and cuprate high-Tc superconductivity are established, Nobel-lineage physics — Kane–Mele / Bednorz–Müller (established)
- T2 Magic-angle twisted bilayer graphene superconducts at $\sim 1.1°$ — Cao et al., Nature 556 (2018) (established)
- T2 REBCO high-Tc tape deployed in fusion magnets (SPARC) — Commonwealth Fusion / MIT, 2021+ (demonstrated, commercial)
- T2/T3 Composition-tuned topological superconductivity in FeTeSe — Nat. Commun. 2025/26 (demonstrated)
- T6 Room-temperature ambient-pressure superconductor — none verified; **LK-99 (2023)** and the **Ranga Dias retractions** are the cautionary tales (contested/refuted)

CONFLICTS / OPEN QUESTIONS

- **C-hightc-mechanism**: after ~40 years there is still no accepted microscopic theory of cuprate superconductivity — the field's standing embarrassment and its biggest open prize. Without it, high-Tc discovery stays empirical.
- **Can topological protection be shown cleanly in a device?** The Majorana debate (`H-topo`) remains contested after multiple retracted/withdrawn zero-bias-peak claims; a decisive, reproducible topological-qubit demonstration would resolve it.
- **Room-temperature superconductivity** is the recurring hype magnet — every few years a claim (LK-99, Dias's hydrides) collapses under replication. Grade any new one T6-contested until independent labs confirm.

THE HONEST CALL

Commercial at the boring end, research at the exciting end. Low-Tc and now high-Tc REBCO are real industrial materials doing real work (magnets, qubits, fusion). Topological materials — the ones that would rewrite computing — remain lab physics with a credibility overhang from retractions. The valuable discipline this node enforces: separate "superconductor that ships" from "topological phase we hope powers a qubit," and treat every room-temperature-superconductor headline as guilty until replicated.

SOURCES

- https://phys.org/news/2026-02-tuning-topological-superconductors-adjusting-ratio.html
- https://arxiv.org/pdf/2102.02644 (2021 Quantum Materials Roadmap)
- Cao et al., "Unconventional superconductivity in magic-angle graphene superlattices," Nature 556 (2018)
- Commonwealth Fusion Systems / MIT PSFC SPARC magnet reports (REBCO $20\ \mathrm{T}$ magnet)

NV-diamond sensing · A-nvsensing

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

The nitrogen-vacancy (NV) center — a nitrogen atom next to a missing carbon in diamond — is an atom-sized quantum sensor whose spin can be initialized, manipulated, and read out optically at **room temperature and ambient pressure**. That combination (no cryostat, nanoscale spatial resolution, vector sensitivity) makes NV-diamond its own sensing platform, spanning magnetometry, electrometry, thermometry, and pressure. This node consolidates NV *sensing* as a discipline; it deliberately cross-links `A-magneto` (where NV competes with SQUID/OPM), `H-nv` (NV as a compute/qubit modality), and `A-sensing` (the broader quantum-sensing market). Its niche is where you need a sensor *at* the sample — scanning-probe imaging, single-cell biology, current mapping inside a chip — not just the highest raw sensitivity.

MATURITY & REAL DEPLOYMENTS (2025–26)

- **Element Six** (De Beers) supplies the quantum-grade CVD diamond (DNV series) underneath most NV sensors. **Bosch + Element Six** formed a JV, "Bosch Quantum Sensing," in 2025 to scale NV sensors toward mass-market (mobility, medical); Element Six holds ~25%.
- **SBQuantum** (Canada) — field-proven diamond vector magnetometers for defense, navigation, and mineral exploration; a handheld unit at ~400 pT/√Hz vector sensitivity, heading errors <5 nT; flying an NGA MagQuest payload with Spire Global to map Earth's field for GPS-free navigation.
- **QDTI** (Quantum Diamond Technologies, Harvard spin-out) — NV "magnetic microscope" for biomarker diagnostics: <1 pg/mL detection from a 5 µL sample in <1 hour.
- Academic: commercial scanning-NV magnetometers in closed-cycle cryostats; fully integrated nanotesla-sensitivity NV chips (arXiv 2508.03237); a 2025 Quantum Diamond Workshop findings report (arXiv 2511.11791) marks the field's coalescence.

KEY GRADED CLAIMS

- T2 NV magnetometry operates at room temperature with nanoscale resolution — established NV literature / NIST (established)
- T4 SBQuantum handheld ~400 pT/√Hz vector sensitivity, <5 nT heading error — company figures (claimed)
- T4 QDTI <1 pg/mL biomarker detection from 5 µL in <1 hr — company claim (claimed)
- T3 Bosch×Element Six JV to industrialize NV sensing (2025) — press + corporate filings (roadmap)

CONFLICTS / OPEN QUESTIONS

NV's ambient operation and spatial resolution are unmatched, but its raw magnetic sensitivity still lags SQUID (~1–5 fT/√Hz) and OPM (~7–15 fT/√Hz) by 2–3 orders — the best NV ensembles reach ~9.4 pT/√Hz for MEG-scale work (see `A-magneto`). So its commercial fit vs the `A-magneto` incumbents is unsettled: NV wins wherever you must put the sensor *at* the sample (current mapping inside a live chip, single-cell biology, high-pressure diamond-anvil metrology) and loses on any application that just wants the lowest field noise in a shielded room.

THE HONEST CALL

A real, shipping specialist instrument — with a mass-market bet still unproven. NV magnetometers and NV "magnetic microscopes" sell today into defense navigation, materials/failure analysis, and biomarker diagnostics; Element Six supplies the diamond and independent groups reproduce the room-temperature physics. What is *claimed* rather than established is the Bosch×Element Six thesis that NV goes to automotive/consumer volume — that is a roadmap, not a deployment. Grade NV as commercial-in-niches, mass-market-aspirational.

SOURCES

e6.com/about/news (Bosch JV; SBQuantum MagQuest); thequantuminsider.com 2025/04/07 (Bosch×Element Six); sbquantum.com; spacenews.com (SBQuantum + Spire); qdti.com/instrument; arXiv:2508.03237; arXiv:2511.11791 (2025 Quantum Diamond Workshop); nist.gov/noac (NV magnetometry).

Assured PNT / GPS-denied navigation · A-pnt

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Assured Positioning, Navigation, and Timing is the *system-level* problem of knowing where and when you are without GPS/GNSS — because GNSS is jammable, spoofable, and unavailable underwater, underground, indoors, and in contested airspace. This node exists because the quantum answer to that problem is not one sensor but a **fusion**: a drift-free timing source (optical/atomic clock, `A-clocks`), a drift-free inertial reference (cold-atom accelerometers/gyros, `A-gravimetry`), and a map-matching field reference (quantum magnetometry, `A-magneto`; NV vector magnetometry, `A-nvsensing`). Previously this was smeared across the gravimetry, magnetometry, and clock cards; A-pnt is the integration layer where they combine into a navigation solution. The core physics payoff: a quantum clock plus a quantum IMU lets you *dead-reckon* accurately for hours, and a magnetic/gravity map lets you *fix* your absolute position against Earth's own signatures — neither of which an adversary can jam.

MATURITY & REAL DEPLOYMENTS (2025–26)

Advanced R&D nearing deployment — well-funded, defense-led, not yet fielded at fleet scale. - **Fusion navigators**: **Q-CTRL's Ironstone Opal** is the most public — magnetic + gravity map-matching with ML, claiming up to **111× better GPS-denied accuracy** than the best conventional alternative, **~4 m over 700 km** airborne, and **144+ hours** unattended at sea (2025–26); TIME Best Inventions 2025. **SandboxAQ's AQNav** does AI + magnetic-anomaly navigation and has flown on multiple aircraft types. - **Quantum IMUs**: **AOSense** cold-atom inertial units target **~5 m/hour** drift unaided (the DARPA PINS lineage) — vs a conventional tactical IMU that can drift kilometers in the same time. - **Timing anchor**: **Infleqtion's Tiqker** optical clock deployed on the **Royal Navy XV Excalibur** uncrewed submarine (Oct 2025) — a holdover timing source for a platform that cannot surface for GPS. - **Programs**: DARPA **Robust Quantum Sensors (RoQS)** (ruggedization; $24.4M to Q-CTRL), DARPA **PINS** (precision inertial nav), the DIU, and a March 2025 **Lockheed Martin + Q-CTRL** prototype quantum-INS contract. A 2036 PNT-stack forecast keeps clocks dominant (~76%) with quantum IMUs reaching ~18% as they progress to platform integration (helicopter, maritime, space).

KEY GRADED CLAIMS

- T4 Ironstone Opal: up to 111× better GPS-denied accuracy, ~4 m over 700 km, 144+ h at sea — Q-CTRL field trials, 2025 (claimed)
- T3 Cold-atom IMU ~5 m/hour unaided navigation drift — AOSense / DARPA PINS (demonstrated, program-level)
- T3 Optical clock (Tiqker) deployed as holdover timing on an uncrewed submarine — Infleqtion / Royal Navy, Oct 2025 (demonstrated)
- T5 Quantum PNT market shifts toward clocks-dominant + ~18% quantum-IMU stack by 2036 — analyst forecast (forecast)

CONFLICTS / OPEN QUESTIONS

- **No independent benchmark against black-program INS.** Every headline number is a vendor field trial against a specified (often unnamed) baseline; classified military ring-laser/FOG INS performance is the real comparator and it is not public.
- **Fusion is the hard part**: individual quantum sensors work, but robustly combining a jittery atom-interferometer IMU, a map-matching magnetometer (which needs a good, current, classified magnetic map), and a clock into one resilient solution on a maneuvering platform is a systems problem, not a physics one.
- **SWaP + dynamics**: launched atom clouds dislike vibration and hard turns; shrinking the whole stack to fit a drone or missile, not just a ship, is unsolved.

THE HONEST CALL

The most strategically important integration story in quantum sensing, and the reason the defense money is flowing — but it is prototyping, not deployment. The pieces (clock, IMU, magnetometer) are individually real; the fused, ruggedized, independently-benchmarked GPS-denied navigator on a fleet of moving platforms does not yet exist in the open literature. Grade A-pnt as "advanced R&D nearing first operational deployments," with the flashiest accuracy multipliers unverified.

SOURCES

- https://spaceinsider.tech/2026/04/10/quantum-sensing-for-pnt-nears-deployment/ (Tiqker on XV Excalibur; market split)
- https://q-ctrl.com/blog/2025-year-in-review-realizing-true-commercial-quantum-advantage-in-the-international-year-of-quantum (Ironstone Opal)
- https://www.darpa.mil/news/features/quantum-sensing-computing (RoQS / PINS)
- https://link.springer.com/article/10.1007/s10291-026-02030-y (quantum sensors for PNT, review)

Post-Quantum Cryptography · A-pqc

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Classical cryptography built on problems believed hard even for quantum computers (structured lattices, hash functions, error-correcting codes, isogenies), replacing the RSA and elliptic-curve schemes that Shor's algorithm (`S-shor`) breaks in polynomial time. The driver is **"harvest now, decrypt later" (HNDL)**: adversaries record encrypted traffic today to decrypt once a cryptographically-relevant quantum computer (CRQC) exists — so any secret with a shelf life past the CRQC date (state secrets, health records, genomic data, long-term IP) is *already* at risk, regardless of when the machine actually arrives. PQC runs on today's classical hardware; it is the pragmatic answer to the quantum threat, and the one all four Five-Eyes-adjacent signals agencies endorse over QKD (`A-qkd`).

MATURITY & REAL DEPLOYMENTS (2025–26)

Commercial and deploying at internet scale now — the most consequential quantum-adjacent technology of the decade. - **Standards**: NIST finalized **FIPS 203 ML-KEM** (Kyber), **FIPS 204 ML-DSA** (Dilithium), and **FIPS 205 SLH-DSA** (SPHINCS+) in **August 2024**. **HQC** (code-based) was selected March 2025 as a backup KEM on a different math assumption, finalization ~2027. - **Live deployment**: hybrid **X25519 + ML-KEM-768** key exchange is default in **Chrome, Firefox, Cloudflare** and major CDNs — a large fraction of TLS 1.3 handshakes on the open web are already post-quantum. **Signal (PQXDH)** and **Apple iMessage (PQ3)** shipped PQC messaging; PQ3 does continuous rekeying for forward secrecy against HNDL. - **Blockchain adoption**: Algorand executed quantum-resistant **Falcon**-signed transactions and enabled Falcon-based accounts on mainnet (Nov 2025) — an early production PQC signature in public-ledger infrastructure. - **Mandated deadlines**: NSA **CNSA 2.0** requires new national-security-system acquisitions to be PQC-capable by **Jan 1 2027**, full NSS migration by **2030–2035**; NIST IR 8547 deprecates quantum-vulnerable algorithms by **2030**, disallows after **2035**.

KEY GRADED CLAIMS

- T2 ML-KEM/ML-DSA/SLH-DSA standardized after multi-year public cryptanalysis — NIST FIPS 203/204/205, Aug 2024 (established)
- T2 Hybrid PQC key exchange live across major browsers/CDNs at internet scale — Cloudflare/Google deployment telemetry (demonstrated)
- T2 Falcon-signed accounts live on Algorand mainnet — Nov 2025 (demonstrated)
- T1/T2 Shor breaks RSA/ECC given an FTQC — Shor 1994 (established); the *date* such a machine exists is T6
- T5 HNDL is economically rational for state actors today — CSA / arXiv analyses (claimed)

CONFLICTS / OPEN QUESTIONS

- **C-pqc-confidence**: lattice hardness has *no* unconditional security proof — a classical or quantum cryptanalytic surprise (cf. **SIKE's 2022 one-afternoon collapse** on a classical laptop) would force re-migration. Hence HQC as a math-diverse hedge, and **hybrid modes** (classical + PQC together) as the default so a break in either half is not catastrophic.
- **Migration inventory is the real bottleneck**: the algorithms are ready, but most organizations cannot enumerate where cryptography lives in their stacks (embedded keys, protocols, certificates, HSMs), so "crypto-agility" and discovery tooling — not new math — is the gating work.

THE HONEST CALL

The single most real, most deployed, most economically-important item in the whole adjacent-tech chapter — and it isn't even quantum hardware. PQC is classical software, shipping now, on a legally-mandated clock. If any node in §04 pays for itself this decade, it is this one. The open risk is cryptanalytic (an assumption breaks), not deployment.

SOURCES

- https://csrc.nist.gov/projects/post-quantum-cryptography
- https://pages.nist.gov/nccoe-migration-post-quantum-cryptography/
- https://algorand.co/technology/post-quantum (Falcon on mainnet, 2025)
- NSA CNSA 2.0 FAQ / timeline; NIST IR 8547

Quantum-secured & quantum-resistant blockchain · A-qblockchain

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Blockchains are unusually exposed to quantum attack: their security rests almost entirely on the elliptic-curve signatures (ECDSA/EdDSA) that Shor's algorithm (`S-shor`) breaks, and every public key ever exposed on-chain is *permanently* harvestable (`A-pqc`'s HNDL threat, but worse — the ciphertext-equivalent is the immutable ledger itself). Two responses, often conflated: 1. **Quantum-resistant blockchain** — swap ECDSA for a PQC signature (lattice-based Dilithium/ML-DSA, or hash/NTRU-based **Falcon**, or stateful hash schemes). This is classical software; the dominant, practical track. 2. **Quantum-secured blockchain** — use QKD (`A-qkd`) or QRNG (`A-qrng`) to protect the *network* links or seed keys. Mostly experimental and niche.

MATURITY & REAL DEPLOYMENTS (2025–26)

Quantum-resistant is deploying now; quantum-secured is a research sideshow. - **Algorand** executed quantum-resistant **Falcon**-signed transactions and, in **November 2025**, enabled **Falcon-based accounts on mainnet** — an early production PQC signature scheme on a mainstream public ledger (via state proofs already using Falcon since 2022). - **QRL (Quantum Resistant Ledger)** has run a hash-based-signature (XMSS) chain since 2018 — a purpose-built PQC blockchain. - **Bitcoin/Ethereum** face the hardest migration: proposals exist (e.g. quantum-resistant address types, a "P2QRH" BIP for Bitcoin; account-abstraction paths for Ethereum) but require contentious hard/soft forks and a plan for the millions of coins in exposed-public-key addresses (including dormant early-Bitcoin wallets that cannot be moved by their owners). Research consolidations (Hyperledger Fabric + Kyber/Dilithium/Falcon; SoK arXiv:2512.13333) map the design space. - **Threat timing**: consensus estimates put a CRQC able to break ECC at **~5–15 years** out, which is *inside* the lifetime of long-lived on-chain keys — the reason migration is being taken seriously now.

KEY GRADED CLAIMS

- T2 Falcon-signed accounts live on Algorand mainnet (Nov 2025); QRL XMSS chain live since 2018 — project releases (demonstrated)
- T2 PQC signatures (ML-DSA/Falcon/SLH-DSA) integratable into blockchains without new math — NIST FIPS 204/205; Hyperledger integrations (established)
- T5 A CRQC threatening ECDSA is ~5–15 years out — analyst/CSA consensus (claimed/forecast)
- T4 QKD/QRNG "quantum-secured" ledgers as a needed product — vendor/academic pilots (claimed, niche)

CONFLICTS / OPEN QUESTIONS

- **Signature bloat**: PQC signatures are large (Dilithium ~2.4 kB, Falcon ~0.7 kB, SPHINCS+ ~8–50 kB) vs ECDSA's ~64 bytes — inflating block size, fees, and validation cost. This is the real engineering tax, and it is why chains pick Falcon (small but complex to implement) over Dilithium where they can.
- **The unmovable-coins problem**: exposed public keys on existing UTXOs (and lost-key wallets) can't be retroactively protected — a fork can require new PQC addresses going forward but cannot rescue the old exposed ones from a future quantum attacker.
- **Quantum-secured (QKD) blockchain** inherits all of QKD's limits (distance, trusted relays, `C-qkd-vs-pqc`) and adds little a PQC signature doesn't — mostly a solution in search of a problem.

THE HONEST CALL

Quantum-resistant blockchain is real, live, and mostly a PQC-migration story (so it's `A-pqc` applied to ledgers) — the interesting nuance is signature-size economics and the unmovable-coins overhang. "Quantum-secured" (QKD-based) blockchain is a research curiosity with weak motivation. The genuine near-term risk is not exotic: it's that Bitcoin/Ethereum are slow to fork and hold trillions in ECDSA-exposed value on a ~5–15-year clock.

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- https://www.chainalysis.com/blog/quantum-computing-crypto-security/ (threat timing)

Quantum-dot displays (boundary case) · A-qdisplay

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node — boundary case)

WHAT IT IS

A deliberately-included **boundary case**, here to police the word "quantum" in the largest-revenue product that carries it. A **quantum dot** is a semiconductor nanocrystal (~2–10 nm) whose emission wavelength is set by quantum confinement — make the dot smaller and the bandgap widens, so size *tunes color* with near-perfect purity. This is real quantum mechanics (particle-in-a-box confinement). But whether a QD *display* belongs in a quantum-technology atlas depends entirely on which of three things you mean: 1. **Photoluminescent QLED (today's "QLED" TVs)** — QDs sit in a film and simply down-convert blue LED backlight into pure red/green. Confinement sets the color, but the device is an ordinary LCD. **Quantum physics, not quantum *technology*** in the atlas's sense — no superposition, entanglement, or single-quantum control is exploited. 2. **Electro-luminescent QD (EL-QD / "NanoLED" / true QLED)** — QDs that conduct electricity and emit their own light, no backlight, no OLED. A real display breakthrough, still QD-as-a-material. 3. **QD single-photon / entangled-photon sources** — the one that *is* quantum technology: individual dots emit one photon at a time on demand (Quandela, `H-photonsource`), a building block for photonic QC and QKD. That belongs to the hardware chapter, cross-linked here.

MATURITY & REAL DEPLOYMENTS (2025–26)

- **Photoluminescent QLED / QD-OLED**: fully commercial, mass-market — Samsung, TCL, Sony ship tens of millions of QD TVs; **Nanosys** (acquired by Shoei Chemical) is the dominant QD-material supplier. This is a multi-billion-dollar consumer market and the reason "quantum" is a household display word.
- **Electroluminescent NanoLED / EL-QD**: prototype stage. Samsung Display showed brighter EL-QD prototypes (2025–26); Nanosys/Shoei repeatedly pushed the "ready ~2025–26" target — and, as of 2026, slipped it to **~2029**. Not yet a product.
- **QD single-photon sources**: research/early-commercial for quantum-photonics labs (Quandela's Prometheus), covered under `H-photonsource`.

KEY GRADED CLAIMS

- T1 Quantum confinement sets QD emission color (size-tunable bandgap) — established solid-state physics (established)
- T2 Photoluminescent QD displays are a mass-market commercial product — Samsung/TCL/Nanosys shipments (established, commercial)
- T4/T6 Electroluminescent EL-QD ("true QLED") displays for consumers ~2029 — Samsung/Nanosys roadmap, repeatedly slipped (roadmap)
- T2 On-demand single-photon emission from individual quantum dots — established QD-photonics literature (established; see `H-photonsource`)

CONFLICTS / OPEN QUESTIONS

- **The definitional one, which is the point of this card**: is a QLED TV "quantum technology"? By the atlas's standard (exploiting a controlled quantum state — superposition/entanglement/single-quantum readout), **no** for photoluminescent QLED — it uses a quantum *material* the way a laser or an LED does, not a quantum *information* resource. **Yes** for QD single-photon sources.
- EL-QD's engineering blockers (blue-dot lifetime/efficiency, cadmium-free formulations, electrical stability) keep slipping the timeline.

THE HONEST CALL

Mostly not a quantum technology in this atlas's sense — included to say so out loud. Today's QD displays are a huge, real consumer market built on a quantum *material*, and calling them "quantum tech" alongside qubits and QKD is a category error the marketing invites. The genuinely quantum-technology thread is the QD *single-photon source* (`H-photonsource`), not the TV. Grade the TV as commercial-but-boundary; grade the photon source in hardware.

SOURCES

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- https://www.flatpanelshd.com/news.php?subaction=showfull&id=1777964273 (Samsung EL-QD/NanoLED prototypes)
- https://en.wikipedia.org/wiki/Quantum_dot_display (PL vs EL QLED distinction)
- Cross-link: `H-photonsource` (QD single-photon sources — the actual quantum-tech thread)

Quantum Internet & Repeaters · A-qinternet

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

A network that distributes *entanglement* between distant nodes — enabling QKD without trusted relays, blind/distributed quantum computing, and networked quantum sensors (`A-sensing`, entanglement-based clock networks). Photon loss in fiber is exponential (~0.2 dB/km) and optical amplification is forbidden by no-cloning (`F-nocloning`), so long links need **quantum repeaters**: quantum memories (`A-qmemory-hw`) that store entanglement in short segments, then fuse segments by entanglement swapping (`F-teleport`) and purification. The figure of merit is delivered **entanglement rate vs distance**, and the field's watershed is the moment a repeater beats direct transmission at that.

MATURITY & REAL DEPLOYMENTS (2025–26)

Research — no repeater has yet beaten direct fiber in delivered entanglement rate. That crossover has not happened; everything below is a building block toward it. - **Metropolitan entanglement network**: QuTech (Delft) linked **Delft–The Hague over ~25 km of deployed fiber** with heralded entanglement between processor nodes (Stolk et al., Sci. Adv. 2024) — the first multi-node network on installed fiber. - **Long-distance memory entanglement**: a preprint reports entangling quantum memories over **420 km** of fiber (arXiv:2504.05660); USTC reported **memory–memory entanglement + device-independent QKD across 100 km** (Feb 2026); long-lived **remote ion–ion entanglement** suited to repeaters appeared in Nature (2026). ICFO Barcelona demonstrated **multiplexed** memory-based distribution (2025), attacking the rate problem. - **Programs & vendors**: Germany's **TD.QR** repeater project (started Jan 2026) targets viable repeater demos by **2028**. Startups sell early network gear — **Qunnect** (room-temperature memories, `A-qmemory-hw`) and **Aliro** (network orchestration/SDN). Testbeds run in Boston, Chicago (the Chicago Quantum Exchange), Amsterdam, and Hefei.

KEY GRADED CLAIMS

- T2 Multi-node heralded entanglement between two cities over ~25 km deployed fiber — Stolk et al./QuTech, Sci. Adv. 2024 (demonstrated)
- T3 Quantum memories entangled over 420 km fiber — arXiv:2504.05660 (claimed, preprint)
- T2/T3 Memory–memory entanglement + DI-QKD across 100 km — USTC, Feb 2026 (demonstrated; verify final journal ref)
- T6 Useful multi-node quantum internet by the early 2030s — various national roadmaps (roadmap)

CONFLICTS / OPEN QUESTIONS

- **Which memory platform wins** is unsettled: atomic ensembles (highly multiplexable, good rate) vs single ions/NV/atoms (processable, can do logic at the node) vs rare-earth crystals (long storage). The network market may fork by use case.
- **Repeater-vs-direct crossover date is unproven**, and some argue **satellite links** (`A-satqkd`) will reach continental scale before fiber repeaters do — making the "quantum internet" partly a space story.
- **The demand question underneath it all**: beyond relay-free QKD (itself contested vs PQC, see `C-qkd-vs-pqc`), the killer app for a quantum internet — distributed QC, blind computing, sensor networks — is real in principle but has no paying customer yet.

THE HONEST CALL

Genuinely research-stage, and honest about it. The milestones are real and accelerating (city links, 100–420 km memory entanglement, a repeater program aiming at 2028), but the defining achievement — a repeater that beats direct transmission — has not been reached, and the economic case rests on applications that don't yet have buyers. This is a decade-plus infrastructure bet, not a near-term product.

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- https://www.innovationnewsnetwork.com/german-quantum-repeater-project-advances-future-quantum-internet/68370/ (TD.QR)

Quantum Key Distribution · A-qkd

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

QKD lets two parties grow a shared secret key whose security rests on physics — the no-cloning theorem (`F-nocloning`) and measurement disturbance — rather than computational hardness. BB84 (Bennett–Brassard 1984) encodes bits in non-orthogonal photon states; E91 (Ekert 1991) derives keys from entangled pairs and certifies security via Bell violation. An eavesdropper unavoidably raises the quantum bit error rate (QBER) and is detected. The security proof is information-theoretic in the ideal model — but every real deployment inherits two hard constraints: fiber loss is exponential (≈ 0.16–$0.2\,\mathrm{dB/km}$) so unamplified reach caps near a few hundred km, and QKD only distributes *keys*, so it still needs classically-authenticated channels and symmetric encryption to do anything useful.

MATURITY & REAL DEPLOYMENTS (2025–26)

Commercial today, in niches — a two-decade-old product line with government and finance as the buyers. **Measured performance envelope:** at inter-city distances ($\sim 200\,\mathrm{km}$) twin-field QKD reaches $\mathbf{111.74}\,\mathrm{kbit/s}$ secure key rate; the same protocol has been pushed to $\mathbf{1002}\,\mathrm{km}$ of ultralow-loss fiber (USTC/Pan, PRL 2023) but at only $\sim \mathbf{0.0034}\,\mathrm{bit/s}$ — a headline distance, not a usable rate. Practical metro QKD (Toshiba, ID Quantique) runs tens-of-kbit/s to Mbit/s over 10–$100\,\mathrm{km}$. - **Toshiba** runs commercial metro QKD, launched a Paris service in 2025, and (with Quantum Corridor, Dec 2025) demonstrated cross-state QKD over live commercial fiber (Chicago→Indiana), including a $21.8\,\mathrm{km}$ production segment carrying ordinary traffic. - **ID Quantique** has sold QKD boxes since the early 2000s; anchors Geneva/Korea (SK Telecom) networks. - **China** operates the $\sim 2000\,\mathrm{km}$ Beijing–Shanghai backbone (trusted-relay), the largest fielded network, plus the space segment (`A-satqkd`). - **EU EuroQCI**: a continent-wide quantum-secure network is being built — Madrid's MadQCI testbed (IDQ, Toshiba, AIT), Hungary's first multi-node net with Magyar Telekom, national arms across most member states.

KEY GRADED CLAIMS

- T1 BB84/E91 offer information-theoretic key security in the ideal model — Bennett & Brassard 1984; Ekert PRL 1991 (established)
- T2 Real devices admit side-channel attacks (detector blinding, Trojan-horse); implementation ≠ ideal proof — Lydersen et al., Nat. Photonics 2010 (established)
- T2 Twin-field QKD over $1002\,\mathrm{km}$ fiber at $\sim 0.0034\,\mathrm{bit/s}$; $111.74\,\mathrm{kbit/s}$ at $202\,\mathrm{km}$ — Liu et al., PRL 130, 210801 (2023) (demonstrated)
- T2 Beijing–Shanghai $2000\,\mathrm{km}$ QKD backbone operational via trusted relays — Chen et al., Nature 589 (2021) (demonstrated)
- T4 Toshiba/Quantum Corridor cross-state QKD over live commercial metro fiber, $21.8\,\mathrm{km}$ segment — Toshiba PR, Dec 2025 (claimed)

CONFLICTS / OPEN QUESTIONS

- **C-qkd-vs-pqc**: NSA, UK NCSC, French ANSSI, and German BSI advise *against* QKD for national-security systems — citing cost, the trusted-relay hole, no protection against store-and-forward at relays, and the fact that authentication still needs classical (ideally PQC) crypto — and recommend PQC (`A-pqc`) instead. QKD vendors and China's program argue physics-based security is worth the cost for the highest-value links. Resolution: field success/failure of large deployments + whether PQC survives cryptanalysis over the next decade.
- Trusted relay nodes break end-to-end security — every backbone today depends on them until repeaters (`A-qinternet`) arrive. This is the single biggest gap between the marketing ("unhackable") and the deployed reality.

THE HONEST CALL

Commercial but structurally niche. QKD sells real boxes and secures real links today, but the addressable market is capped by three things it cannot escape without a quantum internet: distance (repeaterless ~a few hundred km), the trusted-relay compromise, and the four-agency government advice steering critical infrastructure toward PQC. It is a genuine product in a small, contested market — not the coming default for internet security.

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Quantum memories as a component industry · A-qmemory-hw

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

A quantum memory stores a photonic qubit and releases it on demand with its quantum state intact. It is the missing hardware component that turns point-to-point QKD into a real **quantum internet**: repeaters need memories to hold entanglement while neighboring links succeed, defeating fiber loss over distance (see `A-qinternet`, `F-teleport`). Historically memories were cryogenic lab curiosities. The shift being tracked here is memories becoming a **buyable component** — rack-mounted, room-temperature, sold to network builders — the same maturation single-photon detectors and sources went through. Platforms: warm atomic vapor (rubidium/cesium), rare-earth-doped crystals, cold-atom ensembles, and defect centers.

MATURITY & REAL DEPLOYMENTS (2025–26)

- **Qunnect** (Brooklyn, US) — the standout commercializer. Its rubidium-**vapor** approach runs at room temperature with no cryogenics or frequency conversion. Demonstrated entanglement between telecom-wavelength photons and a room-temperature memory at 90.2% fidelity, generating ~1,200 entangled photon-memory pairs/second. Sells the "GothamQ" testbed hardware; closed a $10M Series A extension (Airbus Ventures, Cisco Investments, Quantonation) in June 2025; ~$60M+ raised total.
- **Aliro Quantum** (US) — not a memory maker but the orchestration/SDN layer that manages multi-vendor quantum networks; its platform interoperates with Qunnect, Single Quantum, Cisco, IonQ, Thorlabs, Keysight, and others. Passed 50 supported network devices (Sept 2025).
- Academic memories (rare-earth crystals, cold atoms) still lead on storage time × efficiency but do not ship as products.

KEY GRADED CLAIMS

- T3 Room-temperature telecom-band photon↔memory entanglement at 90.2% fidelity, ~1,200 pairs/s — Qunnect, 2025 (demonstrated)
- T4 Qunnect memory hardware is commercially deployable outside the lab — company positioning (claimed)
- T2 Repeater-based long-distance entanglement distribution requires quantum memories — network theory (established)

CONFLICTS / OPEN QUESTIONS

Room-temperature vapor memories trade storage time and efficiency for practicality; cryogenic rare-earth-doped crystals win on the storage-time × efficiency product (the metric that actually sets repeater performance) but not deployability. Which wins the network market is unsettled. The deeper problem: a repeater needs memory *and* the delivered entanglement rate to beat direct transmission, and that crossover has not been reached anywhere (`A-qinternet`).

THE HONEST CALL

A component industry being built ahead of its own market. Qunnect's room-temperature memories are a real, demonstrated product with fidelity/rate numbers you can cite, and Aliro's orchestration layer treats quantum networks as a systems-integration problem — both are early-commercial. But the whole industry is only worth building if a quantum internet has a paying use case beyond QKD (itself contested vs PQC, `C-qkd-vs-pqc`). This is picks-and-shovels for a gold rush that hasn't been confirmed — well-funded, technically real, demand-unproven.

SOURCES

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Quantum-enhanced MRI / NMR & biomagnetism · A-qmri

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Two distinct quantum leverages on magnetic-resonance and biomagnetic imaging, grouped because both aim at the same clinical/biological target from different physics: 1. **Hyperpolarization** — force nuclear spins far from thermal equilibrium (parahydrogen-induced polarization/SABRE, dynamic nuclear polarization, or brute-force) so the MR signal jumps by **10,000–100,000×**, turning slow *metabolic* MRI (watching, e.g., pyruvate→lactate in a tumor in real time) from impossible to feasible. 2. **Quantum-sensor detection** — read magnetic-resonance signals with **NV-diamond** (`A-nvsensing`) or atomic magnetometers (`A-magneto`) instead of pickup coils, enabling **nano-/micro-scale NMR**, **zero-to-low-field NMR** (no giant superconducting magnet), and biomagnetism (MEG/MCG) — see the OPM-MEG story in `A-magneto`.

MATURITY & REAL DEPLOYMENTS (2025–26)

Hyperpolarization is entering the clinic; quantum-sensor NMR is lab/preclinical. - **NVision Quantum Technologies** (Germany) built the **POLARIS** parahydrogen hyperpolarizer, boosting sugar/metabolite MRI signal **>10,000×** (up to ~100,000× claimed), and in 2025 ran feasibility work with **Memorial Sloan Kettering** toward standardized high-throughput metabolic imaging; its earlier collaboration seeded a Cambridge quantum-MRI hub. Hyperpolarized ^{13}C-pyruvate MRI is already in human oncology/cardiology trials at multiple academic centers (the field predates NVision, but hyperpolarizer commercialization is the 2025 story). - **Diamond/NV NMR**: MCQST (Munich) combined optical microscopy with NV-detected NMR to push MRI toward the microscopic scale, converting MR signals to optical readout on a camera (2025). Zero-/low-field **J-spectroscopy with a diamond magnetometer** (arXiv:2512.05776) and NV nuclear-spin-locking for microscale high-field NMR (arXiv:2504.00887) demonstrate NMR in regimes conventional coils can't reach — no superconducting magnet, no large shield.

KEY GRADED CLAIMS

- T2 Parahydrogen/DNP hyperpolarization raises MR signal 10^4–10^5× enabling real-time metabolic MRI — established MR physics; hyperpolarized ^{13}C human trials (established/demonstrated)
- T3 NVision POLARIS hyperpolarizer feasibility for standardized metabolic imaging with MSK — company + 2025 collaboration (demonstrated, preclinical/translational)
- T3 NV-diamond micro-scale and zero-field NMR without superconducting magnets — MCQST 2025; arXiv:2512.05776, 2504.00887 (demonstrated, lab)
- T2 OPM/atomic-magnetometer biomagnetism (MEG/MCG) — see `A-magneto` (demonstrated, clinical trials)

CONFLICTS / OPEN QUESTIONS

- **Hyperpolarization's clock problem**: hyperpolarized states decay in seconds to a couple of minutes (T_1-limited), so the polarizer must sit beside the scanner and imaging must be fast — a workflow/logistics barrier, not a physics one, and the main thing standing between trials and routine clinical use.
- **NV/atomic NMR's sensitivity–volume tradeoff**: quantum sensors excel at *tiny* samples (nano/micro) but do not replace whole-body clinical MRI; their clinical path is benchtop diagnostics, point-of-care low-field NMR, and research microscopy, not the hospital 3 T magnet.

THE HONEST CALL

The most clinically plausible near-term quantum-biomedical story — but "quantum-enhanced MRI" bundles two very different maturities. Hyperpolarized metabolic MRI is in real human trials and hyperpolarizers are being commercialized (NVision) — this is close. Quantum-sensor (NV/atomic) NMR is exciting lab physics enabling magnet-free micro-NMR, years from clinical routine. Biomagnetism via OPMs is already clinical (`A-magneto`). Grade each strand separately and resist the umbrella hype.

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- https://arxiv.org/pdf/2504.00887 (NV nuclear-spin-locking, microscale high-field NMR)

Quantum Random Number Generation · A-qrng

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Randomness drawn from quantum measurement, where outcomes are unpredictable by physics (`F-measure`) rather than by algorithmic complexity. Three rungs of trust: 1. **Hardware QRNG** — trust the device; measure vacuum-field fluctuations, photon arrival times, or beam-splitter path. Fast and cheap. 2. **Device-independent (DI) randomness** — trust only physics; a Bell-inequality violation certifies genuine unpredictability even with untrusted hardware. Slow, needs loophole-free Bell setups. 3. **Certified randomness via random circuit sampling (RCS)** — a quantum computer produces bits a remote user can verify are fresh and random even while distrusting the server (Aaronson's protocol).

MATURITY & REAL DEPLOYMENTS (2025–26)

The oldest commercial quantum product line — and the fastest-moving on raw throughput. - **Hardware**: ID Quantique's **Quantis** chips have sold for two decades and shipped inside Samsung Galaxy "Quantum" phones; **Quside**, **Toshiba**, **Quantum Dice** sell QRNG hardware. Buyers are lotteries, casinos, HSMs, and key generation. - **Speed race**: a **photonic-integrated QRNG hit 18.8 Gbit/s** real-time output (USTC, 2021) — still a benchmark; **Quantum Dice's VERTEX** PCIe card does 2.66 Gbit/s post-processed; and a 2025 **Toshiba chip-scale QRNG** reached **~3 Gbit/s** while shrinking to a device that could embed in everyday hardware (Optica, 2025). Throughput and integration, not sensitivity, are the competitive axes. - **Certified randomness (the 2025 headline)**: JPMorgan Chase, Quantinuum, Argonne, Oak Ridge, and UT Austin demonstrated certified randomness on Quantinuum's **56-qubit H2** (Nature 640, 2025) — the first claimed commercial-relevant application of a quantum computer beyond classical capability, using Aaronson's RCS protocol with supercomputer verification at **Frontier**. **Quantinuum's Quantum Origin** became the first software QRNG to earn **NIST ESV validation** (April 2025), and a certified-randomness product line followed.

KEY GRADED CLAIMS

- T1 Measurement outcomes on superposed states are irreducibly random — quantum theory + loophole-free Bell tests 2015 (established)
- T2 Chip/photonic hardware QRNG at multi-Gbit/s (18.8 Gbit/s record; ~3 Gbit/s integrated, 2025) — USTC 2021; Toshiba/Optica 2025 (demonstrated)
- T2 Certified randomness expansion on a 56-qubit trapped-ion processor, verified at exascale — Liu et al., Nature 640 (2025) (demonstrated)
- T2 Quantum Origin first software QRNG with NIST ESV validation — Quantinuum/NIST, April 2025 (demonstrated)
- T4 Certified randomness as a commercial product with near-term customer demand — Quantinuum/JPMC (claimed)

CONFLICTS / OPEN QUESTIONS

- **Aaronson himself flags the certified-randomness demo's limits**: certification needed exascale classical verification (only a handful of machines on Earth can do it), and the guarantee is *computational* — it assumes the server can't out-compute Frontier at spoofing, so it weakens as classical simulation improves. It is a beautiful proof-of-principle, not yet a product anyone must buy.
- **Does anyone need it?** Most applications are fine with a $50 hardware chip or a well-seeded CSPRNG. The market question for both certified and device-independent RNG is whether the extra trust guarantee is worth the price — outside high-assurance government/finance niches, the answer today is usually no.

THE HONEST CALL

Hardware QRNG is a mature, real, commodity product (two decades of sales, in your phone). **Certified randomness is a landmark 2025 demonstration** — arguably the first genuinely-useful thing a beyond-classical quantum computer has done — **but not yet a business**; its guarantee is computational and its verification cost is exascale. The gap between "shipping chip" and "verifiable quantum-advantage service" is the whole story of this node.

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Rydberg RF electrometry / quantum antennas · A-rydberg

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

A **Rydberg atom** has an electron excited to a very high principal quantum number (n ~ 50–100), giving it a huge electric dipole moment — which makes it exquisitely sensitive to RF electric fields. Shine two lasers through a room-temperature alkali (rubidium/cesium) vapor cell to create **electromagnetically induced transparency (EIT)**; an incident RF field splits the EIT line (Autler–Townes splitting), and the split is read *optically*. The result is an **RF receiver whose sensing element is atoms, not metal** — self-calibrated (referenced to atomic constants and Planck's constant), physically tiny, and broadband from MHz to sub-THz in one cell. This is quantum sensing pointed at the electromagnetic spectrum instead of at magnetic or gravitational fields, and it is distinct enough from `A-magneto`/`A-nvsensing` to warrant its own node.

MATURITY & REAL DEPLOYMENTS (2025–26)

Early commercial for niche RF sensing; research for general communications. - **Rydberg Technologies** (US, Michigan spin-out) is the furthest along — SI-traceable Rydberg field sensors and a demonstrated over-the-air Rydberg RF communication receiver for defense; DARPA/Army-funded. - **Infleqtion** markets a **quantum spectrum sensing** ("SqyWire") product line using Rydberg atomic sensing to detect, locate, and classify RF signals across a broad band, positioned to replace conventional antenna+receiver front-ends for spectrum awareness. - **ESA** commissioned Rydberg sensor commercialization work beginning 2025 (satellite RF/comms). - **2025 research surge**: an all-optical Rydberg radio antenna (Oct 2025); a metamaterial GRIN lens boosting receiver gain/bandwidth (arXiv:2512.04298); a scalable Rydberg vapor-cell array with a "Stark comb" for arbitrary instantaneous bandwidth (arXiv:2509.26026); quantum-limited microwave electrometry in a Rydberg **atom array** (arXiv:2512.05413); and demonstrations of angle-of-arrival, digital beamforming, and moving-target detection with atomic receivers.

KEY GRADED CLAIMS

- T2 Rydberg EIT gives SI-traceable, self-calibrated RF electric-field measurement — established atomic-physics literature (established)
- T3 Over-the-air Rydberg atomic communication receiver demonstrated — Rydberg Technologies / arXiv (demonstrated)
- T3 Rydberg receiver angle-of-arrival, beamforming, moving-target detection — 2025 arXiv results (demonstrated, lab/field)
- T4 Infleqtion quantum spectrum sensing as a deployable product replacing antenna front-ends — company positioning (claimed)

CONFLICTS / OPEN QUESTIONS

- **Sensitivity vs the best classical receiver**: standard Rydberg receivers have historically *not* beaten a good cryogenic low-noise-amplifier classical receiver on raw sensitivity; the advantage is self-calibration, bandwidth coverage in one device, small size, and immunity to some jamming/damage (no metal to fry). Whether quantum-enhanced (atom-array, spin-squeezed) Rydberg sensing decisively beats classical on sensitivity is the open research question.
- **Instantaneous bandwidth** and dynamic range have been real limits; the 2025 array/Stark-comb work is aimed squarely at them but is not yet productized.

THE HONEST CALL

Genuinely novel, early-commercial for specialized RF/spectrum sensing — not a general antenna replacement yet. Rydberg receivers ship into defense spectrum-awareness and metrology (self-calibrated field standards), which is a real if narrow market. The bigger claims — replacing communications antennas, beating classical receivers on sensitivity — are research with fast 2025 momentum. Grade as "real product in a niche, contested advantage in the mainstream RF use case."

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- https://infleqtion.com/quantum-spectrum/ (Infleqtion quantum spectrum sensing)
- https://arxiv.org/pdf/2507.13111 (Perspective: Practical Atom-Based Quantum Sensors)
- https://arxiv.org/pdf/2512.05413 (quantum-limited Rydberg electrometry in an atom array)

Satellite QKD · A-satqkd

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Distributing quantum keys (and entanglement) via satellite, where photon loss through vacuum beats fiber's exponential attenuation over continental distances. A satellite either acts as a **trusted relay** between two ground stations (it holds both keys and XORs them — secure only if you trust the spacecraft) or **beams entangled pairs** to two stations at once (end-to-end secure, no trust in the satellite, but far lower rate). This is the near-term route to intercontinental quantum-secure links while fiber repeaters (`A-qinternet`) mature.

MATURITY & REAL DEPLOYMENTS (2025–26)

Demonstrated at scale by China; entering an early commercial phase globally. - **Micius** (launched 2016, USTC/CAS) is the flight-heritage benchmark: entanglement distribution over 1,200 km (Yin et al., Science 2017), decoy-state QKD downlink, and the 2017 Beijing–Vienna intercontinental video call secured by satellite-relayed keys. It is a ~600 kg spacecraft needing large, expensive ground stations. - **Jinan-1** (2022, operational) is the practicality milestone. It shrank the payload to **~22.7 kg** and the satellite to **~95.9 kg**, cutting cost roughly **45×** while raising key rates **2–3 orders of magnitude** over Micius. Using an 850 nm source at 625 MHz rep rate (~250 million photons/s transmitted), it generated **up to 1.07 million secure bits in a single pass** and ran **real-time** QKD with portable (~100 kg) ground stations, enabling one-time-pad-encrypted image transfer between China and South Africa over 12,900 km (Nature, March 2025). Real-time on-board key handling — not post-pass processing — is the operational jump. - **Europe**: EAGLE-1 (ESA/SES) is Europe's first end-to-end satellite QKD system, launch slated 2026, feeding EuroQCI's space segment. **Toshiba** announced a satellite QKD transmitter–receiver interoperable with terrestrial fiber QKD (Jan 2026). **SealSQ** began launching a six-satellite QKD/PQC constellation (Jan 2025).

KEY GRADED CLAIMS

- T2 Satellite entanglement distribution over 1,200 km — Yin et al., Science 356 (2017) (established)
- T2 Real-time microsatellite QKD, up to 1.07 Mbit secure key/pass, 12,900 km China–South Africa link, ~45× cheaper spacecraft — Li et al., Nature (2025) (demonstrated)
- T4 EAGLE-1 launch and end-to-end European satellite QKD service by ~2026–27 — ESA/SES, arXiv:2505.20838 (roadmap)
- T4/T6 Chinese multi-satellite quantum constellation / space quantum internet by ~2030 — CAS/USTC statements (roadmap)

CONFLICTS / OPEN QUESTIONS

- **Trust model**: downlink QKD via a single satellite is trusted-node — the spacecraft (and whoever controls it) knows the key — unless it distributes *entanglement* end-to-end, which costs orders of magnitude in rate. Most fielded/announced systems are the trusted-relay kind.
- **Operations**: daytime operation (sky background swamps single photons), weather/cloud outage, and pointing over fast LEO passes limit availability. Whether the key-rate-per-dollar beats just deploying PQC (`A-pqc`) on the same links is the live commercial question.

THE HONEST CALL

Demonstrated and near-commercial, led decisively by China. Jinan-1 turned satellite QKD from a hero experiment into something that looks like a deployable service, and Europe is a launch or two behind. But it inherits QKD's core limitation — it distributes keys, still needs classical authentication, and the cheap version trusts the satellite — so it is a strategic-sovereignty capability for states and critical infrastructure, not a mass-market technology.

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Quantum Sensing & Metrology · A-sensing

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Using quantum systems — atomic transitions, spin states, entanglement, squeezing — as measurement instruments. The same fragility that plagues qubits (`F-decoher`) becomes the product: a state that decoheres at the slightest field perturbation is an exquisite detector of that field. This card is the **overview node** for the sensing cluster: clocks (`A-clocks`), magnetometry (`A-magneto`), gravimetry/inertial (`A-gravimetry`), imaging (`A-imaging`), NV-diamond (`A-nvsensing`), Rydberg RF (`A-rydberg`), and the assured-PNT integration (`A-pnt`). The unifying physics: interferometric phase readout at or below the standard quantum limit, where signal scales with coherence time and entanglement can push sensitivity toward the Heisenberg limit ($\propto 1/N$ rather than $1/\sqrt{N}$).

MATURITY & REAL DEPLOYMENTS (2025–26)

The nearest-term commercial quantum technology — revenue exists today, unlike compute. Chip-scale atomic clocks have shipped for over a decade; LIGO has run squeezed light in production since 2019 (`A-squeezed`); OPM magnetometers sell into neuroimaging (`A-magneto`). The 2024–26 wave is **quantum navigation** for the GPS-denied problem: - **Q-CTRL's Ironstone Opal** (magnetic/gravity map-matching + AI) ran air, land, and maritime defense trials, claiming up to **~100–111× better GPS-denied accuracy** than the best conventional INS and **~4 m positioning over 700 km flights**; a maritime trial logged **144+ hours** unattended at sea (2025–26). It made TIME's Best Inventions 2025. - **SandboxAQ's AQNav** does AI + magnetic-anomaly navigation; both companies hold US DoD contracts (DARPA RoQS — $24.4M to Q-CTRL; Lockheed Martin/Q-CTRL prototype quantum-INS award, March 2025). - **Medical**: OPM-MEG helmets in clinical trials, SandboxAQ/Mayo CardiAQ cardiac sensing (`A-magneto`).

Market: analysts put quantum sensors at roughly **$0.4–0.9B in 2025** growing **~15–23% CAGR** to ~$1.5B by the mid-2030s (Fortune Business Insights and peers). Small but real and mostly defense-led — a 2036 forecast has clocks still dominant (~76% of the PNT-sensing stack) with quantum IMUs reaching ~18%.

KEY GRADED CLAIMS

- T1 Quantum-limited metrology (squeezing, entanglement-enhanced sensitivity) beats classical shot-noise limits — LIGO squeezed-light operation, PRL 2019+ (established)
- T4 Ironstone Opal delivers ~4 m accuracy over 700 km flights, up to 111× better than best conventional alternative — Q-CTRL announcements 2025 (claimed, awaiting independent verification)
- T5 Quantum sensors market ~$435M (2025) → ~$1.5B (2034), ~15% CAGR — Fortune Business Insights (forecast)

CONFLICTS / OPEN QUESTIONS

- **Vendor navigation numbers are self-reported field trials.** No peer-reviewed head-to-head against classified military-grade ring-laser/fiber-optic INS is public — the "111×" is against a specified baseline, not the state of the art in a black program.
- **The label problem**: where is the line between "quantum sensor" and mature atomic physics (MRI, cesium clocks, SQUIDs all predate the "quantum tech" branding)? Marketing stretches "quantum" to anything using atoms — grade accordingly, and see the per-modality cards for the honest split.

THE HONEST CALL

The revenue-bearing corner of quantum technology, and the one most likely to matter this decade — but the fielded wins are concentrated in defense PNT and neuroimaging, and the flashiest navigation claims still lack independent benchmarks. Treat sensing as "real and shipping in niches," with the per-modality cards carrying the exact numbers and the honest maturity call.

SOURCES

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Squeezed light as a cross-cutting resource · A-squeezed

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Squeezed light is a quantum state of an optical field where the uncertainty in one quadrature is pushed below the vacuum (shot-noise) limit, at the cost of more uncertainty in the conjugate one — Heisenberg's bound rearranged, not beaten (see `F-uncertainty`). It is a genuine cross-cutting quantum resource: it sharpens metrology below the standard quantum limit, and it is the *substrate* for continuous-variable photonic quantum computing. Squeezing strength is quoted in decibels below shot noise; more dB means a better resource, bounded in practice by optical loss.

MATURITY & REAL DEPLOYMENTS (2025–26)

- **LIGO / Virgo — the flagship production use.** Squeezed vacuum is injected into the interferometer dark port to cut quantum shot noise every observing run. In O3, Hanford measured ~2.0 dB and Livingston ~2.7 dB of shot-noise reduction, raising the detection rate ~40–50%. O4 uses **frequency-dependent squeezing** (a filter cavity) to squeeze the right quadrature across the band — the first at-scale deployment of a technique that reduces both shot noise and radiation-pressure noise. This is the clearest case of a quantum resource doing paid work daily.
- **Xanadu — squeezing as compute.** Its photonic machines encode GKP bosonic qubits in squeezed states via optical parametric oscillators; the Aurora system (2025) networks 35 photonic chips over 13 km of fiber, and Xanadu demonstrated on-chip GKP-state generation on 300 mm silicon-nitride wafers (June 2025).
- **Integrated squeezing** is advancing fast: 18 dB squeezing / 20 dB anti-squeezing at 1570 nm demonstrated in a thin-film lithium-niobate waveguide — the highest for any integrated photonic platform.

KEY GRADED CLAIMS

- T2 LIGO uses squeezed light in production to boost sensitivity ~40–50% via shot-noise reduction — Phys. Rev. X 13, 041021 (established/demonstrated)
- T3 18 dB on-chip squeezing in TFLN waveguide — arXiv 2025 (demonstrated)
- T3 On-chip GKP states from squeezed light on 300 mm wafers — Xanadu, June 2025 (demonstrated)

CONFLICTS / OPEN QUESTIONS

Squeezing is loss-limited: every lost photon degrades it toward vacuum, so end-to-end optical efficiency caps usable dB — the same wall that gates continuous-variable photonic QC. The gap between the ~18–20 dB generated on-chip and the ~2–3 dB LIGO actually uses in production is *all loss* (mirrors, mode-matching, the km-scale interferometer). Whether squeezing-based (CV/GKP) architectures reach fault tolerance before discrete-photon ones is open (`H-photonic`, `O-roomtemp`).

THE HONEST CALL

The clearest case in the whole manual of a quantum resource doing paid work every day — LIGO/Virgo inject squeezed vacuum on every observing run and it demonstrably raises the detection rate ~40–50%, which is why more gravitational-wave events get caught. That is established, deployed, and reproduced. The compute use (CV/GKP photonic QC, Xanadu) is real hardware progress but still research: on-chip GKP generation and a 35-chip networked machine are demonstrations, not fault tolerance. So: sensing use = production; compute use = frontier.

SOURCES

Phys. Rev. X 13, 041021 (frequency-dependent squeezing at LIGO); arXiv:2202.00847 (Advanced LIGO O4 review); arXiv:2603.02744 (12–18 dB waveguide squeezing); thequantuminsider.com 2025/06/05 (Xanadu Aurora / on-chip GKP).

Quantum/optical time-transfer networks · A-timedist

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Distinct from the clocks themselves (`A-clocks`): this node is the **network problem** of comparing and distributing ultra-stable time and frequency between distant clocks without degrading their accuracy. Optical clocks are now so good (~19 decimal places) that ordinary GPS/satellite time links are the limiting error, not the clocks. So a parallel infrastructure is being built — stabilized optical fiber links, free-space laser links, and frequency-comb transfer — that carries an optical clock's stability across cities and countries. This is the plumbing without which the **redefinition of the SI second** cannot happen: BIPM's roadmap requires multiple optical standards in different countries to agree, which means comparing them, which means time transfer.

MATURITY & REAL DEPLOYMENTS (2025–26)

- **International optical-clock comparison across six countries** reported June 2025 — a coordinated cross-border measurement, the largest step yet toward a global optical timescale.
- **Deployed multicore-fiber transfer**: ultrastable optical signals sent through installed multicore fiber alongside live telecom traffic, reaching fractional instability ~3×10^{-19} over nearly 3 hours (Optica, 2025) — showing time transfer can share commercial fiber.
- **QTF-Backbone** (Germany): proposal for a nationwide optical-fiber backbone for both quantum tech and time/frequency metrology (arXiv 2506.03998).
- **BIPM/CCTF** updated the second-redefinition roadmap in early 2025: optical standards must beat cesium by 100×, ≥5 independent systems running continuously, >1 year of traceable data — every threshold implies robust time-transfer.
- Underpins GPS-independent PNT, financial-transaction timestamping, telecom sync, and VLBI/geodesy.

KEY GRADED CLAIMS

- T2 Optical-frequency transfer over deployed multicore fiber at ~3×10^{-19} instability alongside telecom traffic — Optica, 2025 (demonstrated)
- T2 Six-country optical-clock comparison completed — Optica newsroom, June 2025 (demonstrated)
- T4 SI-second redefinition on track for late-2020s/~2030 — BIPM/CCTF roadmap (roadmap)

CONFLICTS / OPEN QUESTIONS

Most record-setting transfer uses purpose-run fiber; scaling to a routine, always-on continental network sharing commercial infrastructure is unproven. **The naming trap**: "quantum" time transfer (entanglement-based clock synchronization — see the entanglement-based clock-network node `A-entclock`) is proposed and theoretically elegant, but a 2026 critical assessment (arXiv 2604.10243) questions whether it beats classical optical/comb-stabilized methods in practice, since the record-holding links are all classically stabilized. So the near-term network is optical/frequency-comb-stabilized, not entanglement-based — the word "quantum" here mostly refers to the *clocks*, not the *transfer*.

THE HONEST CALL

Real, demonstrated metrology infrastructure — and mostly classical despite the "quantum" framing. Six-country optical-clock comparison and ~3×10^{-19} transfer over shared commercial multicore fiber are done and published; this plumbing is a hard requirement for the SI-second redefinition, so it will get built regardless of market pull. What is *not* established is that entanglement adds anything over comb-stabilized optical links — so treat the transfer layer as advanced classical photonics serving quantum clocks.

SOURCES

optica.org newsroom 2025 (six-country clock network; multicore-fiber transfer); arXiv:2506.03998 (QTF-Backbone); arXiv:2503.13278 (SI-second from multiple optical transitions); arXiv:2604.10243 (quantum time-sync assessment); bipm.org/faq-redefinition-second.

Chapter 5 · The Industry Map

Quantum in Aerospace & Defense · I-aerospace

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Defense is where quantum tech is *closest to fielded* — and the payoff comes from **sensing and navigation**, not computing: GPS-free inertial navigation, quantum magnetometry/gravimetry for detection, secure comms, plus longer-horizon codebreaking (Shor, `S-shor`) and simulation. The strategic driver is GPS-denied/contested environments and the "harvest now, decrypt later" threat to military secrets. (Pure SIGINT/cryptanalysis is split into `I-intelligence`.)

REAL ACTIVITY (NAMED, DATED)

- **DARPA + Q-CTRL** (Aug 2025) — two awards under the **Robust Quantum Sensors (RoQS)** program (~$24.4M) for navigation sensors on moving platforms without shielding; **Lockheed Martin** subcontractor. Q-CTRL's April 2025 flight/maritime trials (incl. Royal Australian Navy MV *Sycamore*, ~180 W package) validated quantum inertial nav against military-adequate thresholds.
- **Boeing + US Space Force** — X-37B spaceplane (launched Aug 2025) carried "the highest-performing quantum inertial sensor to fly in space." Boeing also completed ground testing of its **Q4S** quantum-networking satellite payload (entanglement swapping).
- **PsiQuantum + Lockheed Martin** (Nov 2025) — MoU on quantum computing for national security/aerospace.
- **DARPA Quantum Benchmarking Initiative (QBI)** — vetting whether any vendor can reach industrial-scale FTQC by ~2033; the defense channel's reality check (see `E-defense`).
- **Airbus** — magnetic-navigation research; broad quantum program.

KEY GRADED CLAIMS

- T3 Airborne/maritime quantum inertial navigation field trials — Q-CTRL, April 2025 (demonstrated, military-adequate performance claimed)
- T3/T4 Highest-performing quantum inertial sensor flown in space — Boeing/Space Force X-37B, 2025 (claimed)
- T4 Quantum computing for defense applications — PsiQuantum/Lockheed MoU, Nov 2025 (roadmap)
- T5 Defense quantum-market forecasts — analyst; grade hard

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** quantum inertial navigation has cleared airborne, maritime, and space field trials — the nearest-term real quantum deployment in the atlas.
- **Promise:** quantum magnetometry for submarine/mine detection, space quantum networking — advancing, not fielded.
- **Hype:** "quantum computer breaks enemy crypto now," near-term battlefield quantum computing.

HONEST ASSESSMENT

Aerospace/defense has the **nearest-term real quantum payoff**, and it comes from `A-sensing`/`A-gravimetry`/`A-magneto` rather than computing. Quantum inertial navigation has cleared field and space trials and is on a credible path to deployment this decade. Quantum *computing* for defense (codebreaking, sim) is the same fault-tolerance-gated future as everyone else, plus the classified overhang that adversary capability is unknown. Split the story: sensing/nav is arriving; compute is a bet. This is the strongest "quantum is real now" case in the atlas, and the thing that is real is a sensor.

SOURCES

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- Q-CTRL/Lockheed (GPS World): https://www.gpsworld.com/q-ctrl-lockheed-to-develop-quantum-navigation-for-darpa/
- Aerospace America "A year of acceleration and the rise of quantum sensing"
- Lockheed Martin quantum technology page; War on the Rocks "Genesis Mission"

Quantum in Agriculture & Food Science · I-agri

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Split out of `I-climate`, this node covers two threads. (1) **Food/fertilizer chemistry** — the FeMoco/nitrogenase and Haber-Bosch-replacement story that is really quantum chemistry (`I-chem`, `S-qsim`), plus catalyst and soil-chemistry modeling. (2) **Precision-agriculture optimization** — variational/QUBO methods for crop-yield prediction, irrigation and fertilizer scheduling, plus **quantum sensing** (`A-sensing`) for soil and canopy measurement, which is a nearer-term, more credible thread than compute. (Food & beverage *manufacturing/supply chain* rides `I-manufacturing`/`I-logistics`.)

REAL ACTIVITY (NAMED, DATED)

- **Alice & Bob** (Oct 2025) — FeMoco resource estimate showing ~99,000 physical cat qubits (27x cut vs 2021 Google), explicitly framed for fertilizer/agriculture. An estimate, not a run. (Shared with `I-climate`/`I-chem`.)
- **qHPC-GREEN** (CASUS, Dr. Werner Dobrautz, launched Jan 2025 → 2029) — quantum-HPC hybrid research into energy-efficient nitrogen-fixation simulation. Funded research program, not a deployed tool.
- **Precision-ag method papers** — VQC crop-yield/resource algorithms claiming ~30% better yield prediction and ~25% less water/fertilizer, incl. a "5-hectare wheat" pilot writeup (EPJ Web of Conferences 2025). Small academic trials; treat the percentages skeptically.
- **Quantum-sensing review** — *Potential applications of quantum sensors in agriculture* (Computers and Electronics in Agriculture, 2025) surveys magnetometry/gravimetry/NV soil-and-nutrient sensing as the credible near-term hook.

KEY GRADED CLAIMS

- T3 FeMoco needs ~99k physical qubits with cat qubits — Alice & Bob (demonstrated *estimate*, not execution)
- T3 VQC crop-yield/resource algorithms show ~30%/25% gains on small trials — EPJ 2025 (contested; toy-scale, headline numbers unverified, classical ML matches)
- T5/T6 Quantum precision agriculture at farm scale — analyst/vendor scenarios (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** quantum sensors for soil/nutrient/water measurement are real lab-grade instruments (need no fault-tolerant computer).
- **Promise:** nitrogen-fixation chemistry ($\mathrm{N_2 + 3\,H_2 \longrightarrow 2\,NH_3}$) — the textbook killer app, ~99k physical qubits away.
- **Hype:** "quantum boosts crop yields 30%," farm-scale quantum precision agriculture.

HONEST ASSESSMENT

The fertilizer-chemistry payoff is real as a target and distant as a deliverable — the same ~99k-qubit / 2030s+ verdict as `I-climate`, gated entirely on fault tolerance. The precision-ag optimization papers are the hype-heavy corner: small, lightly-refereed trials with headline percentages that classical ML matches. The one thread with a plausible short horizon is quantum sensing for soil/nutrient/water measurement. Present-day deployed agricultural value from quantum computing is zero.

SOURCES

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Quantum in AI & Machine Learning · I-ai

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Quantum machine learning (QML, `S-qml`) promises quantum kernels, quantum neural networks, faster sampling, and quantum-enhanced optimization inside classical ML pipelines — plus, aspirationally, help training or running large models. It's pitched as the fusion of the two hottest technologies. It is also the **most hype-inflated corner of the entire field** and gets graded hardest (per the map's honest edge).

REAL ACTIVITY (NAMED, DATED)

- Enterprise pilots exist in finance, pharma, logistics, and aerospace, and enterprises test QML as a specialized hybrid-pipeline component rather than replacing classical AI stacks.
- **Quantum-kernel methods** for credit scoring, classification, and generative sampling appear in academic + vendor studies (e.g. credit-scoring arXiv 2308.03575), typically on tiny datasets with no advantage shown.
- Vendors (IBM, IonQ, Quantinuum, Xanadu/PennyLane, Multiverse) ship QML SDKs and demos; most "results" prove *feasibility*, not advantage.
- **The dequantization record** — Tang et al. (2019, quantum recommendation systems) and successors produced classical algorithms inspired by the quantum method that matched the claimed speedups, retroactively erasing several QML "advantages."
- **NVIDIA CUDA-Q / cuQuantum** — the dominant tooling is *classical GPU simulation* of quantum circuits, a tell about where the compute actually runs.

KEY GRADED CLAIMS

- T3/T4 Quantum-kernel classifiers competitive on small datasets — assorted arXiv + vendor studies (no advantage at scale)
- T2 Certified randomness (JPMC/Quantinuum, *Nature* 2025) has downstream ML/privacy uses — established primitive, not "QML" per se
- T6 Quantum acceleration of large-model training/inference — aspirational (speculative)
- T5 "Quantum AI" market-size projections — forecast; among the most inflated in the field

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** QML circuits run and classify toy datasets — feasibility, not advantage. Certified randomness (adjacent) is real.
- **Promise:** quantum-native data (simulating quantum systems for ML) may be the one durable niche.
- **Hype:** "quantum AI accelerates LLMs," quantum-kernel "advantage" on real data — repeatedly dequantized.

HONEST ASSESSMENT

QML is where skepticism should be maximal. Multiple provisional quantum-ML "advantages" have been dequantized — classical algorithms inspired by the quantum method matched them — so the speedup evaporated. Today's hardware (tens–hundreds of noisy qubits, shallow circuits) can't run QML at data scales where it would matter, and loading classical data into quantum states (the I/O bottleneck, `S-qram`) often erases any theoretical gain. Business leaders who bet on "quantum AI" in 2025 largely walked it back. The science is worth following; the near-term product is not. Realistic advantage: unproven, possibly never for mainstream ML, plausibly niche for quantum-native data.

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- Tang, "A quantum-inspired classical algorithm for recommendation systems" (dequantization)

Quantum in Air-Traffic Management · I-atm

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Air-traffic management (ATM) is split from logistics (`I-logistics`) and aerospace (`I-aerospace`) because it is a distinct optimization problem: traffic-flow management, multi-aircraft trajectory deconfliction, arrival-slot/gate scheduling, and airspace-density maximization under weather and congestion constraints. These map cleanly to QUBO/QAOA formulations, and the sector is a favorite for quantum-optimization demos because the combinatorics are brutal and the fuel/delay payoff is quantifiable.

REAL ACTIVITY (NAMED, DATED)

- **Airbus + QC Ware** — Airbus-run quantum optimization challenges report large speedups in airspace/traffic analysis (widely cited "~400% faster analysis vs classical-only" and "up to ~70% higher airspace utilization"). Vendor/challenge results — treat percentages as marketing until refereed.
- **Airbus + BMW Quantum Mobility Quest** (2024) — a broader transport-optimization challenge including traffic-flow problems. Competition, not deployment.
- **Method literature** — *Mini-scale traffic flow optimization: an iterative QUBOs approach* (Nature Scientific Reports s41598-025-04568-2, 2025); *Quantum Computing Applications for Flight Trajectory Optimization* (arXiv 2304.14445); *Exploring Airline Gate-Scheduling Optimization Using Quantum Computers* (arXiv 2111.09472). Small/hybrid-solver studies.
- **SESAR/Eurocontrol & NASA** — exploratory quantum-optimization studies for trajectory/flow management; framing and feasibility, no operational use.

KEY GRADED CLAIMS

- T3 QUBO traffic-flow / trajectory / gate-scheduling solved on quantum + hybrid solvers at small scale — Nature Sci Rep 2025, arXiv 2304.14445 / 2111.09472 (demonstrated on mini instances)
- T4 Airbus/QC Ware "~400% faster" / "~70% higher airspace utilization" — challenge/vendor claims (not independently reproduced)
- T4 Quantum-inspired conflict-resolution speedups on existing HPC/GPU — vendor framing (quantum-inspired = classical; grade accordingly)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** small QUBO deconfliction/scheduling demos run on quantum + hybrid solvers — classically matchable.
- **Promise:** operational traffic-flow optimization — gated on fault tolerance *and* safety certification (a high, slow bar for aviation).
- **Hype:** the Airbus challenge percentages read as operational gains; many are quantum-inspired (classical) or constrained-problem results.

HONEST ASSESSMENT

No air-navigation service provider (FAA, Eurocontrol/SESAR, NATS) runs a quantum computer in operational ATM. The credible near-term wins are frequently quantum-inspired — classical algorithms borrowing quantum structure — honest optimization progress that is not quantum hardware advantage, and the two get blurred in the press. The Airbus challenge percentages are competition results on constrained problems, not certified operational gains. The QUBO demos are real but small and classically matchable. Realistic operational value: gated on fault tolerance and safety certification, late 2020s at the earliest; today it is exploratory.

SOURCES

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Quantum in Automotive · I-auto

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Automotive spans several quantum threads: battery/electrolyte and fuel-cell-catalyst simulation (materials), aerodynamics and crash/metal-forming simulation, production-line scheduling and paint-shop optimization, and autonomous-driving ML. German OEMs are the most active enterprise cohort outside finance and pharma, and automotive holds one of the field's most defensible "quantum in production" claims.

REAL ACTIVITY (NAMED, DATED)

- **Ford Otosan + D-Wave** (2024) — **deployed a hybrid-quantum app in production** on the Ford Transit line, cutting vehicle sequencing/scheduling time ~50% (roughly 30 min → under 5 min per ~1,000-vehicle run). One of the few real production deployments in any industry (annealing-based).
- **BMW** — quantum-enhanced solvers for sheet-metal forming and crash-safety (2024); co-ran the 2024 Airbus/BMW Quantum Challenge; earlier fuel-cell catalyst work.
- **BMW + Airbus + Quantinuum** (2023) — hybrid quantum-classical **fuel-cell catalyst** simulation (oxygen reduction on platinum).
- **Hyundai + IonQ** — battery chemistry (electrolytes, cathode catalysts) for next-gen batteries.
- **Ford + Quantinuum** — lithium-ion battery reaction modeling. **Volkswagen + IQM** — hybrid battery-simulation study (2024–25).
- **Mercedes-Benz + PsiQuantum / IBM** — battery-materials feasibility work.
- **Volkswagen + D-Wave** (earlier program) — paint-shop color-switching optimization, a QUBO scheduling problem framed to cut solvent/flush waste; the archetypal automotive annealing demo classical solvers also handle.
- **Toyota / Toyota Tsusho** — participation in Japanese quantum consortia (Q-STAR) on materials and mobility optimization; exploratory.

KEY GRADED CLAIMS

- T3/T4 Ford Otosan production sequencing, ~50% scheduling-time cut — D-Wave/Ford Otosan, 2024 (demonstrated, in production; annealing not gate-model)
- T4 Fuel-cell catalyst simulation — BMW/Airbus/Quantinuum (exploratory)
- T4 Battery-chemistry pilots — Hyundai/IonQ, Ford/Quantinuum, VW/IQM (claimed)
- T5 Automotive slice of quantum economic-value forecasts — analyst; grade hard

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** Ford Otosan runs an annealing-based scheduling app in a live plant — a legitimate deployed workflow.
- **Promise:** battery/electrolyte and fuel-cell catalyst design — small-molecule exploratory studies short of design-grade accuracy.
- **Hype:** "quantum computer beats classical on scheduling" (contested), quantum-accelerated autonomous driving.

HONEST ASSESSMENT

The Ford Otosan case is one of the most defensible "quantum in production" claims anywhere, and it is *quantum annealing* solving a scheduling problem where D-Wave competes head-to-head with classical solvers (advantage contested — see `H-anneal` / `O-advantage`). Battery and catalyst simulations are small-molecule exploratory studies far from design-grade accuracy. Autonomous-driving "quantum ML" is mostly aspirational. Honest read: optimization/scheduling has real *today* value via annealing and quantum-inspired methods (advantage debated); materials simulation is a **2030s** bet.

SOURCES

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- postquantum.com aerospace & automotive use cases

Quantum in Chemicals & Materials · I-chem

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Like pharma, chemicals is a native quantum-simulation target (`S-qsim`): designing catalysts, battery electrolytes, photochromic and semiconductor materials, and reaction pathways by computing electronic structure directly instead of trial-and-error lab work. McKinsey frames it as compressing decades of chemistry. This is the domain with the clearest *theoretical* quantum advantage — the algorithms (VQE, quantum phase estimation) map onto exactly what quantum computers do naturally.

REAL ACTIVITY (NAMED, DATED)

- **BASF** — collaboration with **SEEQC** on quantum for homogeneous catalysis; with **Kipu Quantum** (2024) on optimization algorithms; broader hybrid catalyst-design program via the German QUTAC consortium.
- **Mitsubishi Chemical + PsiQuantum** — simulating **excited states of photochromic molecules** (smart windows, solar storage, data storage). Also **Mitsubishi + Xanadu** on semiconductor materials; long-running IBM case study.
- **Dow** — quantum chemistry exploration (historical IBM Quantum Network member).
- **Microsoft Azure Quantum Elements** — AI+quantum chemistry/materials platform pitched to chemical companies. **NVIDIA cuEST** (GPU electronic-structure library) — TSMC reports ~50x faster chemistry simulations; a *classical*-acceleration reminder that the near-term competition to quantum chemistry is GPU HPC, not other quantum machines.
- **Merck, Covestro, Evonik** — QUTAC members probing polymer/formulation chemistry.

KEY GRADED CLAIMS

- T1 Molecular electronic-structure simulation is a provably natural quantum-computer task — textbook (established in principle; hardware not yet capable at industrial scale)
- T3/T4 Photochromic excited-state simulation — Mitsubishi Chemical/PsiQuantum (exploratory)
- T4 Homogeneous-catalysis quantum modeling — BASF/SEEQC (research collaboration)
- T5 Chemicals/materials as a leading slice of McKinsey's ~$2.7T-by-2035 value estimate and BCG's $450–850B-by-2040 — analyst forecasts; grade hard, inflation-unadjusted and double-counted with pharma/energy

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** VQE/QPE run on real hardware for tiny molecules (H_2, LiH, small actives) — results classical DFT/coupled-cluster already match or beat.
- **Promise:** catalyst and photochromic design at useful accuracy — the strongest theoretical case in any industry, gated on thousands of logical qubits.
- **Hype:** "quantum is designing industrial catalysts now," trillion-dollar near-term chemicals TAM.

HONEST ASSESSMENT

The theoretical case is the strongest in any industry — simulating quantum chemistry is what quantum computers are *for*. But $FeMoco$-class catalysts and industrial reaction networks need error-corrected machines with thousands of logical qubits; today's devices simulate only tiny molecules classical methods already handle, and GPU-accelerated classical chemistry keeps raising the bar quantum must clear. Every named "pilot" is a research collaboration probing algorithms, not a deployed design tool. Honest timeline: fault-tolerant chemistry advantage is a **2030s** proposition. Hold the distinction — strong theory, essentially zero present-day production advantage.

SOURCES

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- BASF/QUTAC catalysts: https://www.qutac.de/en/basf-how-quantum-computing-can-help-develop-chemical-catalysts/
- Mitsubishi Chemical/IBM: https://www.ibm.com/case-studies/mitsubishi-chemical
- NVIDIA cuEST / TSMC 50x chemistry: nvidianews.nvidia.com (GTC 2025)
- C&EN "Will quantum computing be chemistry's next AI?" (Nov 2025)

Quantum in Climate & Sustainability · I-climate

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Climate is largely a *downstream application* of quantum chemistry (`I-chem`, `S-qsim`): better catalysts for $\mathrm{CO_2}$ capture, more efficient fertilizer synthesis (cracking the FeMoco nitrogenase problem to replace energy-hungry Haber-Bosch), plus grid/ emissions optimization. The prize is enormous — ammonia production alone is up to ~1.8% of global $\mathrm{CO_2}$ and ~1–2% of world energy. (Precision agriculture is split into `I-agri`; numerical weather/climate *modeling* is split into `I-weather`.)

$$\mathrm{N_2 + 3\,H_2 \longrightarrow 2\,NH_3}$$

REAL ACTIVITY (NAMED, DATED)

- **Alice & Bob** (Oct 2025) — published a quantum resource estimate for simulating FeMoco (biological nitrogen fixation) showing a **~27x reduction in physical qubits** vs a 2021 Google study (2.7M → ~99,000 qubits) using cat qubits. A resource estimate, not a run — a notable efficiency milestone that still points at hardware that does not exist.
- **Microsoft / Riverlane / McKinsey** — frame carbon-capture sorbent (MOF) design and nitrogen-cycle modeling as flagship "quantum for the planet" targets.
- **qHPC-GREEN** (CASUS, launched Jan 2025 → 2029) — quantum-HPC hybrid research into energy-efficient fertilizer/ nitrogen-fixation simulation (shared with `I-agri`).

KEY GRADED CLAIMS

- T3 FeMoco simulation needs ~99k physical qubits (27x cut) with cat qubits — Alice & Bob, Oct 2025 (demonstrated *estimate*, not execution)
- T1 Haber-Bosch's carbon/energy footprint — established chemistry
- T5 "Quantum computing just might save the planet" TAM framing — McKinsey; grade hard, downstream of the same chemicals value pool, inflation-unadjusted

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** resource *estimates* have improved (27x fewer qubits for FeMoco) — an engineering datapoint, not a computation.
- **Promise:** carbon-capture and fertilizer catalyst design — the textbook killer app, ~99k physical qubits away, i.e. thousands of logical qubits.
- **Hype:** "quantum will decarbonize industry this decade," present-day climate impact from quantum computers (zero).

HONEST ASSESSMENT

The nitrogen-fixation story is the honest headline and the honest caution at once. FeMoco is the textbook "killer app" for quantum chemistry, and the best current estimate still says ~99,000 physical qubits — many thousands of logical qubits with error correction, hardware that does not exist and is years to a decade+ away. The 27x improvement is worth reporting and a reminder of the distance to payoff. Carbon-capture sorbent design shares the same fault-tolerance wall. Realistic climate impact from quantum computing: **2030s at the earliest**, entirely gated on fault tolerance. Present-day value is essentially zero.

SOURCES

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- McKinsey "Quantum computing just might save the planet": https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/quantum-computing-just-might-save-the-planet
- Riverlane "How quantum computing can help tackle climate change"
- Microsoft Research "What problems will we solve with a quantum computer?"

Quantum in Construction & Built Environment · I-construction

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Construction/built-environment cuts across three quantum threads: (1) **structural materials simulation** — quantum chemistry (`I-chem`, `S-qsim`) for greener cement/concrete and durable materials; (2) **project scheduling and supply-chain optimization** — QUBO/QAOA for construction schedules, resource conflicts, and prefab supply chains, sharing the `I-logistics` template; (3) **urban microclimate and energy modeling** — CFD-style built-environment and HVAC/EV-charging optimization for greener cities.

REAL ACTIVITY (NAMED, DATED)

- **Concordia / UCF / Syracuse / Tongji** — Liangzhu Leon Wang et al., *"Take a BITE for Built Environment and Urban Microclimate Research"* (arXiv 2604.18407) maps quantum methods onto urban microclimate and building-energy problems. Academic survey/framing.
- **Civil-engineering reviews** — *Quantum and quantum-inspired computing in civil engineering* (ScienceDirect 2025) and *Quantum computing in civil engineering: Potentials and Limitations* (arXiv 2402.14556) document QUBO scheduling and structural-analysis demos on small instances.
- **Prefab supply chain** — *Unlocking the potential of quantum computing in prefabricated construction supply chains* (ScienceDirect 2025) — trends/challenges review, not deployment.
- **The honest datapoint** from these reviews: ~42% of "quantum in construction" research is actually **quantum materials** (`A-materials`), and compute/QML applications are "predominantly theoretical and confined to small-scale simulations."

KEY GRADED CLAIMS

- T3 QUBO project scheduling finds near-optimal construction schedules faster than classical on small cases — civil-eng reviews (demonstrated on toy instances; no advantage at scale)
- T3 Quantum methods framed for urban microclimate / building energy — Wang et al. arXiv 2604.18407 (framing/survey)
- T5/T6 Quantum-designed sustainable construction materials — review scenarios (speculative, FTQC-gated)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** small QUBO scheduling and structural demos exist in the literature — classical MILP still beats them.
- **Promise:** greener cement/concrete via quantum materials chemistry — inherits `I-chem`'s fault-tolerance wall.
- **Hype:** "quantum optimizes construction projects" as a near-term product — no industrial pilot runs quantum hardware in the loop.

HONEST ASSESSMENT

Construction is an early, academic node with no named industrial pilot running quantum hardware in the loop. Its own literature admits most activity is either materials science (belongs in `A-materials`) or small-scale QUBO demos that classical MILP still beats. The materials-simulation payoff is real and inherits the fault-tolerance timeline of `I-chem`. Scheduling and microclimate are optimization pitches without demonstrated advantage. Present-day value is zero; realistic horizon is 2030s and gated on FTQC.

SOURCES

- https://arxiv.org/abs/2604.18407 (BITE — built environment & urban microclimate)
- https://www.sciencedirect.com/science/article/pii/S1474034625008535 (quantum in civil engineering)
- https://arxiv.org/html/2402.14556v2 (civil engineering potentials & limitations)
- https://www.sciencedirect.com/science/article/abs/pii/S1566253525001162 (prefab supply chains)

Quantum in Cybersecurity · I-cyber

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Cybersecurity is the industry with the most *certain* quantum impact — and it lands as a **threat before it lands as a tool**. A cryptographically relevant quantum computer (CRQC) running Shor's algorithm (`S-shor`) breaks RSA and elliptic-curve crypto, so the internet's encryption must migrate to post-quantum cryptography (`A-pqc`). "Harvest now, decrypt later" (HNDL) means adversaries are already storing encrypted traffic to decrypt once a CRQC exists. This is a real, funded, deadline-driven market today.

REAL ACTIVITY (NAMED, DATED)

- **NIST** — finalized first PQC standards **FIPS 203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA)** in Aug 2024, triggering the largest mandated crypto migration in history. NIST IR 8547 sets deprecation timelines (2030 deprecate, 2035 disallow).
- **NSA CNSA 2.0** — quantum-safe algorithms required for new national-security systems by **Jan 2027**, full app migration by 2030, infrastructure by 2035.
- **Resource-estimate shift** — work published May 2025–March 2026 cut the estimated qubits to break RSA-2048 from ~20M (Gidney/Ekerå 2019) toward **<1M, potentially ~100k** on newer architectures — sharpening the HNDL urgency without a CRQC yet existing.
- **Enterprise reality** — a May 2025 survey of 1,000+ security managers found only **~5% had quantum-safe encryption deployed**; 81% said libraries/HSMs weren't PQC-ready.

KEY GRADED CLAIMS

- T2 NIST PQC standards final (FIPS 203/204/205), migration mandated — NIST, Aug 2024 (established)
- T1 Shor breaks RSA/ECC given a CRQC — established theory (`S-shor`); the CRQC itself is T6
- T3 RSA-2048 resource estimate reduced toward <1M qubits — 2025–26 preprints (revised estimate, not a demonstration)
- T5 ">$15B migration market by 2030" / 5% deployment rate — surveys + analyst projections (forecast; inflation-unadjusted)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** the threat model and the standards. Shor's math is established; FIPS 203/204/205 are final; deadlines are legally binding for regulated/defense sectors.
- **Promise:** actual enterprise migration — mandated but barely started (~5% deployed).
- **Hype:** "Q-Day is next year," QKD as the answer (PQC software is), and CRQC timelines stated with false precision.

HONEST ASSESSMENT

This is the one industry where quantum's impact is **not** overstated — the threat is mathematically certain, the standards are final, the deadlines are binding. The uncertainty is *timing*: CRQC estimates range 2030 to "not before 2040+" (a first-class conflict, `O-scaling`), and the recent qubit-count reductions tighten but do not resolve it. Because HNDL makes the risk retroactive, migration must start now regardless. The honest caveat here runs opposite to the rest of L4 — the danger is *under*-reacting. QKD is over-marketed; classical-software PQC is the real answer.

SOURCES

- NIST IR 8547 draft: https://nvlpubs.nist.gov/nistpubs/ir/2024/NIST.IR.8547.ipd.pdf
- PQShield on NIST timelines: https://pqshield.com/nist-recommends-timelines-for-transitioning-cryptographic-algorithms/
- "Harvest now, decrypt later" — The Quantum Insider (May 2026), CSA Q-Day clock
- PRNewswire "The $15 Billion Post-Quantum Migration"

Quantum in Energy & Utilities · I-energy

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Two threads: (1) optimization — grid balancing, unit commitment, EV-charging scheduling, demand forecasting, LNG shipping routes; and (2) simulation — battery chemistries, fusion plasma, carbon-capture sorbents. Utilities face combinatorial grid problems classical solvers approximate; energy majors want better materials. (Fusion plasma modeling and nuclear are deep enough to split into `I-nuclear`; oil/gas reservoir work lives in `I-extractive`.)

REAL ACTIVITY (NAMED, DATED)

- **EDF + Pasqal + GENCI** (Jan 2025) — neutral-atom quantum computing demonstrated on **EV smart-charging and energy-demand forecasting** at 100+ scale; explicitly a pilot, not commercially viable.
- **EDF + Alice & Bob + Quandela + CNRS** (July 2024) — collaboration on optimizing the *energy consumption of quantum computing itself* (the fridge/laser power budget).
- **E.ON + IBM** — quantum algorithms for electricity-distribution complexity, energy pricing, risk management.
- **ExxonMobil** — first energy company to join the IBM Quantum Network (2019); quantum algorithms for LNG shipping-route optimization and carbon capture.
- **Iberdrola, Enel, Vattenfall** — European utilities running grid-optimization feasibility studies with vendors.
- **Aramco** — targeting local QCaaS access ~2025; **Shell** exploring reservoir optimization (see `I-extractive`).

KEY GRADED CLAIMS

- T3/T4 EV-charging & demand forecasting on neutral atoms — Pasqal/GENCI/EDF, Jan 2025 (demonstrated pilot, not commercially viable)
- T4 LNG shipping-route optimization — ExxonMobil/IBM (exploratory)
- T4 Grid/pricing quantum algorithms — E.ON/IBM (claimed)
- T5 "Energy transition" TAM lines in analyst reports — forecast; grade hard, entangled with the climate/chemicals figures

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** dated feasibility pilots (EDF/Pasqal EV-charging) run and return results — pre-commercial by the participants' own framing.
- **Promise:** grid dispatch, unit commitment, battery-chemistry simulation — real hard problems, no demonstrated advantage over classical/quantum-inspired solvers yet.
- **Hype:** "quantum will run the grid this decade," trillion-dollar energy-transition quantum TAM.

HONEST ASSESSMENT

Grid optimization is a real, valuable, hard problem, and classical plus quantum-inspired solvers (and D-Wave annealing) currently match or beat gate-model quantum on the instances tried. The EDF/Pasqal work is a credible, dated pilot stated openly as pre-commercial. Battery and carbon-capture simulation share pharma/chem's fault-tolerance wall. No utility runs quantum in dispatch. The nearer-term energy payoff more plausibly comes from quantum *sensing* (grid fault detection, magnetometry) than compute. Realistic optimization payoff: **late 2020s**; simulation payoff: **2030s**.

SOURCES

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- EDF/Alice&Bob/Quandela/CNRS: https://alice-bob.com/newsroom/edf-alice-bob-quandela-cnrs-quantum-computing/
- Hello Tomorrow "Quantum Computing in the Energy Sector"
- postquantum.com energy & utilities use cases

Quantum in Oil & Gas · I-extractive

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Oil & gas pitches quantum for **seismic data processing and subsurface imaging** (faster inversion, pattern-finding in seismic/well-log big data), **reservoir simulation** (multi-week optimization compressed to hours), materials discovery (catalysts, carbon capture), and LNG logistics. Big analyst TAM numbers attach here. (**Mining and mineral exploration** — a distinct, sensing-led, nearer-term story — is split into `I-mining`.)

REAL ACTIVITY (NAMED, DATED)

- **ExxonMobil, bp** — joined the **IBM Quantum Network** (ExxonMobil 2019, first energy major) to explore reservoir simulation, LNG shipping-route optimization, and advanced materials.
- **Shell + D-Wave** — quantum annealing applied to **North Sea reservoir mapping** optimization (reported multi-week → hours framing). Shell has a multi-year quantum research program (also with IBM, Zapata historically).
- **Aramco** — targeting **local QCaaS** access ~2025 for Saudi research; invested in Pasqal (2022) toward an on-premise neutral-atom machine.
- **Seismic method work** — quantum travel-time seismic inversion (arXiv 2208.05794) and QUBO formulations of inversion sub-problems; preprints on small instances.

KEY GRADED CLAIMS

- T4 Shell/D-Wave reservoir-mapping speedup (weeks→hours) — vendor+operator framing (annealing, advantage contested)
- T4 Reservoir sim & materials exploration — ExxonMobil/bp/IBM (exploratory)
- T3 Quantum seismic-inversion formulations — arXiv 2208.05794 (demonstrated on toy instances)
- T5 "$2.6T oil-and-gas quantum impact by 2035 / ~30% CAGR" — market-research projections (forecast; among the most inflated in L4, inflation-unadjusted)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** operators run feasibility studies and publish sub-problem demos — no operator runs quantum in its production geophysics pipeline.
- **Promise:** reservoir optimization and seismic inversion at field scale — gated on fault tolerance; classical HPC dominates today.
- **Hype:** trillion-dollar "impact by 2035" TAM lines — grade hardest in the atlas.

HONEST ASSESSMENT

Oil & gas is heavy on analyst TAM and light on reproduced results. The seismic/reservoir "advantage" claims are annealing or quantum-inspired optimization on subproblems; classical HPC still dominates production geophysics, and no operator runs quantum in its exploration pipeline. The genuinely nearer-term extractive thread is *quantum sensing* (cold-atom gravimeters, NV magnetometers) — covered under `I-mining` and `A-magneto`/`A-gravimetry`. Quantum *computing* for reservoirs/seismic is a fault-tolerance-gated **2030s** bet dressed in trillion-dollar forecasts. Grade the TAM numbers hard.

SOURCES

- JPT (SPE) "Quantum Computing: A Beacon of Transformation for Oil and Gas": https://jpt.spe.org/guest-editorial-quantum-computing-a-beacon-of-transformation-for-the-oil-and-gas-industry
- oilandgasiq.com "Quantum Computing's Impact on Oil & Gas"
- energyconnects.com "The oil and gas industry aims to take a leap with quantum computing" (June 2024)
- Quantum travel-time seismic inversion: https://arxiv.org/pdf/2208.05794

Quantum in Finance · I-finance

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Finance is the flagship "quantum use case": portfolio optimization, Monte Carlo derivative pricing, risk (VaR/CVA), and fraud/anomaly detection all map to problems where classical compute is expensive. Vendors promise quadratic speedups on Monte Carlo (amplitude estimation, see `S-qmc`) and better solutions to NP-hard portfolio selection (QAOA, `S-variational`). Two sub-verticals behave differently: **retail/commercial banking** chases fraud graphs and credit scoring; **capital markets** (trading desks, derivatives, reinsurance-adjacent risk) chases pricing and optimization. Banks were the earliest and heaviest enterprise experimenters, with 15+ global banks running programs by March 2026 (JPMorgan, Goldman, HSBC, BBVA, Barclays, BNP Paribas, Wells Fargo).

REAL ACTIVITY (NAMED, DATED)

- **JPMorganChase** — the most substantive results. With Quantinuum, Argonne, ORNL, UT Austin, published **certified randomness** on the 56-qubit Quantinuum H2-1 (*Nature*, 26 Mar 2025) — a genuine "beyond classical" task (see `S-certrand`). Separately showed a **theoretical QAOA speedup** for constrained optimization (*Science Advances*, 29 May 2025). A JPMorgan+AWS hybrid decomposition (2024) reported a modest **~12% runtime reduction** on a constrained optimization by partitioning. "Up to 100x speed" figures appear in controlled-test PRs and are T4.
- **HSBC + IBM** (Sept 2025) — claimed **up to 34% improvement** predicting whether a corporate bond trade fills at the quoted price vs classical, on real algorithmic bond-trading data using an IBM Heron processor in a hybrid workflow. Framed as "first empirical evidence quantum adds value in production-representative trading." Bank+vendor co-announcement, not independently reproduced.
- **Goldman Sachs** — early leader on Monte Carlo option pricing (with IonQ/QC Ware), then reportedly **scaled back** after concluding practical advantage remains far off — the honest counter-signal to the bullish PRs.
- **Multiverse Computing** — Spanish vendor selling "quantum-inspired" (tensor-network) finance tooling to banks; revenue is classical software, useful context on where the money actually flows.

KEY GRADED CLAIMS

- T2 Certified randomness on H2-1, classically unachievable — JPMC/Quantinuum, *Nature* s41586-025-08737-1 (demonstrated; a cryptography primitive, not trading P&L)
- T3 QAOA constrained-optimization speedup (theoretical) — JPMC, *Science Advances* 2025
- T4 HSBC/IBM 34% bond-fill improvement — co-announcement Sept 2025 (controlled POC)
- T4 Portfolio-optimization / Monte-Carlo "100x" speedups — bank+vendor PRs (no production deployment)
- T5 Finance as a headline share of McKinsey's ~$2.7T-by-2035 economic-value estimate — analyst forecast; grade hard, double-counted across verticals and inflation-unadjusted (2026 dollars)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** certified randomness (peer-reviewed, beyond-classical) — a cryptographic primitive with privacy/audit uses, not alpha.
- **Promise:** portfolio optimization, Monte Carlo pricing, bond-fill prediction — real POCs on tiny instances, awaiting fault tolerance to beat Gurobi/CPLEX at scale.
- **Hype:** "quantum trading edge today," "100x in production," multi-trillion-dollar finance TAM lines.

HONEST ASSESSMENT

No bank runs a quantum computer in live operations as of mid-2026. The certified-randomness result is real and refereed and is a cryptography primitive. The optimization and pricing "advantages" are controlled POCs on small instances that classical methods still beat at scale, and the amplitude-estimation speedup that finance has chased for years needs fault-tolerant depth that erases the near-term gain. Goldman's pullback and JPMorgan's careful language are the honest signals inside the sector. Realistic operational value: late 2020s at the earliest, gated on fault tolerance.

SOURCES

- JPMorgan certified randomness: https://www.jpmorgan.com/technology/news/certified-randomness · Nature: https://www.nature.com/articles/s41586-025-08737-1
- HSBC/IBM bond trading (Sept 2025): reported via The Quantum Insider / FStech
- "15+ global banks probing quantum" — The Quantum Insider, 27 Mar 2026
- "Goldman Sachs, JPMorgan sharply diverge on quantum" — TheStreet

- McKinsey Quantum Technology Monitor 2026 (economic-value figure)

Quantum in Government Services · I-gov

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Government is pitched as a quantum customer distinct from defense (`I-aerospace`) and intelligence (`I-intelligence`): tax administration (real-time audits, anomaly/fraud detection), customs and border flow, census and statistical disclosure, benefits fraud, and general public-sector optimization. The problems map to the same QUBO optimization and anomaly-detection templates sold to finance — combinatorial matching, graph community detection over transaction networks, pattern-finding in large administrative datasets.

REAL ACTIVITY (NAMED, DATED)

- **Academic / preprint** — the substantive work is QUBO-formulated fraud detection via community detection in transaction graphs (*Quantum Computing in Community Detection for Anti-Fraud Applications*, PMC 2025) and unsupervised quantum ML for fraud (arXiv 2208.01203). Method papers on small instances, no agency deployment.
- **Advisory framing** — EY and the Inter-American Center of Tax Administrations (CIAT) publish "how quantum will change tax administration" guides. Consulting scenarios, not pilots.
- **The mislabel trap** — HMRC's £175M 10-year Quantexa contract (2026) to hunt tax fraud is **classical AI/decision-intelligence**, despite the "Quant-" branding. Much "quantum government" press is this pattern: AI relabeled, or PQC/export-control work (covered in `A-pqc`, `E-export`).
- **Real government quantum spend** is on the *defensive* side — national PQC-migration mandates (US OMB M-23-02, NSA CNSA 2.0) — which is `I-cyber`, not offensive optimization.

KEY GRADED CLAIMS

- T3 QUBO community detection improves anti-fraud risk scoring on transaction graphs — PMC/arXiv method papers (demonstrated on toy data)
- T4 Vendor "quantum for public sector" optimization offerings — press/marketing (claimed)
- T5 Tax authorities should prepare for real-time quantum-enabled audits — EY/CIAT advisory (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** nothing offensive is deployed. The genuine government quantum program that exists is defensive PQC migration.
- **Promise:** anti-fraud graph analytics and public-sector optimization — method-paper stage, inheriting finance's caveats.
- **Hype:** headline "quantum government" wins that are actually classical AI (the Quantexa/HMRC pattern).

HONEST ASSESSMENT

There is no known quantum computer running in production inside any tax, customs, or census agency. The genuine near-term government exposure to quantum is defensive — PQC migration and the harvest-now-decrypt-later threat to citizen data — which lives in `A-pqc`/`I-cyber`. Offensive/optimization use is at the method-paper stage and inherits every caveat of `I-finance` optimization (tiny instances, classical methods still win). The honest signal is the naming confusion: most headline "quantum government" wins are classical AI. Realistic operational value: gated on fault tolerance, late 2020s at the earliest; today it is essentially zero.

SOURCES

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- https://arxiv.org/pdf/2208.01203 (unsupervised quantum ML for fraud)
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- https://www.ciat.org/quantum-computer-impacts-on-tax-administration-at-first-glance/?lang=en
- https://thenextweb.com/news/quantexa-hmrc-ai-tax-fraud-sovereignty (the AI-not-quantum contrast)

Quantum in Healthcare Imaging & Diagnostics · I-healthimaging

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from `I-pharma` drug-discovery; this is the nearer-term quantum-*sensing* health story)

THE PITCH

Split from `I-pharma` (molecular simulation), this node is the quantum-**sensing** side of healthcare — and it is far closer to the clinic than quantum computing. The flagship is **OPM-MEG**: optically-pumped magnetometers (a quantum sensor) replacing cryogenic SQUIDs for magnetoencephalography, giving wearable, room-temperature brain imaging. Adjacent threads: NV-diamond and hyperpolarization for enhanced MRI/NMR contrast, quantum-enhanced (sub-shot-noise) optical imaging, and diagnostic ML.

REAL ACTIVITY (NAMED, DATED)

- **Cerca Magnetics** (UK, Nottingham spinout) — commercial **wearable OPM-MEG** helmet systems; a real product shipped to research sites. As of 2025 it is sold as a **research system with no medical/regulatory approval** in any jurisdiction — the honest ceiling on the claim.
- **384-channel OPM-MEG** (arXiv 2509.03107, 2025) — toward whole-head high-channel-count systems; the technical frontier.
- **EU Horizon OPMMEG project** (CORDIS 101099379) — pushing OPM-MEG toward routine clinical practice; epilepsy pre-surgical workup is the most advanced application, with mild-TBI use growing.
- **NIHR Innovation Observatory report** (Jan 2025) — surveys quantum sensing in healthcare; notes only two sensing technologies are beyond **TRL 6** (entering clinical testing), i.e. still early.

KEY GRADED CLAIMS

- T2/T3 OPM-MEG matches SQUID sensitivity without cryogenics, wearable — peer-reviewed OPM literature + PMC epilepsy studies (demonstrated capability)
- T3 384-channel OPM-MEG system — arXiv 2509.03107 (preprint, prototype)
- T4 Cerca commercial OPM-MEG — vendor product (real, research-only; no regulatory approval as of 2025)
- T5 Quantum-healthcare-sensing market forecasts — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** OPM-MEG works — wearable, cryogen-free brain magnetometry with SQUID-class sensitivity, running in research labs today. This is a real quantum sensor in health use.
- **Promise:** routine clinical MEG (epilepsy, TBI), NV/hyperpolarized MRI contrast — advancing through clinical testing and regulatory pathways.
- **Hype:** "quantum computers diagnose disease" — this node is sensing, not compute; and OPM-MEG is not yet cleared for diagnosis.

HONEST ASSESSMENT

This is one of the strongest "quantum is real in health now" stories, and the thing that is real is a *sensor*. OPM-MEG is a shipped quantum-sensing product with genuine clinical promise (epilepsy pre-surgical mapping), held back by regulatory approval and magnetically-shielded-room requirements rather than physics. NV-diamond and hyperpolarization for MRI are earlier. The whole node needs no fault-tolerant computer, which is exactly why it is nearer-term than `I-pharma` drug discovery. Realistic clinical adoption: **this decade** for MEG in specialist centers, contingent on regulatory clearance. Keep it separate from quantum-computing hype.

SOURCES

- NIHR IO "Emerging applications of quantum sensing technology in healthcare" (Jan 2025): https://io.nihr.ac.uk/wp-content/uploads/2025/03/NIHR-IO-Quantum-Sensing-Technology-Report_Jan-2025.pdf
- 384-channel OPM-MEG: https://arxiv.org/pdf/2509.03107
- Cerca Magnetics OPM-MEG: https://www.cercamagnetics.com/
- OPM epilepsy presurgical workup: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10896867/

Quantum in Insurance & Risk · I-insurance

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Insurance is finance's cousin (`I-finance`): Monte Carlo-heavy catastrophe modeling, capital/reserve optimization under reinsurance, actuarial pricing, tail-risk and dependency modeling, and fraud detection. Quantum is pitched for amplitude-estimation speedups on Monte Carlo (`S-qmc`) and better modeling of extreme-event correlations (tail dependence) that classical Gaussian copulas capture poorly.

REAL ACTIVITY (NAMED, DATED)

- **Allstate** — among the first major US insurers to join the **Chicago Quantum Exchange** (2025), partnering with quantum researchers on complex risk modeling.
- **Method / academic work** — quantum-AI copulas for catastrophe and reinsurance capital modeling: *Quantum computational insurance and actuarial science* (Science China Information Sciences, 2025 review); *Quantum Computing in Insurance Capital Modelling under Reinsurance Contracts* (AppliedMath, 2024); World Quantum Summit 2025 sessions on quantum-AI copulas.
- Reinsurers (Swiss Re, Munich Re) and actuarial bodies (IFoA) are watching and running small experiments; no named production deployment has surfaced.

KEY GRADED CLAIMS

- T3 Quantum methods for non-life/life/reinsurance capital modeling — Science China Info Sci 2025, AppliedMath (peer-reviewed numerical demonstrations on small instances)
- T4 Allstate/Chicago Quantum Exchange risk-modeling exploration — membership announcement (early)
- T5 Quantum-AI copulas improve tail-dependence/catastrophe modeling at scale — summit + review framing (forecast)
- T5 Insurance slice of quantum-value forecasts — analyst; tracks finance, lags in adoption

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** small numerical demonstrations of quantum capital/copula methods exist in the literature — no advantage over classical actuarial compute.
- **Promise:** amplitude-estimation Monte Carlo for cat modeling and tail dependence — the same finance bet, one adoption step behind.
- **Hype:** any claim of quantum catastrophe modeling in production (none exists).

HONEST ASSESSMENT

Insurance is one of the *least* mature L4 verticals — mostly academic papers and consortium memberships, few named corporate pilots, no deployments. The theoretical hook (quadratic Monte Carlo speedup via amplitude estimation) is the same one finance has chased for years and never realized in production, because the speedup needs fault-tolerant depth that erases the near-term advantage. Catastrophe modeling is genuinely hard and valuable, so the motivation is real, and nothing here beats classical actuarial compute today. This vertical is almost entirely promise: realistic value tracks finance's **late 2020s–2030s** timeline and lags it in adoption.

SOURCES

- The Actuary Weekly "Quantum Computing in Catastrophe Modeling"
- Science China Information Sciences "Quantum computational insurance and actuarial science": https://link.springer.com/article/10.1007/s11432-024-4411-8
- AppliedMath "Quantum Computing in Insurance Capital Modelling under Reinsurance Contracts": https://doi.org/10.3390/appliedmath3040040
- World Quantum Summit 2025 "Insurance Risk Modelling with Quantum-AI Copulas"

Quantum in Intelligence & Cryptanalysis · I-intelligence

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from `I-aerospace` defense and `I-cyber` defense; the offensive SIGINT/cryptanalysis angle is distinct)

THE PITCH

Distinct from defense hardware (`I-aerospace`) and defensive PQC (`I-cyber`), this node is the **offensive signals-intelligence** dimension: a cryptographically relevant quantum computer (CRQC) running Shor (`S-shor`) would let a state actor decrypt intercepted RSA/ECC traffic. The strategic behavior this *already* drives is **harvest-now-decrypt-later (HNDL)** — bulk collection and storage of today's encrypted traffic for future decryption. Secondary: quantum-enhanced pattern-analysis over massive intercept datasets (the same QML caveats as `I-ai`).

REAL ACTIVITY (NAMED, DATED)

- **Intelligence-community warnings** — NSA, CISA, and NIST have jointly warned that adversaries **may already be harvesting** encrypted data with long-term value to decrypt once a CRQC exists (the HNDL doctrine). Nation-state SIGINT programs (US, China, Russia) operate the bulk-collection infrastructure that makes HNDL feasible.
- **NSA CNSA 2.0** — mandates quantum-resistant algorithms for new national-security systems by **Jan 2027**, full transition by 2035 — an intelligence-driven timeline (defend own secrets against adversary CRQC).
- **Resource-estimate shift (2025–26)** — the RSA-2048 qubit estimate fell from ~20M (2019) toward **<1M, possibly ~100k** on newer architectures — which compresses the intelligence risk horizon without a CRQC existing yet.
- **No CRQC exists.** No public evidence any agency can run Shor at cryptographic scale. Adversary capability is classified — the honest overhang.

KEY GRADED CLAIMS

- T1 Shor breaks RSA/ECC given a CRQC — established theory (`S-shor`); the CRQC itself is T6
- T3 Bulk HNDL collection is technically feasible and warned-of by NSA/CISA/NIST — official advisories (established as doctrine; specific programs classified)
- T3 RSA-2048 resource estimate reduced toward <1M qubits — 2025–26 preprints (revised estimate, not a demonstration)
- T6 Any agency currently possesses cryptanalytic quantum capability — speculative/classified (no public evidence)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** the doctrine and the math. HNDL is a real, warned-of collection strategy; Shor's threat is certain given a CRQC.
- **Promise (or threat):** a future CRQC that decrypts harvested traffic — timing genuinely unknown (2030–2040+, `O-scaling`).
- **Hype:** claims that a nation "already breaks encryption with quantum," stated with false certainty from classified inference.

HONEST ASSESSMENT

Intelligence is the vertical where quantum's impact is most certain in *kind* and most uncertain in *timing*, wrapped in classification. Shor's threat to public-key crypto is established; HNDL collection is real doctrine, which is precisely why CNSA 2.0's deadlines exist. What no one outside classified spaces knows is whether — or how close — any adversary is to a CRQC, and the recent qubit-count reductions tighten the risk window without resolving it. The rational response is the same as `I-cyber`: migrate to PQC now, because HNDL makes the risk retroactive. Treat any "they already have it" claim as unverifiable. Offensive quantum SIGINT-analytics (QML over intercepts) inherits `I-ai`'s dequantization caveats.

SOURCES

- "Harvest now, decrypt later": https://en.wikipedia.org/wiki/Harvest_now,_decrypt_later
- The Quantum Insider "Harvest Now, Decrypt Later — Why Should You Care?" (May 2026)
- NSA CNSA 2.0 (quantum-resistant mandate, 2027/2035)
- MDPI "Harvest-Now, Decrypt-Later: A Temporal Cybersecurity Risk in the Quantum Transition"

Quantum in Logistics & Supply Chain · I-logistics

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Routing, scheduling, bin-packing, and supply-chain optimization are combinatorial (often NP-hard), and vendors pitch QAOA/ annealing for vehicle routing, traffic flow, cargo loading, and network design. This is the most "demo-friendly" industry because small optimization instances run on today's hardware and produce photogenic results. It is also where the **quantum-inspired vs quantum-hardware** conflation is most misleading in the press.

REAL ACTIVITY (NAMED, DATED)

- **Volkswagen** — early real-world **traffic-routing pilot in Lisbon** (2019 WebSummit), near-real-time bus routes via a quantum hybrid solver over live traffic/passenger data. A landmark demo, small fleet.
- **Airbus + IonQ** — cargo/aircraft loading via the **MAL-VQA** algorithm on trapped-ion hardware (arXiv 2504.01567, 2025). Airbus + BMW ran a **2024 Quantum Computing Challenge** including a sustainable-manufacturing logistics task.
- **Deutsche Bahn (Netz AG) + Cambridge Quantum** — **train rescheduling** with the F-VQE algorithm on realistic delayed timetables (part of Digitale Schiene Deutschland). DLR also pursuing quantum rail scheduling.
- **DHL** — quantum-inspired route-optimization work; IBM collaboration factoring customs, weather, fuel.
- **IonQ freight/logistics pilots** (2024–25) — container/freight optimization; IonQ has leaned into logistics as a commercial narrative.

KEY GRADED CLAIMS

- T3 Aircraft cargo loading via MAL-VQA on IonQ hardware — arXiv 2504.01567 (demonstrated at small scale)
- T3 Train rescheduling via F-VQE — Cambridge Quantum/DB Netz (demonstrated, realistic-instance study)
- T4 VW Lisbon traffic routing — press pilot (hybrid solver, small fleet)
- T5 Multi-billion logistics-optimization TAM projections — analyst forecasts (speculative; classical solvers already dominate)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** the workflow *runs* on today's hardware for small instances — a feasibility fact, not an advantage claim.
- **Promise:** vehicle routing, rail rescheduling, cargo loading at production scale — awaits fault tolerance and better encodings.
- **Hype:** "quantum optimizes global supply chains today" — most near-term wins are quantum-*inspired* (classical) methods relabeled.

HONEST ASSESSMENT

Logistics produces the most demos and the least proven advantage. Every result is a small instance where classical solvers (Gurobi, CPLEX, OR-Tools) and quantum-*inspired* methods still win at production scale. The value shown is that the workflow runs, and the genuine near-term wins come from quantum-inspired algorithms that borrow quantum structure while running on classical hardware — honest progress that is not a quantum computer. Genuine routing advantage awaits fault tolerance; realistic: **end of decade**.

SOURCES

- Q-CTRL Airbus/BMW logistics: https://q-ctrl.com/blog/exploring-the-future-of-quantum-powered-logistics-with-airbus-and-bmw-group
- Cambridge Quantum/DB train scheduling: https://www.hpcwire.com/off-the-wire/cambridge-quantum-and-deutsche-bahn-netz-ag-leverage-latest-quantum-algorithms/
- Aircraft loading: https://arxiv.org/html/2504.01567v1
- DHL "The quantum leap in logistics"; IonQ cargo optimization

Quantum in Manufacturing & Industrials · I-manufacturing

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Manufacturing quantum pitches cluster on **combinatorial optimization**: production sequencing, job-shop scheduling, factory-floor layout, paint-shop switching, supply/quality allocation, and quantum-enhanced digital twins that simulate whole production lines to find bottlenecks. Because these are annealing-friendly optimization problems, manufacturing has produced some of the earliest claimed *production* deployments in the whole field.

REAL ACTIVITY (NAMED, DATED)

- **Ford Otosan + D-Wave** (2024) — deployed a **hybrid-quantum application in production** for Ford Transit manufacturing sequencing; reported **~50% reduction in scheduling time** at one site (annealing-based). See `I-auto`. The single most-cited "quantum in production" case.
- **D-Wave** (Qubits 2025 user conference, March 2025) — announced expanded commercial optimization offering and customer growth across manufacturing/scheduling; D-Wave reported its first meaningful optimization *revenue* — real dollars, on annealing whose advantage-vs-classical is contested.
- **Digital-twin case studies** — a "global automotive company" reported ~20% production-time reduction using quantum-enhanced digital-twin scenario simulation (case-study framing, unnamed).
- **WEF "How manufacturing is harnessing quantum technologies"** (Oct 2025) — frames process optimization, quality, and materials as the leading near-term manufacturing targets.

KEY GRADED CLAIMS

- T3/T4 Ford Otosan production sequencing ~50% faster — D-Wave, 2024 (demonstrated, in production; annealing, advantage-vs-classical contested)
- T4 ~20% digital-twin production-time reduction — unnamed automotive case study (claimed)
- T4 D-Wave optimization revenue growth — vendor report, Mar 2025 (real revenue, contested advantage)
- T5 Broad manufacturing process-optimization TAM — WEF/analyst framing (forecast)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** an annealing-based scheduling app runs in a live plant (Ford Otosan) and customers pay for optimization services.
- **Promise:** gate-model optimization and quantum digital twins beating classical HPC — no demonstrated advantage yet.
- **Hype:** "quantum computer beats classical solvers on scheduling" (contested); fully quantum digital twins.

HONEST ASSESSMENT

Manufacturing shares automotive's honest split. Annealing and quantum-inspired optimization deliver real workflow value today (Ford Otosan is a legitimate production case), and whether a *quantum* processor beats a good classical solver on the same scheduling problem is contested — D-Wave's own advantage claims live under `O-advantage`, and classical and quantum-inspired methods frequently match it. Digital twins remain mostly classical HPC with a quantum-optimization add-on. The near-term story is "quantum-flavored optimization services that sometimes help." Gate-model manufacturing advantage: **end of decade**. This is one of the more grounded L4 verticals precisely because it under-claims relative to finance/pharma.

SOURCES

- D-Wave customer growth / Ford Otosan: https://www.dwavequantum.com/company/newsroom/press-release/d-wave-highlights-quantum-optimization-customer-growth-and-introduces-expanded-offering-to-accelerate-adoption-and-deployment/
- WEF "How manufacturing is harnessing quantum technologies" (Oct 2025)
- Augmented Qubit "Definitive Report on Quantum-Enhanced Digital Twins"

Quantum in Media & Entertainment · I-media

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

The thinnest but genuinely pitched industry node: procedural content generation (PCG) for games/worlds seeded by true quantum randomness (`A-qrng`), quantum-assisted rendering and physics simulation, content-recommendation optimization, and ad/creative optimization. The one place quantum offers something classical cannot fake is **certified true randomness** as a generative resource; the rest is optimization/ML reheated for a creative audience. (Sports analytics/betting, sometimes grouped here, is a marginal recommendation/optimization case with no quantum-specific hook.)

REAL ACTIVITY (NAMED, DATED)

- **James Wootton / Qiskit** — the reference thread: "Procedural Generation Using Quantum Computation" (Qiskit blog + IEEE CG&A 2024, *Quantum Wave Function Collapse for PCG*, arXiv 2312.13853). Demonstrates quantum-seeded wave-function-collapse PCG on real hardware/simulators. Research and demos, not shipped games.
- **arXiv 2508.09683** (2025) — *Procedural Generation and Games at the Dawn of Fault-Tolerant Quantum Computing* frames the near-term realistic role as hybrid quantum-classical subroutines injecting variability, not standalone quantum rendering.
- **QRNG in market** — ID Quantique, Quantinuum, and cloud QRNG feeds ship true randomness usable for generative seeds; a component, not a media product.
- **Telefónica Servicios Audiovisuales** and similar trade pieces sketch 1–3 year horizons for recommendation optimization and 3–5 years for rendering/simulation. Vendor/advisory forecasts.

KEY GRADED CLAIMS

- T3 Quantum wave-function-collapse PCG runs on quantum hardware/simulators — Wootton, IEEE CG&A 2024 / arXiv 2312.13853 (demonstrated; research artifact, no advantage)
- T4 Quantum recommendation/ad optimization near-term — vendor framing (claimed)
- T6 Quantum rendering/simulation productized within 3–5 years — trade-press forecast (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** QRNG-seeded generative art/PCG runs in toy form — a novelty resource, no performance advantage (classical PRNGs are indistinguishable for entertainment).
- **Promise:** hybrid quantum-classical PCG subroutines for variability.
- **Hype:** quantum rendering, quantum recommendation engines, quantum-accelerated content pipelines — no deployed pilots.

HONEST ASSESSMENT

This is a curiosity node. The QRNG-as-creativity angle is real and shipped in toy form, and it is a novelty resource rather than a performance advantage — classical PRNGs are indistinguishable for entertainment purposes. Rendering, recommendation, and ad optimization are the usual optimization/ML pitches with no media-specific advantage and no deployed pilots. No studio or platform runs quantum hardware in production. Keep this node for completeness; its honest weight is low and mostly T3-demo plus T6-forecast.

SOURCES

- https://medium.com/qiskit/introducing-procedural-generation-using-quantum-computation-956e67603d95
- https://arxiv.org/pdf/2312.13853 (Quantum Wave Function Collapse for PCG)
- https://arxiv.org/html/2508.09683v1 (PCG at the dawn of FTQC)
- https://www.telefonicaserviciosaudiovisuales.com/en/audiovisual-divulging-articles/quantum-computing-and-its-impact-on-the-media-and-entertainment-industry/

Quantum in Mining & Mineral Exploration · I-mining

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from `I-extractive`; mining's story is sensing-led and distinct from oil & gas reservoir compute)

THE PITCH

Mining's quantum story is led by **sensing**, not computing — the opposite balance from oil & gas (`I-extractive`). Cold-atom **gravimeters** and NV-diamond **magnetometers** (`A-gravimetry`/`A-magneto`/`A-nvsensing`) detect ore bodies and density anomalies from the air, sharpening exploration for critical minerals (copper, lithium, nickel, cobalt, rare earths) needed for the energy transition. Secondary compute threads: mine-plan/haulage optimization and mineral-processing chemistry (froth flotation, leaching catalysts).

REAL ACTIVITY (NAMED, DATED)

- **SBQuantum + Silicon Microgravity — QUAMINEX** (announced March 2024) — a drone-mounted system fusing a **diamond quantum vector magnetometer** with a MEMS gravimeter to map underground deposits in 3D. Funded by Canada (CAD $500k) and the UK (£414k). SBQuantum's magnetometer is miniaturized to a handheld <1 lb, <4 W device; claims ≥30% better resolution than the current magnetic-mapping standard.
- **SBQuantum** (2025) — also pursuing defense/aerospace magnetic navigation; a June 2025 Metal Tech News update tracks continued exploration-tech progress.
- **Rio Tinto** and other majors — using advanced geophysical exploration platforms; quantum sensors are entering the toolchain as a nearer-term option than quantum compute.
- **Mine-optimization compute** — QUBO haulage/scheduling and processing-chemistry pilots exist at the same small-instance, no-advantage stage as `I-logistics`/`I-chem`.

KEY GRADED CLAIMS

- T3/T4 Drone-based diamond-magnetometer + gravimeter mineral mapping, ≥30% resolution gain — SBQuantum/Silicon Microgravity, 2024 (demonstrated dev program; resolution claim vendor-stated)
- T4 Handheld <1 lb diamond quantum magnetometer — SBQuantum (vendor product spec)
- T3 QUBO mine-plan/processing optimization — method papers (toy scale, no advantage)
- T5 Critical-minerals quantum-exploration market — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** diamond quantum magnetometers and cold-atom gravimeters are real, deployable field instruments — mining's genuine near-term quantum tech.
- **Promise:** drone-fused magneto+gravity 3D exploration (QUAMINEX) at production scale, quantum processing-chemistry.
- **Hype:** "quantum computers optimize the mine" — the compute side is the same small-instance no-advantage story as logistics.

HONEST ASSESSMENT

Mining is a sensing-led vertical, and that is what makes it credible. Diamond quantum magnetometers (`A-nvsensing`) and cold-atom gravimeters are real instruments with a plausible short horizon for critical-minerals exploration — a strategically important use given lithium/copper/rare-earth demand. The QUAMINEX program is dated, government-funded, and concrete, though the resolution numbers are vendor-stated and the systems are still commercializing (niche, expensive). Quantum *computing* for mine planning and mineral-processing chemistry is at the small-instance no-advantage stage. Split the story: sensing has a credible near path; compute is a **2030s** bet.

SOURCES

- SBQuantum + Silicon Microgravity QUAMINEX: https://thequantuminsider.com/2024/03/01/sbquantum-and-silicon-microgravity-partner-to-accelerate-mining-exploration-using-quantum-sensing-with-funding-from-uk-canada/
- SBQuantum diamond magnetometry: https://magneticsmag.com/sbquantum-pioneers-quantum-magnetometry-for-defense-aerospace-minerals-exploration/
- Canadian Mining Journal "Quantum technologies in mining exploration"
- Metal Tech News "Quantum leap in mineral exploration tech" (June 2025)

Quantum in Nuclear & Fusion · I-nuclear

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from `I-energy`; fusion/nuclear is deep enough to stand alone)

THE PITCH

Three threads. (1) **Fusion plasma** — magnetized-plasma dynamics in tokamaks/stellarators is a quantum many-body and stiff-PDE problem; quantum simulation and QAOA-for-stability-optimization are pitched to accelerate confinement modeling and coil design. (2) **Nuclear structure & reactions** — lattice/nuclear-physics Hamiltonians are a native quantum-simulation target (the same `S-qsim` logic as chemistry, at the nuclear scale). (3) **Fission engineering** — reactor core optimization, fuel-loading, and materials-under-irradiation modeling as QUBO/chemistry problems.

REAL ACTIVITY (NAMED, DATED)

- **Frontiers in Physics** (March 2025) — *The role of quantum computing in advancing plasma physics simulations for fusion energy and high-energy physics* — a substantive review; QAOA is applied to optimize tokamak plasma-stability criteria, Hamiltonian-simulation to plasma dynamics.
- **PPPL (Princeton Plasma Physics Lab)** — 2025 Kaul Foundation Prize to a team optimizing 3D magnetic fields to control tokamak edge instabilities; the broader lab explores quantum-computing methods for plasma. HPC-classical today, quantum-curious.
- **Nuclear-structure quantum simulation** — VQE/quantum-simulation demos of light-nucleus (deuteron, $^4\mathrm{He}$-scale) binding energies on superconducting/trapped-ion hardware (Oak Ridge and others, ongoing since ~2018) — small but genuine physics results.
- **Fusion startups** (Commonwealth Fusion, TAE, Tokamak Energy) and **ENN** (proton-boron) run heavy classical HPC plasma simulation; quantum is on watch-lists, not in the loop.

KEY GRADED CLAIMS

- T2/T3 VQE/quantum-simulation of light-nucleus binding energies on real hardware — nuclear-physics literature (demonstrated at small scale; landmark physics, not engineering advantage)
- T3 QAOA for tokamak plasma-stability optimization / Hamiltonian-sim for plasma — Frontiers 2025 review (method framing + small demos)
- T4 Quantum-accelerated fusion reactor design — vendor/lab aspiration (roadmap)
- T5 Fusion/nuclear quantum TAM — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** quantum computers have simulated small nuclei (deuteron-scale) — real nuclear-physics results, no engineering payoff.
- **Promise:** plasma-stability optimization and full plasma dynamics — strong theoretical fit, no advantage over classical HPC yet.
- **Hype:** "quantum computing will crack fusion" — fusion's bottleneck is engineering and confinement, and its modeling is elite classical HPC.

HONEST ASSESSMENT

Nuclear/fusion has the best *physics* pedigree of any young L4 node — simulating nuclear and plasma Hamiltonians is exactly what quantum computers are for, and the deuteron/light-nucleus results are genuine science. It also has the widest gap to engineering value: fusion plant design runs on world-class classical supercomputing (M3D-C1, JOREK, NIMROD), and quantum methods are at the review-and-small-demo stage. Plasma-stability QAOA is a promising formulation without demonstrated advantage. Realistic engineering value: **2030s+**, gated on fault tolerance. Near-term, the honest role is accelerating specific nuclear/plasma sub-Hamiltonians as a research tool.

SOURCES

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- PPPL 3D-field / Kaul Prize: interestingengineering.com/energy/3d-magnetic-field-breakthrough-nuclear-fusion
- Nuclear-structure VQE (deuteron) — Oak Ridge quantum-simulation literature
- AZoQuantum "Tokamak Reactors and the Key to Nuclear Fusion"

Quantum in Pharma & Healthcare · I-pharma

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Drug discovery is chemistry, and chemistry is quantum — simulating molecular ground/excited states, protein folding, and binding affinities is the canonical Feynman-era pitch (see `S-qsim`). Quantum promises to model molecules classical methods approximate poorly, shrinking discovery cycles. Pharma is a top-three enterprise adopter. This node covers **drug discovery / molecular simulation**; healthcare *imaging and diagnostics* (a nearer-term quantum-sensing story) is split into `I-healthimaging`.

REAL ACTIVITY (NAMED, DATED)

- **Cleveland Clinic + IBM** — "Discovery Accelerator," first on-site private-sector quantum computer (IBM System One at Lerner Research Institute, live 2023). 50+ joint projects; Heron chip upgrade + HPC coupling Dec 2025. **Algorithmiq + Cleveland Clinic + IBM won the $2M Wellcome Leap Q4Bio prize** (~April 2026) for simulating photodynamic-therapy processes on up to 100 qubits — the most concrete pharma-quantum result to date.
- **IBM "quantum-centric supramolecular interactions"** (arXiv 2410.09209, Oct 2024) — accurate simulation of supramolecular systems on Heron; a refereeable capability demo, still small-molecule scale.
- **Boehringer Ingelheim** — formal quantum program since 2021, partnered **Google Quantum AI** for molecular-dynamics work.
- **AstraZeneca + IonQ** (via AWS/NVIDIA, published 2025) — reported **~20x speedup** on a drug-discovery workflow subroutine; vendor-framed.
- **Qubit Pharmaceuticals + Pasqal** — drug-discovery tasks on neutral-atom hardware. **Menten AI + D-Wave** — peptide design via hybrid annealing. **Roche pRED + QC Ware**; **Moderna + IBM** (mRNA structure).
- **Quantinuum Helios** (48 logical qubits, late 2025) — approaching the scale where quantum chemistry begins to *challenge* classical on specific systems (~20 active orbitals), per 2026 pharma reviews.

KEY GRADED CLAIMS

- T2/T3 Photodynamic-therapy ground/excited-state simulation on up to 100 qubits (Q4Bio) — Algorithmiq/Cleveland Clinic/IBM (competitively judged; demonstrates a *path*, not advantage over DFT/coupled-cluster)
- T3 Quantum-centric supramolecular simulation — IBM, arXiv 2410.09209 (preprint capability demo)
- T4 AstraZeneca/IonQ 20x workflow speedup — vendor+pharma announcement (narrow subroutine)
- T4 Boehringer/Google, Qubit/Pasqal, Menten/D-Wave pilots — partnership PRs (exploratory)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** quantum hardware can now simulate ~20-orbital molecules accurately — matching what classical methods already do well.
- **Promise:** photodynamic-therapy chemistry, peptide design, binding affinity — real refereeable demos at 100-qubit scale, still short of design-grade accuracy on hard targets (FeMoco, cytochrome P450).
- **Hype:** "quantum discovered a drug," end-to-end "20x faster pipelines," near-term clinical impact.

HONEST ASSESSMENT

No approved drug has been discovered by a quantum computer as of mid-2026, and none is close — a fact 2026 pharma reviews state plainly (no quantum computer yet simulates a drug-relevant molecule better than classical chemistry on the same system). The Q4Bio work is the strongest signal: real chemistry circuits at 100-qubit scale, competitively judged. The "20x" claims are narrow subroutines. Pharmaceutically useful accuracy on hard active sites needs fault tolerance (millions of gates). Realistic operational impact: **late 2020s into the 2030s**. Pharma keeps investing because the eventual payoff is enormous.

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- Quantum Computing Watch List for Pharma R&D Leaders 2026 (Sakara Digital)
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Quantum in Retail · I-retail

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Retail's pitch is demand forecasting, dynamic pricing, assortment and shelf-space optimization, inventory/replenishment, and recommendation. All reduce to optimization (QUBO/QAOA over combinatorial assortment and stocking) or ML (quantum-enhanced forecasting, quantum neural networks). The claimed payoff is fewer stockouts and less overstock across huge SKU catalogs — a combinatorial space classical solvers already handle well, which is the tension.

REAL ACTIVITY (NAMED, DATED)

- **Strangeworks / Quantagonia** (Aug 2025) — Strangeworks acquired Quantagonia to fold quantum-enhanced demand forecasting and decision-making into its platform. Vendor consolidation, not a named retailer deployment.
- **Method research** — *Exploring Quantum Neural Networks for Demand Forecasting* (Entropy/MDPI, 2025, vol 27:490): QNN forecasting on benchmark datasets, no advantage over classical shown.
- **Analyst-heavy** — most visible activity is market-report noise ("Quantum-Enhanced Demand Forecasting" TAM projected ~\$2B (2025) → ~\$2.64B (2026), ~32% CAGR). T5 forecasts, not pilots.
- Named marquee retail deployments (Walmart/Amazon/Kroger running a QPU in the loop) do not exist in the public record; retail quantum work sits inside logistics (`I-logistics`) and finance-style optimization pilots.

KEY GRADED CLAIMS

- T3 Quantum neural networks tested for demand forecasting on benchmark data — MDPI Entropy 2025 (demonstrated on toy data; no advantage vs classical)
- T4 Quantum-enhanced forecasting productized (Strangeworks/Quantagonia) — vendor consolidation announcement (claimed)
- T5 "Quantum-enhanced demand forecasting" multi-billion-dollar market — GII/Research&Markets forecasts (speculative; double-counted with logistics/finance, inflation-unadjusted)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** nothing operational — QNN forecasting demos on benchmark data, matched by classical ML.
- **Promise:** assortment/pricing optimization at catalog scale — high bar because classical MILP is strong and cheap.
- **Hype:** the TAM reports themselves — this node is where market-sizing noise outweighs any real pilot.

HONEST ASSESSMENT

Retail is one of the thinner industry pitches. Demand forecasting and assortment are problems where classical ML and modern MILP solvers are strong and cheap, so the quantum bar for advantage is high and unmet. The visible "activity" is dominated by market-sizing reports and one vendor acquisition — a T5-heavy corner with essentially no retailer pilots. No retailer runs quantum hardware operationally. Realistic value is far off and, even then, likely marginal versus classical optimization. Grade the TAM numbers hardest here.

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- https://www.mdpi.com/1099-4300/27/5/490 (QNN demand forecasting)
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Quantum in Semiconductors & EDA · I-semiconductor

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — genuinely new, non-duplicative)

THE PITCH

Two directions, both real. (1) **Quantum FOR chips** — quantum computing applied to semiconductor design: electronic-structure simulation of new device materials (high-k dielectrics, 2D channels, photoresists), computational lithography, and QUBO placement/routing in EDA. (2) **Chips FOR quantum** — the semiconductor industry is quantum's manufacturing base: foundries fabricate transmon and spin-qubit chips, and a "fabless quantum chip" model is emerging (design house + foundry). This node covers the *application* direction; the fab/supply-chain physics lives in `H-fab`/`H-silicon`, cryo-CMOS in `H-cryocmos`.

REAL ACTIVITY (NAMED, DATED)

- **NVIDIA + TSMC** (GTC 2025) — the honest near-term truth is *classical* acceleration: **cuLitho** (20–50% better cost/cycle-time on computational lithography) and **cuEST** (~50x faster electronic-structure chemistry for materials). This is GPU HPC, and it is the bar quantum must clear in this vertical.
- **Q-EDA research stack** — EDA-Q, KQCircuits, QPDK, GDSII-to-wafer flows (arXiv 2606.17956, "Fabless Quantum Chip Design and Commercial Production," 2026) demonstrate automated topology design, layout, DRC/LVS/DFM, and mask-data prep — for designing *quantum* chips, using classical EDA methods.
- **Materials simulation pilots** — Mitsubishi Chemical + Xanadu (semiconductor materials), IBM/Azure Quantum Elements chemistry platforms pitched to fabs and materials houses.
- **Infineon, imec, TSMC** — running spin/superconducting qubit process R&D that doubles as advanced-node process learning.

KEY GRADED CLAIMS

- T4 Classical GPU acceleration of lithography/materials (cuLitho/cuEST) — NVIDIA/TSMC, 2025 (vendor; not quantum, sets the bar)
- T3 Automated quantum-chip EDA flow (topology→mask) — arXiv 2606.17956 (preprint; classical EDA for quantum chips)
- T4 Quantum electronic-structure simulation for device materials — vendor pilots (exploratory)
- T5 Semiconductor-design quantum TAM — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** classical GPU HPC (cuLitho/cuEST) accelerates real fab workflows now — this is the competition, not quantum.
- **Promise:** quantum simulation of next-node device materials and QUBO place-and-route — exploratory, fault-tolerance-gated.
- **Hype:** "quantum computers design chips today."

HONEST ASSESSMENT

The semiconductor industry is central to quantum as its *fab base* — that part is real and load-bearing (`H-fab`, `H-silicon`, `H-cryocmos`). Quantum computing *applied to* chip design is early: materials-simulation pilots inherit `I-chem`'s fault-tolerance wall, and EDA placement/routing is QUBO optimization that classical solvers dominate. The most concrete 2025–26 wins in "AI/compute for semiconductors" are GPU-classical (NVIDIA cuLitho/cuEST), which is a useful reality check on where the value actually flows. Realistic quantum-for-design value: **2030s**, gated on fault tolerance.

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- "Fabless Quantum Chip Design and Commercial Production": https://arxiv.org/pdf/2606.17956
- DCD "Quantum of promise: How to build a quantum chip"
- Mitsubishi Chemical/Xanadu semiconductor materials (IBM case studies)

Quantum in Space & Earth Observation · I-space

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Space is a natural home for quantum *sensing and communication* more than computing: satellite-based QKD (`A-satqkd`) for global secure comms, ultra-stable atomic clocks (`A-clocks`) for PNT and constellation timing, quantum magnetometers/gravimeters and atom interferometers for GPS-free spacecraft navigation and Earth observation, plus optimization for orbit/constellation/mission planning. The harsh, GPS-independent environment rewards quantum's absolute-reference advantages.

REAL ACTIVITY (NAMED, DATED)

- **ESA ACES** (Atomic Clock Ensemble in Space) — installed on the ISS **April 2025**; the most accurate clock ensemble ever flown, targeting fractional stability ~1×10^{-16}, comparing on-board cesium (PHARAO) + H-maser clocks to ground stations — a step toward relativistic geodesy and SI-second work (`A-timedist`).
- **Boeing Q4S** — space-qualified quantum-networking payload passed environmental qualification and entanglement-swapping ground tests; orbital demo planned. Boeing/Space Force flew "the highest-performing quantum inertial sensor in space" on X-37B (**Aug 2025**, see `I-aerospace`).
- **University of Vienna** (2025) — launched a compact **photonic quantum computer into orbit** (~10 W, ~3 L) — a demonstration that quantum hardware survives launch/space, not a useful computation.
- **China Micius** — the landmark satellite-QKD platform (`A-satqkd`), with a follow-on constellation program.
- **NASA CAL** (Cold Atom Lab, ISS) — BEC/atom-interferometry science platform underpinning future space gravimetry.

KEY GRADED CLAIMS

- T2/T3 ACES ~1×10^{-16} optical-clock-class stability in orbit — ESA, 2025 (demonstrated hardware milestone)
- T3 Space-qualified quantum-networking payload (entanglement swapping) — Boeing Q4S (demonstrated ground test; orbital pending)
- T3 Photonic quantum computer operates in orbit — U. Vienna 2025 (demonstrated survival, not useful compute)
- T5 Space-quantum market forecasts — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** ACES clocks, quantum inertial sensors, and satellite QKD are flying and returning data now — mostly `A`-layer sensing/timing/comms.
- **Promise:** operational satellite quantum networking (Q4S), space gravimetry for Earth observation.
- **Hype:** quantum computers doing orbit/constellation optimization in space — classical HPC handles it fine.

HONEST ASSESSMENT

Space quantum is real where it is sensing/timing/comms and aspirational where it is computing. ACES, quantum inertial sensors, and satellite QKD fly and return data now — meaningful, near-term, and mostly `A`-layer tech rather than L4 compute. Orbit/constellation optimization by quantum computers is speculative and well-served by classical HPC. The Vienna orbital "quantum computer" is a survivability demo. Split cleanly: quantum sensors and clocks are arriving in space this decade; quantum computing in/for space is a long-horizon bet. As with aerospace, the near-term space win is a sensor.

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- VisionSpace "Quantum Horizons: Redefining Satellite Operations in 2025"
- aero-defence.tech Boeing Q4S payload integration

Quantum in Telecom & Networking · I-telecom

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Telecom's quantum story is mostly about being the *substrate* for quantum-secure communication — deploying QKD (`A-qkd`) and post-quantum crypto (`A-pqc`) across metro and backbone networks, plus longer-term ambitions to build the quantum internet (`A-qinternet`). Secondary threads: network-optimization and radio-resource scheduling. Carriers monetize "quantum-safe-as-a-service." Two tracks run at very different speeds: PQC is software crypto shipping now; QKD is hardware, deployed but niche.

REAL ACTIVITY (NAMED, DATED)

- **BT** — deployed the **UK's first commercial quantum-secured metro network** in London (2022, with Toshiba; customers included HSBC, EY); trialing long-distance and satellite QKD.
- **SK Telecom** — launched a **hybrid QKD-PQC** encryption solution (Oct 2024) for 5G backbone/VPN/data centers with Nokia and ID Quantique at Equinix Seoul; partnered IonQ (Feb 2025) to embed trapped-ion compute into AI platforms.
- **Verizon** — PQC migration focus; participating in NIST PQC standardization; early QKD experiments.
- **NTT** — Quantum Datacenter Alliance member (2025); long-standing quantum research arm; IOWN all-photonics network program.
- **Telefónica, Orange, KT** — national QKD testbeds (EuroQCI-aligned in Europe).
- Industry-wide: by end-June 2025, ~29 operators had made ~75 quantum-related statements/deployments.

KEY GRADED CLAIMS

- T3/T4 Commercial quantum-secured metro network live with enterprise customers — BT/Toshiba (demonstrated deployment)
- T3/T4 Hybrid QKD-PQC service in production data centers — SK Telecom/Nokia/ID Quantique, Oct 2024 (demonstrated)
- T2 PQC (ML-KEM/ML-DSA) is the NIST-standardized path most carriers actually adopt — see `A-pqc` (established standards)
- T5 "Quantum-safe telecom" TAM projections — forecast

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** PQC migration (classical software crypto) is real, mandated, and underway across carriers now.
- **Promise:** commercial QKD links (BT, SKT) — genuine but small, expensive, distance-limited footprints.
- **Hype:** "the quantum internet is here," QKD as a general-purpose replacement for PQC (security agencies disagree).

HONEST ASSESSMENT

Telecom is a two-track reality. PQC migration is real and happening — software crypto, no quantum hardware, where carriers spend seriously. QKD is deployed and niche — expensive, distance-limited, point-to-point, and NSA/UK NCSC prefer PQC over QKD for general use. The BT/SKT deployments are genuine but small commercial footprints. The quantum internet is a research decade away. Telecom's honest near-term quantum value is defensive crypto, mostly the classical-software PQC kind.

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Quantum in Weather & Climate Modeling · I-weather

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from `I-climate`, which is chemistry/carbon-capture)

THE PITCH

Distinct from `I-climate` (quantum *chemistry* for carbon capture/fertilizer), this node is about **numerical weather prediction (NWP) and climate dynamics** — the fluid-dynamics side. Weather/climate models solve Navier-Stokes and radiative-transfer PDEs on enormous grids; the pitch is quantum linear-solvers (HHL, `S-hhl`) for the linear-algebra core, quantum algorithms for turbulent/CFD flow, and QML for pattern-finding in satellite/sensor data (extreme-event and precipitation nowcasting).

REAL ACTIVITY (NAMED, DATED)

- **Open Quantum Institute (OQI, at CERN, pilot phase 2024–2026)** — runs a **"Weather and Climate Forecasting" project** whose core is a *quantum fluid-dynamics* solver to improve forecast reliability. Multi-institution, pre-commercial, the most concrete organized effort.
- **arXiv 2502.10488** (Feb 2025) — *Opportunities and challenges of quantum computing for climate modelling* — a serious survey mapping HHL/QLSA and Hamiltonian-simulation methods onto dynamical cores, with honest caveats on data loading (`S-qram`) and readout.
- **arXiv 2509.01422** (2025) — *Exploring Quantum Machine Learning for Weather Forecasting* — QNNs for wind-speed prediction on benchmark data; potential-outperformance framing, small scale.
- **BAMS 2023** — *Quantum Computers for Weather and Climate Prediction: The Good, the Bad, and the Noisy* — the reference sober review from the meteorology community.

KEY GRADED CLAIMS

- T3 Quantum linear-solver / Hamiltonian-simulation methods mapped onto NWP dynamical cores — arXiv 2502.10488 (survey; no end-to-end advantage)
- T3 QNN wind-speed/precipitation nowcasting on benchmark data — arXiv 2509.01422 (toy-scale demonstration)
- T4 OQI quantum fluid-dynamics forecasting pilot — CERN/OQI (exploratory, pre-commercial)
- T5 Quantum-weather market projections — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** nothing operational — surveys and toy-scale QNN nowcasting demos, matched by classical ML.
- **Promise:** quantum linear-solvers for the PDE core and quantum CFD — real theoretical hooks, blocked by data-loading and readout bottlenecks.
- **Hype:** "quantum will fix weather forecasting" — today's NWP is world-class classical HPC that quantum cannot yet touch.

HONEST ASSESSMENT

Weather/climate modeling is one of the harder places to get a quantum win: the HHL-style speedup for linear systems is real on paper but dies on the input/output problem (loading a petabyte-scale state and reading out a full field), which the field's own 2025 surveys flag plainly. NWP is a mature, heavily-optimized classical-HPC discipline (ECMWF, NOAA), so the bar is very high. The credible near-term thread is QML *nowcasting* on sensor/satellite data, which is small and classically matchable. The OQI pilot is the honest bright spot: organized, dated, and explicitly pre-commercial. Realistic value: **2030s+**, gated on fault tolerance and a qRAM-class data-loading solution.

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Chapter 6 · Money, Nations, and Standards

China's parallel software stack — fork & soft-power risk · E-china-stack

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk, field-flagged)

SUMMARY

China runs a **full parallel quantum-software stack** that mirrors the Qiskit/Cirq/PennyLane ecosystem (E-oss) end to end, so a Chinese developer, chip, and cloud can operate without touching any US-governed toolchain. The anchors: **Origin Quantum** (Hefei) with **QPanda** (programming framework), **Origin Pilot** (a quantum operating system, open-sourced Feb 2026), **ChemiQ** (chemistry), and **Tianji** (measurement-and-control, v4.0 supporting 500+ qubits, May 2025); **Baidu's Paddle Quantum** (quantum-ML on the PaddlePaddle DL framework); and **Huawei's HiQ** (simulator + algorithm library on Huawei Cloud). This is the software counterpart to the hardware story in E-china. The strategic point is a **fork risk**: whoever's framework developers learn first shapes which hardware they later reach for, and export controls (E-export) make convergence on one global toolchain less likely each year. Origin is notable as arguably the only company anywhere building the *entire* stack — processor, cryogenics, control system (Tianji), OS (Origin Pilot), framework (QPanda), and cloud — in-house.

THE LANDSCAPE (GRADED)

- T2 **Origin QPanda** is China's leading open-source quantum programming framework; Origin operates the Wukong superconducting machine on a public cloud — company/press (established)
- T3 **Origin Pilot** (quantum OS for multi-task scheduling / resource management) was **open-sourced in February 2026** — a deliberate soft-power move to seed adoption around Chinese hardware — postquantum.com (reported)
- T3 **Tianji 4.0** measurement-and-control system supports 500+ qubits and reduces PhD-level bring-up to standard engineering workflows (May 2025) — Origin (claimed; vendor)
- T2 **Baidu Paddle Quantum** — open-source QML toolkit on PaddlePaddle; Baidu built the 36-qubit Qianshi processor (2023) before **donating its quantum lab to BAAI (2024)** — Baidu docs / press (established). The software outlived the hardware program.
- T2 **Huawei HiQ** — cloud quantum simulator + algorithm library integrated with Huawei Cloud — Huawei (established)

KEY GRADED CLAIMS

- T3 The Chinese stack is a **genuine fork**, not a veneer over Western tools — QPanda/Paddle Quantum/HiQ have independent APIs and do not depend on Qiskit/Cirq (reported)
- T3 **Soft-power play:** open-sourcing Origin Pilot/QPanda mirrors IBM's Qiskit funnel strategy — free tooling as a channel to Chinese cloud + hardware, aimed at Belt-and-Road / non-aligned markets (analysis)
- T4 Interop with OpenQASM/QIR (E-oss) exists partially, but a determined bifurcation along the export-control line would strand cross-stack portability (claimed)

CONFLICTS / OPEN QUESTIONS

- **Does the world converge or fork?** OpenQASM/QIR portability (E-oss) vs export-control-driven decoupling (E-export). Resolution: whether Chinese and Western clouds keep honoring each other's IRs through 2027–28.
- Corporate retreat paradox: Baidu/Alibaba exited *hardware* (E-china, E-patents) yet their *software* (Paddle Quantum) persists — soft power can outlast the lab that spawned it.
- Adoption outside China is unmeasured — GitHub stars and cloud-usage numbers for QPanda/HiQ are not independently audited.

SOURCES

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National programs — China · E-china

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

China runs the largest state-directed quantum program in the world, centered on **USTC (Hefei)**, the CAS system, and the **National Laboratory for Quantum Information Sciences** (Hefei; construction reported in the ~$1B+ range). Signature results: the **Micius** QKD satellite (2016), the **Jiuzhang** photonic-advantage series (now Jiuzhang 4.0, 3,050 photons), and the **Zuchongzhi** superconducting line (Zuchongzhi 3.0, 105 qubits, 2025). The famous "$15B commitment" is the least verifiable headline number in the field — cited for years with no primary source, with insiders putting real spending at a fraction of it. The **15th Five-Year Plan** (adopted early 2026) elevated quantum to the top-ranked future industry, ahead of AI and semiconductors. (China's parallel *software* stack — Origin/QPanda, Baidu Paddle Quantum, Huawei HiQ — is broken out as its own node, `E-china-stack`.)

THE NUMBERS (GRADED, WITH CAVEATS)

- T5 "$15B+" (often $15.3B) — the perennial headline commitment. No primary Chinese budget document supports it; postquantum.com traces it to circular citation. A USTC physicist estimated actual money at "maybe 25 percent of that" (~$3.75–5B). Treat as an upper bound on rhetoric, not a budget line.
- T5 >$1B — reported investment in the Hefei National Laboratory for Quantum Information Sciences (CSIS / secondary reporting). Construction-cost figures and research budgets get conflated.
- T5 ~$17.5B — estimated National Guidance Venture Fund backing quantum under the 15th FYP (postquantum.com, 2026). The fund spans multiple deep-tech sectors; quantum's slice is unknown — clear double-counting risk.
- T5 $15B in 2024 alone vs ~$7B combined US public+private — Asia Times framing (May 2026); methodology opaque, likely folds provincial + venture-guidance money.

KEY GRADED CLAIMS

- T2 Micius demonstrated space-to-ground QKD and entanglement distribution — Science/Nature (established). Mar 2025: the **Jinan-1** microsatellite sent a quantum-encrypted image ~13,000 km China → South Africa (USTC) — the longest intercontinental QKD link reported (T3).
- T3 Jiuzhang 4.0 claims sampling advantage ~10^54 over classical — USTC (claimed; boson-sampling advantage claims have historically eroded under classical spoofing — see S-bench)
- T2 Quantum named a frontier priority in the 14th FYP; top-ranked future industry in the 15th FYP — plan texts (established)
- T4 Program secrecy: military-adjacent work (sensing, comms) is unpublished, so open-source estimates undercount capability and possibly overcount civilian spend (claimed)

CONFLICTS / OPEN QUESTIONS

- **C-china-spend:** $15B headline vs ~25% insider estimate. Resolution: an audited budget line, which will likely never surface.
- How much of Jiuzhang/Zuchongzhi advantage survives classical counterattack (see O-advantage)?
- **Corporate retreat vs state ascent:** Baidu **donated** its quantum lab to BAAI and **Alibaba shuttered** its DAMO quantum effort (2023–24, equipment to Zhejiang University) — state labs and Origin Quantum now carry nearly all momentum. This is the inverse of the US private-capital surge (see E-vc).
- China is building domestic dilution-fridge, laser, and cryo supply to route around export controls (see E-export, E-supply-chain).

SOURCES

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Defense funding channel — DARPA QBI & procurement · E-defense

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Defense is a distinct funding channel from the civilian national program (E-us), and its centerpiece is **DARPA's Quantum Benchmarking Initiative (QBI)** — a reframing of the earlier US2QC program whose stated question is whether an **industrially-useful quantum computer is buildable by 2033** ("computational value exceeds its cost"). QBI's design move is unusual: DARPA **does not fund development directly**; it runs an interagency verification-and-validation team to separate hype from reality across every architecture, evaluating ~20 companies since mid-2024. This makes it as much a **national due-diligence instrument** as a grant program — a defense-run reality check the whole field cites. The June 2026 EOs added a second defense-adjacent thread by renaming DoD as the "Department of War" in tasking language and expanding the Quantum Counterintelligence Protection Team.

THE MECHANISM & NUMBERS (GRADED)

- T2 QBI operates in three stages — A (concept), B (yearlong R&D plan + risk-reduction prototypes), C (government V&V build feasibility) — darpa.mil (established)
- T2 Funding ladder: ~$1M (Stage A concept) → up to ~$15M (Stage B roadmap) → up to ~$300M (Stage C build/demo) — DARPA (established)
- T2 As of **6 Nov 2025, 11 companies advanced to Stage B**; ~20 evaluated across modalities since mid-2024 — DARPA / Quantum Insider (established)
- T3 Named performers include Quantinuum (Stage B), with Microsoft and PsiQuantum in the final US2QC phase and Google joining Stage A (Sep 2025) — company/DARPA releases (claimed/reported)

KEY GRADED CLAIMS

- T2 QBI's premise is **verification, not development funding** — its output is independent feasibility assessment, deliberately vendor-neutral — DARPA (established)
- T4 Defense **procurement of operational quantum systems** — essentially none yet; the channel today funds evaluation and prototypes, not fielded machines (claimed/absent). The nearest-term fielded quantum tech for defense is sensing/PNT (A-sensing, A-gravimetry), not compute.
- T5 "Industrially useful by 2033" is DARPA's framing question, not a prediction — a benchmark date, not a roadmap (forecast)

CONFLICTS / OPEN QUESTIONS

- **QBI is a filter, not a bet** — it explicitly is "not a competition," so being cut from a stage signals DARPA's V&V team found the concept unconvincing: a rare public *negative* signal in a hype-heavy field. How much weight the market gives those cuts is open, but it is one of the few credible independent screens (contrast the equity-picking turn in E-us/E-sovereign).
- The broader defense channel (NSA/DoD codebreaking interest — the original HNDL motive behind PQC, see A-pqc — plus service-lab sensing/PNT) is **opaque and classified**; public numbers understate real defense spend.
- **Double-counting risk:** QBI dollars and IARPA/service-lab programs get folded into "US quantum spending" tallies (E-us) — keep the defense channel separate.

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National programs — European Union · E-eu

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The EU's base layer is the **Quantum Technologies Flagship** (2018, €1B over ten years). On top of it sit **EuroQCI** (a quantum-secure communication network across all 27 member states plus ESA's **Eagle-1** satellite), **EuroHPC** quantum-computer procurements co-located with supercomputers, and — since **2 July 2025** — the **Quantum Europe Strategy** (COM(2025) 363), a five-pillar plan to make Europe a "quantum industrial powerhouse" by 2030. A binding **EU Quantum Act** is scheduled for adoption in 2026. Europe's strength is public research and startup density; its acknowledged weakness is private capital and scale-up, which the Act is meant to address. The 2025–26 story is EuroHPC turning procurement pledges into **six machines actually inaugurated on European soil** (detailed in E-sovereign).

THE NUMBERS (GRADED, WITH CAVEATS)

- T5 €1B / 10 yr — Quantum Flagship (digital-strategy.ec.europa.eu). Frequently *added* to member-state programs in "Europe spends €X" claims — the same euros counted at EU and national level.
- T5 ~€11B+ — commonly cited combined EU + member-state public commitment (Qureca tallies). Sums multi-year pledges of different vintages/inflation bases; order-of-magnitude only.
- T5 Member-state headline programs (nationally administered; see E-others for Germany): Germany ~€5B total (€2B stimulus 2020 + €3B action plan 2023), France ~€1.8B national strategy (2021), Netherlands ~€615M Quantum Delta NL.
- T5 Quantum Act 2026 — "potentially billions" in industrialization incentives; nothing appropriated yet (twobirds.com / EC call for evidence). Likely structured as a Joint Undertaking modeled on EuroHPC.

KEY GRADED CLAIMS

- T2 All 27 member states signed the EuroQCI Declaration (since June 2019) — EC (established)
- T2 Quantum Europe Strategy adopted 2 July 2025, five pillars: R&I, infrastructures, ecosystem, space & dual-use, skills — COM(2025) 363 (established)
- T2 **EuroHPC JU procured six quantum computers** now inaugurating across DE, CZ, FR, IT, PL, ES (2025–26) — multiple platforms (superconducting, trapped-ion, photonic, neutral-atom, annealer) integrated with Tier-0 supercomputers — eurohpc-ju.europa.eu (established; see E-sovereign for the roster)
- T3 Quantum Act proposal expected 2026 — EC signaling (roadmap)
- T4 "Europe leads in quantum science, lags in commercialization" — EC's own framing; self-serving for a funding push but consistent with VC data (see E-vc) (claimed)

CONFLICTS / OPEN QUESTIONS

- Will the Quantum Act carry **new money** or restructure existing Horizon Europe funds?
- EuroQCI deployment status per member state is uneven and poorly reported; Eagle-1's launch slipped repeatedly.
- Flagship phase-2 evaluation: what did the first €1B actually produce beyond papers and pilots?
- Europe's scale-up gap is structural — its best startups (IQM, Pasqal, Quandela, Alice&Bob) raise less than US peers and some list via US SPACs (IQM → Nasdaq, 2026; see E-mna).

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Quantum export controls · E-export

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

On **6 September 2024** the US Commerce Department's **Bureau of Industry and Security (BIS)** issued an interim final rule adding export controls on quantum computing items — full quantum computers, related equipment, components, materials, software, and development/maintenance technology — bundled with parallel controls on advanced semiconductors and additive manufacturing (new **ECCNs 3A901, 3D901, 3E901** and related). The defining feature is that it is **plurilateral**: BIS built an Export Administration Regulations framework with a **license exception (IEC — Implemented Export Controls)** authorizing exports to partner countries that adopt **harmonized** controls, so allies coordinating with the US face fewer barriers than non-aligned states. Allied measures landed in lockstep — the **Netherlands** issued its own controls **7 September 2024**, with the **UK, France, Spain, Australia, and Japan** adopting similar rules. Controls affecting exports to certain allies took effect **5 November 2024**; the rest applied from 6 September 2024. This is the first time quantum was made a first-class multilateral export-control category rather than folded into dual-use catch-alls.

THE MECHANISM (GRADED)

- T2 BIS interim final rule, 6 Sep 2024, controls quantum computers + enabling equipment/materials/software/technology — bis.gov / Federal Register (established)
- T2 Plurilateral framework + IEC license exception rewards allies with harmonized controls — Covington / Mayer Brown / Cleary analyses (established)
- T2 Netherlands announced coordinated controls 7 Sep 2024; UK/France/Spain/Australia/Japan aligned — Orrick / APS News (established)
- T2 **"Deemed export" reach:** sharing controlled quantum tech with foreign nationals *inside* the US is itself a controlled export — UChicago/LBNL guidance (established); directly ties this node to E-immigration and E-talent.

KEY GRADED CLAIMS

- T2 The rule is explicitly **coordination-first** — the license exception makes allied harmonization the point, not a byproduct (established)
- T3 BIS simultaneously **expanded CFIUS mandatory-filing scope** to cover quantum-relevant inbound investment — Cleary (claimed/analysis); the June 2026 Quantum EO reinforced the inbound-investment-screening posture.
- T4 Real-world chilling effect on academic collaboration and hiring of foreign nationals — reported concern, not yet measured (contested)

CONFLICTS / OPEN QUESTIONS

- **Effectiveness is unproven:** controls slow but do not stop a determined state, and China is documented building domestic supply to route around them (see E-supplychain, China self-sufficiency).
- **Scope ambiguity** — "quantum computer" thresholds and dual-use materials (cryogenics, lasers) are hard to police; enforcement precedent is thin, and the "logical/physical qubit" line is undefined in standards (see E-standards).
- **Bloc-cohesion risk:** does the plurilateral bloc hold when a member's national champion is disadvantaged? The IonQ acquisition of UK-based **Oxford Ionics** (E-mna) and Australia's bet on US-domiciled PsiQuantum (E-others) both test whether "allied" firms are treated as domestic — echoes of the ASML/Netherlands semiconductor tensions.

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Workforce immigration & clearance policy · E-immigration

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

The workforce *gap* is E-talent; this node is the **policy layer that decides who is allowed to fill it** — immigration/visa rules and security-clearance requirements, which pull in opposite directions. Quantum is one of the few deep-tech fields where domestic supply clearly cannot meet demand, so employers rely on **international hires**; yet the same technology is export-controlled (E-export "deemed exports") and clearance-gated for defense work (E-defense), which **shrinks the eligible pool**. The result is a structural tension: the US and allies need foreign quantum talent to build the workforce, and simultaneously restrict foreign nationals' access to the controlled technology those jobs involve.

THE LANDSCAPE (GRADED)

- T5 **50% of advanced-STEM-degree holders in the US defense industrial base are foreign-born**; 82% of DIB firms report difficulty finding qualified STEM workers — IFP / CSIS (reported)
- T3 Quantum roles are sponsored across **national labs (Argonne, ORNL, LBNL), universities, and IBM/Google/Microsoft/AWS**, plus contractors (Booz Allen, Leidos) — though clearance requirements vary by project and exclude many non-citizens — F1Jobs / migratemate (reported)
- T2 The **22 June 2026 Quantum EO** directs a **domestic** workforce push (apprenticeships, credentials) and **expands the Quantum Counterintelligence Protection Team** — a tightening, not a loosening, of the foreign-access posture — whitehouse.gov (established)

THE TWO-SIDED TENSION (GRADED)

- T2 **Deemed-export rule (E-export):** sharing controlled quantum technology with a foreign national *inside* the US is legally an export — so hiring a non-citizen into a controlled-tech role can itself require a license — BIS / university guidance (established)
- T3 **Clearance bottleneck:** defense quantum work (E-defense) requires clearances that most non-citizens cannot hold, and clearance backlogs further slow staffing — RAND / reporting (reported)
- T3 **Global competition for talent:** other states (Canada, Australia, EU, Gulf) court the same PhDs with faster visa tracks — a talent-attraction race layered on the funding race — Robinson Immigration / CSIS (reported)

KEY GRADED CLAIMS

- T4 US immigration policy volatility (H-1B cost/reform debates, 2025–26) adds hiring uncertainty precisely where the shortage is most acute — CSIS/press (claimed)
- T5 The clearance + export-control filter is **largely absent from the quantum-shortage literature** (E-talent), which counts open roles without netting out who is *eligible* to fill them (analyst gap)

CONFLICTS / OPEN QUESTIONS

- **Self-defeating loop?** Restricting foreign access protects IP (E-export, E-patents) but starves the very workforce the national strategies say is the binding constraint (E-talent). No government has published how it resolves the tradeoff.
- Does the June 2026 EO's domestic-apprenticeship push produce hybrid technicians fast enough to reduce reliance on foreign PhDs, or is it a decade too slow (O-talent-attrition)?
- Measurement gap: no dataset separates "open quantum roles" from "roles a security-cleared citizen could fill" — the true domestic shortfall is unquantified.

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Insurance & liability for quantum risk · E-insurance

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

This node is about **insuring against the quantum threat** — cyber underwriters, D&O liability, and regulators pricing the risk that current cryptography breaks — which is the mirror image of I-insurance (using quantum computing *for* actuarial modeling). The 2026 industry consensus is a split: the **long-run** exposure is real and potentially systemic (a cryptographically-relevant quantum computer could invalidate the confidentiality assumption under most cyber policies at once), but **near-term claims are considered unlikely**, so insurers are folding **PQC-readiness into underwriting questionnaires** rather than repricing books today. The annual cyber-renewal cycle is itself the industry's stated hedge — it lets insurers adjust before quantum becomes a material exposure.

THE LANDSCAPE (GRADED)

- T3 **Lloyd's** analysis: the largest potential insurance impact comes from the **cybersecurity risk of cryptographically-relevant quantum computers**; underwriters describe it as a "long-term issue" managed via annual renewals — Lloyd's / Finadium (reported)
- T3 **Cyber insurers see little near-term claims threat** (Jul 2026) — HNDL was a central conference theme, but insurance lawyers argued it is unlikely to become a major near-term claims source — The Insurer (reported)
- T3 Insurers are **integrating PQC-readiness evaluations into underwriting** to keep portfolios profitable and to nudge clients to begin migration (E-pqcmarket) — Gallagher / Insurance Business (reported)

THE EXPOSURE CATEGORIES (GRADED)

- T3 **Confidentiality/data risk** — customer, claims, health, and underwriting data stored today could be decrypted later (HNDL); a *silent aggregation* risk because one crypto break hits many policies simultaneously — Hogan Lovells / PLUS (reported)
- T3 **Operational & contractual risk** — long-term technology arrangements (outsourcing, PKI, long-dated bonds/records) that assume today's crypto — Lloyd's (reported)
- T3 **D&O / governance liability** — as regulators (see E-us June 2026 PQC EO deadlines) increasingly *expect* firms to plan for quantum, boards that fail to act face governance-failure exposure — PLUS "Quantum Risk Series" (reported)

KEY GRADED CLAIMS

- T4 No dedicated "quantum insurance" product line exists yet; quantum risk is being **absorbed into existing cyber/D&O lines** with questionnaire changes, not new policies (reported/emerging)
- T5 "Q-Day could reshape cyber claims" — insurance-industry framing (PLUS); a scenario, not a priced event (speculative)

CONFLICTS / OPEN QUESTIONS

- **Systemic-correlation problem:** most cyber risk is idiosyncratic, but a crypto break is a **single correlated event across an entire book** — classic uninsurable/catastrophe structure. How (or whether) reinsurance handles this is unresolved.
- Tension with E-pqcmarket: insurers say near-term claims are unlikely, while PQC vendors sell urgency — the truth depends on *data lifetime* vs *Q-day* timing (O-scaling).
- Liability allocation: if crypto breaks, who is at fault — the software vendor, the enterprise that didn't migrate by the EO deadline, or the standards body? No case law exists.

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Academic-lab leaderboard · E-labs

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

Where E-patents ranks corporate IP and E-talent counts the workforce, this node maps the **academic research base** — the university labs that produce the physics, the people, and the spinouts. It matters geopolitically because national programs (E-us..E-others) are downstream of a small set of anchor institutions, and because the leaderboard reveals a **US/China/Europe tripole** with distinct modality specializations. Rankings are bibliometric (Nature Index Share, publication/citation counts) and therefore favor *output volume* over *breakthrough* — read them the way E-patents reads filing counts: informative about mass, silent about which lab produced the load-bearing result.

THE LEADERBOARD (GRADED, WITH CAVEATS)

- T3 **USTC (Hefei)** ranks **first in quantum-physics output** in the Nature Index (2024–25 window) — home of Micius, Jiuzhang, Zuchongzhi, and the 2025 Jinan-1 13,000 km QKD link (E-china) — Nature Index (reported; output-weighted)
- T3 **US anchors:** MIT, Harvard (Lukin — neutral-atom simulation/computing, spun out QuEra), Caltech, Chicago (Pritzker/PME + Argonne/Fermilab), Maryland/JQI (ion trapping — IonQ lineage), Princeton, Berkeley/LBNL — Nature Index / TQI (reported)
- T3 **Chinese Academy of Sciences** system + Tsinghua, plus **USTC**, dominate the China cluster — bibliometric studies (reported)
- T3 **Europe:** TU Delft / **QuTech** (superconducting + spin + networking; Intel/Microsoft ties), ETH Zurich, Oxford (ion traps → Oxford Ionics), Sydney/UNSW (silicon spin — Diraq, SQC), Waterloo/IQC (Canada), Innsbruck (ions), Paris (CNRS, neutral atoms → Pasqal) — Nature Index / TQI (reported)

KEY GRADED CLAIMS

- T3 The field's output grew **exponentially since ~2015**, led by the US, China, and Europe, with MIT, CAS, CNRS, and Waterloo cited as bibliometric leaders — EPJ Quantum Technology mapping study (demonstrated)
- T3 **Labs map to modalities and to spinouts:** Maryland/Innsbruck → ions/IonQ-Quantinuum, Harvard-MIT → neutral atoms/QuEra, UNSW → silicon/Diraq, Delft → spin+networking, Sydney → control (Q-CTRL). The academic map *is* the industry map one generation earlier (analysis)
- T4 Rankings favor volume; a single below-threshold-QEC result (Google Willow, a corporate lab) can outweigh a year of a university's papers — bibliometrics miss this (claimed)

CONFLICTS / OPEN QUESTIONS

- **Output vs breakthrough:** USTC tops output counts, but the field's landmark QEC results came from *corporate* labs (Google, IBM, Quantinuum) — academic leaderboards understate the industrial research frontier (E-patents trade-secret shadow).
- **Brain-drain direction:** do top academic labs feed *domestic* champions or foreign ones (E-immigration, E-mna)? Oxford → IonQ (US) and UNSW → global shows leakage across borders.
- No neutral, methodologically stable "quantum lab ranking" exists — Nature Index, QS, and TQI lists diverge; treat any single ordering as one lens.

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Market forecasts — read with gloves on · E-market

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Every number in this card is **T5 by construction**. The forecast spread is enormous and the biggest inflation trick is **unit-switching**: "economic value created" (trillions) gets quoted as if it were vendor revenue (tens of billions), a 20–50x difference. Current reality check: **actual sector revenue was ~$1B in 2025** (McKinsey's own figure), which the same report projects to as much as $4.4B by 2028. McKinsey's 2026 Monitor declared a "commercial tipping point" and revised its own market numbers sharply upward year-over-year — which says as much about forecast elasticity as about the market. The honest reading is to track each house's **revision history**, not its point estimate.

THE NUMBERS (GRADED, WITH CAVEATS)

- T5 **McKinsey (2026 Monitor):** quantum-technology market **$60–100B by 2035**, of which quantum computing **$43–71B** — revised UP from ~$28B in the prior edition. A >2x self-revision in 12 months is the caveat.
- T5 **McKinsey:** up to **$2.7T "economic value"** by 2035 (prior editions said $1.3–2.7T). Value-created ≠ revenue — do not mix with the line above; most press does.
- T5 **McKinsey:** present quantum-computing revenue **~$1B (2025) → up to $4.4B (2028)**; >300 companies (Airbus, JP-Morgan, Boehringer) now engaged with vendors. Motivated forecaster caveat: McKinsey sells quantum-readiness engagements.
- T5 **BCG:** $450–850B economic value by 2040, supporting a $90–170B hardware+software vendor market. Different horizon (2040 vs 2035) makes it incomparable to McKinsey's headline despite constant press comparison.
- T5 **Gartner** keeps quantum computing on the long-horizon end of its hype cycle (10+ years to plateau); **IDC** has repeatedly cut near-term revenue forecasts — the revision trail beats any single number.
- T5 Normalized spread: 2035 vendor-market estimates ~$40–100B; near-term (2027–28) estimates cluster in single-digit billions.

KEY GRADED CLAIMS

- T5 "Commercial tipping point in 2026" — McKinsey framing; provenance-flagged (claimed)
- T5 Analyst TAMs assume fault-tolerant machines arrive on vendor roadmap schedules — the single load-bearing assumption; every year of FTQC slip pushes the hockey stick right (contested)
- T2 Present-day sector revenue ~$1–1.5B/yr, dominated by hardware R&D contracts, cloud access, and government programs — company filings + McKinsey (established)

CONFLICTS / OPEN QUESTIONS

- **C-market-size:** McKinsey $43–71B by 2035 vs BCG's flatter path to 2040. Resolution: audited vendor revenue through 2030, and whether any commercial advantage case (see O-killerapp) closes.
- No analyst house publishes its **miss rate** on prior quantum forecasts; a backtest of 2019–2022 editions would likely be brutal.
- **Double-counting:** "quantum market" tallies often fold in **PQC migration** spending — a classical cybersecurity market ($1–15B depending on definition, see E-pqcmarket) that exists regardless of quantum hardware progress.

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Metrology standards & SI-redefinition governance · E-metrology-gov

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

Quantum has a **second, quieter standards front** from the PQC/QKD one (E-standards): the governance of the **International System of Units (SI)** itself, run by **BIPM / CIPM / CGPM** and the consultative committees (CCTF for time, CCU for units). The 2019 SI redefinition already anchored the kilogram, ampere, kelvin, and mole to fundamental constants (Planck, elementary charge, Boltzmann, Avogadro), making national metrology institutes able to **realize units from quantum physics** rather than artifacts. The live 2025–26 governance question is the **redefinition of the second** onto an **optical atomic-clock** transition — a decision headed for **CGPM 2026**. This is geopolitically loaded because whoever's clocks and realizations anchor the new second holds metrological soft power, and the outcome underpins timing, PNT (A-clocks, A-timedist), and finance/telecom infrastructure.

THE GOVERNANCE TRACK (GRADED)

- T2 **BIPM operates under CIPM supervision**; the CGPM (member states) is the decision body; the **CCTF** drafts the second-redefinition roadmap — bipm.org (established)
- T2 **2019 SI redefinition** fixed the kilogram to the **Planck constant** (realized via the **Kibble balance** or the X-ray-crystal-density/Avogadro route), the ampere to elementary charge, kelvin to Boltzmann — a quantum-constant-based system (established)
- T3 **Roadmap toward redefining the second:** several optical frequency standards already beat caesium fountains by ~2 orders of magnitude in systematic uncertainty; the CCTF needed a draft ready for its **Sep 2025 meeting** to present at **CGPM 2026** — IOP Metrologia roadmap (demonstrated)
- T3 Multiple NMIs (PTB, NIST, INRIM, NPL, SYRTE, KRISS) are cross-evaluating candidate optical transitions; **>80–90% clock uptime** demonstrations are the readiness signal — Nature Index / Metrologia (reported)

KEY GRADED CLAIMS

- T2 The realization of quantum-based units is **decentralized by design** — any NMI can realize the kilogram/second independently, which is itself a governance principle (no single artifact, no single owner) — BIPM (established)
- T3 Optical-clock redefinition of the second is likely **"within the current decade"** and is the flagship deliverable of the 2025 International Year of Quantum Science & Technology metrology agenda — BIPM (reported)
- T4 Metrological leadership is soft power: the NMIs whose clocks define the ensemble shape traceability chains the whole economy inherits (analysis)

CONFLICTS / OPEN QUESTIONS

- **Which optical transition wins?** Competing candidates (Sr, Yb, Al^+, Yb^+ E3, etc.) across rival NMIs — a standards choice with national prestige attached; resolution at/after CGPM 2026.
- **Realization inequality:** not every member state can build an optical clock or Kibble balance, so "decentralized realization" is decentralized only among wealthy NMIs — a metrology-access gap.
- Overlap with A-timedist: redefinition needs clock **comparison** networks (optical fiber / free-space time transfer) to disseminate the new second — the governance decision is downstream of that hardware.

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The consolidation wave — M&A and public listings 2025–26 · E-mna

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

2025–26 is the field's **consolidation phase**, distinct from primary VC rounds (E-vc). Two forces run in parallel: **roll-ups** — public hardware companies buying components, foundries, and rivals to build full stacks — and an **exit wave** of SPAC mergers and IPO filings that let 2021-era private companies reach public markets. IonQ ran the most aggressive acquisition spree in the sector's history; D-Wave, Xanadu, IQM, and Quantinuum each made defining 2026 moves. The strategic reading: the sector is maturing toward a **small number of vertically integrated players**, which concentrates IP (E-patents) and interacts with export controls when the target is foreign (E-export).

THE DEALS (GRADED, WITH CAVEATS)

- T5 **IonQ 2025 spree:** acquired **Oxford Ionics (~$1.08B, the year's largest deal)**, Lightsynq, Qubitekk, Capella Space, and Vector Atomic; took a **majority stake in ID Quantique** — McKinsey/Crunchbase (reported; some values are stock-heavy)
- T5 **IonQ → SkyWater Technology, ~$1.8B (announced Jan 2026, expected close 2026)** — buying a US semiconductor foundry with government/defense contracts; a vertical-integration + supply-security play (E-supplychain) — Crunchbase
- T5 **D-Wave → Quantum Circuits Inc., $550M** (stock + cash, early 2026) — annealing leader buying a gate-model developer — Crunchbase
- T5 **Xanadu** went public via SPAC on Nasdaq + TSX (~$3.1B enterprise value cited) — photonic; H1-2026 exit — Crunchbase/PitchBook
- T5 **IQM → Real Asset Acquisition Corp. SPAC** — first publicly traded *European* quantum-computing company on Nasdaq; ~$233.5M injected incl. a $145.5M PIPE — press
- T5 **Quantinuum** filed for a **Nasdaq IPO in 2026** after its ~$839M round at $10B pre-money (E-vc) — Crunchbase News
- T5 H1-2026 exits totaled **~$5.7B across four listings** (Xanadu, Infleqtion, Quantum Circuits, Horizon Quantum) — ~15x the prior three years' exit value combined (PitchBook)

KEY GRADED CLAIMS

- T5 M&A + exit activity "accelerated markedly in 2025" and signals market maturation/consolidation — McKinsey 2026 Monitor (analyst framing)
- T3 Vertical integration is the explicit thesis — hardware firms buying foundries (SkyWater), photonic interconnect (Lightsynq), sensing/space (Capella, Vector Atomic) to own the stack (reported)

CONFLICTS / OPEN QUESTIONS

- **Is consolidation strength or distress?** Roll-ups can signal maturation *or* that standalone economics don't work at ~$1B total sector revenue (E-market). Resolution: post-merger revenue at the acquirers through 2027.
- **Sovereignty friction:** IonQ (US) absorbing UK-based Oxford Ionics and Swiss ID Quantique puts allied national champions under US ownership + export-control reach (E-export, E-uk, E-others).
- Stock-heavy deal values inflate headline M&A totals when acquirer shares are themselves richly valued (E-vc) — a circularity worth flagging.

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Open-source ecosystem & soft power · E-oss

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The quantum software layer is dominated by **vendor-backed open-source SDKs**, and whoever's framework a developer learns first shapes which hardware they reach for later — soft power and lock-in dressed as free tooling. The big three are **Qiskit** (IBM), **Cirq** (Google), and **PennyLane** (Xanadu), joined by tket/pytket (Quantinuum), Braket SDK (AWS), Q#/QDK (Microsoft), and D-Wave's Ocean. All are permissively licensed and free, but each is a funnel toward its sponsor's cloud and hardware. The counter-force is interop standards — **OpenQASM** (IBM-originated assembly, now OpenQASM 3) and **QIR** (Microsoft-led LLVM-based intermediate representation under the **QIR Alliance / Linux Foundation**) — which keep code portable across backends and blunt pure lock-in. The open geopolitical question is whether the world converges on this Western toolchain or **forks** along the export-control line into China's parallel stack (broken out as `E-china-stack`).

THE LANDSCAPE (GRADED)

- T2 Qiskit, Cirq, PennyLane are the leading full-stack SDKs, all corporate-maintained (IBM/Google/Xanadu) yet open and free — multiple 2025 surveys (established)
- T2 PennyLane is hardware-agnostic by design (device plugins target Qiskit/Cirq/Forest/Braket) — Xanadu docs (established)
- T2 OpenQASM + QIR provide cross-backend portability; a Cirq circuit can be lowered to OpenQASM and run via Qiskit on IBM hardware — 2025 ecosystem writeups (established)
- T3 The ecosystem is partly siloed — specialized QEC/chemistry tools don't interoperate cleanly; cohesion is an open community effort — surveys (reported)
- T3 NVIDIA **CUDA-Q** is emerging as a de facto hybrid GPU-QPU orchestration layer, giving Nvidia soft-power leverage over the classical side of every quantum workflow — Nvidia (reported)

KEY GRADED CLAIMS

- T3 Framework choice functions as soft-power lock-in — learning Qiskit funnels toward IBM Quantum cloud; governance-by-defaults, not contract lock-in (analysis/claimed)
- T2 Governance is **corporate, not neutral-foundation**, for most SDKs — Qiskit/Cirq/PennyLane steered by their sponsors; the **QIR Alliance (Linux Foundation)** is the main neutral-governance exception (established)
- T3 A 2026 formal-verification study (arXiv 2604.06712) found security vulnerabilities across ~45 open-source quantum simulators — software-supply-chain risk (preprint)

CONFLICTS / OPEN QUESTIONS

- Is portability (OpenQASM/QIR) real enough to defeat lock-in, or do deep tooling layers (transpilers, error mitigation, pulse control) **re-lock** users to one vendor in practice?
- No dominant *neutral* foundation governs the top SDKs — unlike Linux/Apache — so "open source" here is closer to open-source-as-marketing than a community-governed commons. The one durable neutral layer is the IR (QIR), not the SDKs.
- **Fork risk (see E-china-stack):** China's Origin QPanda / Baidu Paddle Quantum / Huawei HiQ are a full parallel stack; export controls (E-export) make convergence less likely each year.

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National programs — the rest of the pack · E-others

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Beyond the US/China/EU/UK quartet, roughly a dozen countries run serious national quantum programs. Common pattern: a multi-year headline pledge (T5, usually mixing new money with relabeled research budgets), one flagship national lab or mission, and a bet on a niche — India on indigenous full-stack hardware, Japan on industrialization, Australia on photonics via PsiQuantum, Canada on a deep startup bench, Israel on defense-adjacent depth, South Korea on late-but-large catch-up, Germany on error-corrected machines within the EU frame. The near-universal caveat: **pledge ≠ disbursement**, and almost none of these countries publish the gap.

THE NUMBERS (GRADED, WITH CAVEATS — ALL T5, ALL MULTI-YEAR PLEDGES, CURRENCIES AS ANNOUNCED)

- T5 **India** — National Quantum Mission, ₹6,003.65 crore (~$730M), 2023–2031 (dst.gov.in). Four thematic hubs (computing at IISc Bengaluru, comms, sensing/metrology, materials/devices). 17 supported startups; QpiAI shipped a 25-qubit "Indus" (Apr 2025) and a 64-qubit "Kaveri" chip (Nov 2025) — both T4 vendor claims. Still imports key lasers/cryogenics (see E-supplychain).
- T5 **Japan** — 2025 declared "first year of quantum industrialization"; ¥1.05T (~$7.4B) headline covers semiconductors **and** quantum (severe double-counting if quoted whole); ~$900M is quantum-specific. Goal: 10M quantum users by 2030. RIKEN (which fielded Japan's first homegrown superconducting machine, 2023) + the Moonshot R&D program underpin the base.
- T5 **Australia** — ~AU$2.3B cumulative, dominated by the ~AU$940M federal+Queensland investment in **PsiQuantum's Brisbane** machine (2024) — a bet on one (US-headquartered) company, politically contested and audited.
- T5 **Canada** — National Quantum Strategy C$360M (2021) + C$334M/5yr defence (2025 budget) + C$92M Quantum Champions Phase 1 (Dec 2025) + C$50M DISH defence R&D (Feb 2026). Deep startup bench (Xanadu, D-Wave origins, Photonic Inc., Nord Quantique) — though Xanadu went public via US SPAC in 2026 (see E-mna).
- T5 **Israel** — NIS 1.2B (~$380M) six-year program; small headline, high density (Quantum Machines, Classiq, Quantum Source), deep defense overlap that hides real totals.
- T5 **South Korea** — ~₩3T (~$2.3B) toward 2035 quantum goals under the 2023 Quantum Science & Technology Act; late entrant, chaebol-backed (SK Telecom in QKD, Samsung/SK in supply chain).
- T5 **Germany** — ~€5B total: €2B (2020 stimulus) + €3B Action Plan (2023) targeting a **universal error-corrected** computer; hosts EuroHPC's Euro-Q-Exa (IQM, 54→150 qubits, LRZ Munich). Counted again inside EU tallies — see E-eu.

KEY GRADED CLAIMS

- T4 QpiAI "India's first full-stack quantum computer" — vendor claim, no independent benchmark (claimed)
- T3 Germany's strategy centers error-corrected systems over NISQ demos — government strategy documents (roadmap)
- T3 Japan/RIKEN operates the first non-US/China domestically built superconducting quantum computer at national-lab scale (demonstrated)

CONFLICTS / OPEN QUESTIONS

- Pledge-vs-disbursement gap is unmeasured for nearly every country here.
- Australia's single-vendor concentration (PsiQuantum) has no public plan B, and the vendor is US-domiciled — a sovereignty wrinkle.
- Still missing / candidates for a future split: **Singapore** (CQT, early mover), **Netherlands** (QuTech/Delft, ASML-adjacent supply leverage), **Taiwan**, **Denmark** (Novo-backed), **Russia** (opaque, sanctioned).

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Patents & the IP landscape · E-patents

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The patent race splits cleanly on a **volume vs quality** axis. China files the most quantum patents by a wide margin (~60% of global *filings* in 2024); the US dominates **International Patent Families** — patents protected in multiple jurisdictions, the proxy for inventions someone expects to monetize globally (~48% US vs ~11% China). Specialization mirrors national strategy: China owns QKD/communications IP (~80% of families in that niche); the US leads computing hardware and algorithms, driven by IBM, Google, and Microsoft. The load-bearing lesson: **which country "leads" depends entirely on the denominator you pick**, and headlines swap weekly using the same underlying data.

THE NUMBERS (GRADED, WITH CAVEATS)

- T5 ~60% China vs ~19% US — share of global quantum patent *filings*, 2024 (patentailab.com). Raw counts are inflated by Chinese subsidy programs that reward filing volume; many are domestic-only and never examined abroad.
- T5 ~48% US vs ~11% China — share of **International Patent Families** (multi-jurisdiction). The better quality proxy, still imperfect: family-counting methods differ across analyses.
- T5 ~80% — China's share of QKD-related patent families (patentailab.com); consistent with its QKD infrastructure build (see E-standards, A-qkd).
- T5 **IBM holds the largest single corporate quantum-computing portfolio** — repeated across landscape studies; exact counts vary by taxonomy (CPC class **G06N 10/00** vs keyword searches disagree by 2–3x).

KEY GRADED CLAIMS

- T3 **Filing-subsidy distortion:** Chinese volume partly reflects state per-filing incentives — patent-policy literature (demonstrated)
- T2 Baidu **donated** its quantum lab to BAAI and Alibaba **shuttered** its quantum effort (2023–24), transferring equipment to Zhejiang University — the corporate patent engines went quiet while state/Origin filings continued (established)
- T6 Whether quantum patents filed 2015–2025 matter commercially before they expire (20-year term vs FTQC timelines) — the quiet joke of the whole race (speculative)

CONFLICTS / OPEN QUESTIONS

- **C-patent-lead:** "China leads quantum IP" vs "US leads quantum IP" — both are printed weekly from the same data; the answer depends on filings-vs-families methodology. Resolution: none needed — always state the denominator.
- **Trade-secret shadow:** error-correction decoders (see S-decoders), control calibration, and fab recipes are increasingly kept as **secrets**, so patent landscapes systematically undercount the real IP frontier. This is growing as the field industrializes.
- **M&A concentrates IP:** IonQ's 2025 acquisition spree (Oxford Ionics, Lightsynq, Qubitekk, ID Quantique stake — see E-mna) folds independent patent estates into single US owners, interacting with export controls (E-export).

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The quantum-safe migration market · E-pqcmarket

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

Post-quantum cryptography migration is an **economic market in its own right**, separate from the crypto science (A-pqc, technical) and from the quantum-computing vendor market (E-market). It is driven not by quantum hardware progress but by **"harvest now, decrypt later" (HNDL)** — adversaries archiving encrypted traffic today to decrypt once a cryptographically-relevant quantum computer exists — and by hard regulatory deadlines. It is the one quantum-adjacent market with **real, mandated near-term revenue**, because it sells classical software/services (crypto discovery, crypto-agility, PKI replacement) that must ship regardless of whether FTQC ever arrives. This is exactly why it inflates "quantum market" tallies (E-market double-counting flag).

THE NUMBERS (GRADED — ALL T5 FORECASTS, WIDE SPREAD BY DEFINITION)

- T5 **Narrow "PQC market":** ~\$0.5–0.8B in 2025 → ~\$1.1B in 2026 (marketsandmarkets / imarc-style). Counts PQC products specifically.
- T5 **"PQC migration market":** ~\$1.9B (2025) → ~\$12.4B (2035), CAGR ~20.6% (Future Market Insights). Adds migration services.
- T5 **"PQC migration solutions":** ~\$15.0B (2025) → ~\$86.5B (2034), CAGR ~22% (MarketIntelo). Broadest definition — folds in consulting + managed services; the outlier high end.
- T5 The **NIST FIPS 203/204/205 finalization (Aug 2024) is estimated to trigger ~\$4.2B in net-new federal + regulated-industry spending in 2025–2026 alone** (industry analysis).
- T5 Nasdaq/press framing: PQC migration is now a **"trillion-dollar imperative"** across the global installed base — motivated-vendor rhetoric; treat as TAM-of-everything, not revenue.

THE REGULATORY ENGINE (GRADED)

- T2 NIST finalized ML-KEM/ML-DSA/SLH-DSA Aug 2024; HQC backup Mar 2025 (E-standards) — the standards that make procurement possible (established)
- T2 **US 22 June 2026 PQC EO:** OMB deadlines — high-value-asset **key establishment by 31 Dec 2030, digital signatures by 31 Dec 2031**; Commerce migration pilot by 31 Dec 2027 (E-us) — whitehouse.gov (established)
- T5 NSA **CNSA 2.0** targets ~2033 for full NSS transition (E-standards) — policy deadline (slips possible)

KEY GRADED CLAIMS

- T3 **HNDL is the demand driver**, assessed by US intelligence to involve China/Russia/DPRK archiving encrypted data now — Cloudflare/Gopher/agency framing (reported)
- T5 This market exists **independent of quantum-hardware timelines** — its risk is set by *data lifetime* vs *Q-day*, so long-lived secrets justify migrating today even under optimistic FTQC-slip scenarios (analyst consensus)

CONFLICTS / OPEN QUESTIONS

- **Definitional 30x spread** (\$0.5B vs \$15B for "2025") makes any single headline meaningless — always name the definition (product vs migration vs solutions/services).
- Does mandated federal migration crowd in private-sector spend, or does the private base wait until a public break (contrast the insurance view, E-insurance, that near-term claims are unlikely)?
- Overlap: is PQC spend "quantum" spend at all? It is classical cryptography — folding it into E-market inflates the sector.

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Sovereign compute & national champions · E-sovereign

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

A distinct 2025–26 pattern cutting across the national-program cards: states are no longer just *funding* quantum research, they are **acquiring sovereign compute** — government-owned or government-hosted machines on national soil, coupled to national supercomputers, plus **equity stakes** in national-champion firms. The thesis is that quantum, like advanced semiconductors, is strategic infrastructure that a country must **own and physically host**, not merely rent from a foreign cloud (E-oss soft-power concern). The clearest instance is **EuroHPC** turning procurement pledges into six machines inaugurated across Europe; the sharpest US instance is the **~$2B Commerce package reportedly taking equity** in nine firms (E-us).

THE MOVES (GRADED)

- T2 **EuroHPC JU procured six quantum computers**, each hosted at a European supercomputing centre and integrated with a Tier-0 machine for hybrid HPC-QC — a deliberate **digital-sovereignty** program — eurohpc-ju.europa.eu (established):
- **Germany (Munich, LRZ):** Euro-Q-Exa — IQM superconducting, **54 qubits, inaugurated Feb 2026**, upgrade to **150 qubits** planned end-2026
- **France (Bruyères-le-Châtel, CEA/GENCI): Lucy** — Quandela **12-qubit photonic**, **80% EU-sourced components**, coupled to Joliot-Curie / Alice Recoque exascale (inaugurated Apr 2026)
- **Poland (Poznań):** PIAST-Q — trapped-ion (Jun 2025); **Czechia (Ostrava):** VLQ (Oct 2025); **Spain (Barcelona, BSC):** EuroQCS-Spain — analog annealer (May 2026); **Italy (Bologna):** SOL — neutral-atom simulator
- T2 **EuroHPC Quantum Grand Challenge** call opened Oct 2025 to drive utilization of the fleet — eurohpc-ju.europa.eu (established)
- T4 **US ~$2B Commerce awards to nine firms, ~half to IBM, reportedly with government equity** (May 2026) — a sovereign-stake model new to US tech policy (E-us) — CNN/Forbes (claimed)

KEY GRADED CLAIMS

- T3 **On-prem hosting is the sovereignty move** — owning the physical machine (not cloud access) is what "sovereign compute" means; Lucy's 80%-EU-component sourcing makes the supply-sovereignty point explicit (E-supplychain) (reported)
- T3 Multi-platform hedging: EuroHPC deliberately bought **five different modalities** (superconducting, ion, photonic, annealer, neutral-atom) rather than betting on one — a portfolio approach to technological uncertainty (reported)
- T4 The equity-stake turn (US) shifts government from **grant-maker to shareholder**, aligning fiscal upside with national champions but risking picking-winners distortion (E-us, E-vc) (claimed)

CONFLICTS / OPEN QUESTIONS

- **Does hosting a 12–54 qubit NISQ machine confer real sovereignty**, or is it symbolic until fault tolerance? The machines inaugurated are small; sovereignty of a pre-utility technology is a bet on the trajectory (O-scaling).
- **Champion capture:** Australia's PsiQuantum (US-domiciled) and IonQ's absorption of Oxford Ionics (E-mna) show "national champion" and "foreign-owned" can collide under export controls (E-export).
- Utilization risk: sovereign machines can sit idle without a user pipeline — hence EuroHPC's Grand Challenge call.

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Standards bodies · E-standards

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Standards are where quantum stops being physics and becomes procurement. The mature front is **post-quantum cryptography**: NIST finalized **FIPS 203 (ML-KEM)**, **FIPS 204 (ML-DSA)**, and **FIPS 205 (SLH-DSA)** in **August 2024**, selected **HQC** as a backup KEM in **March 2025**, and US EO 14144 (Jan 2025), reinforced by the **22 June 2026 PQC EO** (deadlines 2030/2031, see E-us, E-pqcmarket), pushed federal migration. Everything else — terminology, benchmarking, QKD certification — runs years behind, spread across ISO/IEC, ETSI, ITU, IEEE, and the new **ISO/IEC JTC 3** committee dedicated to quantum technologies. Standards racing is itself geopolitical: China dominates **QKD** standardization the way it dominates QKD patents (see E-patents), while the SI-metrology standards war is its own node (E-metrology-gov).

THE NUMBERS (GRADED, WITH CAVEATS)

- T2 3 finalized NIST PQC standards (FIPS 203/204/205, Aug 2024) + HQC selected Mar 2025 — csrc.nist.gov. Published, testable standards — the hardest artifact in this layer.
- T5 Migration deadlines: NSA **CNSA 2.0** targets ~2033 for full national-security-system transition; the June 2026 EO adds OMB high-value-asset deadlines (key establishment 2030, signatures 2031). Deadlines are policy, so they slip.

KEY GRADED CLAIMS

- T2 ISO/IEC JTC 1/WG 14 (renamed "Quantum Information Technology") published ISO/IEC 4879 terminology — jtc1info.org (established)
- T2 ISO/IEC 23837-1/-2 define security requirements + test/evaluation methods for QKD systems — ISO/IEC SC 27 WG 3 (established)
- T2 **ISO/IEC JTC 3** on quantum technologies stood up (kick-off spring 2024) — the first top-level ISO committee dedicated to quantum, a governance milestone — CEN/CENELEC / NPL TQE19 (established)
- T3 **ETSI ISG-QKD** maintains the GS QKD series — the oldest quantum standards effort, running since 2008 (established as activity; deployment adoption thin)
- T3 ITU-T ran **FG-QIT4N** producing terminology for quantum networks/QKD; SG13/SG17 carry follow-on work (demonstrated)
- T4 IEEE efforts (P7130 terminology, P7131 benchmarking) remain drafts with limited industry pull (claimed)
- T5 **NSA's position that QKD is unsuitable for national-security systems** (favoring PQC) is a live standards-war fault line against China's QKD-heavy strategy (contested)

CONFLICTS / OPEN QUESTIONS

- **C-qkd-vs-pqc:** US agencies favor PQC over QKD; China builds QKD backbones. Resolution: a decade of deployment economics and any PQC break.
- **Benchmarking vacuum:** what counts as a "logical qubit" or a "QuOp" remains unstandardized while procurement targets (E-uk ProQure) already cite them — buyers are writing contracts against undefined units.
- Terminology fragmentation: QED-C, CEN/CENELEC JTC 22, ISO WG14, and JTC 3 all issue overlapping vocabularies.

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Quantum supply-chain chokepoints · E-supplychain

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

This is the **geopolitics/supply** angle on the physical inputs a quantum computer needs — distinct from `H-fab` (the technical fabrication node). A short list of components has no elastic supply and few qualified vendors, turning each into a strategic chokepoint: **dilution refrigerators** (superconducting/spin), **helium-3** (their coldest stage), **quantum-grade UV/visible lasers** (trapped-ion, neutral-atom), **^{28}Si enrichment** (spin qubits), **CVD quantum-grade diamond** (NV), **cryo-CMOS foundry access**, **high-density cryogenic connectors/coax**, and **SNSPD** uniformity (photonic/QKD). The NATO Transatlantic Quantum Community study reportedly found **>90% of high-purity material processing for quantum happens outside NATO territory** — the strategic worry that motivates both export controls (E-export) and reshoring. The chokepoint map is **modality-dependent**: no single supply strategy covers all qubit types.

THE CHOKEPOINTS (GRADED)

- T3 **Dilution fridges** — effective build rate ~1 system/day industry-wide (Bluefors, Oxford Instruments dominant) becomes binding when builders re-spin hardware every 12–18 months; the single most critical bottleneck for superconducting/spin qubits — War on the Rocks / PostQuantum 2025 analyses (demonstrated/reported)
- T3 **Helium-3** — not naturally abundant; primary source is **tritium decay from nuclear-weapons stockpiles** (a national-security-controlled feedstock); ~$2,000–15,000/L (per-mass more valuable than gold); structural deficit ~20,000–40,000 L/yr — PostQuantum cryogenics guide (reported estimates)
- T2 **Kiutra** (Germany) raised €13M in 2025 for helium-3-free **solid-state magnetic (ADR) cooling**, explicitly a supply-resilience play — company/press (established funding fact)
- T3 Lasers are a real trapped-ion/neutral-atom constraint, but the broad laser market is huge ($100Bs); the chokepoint is *quantum-grade* specificity (linewidth, stability, specific wavelengths), not volume — PostQuantum (reported)
- T2 **NIST launched a ~$20M quantum manufacturing center** targeting the cryostat/cryogenic bottleneck (mid-2026) — a direct policy response to the fridge chokepoint — Tech Times / NIST (established)

KEY GRADED CLAIMS

- T3 **>90% of high-purity quantum material processing occurs outside NATO territory** — NATO Transatlantic Quantum Community study, via 2025 supply-chain analyses (claimed; primary study not directly read)
- T3 China is building domestic dilution-fridge/laser capacity partly in response to export controls — PostQuantum China self-sufficiency reporting (contested/reported)
- T3 Rare inputs beyond He-3: ^{28}Si enrichment capacity is thin and partly ex-NATO; CVD electronic-grade diamond is near-monopoly (Element Six) — supply-chain reports (reported)

CONFLICTS / OPEN QUESTIONS

- Helium-3 deficit figures vary widely and fold in **non-quantum demand** (neutron detection, medical MRI, security) — treat the L/yr numbers as order-of-magnitude.
- Whether alternative cooling (Kiutra-style ADR, or higher-temp qubits) relieves the fridge/He-3 chokepoint before it binds hard.
- Reshoring economics: a $20M NIST center is small against the capital intensity of the bottleneck — is national supply resilience even affordable without industrial-policy money (see E-us $2B Commerce package, E-sovereign)?

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Talent & workforce · E-talent

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The binding constraint is **hybrid practitioners** — people who can run a dilution fridge, write control software, or translate a chemistry problem into circuits. PhD physicists exist; production engineers, cryo technicians, and quantum-literate domain scientists are scarce. The commonly cited ratio is **one qualified candidate per three open roles**. Every national strategy (E-us through E-others) now carries a workforce pillar, and the university pipeline is growing but still thin at the applied end — most programs mint researchers while industry increasingly hires engineers. The immigration and security-clearance dimension of this — which reshapes *who is eligible* to fill the gap — is broken out as `E-immigration`.

THE NUMBERS (GRADED, WITH CAVEATS)

- T5 ~1 qualified candidate per 3 open roles — widely repeated (WEF/industry surveys); the survey base is small and self-reported by employers with hiring incentives.
- T5 ~30,000 — rough estimate of the global quantum workforce (industry reports); definitional soup (does a PQC migration engineer count?).
- T5 US quantum-skill job postings roughly tripled since 2018; ~180% growth 2020–2024 — MIT Quantum Index Report 2025. Posting counts overstate headcount (reposts, evergreen reqs).
- T5 **176 university quantum programs worldwide; only 29 offer a graduate degree specifically in quantum computing.** Germany leads master's programs (12), UK 10, US 9 (Technical.ly / program censuses).
- T5 82% of US defense-industrial-base companies report difficulty finding qualified STEM workers; **50% of advanced-STEM-degree holders in the DIB are foreign-born** (IFP/CSIS) — the tie to immigration policy.

KEY GRADED CLAIMS

- T3 The gap is concentrated in **hybrid roles** — technicians, control engineers, research software engineers — not PhDs — RAND workforce study (demonstrated)
- T5 Salary inflation for senior quantum engineers ($200k–500k+ at US firms) reflects the shortage — press/recruiter data (claimed)
- T4 Vendor education pushes (IBM Qiskit ecosystem, QuEra/others' programs) double as funnel marketing; completion-to-employment rates unpublished (claimed)
- T2 The **22 June 2026 EO** directs a domestic-workforce push via expanded apprenticeships and credentials — whitehouse.gov (established policy direction)
- T6 Whether AI-assisted development compresses the needed workforce (fewer, more leveraged engineers) is open and rarely modeled (speculative)

CONFLICTS / OPEN QUESTIONS

- **C-talent-size:** "crippling shortage" narrative vs the 2022–23 period when quantum startups ran layoffs. Both were true at once — shortage in niche roles, surplus in generalists. Resolution: longitudinal hiring data, which nobody publishes.
- Security-clearance requirements shrink the eligible pool further for defense quantum work — mostly undiscussed in the shortage literature (see E-immigration).
- Does the field need quantum-native curricula or physics + software retraining? Program design diverges by country.

SOURCES

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- https://ifp.org/stem-immigration-is-critical-to-american-national-security/
- https://www.forbes.com/sites/bernardmarr/2025/05/02/quantum-talent-wars-the-hidden-crisis-threatening-techs-next-frontier/

National programs — United Kingdom · E-uk

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The UK has run a continuous national program since **2014** (the National Quantum Technologies Programme, ~£1B over its first decade) — the earliest sustained national quantum bet among Western states. The **March 2023 National Quantum Strategy** committed **£2.5B over ten years** toward being a "leading quantum-enabled economy by 2033," structured around **five quantum missions** (announced Nov 2023) with the **National Quantum Computing Centre (NQCC, Harwell)** as the compute hub. In 2026 the government layered a **~£2B scaling-and-procurement package** whose centerpiece is **ProQure** — using state procurement as the demand signal (an advanced market commitment to buy large-scale machines) with explicit QuOps milestones. The UK's distinctive move is demand-side pull; most rivals push supply-side grants.

THE NUMBERS (GRADED, WITH CAVEATS)

- T5 £2.5B / 10 yr — National Quantum Strategy (gov.uk, Mar 2023). A pledge across research, innovation, skills; annual appropriations arrive via spending reviews and can lag the headline.
- T5 ~£1B — cumulative NQTP investment 2014–2023 (uknqt.ukri.org). Sometimes added to the £2.5B in "UK has spent £3.5B" claims even though the strategy partly *continues* the same funding streams — double-counting risk.
- T5 £2B (~$2.67B) — 2026 scaling + procurement commitment: ~£1B new R&D over four years + ~£1B advanced market commitment to procure a billion-quantum-operation platform by 2031 (Quantum Computing Report / NQCC).
- T5 **QuOps milestones (NQCC):** 1 MQuOp by 2030, 1 GQuOp by 2033, 1 TQuOp by 2035. Targets set by policy; feasibility is a T6 question and depends on the still-unstandardized "QuOp" unit (see E-standards).

KEY GRADED CLAIMS

- T2 NQCC operational at Harwell; testbed program hosts multiple modalities (superconducting, ion, photonic, neutral-atom) — NQCC (established)
- T3 **ProQure** launched late March 2026, inviting prototype submissions for evaluation-led procurement — NQCC/QCR (roadmap)
- T4 "Billion quantum operations by 2031" procurement target — government/vendor co-produced timeline; no machine today is close (roadmap, contested feasibility)
- T3 Five missions span compute, networks, PNT/navigation, and healthcare imaging (brain scanners via OPM magnetometry — see A-magneto) — gov.uk (established framing)

CONFLICTS / OPEN QUESTIONS

- Is the £2B package **inside or on top of** the £2.5B strategy envelope? Reporting is ambiguous — flag until a budget document settles it.
- **Post-Brexit exclusion** from parts of Horizon Europe quantum calls (the UK re-associated to Horizon in 2024 but on limited terms): how much does it cost the ecosystem?
- Can procurement pull (ProQure) work when no vendor can yet deliver the spec — or does it just subsidize R&D under a purchasing label?
- UK champions (Oxford Ionics, Quantum Motion, ORCA, Riverlane) are strong but small; Oxford Ionics was **acquired by IonQ** (2025, ~$1.08B) — a sovereignty question given US export-control leverage (see E-mna, E-export).

SOURCES

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- https://quantumcomputingreport.com/uk-government-commits-2-billion-2-67-billion-usd-to-national-quantum-scaling-and-procurement/
- https://www.nqcc.ac.uk/blogs/the-uk-has-thrown-down-the-gauntlet-on-quantum/
- https://uknqt.ukri.org/our-programme/

National programs — United States · E-us

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The US program is built on the **National Quantum Initiative (NQI) Act of 2018** (Public Law 115-368), which stood up the National Quantum Coordination Office in OSTP, tasked NIST, NSF, and DOE with QIS Research Centers, and authorized ~$1.2B over its first five years. The authorization **lapsed 30 September 2023**; the field then ran on annual appropriations for over two years. Two 2026 events redefined the posture: a **reauthorization bill (S.3597, Young/Cantwell, Jan 2026)** extending NQI to 2034, and — decisively — **two executive orders signed 22 June 2026** that pivoted federal policy from grant-making toward industrial policy and a hard national build target.

THE NUMBERS & MECHANISMS (GRADED, WITH CAVEATS)

- T5 ~$1.2B — original NQI five-year authorization (2019–2023) (quantum.gov). *Authorized ≠ appropriated*; annual QIS budgets ran ~$900M–1B/yr per the NQI budget supplements, which fold in pre-existing agency programs (double-counting risk).
- T5 ~$1.8B / 5 yr — House NQI Reauthorization figure; Senate S.3597 (Jan 2026) authorizes NIST ~$85M/yr FY26–30, up to three new NIST quantum centers at $18M/yr each, $25M for NASA QIST (congress.gov). Authorization only until appropriated.
- T5 $625M — DOE next-phase funding for the five National QIS Research Centers (energy.gov). Same headline as the 2020 tranche; verify renewal vs recycled 2020 announcement.
- T5 ~$2B — Commerce Dept. awards to nine quantum firms (May 2026), ~half to IBM, reportedly taking equity stakes (CNN/Forbes). Terms not fully public; blurs public/private tallies (see E-vc, E-sovereign).
- T2 **EO "Ushering in the Next Frontier of Quantum Innovation," 22 Jun 2026** — launches **QC-ADDS** (Quantum Computer for Application Development and Discovery Science), a DOE-led effort (launched 23 Jun 2026) to deploy a discovery-capable quantum computer **by 2028**; directs Commerce/War/Energy/NASA capability + workforce plans; expands the Quantum Counterintelligence Protection Team (whitehouse.gov).
- T2 **Companion PQC EO "Securing the Nation Against Advanced Cryptographic Attacks," 22 Jun 2026** — OMB deadlines for high-value-asset PQC: **key establishment by 31 Dec 2030, digital signatures by 31 Dec 2031**; Commerce PQC-migration pilot due 31 Dec 2027 (see A-pqc, E-pqcmarket).

KEY GRADED CLAIMS

- T2 NQI authorization lapsed Sept 2023; programs continued via appropriations — congress.gov / AIP FYI (established)
- T3 S.3597 introduced Jan 2026; extends NQI to Dec 2034 — congress.gov (claimed; passage status stale-checks fast)
- T4 Equity-stake structure of the $2B Commerce package — press reporting on unreleased terms (claimed)
- T4 "Quantum computer by 2028" (QC-ADDS) is a DOE program target, not a peer-reviewed roadmap — treat as policy ambition (roadmap)

CONFLICTS / OPEN QUESTIONS

- Has S.3597 passed either chamber, and does the June 2026 EO framework moot it? Recheck congress.gov.
- How much of the $2B Commerce package is new money vs repackaged CHIPS/DOE funds?
- NQI supplement totals mix new-start and legacy programs — the "US spends $X/yr on quantum" number is soft.
- The 2026 EOs shift the model from vendor-neutral grants (cf. DARPA QBI, E-defense) to picking champions with equity — a strategic turn the NQI Act's authors did not design for (see E-sovereign).

SOURCES

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- https://www.whitehouse.gov/presidential-actions/2026/06/securing-the-nation-against-advanced-cryptographic-attacks/
- https://www.mayerbrown.com/en/insights/publications/2026/06/president-trump-signs-two-executive-orders-on-quantum-computing-and-accelerated-post-quantum-cryptography-migration
- https://www.congress.gov/bill/119th-congress/senate-bill/3597/text
- https://www.energy.gov/articles/energy-department-announces-625-million-advance-next-phase-national-quantum-information
- https://www.cnn.com/2026/05/21/business/ibm-quantum-computing-firms-grants
- https://www.quantum.gov/wp-content/uploads/2024/12/NQI-Annual-Report-FY2025.pdf

Private investment & VC · E-vc

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Private quantum capital ran through three phases: the **2021 SPAC wave** (IonQ, Rigetti, D-Wave, Arqit went public early, most then traded far below listing), a **2022–23 trough**, and a **2024–26 boom** in which generalist mega-capital (BlackRock, Nvidia, Temasek, sovereign funds) replaced specialist quantum VCs. The two most-cited 2025 totals **disagree by 3x because they count different things**: PitchBook's ~$3.9B is pure quantum-*computing* VC; McKinsey's ~$12.6B (6.3x over 2024) counts *all* quantum-technology startup investment including sensing/comms **and** M&A (see E-mna). Either way, government's share of investment fell from ~a third in 2024 to ~3% in 2025 as private capital took over. In H1 2026 the action moved to **exits** (see E-mna).

THE NUMBERS (GRADED, WITH CAVEATS)

- T5 ~$3.9B across 127 deals — 2025 global quantum-**computing** VC record (PitchBook). Narrow definition.
- T5 ~$12.6B — 2025 quantum-**technology** startup investment, 6.3x over 2024, ~90% to computing (McKinsey 2026 Monitor). Broad definition + includes acquisition value. **Do not add to or compare directly with PitchBook's $3.9B** — different denominators.
- T5 $1B at ~$6B pre / ~$7B post — PsiQuantum Series E, BlackRock-led, Sept 2025 (press variously reports $6B pre and $7B valuation — flag the discrepancy).
- T5 ~$838.9M at $10B pre-money — Quantinuum Series B, Nov 2025; Nasdaq IPO filed 2026 (Crunchbase News).
- T5 Government share of quantum investment fell **~33% (2024) → ~3% (2025)** as private funds/capital markets took over (McKinsey).
- T5 Top capital deployers: BlackRock (~$1.7B), Nvidia (~$1.6B), Baillie Gifford, Temasek (Quantum Insider, Jun 2026).

KEY GRADED CLAIMS

- T2 2021 SPAC cohort (IonQ, Rigetti, D-Wave) went public years before product-market fit; all traded >75% below peak during 2022–23 — market record (established)
- T5 Venture-growth share of deal value jumped from ~1% to ~27.5%; sovereign funds and crossover investors now lead the largest rounds (PitchBook) (demonstrated, as data)
- T3 Crunchbase (mid-2026): startup *investment* actually **slowed** in 2026 even as public markets held strong — the money rotated from private rounds to IPOs/M&A (demonstrated)
- T6 Whether 2024–26 valuations survive contact with revenue (sector revenue ~$1B/yr, see E-market) is the open bet (speculative)

CONFLICTS / OPEN QUESTIONS

- **C-vc-bubble:** "capital cycle rationally pricing 2030s FTQC" vs "SPAC-wave redux with bigger names." Resolution: 2027–28 revenue growth at IonQ/Quantinuum/D-Wave versus guidance.
- Double-counting: government equity stakes (E-us $2B Commerce package; see E-sovereign) now blur public/private tallies.
- Keyword pollution: defense-drone maker "Quantum Systems" ($1.2B raise, Jul 2026) is **unrelated** to quantum computing but contaminates naive funding scrapes.
- IonQ +700% trailing-year and peers trade at revenue multiples with no historical analog outside dot-com — a valuation the ~$1B sector revenue cannot yet explain.

SOURCES

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Chapter 7 · How We Got Here

The "second quantum revolution" framing (2003) · T-2ndrev

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

By the early 2000s the field needed a way to explain what was new. The first quantum revolution (1900–1935, see T-old, T-formalism) discovered the *rules* — quantization, superposition, entanglement — and used them passively to *understand* nature: why atoms are stable, why metals conduct, why the sun shines. It gave the 20th century the transistor, the laser, and the atomic clock, all of which exploit quantum mechanics in bulk without controlling individual quantum states. The **second quantum revolution** is the shift from understanding the rules to *engineering with them* — deliberately preparing, controlling, and measuring individual quantum systems (single atoms, photons, spins, superconducting circuits) and harnessing superposition and entanglement as active resources. The phrase was crystallized by Jonathan Dowling and Gerard Milburn in a 2003 Royal Society paper (arXiv 2002) that opened flatly: "We are currently in the midst of a second quantum revolution." It became the standard framing for the whole quantum-technology enterprise — computing, communication, sensing, simulation — and now anchors funding programs (the EU Quantum Flagship explicitly invokes it), popular explanations, and strategy documents worldwide.

WHAT THE FRAMING CAPTURES

The distinction is real, not just rhetorical. The first revolution's technologies (transistor, laser, LED, MRI, GPS clocks) rely on quantum mechanics governing the collective behavior of many particles; you never address a single electron. The second revolution's technologies require *single-quantum control*: trap one ion and watch it, entangle two photons on demand, keep a superconducting qubit coherent long enough to run a gate. That capability is exactly what the 2012 Nobel to Haroche and Wineland recognized — "measuring and manipulating individual quantum systems" — and it is the enabling condition for everything in the atlas's hardware and stack layers (see H-, *S*-). Entanglement moves from a philosophical puzzle (the first revolution's leftover, see T-epr) to an engineering resource.

MILESTONE TIMELINE

- 1980s–1990s — Single-system control becomes real: single trapped ions (Wineland, Dehmelt), cavity QED (Haroche), the first superconducting and ion qubits (see T-early) — the technical precondition for the framing — T1 historical record
- 2002 (Jun) — "Quantum Technology: The Second Quantum Revolution" posted to arXiv — Jonathan P. Dowling & Gerard J. Milburn — T1 arXiv:quant-ph/0206091
- 2003 (Aug) — Published in *Philosophical Transactions of the Royal Society A* — the citation that fixes the term — T1 Phil. Trans. R. Soc. A 361, 1655 (2003)
- 2012 — Nobel Prize to Haroche & Wineland "for ground-breaking experimental methods that enable measuring and manipulation of individual quantum systems" — the revolution's core capability, honored (see T-nobel) — T1 nobelprize.org
- 2016– — The term is adopted into policy: the EU Quantum Flagship (€1B) and subsequent national strategies frame themselves as advancing the second quantum revolution (see E-eu, E-us) — T5 program documents
- 2025 (UN) — "International Year of Quantum Science and Technology," marking 100 years since matrix mechanics — the framing goes fully mainstream — T1 UN/UNESCO designation

WHY IT BELONGS ON THE TIMELINE

Naming an era changes how it is funded and pursued — the same way "NISQ" (see T-nisq) organized the noisy-hardware research program. "Second quantum revolution" gave governments and grant agencies a clean story: the first revolution paid off enormously (transistor, laser), so the second is worth a national bet. It also sets an honest expectation bar. If the first revolution's payoff was the transistor, the second revolution's advocates are implicitly promising something comparably transformative — which is precisely the promise the skeptics (see T-winter, O-hype) say has not yet been delivered.

KEY GRADED CLAIMS

- claim: The term "second quantum revolution" was crystallized by Dowling & Milburn (2002/2003) · tier: T1 · source: arXiv:quant-ph/0206091; Phil. Trans. R. Soc. A 361, 1655 (2003) · status: established (the underlying idea has older roots, e.g. Alain Aspect's usage, but this is the canonical citation)
- claim: The distinction (passive understanding vs. active single-system engineering) is technically substantive, not just branding · tier: T1 · status: established — single-quantum control is a real experimental capability, recognized by the 2012 Nobel
- claim: The second revolution has delivered a transistor-scale economic payoff · tier: T6 · status: speculative/contested — the central open question (see O-killerapp, T-winter)

SOURCES

- Dowling & Milburn, "Quantum Technology: The Second Quantum Revolution," Phil. Trans. R. Soc. A 361, 1655 (2003), arXiv:quant-ph/0206091
- Nobel Prize in Physics 2012 (Haroche & Wineland); EU Quantum Flagship documentation; UN International Year of Quantum 2025

- **Go deeper:** T-nisq (a later era-naming); T-nobel; E-eu, E-us (how the framing drives policy); T-winter (the counter-narrative)

Loophole-free Bell tests (2015) · T-belltests

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

For fifty years after Bell's 1964 theorem, every experiment that violated a Bell inequality left a door open. A determined local-hidden-variable believer could always invoke one of two escape hatches. The **detection loophole**: real detectors miss most particles, so maybe the detected sub-sample is unrepresentative and a "fair sampling" assumption is smuggling in the result. The **locality (communication) loophole**: if the two measurement stations are close enough, or fix their settings early enough, a subluminal signal could in principle coordinate them. Every experiment before 2015 closed one loophole while leaving the other open — Aspect (1982) closed locality with fast-switched analyzers but used inefficient photon detectors; later ion experiments closed detection but sat centimeters apart. The believer could always retreat to the un-closed hatch. In 2015 three groups, on three independent platforms, closed *both loopholes simultaneously* for the first time, finally ruling out local realism with no remaining escape. This is the experimental capstone of the foundations program (see T-foundations) and the work most directly recognized by the 2022 Nobel to Aspect, Clauser, and Zeilinger (see F-bell, T-nobel).

THE TECHNICAL BREAKTHROUGH

Closing both loopholes at once requires a brutal combination: detectors efficient enough (>~2/3, well above the fair-sampling threshold) *and* stations far enough apart that no light-speed signal can traverse the gap during a measurement. Delft solved it with an **event-ready entanglement-swapping scheme** — two distant NV-center electron spins are entangled by having their emitted photons interfere at a midpoint, so only genuinely entangled, heralded events are counted, sidestepping the detection loophole entirely while the 1.3 km separation guarantees locality. The photonic groups (Vienna, NIST) instead used near-unity-efficiency superconducting nanowire single-photon detectors (SNSPDs — see H-detect) with fast quantum-random setting choices over tens of meters.

MILESTONE TIMELINE

- 2015 (Aug submitted, Oct published) — First loophole-free test: two NV-center electron spins in diamond 1.3 km apart, event-ready entanglement swapping, 245 trials, S = 2.42 ± 0.20 — Hensen, Bernien et al. (Hanson group, TU Delft) — closes detection + locality together — T2 Hensen et al., Nature 526, 682 (2015), doi:10.1038/nature15759
- 2015 (Nov, same PRL issue) — Photon experiment with high-efficiency SNSPDs, violation at ~11.5 standard deviations — Giustina, Zeilinger et al. (Vienna) — independent confirmation, photonic platform — T2 PRL 115, 250401 (2015)
- 2015 (Nov, same PRL issue) — Independent photon experiment, ~7σ — Shalm, Nam et al. (NIST Boulder) — third platform, same conclusion — T2 PRL 115, 250402 (2015)
- 2016–2017 — Ion (Rosenfeld/Weinfurter, 398 m) and further photon tests tighten statistics — T2
- 2018 (May) — Cosmic Bell test uses light from stars (and, in the BIG Bell Test, ~100,000 human volunteers generating bits) to choose measurement settings, pushing the settings' causal origin back centuries/light-years and closing the **freedom-of-choice** loophole — Handsteiner et al.; BIG Bell Test Collaboration — T2 PRL 118, 060401 (2017); Nature 557, 212 (2018)

THE HUMAN CONTEXT

The three 2015 papers appeared within months of each other, a near-simultaneous finish after a fifty-year race — the same pattern of independent convergence that marked the birth of the formalism in 1925–26. Delft's event-ready scheme was slower and yielded far fewer events (245 trials over weeks) but was conceptually airtight; the photonic experiments were statistically overwhelming but relied on detector efficiency. Together they left the local-realist with only philosophically desperate positions.

KEY GRADED CLAIMS

- claim: Local hidden-variable theories are experimentally excluded with detection and locality loopholes closed simultaneously · tier: T2 (three independent 2015 experiments; now textbook) · status: established — Hensen (Delft), Giustina (Vienna), Shalm (NIST), three platforms
- claim: The residual "superdeterminism" loophole cannot be closed by any experiment · tier: T1 (logical) · status: established — it denies the free choice of settings and is unfalsifiable in principle; noted, not a live scientific dispute

SOURCES

- Hensen et al., Nature 526, 682 (2015), doi:10.1038/nature15759
- Giustina et al., PRL 115, 250401 (2015); Shalm et al., PRL 115, 250402 (2015)
- Handsteiner et al. (cosmic Bell), PRL 118, 060401 (2017); The BIG Bell Test Collaboration, Nature 557, 212 (2018)

- **Go deeper:** F-bell (the inequality and its meaning); T-foundations (the 1935→1982 lead-up); H-detect (the SNSPDs that made it possible); T-nobel (2022)

Birth of quantum computing (1980–1994) · T-birth

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The idea arrived from several directions at once. Yuri Manin (1980, in a Russian book) and Paul Benioff (1980) asked whether computation itself could be quantum-mechanical — Benioff building an explicit quantum Turing machine to show computation is compatible with unitary evolution. Richard Feynman flipped the question in a now-legendary May 1981 keynote at the first MIT Physics of Computation conference: classical computers choke on simulating quantum systems because the state space grows exponentially, so build a quantum machine to simulate quantum physics natively. "Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical." David Deutsch made it rigorous in 1985 — a universal quantum computer and the first quantum algorithm — motivated partly by wanting to test the many-worlds interpretation. Bennett and Brassard, building on Wiesner's dormant conjugate-coding idea (see T-qinfobirth), showed quantum states could carry provably secure keys (BB84, 1984). The era ends with a detonation: Peter Shor's 1994 algorithm factored integers in polynomial time, threatening RSA and the entire public-key infrastructure. Overnight, quantum computing went from a physicists' curiosity to a national-security priority. Every funding line that exists today traces back to that one result.

MILESTONE TIMELINE

- 1980 — Quantum computation floated in *Computable and Uncomputable* — Yuri Manin — earliest published suggestion (in Russian, so little-noticed in the West for years) — T1 Manin, Sovetskoye Radio (1980)
- 1980 — Quantum-mechanical model of a Turing machine — Paul Benioff (Argonne) — showed computation is compatible with reversible quantum evolution — T1 Benioff, J. Stat. Phys. 22, 563 (1980)
- 1981 (6–8 May) — "Simulating Physics with Computers" keynote, MIT Physics of Computation conference — Richard Feynman — the field's founding motivation: simulate quantum with quantum — T1 published Feynman, Int. J. Theor. Phys. 21, 467 (1982)
- 1982 — No-cloning theorem: an unknown quantum state cannot be copied — Wootters & Zurek; Dieks — the security root of quantum crypto (see F-nocloning) — T1 Nature 299, 802 (1982)
- 1984 — BB84 quantum key distribution protocol — Charles Bennett (IBM) & Gilles Brassard (Montréal) — first quantum information technology; still deployed today — T1 Proc. IEEE Int. Conf. on Computers, Systems and Signal Processing, Bangalore, 175 (1984)
- 1985 — Universal quantum computer; the quantum Turing machine; the first quantum algorithm (Deutsch's problem) — David Deutsch (Oxford) — the theoretical foundation of the whole field — T1 Deutsch, Proc. R. Soc. A 400, 97 (1985)
- 1992 — Deutsch–Jozsa algorithm, first provable exponential quantum–classical separation in the exact query model — David Deutsch & Richard Jozsa — proof the speedups could exist — T1 Proc. R. Soc. A 439, 553 (1992)
- 1993 — Bernstein–Vazirani and the first steps toward complexity-theoretic separation; quantum teleportation protocol — Bernstein & Vazirani (STOC 1993); Bennett, Brassard, Crépeau, Jozsa, Peres, Wootters — entanglement as a communication resource — T1 PRL 70, 1895 (1993)
- 1994 (20 Nov) — Polynomial-time factoring and discrete-log algorithm — Peter Shor (AT&T Bell Labs) — broke public-key crypto in principle; ignited the field and its funding — T1 Shor, Proc. 35th FOCS, 124 (1994); SIAM J. Comput. 26, 1484 (1997)
- 1994 — Simon's problem: the exponential separation that directly inspired Shor — Daniel Simon — the algorithmic template (period-finding) Shor generalized — T1 Proc. 35th FOCS, 116 (1994)

THE HUMAN CONTEXT

Feynman's talk is often called the birth of the field, but he was pitching a simulator, not a general-purpose computer, and he was skeptical the machine could be built. Deutsch's universal model came from a physicist worried about interpretation, not computation. Shor's algorithm reportedly grew out of Simon's problem, which Shor heard about at a talk; he generalized period-finding to factoring within weeks. The result was so consequential that within two years the field went from a few dozen people to a global program — and cryptographers began the migration that is still underway 30 years later (see A-pqc).

KEY GRADED CLAIMS

- claim: Shor's algorithm factors N in polynomial time on an ideal fault-tolerant quantum computer · tier: T1 · source: Shor 1994; SIAM J. Comput. 26, 1484 (1997) · status: established (mathematically proven; large-scale execution still awaits hardware — see S-shor, O-scaling)
- claim: Feynman's simulation pitch is the application most likely to pay off first · tier: T6 · provenance: widely held field belief, unproven at scale · status: speculative — see S-qsim, O-killerapp

- claim: The no-cloning theorem (1982) makes BB84 security information-theoretic, not computational · tier: T1 · status: established — see F-nocloning, A-qkd

SOURCES

- Manin (1980); Benioff, J. Stat. Phys. 22 (1980); Feynman, IJTP 21 (1982); Wootters–Zurek, Nature 299 (1982); Bennett–Brassard, BB84 (1984); Deutsch, Proc. R. Soc. A 400 (1985); Shor, FOCS (1994); Simon, FOCS (1994)

- **Go deeper:** T-qinfobirth (Wiesner → Holevo → Bennett-Brassard lineage); S-shor; F-nocloning; T-early (turning theory into hardware)

Early experiments (1995–2010) · T-early

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

After Shor, the question stopped being "is it useful?" and became "can you actually build one?" This era answered "in principle, yes" — while quietly revealing how hard the engineering would be. Cirac and Zoller published a realistic trapped-ion two-qubit gate in May 1995, and Wineland's NIST group demonstrated a working quantum logic gate on a single beryllium ion within months. The deepest result was theoretical: Shor (1995) and Steane (1996) proved that quantum errors could be corrected *without measuring — and thereby destroying — the encoded data*, removing the single strongest objection to the whole enterprise. The threshold theorem (Aharonov–Ben-Or, Knill–Laflamme–Zurek, Kitaev, 1996–98) then showed that below a critical error rate, arbitrarily long computations are possible. Grover added a second headline algorithm. NMR machines ran the first real algorithms — including Shor's algorithm factoring 15 — before hitting a hard scalability wall. DiVincenzo wrote down the five-item checklist any real platform must satisfy, still the field's scorecard. And superconducting circuits went from the first coherent charge qubit (NEC 1999) through circuit QED (Yale 2004) to the transmon (Yale 2007) — the noise-tolerant design that would dominate the next two decades.

MILESTONE TIMELINE

- 1995 (May) — Trapped-ion CNOT gate proposal using a shared vibrational mode as the bus — Ignacio Cirac & Peter Zoller (Innsbruck) — first realistic hardware blueprint for any platform — T1 Cirac & Zoller, PRL 74, 4091 (1995)
- 1995 (Dec) — First quantum logic gate demonstrated on a single $^{9}Be^{+}$ ion — Monroe, Meekhof, King, Itano, Wineland (NIST) — theory becomes experiment in seven months — T1 Monroe et al., PRL 75, 4714 (1995)
- 1995–96 — Quantum error-correcting codes: the 9-qubit Shor code, then the 7-qubit Steane (CSS) code — Peter Shor; Andrew Steane — errors are correctable; fault tolerance is possible — T1 Shor, Phys. Rev. A 52, R2493 (1995); Steane, PRL 77, 793 (1996)
- 1996 — Threshold theorem: below a critical physical error rate, arbitrarily long computation is possible — Aharonov–Ben-Or; Knill–Laflamme–Zurek; Kitaev — the theoretical license for fault tolerance (see S-qec) — T1 STOC 1997; Science 279, 342 (1998)
- 1996 — Unstructured-search algorithm, quadratic speedup $O(\sqrt{N})$ — Lov Grover (Bell Labs) — the second canonical algorithm — T1 Grover, Proc. 28th STOC, 212 (1996)
- 1997–98 — First NMR quantum algorithms (Deutsch–Jozsa, Grover on 2 qubits) — Chuang, Gershenfeld, Jones et al. — first algorithms run on any hardware — T2 Nature 393, 143 (1998); Chuang et al., PRL 80, 3408 (1998)
- 1999 — First coherent superconducting (charge) qubit; Rabi oscillations in a Cooper-pair box — Nakamura, Pashkin, Tsai (NEC) — solid-state qubits arrive — T2 Nakamura et al., Nature 398, 786 (1999)
- 2000 — DiVincenzo criteria: five requirements (plus two for communication) any quantum computer must meet — David DiVincenzo (IBM) — the checklist every new modality is still graded against — T1 DiVincenzo, Fortschr. Phys. 48, 771 (2000)
- 2001 — Shor's algorithm factors $15 = 3 \times 5$ on 7 NMR qubits — Vandersypen, Steffen, Chuang et al. (IBM Almaden/Stanford) — proof-of-principle for the algorithm that started the field — T2 Nature 414, 883 (2001)
- 2004 — Circuit QED: a superconducting qubit strongly coupled to a single microwave photon in a resonator — Wallraff, Schoelkopf, Girvin et al. (Yale) — the readout-and-coupling architecture of all superconducting QCs — T2 Nature 431, 162 (2004)
- 2007 — Transmon qubit: charge-noise-insensitive by design — Koch, Schoelkopf, Devoret et al. (Yale) — the workhorse used by IBM, Google, Rigetti to this day — T2 Koch et al., Phys. Rev. A 76, 042319 (2007)

THE HUMAN CONTEXT

NMR's early lead was real and then evaporated — Braunstein and colleagues showed in 1999 that room-temperature liquid-state NMR states are so mixed they are arguably never entangled, and the signal decays exponentially with qubit count, so the "factoring 15" demo could not have scaled. Two of the 2007 transmon co-authors, Devoret and Martinis, would later share the 2025 Nobel Prize for the macroscopic-quantum-tunneling experiments (1984–85) that made superconducting qubits conceivable in the first place (see T-nobel). This era is where the modern hardware map was drawn: ions and superconductors as the two front-runners, error correction as the organizing goal.

KEY GRADED CLAIMS

- claim: Quantum error correction works — errors can be detected and reversed without measuring the logical data · tier: T1 · source: Shor (1995), Steane (1996) · status: established (theory); experimentally realized below threshold only in 2024 (see T-ecera)

- claim: Liquid-state NMR does not scale (signal decays exponentially; states arguably unentangled) · tier: T1 · source: Braunstein et al., PRL 83, 1054 (1999) · status: established — why NMR's early lead evaporated
- claim: A working platform must satisfy the DiVincenzo criteria (scalable qubits, initialization, universal gates, long coherence, measurement) · tier: T1 · status: established as the field's standard scorecard

SOURCES

- Cirac–Zoller, PRL 74 (1995); Monroe et al., PRL 75 (1995); Shor (1995); Steane (1996); Grover, STOC (1996); DiVincenzo, Fortschr. Phys. 48 (2000); Vandersypen et al., Nature 414 (2001); Wallraff et al., Nature 431 (2004); Koch et al., PRA 76 (2007)

- **Go deeper:** S-qec (codes and threshold); H-ion, H-supercon (the two platforms this era chose); T-nobel (Devoret/Martinis 2025); T-race (industrial scale-up)

The error-correction era (2020–now) · T-ecera

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The scoreboard metric changed. Through the 2010s the field measured itself in physical qubit counts and one-off "supremacy" stunts; from about 2023 it measures itself in *logical* qubits and error suppression. Two results reset expectations: QuEra/Harvard's 48-logical-qubit neutral-atom demo (late 2023) showed logical qubits could be operated at scale, and Google's Willow (Dec 2024) achieved the first *below-threshold* surface code — the regime where making the code bigger makes the logical error rate exponentially *smaller*, which is the threshold theorem finally realized in hardware rather than proven on paper. Fault tolerance became an engineering trajectory instead of a hope, and vendors began publishing logical-qubit roadmaps with dates attached. That is exactly where the grading discipline earns its keep: this era mixes peer-reviewed landmarks (T2) with contested vendor claims (T4), sometimes inside the same press release, and by 2026 nearly every "advantage" headline draws a classical counterattack within months. The 2025 Nobel Prize — to Clarke, Devoret, and Martinis for the 1984–85 experiments that first showed a macroscopic circuit obeys quantum mechanics — was the establishment's formal blessing of the hardware lineage that made all of this possible (see T-nobel).

MILESTONE TIMELINE

- 2020 (Dec) — Jiuzhang photonic boson-sampling advantage claim — USTC (Pan group) — second beyond-classical claim; non-programmable, so no algorithm runs on it — [T2, contested] Zhong et al., Science 370, 1460 (2020)
- 2021 (Nov) — Eagle, 127 qubits, first processor over 100 — IBM — physical scaling continues — T4 IBM announcement
- 2023 (Feb) — A distance-5 surface code logical qubit outperforms distance-3 for the first time — Google Quantum AI — first hardware sign QEC scaling works, though not yet below threshold — T2 Nature 614, 676 (2023)
- 2023 (Jun) — "Utility" experiment: 127-qubit Eagle + zero-noise extrapolation beats brute-force classical on a 2D Ising spin model — IBM — [T2 experiment, contested] Kim et al., Nature 618, 500 (2023); matched by tensor-network methods within weeks
- 2023 (Oct) — 1,180-qubit neutral-atom array announced — Atom Computing — first 1,000+ physical qubits — T4 company announcement
- 2023 (Dec) — 48 logical qubits, error-corrected operations, on 280 neutral atoms — Bluvstein et al. (Harvard/QuEra/MIT) — logical qubits at scale, first credible demonstration — T2 Nature 626, 58 (2024)
- 2024 (Apr) — 12 logical qubits on trapped-ion H2 at ~800× fewer errors than physical, with active syndrome extraction — Microsoft + Quantinuum — logical-over-physical advantage demonstrated — [T3→T2] arXiv:2404.02280
- 2024 (9 Dec) — Willow, 105 qubits: below-threshold surface code, logical error rate ~halves per code-distance step (d=3→5→7), $\Lambda \approx 2.14$ — Google Quantum AI — the threshold theorem realized in hardware — T2 Nature 638, 920 (2025); the companion RCS claim "under 5 min vs. 10^{25} classical years" is [T4, contested]
- 2025 (Feb) — Majorana 1 "first topological qubit" — Microsoft — [T4, contested] Nature's referees noted the published data "does not, on its own, determine whether Majorana zero modes are present"; strong skepticism (APS Physics, Science News, Aaronson)
- 2025 (Jun) — Large-scale FTQC roadmap: qLDPC codes; "Starling" targeting ~200 logical qubits / 100M gates by 2029 — IBM — [T4, roadmap]
- 2025 (Oct 7) — Nobel Prize in Physics to John Clarke, Michel Devoret, John Martinis for macroscopic quantum tunneling and energy quantization in Josephson-junction circuits (experiments 1984–85) — the superconducting-qubit lineage honored (see T-nobel) — T1 nobelprize.org (2025)
- 2025 (Oct) — "Quantum Echoes" out-of-time-order-correlator experiment claimed as a first *verifiable* quantum advantage — Google (Willow) — [T3/T4, contested by default per SCHEMA rule 2]
- 2025 (5 Nov) — Helios launched: 98 $^{137}Ba^{+}$ ion qubits, 99.921% two-qubit fidelity, up to 48 logical qubits via color codes (2:1 encoding) — Quantinuum — T4 company launch; named customers include JPMorganChase, Amgen, BMW, SoftBank
- 2025 (12 Nov) — Nighthawk (120 qubits, higher connectivity) and Loon (qLDPC-connectivity test chip) delivered; real-time decoder progress — IBM — T4 IBM newsroom
- 2026 (Jan) — 96 logical qubits via a high-rate [[16,6,4]] code on 448 neutral atoms (~4.7:1 encoding), with below-threshold error suppression across all logical qubits — QuEra — the logical-qubit record roughly doubles in ~13 months — T2 peer-reviewed, Nature s41586-025-09848-5 (doi:10.1038/s41586-025-09848-5, Jan 2026) — primary-source verified

- 2026 (Jan) — Quantinuum files confidentially for a US IPO (~$20B valuation reported) — T5 press reports
- 2026 (Jun) — First fully-distributed GHZ (tripartite) state across a three-node network of remote trapped-ion memories linked by photonic interconnects; fidelity 0.841–0.881, Mermin violation with the detection loophole closed — Duke Quantum Center + IonQ (Monroe group) — a step toward modular/networked QC (see H-intercon) — T3 preprint, Goetting et al., arXiv:2606.17173 (2026) — primary-source verified
- 2026 (Jun) — Microsoft reports Majorana 2 progress (lead-based superconducting stack, larger topological gap) — [T4, contested] company claim; extends the disputed 2025 result, independent confirmation still absent. On 24 Jun 2026 Henry Legg (St Andrews) published a formal Nature critique (d41586-026-01788-y) citing flawed tune-up routines, software errors in the analysis code, and omitted transport measurements; Microsoft filed a formal Nature reply (Nayak: "we stand by our results and our roadmap")

THE HONEST READ (MID-2026)

Below-threshold QEC is real and peer-reviewed (Willow). Logical-qubit counts are climbing fast (48 → 96 in about a year) but the headline numbers span the full grading range: some are Nature papers, some are launch-day press. The topological-qubit claim remains the era's sharpest credibility test — two years of Microsoft announcements without independent confirmation. And every "verifiable advantage" so far has drawn a classical response, so per SCHEMA rule 2 they are contested by default. The trajectory is genuine; the specific superlatives are provisional.

KEY GRADED CLAIMS

- claim: Below-threshold QEC is demonstrated in hardware (error suppression grows with code distance, $\Lambda \approx 2.14$) · tier: T2 · source: Nature 638, 920 (2025) · provenance: Google, but peer-reviewed and independently discussed · status: demonstrated
- claim: A topological (Majorana) qubit exists · tier: T4 · provenance: Microsoft (sells Azure Quantum) · status: contested — no independent confirmation as of mid-2026
- claim: Logical-qubit counts are doubling on a ~roughly annual cadence (48→96) · tier: T2 · source: QuEra, Nature s41586-025-09848-5 (2026) · status: demonstrated for the specific 2024→2026 neutral-atom results (both peer-reviewed); not a guaranteed trend
- claim: Utility-scale / verifiable-advantage results resist classical counterattack · tier: T3 · status: contested — every claim so far has drawn a classical response within months

SOURCES

- Bluvstein et al., Nature 626 (2024); Google Quantum AI, Nature 638, 920 (2025); Kim et al., Nature 618 (2023); Zhong et al., Science 370 (2020); Microsoft/Quantinuum arXiv:2404.02280; nobelprize.org Physics 2025; quantinuum.com Helios launch (2025-11-05); newsroom.ibm.com Nighthawk/Loon (2025-11-12); ibm.com/quantum/blog/large-scale-ftqc
- QuEra 96-logical (Jan 2026): primary source confirmed — Nature s41586-025-09848-5 (doi:10.1038/s41586-025-09848-5), peer-reviewed, [[16,6,4]] high-rate code on 448 atoms → promoted to T2. Duke/IonQ networking (Jun 2026): primary source confirmed — Goetting et al., arXiv:2606.17173 (preprint → T3)
- **Go deeper:** S-logical, S-qec, S-decoders; T-nobel (2025 prize); O-scaling, O-hype (the roadmap-vs-reality gap); T-winter (the funding-downturn risk)

The EPR paradox & Bohr's reply (1935) · T-epr

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

In May 1935 Einstein, Boris Podolsky, and Nathan Rosen published four pages in *Physical Review* under a title posed as a question: "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" Their answer was no. Using a pair of particles that had interacted and separated, they argued that measuring one instantly fixes a property of the other; since no signal could travel between them (locality) and the outcome is definite (realism), that property must have been *real all along* — an "element of reality" the wavefunction fails to include. Either quantum mechanics is incomplete, or nature is nonlocal. Bohr replied in the same journal five months later with a subtle, much-debated argument that the two particles form a single indivisible system and that EPR's criterion of reality is ambiguous when applied to complementary quantities. Schrödinger, energized by the paper, coined "entanglement" (*Verschränkung*) to name exactly the correlation EPR had isolated, and wrote his cat into the record the same year. For thirty years the exchange looked like unresolvable philosophy. Then Bell showed EPR's own premises — locality plus realism — make a prediction that quantum mechanics violates, and the argument became an experiment (see T-foundations, T-belltests).

WHY IT EARNS ITS OWN CARD

EPR is usually folded into the 1925–35 formalism, but it deserves standing as a milestone in its own right because it is the single most consequential *question* in the field's history. Every later development in quantum foundations and much of quantum information traces to it: Bell's theorem is a direct answer to EPR; the loophole-free tests of 2015 are the experimental verdict; entanglement — named in response to EPR — is the resource behind teleportation, dense coding, device-independent cryptography, and the speedups of quantum computing. Einstein intended the paper as a critique; it became the field's most fertile seed. The word "paradox" is a misnomer: there is no logical contradiction, only a forced choice between locality and completeness that nature (via Bell) later made for us.

MILESTONE TIMELINE

- 1935 (15 May) — "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" — Einstein, Podolsky, Rosen (Institute for Advanced Study, Princeton) — poses the locality-vs-completeness dilemma; the "EPR criterion of reality" — T1 Phys. Rev. 47, 777 (1935)
- 1935 (Jun–Aug) — Schrödinger's correspondence with Einstein; the three-part "Die gegenwärtige Situation in der Quantenmechanik" coins *Verschränkung* (entanglement) and presents the cat — Erwin Schrödinger — names the phenomenon EPR isolated — T1 Naturwissenschaften 23, 807/823/844 (1935)
- 1935 (15 Oct) — Bohr's reply, same title as EPR — Niels Bohr — argues the system is indivisible and EPR's reality criterion is ambiguous under complementarity — T1 Phys. Rev. 48, 696 (1935)
- 1951 — Bohm recasts EPR in discrete spin-½ variables (the version everyone now teaches) — David Bohm — makes EPR concrete and testable-in-principle — T1 Bohm, *Quantum Theory* (1951)
- 1964 — Bell converts EPR's premises into a testable inequality — John S. Bell — turns the paradox into a measurement (see T-foundations) — T1 Physics 1, 195 (1964)

THE HUMAN CONTEXT

Einstein reportedly disliked the final EPR text — Podolsky drafted it, and Einstein felt the essential point (the tension between locality and completeness) was buried under formalism. He restated it more sharply in later letters, especially to Schrödinger, using the phrase *spukhafte Fernwirkung* — "spooky action at a distance." Podolsky, a Soviet-born physicist, is a footnote to most accounts but is the reason the paper exists in the form it does. Bohr, according to Léon Rosenfeld, dropped everything to respond and spent weeks agonizing over the wording — a sign that EPR had landed a real blow, even if the eventual verdict went against Einstein.

KEY GRADED CLAIMS

- claim: EPR is a valid argument, not a fallacy: *if* locality and realism both hold, QM is incomplete · tier: T1 · status: established as logic; the conditional's premises were later shown false by experiment
- claim: Nature is nonlocal in the EPR/Bell sense — one of locality or realism must go · tier: T1 · source: Bell tests, loophole-free 2015 (see T-belltests) · status: established
- claim: Entanglement, the phenomenon EPR isolated, is a physical resource · tier: T1 · status: established — it underlies teleportation, QKD, and quantum computing (see F-entangle, F-teleport)

SOURCES

- Einstein, Podolsky, Rosen, Phys. Rev. 47, 777 (1935); Bohr, Phys. Rev. 48, 696 (1935); Schrödinger, Naturwissenschaften 23 (1935) and Proc. Camb. Phil. Soc. 31, 555 (1935)
- Bell, Physics 1, 195 (1964); Gilder, *The Age of Entanglement* (2008)

- **Go deeper:** T-solvay (the debate EPR grew out of); T-foundations (Bell's answer); T-belltests (the verdict); F-entangle; F-interp

The formalism (1925–1935) · T-formalism

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

In one astonishing decade the ad-hoc patches of the old quantum theory were replaced by a complete mathematical framework — twice, independently, from opposite instincts. Heisenberg, 23 and recovering from hay fever on the island of Helgoland in June 1925, built matrix mechanics by refusing to talk about anything unobservable (electron orbits) and keeping only measurable transition amplitudes. Schrödinger, 38 and unusually old for a revolution, took de Broglie's matter waves literally and wrote down a wave equation over the Christmas holidays of 1925–26. The two formalisms looked nothing alike and their authors disliked each other's approach — Heisenberg called wave mechanics "disgusting," Schrödinger found matrix mechanics "repellent" — yet within months Schrödinger, Pauli, and later von Neumann proved them mathematically equivalent. Born supplied the interpretation that made the whole thing physics: $|\psi|^2$ is a probability, a claim so uncomfortable that Einstein spent the rest of his life objecting to it. Dirac, the quietest of them, unified the formalism, married it to special relativity, and predicted antimatter from the mathematics alone. The decade closed with the framework's own architects flagging what it left unresolved: EPR named the spooky correlations later called entanglement, and Schrödinger's cat dramatized the unsolved measurement problem. Both puzzles became the raw material of the foundations era and, eventually, of quantum information itself.

MILESTONE TIMELINE

- 1925 (Jul) — Matrix mechanics (the Umdeutung / "reinterpretation" paper), drafted on Helgoland — Werner Heisenberg, formalized by Born & Jordan and then the Dreimännerarbeit (three-man paper) with Born, Heisenberg, Jordan — first complete quantum formalism — T1 Heisenberg, Zeitschrift für Physik 33, 879 (1925)
- 1926 (Jan–Jun) — Wave equation for matter; hydrogen solved analytically in the first paper — Erwin Schrödinger (four-part series) — the workhorse equation of all quantum science — T1 Schrödinger, Annalen der Physik 79–81 (1926)
- 1926 (Jun) — Born rule: $|\psi|^2$ is a probability density — Max Born (added in proof, in a footnote) — probability enters physics at the fundamental level; the interpretive keystone — T1 Born, Zeitschrift für Physik 37, 863 (1926)
- 1927 (Feb–Mar) — Uncertainty principle: $\Delta x \Delta p \gtrsim \hbar/2$ — Heisenberg — conjugate observables cannot be jointly sharp — T1 Heisenberg, Zeitschrift für Physik 43, 172 (1927)
- 1927 — Electron diffraction observed independently — Davisson & Germer (Bell Labs, nickel crystal); G.P. Thomson (Aberdeen, thin films) — de Broglie's matter waves confirmed; father J.J. Thomson had shown the electron was a particle, son showed it was a wave — T1 Phys. Rev. 30, 705 (1927)
- 1927 (Oct 24–29) — Fifth Solvay Conference, Brussels — the Bohr–Einstein debates begin (see T-solvay for the full card) — the interpretation argument that never ended — T1 historical record; Bacciagaluppi & Valentini (2009)
- 1928 — Relativistic electron equation; antimatter implied by the negative-energy solutions — Paul Dirac — QM meets special relativity; the positron was found by Anderson in 1932 — T1 Dirac, Proc. R. Soc. A 117, 610 (1928)
- 1932 — Rigorous Hilbert-space axiomatization and measurement theory; the density operator — John von Neumann — the mathematical foundation still in use, including a (later-critiqued) no-hidden-variables argument — T1 von Neumann, *Mathematische Grundlagen der Quantenmechanik*
- 1935 (May) — EPR paper: if QM is complete, reality is nonlocal — Einstein, Podolsky, Rosen; Bohr's reply followed in October — defined the question Bell made testable 29 years later (see T-epr) — T1 Phys. Rev. 47, 777 (1935)
- 1935 — "Entanglement" (Verschränkung) named; the cat thought experiment published — Erwin Schrödinger, in dialogue with Einstein — the resource quantum computing runs on gets its name — T1 Naturwissenschaften 23, 807/823/844 (1935); Proc. Camb. Phil. Soc. 31, 555 (1935)

THE HUMAN CONTEXT

The speed is the story: from Heisenberg's Helgoland notebook to von Neumann's axioms was seven years. Most of it happened at Göttingen, Copenhagen, and Munich, among people in their twenties. Born, whose interpretation is arguably the deepest single idea of the decade, was not included in Heisenberg and Jordan's 1932 Nobel and had to wait until 1954. Dirac derived a new particle from an equation before anyone had seen one — the modern template for theory leading experiment.

KEY GRADED CLAIMS

- claim: Matrix and wave mechanics are mathematically equivalent formulations of the same theory · tier: T1 · source: Schrödinger (1926); von Neumann (1932) · status: established
- claim: The Born rule (probabilities as squared amplitudes) is a postulate, not yet derived from the other axioms · tier: T1 · status: established — deriving it is an open program (see F-measure, O-foundations)

- claim: EPR exposed a genuine tension between QM and local realism · tier: T1 (as a historical argument) · status: contested-then-resolved — Bell tests ruled for QM (see T-foundations, T-belltests)

SOURCES

- Primary papers above; van der Waerden, *Sources of Quantum Mechanics* (1967); Jammer, *The Conceptual Development of Quantum Mechanics* (1966); Mehra & Rechenberg, *The Historical Development of Quantum Theory* (6 vols)

- **Go deeper:** T-solvay (1927 debates); T-epr (the 1935 exchange as its own milestone); F-uncertainty; F-interp; T-nobel (Heisenberg 1932; Schrödinger & Dirac 1933)

Foundations era (1935–1980) · T-foundations

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

For three decades the EPR question sat in philosophical exile while physicists built QED and the bomb — asking what the wavefunction "really" meant was, as John Clauser later put it, a good way to end a career. Bohm reopened it in 1952 with a working hidden-variable theory (pilot waves) that the orthodoxy had declared impossible, citing von Neumann's 1932 no-go proof; Bohm's counterexample showed the proof assumed too much. That provocation reached John Stewart Bell, a particle physicist at CERN who did foundations "on the side." In 1964 Bell proved that *any* local hidden-variable theory obeys an inequality that quantum mechanics violates — turning a metaphysical dispute into a laboratory measurement. CHSH (1969) recast the inequality into a form real detectors could test. Freedman and Clauser ran the first experiment in 1972 on a shoestring at Berkeley; Aspect's 1981–82 experiments in Orsay added time-varying analyzers that switched settings while photons were in flight, closing the locality loophole of the day. Quantum mechanics won every round. The 2022 Nobel Prize to Clauser, Aspect, and Zeilinger certified this quiet, career-risky arc as the origin of quantum information science: entanglement is real, physical, and usable.

MILESTONE TIMELINE

- 1948–1950 — QED renormalized; the mainstream of physics moves to fields, leaving foundations to a marginalized few — Feynman, Schwinger, Tomonaga, Dyson — the intellectual climate the foundations workers had to fight — T1 historical record (see T-nobel, 1965)
- 1952 — Pilot-wave (hidden-variable) reformulation of QM — David Bohm — proved a deterministic completion is logically possible, refuting the "von Neumann forbids it" folklore and provoking Bell — T1 Bohm, Phys. Rev. 85, 166 & 180 (1952)
- 1957 — Relative-state ("many-worlds") interpretation — Hugh Everett III (Princeton PhD under Wheeler) — collapse-free QM; ignored for a decade, now mainstream — T1 Everett, Rev. Mod. Phys. 29, 454 (1957)
- 1964 — Bell's theorem: local hidden variables bound correlations; QM violates the bound — John S. Bell, on sabbatical from CERN — made metaphysics falsifiable; the pivot of the whole era — T1 Bell, Physics 1, 195 (1964)
- 1966 — Kochen–Specker / Bell's review of no-hidden-variable proofs — Bell (Rev. Mod. Phys. 38, 447), Kochen & Specker (1967) — contextuality as a separate no-go (see F-contextuality) — T1
- 1969 — CHSH inequality, an experimentally testable form usable with real (inefficient) detectors — Clauser, Horne, Shimony, Holt — the version every Bell test since has used — T1 PRL 23, 880 (1969)
- 1970 — Decoherence program begins: environment-induced superselection — H. Dieter Zeh — why the classical world emerges from the quantum; later the enemy of every qubit — T1 Zeh, Found. Phys. 1, 69 (1970)
- 1972 — First experimental Bell test (calcium cascade photons); QM violates CHSH — Stuart Freedman & John Clauser (Berkeley) — first data against local realism, on borrowed equipment — T1 Freedman & Clauser, PRL 28, 938 (1972)
- 1973 — Holevo bound: a qubit carries at most one classical bit of accessible information — Alexander Holevo (Moscow) — a founding theorem of quantum information theory (see T-qinfobirth) — T1 Holevo, Probl. Peredachi Inf. 9, 3 (1973)
- 1981–82 — Bell tests with locality-motivated timing, including analyzers switched while photons are in flight — Alain Aspect, Dalibard, Roger (Orsay) — closed the most glaring locality loophole of the day; $S = 2.697 \pm 0.015$ — T1 PRL 47, 460 (1981); 49, 91 & 1804 (1982)
- 2022 — Nobel Prize in Physics to Aspect, Clauser, Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science" — the era's retroactive coronation — T1 nobelprize.org (2022)

THE HUMAN CONTEXT

Bell called quantum foundations his "hobby" and published his theorem in an obscure, short-lived journal (*Physics*) that paid authors — partly so he could keep his real job doing accelerator physics. Clauser was warned the Bell test would sink his career; he ran it anyway and got the "wrong" answer (QM was right, disappointing his own realist hopes). Everett left academia for defense work after his interpretation was dismissed. The people who built the conceptual foundation of a trillion-dollar field mostly did it against professional incentives.

KEY GRADED CLAIMS

- claim: Nature violates Bell inequalities; local hidden-variable theories are ruled out · tier: T1 · source: Aspect (1982); loophole-free tests 2015 sealed it (see T-belltests) · status: established
- claim: Bohm's 1952 theory shows von Neumann's no-hidden-variables proof was not decisive · tier: T1 · status: established (the proof's flawed assumption was later pinpointed by Bell 1966)

- claim: The measurement problem itself remains unresolved · tier: T1 (as consensus about openness) · status: contested — see O-foundations, F-interp

SOURCES

- Bell, Physics 1, 195 (1964); CHSH, PRL 23, 880 (1969); Freedman–Clauser, PRL 28, 938 (1972); Aspect et al. (1981–82); Zeh, Found. Phys. 1, 69 (1970); NobelPrize.org 2022 scientific background
- Bell, *Speakable and Unspeakable in Quantum Mechanics* (1987) — the collected essays; Gilder, *The Age of Entanglement* (2008)
- **Go deeper:** T-belltests (the 2015 loophole-free capstone); T-epr (the question Bell answered); F-bell; F-decoher; T-nobel

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The NISQ era (Preskill 2018) · T-nisq

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

By 2017 the field needed a word for the machines it was actually building. They were no longer the handful-of-qubit demos of the 2000s, and they were nowhere near the fault-tolerant, error-corrected computers of theory. They were processors with roughly 50–100 physical qubits, imperfect gates, and no error correction — powerful enough to be hard to simulate classically, too noisy to run a deep useful circuit. John Preskill supplied the name in a December 2017 keynote (arXiv 2 January 2018, published in *Quantum* 6 August 2018): **NISQ — Noisy Intermediate-Scale Quantum**. "Intermediate-scale" for a qubit count beyond brute-force classical simulation yet far short of utility; "noisy" because uncorrected gate errors cap achievable circuit depth. The coinage stuck instantly and organized an entire research program — variational algorithms (VQE, QAOA), error mitigation (ZNE, PEC), and near-term "advantage" claims all live under it (see S-nisq, S-variational, O-advantage).

WHY THE WORD MATTERED

Preskill's essay did two things at once, and both were acts of expectation-management. It gave hardware groups a realistic near-term target instead of a fault-tolerant fantasy that was still decades off — you could publish NISQ results now. And it sober-set expectations: his often-quoted line is that NISQ technology "will not change the world by itself" and should be regarded "as a step toward more powerful quantum technologies we will develop in the future." He explicitly named the gap between the NISQ present and the fault-tolerant future that the whole field is now trying to cross (see T-ecera, S-logical, O-scaling). The term became load-bearing in three registers at once: it frames vendor roadmaps, it frames funding pitches, and it frames the honest-skeptic critique — because if the NISQ era ends without a single unambiguous practical advantage, that is itself the verdict.

THE VERDICT, EIGHT YEARS ON

The uncomfortable fact by mid-2026 is that no NISQ algorithm has produced a clear, durable practical advantage. Variational methods hit barren plateaus and optimization walls; every "utility" or "advantage" demo on a noisy device has drawn a classical counterattack within months (see T-race, T-ecera, O-advantage). The field's own answer has been to shift focus from NISQ to early fault tolerance — which is why "post-NISQ" and "the era of logical qubits" now appear in the same roadmaps that once sold NISQ. Preskill framed the practical-advantage question as open in 2018; the honest reading is that it is still open, and increasingly the bet is that the payoff waits on error correction, not on cleverer noisy circuits.

KEY GRADED CLAIMS

- claim: The 2018-onward regime of ~50–100 noisy, uncorrected qubits is the defining near-term phase, and it is named NISQ · tier: T2 · status: established (as terminology) — Preskill, *Quantum* 2, 79 (2018), arXiv:1801.00862
- claim: NISQ devices will yield near-term commercial advantage · tier: T4 · status: contested — Preskill himself framed it as open; eight years on, no unambiguous practical NISQ advantage has survived classical counterattack (see O-advantage, O-killerapp)
- claim: The field is transitioning from NISQ to early fault tolerance · tier: T3 · status: demonstrated in direction (below-threshold QEC, logical-qubit demos — see T-ecera), not yet in delivered applications

SOURCES

- Preskill, "Quantum Computing in the NISQ era and beyond," *Quantum* 2, 79 (2018). arXiv:1801.00862, doi:10.22331/q-2018-08-06-79
- Preskill, Q2B keynote, Dec 2017 — the origin of the term
- **Go deeper:** S-nisq, S-variational (the algorithms of the era); O-advantage (the classical-counterattack pattern); T-ecera (the shift to logical qubits)

Nobel Prizes as the quantum spine (1918→2025) · T-nobel

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The Nobel Prize in Physics is an imperfect but useful backbone for the whole history of quantum science: each prize marks the moment the establishment ratified a result as settled. Reading the prizes in order traces the field's two revolutions. The early prizes (1918–1933) canonized the *discovery of the rules* — Planck's quantum, Einstein's photon, Bohr's atom, the matrix and wave mechanics. A middle cluster (1954–1972) honored the *deep theory and the materials physics* that would later become hardware — Born's probability rule, QED, BCS superconductivity, the Josephson effect. The recent prizes (1997–2025) reward the *engineering of single quantum systems* and the *foundations made experimental* — laser cooling, Bose–Einstein condensation, individual-quantum control, the Bell-test verdict, and finally macroscopic quantum circuits, the direct ancestors of the superconducting qubit. Two centuries-apart bookends define the span: Planck in 1918 for the constant that started it, and Clarke, Devoret, and Martinis in 2025 for showing an electrical circuit big enough to hold in your hand still obeys quantum mechanics — the physics every superconducting quantum computer runs on.

THE SPINE (PHYSICS PRIZES CENTRAL TO QUANTUM)

- 1918 — Max Planck — energy quanta (the constant h) — first revolution begins — T1
- 1921 (awarded 1922) — Albert Einstein — the photoelectric effect (explicitly, not relativity) — light is quantized — T1
- 1922 — Niels Bohr — the structure of atoms and their radiation — T1
- 1929 — Louis de Broglie — the wave nature of electrons — T1
- 1932 (announced 1933) — Werner Heisenberg — the creation of quantum mechanics — T1
- 1933 — Erwin Schrödinger & Paul Dirac — new productive forms of atomic theory — T1
- 1945 — Wolfgang Pauli — the exclusion principle — T1
- 1954 — Max Born — the statistical interpretation of the wavefunction (Born rule), a 28-year wait; shared with Walther Bothe — T1
- 1956 — Shockley, Bardeen, Brattain — the transistor, the first quantum revolution's defining technology — T1
- 1965 — Sin-Itiro Tomonaga, Julian Schwinger, Richard Feynman — quantum electrodynamics (QED) — T1
- 1972 — John Bardeen, Leon Cooper, Robert Schrieffer — BCS theory of superconductivity (Bardeen's second physics Nobel) — the physics under every superconducting qubit — T1
- 1973 — Brian Josephson (shared) — the Josephson effect, tunneling supercurrent across a junction — the nonlinear element of the transmon (see H-supercon) — T1
- 1997 — Chu, Cohen-Tannoudji, Phillips — laser cooling and trapping of atoms — enables ion/neutral-atom qubits — T1
- 2001 — Cornell, Ketterle, Wieman — Bose–Einstein condensation — ultracold quantum matter — T1
- 2012 — Serge Haroche & David Wineland — measuring and manipulating *individual* quantum systems — the second revolution's core capability (see T-2ndrev) — T1
- 2022 — Alain Aspect, John Clauser, Anton Zeilinger — Bell-inequality violation with entangled photons, "pioneering quantum information science" (see T-foundations, T-belltests) — T1
- 2023 — Agostini, Krausz, L'Huillier — attosecond light pulses (quantum-adjacent; probes electron dynamics) — T1
- 2025 — John Clarke, Michel Devoret, John Martinis — macroscopic quantum tunneling and energy quantization in Josephson-junction circuits (experiments 1984–85) — the superconducting-qubit lineage (see T-early, T-ecera) — T1 nobelprize.org (7 Oct 2025)

WHAT THE SPINE REVEALS — AND HIDES

Three patterns stand out. **The long lag**: Born waited 28 years, Aspect/Clauser/Zeilinger about 40, Clarke/Devoret/Martinis about 40 — the prize ratifies, it does not predict. **The three-person cap distorts credit**: teams and unmentioned collaborators vanish (Rosalind Franklin's chemistry analogue haunts physics too — Freeman Dyson, who made QED calculable, was excluded in 1965). **The hardware bias toward the past**: the 2025 prize honored circuit experiments from 1984, and no gate-model quantum *computer* has yet produced a Nobel-worthy verified advantage, which is itself an honest signal about where the field really is versus where the press releases say it is (see T-ecera, O-hype). The spine is a reliable record of what is *established*; by construction it lags the frontier by decades.

KEY GRADED CLAIMS

- claim: The 2025 Physics Nobel went to Clarke, Devoret, Martinis for macroscopic quantum tunneling in superconducting circuits · tier: T1 · source: nobelprize.org (2025) · status: established

- claim: The 2022 Physics Nobel recognized Bell-test experiments as the origin of quantum information science · tier: T1 · source: nobelprize.org (2022) scientific background · status: established
- claim: No quantum *computer* result has yet earned a Nobel; the prizes track established physics, lagging the hardware frontier by decades · tier: T1 (observation) · status: established

SOURCES

- NobelPrize.org prize records and scientific-background documents, 1918–2025 (especially 2012, 2022, 2025)
- Physics World, C&EN, Nature Physics coverage of the 2025 prize (Clarke/Devoret/Martinis)
- **Go deeper:** every era card in §07 cross-references its prize; T-2ndrev (the 2012 capability); T-ecera (the 2025 lineage); O-hype (prizes as a lag indicator)

Old quantum theory (1900–1925) · T-old

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

Classical physics broke on two experimental fronts — blackbody radiation and the photoelectric effect — and the patches applied between 1900 and 1925 became the "old quantum theory": Newtonian mechanics and Maxwell's electromagnetism with ad-hoc quantization rules bolted on. It was a quarter-century of brilliant guessing without a governing equation. Max Planck, a conservative 42-year-old thermodynamicist who disliked his own result, quantized energy exchange to fit the blackbody spectrum and later called it "an act of desperation." Einstein — then a patent clerk in Bern — took the quantum literally as a particle of light in his 1905 *annus mirabilis*, a step so radical that Planck himself resisted it for a decade. Bohr quantized the atom's orbits and, in a single stroke, computed the hydrogen spectrum's Rydberg constant from first principles. De Broglie closed the loop in his 1924 doctoral thesis by giving *matter* a wavelength, an idea his examiners found so strange they forwarded it to Einstein for a verdict (he approved). By 1925 the patchwork was visibly failing — it could not handle helium's two electrons or the anomalous Zeeman effect, and the Bohr–Sommerfeld quantization rules had no principled justification. That failure is exactly what forced the full formalism of the next era.

MILESTONE TIMELINE

- 1900 (14 Dec) — Quantum hypothesis: energy exchanged in units $E = h\nu$, fitting the blackbody spectrum — Max Planck, presented to the German Physical Society — introduced h, the constant the whole field is named after; Planck saw it as a mathematical device, not a physical quantum — T1 Planck, Verhandlungen der DPG 2, 237 (1900)
- 1905 — Photoelectric effect explained via light quanta (later "photons," a term coined by Gilbert Lewis in 1926) — Albert Einstein — light itself is quantized; the specific work later cited for the 1921 Nobel — T1 Einstein, Annalen der Physik 17, 132 (1905)
- 1913 — Quantized atomic orbits explaining the hydrogen spectrum (the Bohr trilogy) — Niels Bohr, then 27, working from Rutherford's Manchester lab — first quantum model of matter's structure; derived the Rydberg constant — T1 Bohr, Philosophical Magazine 26, 1 (1913)
- 1916–1919 — Sommerfeld's elliptical orbits and relativistic fine-structure; the Bohr–Sommerfeld quantization $\oint p\, dq = nh$ — Arnold Sommerfeld — the high-water mark of old quantum theory, and the model that would soon break — T1 Sommerfeld, *Atombau und Spektrallinien* (1919)
- 1922 — Space quantization of angular momentum observed — Otto Stern & Walther Gerlach — direct evidence quantization is real; later read as the prototype of a projective qubit measurement — T1 Zeitschrift für Physik 9, 349 (1922)
- 1923 — X-ray scattering off electrons shows photon momentum $p = h/\lambda$ — Arthur Compton — sealed the particle nature of light against still-widespread skepticism — T1 Phys. Rev. 21, 483 (1923)
- 1924 — Bose–Einstein statistics for indistinguishable particles — Satyendra Nath Bose & Einstein — the first quantum statistics, later the basis of BEC and bosonic modes — T1 Bose, Zeitschrift für Physik 26, 178 (1924)
- 1924 — Matter waves: $\lambda = h/p$ for any particle — Louis de Broglie (doctoral thesis, Sorbonne) — the conceptual bridge to wave mechanics, vindicated by electron diffraction three years later — T1 de Broglie, *Recherches sur la théorie des quanta* (1924)
- 1925 (Jan) — Exclusion principle: no two electrons share a quantum state — Wolfgang Pauli — why atoms have shell structure; the seed of fermionic statistics — T1 Zeitschrift für Physik 31, 765 (1925)

THE HUMAN CONTEXT

This was physics done by very young people under an old guard's skeptical eye. Bohr was 27, Heisenberg would soon be 23, Pauli 25. Planck, the reluctant revolutionary, spent years trying to reconcile his quantum with classical continuity. Einstein's light-quantum was so heterodox that when Planck and others nominated him for the Prussian Academy in 1913, they added a note excusing his "occasional" overreach on the light-quantum. The field's founders did not believe their own equations meant what they said — which is precisely why the 1925–27 formalism landed like a shock.

KEY GRADED CLAIMS

- claim: Energy quantization was forced by experiment (the ultraviolet failure of the classical blackbody law), not chosen for elegance — Planck's own "act of desperation" · tier: T1 · source: Planck 1900; Kuhn, *Black-Body Theory* (1978) · provenance: historical record · status: established
- claim: De Broglie's matter waves were confirmed by electron diffraction (Davisson–Germer 1927; G.P. Thomson 1927, covered in T-formalism) · tier: T1 · status: established

- claim: The "old quantum theory" was a genuine research program, not merely a set of mistakes — it produced correct spectra while lacking a governing equation · tier: T1 · status: established

SOURCES

- Planck (1900), Verhandlungen der DPG 2, 237; Einstein (1905), Annalen der Physik 17, 132; Bohr (1913), Phil. Mag. 26, 1; de Broglie thesis (1924)
- Kuhn, *Black-Body Theory and the Quantum Discontinuity* (1978); Pais, *Subtle is the Lord* (1982); Jammer, *The Conceptual Development of Quantum Mechanics* (1966) — secondary T1 histories

- **Go deeper:** T-formalism (the equation that replaced the patchwork); T-nobel (Planck 1918, Einstein 1921, Bohr 1922); F-statistics (Bose/Fermi–Dirac lineage)

Birth of quantum information theory (1970–1984) · T-qinfobirth

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

Before there was quantum *computing*, there was quantum *information* — the recognition that a quantum state is a carrier of information governed by its own rules, distinct from Shannon's classical theory. The idea's patient zero is Stephen Wiesner, a Columbia graduate student who around 1970 wrote a manuscript called "Conjugate Coding" describing how to encode two messages in conjugate observables (like the linear and circular polarization of light) so that a receiver can read one or the other but never both — and, as a bonus, how to make banknotes that cannot be counterfeited because an unknown quantum state cannot be copied. It was so far ahead of its time that it was rejected (reportedly by IEEE Transactions on Information Theory) and sat unpublished for thirteen years, circulating only as a curiosity among a few physicists. One of the few who took it seriously was Charles Bennett, Wiesner's friend; another was Gilles Brassard. In 1984 they turned Wiesner's conjugate coding into BB84, the first quantum key distribution protocol — and quantum information had its first practical technology. In parallel, Alexander Holevo (Moscow, 1973) proved the fundamental limit on how much classical information a quantum state can carry, and Wootters, Zurek, and Dieks (1982) proved the no-cloning theorem that makes the whole edifice secure. By 1984 the conceptual toolkit — qubits as information carriers, no-cloning as a security primitive, channel capacities, conjugate coding — was in place, waiting for Deutsch and Shor to add computation (see T-birth).

MILESTONE TIMELINE

- ~1970 — "Conjugate Coding" manuscript: quantum money and multiplexed messages via conjugate observables — Stephen Wiesner — the founding idea of quantum cryptography, rejected and unpublished for over a decade — T1 Wiesner, eventually in SIGACT News 15(1), 78 (1983)
- 1973 — Holevo bound: a qubit conveys at most one bit of accessible classical information; the general capacity limit for quantum channels — Alexander Holevo — a founding theorem of quantum information theory (later completed by the HSW theorem, 1996–98) — T1 Holevo, Probl. Peredachi Inf. 9(3), 3 (1973)
- 1975–1976 — Early quantum detection and estimation theory; distinguishability limits — Helstrom; Holevo — the quantum analogue of statistical inference — T1 Helstrom, *Quantum Detection and Estimation Theory* (1976)
- 1982 — No-cloning theorem: an unknown quantum state cannot be perfectly copied — Wootters & Zurek (Nature); Dieks (Phys. Lett. A) — the security root of quantum cryptography; makes eavesdropping detectable (see F-nocloning) — T1 Nature 299, 802 (1982)
- 1983 — Wiesner's "Conjugate Coding" finally appears in print — SIGACT News — thirteen years late — T1 SIGACT News 15(1), 78–88 (1983)
- 1984 — BB84 quantum key distribution: Wiesner's conjugate coding turned into a working key-exchange protocol — Charles Bennett & Gilles Brassard — quantum information's first technology; still deployed (see A-qkd) — T1 Proc. IEEE Int. Conf. Computers, Systems and Signal Processing, Bangalore, 175 (1984)
- 1991 — Entanglement-based QKD (E91), grounding security in Bell-inequality violation — Artur Ekert — links quantum crypto to the foundations program (see T-foundations) — T1 Ekert, PRL 67, 661 (1991)

WHY IT EARNS ITS OWN CARD

The birth-of-computing card (T-birth) starts the clock at 1980 with Manin, Benioff, and Feynman, but that framing hides an older lineage: the information-theoretic roots that predate the computational ones. Wiesner's 1970 insight — that conjugate observables plus no-cloning give you security for free — is arguably the first genuinely quantum *application* ever conceived, and it arrived before anyone was thinking about computation at all. Giving it a dedicated card corrects the "Feynman started everything" simplification and makes visible the crypto→computing sequence: the field learned to *protect* information with quantum mechanics before it learned to *process* it.

THE HUMAN CONTEXT

Wiesner (1942–2021) never held a conventional physics career commensurate with his ideas; Scott Aaronson's memorial notes that he spent years as a construction laborer in Israel while the field he seeded became a global enterprise. Bennett has said BB84 was essentially Wiesner's idea with a use case attached. The pattern — a foundational insight, rejected and dormant for a decade, then activated by someone who happened to know the originator — recurs throughout quantum history (Everett's many-worlds, Bell's theorem in an obscure journal).

KEY GRADED CLAIMS

- claim: Quantum key distribution (BB84) derives its security from the no-cloning theorem and measurement disturbance, making it information-theoretic rather than computational · tier: T1 · status: established — see A-qkd, F-nocloning
- claim: The Holevo bound caps accessible classical information in a quantum state at one bit per qubit · tier: T1 · status: established — a founding result of quantum Shannon theory

- claim: Wiesner's ~1970 conjugate coding is the earliest conceived quantum information application · tier: T1 (historical) · status: established, though "earliest" is a claim about priority and framing

SOURCES

- Wiesner, "Conjugate Coding," SIGACT News 15(1), 78 (1983, written ~1970); Holevo, Probl. Peredachi Inf. 9 (1973); Wootters & Zurek, Nature 299 (1982); Bennett & Brassard, BB84 (1984); Ekert, PRL 67 (1991)
- Aaronson, "Stephen Wiesner (1942–2021)," scottaaronson.blog; Wikipedia "Quantum money," "Holevo's theorem"

- **Go deeper:** F-qinfo (the modern theory); F-nocloning; A-qkd; T-birth (where computation joins); T-nobel

The engineering race (2011–2019) · T-race

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

Quantum computing left the physics lab and became an industrial race. D-Wave sold the first commercial machine in 2011 — an annealer, not a gate-model computer — kicking off a decade-long argument about whether it delivered any quantum speedup at all. IBM put a real processor on the open cloud in 2016 and normalized the practice of publishing qubit counts as press events. Google hired John Martinis's UCSB group in 2014 and pointed it at a single target: beat a classical supercomputer at *something*. Governments answered with billion-dollar national programs (EU Flagship 2016, US NQI Act 2018 — see E-eu, E-us). The era peaked in October 2019 with Google's Sycamore "quantum supremacy" claim — a 53-qubit chip sampling random circuits in 200 seconds against an estimated 10,000 classical years. IBM contested the estimate within days (arguing a well-configured Summit supercomputer could do it in 2.5 days), and over the next three years classical algorithms shrank the gap to hours. That claim-and-counterattack cycle — a headline number, a rapid classical response, a milestone that survives while its superlative does not — became the template governing every advantage announcement since (see SCHEMA rule 2, O-advantage).

MILESTONE TIMELINE

- 2011 (May) — First commercial quantum computer sold: D-Wave One (128-qubit annealer) to Lockheed Martin — D-Wave — a market exists; the underlying Nature paper shows quantum annealing in the hardware, but generic speedup stays contested for years — [T4 sale → T2 physics] D-Wave; Johnson et al., Nature 473, 194 (2011)
- 2013 — Google, NASA, and USRA buy a D-Wave Two for the Quantum AI Lab at Ames — big tech enters, betting on annealing — T4 press record
- 2014 — Martinis group (UCSB) joins Google; superconducting fidelity crosses the surface-code threshold in the lab — Google — the start of the supremacy program — T2 Barends et al., Nature 508, 500 (2014)
- 2014 — No generic quantum speedup found in D-Wave benchmarks — Rønnow, Troyer et al. — the defining skeptical result on annealing — T2 Rønnow et al., Science 345, 420 (2014)
- 2016 (May) — IBM Quantum Experience: a 5-qubit processor free on the public cloud — IBM — anyone can run a circuit; the cloud-QC business model is born (see S-cloud) — T2 IBM + subsequent literature
- 2016 — EU announces the €1B Quantum Flagship — European Commission — governments commit at scale (see E-eu) — T5 EC announcement
- 2017–2018 — 50-qubit IBM prototype; 72-qubit Bristlecone (Google); 49-qubit Tangle Lake (Intel) — the qubit-count arms race in full swing — T4 vendor announcements
- 2018 (21 Dec) — US National Quantum Initiative Act signed into law ($1.2B over 5 years) — US Congress — quantum becomes a statutory national priority (see E-us) — T2 Public Law 115-368
- 2019 (Jan) — IBM Q System One, the first integrated, rack-and-chandelier commercial quantum system — IBM — productization and industrial design begin — T4 IBM announcement
- 2019 (23 Oct) — "Quantum supremacy" on the 53-qubit Sycamore chip: random circuit sampling in ~200 s vs. an estimated 10,000 classical years — Google (Martinis et al.) — first claimed beyond-classical computation — [T2 experiment] Arute et al., Nature 574, 505 (2019)

THE HUMAN CONTEXT

The supremacy paper leaked early — a draft appeared on a NASA server in September 2019 before the embargo — and IBM published its rebuttal the same week the Nature paper ran, a public spat between the two biggest players that set the tone for the field's credibility wars. Martinis left Google in 2020 after a reorganization, a reminder that the era's central figures were people, not logos. D-Wave's long speedup argument never resolved cleanly; the honest summary is that annealing found niche uses without ever demonstrating the general exponential speedup its early marketing implied.

KEY GRADED CLAIMS

- claim: Sycamore performed a sampling task infeasible for any classical computer in 2019 · tier: T2 (experiment) · status: contested — IBM's 2.5-day Summit estimate (Oct 2019); tensor-network methods later reproduced the sampling in hours-to-days (Pan, Chen, Zhang, PRL 129, 090502, 2022). The demo stands; the "10,000 years" figure does not.
- claim: D-Wave annealers deliver a generic quantum speedup on practical problems · tier: T4 · status: contested — Rønnow et al., Science 345, 420 (2014) found no generic speedup; niche advantages remain debated
- claim: Publishing qubit counts as milestones tracks progress · tier: T4 · status: contested — the field itself later shifted the scoreboard to logical qubits and error rates (see T-ecera)

SOURCES

- Johnson et al., Nature 473 (2011); Rønnow et al., Science 345 (2014); Barends et al., Nature 508 (2014); Arute et al., Nature 574 (2019); IBM "On quantum supremacy" blog (Oct 2019); Pan et al., PRL 129, 090502 (2022)

- **Go deeper:** S-bench (how "advantage" is measured); T-ecera (the pivot to logical qubits); E-us, E-eu (the national programs); O-advantage

The 1927 Solvay Conference & the Bohr–Einstein debates · T-solvay

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The Fifth Solvay Conference, held in Brussels 24–29 October 1927 on the theme "Electrons and Photons," is the most famous meeting in the history of physics. Twenty-nine physicists attended; seventeen were or would become Nobel laureates. The group portrait — Einstein front-center, Curie the only woman, Bohr, Planck, Dirac, Heisenberg, Schrödinger, de Broglie, Born, Pauli arrayed around them — is often called "the most intelligent photo ever taken." It marked the moment the new quantum mechanics, barely two years old, was presented as a finished theory, and the moment its interpretation split the field. Bohr and Heisenberg's Copenhagen view — that the wavefunction is a complete description and that a quantum system has no definite properties until measured — was challenged head-on by Einstein, who could not accept that physics had become irreducibly probabilistic. Their exchange did not settle anything in 1927; it opened the interpretation argument that runs to this day and that eventually became testable through Bell's theorem and the loophole-free experiments (see T-foundations, T-belltests, F-interp).

WHAT ACTUALLY HAPPENED

The popular story compresses a week of hard argument into one quip. Einstein's real attack came through a series of ingenious thought experiments — variants of the double slit with movable diaphragms and recoiling screens — each designed to show that one could, in principle, beat the uncertainty relation and pin down both a particle's path and its interference pattern. Bohr answered each over breakfast and dinner, showing that the very apparatus needed to measure the particle's momentum recoil would blur its position by exactly enough to preserve the uncertainty bound. Einstein conceded the internal consistency of quantum mechanics but never its completeness. The famous lines — Einstein's "God does not play dice" and Bohr's "Einstein, stop telling God what to do" — capture the temperature of the debate, though both are paraphrases, not verbatim transcripts of a single moment.

MILESTONE TIMELINE

- 1927 (24–29 Oct) — Fifth Solvay Conference, "Electrons and Photons," Brussels; Bohr, Born, Heisenberg present the new mechanics; Einstein and Schrödinger dissent — the public debut of the interpretation debate — T1 proceedings; Bacciagaluppi & Valentini, *Quantum Theory at the Crossroads* (2009), a full reconstruction from the original French/German records
- 1930 (Oct) — Sixth Solvay Conference: Einstein's "photon box" (light-box weighed to defeat the energy–time uncertainty relation); Bohr rebuts using Einstein's *own* general relativity (gravitational time dilation) — the debate's second round — T1 historical record
- 1935 — Unable to defeat consistency, Einstein shifts ground to *completeness* and *locality*: the EPR paper (see T-epr) — the debate's most productive turn — T1 Phys. Rev. 47, 777 (1935)

WHY IT BELONGS ON THE TIMELINE

Solvay 1927 is where quantum mechanics stopped being an equation and started being a worldview problem. The Copenhagen interpretation became the working default not because it won an argument but because it was operationally sufficient and its rivals were dismissed. Einstein's refusal looks, in hindsight, less like stubbornness and more like the first statement of the questions — locality, realism, completeness — that the second half of the 20th century turned into experiments and the 21st turned into technology. Entanglement, the thing Einstein was circling at Solvay, is the resource quantum computers run on.

KEY GRADED CLAIMS

- claim: The Solvay debates concerned the *interpretation* and *completeness* of QM, not its predictive accuracy, which was never in dispute · tier: T1 · source: Bacciagaluppi & Valentini (2009) · status: established
- claim: Bohr successfully defended the uncertainty principle against each of Einstein's thought experiments · tier: T1 (as the historical consensus) · status: established; the deeper completeness question Einstein raised was resolved only much later, and against him (Bell tests)
- claim: "God does not play dice" / "stop telling God what to do" are paraphrases, not a verbatim single exchange · tier: T1 · status: established

SOURCES

- Bacciagaluppi & Valentini, *Quantum Theory at the Crossroads: Reconsidering the 1927 Solvay Conference* (Cambridge, 2009), arXiv:quant-ph/0609184
- Bohr, "Discussion with Einstein on Epistemological Problems in Atomic Physics" (1949); Wikipedia "Bohr–Einstein debates"

- **Go deeper:** T-epr (the 1935 sequel); T-formalism (the theory being debated); F-interp; F-uncertainty

The "quantum winter" question (AI-winter analogy) · T-winter

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The term "quantum winter" borrows directly from the "AI winters" — the two multi-year collapses of funding and interest (roughly 1974–1980 and 1987–1993) that followed cycles of over-promising in artificial intelligence. A quantum winter would be the same pattern: a period of stalled progress and, more sharply, a withdrawal of capital and attention when the field's grand promises fail to materialize on the advertised schedule. It is not (yet) a historical event — it is a live risk and a recurring debate. The bull case says quantum has never been advancing faster (below-threshold QEC, logical-qubit records — see T-ecera). The bear case, argued by longtime skeptics like Mikhail Dyakonov and by cautious investors, is that quantum advantage has so far appeared only in contrived, useless tasks; that no application pays for itself; and that a downturn could drain talent and funding before fault-tolerant machines arrive (see O-talent-attrition, O-killerapp). Including this card is a matter of intellectual honesty: the atlas grades vendor roadmaps as marketing until peer-reviewed (SCHEMA rule 1), and a field that runs on hype is a field exposed to a winter.

THE EVIDENCE FOR THE RISK

- **Funding volatility.** Private quantum investment reportedly fell sharply in 2023 (widely cited as ~50% globally, ~80% in the US year-over-year) before recovering — a swing consistent with hype-cycle dynamics, not steady industrial build-out. Figures vary by source and are T5.
- **AI cannibalization.** From 2023 the generative-AI boom pulled capital, talent, and executive attention toward a technology with immediate revenue, leaving quantum — still pre-revenue for computing — competing for a shrinking share of "frontier tech" budgets.
- **The advantage gap.** Every beyond-classical claim (Sycamore 2019, Jiuzhang 2020, IBM utility 2023, Google Quantum Echoes 2025) has drawn a classical counterattack (see T-race, T-ecera, O-advantage). A decade of NISQ produced no durable practical advantage (see T-nisq). If the gap persists, patience — and money — can run out.
- **Public-market fragility.** Several pure-play quantum stocks are richly valued relative to near-zero revenue; a single high-profile stumble or accounting controversy could reprice the sector and chill private funding (the "QCI saga" cited in the postquantum.com "quantum winter warning" is the archetype).

THE EVIDENCE AGAINST

- **Genuine technical acceleration.** Below-threshold QEC (Willow, Nature 2025) and logical-qubit counts doubling (48→96 in ~13 months) are peer-reviewed or near-peer-reviewed, not press-release physics (see T-ecera).
- **Deep-pocketed patrons.** National programs (US NQI, EU Flagship, China's multi-billion commitment — see E-us, E-eu, E-china) provide funding floors decoupled from VC sentiment; governments do not exit on a bad quarter.
- **The 2025 Nobel** (Clarke/Devoret/Martinis) and the UN International Year of Quantum reinforce institutional legitimacy at exactly the moment a winter narrative might take hold.

MILESTONE / DISCOURSE TIMELINE

- 2019 — Early "quantum winter is not coming" essays push back on premature doom (Ryan Shaffer and others) — [T5/opinion]
- 2022 (Jan) — Mainstream "what if quantum faces a winter?" coverage (Slate) — the analogy enters public discourse — T5
- 2023 — Reported ~50% global / ~80% US drop in private quantum funding; simultaneous generative-AI boom — the sharpest data point for the bear case — T5
- 2024 — "No quantum winter in 2024" counter-narratives cite rapid hardware advances — T5
- 2025 — "Quantum winter warning" analyses tie the risk explicitly to overhype and specific public-company controversies — T5

KEY GRADED CLAIMS

- claim: A "quantum winter" analogous to the AI winters is a plausible risk, driven by an advantage gap plus hype-cycle funding · tier: T5/T6 · provenance: analysts, skeptics, some investors · status: contested — a scenario, not a fact
- claim: Quantum advantage has so far been shown only in contrived tasks with no practical utility · tier: T2 (about the record to date) · status: established as of mid-2026 — every practical-advantage claim has been contested (see O-advantage)
- claim: National-program funding insulates the field from a purely VC-driven downturn · tier: T5 · status: contested — floors exist, but budgets can still be cut

SOURCES

- postquantum.com "Quantum Winter Warning"; Slate (2022) "What if quantum computing faces a winter?"; Cointelegraph (2024) "no quantum winter"; Infosys "Quantum Winter: Reality or Myth?"
- Dyakonov, *Will We Ever Have a Quantum Computer?* (2020) — the standing skeptical case
- AI-winter history: standard accounts of the 1974–80 and 1987–93 downturns
- **Go deeper:** O-hype, O-killerapp, O-advantage, O-talent-attrition; E-vc (the capital cycle); T-nisq (the unmet-promise precedent)

Chapter 8 · The Honest Frontier

Quantum Advantage in Practice — Which Speedups Survive · O-advantage

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

"Quantum advantage" is a moving target: every claim faces a classical counterattack, and the record so far favors the attackers. Google's 2019 random-circuit-sampling supremacy ("10,000 years") was matched within ~3 years by tensor-network methods on GPUs. The sharp question splits in two: (1) is there any *useful* task with a speedup that survives sustained classical attack, and (2) how much of the theoretical speedup catalog was ever real once loading, readout, and overhead are counted? The 2025–26 record is a live back-and-forth on both spin-glass simulation and OTOC-based "quantum echoes."

WHERE THE DISAGREEMENT IS

- **Advantage-is-real camp.** Google's **Quantum Echoes** (Oct 2025, 105-qubit Willow) claims the first *verifiable* quantum advantage — an OTOC-based algorithm run **13,000× faster** than the best known classical method on a top supercomputer, with results reproducible and cross-checkable on another quantum device [T4→T3; vendor-led, published, young]. A March 2026 preprint, "Tensor Networks with Belief Propagation Cannot Feasibly Simulate Google's Quantum Echoes Experiment" (arXiv:2604.15427), argues the leading classical attack *fails* here — an early sign the claim may hold T3. D-Wave's March 2025 *Science* paper claimed beyond-classical spin-glass dynamics on 5,000+ qubits [T2 paper, contested claim], and in May 2026 D-Wave **rebutted** the Flatiron follow-up, holding that the classical framework fails to scale across the most complex topologies and measurements T4. Shor's and quantum-simulation speedups remain provably superpolynomial given a fault-tolerant machine [T1/T2 theory].
- **Attrition camp. Ewin Tang's dequantization** (2018→) erased the exponential speedup of quantum recommendation systems and much of the QML linear-algebra family — classical sampling matches them up to polynomial factors T2. HHL's fine print (state preparation, condition number, readout) hollows out most quantum-linear-algebra applications T2. On D-Wave, the Flatiron Institute (Joseph Tindall et al., belief-propagation tensor networks) published a peer-reviewed *Science* result (May 2026) computing large-scale annealing dynamics classically, and EPFL's Dries Sels ran comparable instances on a laptop; critics showed the original comparison used weak classical baselines T2/T3. No neutral third party has adjudicated the spin-glass dispute as of mid-2026: D-Wave's May-2026 rebuttal holds that the belief-propagation method never attempted the hardest 3D lattice geometries, the largest 3D simulations, the low-precision ensembles where correlations grow fastest, or the full-state / fourth-order observables of the original *Science* paper — and those specific instances remain unmatched classically, so the claim stays contested rather than overturned. A 2025 result gives quasi-polynomial classical sampling of noisy circuits under approximate Markovianity, squeezing noisy-sampling advantage generally [T3, arXiv:2510.06324]. Grover's quadratic speedup is widely expected to be consumed by QEC overhead [T2/T3, "beyond quadratic speedups"].

WHAT WOULD RESOLVE IT

A verifiable advantage claim that stands **~3+ years** against motivated classical algorithm hunters. Quantum Echoes is the live test case: if belief-propagation and other tensor-network attacks keep failing on it through 2027–28, it becomes the first durable useful-ish advantage; if a classical method matches it (as happened to 2019 supremacy), the pattern of attrition holds. On the theory side, dequantization of the remaining chemistry/simulation candidates would collapse the practical case to fault-tolerant Shor and quantum simulation alone.

SOURCES

- Google Research, "A verifiable quantum advantage" (Quantum Echoes, Oct 2025) T4/T3
- arXiv:2604.15427 — belief-propagation tensor networks cannot feasibly simulate Quantum Echoes (2026) T3
- D-Wave *Science* (Mar 2025) vs Flatiron *Science* (adx2728, May 2026) + D-Wave rebuttal — quantumcomputingreport.com [T2/T3/T4, contested]
- Tang, dequantization line — ewintang.com; Nature Rev. Physics s42254-022-00511-w T2
- arXiv:2510.06324 (classical sampling of noisy circuits); arXiv:2508.05720 ("The vast world of quantum advantage") T3

The Benchmarking Standardization Crisis · O-benchmark-standard

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

There is no agreed way to say one quantum computer is better than another. Vendors report different, self-favoring metrics — IBM's **Quantum Volume** and CLOPS (speed), IonQ/QED-C's **Algorithmic Qubits (#AQ)**, raw physical-qubit count, gate fidelities, "logical qubits" under vendor-chosen definitions — and each metric can be gamed or is sensitive to different subsystems. Unlike classical computing, which converged on SPEC, LINPACK, and MLPerf, quantum has no common yardstick, which makes cross-vendor comparison, procurement, and even honest progress-tracking hard. The sharp question: can the field standardize application-oriented, spoof-resistant benchmarks before the metric zoo lets marketing outrun physics? This is the measurement layer under O-hype and O-advantage — if you can't benchmark honestly, you can't grade the claims.

WHERE THE DISAGREEMENT IS

- **Standardization-is-arriving camp.** The **QED-C** (Quantum Economic Development Consortium) ships an open-source **application-oriented benchmark suite** (15+ applications: Grover, Monte Carlo, Shor period-finding, VQE, etc.) using volumetric benchmarks, and runs a Standards & Performance Metrics committee convening vendors and labs; its 2026 plan puts standards work front and center T3/T4. Quantum Volume is a genuine full-system metric sensitive to qubit number, fidelity, and connectivity together. New suites (QuSquare, QPack, the QuantumBenchmarkZoo aggregation) push toward quality-oriented, reproducible measurement. ISO/IEC JTC1, IEEE, and ETSI have quantum working groups building toward formal standards.
- **Metrics-are-a-mess camp.** "There is not yet a common understanding of standardized metrics in quantum computing" [T3, direct]. Vendor metrics are chosen to flatter the vendor's architecture — #AQ favors high-fidelity low-qubit ion machines, raw counts favor superconducting, "logical qubits" often means memory not a full gate set. Cross-entropy benchmarking, used for supremacy claims, is not a proof and has been spoofed classically (O-advantage, O-verification). Application benchmarks depend heavily on compilation and error-mitigation choices, so the same hardware scores differently under different software, and results are rarely independently reproduced. Quantum Volume saturates and stops discriminating at the top of the field. The absence of a neutral, funded, adversarial benchmarking body (a "quantum NIST-benchmark") is a structural gap.

WHAT WOULD RESOLVE IT

Broad adoption of a small set of **application-oriented, independently reproduced, spoof-resistant** benchmarks — reported by an independent body, not the vendor — that procurement and peer review actually use, the way MLPerf disciplined ML hardware claims. Concretely: a standard that reports *logical* performance with a defined universal gate set at stated code distance, plus an end-to-end application score run by a third party. Until then, the atlas treats every single-metric vendor headline as T4 and asks which subsystem the number really measures.

SOURCES

- QED-C, Standards & Performance Metrics committee + application-oriented benchmark suite — quantumconsortium.org/tac/standards T3/T4
- SRI/QED-C, QC-App-Oriented-Benchmarks (open source, 15+ applications) T3
- IBM Quantum Volume / CLOPS; IonQ + QED-C Algorithmic Qubits (#AQ) [T4, vendor metrics]
- arXiv:2512.19665 — "QuSquare: Scalable Quality-Oriented Benchmark Suite" (2025) T3
- QuantumBenchmarkZoo benchmarking-initiatives aggregation T3

The Classical-Simulation Frontier — Where the Boundary Actually Is · O-classical-sim

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Every quantum-advantage claim implicitly draws a line: on *this* side, classical computers can reproduce the result; on *that* side, they can't. The line moves. It is set by the best classical algorithm anyone has written, and clever people keep writing better ones — so a claim that looked beyond-classical in year N is routinely pulled back inside the boundary in year N+2. The sharp question is not "can a quantum computer beat brute-force state-vector simulation" (it can, at ~50 qubits) but "where does the boundary sit against the *best* classical method — tensor networks, belief propagation, neural quantum states, Clifford-plus-few-T sparsification — for the specific structured problems quantum machines actually run?" Getting this boundary right is what separates real advantage (O-advantage) from a benchmark artifact.

WHERE THE DISAGREEMENT IS

- **Boundary-is-close-and-moving camp.** Classical simulation has repeatedly caught up: Google's 2019 "10,000 years" fell to days on tensor-network/GPU methods within ~3 years; IBM's 2023 utility experiment was matched within days by 2D tensor networks (gPEPS, Tindall et al.) and sparse Pauli methods (Sandia, Caltech). The Flatiron Institute's belief-propagation tensor networks reproduced D-Wave's flagship spin-glass dynamics (*Science*, May 2026). Neural quantum states and matrix-product-state methods keep extending reach. The lesson: most "advantage" demos sit *near* the boundary, on structured problems (low entanglement, shallow depth, geometric locality) that classical heuristics exploit, and headline speedups often used weak classical baselines T2/T3.
- **Boundary-is-being-crossed-for-real camp.** Not every problem yields to classical heuristics. The March 2026 result "Tensor Networks with Belief Propagation Cannot Feasibly Simulate Google's Quantum Echoes" (arXiv:2604.15427) argues the leading classical attack *fails* on that OTOC problem — the first advantage claim in a while that resists the usual counterattack T3. Volume-law-entangled, deep, non-geometrically-local circuits are believed hard for all known classical methods; fault-tolerant Shor and genuine quantum simulation live provably beyond the boundary given a working FTQC machine [T1/T2 theory]. The camp argues the boundary, while it moves, is *converging* on a stable frontier that quantum machines are now starting to clear.

WHAT WOULD RESOLVE IT

For any specific claim: a sustained failure of *all* classical methods (tensor networks with belief propagation, neural quantum states, Clifford perturbation theory, MPS/PEPS, quantum Monte Carlo) to match it over ~3 years of motivated attack. Quantum Echoes is the current live probe — if it holds against classical attack through 2027–28 it marks a real, stable boundary crossing; if a classical method matches it, the "near-the-boundary" reading wins again. Field-wide, a rigorous complexity-theoretic characterization of *which structured problems* admit efficient classical simulation would convert this from an empirical arms race into settled theory.

SOURCES

- Tindall et al., "Efficient tensor network simulation of IBM's largest quantum processors," arXiv:2309.15642 / PRX Quantum 5, 010308 T2
- Flatiron/BU classical annealing-dynamics algorithm — *Science* adx2728 (May 2026) T2
- arXiv:2604.15427 — belief-propagation tensor networks cannot feasibly simulate Quantum Echoes (2026) T3
- arXiv:2510.06324 — classical sampling of noisy circuits under approximate Markovianity (2025) T3
- arXiv:2508.05720 — "The vast world of quantum advantage" T3

The Cryptographically-Relevant Quantum Computer (CRQC) Timeline · O-crqc-timeline

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

A **cryptographically-relevant quantum computer** is one that can run Shor's algorithm at scale — factoring RSA-2048 or solving the discrete logarithm behind ECC, breaking the public-key crypto that secures nearly all internet traffic, finance, and government data. This is a *specific* fault-tolerant milestone, distinct from the general "useful FTQC" question (O-scaling): it needs thousands of logical qubits sustained through ~10^9 operations. It carries its own urgency because of **harvest-now-decrypt-later (HNDL)**: adversaries can record encrypted traffic *today* and decrypt it once a CRQC exists, so any secret that must stay confidential for a decade is already at risk. The sharp question: when does the CRQC arrive, and is PQC migration (A-pqc) outrunning it? This is where **Mosca's inequality** bites — if (migration time) + (secrecy shelf-life) > (time to CRQC), you are already exposed.

WHERE THE DISAGREEMENT IS

- **Sooner / resources-are-falling camp.** Craig Gidney (Google, arXiv:2505.15917, 2025) cut the RSA-2048 estimate ~20×, to **<1M noisy physical qubits at ~1 week runtime** — down from ~20M (2019) — via approximate residue arithmetic, yoked surface codes, and magic-state cultivation T3. Expert-elicitation surveys (Global Risk Institute) put meaningful probability on a CRQC within a decade: roughly **22.7% of surveyed cryptographers expect RSA-2048 to fall by 2030, and ~50% by 2035**; point estimates cluster around **2030 ± 3 years** T5. NIST has already standardized PQC (ML-KEM/Kyber, ML-DSA/Dilithium) precisely because the threat is treated as near enough to act on now [T2, standards].
- **Later / hardware-isn't-close camp.** A <1M-qubit *estimate* is not a <1M-qubit *machine* — the largest devices are ~1,000–6,000 physical qubits, so the gap is 2–3 orders of magnitude in count while *holding* the fidelity Shor needs, which nobody has shown at scale (O-scaling, O-materials). Broad expert consensus places the *heaviest* weight in the mid-2030s, and Gil Kalai's minority position says a CRQC never arrives at all (O-hype). Recent toy Shor demonstrations factored only tiny numbers and do not extrapolate. The energy cost of the run is itself nontrivial (O-energy). Skeptics stress that the resource *estimates* keep improving on paper faster than the *hardware* improves in the lab.

WHAT WOULD RESOLVE IT

A quantum computer factoring a cryptographically meaningful RSA modulus (even RSA-1024, then 2048) with fault tolerance — or credible interim milestones: factoring a 100+-bit semiprime with Shor (not a special-form or annealing shortcut), then scaling. On the defensive side, completion of PQC migration across critical infrastructure before any CRQC appears would neutralize the threat regardless of the exact date. The dials: physical-qubit count *at Shor-relevant fidelity*, published Shor demonstrations on growing moduli, and the pace of NIST-PQC deployment versus HNDL exposure windows.

SOURCES

- Gidney, "How to factor 2048 bit RSA integers with less than a million noisy qubits," arXiv:2505.15917 (2025) T3
- Global Risk Institute Quantum Threat Timeline expert survey (2030 ± 3; ~50% by 2035) T5
- NIST FIPS 203/204/205 (ML-KEM, ML-DSA, SLH-DSA) post-quantum standards (2024) T2
- quantumzeitgeist.com, "Cryptographically Relevant Quantum Computer: Complete 2026 Guide"; postquantum.com HNDL + RSA-2048 energy analyses T3/T5
- Mosca, "Cybersecurity in an era with quantum computers" (Mosca's inequality) T3

Real-Time QEC Decoding Throughput · O-decoder

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Error correction only works if a classical co-processor decodes each round of syndrome measurements and tells the quantum hardware what correction to apply *before the next round arrives*. Superconducting qubits emit a syndrome roughly every microsecond, so the decoder must sustain about **1 MHz per logical qubit** at sub-microsecond latency — for thousands of logical qubits at once. If decoding lags, syndromes pile up (the "backlog problem," Terhal 2015): the effective clock speed collapses exponentially in the backlog. That makes the classical decoder a first-class bottleneck on the road to fault tolerance. A machine can have perfect qubits and still be useless if it can't decode fast enough — and the raw syndrome-data rate off a large cryostat runs to terabytes per second, which is itself an open cryo-electronics problem (see H-cryocmos, O-interconnect-loss).

WHERE THE DISAGREEMENT IS

- **Tractable-engineering camp.** Riverlane has built dedicated decoders — the Local Clustering Decoder targets <1 µs per round, and a Dec 2025 *Nature Communications* result reports hitting a **MegaQuOp** (10^6 reliable operations) with ~4× fewer qubits via better decoding T2. Riverlane's roadmap frames the path from MegaQuOp (late 2020s) to **TeraQuOp** (10^{12} operations, early 2030s) as an engineering scaling problem, and flags that commodity PCIe/GPU interfaces bottleneck beyond ~300 physical qubits, motivating purpose-built interconnects (QECi, <400 ns round-trip) T3/T4. Google has run real-time decoding on Willow's distance-5/7 codes keeping pace with the syndrome stream; FPGA and ASIC decoders (union-find, sliding-window) are demonstrated at small distance T2/T3.
- **Unproven-at-scale camp.** The accuracy/latency/bandwidth *trilemma* is unsolved at scale: high-accuracy decoders (full minimum-weight matching, neural decoders) are slow; fast decoders are less accurate, and lower accuracy silently raises the logical error rate. Nobody has demonstrated real-time decoding across the **thousands-of-logical-qubit** regime a useful computation needs, and qLDPC codes (which IBM now bets on) are *harder* to decode fast than surface codes because their checks are non-local. Transversal-gate architectures need correlated decoding across logical qubits, which is more expensive still (arXiv:2505.13587). Off-chip syndrome bandwidth at million-qubit scale has no demonstrated solution T3.

WHAT WOULD RESOLVE IT

A machine running many logical qubits through millions of error-corrected cycles with decoding kept ahead of the syndrome stream, at demonstrated MHz throughput and **constant, non-growing backlog**, reproduced independently — ideally on a qLDPC code, not just surface codes. Riverlane's TeraQuOp milestone hitting spec, or IBM Kookaburra/Starling shipping with an integrated real-time decoder that holds pace, would settle the direction. The negative signal: decoder latency that grows with qubit count and forces the quantum clock to slow down.

SOURCES

- Riverlane, "QEC Technology Roadmap" (2026) + Local Clustering Decoder papers — riverlane.com T4
- Riverlane et al., *Nature Communications* (Dec 2025), MegaQuOp with ~4× fewer qubits T2
- Terhal, "Quantum error correction for quantum memories," RMP 87, 307 (2015) — backlog problem T2
- arXiv:2605.30765 — real-time QEC system stack (architecture/algorithms/engineering, 2026) T3
- arXiv:2505.13587 — fast correlated decoding of transversal logical algorithms T3

Energy Consumption — Quantum vs Classical · O-energy

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

A quantum computer's headline runtime is only half of a fair comparison; the other half is *energy*. A dilution refrigerator draws roughly **10 kW continuously**, dominated by the mechanical compressor on the 3 K stage, and a full quantum system with control electronics runs on the order of **~18 kW** — a fixed cost paid whether the machine is computing or idle. A classical supercomputer like Frontier or Summit draws megawatts. The sharp question: for the specific tasks quantum machines win on, does the quantum system deliver the answer at *lower total energy* than the classical alternative — and does that "energy advantage" arrive earlier or later than the runtime advantage? This is a distinct axis from speed, and it becomes a scaling argument as classical data-center power grows 20–40% per year.

WHERE THE DISAGREEMENT IS

- **Quantum-is-energy-efficient camp.** For sampling-type tasks, Google's supremacy analysis found the classical simulation on Summit would consume **~7 orders of magnitude more energy** than the quantum processor's run T3. A quantum system at ~18 kW is orders of magnitude below a supercomputer; when *idle* the gap is even larger (~432 kWh/day vs a supercomputer's draw) T5. Recent theory identifies an **"energetic advantage before computational advantage"**: in boson sampling and in superconducting cat-qubit computation, the quantum machine can win on energy at a problem size *smaller* than where it wins on wall-clock time (arXiv:2601.08068; arXiv:2605.19854, 2026) T3. As classical compute's energy demand climbs 20–40%/year, quantum looks comparatively sustainable at scale.
- **Cryo-overhead-dominates camp.** The cryogenic and classical-control overhead is *fixed and large*, so for any small or short computation the quantum machine's energy-per-useful-answer is terrible — you pay 18 kW to run a circuit a laptop finishes for watts. Full-system energy accounting (arXiv:2605.09580, 2026) stresses that control electronics, the classical co-processor (decoders — O-decoder), and the dilution plant must all be counted, and at *fault-tolerant scale* the millions of physical qubits plus their real-time decoding could push total draw far above today's single-fridge figure. The RSA-2048 energy cost, even at ~1M qubits over a week, is nontrivial. The favorable comparisons are for cherry-picked sampling problems with no practical use (O-advantage); no useful workload has yet demonstrated a *net* energy win.

WHAT WOULD RESOLVE IT

A full-system, end-to-end energy measurement — fridge + control + classical co-processor — for a *useful* computation, compared against the best classical method's energy for the same answer done in good faith. A published joules-per-useful-result that undercuts classical would establish a real energy advantage; full-stack accounting showing cryo + decoding overhead swamps the benefit at fault-tolerant scale would deflate it. The theory prediction to test: does the energetic-advantage crossover really precede the computational-advantage crossover on a problem someone cares about?

SOURCES

- arXiv:2605.09580 — "Estimating the Energy Consumption of Quantum Computing from a Full System Aspect" (2026) T3
- arXiv:2601.08068 — "Quantum Energetic Advantage before Computational Advantage in Boson Sampling" (2026) T3
- arXiv:2605.19854 — energetic advantage in superconducting cat-qubit computation (2026) T3
- Google supremacy energy comparison (Summit, ~7 orders of magnitude) T3
- postquantum.com, "The Enormous Energy Cost of Breaking RSA-2048"; Springer, "Energy Efficiency for Quantum Computers" T3/T5

Interpretational Openness — The Measurement Problem, Still Unsolved · O-foundations

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

A century after the formalism was written down (1925 → 2025 centenary), physics still lacks an agreed answer to its central question: what happens during measurement, and what the quantum state *is*. The dual evolution of quantum states — unitary and reversible between measurements, discontinuous and probabilistic at them — remains the field's oldest open problem. Quantum computing sharpens it: every working quantum computer is an experiment in maintaining coherent superposition at ever-larger scales, probing exactly the regime where interpretations differ, and a large-scale fault-tolerant machine would be the most macroscopic superposition ever engineered. The sharp question is whether the problem is *empirically* open (collapse models make testable predictions) or *permanently* open (the surviving no-collapse interpretations agree on all observations).

WHERE THE DISAGREEMENT IS

- **Resolvable / already-resolved camp.** Everettians argue unitarity-all-the-way-down dissolves the problem; David Deutsch famously framed a scalable quantum computer as evidence for many-worlds — "where was the computation performed?" [T3/philosophical]. Objective-collapse theorists (GRW, CSL, Diósi–Penrose) make the strongest claim: collapse is *physical and testable*, and their parameter space is being steadily squeezed — underground experiments (Gran Sasso, using the spontaneous X-ray emission collapse models predict) and space-based data have excluded large regions of Diósi–Penrose and CSL parameter space T2. Decoherence theory rigorously explains the *appearance* of collapse T1, which some read as most of the answer.
- **Stays-open camp.** A 2025 *Nature* centenary survey of 1,100+ physicists found **no consensus** — Copenhagen leading at ~36%, epistemic/QBist views next, then many-worlds, with pilot-wave, spontaneous-collapse, and relational holding steady minorities; respondents split even on whether the question is physics or philosophy [T3/survey]. Recent reviews (arXiv:2502.19278) conclude the measurement problem is *clarified but unsolved* T3. Decoherence explains suppressed interference and leaves the single-outcome ("why this result") question untouched — a point near-universal among foundations specialists T2/T3.

WHAT WOULD RESOLVE IT

An experimental deviation from unitarity — detection of the spontaneous radiation or heating that collapse models predict — would end the debate in favor of objective collapse. Continued null results, plus verified coherence at ever-more-macroscopic scales (large-mass matter-wave interferometry pushing toward 10^6–10^9 amu, and fault-tolerant quantum computers themselves), keep squeezing collapse models toward irrelevance *without* adjudicating between Copenhagen, QBism, and many-worlds. Distinguishing those three may require no experiment at all — which is precisely why the problem may be permanent rather than merely hard. A working million-qubit FTQC machine would be a data point for the no-collapse camps and against any collapse model whose threshold it exceeds.

SOURCES

- arXiv:2502.19278 — "The Quantum Measurement Problem: A Review of Recent Trends" (2025) T3
- Nature centenary survey (2025): physicists disagree on what QM says about reality [T3/survey]
- Diósi–Penrose / CSL experimental constraints — Gran Sasso spontaneous-X-ray tests T2
- CERN Courier, "The measurement problem, measured"; MDPI *Quantum Reports* 7(2):28 T3
- Deutsch, *The Fabric of Reality* (many-worlds / quantum computing argument) [T3/philosophical]

Hype vs Reality — Timelines, Quantum-Washing, Valuations · O-hype

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

The industry's market capitalization runs decades ahead of its revenue. As of May 2026, IonQ, Rigetti, and D-Wave traded at price-to-sales ratios of roughly **109, 836, and 791**; IonQ carried a **$21B+ market cap on ~$187M trailing revenue and a $711M EBITDA loss**. Total industry revenue sits near **$1.4B** (QED-C 2026, computing segment). Every vendor roadmap promises early fault tolerance by 2029–2030 [all T4/roadmap]. The open question is which part of the enthusiasm is early-stage conviction and which part is a bubble. This manual grades every vendor timeline T4/T6 on principle and the valuation debate T5.

WHERE THE DISAGREEMENT IS

- **Substance camp.** IBM, Google, Quantinuum, and Microsoft roadmaps independently converge on 2029–2030 for early fault tolerance T4; real technical milestones keep landing — below-threshold QEC confirmed by four teams (Google Willow, Quantinuum, Harvard/QuEra, USTC) T2, the 96-logical-qubit QuEra record (Jan 2026), Quantum Echoes (Oct 2025). National programs and hyperscalers keep funding. Scott Aaronson (March 2026) said 2025 "met or exceeded my expectations on hardware" and updated toward taking 2028–29 pronouncements more seriously T3. Analysts project tens of billions in value creation by the 2030s [T5, BCG/McKinsey TAM — flagged for optimism].
- **Bubble camp. Sabine Hossenfelder** predicted the hype would crash and argues claimed advantages in logistics and finance dissolve on contact with real-world constraints T5. **Gil Kalai** holds the strongest position: FTQC is impossible because noise correlations kill the threshold theorem — a published minority view; Aaronson (2026) counters that Kalai "keeps saying [the 2019] experiment was fallacious, but meanwhile a dozen other experiments by other companies have reached the same conclusion, so he's fighting a losing battle," adding that Kalai's "path has been getting narrower and narrower" [T3/contested]. Kalai's direct reply (gilkalai.wordpress.com, 10 Mar 2026) concedes he "starts with quantum computation being impossible as [an] axiom" but disputes the interpretation, holding that the observed exponential error decay "does not contradict my conjectures regarding correlated errors" and that his predicted correlations show up at "hundreds… and thousands of gates," not millions [T3/contested]. **Jensen Huang** (Jan 2025): useful machines 15–30 years away; quantum indices fell double digits on the remark, recovering when he walked it back at GTC Quantum Day T5. Insiders at IonQ/Rigetti/D-Wave were net sellers of **~$931M** of stock over five years — a signal analysts flag T5. "Quantum-washing" — rebranding classical or hybrid products as quantum — is tracked as a distinct risk; Quantum Computing Inc.'s Q1 2026 "9,000% revenue surge" to $3.6M came largely from acquisitions and peripheral products T5. The 2026 stress test, per multiple analysts: "Cool story. Where's the revenue?" T5

WHAT WOULD RESOLVE IT

Revenue and reproduction. A commercial customer paying for a quantum computation because it beats the classical alternative on cost would end the debate one way; a funding winter after missed 2029–2030 roadmap dates would end it the other. Kalai's in-principle claim dies the day a large machine sustains fault tolerance; it strengthens with every correlated-noise surprise at scale. Distinguish the *technical* roadmap question (O-scaling) from the *valuation* question — a company can be overpriced while the physics is on track.

SOURCES

- Kalai, "Scott Aaronson's View of my View About Quantum Computing," gilkalai.wordpress.com/2026/03/10/ (Mar 2026); Aaronson, "More on whether useful quantum computing is 'imminent'," scottaaronson.blog/?p=9425 (2026) — the Mar-2026 exchange, both sides [T3/contested]
- Hossenfelder public commentary + UK parliamentary written evidence 121097 T5/T3
- The Motley Fool / Nasdaq / 24-7 Wall St., IonQ-Rigetti-D-Wave valuation + insider-selling coverage (2026) T5
- QED-C, "State of the Global Quantum Industry 2026" (market $1.9B, computing $1.4B → $3B by 2028) T5

Chip-to-Chip Entanglement Fidelity · O-interconnect-loss

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

No single chip will hold the millions of qubits a useful computer needs, so every serious roadmap is **modular**: many smaller processors linked by quantum interconnects. The catch is that gates *across* a link are slower and lower-fidelity than gates on a chip. Long-range links (fridge-to-fridge, via microwave-to-optical transduction and photons) are lossy and probabilistic; even short on-substrate couplers add error. The sharp question is whether inter-module entanglement can be made good enough — high enough fidelity, high enough rate — that error correction still works *across* module boundaries, or whether the interconnect becomes the dominant error source that caps the whole machine (see H-intercon, H-transduce, F-teleport, A-qmemory-hw).

WHERE THE DISAGREEMENT IS

- **Modular-is-inevitable camp.** IBM's 2024 Flamingo demonstrated two Heron R2 chips linked over ~1 m couplers, and the entire roadmap (Cockatoo 2027 establishes module-to-module entanglement via l-couplers; Starling 2028–29 does magic-state injection across modules) is built on chip-to-chip and fridge-to-fridge links. Trapped-ion and neutral-atom platforms shuttle qubits or photons between zones with high fidelity; Quantinuum uses photonic links between QCCD units and reports high-fidelity entanglement across them. Codes tolerant of noisy inter-module links (lattice surgery across boundaries, qLDPC blocks connected by comparatively few long-range checks) exist on paper, and proponents argue distributed/networked QC (see A-qinternet) makes modularity a feature that also enables scaling past a single cryostat T3/T4.
- **Interconnect-caps-the-machine camp.** IBM's own Flamingo link reported roughly **3.5% error** across the coupler in 2024 — about two orders of magnitude worse than the best on-chip two-qubit gates T4. Photon-loss-limited remote-entanglement rates are slow relative to qubit coherence, and if inter-module error correction demands many rounds of noisy entanglement purification, the overhead could swamp the benefit. Microwave-to-optical transduction (the fridge-to-fridge path) still sits at low conversion efficiency with added noise, an unsolved component problem (H-transduce). Nobody has yet shown *fault-tolerant* logical operations spanning modules at scale T3.

WHAT WOULD RESOLVE IT

A logical qubit whose stabilizer measurements *straddle two modules*, kept below threshold across the interconnect, sustained over many rounds — reproduced independently. IBM Cockatoo (2027) demonstrating below-threshold cross-module operation, or a trapped-ion/atom system running a logical gate across a photonic link at on-chip-comparable fidelity, would settle the direction. The negative signal: cross-link error rates that stay stuck near percent-level while on-chip gates keep improving, forcing architectures back toward monolithic chips they can't actually build.

SOURCES

- IBM Quantum roadmap + Flamingo (Heron R2 link demo, 2024; ~3.5% coupler error) — ibm.com/quantum/blog T4
- Monroe et al., "Large-scale modular quantum-computer architecture with atomic memory and photonic interconnects," PRA 89, 022317 (2014) T2
- Benchmarking cavity-mediated interconnects, arXiv:2407.15651 (2024) T3
- Quantinuum QCCD photonic-link entanglement reports (2025–26) T3/T4

The Killer-App Question — What Pays For the Machine · O-killerapp

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Every platform technology needed one application valuable enough to fund the buildout. For quantum computing the candidates are chemistry/materials simulation, optimization, and codebreaking — and each has a problem. Chemistry needs hundreds-to-thousands of error-corrected logical qubits that don't exist yet. Optimization speedups are mostly quadratic and likely to be eaten by error-correction overhead. Shor's algorithm destroys value (breaking RSA) once, after which the world migrates to post-quantum cryptography and that market evaporates. The sharp question: is there any application whose paying demand arrives *before* investor patience runs out? This node is the technical-feasibility twin of O-roi-business (who pays) and shares the chemistry resource numbers with O-crqc-timeline's cousin problem.

WHERE THE DISAGREEMENT IS

- **Chemistry-is-the-app camp.** The Aspuru-Guzik-line argument that "chemistry is quantum computing's killer app" [T2 theory], with Quantinuum and pharma partners (Boehringer, Roche) building toward scalable chemistry-on-quantum workflows T4. Commonly cited commercial timeline: quantum molecular simulation contributing to drug discovery in **5–10 years**, contingent on hundreds of logical qubits T5. A June 2025 perspective maps applications reachable with **25–100 logical qubits** (arXiv:2506.19337), a nearer target than the flagship FeMoco problem T3. Quantum sensing already produces revenue today and de-risks the supply chain T3.
- **Feasibility-skeptic camp.** The canonical FeMoco (nitrogen-fixation cofactor) target keeps needing more, not fewer, resources at the physical level: modern estimates put it at **~1,137 logical qubits and ~3.4×10⁸ logical T-gates**, and "first useful QPE chemistry" is often pegged near **~1,500 logical qubits** — far beyond 2026 hardware T3. Google once estimated ~4M physical qubits, ~4 days for FeMoco T3. 2025–26 feasibility studies find quantum computers face bigger hurdles in quantum chemistry than advertised [T3; New Scientist: "chemistry may not be the killer application"]. Optimization/logistics/finance value is the softest: Hossenfelder's critique that real operations are constrained by everything except raw compute T5, plus Babbush et al.'s "focus beyond quadratic speedups" showing quadratic advantages evaporate under QEC overhead T2/T3. Scott Aaronson's standing honest answer: the killer app may be **simulating quantum physics itself**, a market whose dollar size nobody knows T3.

WHAT WOULD RESOLVE IT

One paying, repeated commercial workload where the quantum result is cheaper or better than the best classical alternative — audited, with the classical baseline done in good faith (density-matrix renormalization, coupled cluster, tensor networks all tried honestly). A pharma company reordering compute budget toward QPU time, or a materials firm shipping a product whose design used a quantum simulation no classical method could do, would settle it. Ten more years of pilot projects that never convert would settle it the other way. The nearest real test: whether the 25–100-logical-qubit chemistry targets produce a *decision-changing* answer once that hardware arrives.

SOURCES

- Aspuru-Guzik et al., "Chemistry is quantum computing's killer app" line — C&EN (2017) T2
- arXiv:2506.19337 — "A Perspective on Quantum Computing Applications...using 25–100 Logical Qubits" (2025) T3
- FeMoco resource estimates (~1,137 logical qubits / 3.4×10⁸ T-gates; Google ~4M physical) T3
- New Scientist / Ground News chemistry-feasibility skepticism (2025–26) T3/T5
- Babbush et al., "Focus beyond quadratic speedups for error-corrected quantum advantage" T2

Decoherence & Materials Defects — The TLS Bottleneck · O-materials

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

The dominant coherence killer in superconducting qubits is a materials problem: **two-level systems (TLS)** — microscopic defects in amorphous oxides, interfaces, and Josephson-junction tunnel barriers — that couple to qubits, drift in frequency, and cause both decoherence and readout failures. Coherence times have climbed from nanoseconds (1999) to the millisecond range, yet TLS make individual qubits *unpredictable*: a qubit fine today can be lossy tomorrow as a defect diffuses onto resonance. The sharp question is whether materials science suppresses TLS density fast enough for million-qubit processors, or whether the defect burden scales with the system so that some fraction of a large chip is always out of spec. This is where the threshold theorem's iid-noise assumption meets a physical process that may not obey it.

WHERE THE DISAGREEMENT IS

- **Suppressible-engineering camp.** Fabrication research is finding handles. 2026 high-throughput junction studies correlate TLS density with Al electrode thickness and grain size, demonstrating a **two-thirds reduction in TLS density** through fabrication changes alone [T3, arXiv:2602.11469]. Site-specific TLS frequency tuning in qubit arrays lets operators steer defects off resonance [T3, arXiv:2503.04702], and ML-accelerated TLS characterization speeds the search [T2/T3, Adv. Quantum Technol. 2026]. The camp's analogy: semiconductor yield problems looked fatal in the 1960s and became a controlled engineering discipline. Tantalum and titanium-nitride films, improved surface treatments, and encapsulation have each bought coherence-time gains.
- **Scaling-wall camp.** TLS-induced dropouts **scale with system size** — on a 1,000+ qubit chip some fraction of qubits is always out of spec, and QEC decoders assume error rates that TLS spectral diffusion silently violates during a run [T3, arXiv:2605.02755]. Correlated error *bursts* — quasiparticle poisoning and cosmic-ray/ionizing-radiation impacts that knock out many qubits at once — add a second materials-linked failure mode that surface codes handle badly T2. Gil Kalai's deeper claim generalizes this: noise in engineered many-body systems is inherently correlated, so the iid assumption underlying the threshold theorem never holds at scale [T3/contested]. Even friendly reviews call TLS "a critical bottleneck for scalable quantum processors" [T3, arXiv:2602.04831].

WHAT WOULD RESOLVE IT

A junction process with TLS density low enough — and stable enough — that a 10,000-qubit chip holds every qubit within QEC spec across weeks of continuous operation, with published defect statistics showing the out-of-spec fraction does *not* grow with chip area. Alternatively, a demonstration that decoders plus real-time TLS-tuning absorb realistic defect drift and cosmic-ray bursts at scale (leakage-aware decoding, gap engineering against quasiparticles). Either result converts the materials bottleneck from open problem to engineering roadmap. The negative signal: persistent out-of-spec fractions that scale with system size, exactly Kalai's predicted regime.

SOURCES

- arXiv:2602.11469 — structural control of TLS density in Josephson junctions (2/3 reduction) T3
- arXiv:2605.02755 — readout failures from junction TLS defects, dropouts scaling with size T3
- arXiv:2503.04702 — scalable site-specific TLS frequency tuning T3
- arXiv:2602.04831 — review of superconducting qubit large-scale integration T3
- Science Advances abc5055 — TLS from trapped quasiparticles; cosmic-ray correlated-error literature T2

Error-Correction Overhead — The Physical:Logical Ratio · O-overhead

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Error correction is a tax: every logical qubit is paid for in physical qubits, and every non-Clifford (T) gate is paid for in magic states. Standard surface codes at useful code distance imply roughly **1,000 physical qubits per logical qubit**, and magic-state distillation factories have historically consumed a large fraction of the total footprint. The sharp question is how far new code families and distillation tricks compress that tax — and, critically, whether the headline low ratios from recent demos survive at *algorithm-relevant code distance and gate depth*. A 4.7:1 ratio at distance 4 on a hundred-qubit demo and a 1,000:1 ratio at distance 25 for a billion-gate algorithm are both true; the atlas must say which regime a number lives in.

WHERE THE DISAGREEMENT IS

- **Optimist camp.** qLDPC codes cut overhead an estimated **~10–20× versus surface codes** T3, and IBM's public Starling accounting claims up to **90% overhead reduction** with its bivariate-bicycle qLDPC codes T4. QuEra demonstrated **96 logical qubits from 448 physical atoms** (~4.7:1) with a high-rate [[16,6,4]] code, running error-corrected gates across all 96 at once, below threshold (Nature, Jan 2026) T2/T3. Quantinuum reported **94 logical qubits at ~2:1** on H-series with logical error below 10^{-4} (March 2026), and Microsoft+Quantinuum logged an **~800× error-rate improvement** (Nature, June 2026) T2/T4. GKP bosonic codes reach effectively 1:1 with the logical qubit outliving its physical parts T2/T3. Gidney's magic-state **cultivation** (2024–25) collapses distillation cost enough to help drive the 20× drop in RSA-2048 estimates; Litinski's line already argued "magic state distillation is cheaper than you think" T3.
- **Skeptic camp.** The flattering ratios come from **low-distance, high-rate codes on small systems** — a [[16,6,4]] code has distance 4, far from the distance ~25 needed to hit 10^{-12} logical error for a billion-gate algorithm. At those target error rates, realistic architectures still project **hundreds-to-1,000:1 all-in** once routing, ancillas, and T-factories are counted T3. qLDPC's long-range connectivity is natural for neutral atoms and ions and remains awkward for fixed-lattice superconducting chips, which is why IBM needs new c-/l-couplers to use it at all T3. Vendor "logical qubit" counts use vendor-chosen definitions and often report memory rather than a full logical *gate set* including T-gates [T4 — grade hard].

WHAT WOULD RESOLVE IT

An end-to-end fault-tolerant algorithm — thousands of logical operations including T-gates, with the full physical budget published — at a measured all-in ratio. If a qLDPC machine sustains **<100:1 at distance ≥ 11** with a working magic-state supply, the overhead pessimists lose. If every deep-circuit demo quietly reverts to ~1,000:1 surface-code accounting once T-gates and routing enter, they win. The tell is whether the low ratios are reported *with* a universal gate set at algorithm-relevant distance, or only for memory/Clifford operation.

SOURCES

- QuEra, 96 logical qubits from 448 atoms, [[16,6,4]] qLDPC, below threshold — Nature (Jan 2026) T2/T3
- Quantinuum 94 logical qubits (Mar 2026); Microsoft+Quantinuum 800× improvement, Nature (Jun 2026) T2/T4
- Gidney, arXiv:2505.15917 (yoked surface codes, magic-state cultivation) T3
- IBM Starling qLDPC overhead accounting — ibm.com/quantum/blog/large-scale-ftqc T4
- postquantum.com qLDPC overview; arXiv:2410.07327 (low-overhead magic states via code switching); arXiv:2606.20263 (Vine Codes, planar qLDPC) T3

The Business-Model & ROI Question — Who Actually Pays · O-roi-business

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

O-killerapp asks whether a quantum computer *can* do something useful. This node asks the separate question: even granting a technical use case, does the *unit economics* work — who pays, at what price, for what margin, on what horizon? A technically real speedup can still be a bad business if the QPU-hour costs more than the classical alternative it beats, if the customer would rather wait for the cloud price to fall, or if the value accrues to a one-time result rather than a recurring service. The sharp question: is there a repeatable commercial workload where a customer chooses quantum on cost/performance grounds *and* the provider earns a margin — before the capital that funds the buildout runs out (O-talent-attrition, O-hype)?

WHERE THE DISAGREEMENT IS

- **Revenue-is-forming camp.** Quantum-as-a-service is monetizing now: cloud access (Amazon Braket pay-as-you-go, Azure Quantum, IBM), hardware-as-a-service, on-prem sovereign installs, and post-quantum-security consulting. D-Wave has a real commercial book — a $20M Florida Atlantic system sale and a ~$10M QCaaS deal with a Fortune 100 buyer in 2026 — and its survey found businesses expecting up to **20× ROI** on quantum-optimization projects, with a claimed **$51.5B** aggregate potential impact [T5, vendor survey — grade hard]. IonQ guided to **$225–245M revenue for 2026** (from ~$130M in 2025). The global quantum-computing market is put near $1.4B (2025), projected to ~$3B by 2028 (QED-C) T5.
- **Economics-don't-close-yet camp.** Most reported "quantum revenue" comes from **acquisitions, government contracts, and peripheral hardware/consulting**, not customers paying for quantum computations that beat classical. Quantum Computing Inc.'s Q1 2026 "9,000% revenue surge" to $3.6M was largely M&A and peripheral products T5. Classical IT investments return in 12–36 months; quantum ROI horizons often exceed a decade, which strains any normal capital-budgeting case T5. Today's machines remain "too fragile for broad, repeatable economic advantage across mainstream enterprise workloads" T5. Annealing/hybrid workflows generate *some* real revenue, but whether they beat a good classical solver on the customer's actual problem is exactly the contested O-advantage question. Extreme valuations (P/S of 100–800+) price in a business that doesn't exist yet.

WHAT WOULD RESOLVE IT

An audited, repeated commercial workload where a customer pays for QPU time *because* it is cheaper or better than the best classical alternative, and the provider books it at a margin — not a pilot, not a grant, not a bundled-services deal. A published cost-per-useful-answer that undercuts classical would settle it positively; a decade of pilots that never convert to recurring paid production, plus valuations compressing toward revenue, would settle it negatively. The dial to watch: the fraction of vendor revenue that is *recurring quantum-compute consumption* versus one-time hardware sales, grants, and consulting.

SOURCES

- QED-C, "State of the Global Quantum Industry 2026" (market $1.9B; computing $1.4B → $3B by 2028) T5
- D-Wave, "Quantum Means Business" ROI survey (up to 20×; $51.5B potential) [T5, vendor]
- IonQ 2026 revenue guidance ($225–245M); D-Wave 2026 commercial deals T5
- Cryptobriefing, "Quantum Computing revenue surges 9,000%...the fine print" (Q1 2026) T5
- The Strategy Institute / entangledfuture.com quantum-ROI analyses T5

Room-Temperature & Photonic Scaling Paths · O-roomtemp

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Superconducting and spin qubits live at millikelvin temperatures inside dilution refrigerators — a cost, wiring, and I/O ceiling that gets brutal at a million qubits. Photonic qubits run gates at room temperature, ship through ordinary fiber, and can be fabbed in semiconductor foundries; NV-diamond systems also operate warm. If either path scales, the economics change category: data-center racks instead of cryostats, networking for free, CMOS-foundry manufacturing. The sharp question is whether photonics' *own* tax — photon loss, probabilistic entangling gates, and the cryogenic single-photon detectors it still needs — comes in cheaper than the dilution-refrigerator tax it removes. "Room temperature" is a claim about *where the hard part moves*, not about eliminating it.

WHERE THE DISAGREEMENT IS

- **Path-changes-the-economics camp. Xanadu's Aurora** (Jan 2025, *Nature*): a modular, networked photonic machine — 4 server racks, 35 photonic chips, 13 km of fiber, 12 physical qubits — presented as scalable in principle to thousands of racks and millions of qubits; Xanadu targets ~1,000 logical qubits by 2029 at ~100:1 [T2 for the demo; T4/T6 for the extrapolation]. **PsiQuantum** partners with GlobalFoundries to fab million-qubit photonic systems, with utility-scale sites announced in Brisbane and Chicago and a multi-hundred-million-dollar raise T4. Fusion-based-QEC papers argue photonic loss thresholds are reachable T3. On the warm-hardware side, Quantum Brilliance's NV-diamond modules run fully cryo-free.
- **Extrapolation-is-huge camp.** "Room temperature" carries an asterisk — the superconducting-nanowire single-photon detectors (SNSPDs) at the heart of every photonic scheme are cryogenic, so the cryoplant shrinks rather than disappears [T3, CACM]. Photon loss plays the role decoherence plays elsewhere, and current per-component losses sit orders of magnitude above fusion thresholds; **Aurora's 12 physical qubits against a million-qubit roadmap is the largest extrapolation ratio in the industry** T3/T6. PsiQuantum has published no full-system demo at any qubit count, making its million-qubit target the field's biggest single T4 claim. NV-diamond is genuinely cryo-free but sits far behind on qubit count and two-qubit gate fidelity T3. Room-temperature *superconductivity* — which would remove the fridge for the transmon path entirely — remains unconfirmed after the retracted LK-99 and Ranga Dias episodes T3.

WHAT WOULD RESOLVE IT

A photonic machine demonstrating error-corrected logical qubits with **published end-to-end loss budgets** — even ~10 logical qubits would validate the architecture and move it from T4 to T2. For the economics claim: a cost-per-logical-qubit-hour comparison against a superconducting system once both exist at comparable capability. PsiQuantum's first full-stack public benchmark is the single most information-rich pending event on this node; Aurora scaling from 12 to hundreds of physical qubits with loss held constant would be the second.

SOURCES

- Xanadu Aurora — *Nature* s41586-024-08406-9 (2025) T2; Xanadu roadmap T4
- PsiQuantum / GlobalFoundries partnership + Brisbane/Chicago site announcements T4
- CACM, "Making Qubits with Photons" (SNSPD-cryogenics caveat) T3
- Springer J. Supercomputing photonic FTQC review (2025); Tom's Hardware photonics roadmap T3/T5

Fault-Tolerant Scaling — How Far Off, Really · O-scaling

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

A cryptographically or chemically useful quantum computer needs thousands of logical qubits, which under current architectures means hundreds of thousands to millions of physical qubits running billions of gates. As of mid-2026 the largest devices sit at roughly 1,000–6,000 physical qubits; the best *verified* logical-qubit demonstrations count 96 (QuEra, Nature Jan 2026). The distance between "dozens of logical qubits, a few thousand operations" and "thousands of logical qubits, a billion gates" is the whole game. The estimates on the table differ by an order of magnitude in both qubit count and calendar time, and the atlas grades every roadmap date T4/T6 on principle.

WHERE THE DISAGREEMENT IS

- **Optimist camp.** IBM published a modular, milestone-by-milestone path in June 2025: **Loon** (2025, qLDPC architectural elements + c-couplers linking distant on-chip qubits), **Kookaburra** (2026, the first fault-tolerant module integrating logic + memory), **Cockatoo** (2027, entanglement between modules via l-couplers), and **Starling** (2028–29): ~200 logical qubits, 100 million gates, "large-scale fault tolerance." **Blue Jay** (2033) targets 2,000+ logical qubits and a billion gates. IBM's move to qLDPC claims up to a **90% reduction in physical-qubit overhead** versus surface codes [all T4/roadmap]. Google targets a useful error-corrected machine on a similar 2029–2030 horizon T4. Craig Gidney (Google, arXiv:2505.15917, 2025) cut the RSA-2048 estimate ~20×, from ~20M physical qubits (2019) to **<1M noisy qubits at ~1 week runtime** T3. Scott Aaronson (March 2026) called 2025 a year that "met or exceeded my expectations on hardware, with multiple platforms now boasting >99.9% fidelity two-qubit gates," and said he had updated "in favor of taking more seriously the aggressive pronouncements...about where they could be in 2028 or 2029" [T3, expert commentary].
- **Skeptic camp.** Jensen Huang (Nvidia, Jan 2025) put "very useful" quantum computers **15–30 years out** (~20 central), a remark that cratered quantum stocks before he softened it at GTC's Quantum Day T5. Gil Kalai argues fault-tolerant scaling is impossible in principle: correlated noise defeats the threshold theorem before you reach a million qubits, and he holds the correlations are observable in straightforward simulations of the noisy circuits that build highly entangled states [T3/contested]. The field has demonstrated well under 1% of the physical-qubit count Gidney's optimized RSA estimate needs, at anything like the required fidelity — closing five orders of magnitude in qubit count while *holding* fidelity is a claim, not a datum. Aaronson keeps a "little voice...that asks 'but what if Gil Kalai turned out to be right'," while adding that in 2026 it "can barely be heard," and that Kalai's "path has been getting narrower and narrower" T3. Kalai's March-2026 reply (gilkalai.wordpress.com, 10 Mar 2026) grants that he "starts with quantum computation being impossible as [an] axiom" but insists the measured exponential error decay "does not contradict my conjectures regarding correlated errors," which he expects to surface at "hundreds… and thousands of gates" [T3/contested].

WHAT WOULD RESOLVE IT

A single machine running hundreds of logical qubits through millions of error-corrected gates on a problem with a checkable answer — IBM Kookaburra (2026) and then Starling (2028–29) hitting spec, verified independently, would settle the direction. Each vendor deadline that holds or slips recalibrates the distribution. Kalai's in-principle claim dies the day any large machine sustains fault tolerance; it strengthens with every correlated-noise surprise at scale. Watch: does fidelity hold as qubit count grows an order of magnitude, or degrade?

SOURCES

- IBM, "IBM lays out clear path to fault-tolerant quantum computing" — ibm.com/quantum/blog/large-scale-ftqc (Jun 2025) T4
- Gidney, "How to factor 2048 bit RSA integers with less than a million noisy qubits," arXiv:2505.15917 (2025) T3
- Aaronson, "More on whether useful quantum computing is 'imminent'," scottaaronson.blog/?p=9425 (2026) [T3, expert]; Kalai reply, gilkalai.wordpress.com/2026/03/10/ (Mar 2026) [T3, contested]
- Google Quantum AI roadmap — quantumai.google/roadmap T4
- IEEE Spectrum, "IBM's Starling: Quantum Computing's Future"; DataCenterDynamics roadmap coverage (2025) T4/T5

Funding-Winter Risk & Talent Attrition · O-talent-attrition

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Fault-tolerant timelines run to 2029–2033 and beyond (O-scaling), but venture funding, public patience, and skilled people operate on much shorter cycles. The open risk: a "quantum winter" — a stretch of disappointing milestones and cheap capital drying up — could drain talent and shutter startups *before* the long roadmaps mature, the way the AI winters of the 1970s and late 1980s set that field back years. Quantum is unusually exposed: capital-intensive, pre-revenue, and its "advantage" claims keep getting matched classically (O-advantage, O-hype). Human capital is thin and slow to grow — quantum PhDs take years, and a few big players absorb most of the trained pool, so a shock disperses expertise fast and refills it slowly.

WHERE THE DISAGREEMENT IS

- **Winter-is-a-real-risk camp.** Q1 2026 quantum-startup funding fell to **~$240M, down 51% quarter-over-quarter and ~68% year-over-year** (Crunchbase), and funding concentrates hard — in 2024 two firms (PsiQuantum, Quantinuum) took ~half of all VC; in Q3 2025 three firms took >80%. That concentration means a downturn strands the long tail of startups and disperses their teams into AI and quant finance, where they rarely return. A widely cited workforce gap — roughly one qualified candidate per three open quantum roles, and ~250k professionals needed by 2030 (McKinsey) — means attrition can't be easily backfilled. Insider stock sales (~$931M across IonQ/Rigetti/D-Wave over five years) suggest even operators are hedging T5.
- **No-winter camp.** Government programs (US NQI, EU Flagship, UK, China, others — E-us/E-eu/E-china/E-others) provide counter-cyclical funding that private downturns don't touch, and public-market quantum names held up into 2026 even as private rounds cooled. Aggregate 2025 equity funding still roughly doubled year-over-year (~$3.8B by September). The real technical milestones landing in 2025–26 (four teams below threshold, QuEra's 96 logical qubits, Quantum Echoes) argue the field is *earning* its capital, not just burning it. Optimists hold that a base of committed national money plus a handful of well-capitalized leaders carries the field through any private-capital lull T5.

WHAT WOULD RESOLVE IT

A 2–3 year stretch that goes one of two ways: (a) a genuinely useful, revenue-generating application appears (killing winter risk — O-killerapp/O-roi-business), or (b) sustained funding collapse plus visible startup closures and a net outflow of researchers into other fields. The observable dials are quarterly VC totals, headcount at the pure-plays, PhD-pipeline enrollment, and whether national programs *sustain* commitments through a private downturn or trim them in sympathy. The 2026 funding drop is the first real stress test; whether it is a blip or a trend is the live question.

SOURCES

- Crunchbase, "Quantum Computing Startup Investment Slows in 2026" (Q1 2026: ~$240M, −51% QoQ) T5
- McKinsey, "Quantum Technology Monitor" / talent-gap analyses (2024–2025, ~250k needed by 2030) T5
- The Quantum Insider, funding-concentration data (2024–2025) T5
- The Motley Fool / Nasdaq, IonQ-Rigetti-D-Wave insider-selling coverage (2026) T5

The "Quantum Utility" Definitional Debate · O-utility-definition

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Words do heavy lifting in this field, and "utility," "advantage," and "supremacy" are not interchangeable — yet marketing blurs them constantly. IBM's June 2023 *Nature* paper, "Evidence for the utility of quantum computing before fault tolerance," coined **quantum utility**: a noisy 127-qubit processor producing *reliable* results at a scale beyond brute-force classical simulation, using error mitigation. The claim was carefully *not* "advantage" (beating all classical methods) — utility meant "useful for scientific exploration, competitive with existing classical approximations." The sharp question: is "utility" a meaningful, defensible category, or a rhetorical hedge that lets vendors claim a win while conceding the harder "advantage" bar? Getting the vocabulary honest is prerequisite to grading everything else in this chapter.

WHERE THE DISAGREEMENT IS

- **Utility-is-a-real-milestone camp.** IBM and allies argue "utility" names a genuine transition: quantum processors now operate at a scale where they can *contribute to research*, run circuits classical brute force can't, and serve as a scientific tool even without asymptotic advantage. The 2023 kicked-Ising experiment on Eagle was, at minimum, a demonstration that error-mitigated NISQ machines can produce trustworthy expectation values at 127 qubits. IBM later reframed toward "the dawn of quantum advantage" as hardware improved T4. Proponents note the honest hedge is a *feature* — the field distinguishing "useful now" from "provably beyond classical later" is exactly the neutrality the atlas asks for.
- **Utility-is-a-goalpost-move camp.** The 2023 utility claim was **classically simulated within days** — 2D tensor networks (gPEPS, Tindall et al.) and sparse Pauli-dynamics methods (Sandia, Caltech) reproduced or beat the quantum results on laptops-to-clusters, showing the specific problem was *not* beyond classical reach T2. Critics argue "utility before fault tolerance" became a movable goalpost: when "supremacy" claims got matched, the bar quietly relaxed to "utility," a term with no agreed operational definition and no independent adjudicator (see O-benchmark-standard). If a result is reproduced classically within a week, calling it "utility" risks quantum-washing (O-hype). The word invites exactly the ambiguity that lets a T4 press release read like a T2 result.

WHAT WOULD RESOLVE IT

A precise, community-adopted operational definition — a taxonomy that fixes "utility" (useful/competitive, may be classically reproducible), "advantage" (beats the best classical method on a task people want), and "supremacy" (beats all classical methods on *any* task, useful or not), with independent adjudication of which bar a given demo clears. Failing a standard, the atlas applies its own: "utility" claims are T4 until an independent party confirms the classical baseline was strong and the result competitive; a claim that is classically reproduced within weeks is graded a *classical-simulation-frontier* data point (O-classical-sim), not an advantage.

SOURCES

- IBM et al., "Evidence for the utility of quantum computing before fault tolerance," *Nature* 618, 500 (2023) [T2 paper / T4 framing]
- Tindall et al., "Efficient tensor network simulation of IBM's Eagle kicked Ising experiment," PRX Quantum 5, 010308 (2024) T2
- IBM, "The dawn of quantum advantage" — ibm.com/quantum/blog/quantum-advantage-era T4
- arXiv:2306.14887 (Eagle kicked-Ising tensor network); sparse Pauli-dynamics classical rebuttals (Sandia/Caltech) T2/T3

Verifying a Quantum Computation Too Large to Check · O-verification

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Once a quantum computer solves a problem no classical machine can redo, how do you know the answer is *right*? For factoring you multiply the factors back — an easy classical check. For quantum simulation, sampling, or optimization there may be no efficient classical way to verify the output, and the whole point is that you can't re-run it classically. This is the verification problem: can a *limited* verifier (classical, or a small trusted quantum device) certify the result of an untrusted, powerful quantum computer? It matters commercially — why pay a cloud QPU if you can't trust its output (S-cloud) — and scientifically: an unverifiable "advantage" is a weak claim (O-advantage, S-bench). Quantum Echoes' selling point is precisely that it is *cross-checkable on another quantum device*, a weaker but real form of verification.

WHERE THE DISAGREEMENT IS

- **Solvable-in-principle camp.** Urmila Mahadev (2018, FOCS Best Paper) proved a *classical* verifier can verify an arbitrary quantum computation using a single quantum prover, assuming the device can't break post-quantum cryptography (Learning With Errors) T2. This is a landmark: interactive classical verification of all of BQP under a standard computational assumption. Related "tests of quantumness" and certified-randomness protocols (Brakerski–Christiano–Mahadev–Vazirani–Vidick) have run on real hardware — Quantinuum's 56-qubit H2 generated *certified* random bits verified classically (Nature 2025, arXiv:2503.20498), the first delivered verification-flavored application. Blind/verifiable observable-estimation protocols advanced in 2025–26 (arXiv:2510.08548). Self-testing, cross-entropy benchmarking, and classical spot-checks give partial verification today T2/T3.
- **Impractical-where-it-matters camp.** Mahadev's protocol is far from usable — it needs a fault-tolerant device well beyond today's, deep circuits, and many interactive rounds. Cross-entropy benchmarking (the metric behind supremacy claims) is *not a proof* and has itself been spoofed by classical spoofing algorithms. For the near-term applications people actually want — chemistry ground-state energies — there is no efficient rigorous verifier; you fall back to trusting variational convergence heuristics or loose classical bounds [T3/contested]. Certified randomness verifies *randomness*, not a chosen useful computation, so the gap between "we can certify random bits" and "we can certify a drug-discovery answer" is wide open.

WHAT WOULD RESOLVE IT

A demonstrated, resource-realistic protocol that verifies a *beyond-classical, useful* computation on a real device with a classical (or exponentially weaker) verifier — going past certified randomness to a chosen task. Failing full Mahadev-style verification, a widely accepted intermediate (device-independent self-testing at useful scale, or robust quantum-checks-quantum cross-validation of an advantage claim) would move the node forward. The near-term test: whether Quantum Echoes-style quantum-verifies-quantum cross-checks become a trusted standard, or whether the community holds that only classical-verifier protocols count.

SOURCES

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- Brakerski, Christiano, Mahadev, Vazirani, Vidick, "Certifiable Randomness from a Single Quantum Device," JACM (2021) T2
- Certified randomness on Quantinuum H2, Nature 645 (2025), arXiv:2503.20498 T2
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- Gheorghiu, Kapourniotis, Kashefi, "Verification of quantum computation: an overview," Theory Comput. Syst. (2019) T3

APPENDIX H

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